

Ubuntu Server

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Ubuntu Server is a version of the Ubuntu operating system designed and engineered as a backbone for the internet.

Ubuntu Server brings economic and technical scalability to your data center, public or private. Whether you want to deploy an OpenStack cloud, a Kubernetes cluster or a 50,000-node render farm, Ubuntu Server delivers the best value scale-out performance available.

This documentation always targets the latest LTS version of Ubuntu. When there are differences towards older LTS releases, they are noted in the guide.

1. In this documentation

1.1. Getting started

The tutorial walks you through a fresh installation and the first steps of operating an Ubuntu Server.

- **Installation:** [System requirements](#) (page 850) • [Basic installation](#) (page 8) • [Ubuntu Pro subscription](#) (page 73)
- **System basics:** [Managing software](#) (page 47) • [Customizing package files](#) (page 982) • [Third party repository usage](#) (page 976)
- **The Ubuntu command line:** [Welcome to the terminal](#) (page 12) • [The command line in depth](#) (page 32) • [Command-line cheat sheet](#) (page 843)

1.2. Networking

Configure network interfaces, provide name and address services, share resources, and synchronize time.

- **Understanding networks:** [Key concepts](#) (page 939) • [About Netplan](#) (page 943) • [Time synchronization](#) (page 963)
- **Configuration:** [Using Netplan](#) (page 943) • [Domain Name System \(DNS\)](#) (page 306) • [Dynamic Host Configuration Protocol \(DHCP\)](#) (page 338)
- **Securing your network:** [Install OpenSSH](#) (page 235) • [Firewalls](#) (page 163) • [DNS Security Extensions \(DNSSEC\)](#) (page 874)
- **High throughput:** [Open vSwitch with DPDK](#) (page 329) • [About DPDK](#) (page 955)
- **Sharing:** [Files with NFS](#) (page 357) • [Printers with CUPS](#) (page 419)
- **With Windows systems:** [Samba](#) (page 366) • [Windows Active Directory](#) (page 402)

1.3. Security

Protect your system with access controls, firewalls, encrypted communications, and network authentication. For more general information about security in Ubuntu, refer also to the [Ubuntu Security documentation](#)¹.

- **Understanding security:** [Introduction](#) (page 867) • [Security suggestions](#) (page 868)
- **Basic security:** [User management](#) (page 154) • [AppArmor](#) (page 170)
- **Authentication:** [Install Kerberos](#) (page 183) • [Set up SSSD with LDAP](#) (page 214) • [Smart cards](#) (page 223)
- **Cryptography:** [Cryptographic libraries](#) (page 884) • [About certificates](#) (page 927) • [OpenSSL](#) (page 890)
- **VPNs:** [OpenVPN](#) (page 261) • [WireGuard VPN](#) (page 280)

¹ <https://documentation.ubuntu.com/security/>

1.4. Managing your system

Install, update, upgrade, and troubleshoot the software packages running on your system.

- **Managing software:** [Package management](#) (page 423) • [etckeeper](#) (page 565) • [Get older package versions using snapshots](#) (page 446)
- **Update considerations:** [Automatic updates](#) (page 430) • [Phased updates](#) (page 990) • [Testing updates in advance](#) (page 993)
- **Upgrade Ubuntu version:** [Release upgrades](#) (page 443)
- **Troubleshooting:** [Report a bug](#) (page 453) • [Debugging with debuginfod](#) (page 770) • [Kernel crash dump](#) (page 776)

1.5. Data and storage

Store and manage data using directory services, databases, logical volumes, and backup tools.

- **Databases:** [MySQL](#) (page 506) • [PostgreSQL](#) (page 513)
- **Directory services:** [Introduction to OpenLDAP](#) (page 996) • [Install OpenLDAP](#) (page 457) • [OpenLDAP replication](#) (page 472)
- **Device mapper multipathing:** [Intro to multipathing](#) (page 1000) • [Multipath configuration options and overview](#) (page 1005)
- **Storage:** [About LVM](#) (page 998) • [Manage logical volumes](#) (page 518) • [Sharing data storage with iSCSI](#) (page 521)
- **Version control:** [gitolite](#) (page 567)

1.6. Web and mail services

Serve web content, proxy traffic, write server-side applications, and handle email.

- **Web servers:** [About web servers](#) (page 1028) • [Apache2](#) (page 597) • [nginx](#) (page 606)
- **Squid proxy server:** [About Squid proxy](#) (page 1029) • [Install a Squid proxy server](#) (page 592)
- **Mail services:** [Postfix](#) (page 572) • [Exim4](#) (page 580) • [Dovecot](#) (page 589)

1.7. Virtualisation and containers

Run workloads in virtual machines and containers using lightweight and full-virtualisation tooling.

- **Understand virtualization:** [Intro to virtualisation](#) (page 1031)
- **Virtual machines:** [Multipass](#) (page 641) • [QEMU](#) (page 649) • [QEMU microvm](#) (page 1034)
- **VM tooling:** [libvirt and virsh](#) (page 668) • [virt-manager](#) (page 683) • [Nested virtualization](#) (page 690)
- **Containers:** [Docker for system admins](#) (page 697) • [Docker storage, networking, and logging](#) (page 1050) • [Run rocks on your server](#) (page 713)

- **Deployment:** [Find cloud images](#) (page 1059) • [About cloud-init](#) (page 1057) • [LXD](#) (page 633)
- **Confidential computing:** [AMD SEV](#) (page 655) • [Intel TDX](#) (page 657)

1.8. Performance

Set up GPU resources and optimize system performance.

- **GPU resources:** [Install Nvidia drivers](#) (page 619) • [GPU virtualization](#) (page 626)
- **Performance:** • [CPU P-states](#) (page 1090) • [hwloc](#) (page 1085)
- **Optimization:** [CPU power management](#) (page 1094) • [TuneD](#) (page 1100) • [Profile-Guided Optimization](#) (page 1079)
- **High availability:** [Introduction to High Availability](#) (page 1060) • [Set up DRBD](#) (page 727)
 - [Migrate from crmsh to pcs](#) (page 851)

2. How this documentation is organized

This documentation uses the [Diátaxis documentation structure](https://diataxis.fr/)².

- The [Tutorial](#) (page 7) takes you step-by-step through installing and setting up your first Ubuntu Server system.
- [How-to guides](#) (page 75) assume basic familiarity with Ubuntu Server and walk you through specific tasks.
- [Reference](#) (page 792) provides system requirements, a glossary, and other technical specifications.
- [Explanation](#) (page 866) includes topic overviews, background and context, and detailed discussion of key concepts.

² <https://diataxis.fr/>

3. Project and community

Ubuntu Server is a member of the Ubuntu family. It's an open source project that welcomes community projects, contributions, suggestions, fixes and constructive feedback.

- [Read our Code of Conduct³](#)
- [Get community support⁴](#)
- [Join the Discourse forum⁵](#)
- [Chat to us on Matrix⁶](#)
- [Contribute to this documentation](#) (page 1114)

If you find any errors or have suggestions for improvements, please use the “Give feedback” button at the top of every page. This will take you to GitHub where you can share your comments or let us know about bugs with any page.

Releases

[Download Ubuntu Server⁷](#)

[Ubuntu Release Notes⁸](#)

PDF versions of this documentation

[Current PDF⁹](#) (for Ubuntu 20.04 LTS onward)

[Ubuntu 18.04 LTS PDF¹⁰](#) (and earlier)

Commercial support

Thinking about using Ubuntu Server for your next project? [Get in touch!¹¹](#)

3.1. Ubuntu Server tutorial

This tutorial should be a good place to start learning about Ubuntu Server in general, how it works, and what it's capable of.

In our “Getting started” tutorial you will learn how to set up an Ubuntu Server; from installing using a bootable USB device, to navigating the Server installer menu.

³ <https://ubuntu.com/community/docs/ethos/code-of-conduct>

⁴ <https://ubuntu.com/support/community-support>

⁵ <https://discourse.ubuntu.com/c/project/server/17>

⁶ <https://matrix.to/#/#server:ubuntu.com>

⁷ <https://ubuntu.com/download/server>

⁸ <https://documentation.ubuntu.com/release-notes/>

⁹ https://ubuntu.com/server/docs/_/downloads/en/latest/pdf/

¹⁰ <https://assets.ubuntu.com/v1/8f8ea0cf-18-04-serverguide.pdf>

¹¹ <https://ubuntu.com/server/contact-us?product=server>

3.1.1. Getting started

- [Basic installation](#) (page 8)

The Server installer

The Ubuntu installer has its own documentation. For more guidance on installing Ubuntu Server with the installer, you can also see these guides in the Ubuntu installer documentation (note: these pages will redirect you).

- [Operating the Server installer](#)¹²
- [Screen-by-screen installer guide](#)¹³
- [Configuring storage](#)¹⁴
- [Troubleshooting the installer](#)¹⁵

3.1.2. Next steps

Once your Ubuntu Server is up and running, you may be interested in this collection of related tutorials and topics that will help you learn more about how it works and what's available to you. These tutorials can be viewed in any order.

- [Welcome to the terminal](#) (page 12)
- [The command line in depth](#) (page 32)
- [Managing software in Ubuntu Server](#) (page 47)
- [Attach your Ubuntu Pro subscription](#) (page 73)

If you have a specific goal, and are already familiar with Ubuntu Server, take a look at our [How-to guides](#) (page 75). These have more in-depth detail and can be applied to a broader set of applications.

Take a look at our [Reference section](#) (page 792) when you need to determine what commands are available, and how to interact with various tools.

Finally, for a better understanding of how Ubuntu Server works and how it can be used and configured, our [Explanation section](#) (page 866) enables you to expand your knowledge of the operating system and additional software.

Basic installation

This chapter provides an overview of how to install Ubuntu Server Edition. You can also refer to this guide on [how to operate the installer](#)¹⁶ for more information on using the installer, and to this [screen-by-screen reference guide](#)¹⁷ for more information about each of the installer screens.

¹² <https://canonical-subiquity.readthedocs-hosted.com/en/latest/tutorial/operate-server-installer.html>

¹³ <https://canonical-subiquity.readthedocs-hosted.com/en/latest/tutorial/screen-by-screen.html>

¹⁴ <https://canonical-subiquity.readthedocs-hosted.com/en/latest/howto/configure-storage.html>

¹⁵ <https://canonical-subiquity.readthedocs-hosted.com/en/latest/howto/report-bugs.html>

¹⁶ <https://canonical-subiquity.readthedocs-hosted.com/en/latest/tutorial/operate-server-installer.html>

¹⁷ <https://canonical-subiquity.readthedocs-hosted.com/en/latest/tutorial/screen-by-screen.html>

Preparing to install

This section explains various aspects to consider before starting the installation.

System requirements

Ubuntu Server Edition provides a common, minimalist base for a variety of server applications, such as file/print services, web hosting, email hosting, etc. This version supports four 64-bit architectures:

- amd64 (Intel/*AMD* 64-bit)
- arm64 (64-bit ARM)
- ppc64el (POWER8 and POWER9)
- s390x (IBM Z and LinuxONE)

The recommended minimal system requirements for this tutorial are:

- RAM: 2 *GB* or more
- Disk: 5 GB or more

If you are looking for more general system requirements, *refer to this page* (page 850).

Perform a system backup

Before installing Ubuntu Server Edition you should make sure all data on the system is backed up.

If this is not the first time an operating system has been installed on your computer, it is likely you will need to re-partition your disk to make room for Ubuntu.

Any time you partition your disk, you should be prepared to lose all data on the disk should you make a mistake or something goes wrong during partitioning. The programs used in installation are quite reliable, most have seen years of use, but they also perform destructive actions.

Download the server ISO

You can obtain the amd64 server download from <https://releases.ubuntu.com/>. Select the version you wish to install and select the “server install image” download. Note that the server download includes the installer.

There are platform specific how-to guides for installations on:

- *s390x LPAR* (page 124)
- *z/VM* (page 99)
- *ppc64el* (page 92)

Create a bootable USB

There are many ways to boot the installer but the simplest and most common way is to [create a bootable USB stick](#)¹⁸ to boot from ([tutorials for other operating systems](#)¹⁹ are also available).

Boot the installer

Plug the USB stick into the system to be installed and start it.

Most computers will automatically boot from USB or DVD, though in some cases manufacturers disable this feature to improve boot times. If you don't see the boot message and the "Welcome" screen which should appear after it, you will need to set your computer to boot from the install media.

There should be an on-screen message when the computer starts telling you what key to press for settings or a boot menu. Depending on the manufacturer, this could be Escape, F2, F10 or F12. Simply restart your computer and hold down this key until the boot menu appears, then select the drive with the Ubuntu install media.

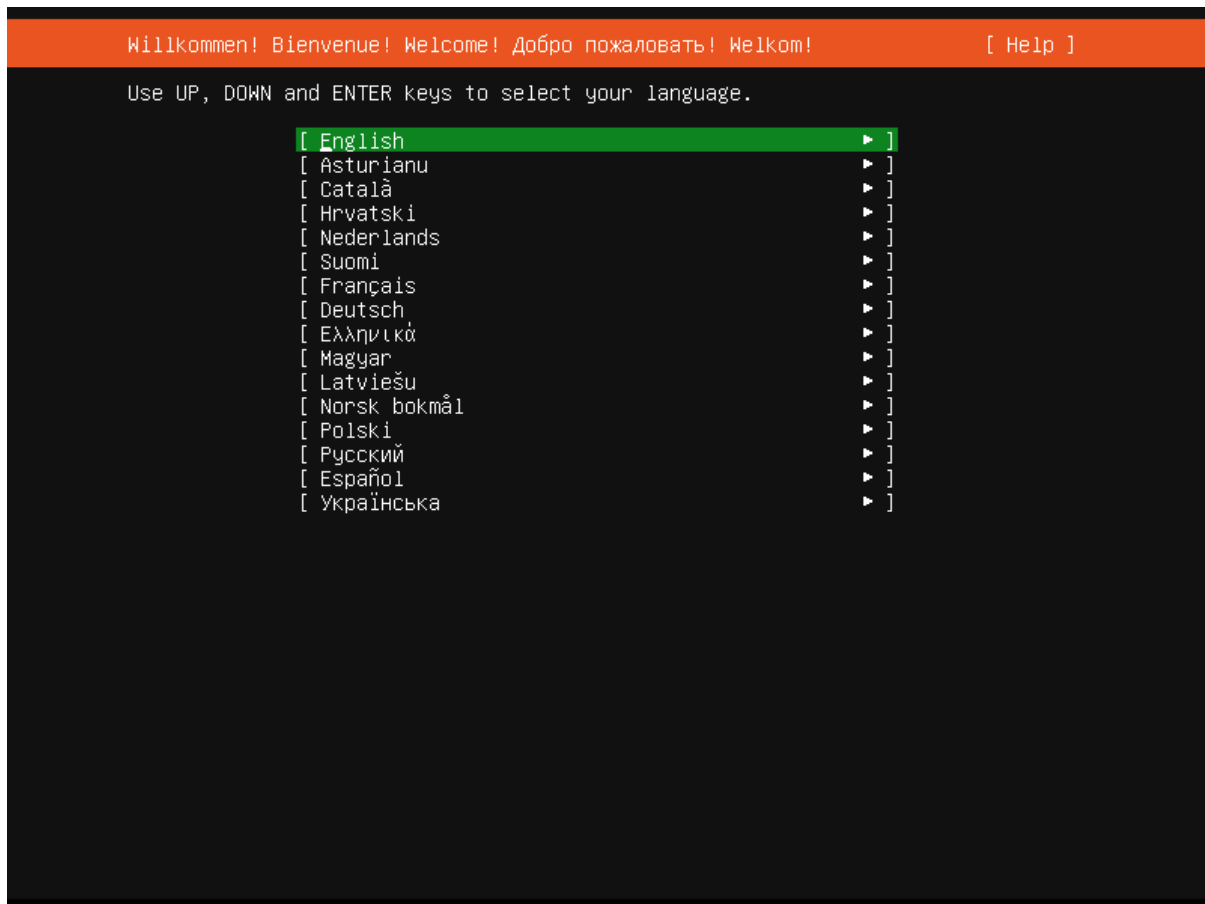
If you are still having problems, check out the [Ubuntu Community documentation on booting from CD/DVD](#)²⁰.

After a few moments, the installer will start in its language selection screen.

¹⁸ <https://documentation.ubuntu.com/desktop/en/latest/how-to/create-a-bootable-usb-stick/>

¹⁹ <https://ubuntu.com/search?q=%22create+a+bootable+USB+stick%22>

²⁰ <https://help.ubuntu.com/community/BootFromCD>



Using the installer

The installer is designed to be easy to use and have sensible defaults so for a first install you can mostly just accept the defaults for the most straightforward install:

- Choose your language
- Update the installer (if offered)
- Select your keyboard layout
- Do not configure networking (the installer attempts to configure wired network interfaces via [DHCP](#), but you can continue without networking if this fails)
- Do not configure a proxy or custom mirror unless you have to in your network
- For storage, leave “use an entire disk” checked, and choose a disk to install to, then select “Done” on the configuration screen and confirm the install
- Enter a username, hostname and password
- On the SSH and snap screens, select “Done”
- You will now see log messages as the install is completed
- Select restart when this is complete, and log in using the username and password provided

If you need more details, however, the Ubuntu installer has [its own documentation](#)²¹, including a [screen-by-screen guide](#)²².

Welcome to the terminal

On Ubuntu Server, the terminal is the primary way of interacting with your system. It's easier to use than you might think, and learning it gives you much more control over your system (including on your Ubuntu Desktop). In this tutorial, you will learn how to use the terminal to navigate your Ubuntu system, find and work with files, and where to find help if you get stuck.

Set up the virtual machine

To avoid making changes to your computer we will set up a virtual machine (VM), which provides us with a safe environment to run the commands in. [Multipass](#)²³ is great for quickly creating Ubuntu virtual machines, so we will use that. It is available on Windows, Mac and Linux, so you don't need to be already running Ubuntu on your computer to follow this tutorial.

Prerequisites

- The Multipass VM needs **5 GB of disk space**, and **1 GB of memory** available.

Install Multipass

On Ubuntu

You can install Multipass directly from [the Multipass page](#)²⁴ in the online snap store (make sure to select the "latest/stable" version from the dropdown menu next to the install button).

On other operating systems

If you're on Windows, Mac, or another Linux operating system, Multipass can be installed [using these instructions](#)²⁵.

²¹ <https://canonical-subiquity.readthedocs-hosted.com/en/latest/index.html>

²² <https://canonical-subiquity.readthedocs-hosted.com/en/latest/tutorial/screen-by-screen.html>

²³ <https://documentation.ubuntu.com/multipass/latest/>

²⁴ <https://snapcraft.io/multipass>

²⁵ <https://documentation.ubuntu.com/multipass/stable/how-to-guides/install-multipass/>

Open a terminal window

Now that Multipass is installed, we need a terminal window to run our commands in. How you open one depends on your operating system.

On Ubuntu

Press `Ctrl + Alt + T` together and a terminal window will open. You can also search for “Terminal” in your application launcher.

On Windows

PowerShell is the most reliable option for working with Multipass on Windows.

1. Press the Windows key.
2. Type PowerShell.
3. Right-click it and select **Run as Administrator** — this helps avoid permission issues when managing virtual machines.

If you have Windows Terminal installed from the Microsoft Store, that works just as well and offers a nicer experience with tab support.

To confirm everything is working, type `multipass version` and press Enter. If you see version numbers, you’re ready to continue.

On macOS

1. Press `⌘ + Space` to open Spotlight.
2. Type Terminal and press Enter.

If you use iTerm2, that works perfectly well too.

To confirm everything is working, type `multipass version` and press Enter. If you see version numbers, you’re ready to continue.

Create the virtual machine

Once you have installed Multipass and opened your terminal, we can launch a new VM using the Ubuntu 26.04 LTS release (codename “resolute”), and let’s give our VM the name `tutorial` using the following command in our terminal window:

```
$ multipass launch resolute --name tutorial
```

It may take a little time, depending on the speed of your internet connection. Multipass downloads the most recent daily **image** and creates the VM for us.

Note:

An Ubuntu **image** is a collection of files we need to install and run Ubuntu. We don't need to specify "server" or "desktop" anywhere in our command, because the image is the same for both. The only difference between Ubuntu Server and Ubuntu Desktop is the subset of software packages we use from the Ubuntu Archive.

Once the above command has finished running, we can access the VM by running:

```
$ multipass shell tutorial
```

We will get a "Welcome to Ubuntu" message. Notice that when we run this command, the terminal username changes to ubuntu and the *hostname* changes to tutorial:

```
ubuntu@tutorial:~ $
```

This shows that we are inside the VM, and this is where we will run all our commands. Now that our setup is complete, let's carry on with the tutorial!

Understanding the terminal

In our new terminal window, we see a **shell prompt** (sometimes called a command prompt), which is constructed as your-username@your-hostname and followed by a dollar sign (\$). This is where we input commands to the computer.

```
your-name@your-machine-name:~$ Input: The commands you type
```

```
Output: The results after running the command
```

The output – the results of running your commands – is shown separately, underneath the shell prompt.

Tip:

You can use your mouse to copy text *into* the terminal by holding the *left mouse button* and dragging your mouse to highlight the text you want to copy. Then, release the left button.

Move your mouse pointer to the terminal window and press the *middle mouse button* (or click the *right mouse button* and select *paste*).

The **Command Line Interface (CLI)** is the language we use in our terminal to interact with our system. The CLI gives you far more control over your system than the **Graphical User Interface (GUI)** that you may be more familiar with.

In software documentation, you will often see instructions to "run" or "execute" a command (or a set of commands). What this means is: type the command into your terminal at the **shell prompt** and press Enter on your keyboard. We did this already when we ran the multipass launch command to create our VM.

The command is sent to the **shell**, which interprets the command and sends it to the operating system, which runs the command. It took a little time for the tutorial instance to be

created, but eventually the shell showed output like this:

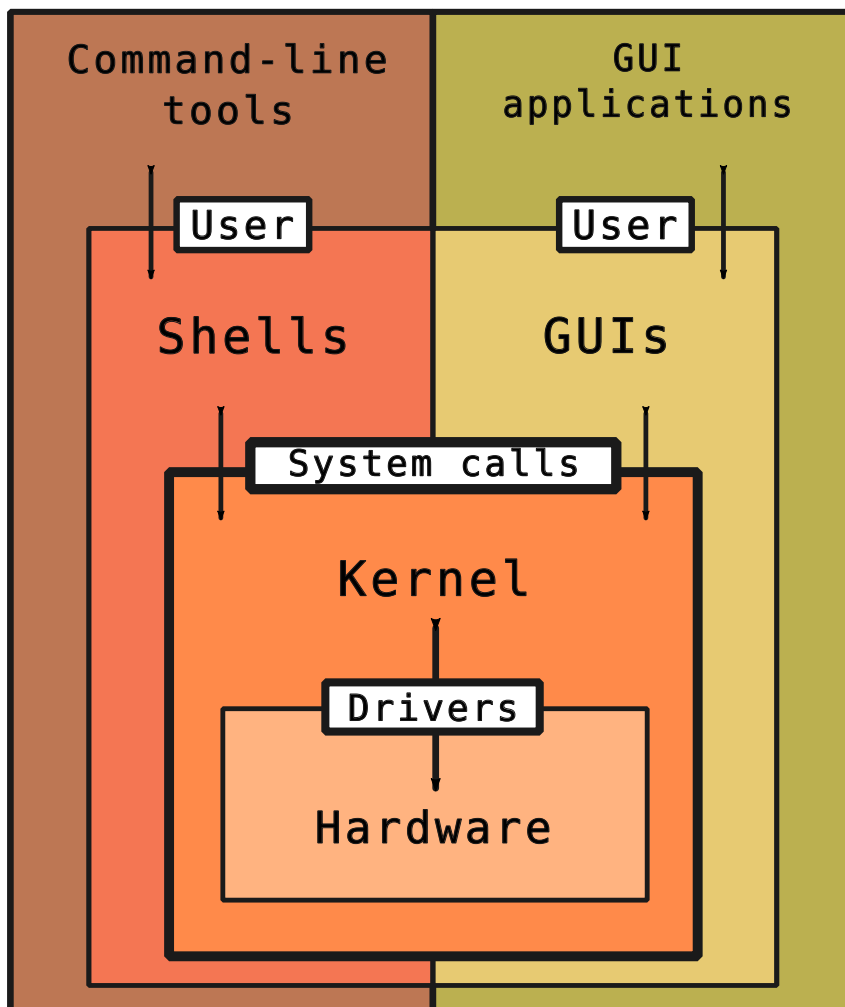
```
user@computer:~$ multipass launch --name tutorial
Launched: tutorial
```

So we know that the VM called tutorial was created successfully!

The shell

You may be wondering: “why we need a shell at all? Why can’t we just tell the operating system what to do using our Desktop environment?”

This is a good question. The simple answer is that while you *can* use the Desktop’s Graphical User Interface (GUI) to interact with your computer, the shell is much more resource-efficient and faster, since it doesn’t need to handle computationally expensive graphics. Servers rarely even have a GUI installed for precisely this reason.



For simple tasks, like renaming a single file, there’s little difference between using your GUI and the CLI to complete it. However, if you need to rename 1,000 files to include the date they were created in the name, this would be a tiresome and time-consuming task to do in the GUI – lots of clicking and typing. Using the CLI, you can pull in the date automatically, set the pattern you want the names to follow, and have all 1,000 files renamed instantly.

The real power and efficiency of the shell can be felt when you need to do something many times. You can even automate routine or large/complex tasks with shell scripts and command chaining – more on that in the next tutorial, though!

The virtual machine provided by Multipass has no Graphical User Interface (GUI). It **abstracts** away all the details of virtualization, which is quite complicated, so we don't have to worry about them. This means we can just open a shell inside our Multipass VM **instance**, which we did when we ran the `multipass shell tutorial` command:

After we ran the command, our terminal window shows us a welcome message like this:

```
ubuntu@tutorial:~$ multipass shell tutorial

Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-15-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri May 29 17:22:28 BST 2026

System load:  0.17               Processes:            110
Usage of /:   56.8% of 3.70GB    Users logged in:     0
Memory usage: 22%               IPv4 address for ens3: 10.234.118.66
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

26 updates can be applied immediately.
10 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
```

You've probably noticed that your shell prompt has changed, and now says `ubuntu@tutorial:~$` – `tutorial` is the name of our virtual machine (the hostname), and `ubuntu` is the default username in a Multipass virtual machine. This is how we know we're now playing in our virtual sandbox, and not in our physical computer.

Other ways to use commands

As we've now seen, CLI commands are the instructions that we give the computer by typing them into the terminal. In our documentation, commands are formatted like this.

Sometimes, we might want to run a whole set of commands at once. For convenience, we can save them in a file called a **shell script**, in the order we want the shell to execute them. Then, we just need to tell the shell to run the script, and it executes the commands in the script by itself.

As you become more familiar with the command line, you can also begin to define your own commands built from other commands. These are called **aliases**, and they can be very handy shortcuts for small sets of commands you want to run often.

Navigating the filesystem

The **filesystem** is the structure your operating system uses to organize all files and directories on the computer. It's like the folder structure you see on a desktop computer's file manager, except we'll be navigating it using text commands in the CLI.

There are some CLI commands that are so useful, and that we'll use so often, they'll become baked into our memory. First, let's try this one:

```
ubuntu@tutorial:~$ ls
```

Nothing happened, right?

The `ls` command ("list show") shows us a list of all the files in our current folder. This is a brand new virtual machine, though, and we haven't created anything yet, so there's nothing to show! Let's fix that by creating a few files using the `touch` command:

```
ubuntu@tutorial:~$ touch my-first-file.txt my-second-file.txt  
my-third-file.txt
```

Now if we run `ls` again it should show them to us:

```
ubuntu@tutorial:~$ ls  
  
my-first-file.txt my-second-file.txt my-third-file.txt
```

We can also create folders to help keep all our files organized using the `mkdir` command:

```
ubuntu@tutorial:~$ mkdir my-folder
```

Let's run `ls` again:

```
ubuntu@tutorial:~$ ls  
  
my-first-file.txt my-folder my-second-file.txt my-third-file.txt
```

There is our new folder!

The `mkdir` command is short for "make directory". A directory is exactly the same as a folder, but it abbreviates to the conveniently short `dir` so it's used more often. We'll use them interchangeably.

Let's now move the three files we created into `my-folder`, so we can keep our current folder nice and tidy. We can do this using the `mv` command.

Since it doesn't know which files to move, we must give some **arguments** to the move (`mv`) command. Actually, we already did this with our `touch` and `mkdir` commands earlier:

```
touch FILE(S)  
touch <file-name> # Create one file  
touch <file-name-1> <file-name-2> # Create more than one file  
  
mkdir FOLDER(S)
```

(continues on next page)

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```
mkdir <folder-name> # To create one folder
mkdir <folder-name-1> <folder-name-2> # To create more than one folder
```

The `mv` command is similar, but takes more arguments:

```
mv FILE(S) FOLDER
mv <file-name> <folder-name> # Move specific file to this folder
mv <file-name-1> <file-name-2> <folder-name> # Move all the named files to this folder
```

You can move any number of files you want. The names of the files you want to move always come right after the `mv` command. The folder you want to move those files to is *always* at the end. That's to say, these are **positional arguments** – the order of where they appear in the command is important.

Note:

It's also worth highlighting the common convention command `[foo] <bar>` for arguments that you'll see in most CLI documentation.

- Square brackets (`[ ]`): anything inside is *optional*, so the command will run without it. This is common for **flags** that change or extend the behaviour of a command, as we'll see shortly.
- Side angle brackets (`< >`): this is a *required* placeholder, where you should substitute your own information.

In both cases, you exclude the brackets when you run the command – the brackets exist to show you what's required and what's optional.

If you have a lot of files to move, though, there's a handy shortcut.

```
ubuntu@tutorial:~$ mv *.txt my-folder
```

The wildcard (`*`) symbol tells the `mv` command to look for *all files* whose name matches the pattern. In this case, files with *any* name, but ending with the `.txt` file extension. Let's check this did what we expected:

```
ubuntu@tutorial:~$ ls

my-folder
```

Nice and tidy! But how do we know that the files have been moved, and not just deleted?

Here, we can use the `cd` command ("change directory"), to move to `my-folder` and take a peek inside:

```
ubuntu@tutorial:~$ cd my-folder/
```

```
ubuntu@tutorial:~/my-folder$ ls

my-first-file.txt my-second-file.txt my-third-file.txt
```

Did you notice how the shell prompt changed to `ubuntu@tutorial:~/my-folder$` after we

used the `cd` command? This is our biggest navigation help! The tilde (`~`) symbol signifies our home directory (located at `/home/your_user_name`, which was our starting point. All the directories and subdirectories we create can trace a path back to this home directory.

Let's illustrate how this works by moving around a bit more. Start by making a subdirectory:

```
ubuntu@tutorial:~/my-folder$ mkdir second-level
```

Another helpful feature of the command line is "tab completion". If a command, directory, or file already exists, we can start typing it and then press the `TAB` key on the keyboard to complete it. Let's navigate to the directory we just made by typing `cd s`, and then press `TAB`. The shell automatically fills in the rest!

```
ubuntu@tutorial:~/my-folder$ cd second-level/
```

And then let's make another subdirectory, and change to it:

```
ubuntu@tutorial:~/my-folder/second-level$ mkdir third-level
```

```
ubuntu@tutorial:~/my-folder/second-level$ cd third-level
```

We're getting quite far into the subdirectory structure now. Let's use the `pwd` command to double check exactly where we are:

```
ubuntu@tutorial:~/my-folder/second-level/third-level$ pwd
/home/ubuntu/my-folder/second-level/third-level
```

This command shows us our "present working directory" (sometimes people refer to it as the "current working directory"), and shows us the path we took to get to where currently are!

So, now we're here, how do we get back again?

You guessed it – with our `cd` command:

```
ubuntu@tutorial:~/my-folder/second-level/third-level$ cd ..
```

Using `cd ..` will take us back up one directory – in our case, from `third-level` back to the `second-level`. The `..` dots represent one folder level, and we could go back any number of levels we like using forward slashes (`/`) as separators between the dots. For example, starting from `third-level`, using `cd ../../` would take us back two levels, to `my-folder`, and `cd ../../../` takes us back three levels, to our home directory.

If we know we want to go back to the home directory, though, we could also use `cd ~` to skip back there directly.

```
ubuntu@tutorial:~/my-folder/second-level$ cd ~
```

```
ubuntu@tutorial:~$ pwd
```

```
/home/ubuntu
```

And now we're back home again!

To keep things tidy, let's remove the directories we created using the `rmdir` ("remove directory") command:

```
ubuntu@tutorial:~$ rmdir my-folder
```

But wait! This returns an error message: `rmdir: failed to remove 'my-folder/': Directory not empty`. It turns out that `rmdir` is a safe option, because it will *only* let you delete a directory if it is empty. This prevents you from accidentally deleting files you didn't know were in the folder.

One option we have is to go back to our second-level directory, and from there, run `rmdir` third-level. Then, change to the `my-folder` directory and delete the second-level directory using `rmdir second-level`. Then, finally, we can return home and run the `rmdir my-folder` command again, and it will work.

A faster (but less safe) option, is to use `rm` ("remove") instead.

Since we're in a sandbox, and we know there's nothing in this entire tree of folders we want to save, we can go ahead. Here, we're going to use the `-r` flag to *recursively* delete the `my-folder` dir, and all its subdirectories.

```
ubuntu@tutorial:~$ rm -r my-folder
```

Now if we run `ls` again, we'll see nothing gets returned, because our home directory is empty once more!

The `rm` command won't let you delete a directory without the `-r` flag, but it also won't stop you (or ask if you're *really, truly* sure) when deleting a folder and all its contents – so you should be cautious when using this command on a real system, to ensure you're really sure you're not going to delete something by accident that you really want to keep.

Working with files

We've already seen how to create empty files. Let's look at how to copy, move, and remove files.

Since our directory is empty again, let's create a new file to play with using the `touch` command:

```
ubuntu@tutorial:~$ touch file1.txt
```

If we run `ls`, we'll see it is there:

```
ubuntu@tutorial:~$ ls  
file1.txt
```

To copy a file, we use `cp` ("copy"), followed by the original file's name, and then the new file's name. This is another case of using *positional arguments*:

```
ubuntu@tutorial:~$ cp file1.txt file2.txt
```

Since we're not renaming or otherwise touching `file1.txt`, we should now have two files:

```
ubuntu@tutorial:~$ ls
file1.txt  file2.txt
```

Before we can see how to move a file, we need to have somewhere to move the file to, so let's quickly create a directory to test this out using `mkdir new-folder`.

Now we can move `file2.txt` to our `new-folder` using the `mv` ("move") command, which takes two arguments:

```
mv <original-location/file-name> <new-location/file-name>
```

In our case, that translates to:

```
ubuntu@tutorial:~$ mv file2.txt new-folder/file2.txt
```

Both of the arguments here can take a **path**, but if we're in the same directory as the file is located in, we don't need to specify it. This becomes really powerful when you have to handle a lot of files in subdirectories.

```
ubuntu@tutorial:~$ ls
file1.txt  new-folder
```

If we `cd` to our `new-folder` we can see our file has been moved with the same name. But – what if we don't want to keep the same name? It turns out `mv` can help us with that too.

```
ubuntu@tutorial:~/new-folder$ mv file2.txt file2-renamed.txt
```

We can also combine both moving the file and renaming it into one command. This time, instead of moving the file down a level by specifying the name of a sub-directory, we can use `../` to indicate we want to move the file up to the previous level:

```
ubuntu@tutorial:~/new-folder$ mv file2-renamed.txt ../file2.txt
```

We could achieve the same by specifying the name of the next directory up (instead of `../`), but this shorthand is much more convenient! Then if we `cd ../` back to the home directory, and run `ls` again, we'll see that our file has been moved back:

```
ubuntu@tutorial:~$ ls
file1.txt  file2.txt  new-folder
```

So what if we want more copies of the same file, but in our new-folder? For that, we can use `cp` again, but give the directory as the destination:

```
ubuntu@tutorial:~$ cp file1.txt new-folder/
```

Double check with `ls` and `cd` that this has worked. You should now have `file1.txt` in both the home directory, and in `new-folder`.

Note:

Be aware that the terminal will not check for, or warn you about, collisions between names. If you tell it to rename a file where a file of that name already exists, it will gleefully overwrite the original!

The `cp` command is not just for files. We can make backup copies of a whole directory (and all its contents) using the `-a` (or `--archive`) flag. This flag preserves all the file permissions:

```
ubuntu@tutorial:~$ cp -a new-folder/ new-folder-backup/
```

Take a look around your `new-folder-backup` directory to make sure it all matches the original, then `cd ..` back to the home directory.

Let's say we no longer want `file.txt`. We can use the `rm` command to remove it, the way we did before. This time, we don't need the `-r` flag because we're deleting a single file.

```
ubuntu@tutorial:~$ rm file1.txt
```

Warning:

Remember, there is no recycle bin in the Linux terminal – once you `rm` a file, it is gone. On your real system, double-check what you are deleting before running this command.

Understanding file attributes

In Linux, everything is represented as a file – including directories, devices, and network connections. When we use `ls -l`, the very first character of each line tells us what type of file it is.

Let's look at a few examples:

```
ubuntu@tutorial:~$ ls -l
-rw-rw-r-- 1 ubuntu ubuntu    0 Apr 21 17:46 file2.txt
drwxrwxr-x 2 ubuntu ubuntu 4096 Apr 21 18:00 new-folder
drwxrwxr-x 2 ubuntu ubuntu 4096 Apr 21 18:00 new-folder-backup
```

The output looks quite complicated at first, so let's break it down using the first row of the output.

Column	What it is	Description
-rw-rw-r--	Permissions	The type of file and who can read/write/execute it
1	Links	The number of distinct filesystem paths that refer to this file or directory (called “hard links”)
ubuntu	Owner	The username of the person who owns the file
ubuntu	Group	The name of the group that has specific permissions
0	Size	The size of the file in bytes
Apr 21 17:46	Times-tamp	The date and time the file was last modified
file2.txt	Name	The actual name of the file or directory

Fields like the file name, size, and timestamp, are quite intuitive once you know that this is what the fields represent. Permissions, owner and group are a bit less intuitive, so let's examine those in more detail.

Users, groups, and permissions

Linux is a **multi-user system** – it is designed to be used by multiple people at once, and it carefully controls who can read, modify, or run any given file. Understanding users, groups, and permissions is essential for working safely on a Linux system.

Every file on a Linux system has an **owner** (a user account) and a **group** associated with it. Permissions grant different types of access to owners and groups.

The nine characters in the permissions field show the different levels of access granted for each file. They are split into three groups of three:

```

-  rwx  r-x  r-x
|      |      |
|      |      |  Other (everyone else): read + execute
|      |      |
|      |      |  Group: read + execute
|      |      |
|      |      |  Owner: read + write + execute
|      |      |
|      |      |  File type: regular file

```

Each group of three characters can contain:

- **r – Read:** view the file’s contents
- **w – Write:** modify or delete the file
- **x – Execute:** run the file as a program (or, for directories, enter them with `cd`)
- **-** – the permission is not granted

In our own example, we can see this means that for `file2.txt`, the permissions are:


```
-  rw-  rw-  r--
|  |    |    |
|  |    |    |----- Other (everyone else): read only
|  |    |    |----- Group: read + write (no execute)
|  |    |    |----- Owner: read + write (no execute)
|  |    |    |----- File type: regular file
```

The first character on each line is the **file type indicator**, which tells us what type of file it is:

Sym- bol	Type	Description
-	Regular file	An ordinary file: documents, scripts, binaries, images, etc.
d	Directory	A file that is a list of other files
l	Symbolic link (or symlink)	A pointer (shortcut) to another file
c	Character device	A device file for hardware that handles data one character at a time (e.g., a keyboard)
b	Block device	A device file for hardware that handles buffered data in blocks (e.g., a hard disk)
s	Socket	A file used for inter-process communication
p	Named pipe	Allows processes to communicate without network socket semantics

Changing file permissions

We can change a file's permissions using `chmod` ("change file mode"). There are two ways to use it.

Symbolic notation

This method uses letters to specify who to change:

- u – user (owner)
- g – group
- o – other

And what to change:

- + – add a permission
- - – remove a permission

Let's test method out first by creating a script file:

```
ubuntu@tutorial:~$ touch symbolic-test.sh
```

```
ubuntu@tutorial:~$ ls -l symbolic-test.sh
-rw-rw-r-- 1 ubuntu ubuntu 0 Apr 23 10:52 symbolic-test.sh
```

And then let's make it executable by us (the owner):

```
ubuntu@tutorial:~$ chmod u+x symbolic-test.sh
```

```
ubuntu@tutorial:~$ ls -l symbolic-test.sh
-rwxrw-r-- 1 ubuntu ubuntu 0 Feb 12 2024 myscript.sh
```

Note how the permissions for the owner have change from "read and write only" (rw-) to "read, write and execute" (rwx).

Numeric notation

This method uses a three-digit number where each digit is calculated by adding: read (4) + write (2) + execute (1):

```
ubuntu@tutorial:~$ touch numeric-test.sh
```

```
ubuntu@tutorial:~$ ls -l numeric-test.sh
-rw-rw-r-- 1 ubuntu ubuntu 0 Apr 23 15:58 numeric-test.sh
```

Then let's use the numeric notation to change the permissions:

```
ubuntu@tutorial:~$ chmod 755 numeric-test.sh
```

```
ubuntu@tutorial:~$ ls -l numeric-test.sh
-rwxr-xr-x 1 ubuntu ubuntu 0 Apr 23 15:58 numeric-test.sh
```

From this we can see that 755 means: the owner has read, write and execute (7) privileges, the group has read and execute (5), while all others have read and execute (5).

Some common permission combinations:

Num ber	Sym- bolic	Meaning
644	rw- r-- r--	Owner can read/write; others can only read. Suitable for most files.
755	rxr- xr- x	Owner can do everything; others can read and execute. Suitable for scripts and directories.
700	rw- --- --	Only the owner has any access.
600	rw- --- ---	Owner can read/write; no one else has any access.
777	rxr- wxrw	Full access to <i>everyone</i> . This is a dangerous permission to grant , but may be useful for debugging permission-related problems or on temporary files. Please avoid using this one unless you really know what you're doing, and if you do see it, you should probably change it to one of the less permissive combinations.

When we create a new file, it gets a default set of permissions. The `umask` ("user file-creation mode mask") controls what those defaults are. We can check the current `umask` with:

```
ubuntu@tutorial:~$ umask  
  
0002
```

The `umask` works by subtracting permissions from the maximum (777 for directories, 666 for files). A `umask` of 022 results in files being created with permissions of 644 and directories with 755 by default.

Temporary admin privileges with `sudo`

Most day-to-day work on Ubuntu is done as a regular user, without administrative privileges. When we do need to run a command as an administrator, we prefix it with `sudo` ("superuser do"), which temporarily grants elevated privileges for that one command. For example, if we want to update the software on our computer, we would run these two commands in sequence:

```
ubuntu@tutorial:~$ sudo apt update
```

```
ubuntu@tutorial:~$ sudo apt upgrade
```

When we run a `sudo` command, we are asked for our password to confirm it is us.

The sudo group on Ubuntu is special: any user account added to it gains the ability to run commands with admin privileges. Groups are listed in `/etc/group`, and a user can belong to many groups at once. We can inspect who is in the sudo group with:

```
ubuntu@tutorial:~$ grep sudo /etc/group  
  
sudo:x:27:ubuntu
```

Each line of the `/etc/group` file follows this format:

```
group_name:password:GID:user_list
```

Where:

- `group_name` is the human-readable name of the group (what you see when you run `ls -l`)
- `password` is always shown as `x` in modern Linux systems - the actual (encrypted) group password is stored in the secure `/etc/passwd` file.
- `GID` is the group ID, which is a unique numerical integer assigned to the group.
- `user_list` is a comma-separated list of usernames that are members of the group.

The Filesystem Hierarchy Standard

Now that we know how to move around the filesystem, it's worth understanding how it is organised. Most Linux distributions, including Ubuntu, follow the **Filesystem Hierarchy Standard** (FHS), which defines where different types of files and directories belong. This is maintained by the Linux Foundation, and it means the structure of any conforming system is predictable and consistent.

All files and directories live under the **root directory**, represented by a single forward slash `/`. Even files stored on separate physical disks appear as part of this single tree.

Let's take a look at the top level:

```
ubuntu@tutorial:~$ ls /  
  
bin  dev  home  lib64  media  opt   root  sbin  srv  tmp  usr  var  
boot  etc  lib   mnt    proc  run   snap  sys
```

Here is what the most important directories contain:

Directory	Purpose
/bin	Essential command binaries (e.g., <code>ls</code> , <code>cp</code> , <code>mv</code>)
/sbin	System binaries (e.g. <code>iptables</code> , <code>reboot</code>)
/boot	Static files needed to boot the system, including the Linux kernel and boot-loader files
/dev	Device files (e.g. <code>/dev/sda</code> and <code>/dev/null</code>)
/etc	System configuration files (e.g. <code>/etc/group</code> and <code>/etc/passwd</code>)
/home	User home directories (optional, but standard)
/lib	Essential shared libraries (needed by the binaries in <code>/bin</code> and <code>/sbin</code>)
/media	Mount points for temporarily mounted removable media (like USBs or CDs)
/mnt	Mount points for manually mounted filesystems (kept separate to avoid clashes)
/root	The home directory for the root user specifically
/srv	Service data, for services provided by the system
/tmp	Temporary files (cleared on reboot)
/usr	Most user programs, libraries, and documentation
/var	Variable data: logs, caches, spool files

This is not a complete list, by any means! If you'd like more information about the Standard and other directories contained within it, take a look at [the official specification](#)²⁶.

Exploring deeper

Let's browse a few of these. The `/etc` directory contains configuration files for the system and its services. Some important files there include:

- `/etc/passwd` – the user account database
- `/etc/shadow` – encrypted passwords (readable only by root)
- `/etc/fstab` – filesystems to be mounted automatically at startup
- `/etc/group` – group account information

The `/var/log` directory holds the system's log files – a useful place to look when diagnosing problems:

```
ubuntu@tutorial:~$ ls /var/log
```

The `tree` command can be used to visualise directory structures in a tree-style listing. It's not installed by default, so let's install it:

²⁶ <https://specifications.freedesktop.org/fhs/latest/>

```
ubuntu@tutorial:~$ sudo apt install tree -y
```

The `-L` flag can be used to set the depth you want to view. To see the top level (or level 1) of the root filesystem (`/`):

```
ubuntu@tutorial:~$ tree -L 1 /
```

To go one level deeper, to level 2:

```
ubuntu@tutorial:~$ tree -L 2 /
```

What happens if you just run `tree` by itself, with no arguments?

```
ubuntu@tutorial:~$ tree

.
├── file2.txt
├── new-folder
│   └── file1.txt
├── new-folder-backup
│   └── file1.txt
├── numeric-test.sh
└── symbolic.sh

3 directories, 5 files
```

This is a much more convenient way to see what files each directory contains.

A note about flags

So far we've been using short flags, with a single dash (`-`) followed by a single letter, like with `ls -l`.

Most flags also have a long version, where you'll see a double dash (`--`) with a full word. If a flag has both a long and short version, they can be used interchangeably.

It's possible to combine flags together. For example, the `-l` and `-a` flags ("long list", "all") would be combined into `ls -la` gives us a long listing that also includes hidden files, or those whose names start with a `.` character:

```
ubuntu@tutorial:~$ ls -la
```

```
drwxr-x--- 6 ubuntu ubuntu 4096 Apr 23 17:26 .
drwxr-xr-x 3 root root 4096 Apr 16 14:15 ..
-rw----- 1 ubuntu ubuntu 936 Apr 23 13:00 .bash_history
-rw-r--r-- 1 ubuntu ubuntu 220 Mar 31 2024 .bash_logout
-rw-r--r-- 1 ubuntu ubuntu 3771 Mar 31 2024 .bashrc
drwx----- 2 ubuntu ubuntu 4096 Apr 16 14:15 .cache
-rw----- 1 ubuntu ubuntu 20 Apr 23 17:26 .lessht
-rw-r--r-- 1 ubuntu ubuntu 807 Mar 31 2024 .profile
drwx----- 2 ubuntu ubuntu 4096 Apr 16 14:15 .ssh
-rw-r--r-- 1 ubuntu ubuntu 0 Apr 23 11:05 .sudo_as_admin_successful
-rw-rw-r-- 1 ubuntu ubuntu 0 Apr 23 15:47 file2.txt
drwxrwxr-x 2 ubuntu ubuntu 4096 Apr 23 15:53 new-folder
drwxrwxr-x 2 ubuntu ubuntu 4096 Apr 23 15:53 new-folder-backup
-rwxr-xr-x 1 ubuntu ubuntu 0 Apr 23 15:58 numeric-test.sh
-rwxrw-r-- 1 ubuntu ubuntu 0 Apr 23 15:54 symbolic.sh
```

This brings us onto one of the most important skills when you are learning the terminal – how to find help.

Getting help

Linux comes with a comprehensive built-in manual, and learning to use it helps us explore on our own. Every command in Linux has a **manual page** that describes what it does, and lists all the option flags available.

Most documentation uses the short version of a flag when demonstrating a command – mostly for convenience. However, if you are ever unsure what a flag does, you can look up the command’s manual page (often called a “manpage” for short).

For example, you can invoke the manual page for `ls` using the `man ls` command.

```
ubuntu@tutorial:~$ man ls
```

This opens the manual page for `ls`. We can navigate it with the following keys on our keyboard:

Key	Action
Enter	Move down one line
Space	Move down one page
g	Move to the top
G	Move to the bottom
h	Display navigation help
q	Quit

You can also scroll up and down using the mouse wheel, or the up and down keys.

Let’s try reading the man page for `man` itself:

```
ubuntu@tutorial:~$ man man
```

The Linux manual is divided into numbered sections – the number appears in the page title, e.g., LS(1). Section 1 means “executable programs and shell commands”. We can search within a man page by pressing / followed by a search term, then Enter.

Manual pages are also rendered online and can be [viewed in your browser](https://manpages.ubuntu.com/)²⁷.

Quick help with `-help`

Most commands also accept a `--help` flag that prints a brief summary of the available options:

```
ubuntu@tutorial:~$ ls --help
```

This is often quicker than opening the full manual page when we just want to check a specific flag.

Searching for a command

If we know what we want to do but don’t know which command to use, `man -k` searches the man page descriptions for a keyword:

```
ubuntu@tutorial:~$ man -k copy
```

This lists every command whose description mentions “copy”, which can help us find the right tool for the job.

Next steps

This is the end of our introduction to the terminal! We have covered a lot of ground: launching a virtual machine, navigating the filesystem, working with files and directories, understanding file types and the filesystem hierarchy, managing permissions, and finding help. All the commands we covered here (as well as some extra ones) can be found in our reference [Command-line cheat sheet](#) (page 843).

If you wish, you can continue to explore the file system on your virtual machine, or proceed straight to the next tutorial where you can use the same VM.

However, if you’re done for now and want to exit the VM, just type “exit”.

```
ubuntu@tutorial:~$ exit
```

This will take you out of the VM and back to your live machine. Then, you can run the following commands to delete the VM and remove it completely from your machine:

```
ubuntu@tutorial:~$ multipass delete tutorial > multipass purge
```

²⁷ <https://manpages.ubuntu.com/>

Next steps

If you wish to continue learning, we recommend following the [The command line in depth](#) (page 32) tutorial next. This tutorial will teach you the more advanced ways to interact with the Command Line Interface; how data flows, how input and output can be used more efficiently, and how to use variables.

The command line in depth

In [Welcome to the terminal](#) (page 12), we learned how to navigate the filesystem, work with files and directories, and manage permissions on the command line. Now we'll go deeper into how to handle data flowing between programs, how to redirect that data to files and other programs, and how **shell variables** and **expansion** make our commands *even more* powerful.

If you are re-using a virtual machine from a different tutorial, skip directly to [Standard streams](#) (page 34).

Set up the virtual machine

To avoid making changes to your computer we will set up a virtual machine (VM), which provides us with a safe environment to run the commands in. [Multipass²⁸](#) is great for quickly creating Ubuntu virtual machines, so we will use that. It is available on Windows, Mac and Linux, so you don't need to be already running Ubuntu on your computer to follow this tutorial.

Prerequisites

- The Multipass VM needs **5 GB of disk space**, and **1 GB of memory** available.

Install Multipass

On Ubuntu

You can install Multipass directly from [the Multipass page²⁹](#) in the online snap store (make sure to select the "latest/stable" version from the dropdown menu next to the install button).

²⁸ <https://documentation.ubuntu.com/multipass/latest/>

²⁹ <https://snapcraft.io/multipass>

On other operating systems

If you're on Windows, Mac, or another Linux operating system, Multipass can be installed using [these instructions](#)³⁰.

Open a terminal window

Now that Multipass is installed, we need a terminal window to run our commands in. How you open one depends on your operating system.

On Ubuntu

Press `Ctrl + Alt + T` together and a terminal window will open. You can also search for "Terminal" in your application launcher.

On Windows

PowerShell is the most reliable option for working with Multipass on Windows.

1. Press the Windows key.
2. Type PowerShell.
3. Right-click it and select **Run as Administrator** — this helps avoid permission issues when managing virtual machines.

If you have Windows Terminal installed from the Microsoft Store, that works just as well and offers a nicer experience with tab support.

To confirm everything is working, type `multipass version` and press Enter. If you see version numbers, you're ready to continue.

On macOS

1. Press `⌘ + Space` to open Spotlight.
2. Type Terminal and press Enter.

If you use iTerm2, that works perfectly well too.

To confirm everything is working, type `multipass version` and press Enter. If you see version numbers, you're ready to continue.

³⁰ <https://documentation.ubuntu.com/multipass/stable/how-to-guides/install-multipass/>

Create the virtual machine

Once you have installed Multipass and opened your terminal, we can launch a new VM using the Ubuntu 26.04 LTS release (codename “resolute”), and let’s give our VM the name `tutorial` using the following command in our terminal window:

```
$ multipass launch resolute --name tutorial
```

It may take a little time, depending on the speed of your internet connection. Multipass downloads the most recent daily **image** and creates the VM for us.

Note:

An Ubuntu **image** is a collection of files we need to install and run Ubuntu. We don’t need to specify “server” or “desktop” anywhere in our command, because the image is the same for both. The only difference between Ubuntu Server and Ubuntu Desktop is the subset of software packages we use from the Ubuntu Archive.

Once the above command has finished running, we can access the VM by running:

```
$ multipass shell tutorial
```

We will get a “Welcome to Ubuntu” message. Notice that when we run this command, the terminal username changes to `ubuntu` and the *hostname* changes to `tutorial`:

```
ubuntu@tutorial:~ $
```

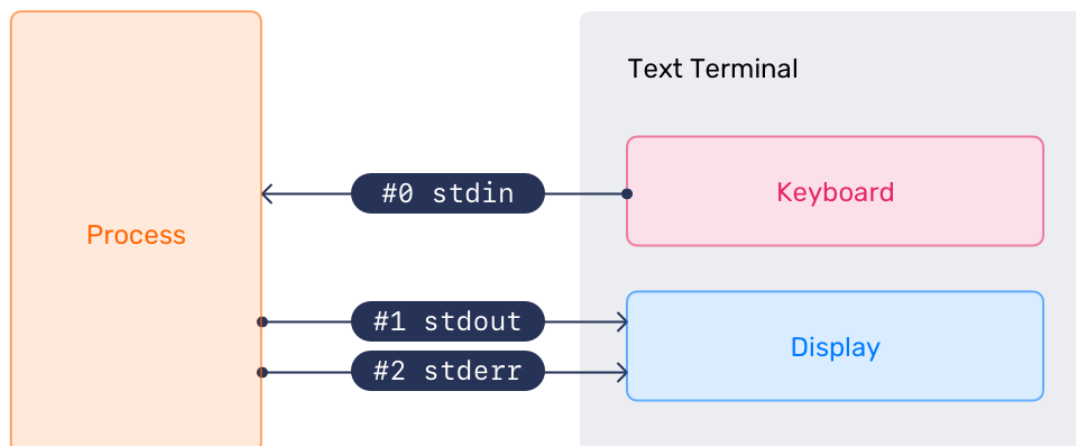
This shows that we are inside the VM, and this is where we will run all our commands. Now that our setup is complete, let’s carry on with the tutorial!

Standard streams

Now that we can work with files, we’re ready to understand something fundamental about how programs work in the terminal: **standard streams**.

Every program that runs has three open communication channels by default:

- **Standard input (stdin)** – the channel through which the program receives input. By default, this is your keyboard.
- **Standard output (stdout)** – the channel through which the program sends its results. By default, this is your terminal screen.
- **Standard error (stderr)** – the channel through which the program sends error messages. This also appears on your terminal screen by default, but it is a separate channel from stdout.



Each stream has a numerical identifier called a **file descriptor**:

- 0 – stdin
- 1 – stdout
- 2 – stderr

You may be wondering: “why three separate channels? Why not just one?”.

The answer is that separating normal output from error output gives us much more flexibility. We can save the output of a command to a file while still seeing errors on screen – or silence errors entirely while saving results. We’ll see exactly how in the next sections.

Redirecting output

Normally, a command’s output goes straight to our terminal screen. We can tell the shell to send it somewhere else instead – like a file.

To redirect output to a file, we use the `>` operator followed by a filename. Let’s redirect the output of `ls` to a file called `file-list.txt`:

```
ubuntu@tutorial:~$ ls -la > file-list.txt
```

No output appears on screen – it has all gone into a new file called `file-list.txt` instead.

Viewing file contents

Now that `file-list.txt` exists and has some content in it, let’s look at the commands we can use to see what the file contains. There are a few to choose from, and which one is most convenient depends on how much of the file we want to see.

To print the full contents of a file to the screen, we can use `cat` (“concatenate”):

```
ubuntu@tutorial:~$ cat file-list.txt

total 36
drwxr-x--- 4 ubuntu ubuntu 4096 May  1 09:25 .
drwxr-xr-x 3 root  root  4096 Apr 16 14:15 ..
-rw----- 1 ubuntu ubuntu 1623 Apr 23 17:33 .bash_history
-rw-r--r-- 1 ubuntu ubuntu  220 Mar 31  2024 .bash_logout
-rw-r--r-- 1 ubuntu ubuntu 3771 Mar 31  2024 .bashrc
drwx----- 2 ubuntu ubuntu 4096 Apr 16 14:15 .cache
-rw----- 1 ubuntu ubuntu   20 Apr 23 17:33 .lessht
-rw-r--r-- 1 ubuntu ubuntu  807 Mar 31  2024 .profile
drwx----- 2 ubuntu ubuntu 4096 Apr 16 14:15 .ssh
-rw-r--r-- 1 ubuntu ubuntu    0 Apr 23 11:05 .sudo_as_admin_successful
-rw-rw-r-- 1 ubuntu ubuntu    0 May  1 09:25 file-list.txt
```

You may see more lines or fewer lines here, depending on whether you ran through the [Welcome to the terminal](#) (page 12) tutorial with this VM or not.

For longer files, `less` is more comfortable – with `less` we can scroll through the outputs one page at a time. Let's look at `/etc/passwd`, which stores user account information and is long enough to scroll through:

```
ubuntu@tutorial:~$ less /etc/passwd
```

Press Space to move down a page, and Q to quit when you are done.

If we only want to see the first lines of a file, we can use `head`:

```
ubuntu@tutorial:~$ head /etc/passwd

root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/spool/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
```

By default, `head` shows the first 10 lines but we can request a different number using the `-n` flag. For example, `head -n 5 /etc/passwd` shows only the first 5 lines.

Or we can see the last few lines with `tail`, which is used the same way as `head`:

```
ubuntu@tutorial:~$ tail -n 3 /etc/passwd

fwupd-refresh:x:990:990:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
polkitd:x:989:989:User for polkitd:/:/usr/sbin/nologin
ubuntu:x:1000:1000:Ubuntu:/home/ubuntu:/bin/bash
```

`tail` is especially useful for examining log files, where the most recent entries are always at the bottom.

Warning:

One thing to be careful about: we know that `file-list.txt` is empty because it didn't exist before, but if `file-list.txt` had already contained something, `>` would have overwritten it completely. To *add* things to the end of an existing file rather than replacing it, we use `>>` (**append** mode):

Let's make our file a bit longer by using `>>` to add extra contents:

```
ubuntu@tutorial:~$ ls -l /tmp >> file-list.txt
```

Now `file-list.txt` contains the original home directory listing followed by the `/tmp` listing. We can check on this by using `cat file-list.txt` again.

Sometimes we want a command to run **silently**, i.e. with its output discarded entirely. We can redirect stdout to `/dev/null`, a special file that throws away everything written to it:

```
ubuntu@tutorial:~$ ping -c 2 localhost > /dev/null
```

We can also redirect stderr (file descriptor 2) independently from stdout. This is useful for capturing errors to a log file while still seeing normal output on screen:

```
ubuntu@tutorial:~$ ls /nonexistent 2> errors.txt
```

Checking on this file shows us the error that would otherwise have been printed to the screen:

```
ubuntu@tutorial:~$ cat errors.txt
ls: cannot access '/nonexistent': No such file or directory
```

We can also merge stderr into stdout using `2>&1`, which means "send file descriptor 2 to wherever file descriptor 1 is currently going":

```
ubuntu@tutorial:~$ ls /nonexistent >> output.txt 2>&1
```

By doing this, both normal output and error messages will now appear in `output.txt`.

Redirecting input

We can go the other direction too! Instead of a command reading from the keyboard, we can feed it input from a file using the `<` operator.

Let's use `wc` ("word count"), which counts lines, words, and characters in its input. Without redirection, we would type text and press `Ctrl + D` to signal the end of input. Instead, we can feed it a file directly:

```
ubuntu@tutorial:~$ wc -l < /etc/passwd
```

```
33
```

Once again, you might see a different value here, depending on whether this is a new VM or you're reusing the one from the previous tutorial.

The `-l` flag tells `wc` to count lines only. When we use `<`, the file name does not appear in the output – `wc` sees the file's contents as if they had been typed at the keyboard.

Here-documents

There is a related form called a **here-document**, written as `<<`. It lets us type multi-line input directly in the terminal, ending when we type a delimiter we choose ourselves (often EOF is used as this marker – short for “end of file”). Let's test this by sending a greeting into a new file called `greeting.txt`:

```
ubuntu@tutorial:~$ cat > greeting.txt << EOF > Hello from the terminal! >
This text is going into a file. > EOF
```

The shell reads everything we type until it sees EOF on a line by itself, then passes it all as input to the command on the left. Let's confirm the file was created correctly:

```
ubuntu@tutorial:~$ cat greeting.txt
Hello from the terminal!
This text is going into a file.
```

Pipes

Pipes are one of the most powerful ideas in the Unix philosophy: the output of one command becomes the input of the next. We connect them with the `|` operator (the “pipe” character).

For example, `ls -la /usr/bin` produces many lines of output – more than can fit on screen, and you can only see the end of the output. Scrolling back up is often very inconvenient, if you can even do it at all (some terminals can't!). To make it more convenient, we can pipe the output to `less` so we can page through it with the space key:

```
ubuntu@tutorial:~$ ls -la /usr/bin | less
```

Press `Q` to quit `less` when you are done.

We can also pipe to `grep` to filter the output, keeping only lines that match a pattern. Let's find all currently set environment variables whose names contain the word `LANG`:

```
ubuntu@tutorial:~$ env | grep LANG
LANG=C.UTF-8
```

Pipes can be chained together to build a **pipeline** – each command passes its output along to the next. For example, to see the 3 most recently modified files in `/usr/bin`, we can sort by modification time and then cut the output down with `head`:

```
ubuntu@tutorial:~$ ls -lt /usr/bin | head -n 3

total 121540
-rwxr-xr-x 1 root root      174328 Apr 30 13:32 kmod
lrwxrwxrwx 1 root root         4 Apr 30 13:32 lsmod -> kmod
```

Each command in the pipeline only ever sees the output of the command immediately before it – the shell connects them all behind the scenes.

Chaining commands

Pipes pass data from one command to another and run all of their commands in parallel. This allows commands to begin working on long streams of data while their source is still producing more. However, there are scenarios where you might want a serial process, in which case you would use **command chaining** instead. Chaining is different: it controls the *order* in which commands run, and whether the next command runs at all.

The simplest way to chain commands is to separate them with a semicolon (;). The shell runs each command in turn, regardless of whether the previous one succeeded, with the output of each command shown on separate lines:

```
ubuntu@tutorial:~$ ls; pwd; whoami

errors.txt  file-list.txt  greeting.txt  output.txt
/home/ubuntu
ubuntu
```

Note:

On Linux, processes declare whether they have “succeeded” or not with an **exit code**. An exit code (or “exit status”) is a number from 0–255 that a process gives to its parent upon completion. “Success” is defined by exiting with code 0, with *all* other values indicating different types of failures. These might be specific to a particular program, but the generic failure status will usually be exit code 1. You can use `exit n` to immediately exit with code `n`.

Often we want the second command to run *only* if the first one succeeded. We use `&&` for this:

```
ubuntu@tutorial:~$ cd /tmp && ls

snap-private-tmp  systemd-private-...
```

If `cd /tmp` had failed (which can happen in a script when the directory does not exist), `ls` would not run at all.

The opposite of `&&` is `||`, which runs the second command only if the first one *failed*:


```
ubuntu@tutorial:~/tmp$ cd /nonexistent || echo "Could not change directory"

-bash: cd: /nonexistent: No such file or directory
Could not change directory
```

We can combine `&&` and `||` to produce a simple success-or-failure message:

```
ubuntu@tutorial:~/tmp$ ping -c1 localhost && echo "Host is up" || echo "Host
is unreachable"

PING localhost (:::1) 56 data bytes
[...]
Host is up
```

A few other operators are useful to know. Try each of these out, both with and without the operators, to see how they change the behaviour for yourself:

Background execution; `&`

Runs a command in the background. The shell prompt returns immediately, and the command continues running. Let's compare the difference in running `sleep 10` and then `sleep 10 &`. The `sleep` command doesn't actually *do* anything, it just makes the system wait for the specified number of seconds. So, in this example our prompt disappears for 10 seconds and you can't interact with the terminal during that time. If we *background* it, using `sleep 10 &`, our command prompt returns immediately and we can continue to use the terminal.

When we use the ampersand (`&`) to background the process, it returns something like:

```
[1] 168658
```

This number is the **process ID** (PID); a unique number identifying every process running on your machine. When the background process has finished running, you'll see a message returned that looks like this, so you know it's done:

```
[1]+  Done                  sleep 10
```

Command grouping; `{ }`

Groups commands together so they share the same output redirection or run as a unit. E.g. to create a quick log file you could run:

```
ubuntu@tutorial:~/tmp$ { echo "System Report"; date; uptime; } >
report.txt
```

Subshell execution; `( )`

Run commands in a subshell, where the commands are forgotten the moment they finish running. This is useful for isolating side effects like a `cd` – the directory change does not affect the current shell.

```
ubuntu@tutorial:/tmp$ (export NAME="Alice"; echo "Hello $NAME") > echo  
"Name is: $NAME"
```

Line continuation; \

Continue a long command on the next line, which helps a lot with command readability. You'll see this often in documentation.

Let's return home before continuing:

```
ubuntu@tutorial:/tmp$ cd ~
```

Shell environment variables

Before we look at shell expansion, we need to understand **shell variables**, because expansion builds heavily on them.

Think of a variable as a labelled box: we give it a name, put a value inside, and later retrieve that value using the name of the box. The shell uses variables to store all kinds of information – including a great deal about the environment we're working in.

Built-in variables

The shell already has a number of variables set up for us. To see every variable currently set in our environment, we can run:

```
ubuntu@tutorial:~$ env
```

We can print the value of any variable using echo with a \$ in front of the variable name:

```
ubuntu@tutorial:~$ echo $SHELL  
/bin/bash
```

```
ubuntu@tutorial:~$ echo $USER  
ubuntu
```

```
ubuntu@tutorial:~$ echo $HOME  
/home/ubuntu
```

One particularly important variable is PATH. It tells the shell which directories to search when we type a command name:

```
ubuntu@tutorial:~$ echo $PATH
```

```
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin
```

Each directory is separated by a colon (:). When we type a command like `ls`, the shell searches through each of these directories in order until it finds a program with that name. This is why we can run commands without typing their full path. If a command lives outside these directories, we have to give its full path explicitly.

Setting and unsetting variables

We can create our own variables by assigning a name and a value. By convention, user-defined variable names are uppercase:

```
ubuntu@tutorial:~$ GREETING="Hello, world!"
```

```
ubuntu@tutorial:~$ echo $GREETING
Hello, world!
```

To remove a variable, we use `unset`:

```
ubuntu@tutorial:~$ unset GREETING
```

```
ubuntu@tutorial:~$ echo $GREETING
```

This time we get no output, because the variable no longer exists.

Exporting variables to subshells

Variables we create in the shell are only available in the current shell session. If a new shell process is started (for example, when running a script), it will not have access to our variables unless we **export** them.

Let's demonstrate. First, we set a variable and try to access it from a subshell:

```
ubuntu@tutorial:~$ MY_VAR="hello"
```

```
ubuntu@tutorial:~$ bash -c 'echo $MY_VAR'
```

Nothing is printed – the subshell has no knowledge of `MY_VAR`. Now let's export it first:

```
ubuntu@tutorial:~$ export MY_VAR="hello"
```

```
ubuntu@tutorial:~$ bash -c 'echo $MY_VAR'
```

```
hello
```

`export` makes the variable part of the environment that is passed down to any child processes we start.

Special read-only variables

Some variables are set by the shell itself and can only be read, never assigned. Two of the most useful are:

- `$?` – the exit status of the last command. 0 means success; any non-zero value means something went wrong.
- `$$` – the Process ID (PID) of the current shell.

Let's see `$?` in action. A successful command returns 0:

```
ubuntu@tutorial:~$ ls /etc/passwd
/etc/passwd
```

```
ubuntu@tutorial:~$ echo $?
0
```

And a failing command returns a non-zero value:

```
ubuntu@tutorial:~$ ls /nonexistent
ls: cannot access '/nonexistent': No such file or directory
```

```
ubuntu@tutorial:~$ echo $?
2
```

This is especially useful when we build scripts, where we can check whether the previous command succeeded before deciding what to do next.

Shell expansion

When we press Enter on a command, the shell does not send our text directly to the program. First, it processes the command line and **expands** certain patterns into their final values. Understanding expansion explains a lot of seemingly “magic” shell behaviour.

Tilde expansion

We have already used tilde expansion without thinking about it. When the shell sees ~ at the start of a word, it replaces it with the value of \$HOME:

```
ubuntu@tutorial:~$ echo ~  
/home/ubuntu
```

```
ubuntu@tutorial:~$ echo ~/my-file.txt  
/home/ubuntu/my-file.txt
```

Variable expansion

We have seen this already: \$VARIABLE_NAME (or \${VARIABLE_NAME}) is replaced with the variable's value before the command runs. The braces form is important when the variable name might be confused with the characters that follow it:

```
ubuntu@tutorial:~$ DAY="Monday"
```

```
ubuntu@tutorial:~$ echo "Today is ${DAY}!"  
Today is Monday!
```

Without the braces, the shell would try to expand \$DAY! as a single variable name and find nothing.

It's also important to note that this variable expansion applies *everywhere* in the line, even with the command name itself:

```
ubuntu@tutorial:~$ CMD="echo"
```

```
ubuntu@tutorial:~$ $CMD "It's dynamic!"  
It's dynamic!
```

This can be useful in, say, a script that runs one of two very similar commands with the same argument, so you don't have to re-type everything twice in your conditional.

Command substitution

Command substitution lets us use the *output of a command* as part of another command. We wrap the inner command in `$(...)`:

```
ubuntu@tutorial:~$ echo "Today is $(date)"
Today is Fri Apr 24 10:00:00 BST 2026
```

The shell runs `date` first, captures its output, and substitutes it into the outer command before running it.

Arithmetic expansion

For integer arithmetic, we use `$((...))`:

```
ubuntu@tutorial:~$ echo $((5 + 3))
8
```

```
ubuntu@tutorial:~$ X=10
```

```
ubuntu@tutorial:~$ echo $((X * 2))
20
```

The available Arithmetic operators are:

- Addition (+)
- Subtraction (-)
- Multiplication (*)
- Integer division (/)
- Modulo/remainder (%)
- Exponentiation (**)

Brace expansion

Brace expansion generates a list of strings from a pattern. Unlike the other types of expansion, it does not involve variables or filenames – it is a purely text-based generation mechanism:

```
ubuntu@tutorial:~$ echo sp{in,an,ar}
spin span spar
```

Ranges work too, using `..`:

```
ubuntu@tutorial:~$ echo {1..10}

1 2 3 4 5 6 7 8 9 10
```

Brace expansion is particularly useful for creating multiple files or directories in one command. For example, to create a directory for each month of the year:

```
ubuntu@tutorial:~$ mkdir {01..12}-reports
```

```
ubuntu@tutorial:~$ ls

01-reports 02-reports 03-reports 04-reports 05-reports 06-reports
07-reports 08-reports 09-reports 10-reports 11-reports 12-reports
```

We can even clean them up in one command, too:

```
ubuntu@tutorial:~$ rm -r {01..12}-reports
```

Wildcard expansion

Wildcard expansion (also called **globbing**) lets us match filenames by pattern. We touched on this in [Welcome to the terminal](#) (page 12) when we used `mv *.txt my-folder` to move all text files at once.

The three main wildcards are:

*

Matches any sequence of characters (including none), so as we saw previously, `*.txt` matches all files ending in `.txt`.

?

Matches any single character. For example, `file?.txt` matches `file1.txt` and `fileA.txt`, but not `file10.txt`.

[...]

Matches any one of the characters inside the brackets. `file[123].txt` matches `file1.txt`, `file2.txt`, and `file3.txt` only.

Character classes like `[:digit:]` and `[:alpha:]` can be used inside brackets to match whole categories of characters.

Let's see wildcards in action. To list every file in `/usr/bin` whose name starts with `m`:

```
ubuntu@tutorial:~$ ls /usr/bin/m*

/usr/bin/make /usr/bin/man /usr/bin/md5sum [...]
```

Wildcards are expanded by the shell *before* the command runs, so the program receives the resulting list of matching filenames – it never sees the `*` character itself.

Success!

This is the end of our journey around the terminal, and some of the powerful things we can do with it, but it's certainly not all there is to know!

If you wish, you can continue to explore the file system on your virtual machine.

Once you are finished and want to exit the VM, just type "exit".

```
ubuntu@tutorial:~$ exit
```

This will take you out of the VM and back to your live machine. Then, you can run the following commands to delete the VM and remove it completely from your machine:

```
ubuntu@tutorial:~$ multipass delete tutorial
```

```
ubuntu@tutorial:~$ multipass purge
```

Next steps

If you wish to continue learning, we recommend following the [Managing your software](#) (page 47) tutorial next. This tutorial will teach you all the different ways to handle installing, removing and customizing software on your Ubuntu machine, as well as how to safely keep it up-to-date.

Managing your software

If you are new to Ubuntu, you may be wondering what to do after installation. After all, Ubuntu is endlessly customizable according to your needs. There are two types of software found in Ubuntu: **Debian packages** (debs) and **snaps** – we will focus mainly on debs in this tutorial!

To help you get the most from your Ubuntu experience, this tutorial will walk you through managing the software on your Ubuntu machine. This tutorial can be completed using either Ubuntu Server or Ubuntu Desktop.

If you are re-using a virtual machine from a different tutorial, skip directly to [Updating the system with APT](#) (page 49).

Set up the virtual machine

To avoid making changes to your computer we will set up a virtual machine (VM), which provides us with a safe environment to run the commands in. [Multipass](#)³¹ is great for quickly creating Ubuntu virtual machines, so we will use that. It is available on Windows, Mac and Linux, so you don't need to be already running Ubuntu on your computer to follow this tutorial.

³¹ <https://documentation.ubuntu.com/multipass/latest/>

Prerequisites

- The Multipass VM needs **5 GB of disk space**, and **1 GB of memory** available.

Install Multipass

On Ubuntu

You can install Multipass directly from [the Multipass page³²](#) in the online snap store (make sure to select the “latest/stable” version from the dropdown menu next to the install button).

On other operating systems

If you’re on Windows, Mac, or another Linux operating system, Multipass can be installed [using these instructions³³](#).

Open a terminal window

Now that Multipass is installed, we need a terminal window to run our commands in. How you open one depends on your operating system.

On Ubuntu

Press **Ctrl + Alt + T** together and a terminal window will open. You can also search for “Terminal” in your application launcher.

On Windows

PowerShell is the most reliable option for working with Multipass on Windows.

1. Press the Windows key.
2. Type PowerShell.
3. Right-click it and select **Run as Administrator** — this helps avoid permission issues when managing virtual machines.

If you have Windows Terminal installed from the Microsoft Store, that works just as well and offers a nicer experience with tab support.

To confirm everything is working, type `multipass version` and press Enter. If you see version numbers, you’re ready to continue.

³² <https://snapcraft.io/multipass>

³³ <https://documentation.ubuntu.com/multipass/stable/how-to-guides/install-multipass/>

On macOS

1. Press `⌘ + Space` to open Spotlight.
2. Type `Terminal` and press `Enter`.

If you use `iTerm2`, that works perfectly well too.

To confirm everything is working, type `multipass version` and press `Enter`. If you see version numbers, you're ready to continue.

Create the virtual machine

Once you have installed Multipass and opened your terminal, we can launch a new VM using the Ubuntu 26.04 LTS release (codename "resolute"), and let's give our VM the name `tutorial` using the following command in our terminal window:

```
$ multipass launch resolute --name tutorial
```

It may take a little time, depending on the speed of your internet connection. Multipass downloads the most recent daily **image** and creates the VM for us.

Note:

An Ubuntu **image** is a collection of files we need to install and run Ubuntu. We don't need to specify "server" or "desktop" anywhere in our command, because the image is the same for both. The only difference between Ubuntu Server and Ubuntu Desktop is the subset of software packages we use from the Ubuntu Archive.

Once the above command has finished running, we can access the VM by running:

```
$ multipass shell tutorial
```

We will get a "Welcome to Ubuntu" message. Notice that when we run this command, the terminal username changes to `ubuntu` and the `hostname` changes to `tutorial`:

```
ubuntu@tutorial:~ $
```

This shows that we are inside the VM, and this is where we will run all our commands. Now that our setup is complete, let's carry on with the tutorial!

Updating the system with APT

The first thing we always want to do with a new system (whether a VM, container, bare metal, or cloud instance) is to make sure we have the latest versions of all the pre-installed software.

Debian packages, commonly referred to as **debs**, are the standard software package format in Ubuntu. They can be identified by the `.deb` file extension.

Every Linux distribution has their own preferred **package manager** for installing, updating and removing packages. In Ubuntu, the default package manager is [Advanced Packaging](#)

Tool³⁴ (or APT, for short).

APT handles all of your system software (and other deb packages). It provides an interface to the Ubuntu Archive *repository*, so it can access both the **database** of all the packages available in Ubuntu, and the means to handle the **packages** themselves.

There are two APT commands we need to update our system: update and upgrade, which we will always run in that order.

apt update

The apt update command is about the **database**. Any bug fixes in a package (or new versions since your last update) will be stored in the metadata about that package in the database (the **package index**).

When we run the update command it updates the APT database on our machine, fetching the newest available metadata from the package index:

```
$ sudo apt update

Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
19 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

As we can see, it checks (“hits”) the various archives (**pockets**) that updates can come from for the 26.04 LTS release (resolute, resolute-updates, resolute-backports, and resolute-security – remember these, as we’ll come back to them later). It has found some packages that can be upgraded to newer versions. If we want to see which packages those are, we can run the command hinted in the output:

```
$ apt list --upgradable
```

The specific packages included in this list change over time, so the exact packages shown will be different, but the output will be structured like this:

```
bind9-dnswutils/resolute-updates,resolute-security 1:9.20.18-1ubuntu2.1 amd64
[upgradable from: 1:9.20.18-1ubuntu2]
bind9-host/resolute-updates,resolute-security 1:9.20.18-1ubuntu2.1 amd64 [upgradable
from: 1:9.20.18-1ubuntu2]
bind9-libs/resolute-updates,resolute-security 1:9.20.18-1ubuntu2.1 amd64 [upgradable
from: 1:9.20.18-1ubuntu2]
bpftrace/resolute-updates 7.7.0+7.0.0-22.22 amd64 [upgradable from: 7.7.0+7.0.0-15.
15]
libgcrypt20/resolute-updates,resolute-security 1.12.0-2ubuntu0.1 amd64 [upgradable
from: 1.12.0-2]
[...]
```

The output tells us:

³⁴ <https://wiki.debian.org/AptCLI>

- the package name and where the update will come from (e.g. bind9-dnsutils package from resolute-updates and resolute-security),
- the most up-to-date package version available (e.g. 1:9.20.18-1ubuntu2.1),
- the hardware version the update is for (e.g. amd64), and
- what package version is currently installed (e.g. 1:9.20.18-1ubuntu2)

apt upgrade

The `apt upgrade` command is about the **packages** on your system. It looks at the metadata in the package index we just updated, finds the packages with available upgrades, and lists them for us. Once we've checked the proposed upgrade and are happy to proceed, it will then install the newer versions for us.

After we have updated the database (which we did by running `apt update`) we can then upgrade the packages to their newest versions by running:

```
$ sudo apt upgrade
```

When we run this command, it will ask us to confirm if the summary of proposed changes that will be made to our system is what we want.

Let's type `Y`, then press `Enter` to confirm that yes, we do want that, and then the upgrade will proceed. This may take a few minutes.

Tip:

You can use the `-y` flag, which is a shorthand for `--assume-yes`. If we ran the command `sudo apt upgrade -y` it would proceed with the upgrade without asking us to confirm. Shorthand versions of flags are common – for most packages, you can check which flags are equivalent using [the manual pages](https://manpages.ubuntu.com/)³⁵ or using the `man` command, as we'll see later.

³⁵ <https://manpages.ubuntu.com/>

In the output, we'll see where `apt upgrade` is fetching the upgrade from for each package. For example:

```
Get:1 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libgcrypt20 amd64
1.12.0-2ubuntu0.1 [670 kB]
```

APT combines the various elements; the package name (`libgcrypt20`), version (`1.12.0-2ubuntu0.1`), source (`resolute-updates/main`), etc into a single URL that it can use for the download. The package is then unpacked, and the upgrade applied to the system.

Note:

These commands only upgrade the packages for the release of Ubuntu that we are using (26.04 LTS). If we wanted to upgrade the entire system to the next release of Ubuntu (e.g. from 22.04 LTS to 24.04 LTS), we would use the `do-release-upgrade` command. See this guide on [how to upgrade your release](#) (page 443) for more information.

It's important to know that `apt upgrade` will only handle packages that can be straightforwardly upgraded. If the package has **dependency** issues (i.e., the version you have “depends” on other packages that also need to be added, upgraded or removed), you would need to use `sudo apt dist-upgrade` instead. The `dist-upgrade` command is able to resolve conflicts between package versions, but it *could* end up removing some packages. So while `apt upgrade` is safe to use unattended (in a script, for example), you should only use `dist-upgrade` when you can review which packages may be removed.

Searching with APT

Now we're up-to-date, we can start exploring! As with any other database, we can search the list of available packages using APT in order to find software. Let's say that we want to find a web server, for example. We can run the following command:

```
$ apt search webserver
```

This will return us a long list of all “webserver” packages it can find. But some of the descriptions don't actually contain the text “webserver” – like in this section of the list:

```
inotify-tools/resolute 4.25.9.0-1 amd64
  command-line programs providing a simple interface to inotify

ipcalc/resolute 0.51-1build1 all
  parameter calculator for IPv4 addresses

iwatch/resolute 0.2.2-12 all
  realtime filesystem monitoring program using inotify
```

We can use `apt show` to inspect the description and summary details of any package, so let's take a closer look at `ipcalc` from our list:

```
$ apt show ipcalc
```

The summary has been replaced with [...] for brevity, but we can see that the text “webserver” is in the long description of the “Description” field.

```
Package: ipcalc
Version: 0.51-1build1
[...]
APT-Sources: http://archive.ubuntu.com/ubuntu resolute/universe amd64 Packages
Description: parameter calculator for IPv4 addresses
  ipcalc takes an IPv4 address and netmask and calculates the resulting
  broadcast, network, Cisco wildcard mask, and host range. By giving a
  second netmask, you can design sub- and supernetworks. It is also
  intended to be a teaching tool and presents the results as
  easy-to-understand binary values.
.
Originally, ipcalc was intended for use from the shell prompt, but a
CGI wrapper is provided to enable colorful HTML display through a
webserver.
```

While we used `apt` in the `apt show` command, in many places you will see reference to `apt-get` and `apt-cache` instead of `apt`. Historically, the *database* part of APT was accessed using `apt-cache` (e.g. `apt-cache show ipcalc`), and the *packages* part of APT used `apt-get` (e.g. `apt-get install ipcalc`).

APT has recently been streamlined, so although it uses `apt-get` and `apt-cache` “behind the scenes” (and these commands do still work), we don’t need to worry about remembering which command to use – we can use the more convenient `apt` directly. To find out more about these packages and how to use them (or indeed, any package in Ubuntu!) we can refer to the manual pages.

Run `man apt`, `man apt-get` or `man apt-cache` in the terminal to access the manuals for these packages on the command line, or view the same content in the [online manual pages](#)³⁶.

Installing deb packages

For the examples for this section, we’re going to use the popular web server package, [Apache2](#)³⁷.

APT gives us a lot of details about what will be included in the installation, and it’s always important to understand the implications of a command *before* we run it. We’ll be taking a close look at the details APT gives us, so we need to be careful in this section.

When we run a command that asks us “Do you want to continue? [Y/n]”, make sure to type `N` for “no” and then press `Enter` unless instructed otherwise – this will let us see the output of the commands without making changes that then need to be undone.

Installing deb packages using APT is done using the `apt install` command. We can install either a single package, or a list of packages at once, by including their names in a space-separated list after the `install` command, in this format:

```
sudo apt install <package 1> <package 2> <package 3>
```

³⁶ <https://manpages.ubuntu.com>

³⁷ <https://httpd.apache.org/>

About sudo

We've seen the `sudo` prefix in a couple of commands already, and you may be wondering what that's about. In Linux, system tasks (like installing software) need elevated administrator permissions. These permissions are often called "root access", and a user with root access is called a "root user".

However, it can be dangerous to operate your machine as a root user – since root access gives you full system control the whole time, it allows you to change or delete important system files. It's very easy to accidentally break your system in root mode!

Instead, we use `sudo` (which is short for superuser do). This command is a safety feature that grants regular users *temporary* (per command) admin privileges to make system changes. It's still important for us to always understand what a command does before we run it, but using `sudo` means we purposefully limit any potential mistakes to a single command.

About dependencies

As we hinted earlier, packages often come with **dependencies** – other packages that *your* package needs so it can function. Sometimes, a package might depend on a specific version of another package. If a package has dependencies, then installing a package via `apt` will also install any dependencies, which ensures the software can function properly.

APT tells us how it will resolve any dependency conflicts or issues when we run the `install` command. Let's try this for ourselves, but remember, we **don't** want to proceed with the install yet, so let's type `N` when it asks us if we want to continue:

```
$ sudo apt install apache2
```

The output should be similar to the below.

```
Installing:
  apache2

Installing dependencies:
  apache2-bin      libapr1t64      libaprutil1t64
  apache2-data     libaprutil1-dbd-sqlite3  liblua5.4-0
  apache2-utils    libaprutil1-ldap  ssl-cert

Suggested packages:
  apache2-doc  apache2-suexec-pristine | apache2-suexec-custom  www-browser

Summary:
  Upgrading: 0, Installing: 10, Removing: 0, Not Upgrading: 0
  Download size: 2116 kB
  Space needed: 8218 kB / 1328 MB available

Continue? [Y/n]
```

It tells us:

- which packages we have but don't need (not shown above because our Apache2 example has none),

- additional packages that will be installed (these are our dependencies),
- suggested packages, and
- a summary of which *new* packages will be present on the system after the install is done (which in this case is `apache2` itself, and all its dependencies).

Let's try and make sense of this output.

Types of dependencies

The relationship between a package and any other packages follows the [Debian policy on binary dependencies](#)³⁸, which we'll briefly look at here. The most common ones you might come across are: `depends`, `recommends`, and `suggests` (although there are others!), so we'll take a look at these three.

- **depends:** Absolutely required, the package won't work without it. If we try to remove a package that is depended on by another, both will be removed!
- **recommends:** Strongly dependent, but not absolutely necessary (which means the package will work better with it, but can still function without it)
- **suggests:** Not needed, but may enhance the usefulness of the package in some way.

We can see, using `apt show`, exactly which packages fall into each of these categories. Let's use `Apache2` as our example again:

```
$ apt show apache2

[...]
Depends: apache2-bin (= 2.4.66-2ubuntu2.4), apache2-data (= 2.4.66-2ubuntu2.4),
apache2-utils (= 2.4.66-2ubuntu2.4), media-types, procps, perl:any
Recommends: ssl-cert
Suggests: apache2-doc, apache2-suexec-pristine | apache2-suexec-custom, ufw, www-
browser
[...]
```

We see that `ssl-cert` is a recommended package. In Ubuntu, the default configuration of `apt` `install` is set to install recommended packages alongside `depends`, so when we ran the `apt install apache2` command, `ssl-cert` was included in the proposed packages to be installed (even though it's only recommended, not strictly needed).

```
[...]
Installing dependencies:
  apache2-bin      libapr1t64          libaprutil1t64
  apache2-data     libaprutil1-dbd-sqlite3  liblua5.4-0
  apache2-utils    libaprutil1-ldap       ssl-cert
[...]
```

We can override this behavior by passing the `--no-install-recommends` flag to our command, like this:

³⁸ <https://www.debian.org/doc/debian-policy/ch-relationships.html#binary-dependencies-depends-recommends-suggests-enh>


```
$ sudo apt install apache2 --no-install-recommends
```

Then the output becomes the following (type N at the prompt again to avoid installing for now):

```
[...]
Installing dependencies:
  apache2-bin  apache2-utils  libaprutil1-dbd-sqlite3  libaprutil1t64
  apache2-data  libapr1t64      libaprutil1-ldap        liblua5.4-0

Suggested packages:
  apache2-doc  apache2-suexec-pristine | apache2-suexec-custom  www-browser

Recommended packages:
  ssl-cert

Summary:
  Upgrading: 0, Installing: 9, Removing: 0, Not Upgrading: 0
  Download size: 2098 kB
  Space needed: 8149 kB / 1328 MB available
[...]
```

Now, we see that `ssl-cert` is only mentioned as a recommended package, and is excluded from the list of packages to be installed.

There is a second flag we could pass – the `--install-suggests` flag. This will not only install the strict dependencies and recommended packages, but *also* the suggested packages. From our previous output, it doesn't look like too much, right? It's only four additional packages.

But actually, if we run this command:

```
$ sudo apt install apache2 --install-suggests

[...]
Installing dependencies:
  apache2-bin      apache2-utils      libaprutil1-ldap
  apache2-data     chromium-browser   libaprutil1t64
  apache2-doc      libapr1t64         liblua5.4-0
  apache2-suexec-pristine  libaprutil1-dbd-sqlite3  ssl-cert

Summary:
  Upgrading: 0, Installing: 13, Removing: 0, Not Upgrading: 0
  Download size: 6078 kB
  Space needed: 34.1 MB / 1328 MB available
```

By comparing the amount of space needed with “suggests” included, to the previous output, we can see that there's a considerable increase! Sometimes, including suggested packages will cause the package to not be installed due to a lack of space on the system.

This is because each of these suggested packages also comes with their own lists of dependencies, including suggested packages, all of which would *also* be installed. It's perhaps clear to see why this is not the default setting!

What if we remove a dependency?

We'll go into more detail about removing packages later, but for now, let's see what happens if we remove a required *dependency*. First, we should (finally!) install the `apache2` package. Let's run the following command again, but this time when we are asked whether we want to continue, let's press Y and then Enter to confirm, and APT will install the package:

```
$ sudo apt install apache2
```

One of the required dependencies is the `apache2-data` package. Let's try to remove it using `apt remove`:

```
$ sudo apt remove apache2-data
```

Once again, `apt` won't proceed without confirmation, so we get the following output – let's take a look before we choose anything:

```
[...]
The following packages were automatically installed and are no longer required:
  apache2-bin      libapr1t64          libaprutil1-ldap    liblua5.4-0
  apache2-utils    libaprutil1-dbd-sqlite3  libaprutil1t64      ssl-cert
Use 'sudo apt autoremove' to remove them.

REMOVING:
  apache2  apache2-data

Summary:
  Upgrading: 0, Installing: 0, Removing: 2, Not Upgrading: 0
  Freed space: 1349 kB

Continue? [Y/n]
```

Let's break this down a little bit, because there are some subtle differences here that we want to understand before we proceed.

- “The following packages were automatically installed and are no longer required”
These were other dependencies that `apache2` needed, but none of them depend upon `apache2-data`, so even if we remove `apache2` and `apache2-data` they would still be functional – they just aren't used by any other installed packages...and so have no reason to be there anymore. They won't be removed, APT is helpfully telling us so we're aware of them.
- “The following packages will be REMOVED”
These are the packages that will be removed directly - we've told APT we want to remove `apache2-data`, so we expect that to be included, but it will also remove `apache2` itself! This is because `apache2-data` is a required dependency, and `apache2` won't function *at all* without it.

Let's now choose Y to confirm we want to remove this dependency.

Warning:

Removing dependencies can, at worst, cause a system to become unusable – you should always be careful when doing so. If you remove a dependency that is part of a chain, the removals will cascade up the chain as each dependency and the package that depends on it are removed. You can end up removing more than you originally anticipated!

Auto-remove dependencies

So, we have removed the `apache2` and `apache2-data` packages, but the other dependencies that were installed alongside `apache2` are still there. The output of our `remove` command gave us the hint about how to deal with these redundant packages – the `autoremove` command:

```
$ sudo apt autoremove
```

When we run this command, `apt` once again gives us a summary of the operation we requested, but let's choose `N` for now when it asks if we want to continue:

```
[...]
REMOVING:
  apache2-bin      libapr1t64          libaprutil1-ldap    liblua5.4-0
  apache2-utils    libaprutil1-dbd-sqlite3  libaprutil1t64      ssl-cert

Summary:
  Upgrading: 0, Installing: 0, Removing: 8, Not Upgrading: 0
  Freed space: 6869 kB

Continue? [Y/n]
```

You may be wondering why we don't need to specify any packages when we call the `autoremove` command – after all, we've just been dealing with packages related to `apache2`. This is because `apt` will check all the packages on your system. It examines the dependency tree, and if the original reason for the package to be installed no longer exists (i.e., it isn't needed by anything), it will be flagged for auto-removal.

But!

We might, in the future, uninstall `Apache2` without uninstalling the redundant packages at the time. We might have found another use for `ssl-cert`, perhaps in a script that makes use of SSL certificates. So how can we keep the `ssl-cert` package, even though it's flagged for auto-removal?

We can solve this problem, and un-flag the `ssl-cert` package for removal, by *manually* installing it:

```
$ sudo apt install ssl-cert
```

This sets `ssl-cert` to **manually installed**. We might well wonder “why didn't APT didn't ask us to confirm anything this time?”. In this case, it's because `ssl-cert` is already present on the system so APT doesn't need to install anything new.

```
[...]
ssl-cert is already the newest version (1.1.3ubuntu2).
ssl-cert set to manually installed.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0
```

If the `ssl-cert` package is manually installed on our system, by us, then `apt` knows the package is wanted, and we can see that it has been removed from the auto-remove list so our next autoremove will not uninstall it. Let's test this, just to make sure!

```
$ sudo apt autoremove
```

This time we'll select `Y` when prompted, and then we can run `apt list ssl-cert` to quickly see if our `ssl-cert` package is still on the system:

```
$ apt list ssl-cert

ssl-cert/resolute,now 1.1.3ubuntu2 all [installed]
```

If you're curious, you can also run `apt list apache2` to see how the output differs for a package that was once installed and then removed!

Anyway, we're not quite finished with the Apache2 package, so let's reinstall it:

```
$ sudo apt install apache2
```

And this time select `Y` to confirm when it asks.

Customize configuration

In general, the default package configuration should just work well, and work "out of the box" when it's installed. But it's almost inevitable that, sooner or later, we'll want to customize the package so that it better fits our own purposes.

Before we try to customize the package, we should probably look at what files are included in it. We can check this using `dpkg(1)`³⁹, which is the Debian package manager. Although APT is now more commonly used for basic package handling, `dpkg` retains some really helpful commands for examining files and finding out package information. It's installed by default on Ubuntu systems so we can use it directly:

```
$ dpkg --getfiles ssl-cert
```

This gives us the following list of files and their directory structure (the end of the list is truncated for brevity):

³⁹ <https://manpages.ubuntu.com/manpages/resolute/man1/dpkg.1.html>

```
/.  
/etc  
/etc/ssl  
/etc/ssl/certs  
/etc/ssl/private  
/usr  
/usr/lib  
/usr/lib/systemd  
[...]  
/usr/share/man  
/usr/share/man/man8  
/usr/share/man/man8/make-ssl-cert.8.gz  
/usr/share/ssl-cert  
/usr/share/ssl-cert/sslkey.cnf  
[...]
```

If we find a file but we're not sure what package it comes from, `dpkg` can help us there too! Let's use the example of one of the configuration files from the previous output: `/usr/share/ssl-cert/sslkey.cnf` and do a search for it using `dpkg`:

```
$ dpkg --search /usr/share/ssl-cert/sslkey.cnf  
  
ssl-cert: /usr/share/ssl-cert/sslkey.cnf
```

Although this is an obvious case, because we already know the source of this file, the `dpkg` search function is really useful for tracking down the sources of files we don't know about!

Conffiles

Most of a package's configuration is handled through [configuration files](#)⁴⁰ (often known as **conffiles**). Conffiles often contain things like file paths, logs and debugging configuration, kernel parameters (which can be changed to optimize system performance), access control, and other configuration settings. The actual parameters available will vary from one package to another.

Package conffiles are different from all other files delivered in a package. A package may have any number of conffiles (including none!). Conffiles are explicitly marked by the package maintainer during development to protect local configuration from being overwritten during upgrades so that your changes are saved. This is not the case for any other types of files – changes you make to regular files in that package *will be overwritten* during an upgrade.

⁴⁰ <https://www.debian.org/doc/debian-policy/ap-pkg-conffiles.html#automatic-handling-of-configuration-files-by-dpkg>

How upgrades are handled

Since a conffile can be changed by us, we might end up with conflicts when the package maintainer changes those same files. Therefore, it's important to understand how such conflicts are handled.

We can show the four possible upgrade scenarios using the following table. What happens during an upgrade depends on whether the conffile on our system has been changed by us ("changed/not changed by user"), and whether the version's default content has been changed by the package maintainer ("changed/not changed by maintainer"):

The conffile is...	not changed by maintainer	changed by maintainer
...changed by user	Keep user's changes	Ask user
...not changed by user	No changes to make	Apply changes from update

So we can see that if we do make changes to a conffile, APT will never overwrite our changes without asking us first.

Identifying conffiles

Out of the list of files in a package, how do we know which ones are the conffiles?

After all, they are not marked by any particular file extension, and although they are often found in the `/etc/` directory, they don't *have* to be there. As we saw before, the only thing conffiles have in common is that the package maintainer decided to mark them as such.

But that's our clue! So once more, `dpkg` can come to our rescue. The following command will show us (`--show`) the subset of files in the `apache2` package that have been marked as "Conffiles" (`-f='${Conffiles}\n'`) by the maintainer and shows each on a new line (`\n`) in the output:

```
$ dpkg-query --show -f='${Conffiles}\n' apache2
```

If you want to understand more about what this command does, you can refer to the manual page by typing `man dpkg-query --show`, and it will talk you through all the options.

Unlike `dpkg --getfiles`, `dpkg-query` *also* gives us a string of letters and numbers. This string is known as the "**MD5 checksum**" or "**MD5 hash**".

```
/etc/apache2/apache2.conf 354c9e6d2b88a0a3e0548f853840674c
/etc/apache2/conf-available/charset.conf e6fbb8adf631932851d6cc522c1e48d7
/etc/apache2/conf-available/localized-error-pages.conf
f542d267bfce7815f9453eb1476e5f73
/etc/apache2/conf-available/other-vhosts-access-log.conf
2cad303fc4221d6b0068a8b37597b9fb
/etc/apache2/conf-available/security.conf 332668933023a463046fa90d9b057193
[...]
```

We can see the checksum of a specific file by running this command, which returns us the checksum followed by the file and its location:

```
$ md5sum /etc/apache2/apache2.conf
354c9e6d2b88a0a3e0548f853840674c /etc/apache2/apache2.conf
```

You might well be wondering “why do we care about that?” since they match (in this example). The checksum is like a fingerprint - it’s unique for every *version* of a file, so any time the file is changed it will get a new checksum – which allows us to see **if a file has been changed** from the default.

Verifying checksums

Let’s set up a situation so we can poke a bit at this idea. We can start by making some changes to a conf file. In Apache2, the main conf file is `/etc/apache2/apache2.conf`, so let’s use that. In a situation where we are setting up a new web server, we might reasonably want to increase the `LogLevel` from “warn” to “debug” to get more debugging messages, so let’s run this command and use `sed` to make that change in the conf file:

```
$ sudo sed -e 's/LogLevel warn/LogLevel debug/' -i /etc/apache2/apache2.conf
```

We won’t be prompted to confirm if we want to make these changes – but we do need root access so we use `sudo` in our command. As we hinted in the section about `sudo`, the fact that we can make these changes without needing to confirm is why it can be so easy to break your system when you’re operating as root! Try running the command without the `sudo`, and you will get a “permission denied” error.

Next, we’ll restart our Apache2 server so that we can activate our configuration changes:

```
$ sudo systemctl restart apache2
```

Now if we run the `md5sum` command again, we can see the hash changed:

```
ubuntu@tutorial:~$ md5sum /etc/apache2/apache2.conf
1109a77001754a836fb4a1378f740702 /etc/apache2/apache2.conf
```

This works great if we know that there’s a file we changed, but what about if someone else tampered with a file, and we don’t know which one? In that case, we can use:

```
$ dpkg --verify apache2
```

This will verify the checksums of the files on our system against those held in the package index for `apache2`, and return a rather strange looking result if (or when) it finds a mismatch:

```
??5????? c /etc/apache2/apache2.conf
```

Which is exactly what we were expecting to see, since we know we changed this file.

But what if something else was messed with...something that shouldn't be, and something not changed by us? Let's make a "silly" change to a different file to test this – in this case, changing all instances of the word "warning" to "silly" in a random package file:

```
$ sudo sed -e 's/warning/silly/' -i /usr/sbin/a2enmod
```

And then run the verification again with:

```
$ dpkg --verify apache2
??5????? c /etc/apache2/apache2.conf
??5????? /usr/sbin/a2enmod
```

Note:

You might have noticed there's a "c" next to the top line but not the bottom – the "c" shows the file is a conffile.

dpkg can tell that the file has been changed, but won't tell us what the change was. However, since the file in question is not a conffile, we know that the change *won't be preserved* if we upgrade the package. This means that we can overwrite the changes and restore the default package content by "reinstalling" Apache2. We can use the `--reinstall` flag to force apt to re-unpack all of the default content:

```
$ sudo apt install --reinstall apache2
```

If we then verify once more...

```
$ dpkg --verify apache2
??5????? c /etc/apache2/apache2.conf
```

...so we can see that our change to the conffile has been preserved because the checksums are different, but the `a2enmod` file isn't listed anymore because it has been restored to the default. Phew!

Note:

We can use `sudo apt install <package>` to upgrade an installed package, but this will only upgrade to the latest version. In our case, we were already on the latest version of Apache2, so we needed to force APT to re-unpack the content to overwrite our "silly" changes.

Removing packages

Since we have just reinstalled the Apache2 package, we know it is in good shape. But what if we decide we're done with it and just want to remove it? Then we can run:

```
$ sudo apt remove apache2

The following packages were automatically installed and are no longer required:
  apache2-bin  apache2-utils  libaprutil1-dbd-sqlite3  libaprutil1t64
  apache2-data  libapr1t64      libaprutil1-ldap        liblua5.4-0
Use 'sudo apt autoremove' to remove them.

REMOVING:
  apache2

Summary:
  Upgrading: 0, Installing: 0, Removing: 1, Not Upgrading: 0
  Freed space: 471 kB

Continue? [Y/n]
```

Let's type Y to proceed.

As before, we see that the dependencies will still be there even when apache2 has been removed. Let's check with dpkg...

```
$ dpkg --getfiles apache2
```

...and see what else might be left behind...

```
/etc
/etc/apache2
/etc/apache2/apache2.conf
/etc/apache2/conf-available
/etc/apache2/conf-available/charset.conf
/etc/apache2/conf-available/localized-error-pages.conf
/etc/apache2/conf-available/other-vhosts-access-log.conf
/etc/apache2/conf-available/security.conf
/etc/apache2/conf-available/serve-cgi-bin.conf
/etc/apache2/conf-enabled
/etc/apache2/envvars
/etc/apache2/magic
/etc/apache2/mods-available
/etc/apache2/mods-available/access_compat.load
/etc/apache2/mods-available/actions.conf
[...]
```

This looks suspiciously like the list of conffiles we saw earlier, right?

Also removing configuration

As it turns out, removing a package doesn't automatically remove the conffiles. But – this is intentional, for our convenience.

By leaving the conffiles in place, if we decide to reinstall `apache2` again in the future, we don't need to spend time setting up all our configuration again.

Let's see the difference in installing `apache2` after it has been installed (and removed) compared to the first time we installed it:

```
$ sudo apt install apache2
```

Notice that it did not ask us to confirm if we wanted to proceed this time. Why not? As we saw earlier, the "Y/n" confirmation is shown when there are dependencies, and we know that `Apache2` *has* dependencies.

...Ah! But this time, we didn't run `autoremove` when we uninstalled `Apache2`, so the dependencies are still installed on our system. This means that when we ask `apt` to install `apache2` now, there is nothing missing and we are getting *exactly* what we are asking for.

Since the dependencies and conffiles are still there, we can use our former config immediately. It even retains the changes we made before, which we can verify by looking at the checksum again:

```
$ md5sum /etc/apache2/apache2.conf
```

Removing and purging

What if we decide that we don't want the changed conffiles? Perhaps we want to go back to the default installation, or we know we won't want to use the package ever again – how can we ensure that all the conffiles are removed at the same time as we remove the package?

In that case, we can use the `--purge` option of the `remove` command:

```
$ sudo apt remove --purge apache2

[...]
The following packages were automatically installed and are no longer required:
  apache2-bin  apache2-utils  libaprutil1-dbd-sqlite3  libaprutil1t64
  apache2-data  libapr1t64      libaprutil1-ldap        liblua5.4-0
Use 'sudo apt autoremove' to remove them.

REMOVING:
  apache2*

Summary:
  Upgrading: 0, Installing: 0, Removing: 1, Not Upgrading: 0
  Space needed: 0 B / 1318 MB available

Continue? [Y/n]
```

If we look very carefully, we see a little asterisk (*) in the output.

```
REMOVING:  
apache2*
```

This tiny indicator tells us that the package will be removed AND purged. However, it still does not remove the dependencies (or the conffiles of those dependencies).

Let's type Y again to confirm we want to proceed. Then, once the removal is complete, we can check the list once more:

```
$ dpkg --get-selections | grep apache2  
  
dpkg-query: package 'apache2' is not installed  
Use dpkg --get-architecture (= dpkg-query --get-architecture) to list archive files contents.
```

This time the output is very different!

Note:

We could also use the `dpkg-query --show -f='${Conffiles}\n' apache2` command from earlier, and `dpkg-query` will find no packages matching `apache2`.

There are other ways to change package files. If you would like to read more, check out our [guide to changing package files](#) (page 982).

What else is on our system?

As we saw earlier, we can search the APT package database for keywords using `apt search <keyword>` to find software we might want to install. We can also see all the packages we already have using `apt list`, although it can be easier to navigate and more informative if we use `dpkg -l` instead – then we can use the up and down arrow keys on our keyboard to scroll (or press Q to return to our terminal prompt).

For every package, we can see what versions of it exist in the database:

```
$ apt policy apache2
```

This will return a summary of all the versions that exist on our particular Ubuntu release, ordered by “most recent” first:

```
apache2:  
  Installed: (none)  
  Candidate: 2.4.66-2ubuntu2.1  
  Version table:  
    2.4.66-2ubuntu2.1 500  
        500 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages  
        500 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages  
    2.4.66-2ubuntu2 500  
        500 http://archive.ubuntu.com/ubuntu resolute/main amd64 Packages
```

We know that Apache2 isn't installed right now, because we removed and purged it, which is why the installed version shows as "none":

```
Installed: (none)
```

If we were to install the default package, we would get this one:

```
Candidate: 2.4.66-2ubuntu2.1
```

Under each version we are also shown the **source**. The newest version (2.4.66-2ubuntu2.1) comes from `resolute-updates (main)` and `resolute-security (main)`. The *original* version (2.4.66-2ubuntu2) comes from `resolute (main)`. This tells us that this was the version released with the 26.04 LTS (Resolute Raccoon).

Installing older package versions

We can install specific older versions if we want to, for example, to satisfy dependency requirements of another package. We can do that by specifying the package name and version:

```
sudo apt install <package>=<version>
```

However, this can be tricky and often leads to conflicts in dependency versions as APT always wants to install the most recent version. We can see an example of this if we run the following command:

```
$ sudo apt install apache2=2.4.66-2ubuntu2
```

APT warns us that the version of `apache2` we want to install depends on earlier versions of the dependencies, but it helpfully tells us which dependency versions we need to successfully install the package we want.

```
[...]
Solving dependencies... Error!
Some packages could not be installed. This may mean that you have
requested an impossible situation or if you are using the unstable
distribution that some required packages have not yet been created
or been moved out of Incoming.
The following information may help to resolve the situation:

Unsatisfied dependencies:
  apache2 : Depends: apache2-bin (= 2.4.66-2ubuntu2) but 2.4.66-2ubuntu2.1 is to be
installed
              Depends: apache2-data (= 2.4.66-2ubuntu2) but 2.4.66-2ubuntu2.1 is to be
installed
              Depends: apache2-utils (= 2.4.66-2ubuntu2) but 2.4.66-2ubuntu2.1 is to be
installed
Error: Unable to satisfy dependencies. Reached two conflicting assignments:
  1. apache2:amd64=2.4.66-2ubuntu2 is selected for install
  2. apache2:amd64=2.4.66-2ubuntu2 Depends apache2-bin (= 2.4.66-2ubuntu2)
    but none of the choices are installable:
    - apache2-bin:amd64=2.4.66-2ubuntu2 is not selected for install
```

So, all we need to do is first install the dependencies, and then run the install command again. Remember that we can install multiple packages at once by separating them with spaces. We can also break the command over multiple lines using backslashes (\) to make it easier to read, but it will still be run as a single command:

```
$ sudo apt install apache2-bin=2.4.66-2ubuntu2 \ >
  apache2-data=2.4.66-2ubuntu2 \ > apache2-utils=2.4.66-2ubuntu2 \ >
  apache2=2.4.66-2ubuntu2
```

APT will warn us that we are downgrading the package, but let us press Y to confirm (when prompted), and it will go ahead and downgrade us anyway. Let's run the following command again to get confirmation that we're running on an older version:

```
$ apt policy apache2

apache2:
  Installed: 2.4.66-2ubuntu2
  Candidate: 2.4.66-2ubuntu2.1
  Version table:
     2.4.66-2ubuntu2.1 500
        500 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packag>
        500 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Pack>
 *** 2.4.66-2ubuntu2 500
        500 http://archive.ubuntu.com/ubuntu resolute/main amd64 Packages
        100 /var/lib/dpkg/status
lines 1-10/10 (END)
```

Where do packages come from?

You may be wondering by now “where exactly do all these packages come from?”. We’ve spotted a few sources very briefly throughout this tutorial, but haven’t paid direct attention to them yet. Let’s take a little time now to define what we mean by all these different sources that APT can pull packages from.

The source behind APT is the **Ubuntu Package Archive**. This Archive splits into many layers, each with its own terminology. The different terminology is quite confusing at first, but we’ve seen a few of the terms already. So if we take a look, layer-by-layer, we’ll see not just what all the terms mean, but how they all fit together.

Let’s have a quick overview with this diagram. The general flow is that the Archive splits into **Ubuntu series**. Each series is split up into **pockets**, and then each pocket contains four **components**. If we tried to show all of this on one diagram, it would be quite extensive, so let’s take a look through a single path.

Series

The **series** is a set of packages that are released with a specific version of Ubuntu – they’re usually referred to by their codename (e.g., mantic, noble and oracular in our diagram). Each version of Ubuntu may have multiple releases. For example, an LTS will have an initial release when it launches (e.g. 24.04 LTS), and then “subsequent point releases” (e.g. 24.04.1 LTS) – these are all part of the same series (noble).

In practice, people often use the term “Ubuntu release” and “Ubuntu series” interchangeably.

Pockets

Every Ubuntu series (noble, jammy, etc) is split into **pockets**, which are related to the software development/release lifecycle:

- **-release** contains the packages as they are at release time.
- **-proposed** contains package updates while they are being tested.
- Once an update is released, they come from either **-security** or **-updates** depending on whether they are a security-related update or not.
- And **-backports**, which contains packages that were not available at release time.

This is why earlier, we saw that some updates came from resolute-updates or resolute-security. These refer to updates and security updates, respectively, from the resolute series. Pockets are usually appended to the end of the series, and it’s quite common to see the hyphen (-) included when referring to pockets.

Remember – the original version of the apache2 package we saw came from resolute. The -release pocket only includes the software that was part of the original LTS release, and so it takes the name of the Ubuntu series by default (i.e., the -release pocket is implied).

Components

Each pocket is split into four **components**, depending on whether the packages they contain are *open source* or *closed source*, and whether they are officially supported by Canonical or are maintained by the Ubuntu Community:

	Open source	Closed source
Officially supported	main	restricted
Community supported	universe	multiverse

- **main** contains the open-source packages that are officially supported by Canonical. These packages are either installed on every Ubuntu machine, or are very widely used for various types of systems and use-cases.
- **universe** holds all other open-source packages in Ubuntu, which are typically maintained by the Debian and Ubuntu communities, but may also include additional security coverage from Canonical under [Ubuntu Pro](https://ubuntu.com/pro)⁴¹, which is available free for personal use on up to five machines.
- **restricted** contains the packages that are officially supported by Canonical but are not available under a completely free license.
- **multiverse** contains community-maintained proprietary software – these packages are completely unsupported by Canonical.

If you would like more information about the Ubuntu release process, how packages are produced, or to learn more about the sort of terminology you might come across, you may be interested in the [Ubuntu Project documentation](https://documentation.ubuntu.com/project/)⁴², which is a great resource containing all this information (and much more!).

Installing a .deb file

Although APT is the preferred way to install packages on your system, due to its ability to handle dependencies and keep software up-to-date, not every package is available in the APT repository – especially if they are so old they are no longer maintained, or conversely, are the newest version still in development!

We can install .deb files that aren't in the APT repository using `dpkg` – all we need is to download the .deb file, and we can run a command like this to install it:

```
sudo dpkg -i <file-name.deb>
```

But – APT is helpful here too. Even if we get a .deb file that isn't from the Ubuntu Archive, we can still install it with APT so that if there are dependencies that can be resolved automatically from the Archive – they will be!

⁴¹ <https://ubuntu.com/pro>

⁴² <https://documentation.ubuntu.com/project/>

```
sudo apt install ./file-name.deb
```

If we ever do want to install a .deb file, APT is definitely the most convenient way to do it. We may still need to handle *some* dependencies manually, but now we have the knowledge to be able to do that.

Luckily, most of the packages you will ever need are likely to be found through APT. If it's not, it's worth checking if the software is available as a **snap** instead.

Snaps

Snaps are a newer, self-contained software format that were developed to be a more portable and easy-to-use alternative to debs. They come with all their dependencies pre-bundled so that there is no need for a package management tool to track dependencies, and they run inside sandboxed environments that limit their interactions with the rest of the system.

Instead of **versions** as we have them in debs, snaps use the concept of **channels**⁴³ to define which release of a snap is installed.

By default, snaps are kept automatically up-to-date, so we don't need to remember to update and upgrade them. There are times on a live system, such as a server in a production environment, where we might not want to have updates automatically applied. In those cases, we can **turn off automatic updates**⁴⁴ and refresh the system snaps when it's convenient (for example, during a maintenance window).

If you would like to try out snaps, we recommend the excellent **quickstart tour**⁴⁵ tutorial in the snap documentation. Feel free to continue using the VM we've been using in this tutorial while exploring!

Completion!

Once you are finished and want to leave the tutorial, you can run:

```
$ exit
```

This will take you out of the VM and back to your live machine. Then, you can run the following commands to delete the VM and remove it completely from your machine:

```
$ multipass delete tutorial
```

```
$ multipass purge
```

⁴³ <https://snapcraft.io/docs/channels>

⁴⁴ <https://snapcraft.io/docs/managing-updates>

⁴⁵ <https://snapcraft.io/docs/get-started>

Summary

Congratulations, we made it to the end! We've covered a lot of material in this tutorial, so let's do a quick recap of what we've learned:

Finding, installing and removing packages

- How to update and upgrade all our system's software with APT:
 - `sudo apt update && sudo apt upgrade`
- How to search for software using keywords or strings:
 - `apt search <keyword>` or `apt search "some content string"`
- How to see the description of a package, including what dependencies it has:
 - `apt show <package name>`

Or how to check what package versions are available:

- `apt policy <package>`
- How to install packages...
 - `sudo apt install <package1> <package2>`
- How to see all the files a package contains
 - `dpkg --getfiles <package>`
- How to find out what package a file belongs to:
 - `dpkg --get-selections <path/to/file>`
- ...And how to remove packages again! As well as the difference between removing and purging.
 - `sudo apt remove <package>`
- We even learned how to downgrade to older versions of APT packages, and all about APT sources.

Customizing package configuration

- How to find the conffiles in a package:
 - `dpkg-query -f='${Conffiles}\n' <package>`
- How to see if package files have been changed:
 - `dpkg --get-selections <package>`
- ...And if a non-conffile has been changed by accident, we can fix it with:
 - `sudo apt install --reinstall <package>`
- We know that our changes to conffiles are always safely preserved, while changes to non-conffiles are reverted at the next upgrade or security fix.
- Importantly, we know how to verify checksums with `md5sum` or similar tools, which helps us to more safely build packages from source.
- And finally, we learned about snaps!

Attach your Ubuntu Pro subscription

Attaching the [Ubuntu Pro](#)⁴⁶ subscription to Ubuntu brings you the [enterprise lifecycle](#)⁴⁷, including [Linux kernel livepatching](#)⁴⁸, access to [FIPS-validated packages](#)⁴⁹, and compliance with [security profiles](#)⁵⁰ such as CIS or DISA-STIG. This is not required for Ubuntu Pro instances through public clouds⁵¹ such as [AWS](#)⁵², [Azure](#)⁵³ or [GCP](#)⁵⁴, since these are automatically attached from launch.

Note:

Subscriptions are not just for enterprise customers. Anyone can get a [personal subscription](#)⁵⁵ for free on up to 5 machines, or 50 if you are an [official Ubuntu Community member](#)⁵⁶.

⁵⁵ <https://ubuntu.com/pro>

⁵⁶ <https://wiki.ubuntu.com/Membership>

The following instructions explain how to attach your subscription to your Ubuntu systems.

Step 1: Install the Ubuntu Pro Client

This step is necessary for Ubuntu Pro users or holders of personal subscriptions. If you are an Ubuntu Pro user through a public cloud offering, your subscription is already attached and you can skip these instructions.

First, make sure that you have the latest version of the Ubuntu Pro Client running. The package used to access the Pro Client (pro) is `ubuntu-pro-client`:

```
$ sudo apt update && sudo apt install ubuntu-pro-client
```

Most systems will have `ubuntu-pro-client` installed by default. If that is the case, this install command will upgrade the package to the latest version.

Step 2: Attach your subscription

To attach your machine to a subscription, run the following command in your terminal:

```
$ sudo pro attach
```

⁴⁶ <https://ubuntu.com/pro>

⁴⁷ <https://ubuntu.com/about/release-cycle>

⁴⁸ <https://ubuntu.com/security/livepatch>

⁴⁹ <https://documentation.ubuntu.com/security/compliance/fips/>

⁵⁰ <https://ubuntu.com/security/security-standards>

⁵¹ <https://ubuntu.com/cloud/public-cloud>

⁵² <https://ubuntu.com/aws/pro>

⁵³ <https://ubuntu.com/azure/pro>

⁵⁴ <https://ubuntu.com/gcp/pro>

Note:

For users who need to attach many systems or automate the process, as in many enterprise scenarios, there are parameters to the attach command to help you. Please check the Ubuntu Pro documentation for [attaching your system with a configuration file](https://documentation.ubuntu.com/pro-client/en/latest/howtoguides/how_to_attach_with_config_file/)⁵⁷

⁵⁷ https://documentation.ubuntu.com/pro-client/en/latest/howtoguides/how_to_attach_with_config_file/

After running the command, you should see output like this, giving you a link and a code:

```
ubuntu@test:~$ sudo pro attach

Initiating attach operation...

Please sign in to your Ubuntu Pro account at this link:
https://ubuntu.com/pro/attach
And provide the following code: H31JIV
```

Open the link without closing your terminal window.

In the field that asks you to enter your code, copy and paste the code shown in the terminal. Then, choose which subscription you want to attach to. By default, the Free Personal Token will be selected.

If you have a paid subscription and want to attach to a different token, you may want to log in first so that your additional tokens will appear.

Once you have pasted your code and chosen the subscription you want to attach your machine to, click on the “Submit” button.

The attach process will then continue in the terminal window, and you should eventually be presented with the following message:

```
Attaching the machine...
Enabling Ubuntu Pro: ESM Apps
Ubuntu Pro: ESM Apps enabled
Enabling Ubuntu Pro: ESM Infra
Ubuntu Pro: ESM Infra enabled
Enabling Livepatch
Livepatch enabled
This machine is now attached to 'Ubuntu Pro - free personal subscription'
```

When the machine has successfully been attached, you will also see a summary of which services are enabled and information about your subscription.

Available services can be enabled or disabled on the command line with `pro enable <service name>` and `pro disable <service name>` after you have attached.

Next steps

- For more information about the Ubuntu Pro Client, you can [read our documentation](#)⁵⁸.
- For a guided tour through the most commonly-used commands available through the Ubuntu Pro Client, [check out this tutorial](#)⁵⁹.

3.2. Ubuntu Server how-to guides

If you have a specific goal, but are already familiar with Ubuntu Server, our how-to guides have more in-depth detail than our tutorials and can be applied to a broader set of applications. They'll help you achieve an end result but may require you to understand and adapt the steps to fit your specific requirements.

3.2.1. Server installation

Our list of [installation guides](#) (page 75) contains installation instructions for a variety of architecture-specific and advanced setups. For a general installation, or if you're just getting started with Ubuntu, you may prefer to use our [basic installation](#) (page 8) tutorial.

Server installation

If you are new to Ubuntu, we recommend our [basic installation](#) (page 8) tutorial to get you started.

Automatic install

The Ubuntu Installer has its own documentation for automatic (or “hands off”) installations. These guides from the Ubuntu Installer documentation are available for automatic installations.

- [Introduction to Automated Server installer](#)⁶⁰
- [Autoinstall quickstart](#)⁶¹
- [Autoinstall quickstart on s390x](#)⁶²

⁵⁸ <https://documentation.ubuntu.com/pro-client/en/latest/>

⁵⁹ https://documentation.ubuntu.com/pro-client/en/latest/tutorials/basic_commands/

⁶⁰ <https://canonical-subiquity.readthedocs-hosted.com/en/latest/intro-to-autoinstall.html>

⁶¹ <https://canonical-subiquity.readthedocs-hosted.com/en/latest/howto/autoinstall-quickstart.html>

⁶² <https://canonical-subiquity.readthedocs-hosted.com/en/latest/howto/autoinstall-quickstart-s390x.html>

Advanced install

This list of guides contains installation instructions for architecture-specific and more advanced setups. Select your preferred architecture to see which guides are available.

amd64

- [Netboot install](#) (page 76)

arm64

- [Netboot install](#) (page 80)
- [Choose between the arm64 and arm64+largemem installer options](#) (page 85)

ppc64el

- [Netboot install](#) (page 86)
- [Virtual CD-ROM and Petitboot install](#) (page 92)

s390x

- [Install via z/VM](#) (page 99)
- [Non-interactive IBM z/VM autoinstall](#) (page 116)
- [Install via LPAR](#) (page 124)
- [Non-interactive IBM Z LPAR autoinstall](#) (page 142)

See also

- Reference: [System requirements](#) (page 850)

How to netboot the server installer on amd64

[amd64](#) systems boot in either UEFI or legacy [BIOS](#) mode, and many systems can be configured to boot in either mode. The precise details depend on the system [firmware](#), but both modes usually support the “Preboot eXecution Environment” (PXE) specification, which allows the provisioning of a [bootloader](#) over the network.

Steps needed

The process for network booting the live server installer is similar for both modes and goes like this:

1. The to-be-installed machine boots, and is directed to network boot.
2. The *DHCP*/BOOTP server tells the machine its network configuration and where to get the bootloader.
3. The machine's firmware downloads the bootloader over TFTP and executes it.
4. The bootloader downloads configuration, also over TFTP, telling it where to download the kernel, RAM Disk and kernel command line to use.
5. The RAM Disk looks at the kernel command line to learn how to configure the network and where to download the server ISO from.
6. The RAM Disk downloads the ISO and mounts it as a loop device.
7. From this point on the install follows the same path as if the ISO were on a local block device.

The difference between UEFI and legacy modes is that in UEFI mode the bootloader is an *EFI* executable, signed so that it is accepted by Secure Boot, and in legacy mode it is *PXELINUX*⁶³. Most DHCP/BOOTP servers can be configured to serve the right bootloader to a particular machine.

Configure DHCP/BOOTP and TFTP

There are several implementations of the DHCP/BOOTP and TFTP protocols available. This document describes how to configure *dnsmasq* to perform both of these roles.

- Install *dnsmasq* with:

```
$ sudo apt install dnsmasq
```

- Put something like this in `/etc/dnsmasq.d/pxe.conf`:

```
interface=<your interface>,lo
bind-interfaces
dhcp-range=<your interface>,192.168.0.100,192.168.0.200
dhcp-boot=pxelinux.0
dhcp-match=set:efi-x86_64,option:client-arch,7
dhcp-boot=tag:efi-x86_64,bootx64.efi
enable-tftp
tftp-root=/srv/tftp
```

Note:

This assumes several things about your network; read `man dnsmasq` or the default `/etc/dnsmasq.conf` for many more options.

⁶³ <https://wiki.syslinux.org/wiki/index.php?title=PXELINUX>

- Restart dnsmasq with:

```
$ sudo systemctl restart dnsmasq.service
```

- Install TFTP

```
$ sudo apt install tftpd-hpa
```

If the installation is successful, check that the corresponding TFTP service is active using this command:

```
$ systemctl status tftpd-hpa.service
```

It is expected to show **active (running)** in the output messages. We will also assume your TFTP root path is `/srv/tftp` for the remainder of this guide.

Mode-independent set up

- Download the latest live server ISO for the release you want to install:

```
$ wget http://cdimage.ubuntu.com/ubuntu-server/noble/daily-live/current/noble-live-server-amd64.iso
```

- Mount it:

```
$ sudo mount noble-live-server-amd64.iso /mnt
```

- Copy the kernel and initrd from it to the TFTP directory served by dnsmasq:

```
$ sudo cp /mnt/casper/{vmlinuz,initrd} /srv/tftp/
```

Set up the files for UEFI booting

- Copy the signed shim binary into place:

```
$ apt download shim-signed > dpkg-deb --fsys-tarfile shim-signed*deb |  
tar x ./usr/lib/shim/shimx64.efi.signed.latest -0 | sudo tee  
/srv/tftp/bootx64.efi >/dev/null
```

- Copy the signed GRUB binary into place:

```
$ apt download grub-efi-amd64-signed > dpkg-deb --fsys-tarfile  
grub-efi-amd64-signed*deb | tar x  
./usr/lib/grub/x86_64-efi-signed/grubnetx64.efi.signed -0 | sudo tee  
/srv/tftp/grubx64.efi >/dev/null
```

- GRUB also needs a font to be available over TFTP:

```
$ apt download grub-common > dpkg-deb --fsys-tarfile grub-common*deb |  
tar x ./usr/share/grub/unicode.pf2 -0 | sudo tee /srv/tftp/unicode.pf2  
>/dev/null
```

- Create /srv/tftp/grub/grub.cfg that contains:

```
sudo mkdir -p /srv/tftp/grub/  
  
sudo tee /srv/tftp/grub/grub.cfg >/dev/null <<EOF  
set default="0"  
set timeout=-1  
  
if loadfont unicode ; then  
    set gfxmode=auto  
    set locale_dir=$prefix/locale  
    set lang=en_US  
fi  
terminal_output gfxterm  
  
set menu_color_normal=white/black  
set menu_color_highlight=black/light-gray  
if background_color 44,0,30; then  
    clear  
fi  
  
function gfxmode {  
    set gfxpayload="${1}"  
    if [ "${1}" = "keep" ]; then  
        set vt_handoff=vt.handoff=7  
    else  
        set vt_handoff=  
    fi  
fi  
}  
  
set linux_gfx_mode=keep  
  
export linux_gfx_mode  
  
menuentry 'Ubuntu 24.04' {  
    gfxmode $linux_gfx_mode  
    linux /vmlinuz $vt_handoff quiet splash root=/dev/ram0 ramdisk_size=1500000 cloud-  
config-url=/dev/null ip=dhcp url=http://cdimage.ubuntu.com/ubuntu-server/noble/daily-  
live/current/noble-live-server-amd64.iso  
    initrd /initrd  
}  
EOF
```


Set up the files for legacy boot

- Download pxelinux.0 and put it into place:

```
$ wget http://archive.ubuntu.com/ubuntu/dists/focal/main/installer-  
amd64/current/legacy-images/netboot/pxelinux.0 > mkdir -p /srv/tftp > mv  
pxelinux.0 /srv/tftp/
```

- Ensure the syslinux-common package is installed, then:

```
$ cp /usr/lib/syslinux/modules/bios/ldlinux.c32 /srv/tftp/
```

- Create /srv/tftp/pxelinux.cfg/default containing:

```
DEFAULT install  
LABEL install  
    KERNEL vmlinuz  
    INITRD initrd  
    APPEND root=/dev/ram0 ramdisk_size=1500000 cloud-config-url=/dev/null ip=dhcp  
url=http://cdimage.ubuntu.com/ubuntu-server/noble/daily-live/current/noble-live-  
server-amd64.iso
```

Note:

Setting `cloud-config-url=/dev/null` on the kernel command line prevents cloud-init from downloading the ISO twice.

As you can see, this downloads the ISO from Ubuntu's servers. You may want to host it somewhere on your infrastructure and change the URL to match.

This configuration is very simple. PXELINUX has many, many options, and you can [consult its documentation](#)⁶⁴ for more.

Netboot the server installer via UEFI PXE on ARM (aarch64, arm64) and x86_64 (amd64)

This document describes the steps to install a system via netbooting and the live server installer (Subiquity) in UEFI mode with Ubuntu 20.04 LTS or later.

This process applies to both arm64 and [amd64](#) architectures and is based on [this Ubuntu Discourse post for legacy mode netbooting](#)⁶⁵ (UEFI's predecessor). Focal (20.04, 20.04.5) and Groovy (20.10) have been tested with this method.

⁶⁴ <https://wiki.syslinux.org/wiki/index.php?title=PXELINUX>

⁶⁵ <https://discourse.ubuntu.com/t/netbooting-the-live-server-installer/14510>

Configure TFTP

This article assumes that you have set up your Trivial File Transfer Protocol (TFTP) (and/or *DHCP*/bootp if necessary, depending on your LAN configuration) by following [the netbooting method for legacy mode](#)⁶⁶. Alternatively, you can set up your own TFTP server as follows if your *DNS* and DHCP are already configured:

```
$ sudo apt install tftpd-hpa
```

After installation, check that the TFTP service is active:

```
$ systemctl status tftpd-hpa.service
```

The output should show **active (running)**. This article assumes your TFTP root path is `/srv/tftp`.

How to serve files

You can **skip this entire section** by using [this non-official tool](#)⁶⁷. The tool will set up your TFTP server to serve the necessary files for netbooting.

Necessary files

The following files are needed for this process:

- Ubuntu live server image:
 - For arm64 architectures, the image name has the suffix `-arm64`. For example, `ubuntu-20.04.5-live-server-arm64.iso`.
 - For amd64 architectures, the image name has the suffix `-amd64`. For example, `ubuntu-20.04.5-live-server-amd64.iso`.
- GRUB *EFI* binary (and the corresponding `grub.cfg` text file):
 - For arm64 architectures, this is called `grubnetaa64.efi.signed`.
 - For amd64 architectures, this is called `grubnetx64.efi.signed`.
- `initrd` extracted from your target Ubuntu live server image (use `hwe-initrd` instead if you want to boot with the HWE kernel).
- `vmlinux` extracted from your target Ubuntu live server image (use `hwe-vmlinux` instead if you want to boot with the HWE kernel).

⁶⁶ <https://discourse.ubuntu.com/t/netbooting-the-live-server-installer/14510>

⁶⁷ <https://github.com/dannf/ubuntu-server-netboot>

Examples

In the following sections, we will use an arm64 image as an example. This means the following files are used:

- Ubuntu 20.04.5 live server image [ubuntu-20.04.5-live-server-arm64.iso](https://cdimage.ubuntu.com/ubuntu/releases/20.04.5/release/ubuntu-20.04.5-live-server-arm64.iso)⁶⁸
- GRUB EFI binary [grubnetaa64.efi.signed](https://ports.ubuntu.com/ubuntu-ports/dists/focal/main/uefi/grub2-arm64/current/grubnetaa64.efi.signed)⁶⁹
- `initrd` extracted from `ubuntu-20.04.5-live-server-arm64.iso`
- `vmlinux` extracted from `ubuntu-20.04.5-live-server-arm64.iso`

Replace the corresponding files if you want to work on an amd64 image. For example, your files may be:

- Ubuntu 20.04.5 live server image [ubuntu-20.04.5-live-server-amd64.iso](https://old-releases.ubuntu.com/releases/20.04.5/ubuntu-20.04.5-live-server-amd64.iso)⁷⁰
- GRUB EFI binary [grubnetx64.efi.signed](http://archive.ubuntu.com/ubuntu/dists/focal/main/uefi/grub2-amd64/current/grubnetx64.efi.signed)⁷¹
- `initrd` extracted from `ubuntu-20.04.5-live-server-amd64.iso`
- `vmlinux` extracted from `ubuntu-20.04.5-live-server-amd64.iso`

Download and serve the GRUB EFI binary

The GRUB binary redirects the download path to the target files via `grub.cfg`. For more information about the Preboot Execution Environment (PXE) process and why this binary is required, see [the netbooting instructions for legacy mode](https://discourse.ubuntu.com/t/netbooting-the-live-server-installer/14510)⁷².

```
$ sudo wget https://ports.ubuntu.com/ubuntu-ports/dists/focal/main/uefi/
grub2-arm64/current/grubnetaa64.efi.signed -O
/srv/tftp/grubnetaa64.efi.signed
```

Note:

Change **the archive distribution's name** from `Focal` to your target distribution name if needed.

⁶⁸ <https://cdimage.ubuntu.com/ubuntu/releases/20.04.5/release/ubuntu-20.04.5-live-server-arm64.iso>

⁶⁹ <https://ports.ubuntu.com/ubuntu-ports/dists/focal/main/uefi/grub2-arm64/current/grubnetaa64.efi.signed>

⁷⁰ <https://old-releases.ubuntu.com/releases/20.04.5/ubuntu-20.04.5-live-server-amd64.iso>

⁷¹ <http://archive.ubuntu.com/ubuntu/dists/focal/main/uefi/grub2-amd64/current/grubnetx64.efi.signed>

⁷² <https://discourse.ubuntu.com/t/netbooting-the-live-server-installer/14510>

Download and serve more files

Download an Ubuntu ARM server ISO to fetch the installer, such as the [20.04.5 live server arm64 ISO⁷³](https://cdimage.ubuntu.com/ubuntu/releases/20.04.5/release/ubuntu-20.04.5-live-server-arm64.iso). The “live” prefix is required because the files needed are only available in the live version.

Mount the ISO and copy the target files we need over to the TFTP folder:

```
$ sudo mount ./ubuntu-20.04.5-live-server-arm64.iso /mnt > sudo mkdir /srv/tftp/grub /srv/tftp/casper > sudo cp /mnt/boot/grub/grub.cfg /srv/tftp/grub/ > sudo cp /mnt/casper/initrd /srv/tftp/casper/ > sudo cp /mnt/casper/vmlinuz /srv/tftp/casper/
```

So, the TFTP root folder should look like this now:

```
$ find /srv/tftp/

/srv/tftp/
/srv/tftp/grub
/srv/tftp/grub/grub.cfg
/srv/tftp/grubnetaa64.efi.signed
/srv/tftp/casper
/srv/tftp/casper/initrd
/srv/tftp/casper/vmlinuz
```

Customize the GRUB menu to install the target image by fetching it directly over the internet:

```
$ sudo chmod +w /srv/tftp/grub/grub.cfg
```

```
$ sudo vi /srv/tftp/grub/grub.cfg
```

Add a new boot entry at the top-level menu (not in any submenu):

```
menuentry "Install Ubuntu Server (Focal 20.04.5) (Pull the iso from web)" {
    set gfxpayload=keep
    linux /casper/vmlinuz url=http://cdimage.ubuntu.com/ubuntu/releases/20.04.5/
release/ubuntu-20.04.5-live-server-arm64.iso only-ubiquity ip=dhcp ---
    initrd /casper/initrd
}
```

In this configuration:

- `ip=dhcp` configures DHCP management for the lab environment
- `url` points to your target image download URL

Change these values according to your scenario.

When successful, you will enter the GRUB menu of the ephemeral live prompt. Select the entry you added to `grub.cfg` (Install Ubuntu Server (Focal 20.04.5) (Pull the ISO from web) in this example). After the ISO downloads, the Subiquity welcome message will appear.

⁷³ <http://cdimage.ubuntu.com/ubuntu/releases/20.04.5/release/ubuntu-20.04.5-live-server-arm64.iso>

Appendix

Check the serving file names

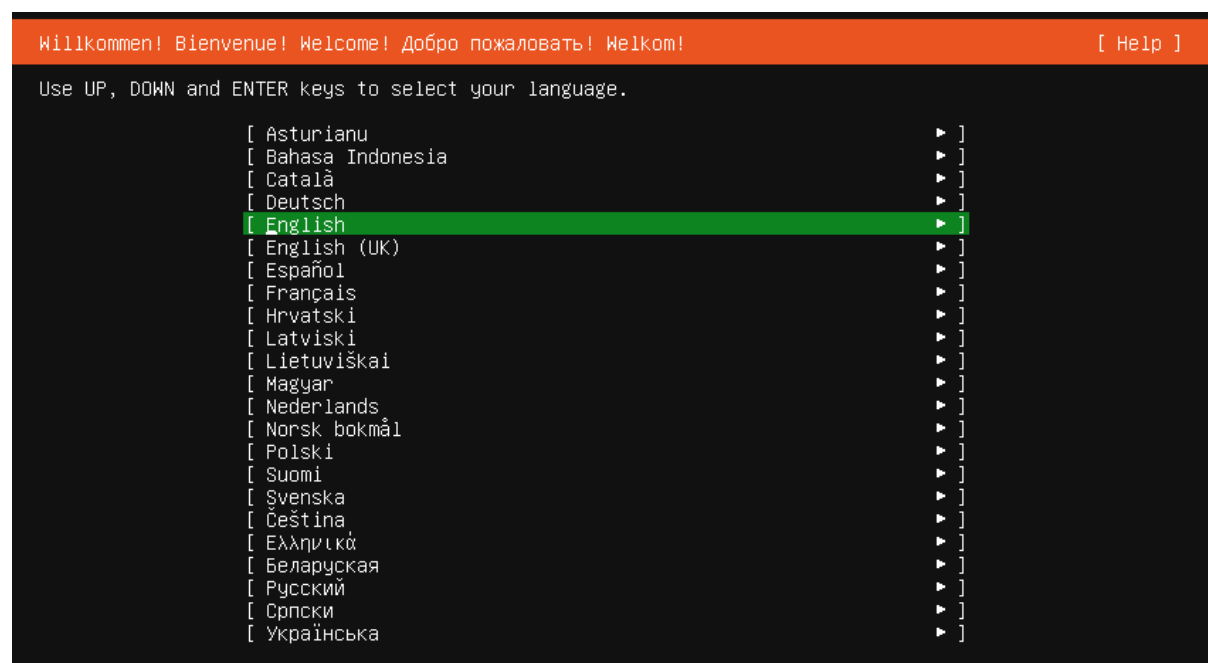
Verify that the target file name for linux and initrd is correct. The default initrd binary file name for 20.04.5 is `initrd`, while for 20.10 it is `initrd.lz`. Serving incorrect file names is a common troubleshooting issue.

Booting screenshots

With a correct setup, `grub.cfg` redirects the process to an ephemeral environment where your target image is downloaded and assigned in the GRUB entry. When you access the console or monitor of your target machine, you will see a screen like this:

```
Listening on LPF/ens3/02:56:e6:08:2d:e3
Sending on LPF/ens3/02:56:e6:08:2d:e3
Sending on Socket/fallback
DHCPDISCOVER on ens3 to 255.255.255.255 port 67 interval 3 (xid=0x83e74430)
DHCPOFFER of 10.228.68.73 from 10.228.68.5
DHCPREQUEST for 10.228.68.73 on ens3 to 255.255.255.255 port 67 (xid=0x3044e783)
DHCPACK of 10.228.68.73 from 10.228.68.5 (xid=0x83e74430)
bound to 10.228.68.73 -- renewal in 286 seconds.
Connecting to 10.228.68.91 (10.228.68.91:80)
ubuntu-20.04.1-live- 0% | 462k 0:34:48 ETA
ubuntu-20.04.1-live- 2% | 22.2M 0:01:20 ETA
ubuntu-20.04.1-live- 4% * 39.7M 0:01:05 ETA
ubuntu-20.04.1-live- 6% ** 59.6M 0:00:57 ETA
ubuntu-20.04.1-live- 8% ** 77.7M 0:00:53 ETA
ubuntu-20.04.1-live- 10% *** 97.7M 0:00:50 ETA
ubuntu-20.04.1-live- 12% *** 114M 0:00:49 ETA
ubuntu-20.04.1-live- 14% **** 132M 0:00:47 ETA
ubuntu-20.04.1-live- 16% ***** 149M 0:00:45 ETA
ubuntu-20.04.1-live- 18% ***** 166M 0:00:45 ETA
ubuntu-20.04.1-live- 20% ***** 183M 0:00:43 ETA
ubuntu-20.04.1-live- 21% ***** 200M 0:00:42 ETA
ubuntu-20.04.1-live- 23% ***** 214M 0:00:42 ETA
```

Wait for the download to complete. If you see this Subiquity welcome page, the installer successfully launched via your UEFI PXE setup. Congratulations!



Choose between the arm64 and arm64+largemem installer options

From 22.04.4 onward, Ubuntu will provide both 4k and 64k page size kernel ISOs for ARM servers.

The default **arm64** ISO will still use a 4k page size kernel, while the new 64k page size kernel ISO is named **arm64+largemem**.

- [arm64 4k ISO download](#)⁷⁴
- [arm64+largemem ISO download](#)⁷⁵

The default arm64 (4k) option

The 4k page size is the default in our arm64 ISO. It is suitable for workloads with many small processes, or environments with tight memory constraints. Typical use cases include (but are not limited to):

- Web servers
- Embedded devices
- General purpose/build systems

The arm64+largemem (64k) option

Our new **arm64+largemem** ISO includes a kernel with 64k page size. A larger page size can increase throughput, but comes at the cost of increased memory use, making this option more suitable for servers with plenty of memory. Typical use cases for this ISO include:

- Machine learning
- Databases with many large entries
- *High performance computing*
- etc.

Note:

It is possible to switch between these kernel options after installation by installing the other kernel alternative, rebooting, and selecting the new kernel from the GRUB menu.

⁷⁴ <https://cdimage.ubuntu.com/releases/22.04/release/ubuntu-22.04.5-live-server-arm64.iso>

⁷⁵ <https://cdimage.ubuntu.com/releases/22.04/release/ubuntu-22.04.5-live-server-arm64+largemem.iso>

Switching kernels post-installation

To switch between the two kernels after the initial installation you can run the following commands, replacing `<desired-kernel>` with `linux-generic-64k` when swapping to 64k, or `linux-generic` when swapping to the default 4k kernel:

```
$ sudo apt update
```

```
$ sudo apt install <desired-kernel>
```

```
$ sudo reboot
```

Upon reboot you will be greeted with the GRUB menu (you may need to hold down the Shift key during the reboot for it to appear). Select "Advanced Options for Ubuntu", then select your desired kernel to boot into Ubuntu.

To permanently change the default to your `<desired-flavour>`, replace `<desired-flavour>` with `generic` or `generic-64k` and then run the following command:

```
$ echo "GRUB_FLAVOUR_ORDER=<desired-flavour>" | sudo tee  
/etc/default/grub.d/local-order.cfg
```

To apply your change run:

```
$ sudo update-grub
```

Future boots will automatically use your new desired kernel flavour. You can verify this by rebooting using:

```
$ sudo reboot
```

And then running the following command to display the active kernel:

```
$ uname -r
```

Netboot the live server installer on IBM Power (ppc64el) with Petitboot

Open a terminal window on your workstation and make sure the `ipmitool` package is installed.

Verify if you can reach the *BMC* of the IBM Power system via `ipmitool` with a simple `ipmitool` call like:

```
$ ipmitool -I lanplus -H Power9Box -U <user> -P <password> power status  
  
Chassis Power is off
```

or:

```
$ ipmitool -I lanplus -H Power9Box -U <user> -P <password> fru print 47
```

```
Product Name       : OpenPOWER Firmware
Product Version    : open-power-SUPERMICRO-P9DSU-V2.12-20190404-prod
Product Extra      : op-build-1b9269e
Product Extra      : buildroot-2018.11.3-12-g222837a
Product Extra      : skiboot-v6.0.19
Product Extra      : hostboot-c00d44a-pb1307d7
Product Extra      : occ-8fa3854
Product Extra      : linux-4.19.30-openpower1-p22d1df8
Product Extra      : petitboot-v1.7.5-p8f5fc86
```

or:

```
$ ipmitool -I lanplus -H Power9Box -U <user> -P <password> sol info
```

```
Set in progress    : set-complete
Enabled            : true
Force Encryption   : false
Force Authentication : false
Privilege Level     : OPERATOR
Character Accumulate Level (ms) : 0
Character Send Threshold : 0
Retry Count        : 0
Retry Interval (ms) : 0
Volatile Bit Rate (kbps) : 115.2
Non-Volatile Bit Rate (kbps) : 115.2
Payload Channel     : 1 (0x01)
Payload Port        : 623
```

Activate serial-over-LAN

Open a second terminal and activate serial-over-LAN (SOL), so that you have two terminal windows open:

- 1) to control the BMC via IPMI
- 2) for the serial-over-LAN console

Activate serial-over-LAN:

```
$ ipmitool -I lanplus -H Power9Box -U <user> -P <password> sol activate
```

```
...
```

Power the system on in the 'control terminal' and watch the SOL console:

```
$ ipmitool -I lanplus -H Power9Box -U <user> -P <password> power on
```

```
...
```

It takes some time to see the first lines in the SOL console:


```
[SOL Session operational. Use ~? for help]
--== Welcome to Hostboot ==--

2.77131|secure|SecureROM valid - enabling functionality
3.15860|secure|Booting in secure mode.
5.59684|Booting from SBE side 0 on master proc=00050000
5.60502|ISTEP 6. 5 - host_init_fsi
5.87228|ISTEP 6. 6 - host_set_ipl_parms
6.11032|ISTEP 6. 7 - host_discover_targets
6.67868|HWAS|PRESENT> DIMM[03]=A0A0000000000000
6.67870|HWAS|PRESENT> Proc[05]=8800000000000000
6.67871|HWAS|PRESENT> Core[07]=3FFF0C33FFC30000
6.98988|ISTEP 6. 8 - host_update_master_tpm
7.22711|SECURE|Security Access Bit> 0xC000000000000000
7.22711|SECURE|Secure Mode Disable (via Jumper)> 0x0000000000000000
7.22731|ISTEP 6. 9 - host_gard
7.43353|HWAS|FUNCTIONAL> DIMM[03]=A0A0000000000000
7.43354|HWAS|FUNCTIONAL> Proc[05]=8800000000000000
7.43356|HWAS|FUNCTIONAL> Core[07]=3FFF0C33FFC30000
7.44509|ISTEP 6.10 - host_revert_sbe_mcs_setup
...
```

After a moment the system reaches the Petitboot screen:

```
Petitboot (v1.7.5-p8f5fc86) 9006-12P 1302NXA

[Network: enP2p1s0f0 / 0c:c4:7a:87:04:d8]
Execute
netboot enP2p1s0f0 (pxelinux.0)
[CD/DVD: sr0 / 2019-10-17-13-35-12-00]
Install Ubuntu Server
[Disk: sda2 / 295f571b-b731-4ebb-b752-60aad80fc1b]
Ubuntu, with Linux 5.4.0-14-generic (recovery mode)
Ubuntu, with Linux 5.4.0-14-generic
Ubuntu

System information
System configuration
System status log
Language
Rescan devices
Retrieve config from URL
Plugins (0)

*Exit to shell

Enter=accept, e=edit, n=new, x=exit, l=language, g=log, h=help
```

Select “*Exit to shell”.

Note:

Make sure you really watch the SOL, since the Petitboot screen (above) has a time out (usually 10 or 30 seconds) and afterwards it automatically proceeds and tries to boot from the configured devices (usually disk). This can be prevented by just navigating in Petitboot. The petitboot shell is small Linux based OS:

```
...
Exiting petitboot. Type 'exit' to return.
You may run 'pb-sos' to gather diagnostic data
```

Note:

In case one needs to gather system details and diagnostic data for IBM support, this can be done here by running 'pb-sos' (see msg).

Download the ISO

Now download the 'live-server' ISO image (notice that 'focal-live-server-ppc64el.iso' uses the liver server installer, Subiquity, while 'focal-server-s390x.iso' uses d-i).

Again, for certain web locations a proxy needs to be used:

```
/ # export http_proxy=http://squid.proxy:3128 # in case a proxy is required
/ #
/ # wget http://cdimage.ubuntu.com/ubuntu-server/daily-live/current/focal-live-server-ppc64el.iso
Connecting to <proxy-ip>:3128 (<proxy-ip>:3128)
focal-live-server-pp 100% |.....| 922M 0:00:00 ETA
```

Next we need to loop-back mount the ISO:

```
/ # mkdir iso
/ # mount -o loop focal-live-server-ppc64el.iso iso
```

Or, in case auto-detect of type iso9660 is not supported or not working, you should explicitly specify the 'iso9660' type:

```
/ # mount -t iso9660 -o loop focal-live-server-ppc64el.iso iso
```

Now load kernel and initrd from the loop-back mount, specify any needed kernel parameters and get it going:

```
/ # kexec -l ./iso/casper/vmlinuz --initrd=./iso/casper/initrd.gz --append="ip=dhcp
url=http://cdimage.ubuntu.com/ubuntu-server/daily-live/current/focal-live-server-ppc64el.
iso http_proxy=http://squid.proxy:3128 --- quiet"
/ # kexec -e
The system is going down NOW!
Sent SIGTERM to all processes
```

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```
Sent SIGKILL to all processes
...
```

Note:

In order to boot with and install the HWE kernel (if available), substitute `vmlinux` with `vmlinux-hwe` in the first `kexec` line.

The system now performs the initial boot of the installer:

```
[ 1200.687004] kexec_core: Starting new kernel
[ 1277.493883374,5] OPAL: Switch to big-endian OS
[ 1280.465061219,5] OPAL: Switch to little-endian OS
ln: /tmp/mountroot-fail-hooks.d//scripts/init-premount/lvm2: No such file or directory
Internet Systems Consortium DHCP Client 4.4.1
Copyright 2004-2018 Internet Systems Consortium.
All rights reserved.
For info, please visit https://www.isc.org/software/dhcp/
Listening on LPF/enP2p1s0f3/0c:c4:7a:87:04:db
Sending on LPF/enP2p1s0f3/0c:c4:7a:87:04:db
Listening on LPF/enP2p1s0f2/0c:c4:7a:87:04:da
Sending on LPF/enP2p1s0f2/0c:c4:7a:87:04:da
Listening on LPF/enP2p1s0f1/0c:c4:7a:87:04:d9
Sending on LPF/enP2p1s0f1/0c:c4:7a:87:04:d9
Listening on LPF/enP2p1s0f0/0c:c4:7a:87:04:d8
Sending on LPF/enP2p1s0f0/0c:c4:7a:87:04:d8
Sending on Socket/fallback
DHCPDISCOVER on enP2p1s0f3 to 255.255.255.255 port 67 interval 3
(xid=0x8d5704c)
DHCPDISCOVER on enP2p1s0f2 to 255.255.255.255 port 67 interval 3
(xid=0x94b25b28)
DHCPDISCOVER on enP2p1s0f1 to 255.255.255.255 port 67 interval 3
(xid=0x4edd0558)
DHCPDISCOVER on enP2p1s0f0 to 255.255.255.255 port 67 interval 3
(xid=0x61c90d28)
DHCPOFFER of 10.245.71.102 from 10.245.71.3
DHCPREQUEST for 10.245.71.102 on enP2p1s0f0 to 255.255.255.255 port 67
(xid=0x280dc961)
DHCPACK of 10.245.71.102 from 10.245.71.3 (xid=0x61c90d28)
bound to 10.245.71.102 -- renewal in 236 seconds.
Connecting to 91.189.89.11:3128 (91.189.89.11:3128)
focal-live-server-pp 1% | 14.0M 0:01:04 ETA
focal-live-server-pp 4% | * 45.1M 0:00:38 ETA
focal-live-server-pp 8% | ** 76.7M 0:00:33 ETA
focal-live-server-pp 11% | *** 105M 0:00:31 ETA
focal-live-server-pp 14% | **** 133M 0:00:29 ETA
focal-live-server-pp 17% | ***** 163M 0:00:27 ETA
focal-live-server-pp 20% | ***** 190M 0:00:26 ETA
focal-live-server-pp 24% | ***** 222M 0:00:25 ETA
focal-live-server-pp 27% | ***** 253M 0:00:23 ETA
focal-live-server-pp 30% | ***** 283M 0:00:22 ETA
focal-live-server-pp 34% | ***** 315M 0:00:21 ETA
focal-live-server-pp 37% | ***** 343M 0:00:20 ETA
focal-live-server-pp 39% | ***** 367M 0:00:19 ETA
focal-live-server-pp 42% | ***** 392M 0:00:18 ETA
```

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```
focal-live-server-pp 45% |*****| 420M 0:00:17 ETA
focal-live-server-pp 48% |*****| 451M 0:00:16 ETA
focal-live-server-pp 52% |*****| 482M 0:00:15 ETA
focal-live-server-pp 55% |*****| 514M 0:00:14 ETA
focal-live-server-pp 59% |*****| 546M 0:00:13 ETA
focal-live-server-pp 62% |*****| 578M 0:00:11 ETA
focal-live-server-pp 65% |*****| 607M 0:00:10 ETA
focal-live-server-pp 69% |*****| 637M 0:00:09 ETA
focal-live-server-pp 72% |*****| 669M 0:00:08 ETA
focal-live-server-pp 75% |*****| 700M 0:00:07 ETA
focal-live-server-pp 79% |*****| 729M 0:00:06 ETA
focal-live-server-pp 82% |*****| 758M 0:00:05 ETA
focal-live-server-pp 85% |*****| 789M 0:00:04 ETA
focal-live-server-pp 88% |*****| 817M 0:00:03 ETA
focal-live-server-pp 91% |*****| 842M 0:00:02 ETA
focal-live-server-pp 93% |*****| 867M 0:00:01 ETA
focal-live-server-pp 97% |*****| 897M 0:00:00 ETA
focal-live-server-pp 100% |*****| 922M 0:00:00 ETA
mount: mounting /cow on /root/cow failed: No such file or directory
Connecting to plymouth: Connection refused
passwd: password expiry information changed.
[ 47.202736] /dev/loop3: Can't open blockdev
[ 52.672550] cloud-init[3759]: Cloud-init v. 20.1-10-g71af48df-0ubuntu1 running
'init-local' at Wed, 18 Mar 2020 15:18:07 +0000. Up 51.87 seconds.
...
```

Once it has completed, you will reach the initial Subiquity installer screen:

```

Willkommen! Bienvenue! Welcome! Добро пожаловать! Welkom
Use UP, DOWN and ENTER keys to select your language.

[ English ]
[ Asturianu ]
[ Català ]
[ Hrvatski ]
[ Nederlands ]
[ Suomi ]
[ Français ]
[ Deutsch ]
[ Ελληνικά ]
[ Magyar ]
[ Latviešu ]
[ Norsk bokmål ]
[ Polski ]
[ Русский ]
[ Español ]
[ Українська ]
```

From this point, you can follow the normal Subiquity installation. For more details, refer to the [Subiquity installer documentation](https://canonical-subiquity.readthedocs-hosted.com/en/latest/index.html)⁷⁶.

⁷⁶ <https://canonical-subiquity.readthedocs-hosted.com/en/latest/index.html>

How to start a live server installation on IBM Power (ppc64el) with a virtual CD-ROM and Petitboot

Note:

Not all IBM Power machines come with the capability of installing via a virtual CD-ROM! However, it is also possible to *boot the installer over the network* (page 86).

A separate system (ideally in the same network, because of `ipmitool`) is needed to host the ppc64el ISO image file, which is later used as the virtual CD-ROM.

Install `ipmitool` and Samba

Log in to this separate host and make sure that the `ipmitool` package is installed:

```
$ sudo apt install ipmitool
```

as well as Samba:

```
$ sudo apt install samba
```

Configure Samba

Next, set up and configure Samba:

```
$ sudo touch /etc/samba/smb.conf && sudo tee -a /etc/samba/smb.conf <<EOF
[winshare]
path=/var/winshare
browseable = yes
read only = no
guest ok = yes
EOF
```

And do a quick verification that the required lines are present:

```
$ tail -n 5 /etc/samba/smb.conf
[winshare]
path=/var/winshare
browseable = yes
read only = no
guest ok = yes
```

Optional step

For downloading the image you may have to use a proxy server:

```
$ sudo touch ~/.wgetrc && sudo tee -a ~/.wgetrc <<EOF > use_proxy=yes >  
http_proxy=squid.proxy:3128 > https_proxy=squid.proxy:3128 > EOF
```

Download the ISO image

The ISO image needs to be downloaded now:

```
$ wget http://cdimage.ubuntu.com/ubuntu/releases/focal/release/ubuntu-20.04-  
live-server-ppc64el.iso --directory-prefix=/var/winshare
```

The proxy can also be passed over as a wget argument, like this:

```
$ wget -e use_proxy=yes -e http_proxy=squid.proxy:3128 http://cdimage.  
ubuntu.com/ubuntu/releases/focal/release/ubuntu-20.04-live-server-ppc64el.iso  
--directory-prefix=/var/winshare
```

Change the file mode of the ISO image file:

```
$ sudo chmod -R 755 /var/winshare/
```

```
$ ls -l /var/winshare/  
-rwxr-xr-x 1 ubuntu ubuntu 972500992 Mar 23 08:02 focal-live-server-ppc64el.iso
```

Restart and check Samba

Next we need to restart and check the Samba service:

```
$ sudo service smbd restart
```

```
$ sudo service smbd status
```

```
❏ smbd.service - Samba SMB Daemon
   Loaded: loaded (/lib/systemd/system/smbd.service; enabled; vendor
   preset: ena
   Active: active (running) since Tue 2020-02-04 15:17:12 UTC; 4s ago
     Docs: man:smbd(8)
           man:samba(7)
           man:smb.conf(5)
  Main PID: 6198 (smbd)
    Status: "smbd: ready to serve connections..."
     Tasks: 4 (limit: 19660)
    CGroup: /system.slice/smbd.service
            └─6198 /usr/sbin/smbd --foreground --no-process-group
            └─6214 /usr/sbin/smbd --foreground --no-process-group
            └─6215 /usr/sbin/smbd --foreground --no-process-group
            └─6220 /usr/sbin/smbd --foreground --no-process-group
Feb 04 15:17:12 host systemd[1]: Starting Samba SMB Daemon...
Feb 04 15:17:12 host systemd[1]: Started Samba SMB Daemon.
```

Test Samba share:

```
ubuntu@host:~$ smbclient -L localhost

WARNING: The "syslog" option is deprecated
Enter WORKGROUP\ubuntu's password:
  Sharename      Type            Comment
  -----
  print$         Disk            Printer Drivers
  winshare        Disk
  IPC$           IPC             IPC Service (host server (Samba, Ubuntu))
Reconnecting with SMB1 for workgroup listing.
  Server          Comment
  -----
  Workgroup        Master
  -----
  WORKGROUP        host
```

Get the IP address of the Samba host:

```
$ ip -4 -brief address show

lo                UNKNOWN         127.0.0.1/8
ibmveth2          UNKNOWN         10.245.246.42/24
```

Optional step

Additional testing to make certain the Samba share is accessible from remote:

```
user@workstation:~$ mkdir -p /tmp/test
```

```
user@workstation:~$ sudo mount -t cifs -o
```

```
username=guest,password=guest //10.245.246.42/winshare /tmp/test/
```

```
user@workstation:~$ ls -la /tmp/test/
```

```
total 1014784
drwxr-xr-x  2 root root      0 May  4 15:46 .
drwxrwxrwt 18 root root    420 May  4 19:25 ..
-rwxr-xr-x  1 root root 1038249984 May  3 19:37 ubuntu-20.04-live-server-ppc64el.iso
```

Now use a browser and navigate to the [BMC](#) of the Power system, which should be installed (let's assume the BMC's IP address is 10.245.246.247):

```
firefox http://10.245.246.247/
```

Log into the BMC, then find and select:

Virtual Media → CD-ROM

Enter the IP address of the Samba share:

10.245.246.42

and the path to the Samba share:

\winshare\focal-live-server-ppc64el.iso

Click "Save and Mount". Make sure that the virtual CD-ROM is really properly mounted!

CD-ROM Image:

This option allows you to share a CD-ROM image over a Windows Share with a maximum size of 4.7GB. This image will be emulated to the host as USB device.

Device 1 There is an iso file mounted.

Device 2 No disk emulation set.

Device 3 No disk emulation set.

<Refresh Status>

Share host: 10.245.246.42

Path to image: \winshare\focal-live-server-ppc64el.iso

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```
User (optional):
Password (optional):
<Save> <Mount> <Unmount>
```

Note:

It's important that you see a status like this:

```
Device 1 There is an iso file mounted
```

Then the virtual CD-ROM is properly mounted and you will see the boot/install from CD-ROM entry in Petitboot:

```
[CD/DVD: sr0 / 2020-03-23-08-02-42-00]
Install Ubuntu Server
```

Boot into the Petitboot loader

Now use `ipmitool` to boot the system into the Petitboot loader:

```
$ ipmitool -I lanplus -H 10.245.246.247 -U ADMIN -P <password> power status
```

```
$ ipmitool -I lanplus -H 10.245.246.247 -U ADMIN -P <password> sol activate
```

```
$ ipmitool -I lanplus -H 10.245.246.247 -U ADMIN -P <password> power on

Chassis Power Control: Up/On
```

And reach the Petitboot screen:

```
Petitboot (v1.7.5-p8f5fc86)                                9006-12C B0S0026
-----
[Network: enP2p1s0f0 / ac:1f:6b:09:c0:52]
  execute
  netboot enP2p1s0f0 (pxelinux.0)

System information
System configuration
System status log
Language
Rescan devices
Retrieve config from URL
*Plugins (0)
Exit to shell
-----
Enter=accept, e=edit, n=new, x=exit, l=language, g=log, h=help
Default boot cancelled
```

Make sure that booting from CD-ROM is enabled:

Petitboot (v1.7.5-p8f5fc86)

9006-12C B0S0026

```
[Network: enP2p1s0f0 / ac:1f:6b:09:c0:52]
Execute
netboot enP2p1s0f0 (pxelinux.0)
[Disk: sda2 / ebdb022b-96b2-4f4f-ae63-69300ded13f4]
  Ubuntu, with Linux 5.4.0-12-generic (recovery mode)
  Ubuntu, with Linux 5.4.0-12-generic
  Ubuntu

System information
System configuration
System status log
Language
Rescan devices
Retrieve config from URL
*Plugins (0)
Exit to shell
```

```
Enter=accept, e=edit, n=new, x=exit, l=language, g=log, h=help
[sda3] Processing new Disk device
```

Petitboot System Configuration

```
Autoboot:      ( ) Disabled
               (*) Enabled

Boot Order:    (0) Any CD/DVD device
               (1) disk: sda2 [uuid: ebdb022b-96b2-4f4f-ae63-69300ded13f4]
               (2) net:  enP2p1s0f0 [mac: ac:1f:6b:09:c0:52]

               [   Add Device   ]
               [ Clear & Boot Any ]
               [      Clear      ]

Timeout:       30    seconds

Network:       (*) DHCP on all active interfaces
               ( ) DHCP on a specific interface
               ( ) Static IP configuration
```

```
tab=next, shift+tab=previous, x=exit, h=help
```

Petitboot System Configuration

```
Network:       (*) DHCP on all active interfaces
               ( ) DHCP on a specific interface
               ( ) Static IP configuration

DNS Server(s):                               (eg. 192.168.0.2)
               (if not provided by DHCP server)
```

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tab=next, shift+tab=previous, x=exit, h=help

```
[CD/DVD: sr0 / 2020-03-23-08-02-42-00]
* Install Ubuntu Server
```

```
Sent SIGKILL to all processes
[ 119.355371] kexec_core: Starting new kernel
[ 194.483947394,5] OPAL: Switch to big-endian OS
[ 197.454615202,5] OPAL: Switch to little-endian OS
```

```

Willkommen! Bienvenue! Welcome! Добро пожаловать! Welkom
Use UP, DOWN and ENTER keys to select your language.

```

From this point, you can follow the normal Subiquity installation. For more details, refer to the [Subiquity installer documentation](#)⁷⁷.

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Interactive live server installation on IBM z/VM (s390x)

Doing an interactive (manual) live installation as described here - meaning without specifying a `parmfile` - has been supported in Ubuntu Server since LTS 20.04.5 ("Focal").

The following guide assumes that a z/VM guest has been defined, and that it is able to either reach the public `cdimage.ubuntu.com` server or an internal FTP or HTTP server that hosts an Ubuntu Server 20.04 installer image, like this [20.04 \(a.k.a. Focal\) daily live image](http://cdimage.ubuntu.com/ubuntu/releases/20.04.5/release/ubuntu-20.04.5-live-server-s390x.iso)⁷⁸

Find a place to download the installer image:

```
user@workstation:~$ wget http://cdimage.ubuntu.com/ubuntu/releases/20.04.5/
release/ubuntu-20.04.5-live-server-s390x.iso

--2020-08-08 16:01:52--
http://cdimage.ubuntu.com/ubuntu/releases/20.04.5/release/ubuntu-20.04.5-live-
server-s390x.iso
Resolving cdimage.ubuntu.com (cdimage.ubuntu.com)... 2001:67c:1560:8001::1d,
2001:67c:1360:8001::27, 2001:67c:1360:8001::28, ...
Connecting to cdimage.ubuntu.com
(cdimage.ubuntu.com)|2001:67c:1560:8001::1d|:80... connected.
HTTP request sent, awaiting response... 200 OK
Length: 705628160 (673M) [application/x-iso9660-image]
Saving to: 'ubuntu-20.04.5-live-server-s390x.iso'

ubuntu-20.04.5-live 100%[=====] 672.94M  37.1MB/s   in
17s

2020-08-08 16:02:10 (38.8 MB/s) - 'ubuntu-20.04.5-live-server-s390x.iso' saved
[705628160/705628160]
```

Now loop-back mount the ISO to extract four files that are needed for a z/VM guest installation:

```
user@workstation:~$ mkdir iso
```

```
user@workstation:~$ sudo mount -o loop ubuntu-20.04.5-live-server-s390x.iso
iso
```

```
user@workstation:~$ ls -1
./iso/boot/{ubuntu.exec,parmfile.*,kernel.u*,initrd.u*}

./iso/boot/initrd.ubuntu
./iso/boot/kernel.ubuntu
./iso/boot/parmfile.ubuntu
./iso/boot/ubuntu.exec
```

Now transfer these four files to your z/VM guest (for example to its 'A' file mode), using either the 3270 terminal emulator or FTP.

Then log on to the z/VM guest that you want to use for the installation. In this example it will be guest '10.222.111.24'.

⁷⁸ <http://cdimage.ubuntu.com/ubuntu/releases/20.04.5/release/ubuntu-20.04.5-live-server-s390x.iso>

Execute the ubuntu REXX script to kick-off the installation:

```
ubuntu
00: 0000004 FILES PURGED
00: RDR FILE 0125 SENT FROM 10.222.111.24 PUN WAS 0125 RECS 101K CPY 001 A NOHOLD NO
KEEP
00: RDR FILE 0129 SENT FROM 10.222.111.24 PUN WAS 0129 RECS 0001 CPY 001 A NOHOLD NO
KEEP
00: RDR FILE 0133 SENT FROM 10.222.111.24 PUN WAS 0133 RECS 334K CPY 001 A NOHOLD NO
KEEP
00: 0000003 FILES CHANGED
00: 0000003 FILES CHANGED
01: HCPGSP2627I The virtual machine is placed in CP mode due to a SIGP initial CPU reset
from CPU 00.
02: HCPGSP2627I The virtual machine is placed in CP mode due to a SIGP initial CPU reset
from CPU 00.
03: HCPGSP2627I The virtual machine is placed in CP mode due to a SIGP initial CPU reset
from CPU 00.
- 0.390935| Initramfs unpacking failed: Decoding failed
Unable to find a medium container a live file system
```

In the usual case that no parmfile was configured, the installation system now offers to interactively configure the basic network:

```
Attempt interactive netboot from a URL?
yes no (default yes): yes
Available qeth devices:
0.0.0600 0.0.0603
zdev to activate (comma separated, optional): 0600
QETH device 0.0.0600:0.0.0601:0.0.0602 configured
Two methods available for IP configuration:
* static: for static IP configuration
* dhcp: for automatic IP configuration
static dhcp (default 'dhcp'): static
ip: 10.222.111.24
gateway (default 10.222.111.1): .
dns (default .):
vlan id (optional):
http://cdimage.ubuntu.com/ubuntu/releases/20.04.5/release/ubuntu-20.04.5-live-server-
s390x.iso (default)
url: ftp://10.11.12.2:21/ubuntu-live-server-20.04.5/ubuntu-20.04.5-live-server-s390x.iso
http_proxy (optional):
```

Ensure that the same version of the ISO image that was used to extract the installer files – kernel and initrd – is referenced at the ‘url:’ setting. It can be at a different location, for example directly referencing the public `cdimage.ubuntu.com` server: `http://cdimage.ubuntu.com/ubuntu/releases/20.04.5/release/ubuntu-20.04.5-live-server-s390x.iso`

The boot-up of the live-server installation now completes:

```
Configuring networking...
QETH device 0.0.0600:0.0.0601:0.0.0602 already configured
IP-Config: enc600 hardware address 02:28:0a:00:00:39 mtu 1500
IP-Config: enc600 guessed broadcast address 10.222.111255
IP-Config: enc600 complete:
address: 10.222.111.24 broadcast: 10.222.111255 netmask: 255.255.255.0
```

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```
gateway: 10.222.111.1      dns0      : 10.222.111.1      dns1      : 0.0.0.0

rootserver: 0.0.0.0 rootpath:
filename :
Connecting to 10.11.12.2:21 (10.11.12.2:21)
focal-live-server-s 5% !*                                ! 35.9M 0:00:17 ETA
focal-live-server-s 19% !*****                          ! 129M 0:00:08 ETA
focal-live-server-s 33% !*****                          ! 225M 0:00:05 ETA
focal-live-server-s 49% !*****                          ! 330M 0:00:04 ETA
focal-live-server-s 60% !*****                          ! 403M 0:00:03 ETA
focal-live-server-s 76% !*****                          ! 506M 0:00:01 ETA
focal-live-server-s 89% !*****                          ! 594M 0:00:00 ETA
focal-live-server-s 100% !*****                          ! 663M 0:00:00 ETA
passwd: password expiry information changed.
QETH device 0.0.0600:0.0.0601:0.0.0602 already configured
no search or nameservers found in /run/net-enc600.conf /run/net-*.conf /run/net6
-*.conf
- 594.766372| /dev/loop3: Can't open blockdev
- 595.610434| systemd-1|: multi-user.target: Job getty.target/start deleted to
break ordering cycle starting with multi-user.target/start
- 0;1;31m SKIP -0m| Ordering cycle found, skipping -0;1;39mLogin Prompts -0m
- 595.623027| systemd-1|: Failed unmounting /cdrom.
- 0;1;31mFAILED -0m| Failed unmounting -0;1;39m/cdrom -0m.

- 598.973538| cloud-init-1256|: Cloud-init v. 20.2-45-g5f7825e2-0ubuntu1 runnin
g 'init-local' at Thu, 04 Jun 2020 12:06:46 +0000. Up 598.72 seconds.
- 599.829069| cloud-init-1288|: Cloud-init v. 20.2-45-g5f7825e2-0ubuntu1 runnin
g 'init' at Thu, 04 Jun 2020 12:06:47 +0000. Up 599.64 seconds.
- 599.829182| cloud-init-1288|: ci-info: ++++++Net device info+++++
- 599.829218| cloud-init-1288|: ci-info: +-----+-----+-----+
+-----+-----+-----+
- 599.829255| cloud-init-1288|: ci-info: ! Device ! Up !           Address
! Mask ! Scope ! Hw-Address !
- 599.829292| cloud-init-1288|: ci-info: +-----+-----+-----+
+-----+-----+-----+
- 599.829333| cloud-init-1288|: ci-info: ! enc600 ! True !           10.222.111.24
! 255.255.255.0 ! global ! 02:28:0a:00:00:39 !
- 599.829376| cloud-init-1288|: ci-info: ! enc600 ! True ! fe80::28:aff:fe00:3
/64 ! . ! link ! 02:28:0a:00:00:39 !
- 599.829416| cloud-init-1288|: ci-info: ! lo ! True !           127.0.0.1
! 255.0.0.0 ! host ! . !
- 599.829606| cloud-init-1288|: ci-info: ! lo ! True !           ::1/128
! . ! host ! . !
- 599.829684| cloud-init-1288|: ci-info: +-----+-----+-----+
+-----+-----+-----+
- 599.829721| cloud-init-1288|: ci-info: ++++++Route IP
v4 info+++++
- 599.829754| cloud-init-1288|: ci-info: +-----+-----+-----+
+-----+-----+-----+
- 599.829789| cloud-init-1288|: ci-info: ! Route ! Destination ! Gateway
! Genmask ! Interface ! Flags !
- 599.829822| cloud-init-1288|: ci-info: +-----+-----+-----+
+-----+-----+-----+
- 599.829858| cloud-init-1288|: ci-info: ! 0 ! 0.0.0.0 ! 10.222.111.1
! 0.0.0.0 ! enc600 ! UG !
```

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```

~ 599.829896| cloud-init~1288|: ci-info: ! 1 ! 10.222.1110 ! 0.0.0.0
! 255.255.255.0 ! enc600 ! U !
~ 599.829930| cloud-init~1288|: ci-info: +-----+-----+-----+-----+
+-----+-----+-----+-----+
~ 599.829962| cloud-init~1288|: ci-info: ++++++Route IPv6 info++++
+++++
~ 599.829998| cloud-init~1288|: ci-info: +-----+-----+-----+-----+
-----+-----+
~ 599.830031| cloud-init~1288|: ci-info: ! Route ! Destination ! Gateway ! Inte
rface ! Flags !
~ 599.830064| cloud-init~1288|: ci-info: +-----+-----+-----+-----+
-----+-----+
~ 599.830096| cloud-init~1288|: ci-info: ! 1 ! fe80::/64 ! :: ! en
c600 ! U !
~ 599.830131| cloud-init~1288|: ci-info: ! 3 ! local ! :: ! en
c600 ! U !
~ 599.830164| cloud-init~1288|: ci-info: ! 4 ! ff00::/8 ! :: ! en
c600 ! U !
~ 599.830212| cloud-init~1288|: ci-info: +-----+-----+-----+-----+
-----+-----+
~ 601.077953| cloud-init~1288|: Generating public/private rsa key pair.
~ 601.078101| cloud-init~1288|: Your identification has been saved in /etc/ssh/
ssh_host_rsa_key
~ 601.078136| cloud-init~1288|: Your public key has been saved in /etc/ssh/ssh
host_rsa_key.pub
~ 601.078170| cloud-init~1288|: The key fingerprint is:
~ 601.078203| cloud-init~1288|: SHA256:kHtkABZwk8AE80fy0KPzTRcYpht4iXdZmJ37Cgi3
fJ0 root@ubuntu-server
~ 601.078236| cloud-init~1288|: The key's randomart image is:
~ 601.078274| cloud-init~1288|: +----RSA 3072|----+
~ 601.078307| cloud-init~1288|: !o+*B++*.. !
~ 601.078340| cloud-init~1288|: ! o.X+=+=+ !
~ 601.078373| cloud-init~1288|: ! +.0.= oo !
~ 601.078406| cloud-init~1288|: ! ++.+.=o !
~ 601.078439| cloud-init~1288|: ! *.=.oSo !
~ 601.078471| cloud-init~1288|: ! = +.E . !
~ 601.078503| cloud-init~1288|: ! . . . !
~ 601.078537| cloud-init~1288|: ! . !
~ 601.078570| cloud-init~1288|: ! !
~ 601.078602| cloud-init~1288|: +-----SHA256|-----+
~ 601.078635| cloud-init~1288|: Generating public/private dsa key pair.
~ 601.078671| cloud-init~1288|: Your identification has been saved in /etc/ssh/
ssh_host_dsa_key
~ 601.078704| cloud-init~1288|: Your public key has been saved in /etc/ssh/ssh_
host_dsa_key.pub
~ 601.078736| cloud-init~1288|: The key fingerprint is:
~ 601.078767| cloud-init~1288|: SHA256:ZBNykvVYZVhKJeL+PWKpsdUcn21yiceX/DboXQd
Pq0 root@ubuntu-server
~ 601.078800| cloud-init~1288|: The key's randomart image is:
~ 601.078835| cloud-init~1288|: +----DSA 1024|----+
~ 601.078867| cloud-init~1288|: ! o++...+=+o !
~ 601.078899| cloud-init~1288|: ! .+...+. .!
~ 601.078932| cloud-init~1288|: ! +. + o o!
~ 601.078964| cloud-init~1288|: ! o..o = oo.!
~ 601.078996| cloud-init~1288|: ! S. =..o++!
~ 601.079029| cloud-init~1288|: ! o *.*=!

```

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```

- 601.079061| cloud-init-1288|: !          o .o.B.*!
- 601.079094| cloud-init-1288|: !          = .oEo.!
- 601.079135| cloud-init-1288|: !          .+          !
- 601.079167| cloud-init-1288|: +-----SHA256|-----+
- 601.079199| cloud-init-1288|: Generating public/private ecdsa key pair.
- 601.079231| cloud-init-1288|: Your identification has been saved in /etc/ssh/
ssh_host_ecdsa_key
- 601.079263| cloud-init-1288|: Your public key has been saved in /etc/ssh/ssh_
host_ecdsa_key.pub
- 601.079295| cloud-init-1288|: The key fingerprint is:
- 601.079327| cloud-init-1288|: SHA256:Bitar9fVHUH2FnYVSJJnldprdAcM5Est0dmRWFTU
i8k root@ubuntu-server
- 601.079362| cloud-init-1288|: The key's randomart image is:
- 601.079394| cloud-init-1288|: +----ECDSA 256|---+
- 601.079426| cloud-init-1288|: !          o**0%&!
- 601.079458| cloud-init-1288|: !          o.OB+=!
- 601.079491| cloud-init-1288|: !          .      B *o+!
- 601.079525| cloud-init-1288|: !          o      . E.=o!
- 601.079557| cloud-init-1288|: !          o . S      ....+!
- 601.079589| cloud-init-1288|: !          o o .      . . .o !
- 601.079621| cloud-init-1288|: !          .      ..      . !
- 601.079653| cloud-init-1288|: !          ..      .      !
- 601.079685| cloud-init-1288|: !          ..      .      !
- 601.079717| cloud-init-1288|: +-----SHA256|-----+
- 601.079748| cloud-init-1288|: Generating public/private ed25519 key pair.
- 601.079782| cloud-init-1288|: Your identification has been saved in /etc/ssh/
ssh_host_ed25519_key
- 601.079814| cloud-init-1288|: Your public key has been saved in /etc/ssh/ssh_
host_ed25519_key.pub
- 601.079847| cloud-init-1288|: The key fingerprint is:
- 601.079879| cloud-init-1288|: SHA256:yWsZ/5+7u7D3SIcd7HYnyajXyewnt5nQ+ZI3So3b
eN8 root@ubuntu-server
- 601.079911| cloud-init-1288|: The key's randomart image is:
- 601.079942| cloud-init-1288|: +---ED25519 256|---+
- 601.079974| cloud-init-1288|: !          !
- 601.080010| cloud-init-1288|: !          !
- 601.080042| cloud-init-1288|: !          !
- 601.080076| cloud-init-1288|: !          . .      . !
- 601.080107| cloud-init-1288|: !          S          o !
- 601.080139| cloud-init-1288|: !          =      o=++!
- 601.080179| cloud-init-1288|: !          + . o**§=!
- 601.080210| cloud-init-1288|: !          .      oo+&B%!
- 601.080244| cloud-init-1288|: !          ..o*/E!
- 601.080289| cloud-init-1288|: +-----SHA256|-----+
- 612.293731| cloud-init-2027|: Cloud-init v. 20.2-45-g5f7825e2-0ubuntu1 runnin
g 'modules:config' at Thu, 04 Jun 2020 12:06:59 +0000. Up 612.11 seconds.
- 612.293866| cloud-init-2027|: Set the following 'random' passwords
- 612.293940| cloud-init-2027|: installer:wgYsAPzYQbFYqU2X2hYm
ci-info: no authorized SSH keys fingerprints found for user installer.
<14>Jun  4 12:07:00 ec2:
<14>Jun  4 12:07:00 ec2: #####
#####
<14>Jun  4 12:07:00 ec2: -----BEGIN SSH HOST KEY FINGERPRINTS-----
<14>Jun  4 12:07:00 ec2: 1024 SHA256:ZBNyksVVYVZhKJeL+PWKpsdUcn21yiceX/DboXQdPq0
root@ubuntu-server (DSA)
<14>Jun  4 12:07:00 ec2: 256 SHA256:Bitar9fVHUH2FnYVSJJnldprdAcM5Est0dmRWFTUi8k

```

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```

root@ubuntu-server (ECDSA)
<14>Jun  4 12:07:00 ec2: 256 SHA256:yWsz/5+7u7D3SIcd7HYnyajXyeWnt5nQ+ZI3So3beN8
root@ubuntu-server (ED25519)
<14>Jun  4 12:07:00 ec2: 3072 SHA256:kHtKABZwk8AE80fy0KPzTrCYpht4iXdZmJ37Cgi3fJ0
root@ubuntu-server (RSA)
<14>Jun  4 12:07:00 ec2: -----END SSH HOST KEY FINGERPRINTS-----
<14>Jun  4 12:07:00 ec2: #####
#####
-----BEGIN SSH HOST KEY KEYS-----
ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoYTItbmlzdHAyNTYAAAAIbmlzdHAyNTYAAABBBIXM6t1/
35ot/aPI59ThIJBzg+qGJJ17+1ZVHfzMEDbsTwpM7e9pstPZUM7W1IHWqDvLQDBm/hGg4u8ZGEqmIMI=
root@ubuntu-server
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIN7QtU+en+RGruj2zuxWgkMqLmh+35/GR/OEOD16k4nA
root@ubuntu-server
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQGDjDkt7iUAvSjkUqI1l3fHysE+Gj7uLwGgGjYh639px
kcHEbbS3V48eROY9BmDISEHfjYXGY2wEH0tGJjNRR0GJhZJVNR+qAqJBioj9d/TwXEgwLP8eAy9aVtJB
K1rIylnMQltx/SIhgiymjHLCtKlVoIS4l0frT9FiF54Qi/JeJlwGJIW3W2XgcY90DT0Q5g3PSmLZ8KTR
imTf9Fy7WJEPa08b3fimYwsz9enuS/gECEUGV3M1MvrzpaQju27NUEOpSMZHR62IMxGvIjYIu3dUkAzM
MBdwxHdLMQ8rI8PehyHdiFr6g2Ifxoy5QLmb3hISKlq/R6pLLeXbb748gN2i8WCvK0AEGfa/kJDW3RNU
VYd+ACBBzyhVbiw7W1CQW/ohik3wyosUyi9nJq2Iq0A7kkGH+1XoYq/e4/MoqxhIK/oaiudYAKaCWmP1
r/fBa3hlf0f7mVhvxA3tWZc2wYUxFPTmePvpydP2PScTHMhgboaHrGIY2CdSgg8SudPKrOE= root@ub
untu-server
-----END SSH HOST KEY KEYS-----
- 612.877357| cloud-init-2045|: Cloud-init v. 20.2-45-g5f7825e2-0ubuntu1 runnin
g 'modules:final' at Thu, 04 Jun 2020 12:07:00 +0000. Up 612.79 seconds.
- 612.877426| cloud-init-2045|: ci-info: no authorized SSH keys fingerprints fo
und for user installer.
- 612.877468| cloud-init-2045|: Cloud-init v. 20.2-45-g5f7825e2-0ubuntu1 finish
ed at Thu, 04 Jun 2020 12:07:00 +0000. DataSource DataSourceNoCloud --seed=/var/l
ib/cloud/seed/nocloud|--dsmode=net|. Up 612.87 seconds
- 612.877509| cloud-init-2045|: Welcome to Ubuntu Server InstallerÜ
- 612.877551| cloud-init-2045|: Above you will find SSH host keys and a random
password set for the `installer` user. You can use these credentials to ssh-in a
nd complete the installation. If you provided SSH keys in the cloud-init datasou
rce, they were also provisioned to the installer user.
- 612.877634| cloud-init-2045|: If you have access to the graphical console, li
ke TTY1 or HMC ASCII terminal you can complete the installation there too.

```

It is possible to connect to the installer over the network, which might allow the use of a more capable terminal.

To connect, SSH to installer@10.222.111.24.

The password you should use is "KRuXtz5dURAYPkcjcUvA".

The host key fingerprints are:

```

RSA      SHA256:3IvYmKU05LQSKBx0VZUJMzdtXpz3RJL3dEQsg3UWc54
ECDSA    SHA256:xd1xnkBpn49DUbuP8uWro2mu1GM4MtnqR2WEWg1fS3o
ED25519  SHA256:Hk3+/4+X7NJBHL6/e/6xHhNXsbHBsOvt6i8YEFUepko

```

Ubuntu Focal Fossa (development branch) ubuntu-server ttyS0

The next step is to remotely connect to the install system and to proceed with the Subiquity installer.

Notice that at the end of the installer boot-up process, all necessary data is provided to pro-

ceed with running the installer in a remote SSH shell. The command to execute locally is:

```
user@workstation:~$ ssh installer@10.222.111.24
```

A temporary random password for the installation was created and shared as well, which should be used without the leading and trailing double quotes:

```
"KRuXtz5dURAYPkcjcUvA"
```

```
user@workstation:~$ ssh installer@10.222.111.24
```

```
The authenticity of host '10.222.111.24 (10.222.111.24)' can't be established.  
ECDSA key fingerprint is  
SHA256:xd1xnkBpn49DUbuP8uWro2mu1GM4MtnqR2WEWg1fS3o.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '10.222.111.24' (ECDSA) to the list of known hosts.  
installer@10.222.111.24's password: KRuXtz5dURAYPkcjcUvA
```

You may now temporarily see some login messages like these:

```
Welcome to Ubuntu Focal Fossa (development branch) (GNU/Linux 5.4.0-42-generic s390x)
```

```
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:        https://ubuntu.com/pro
```

```
System information as of Wed Jun  3 17:32:10 UTC 2020
```

```
System load:   0.0      Memory usage: 2%   Processes:      146  
Usage of /home: unknown  Swap usage:   0%   Users logged in: 0
```

```
0 updates can be installed immediately.  
0 of these updates are security updates.
```

```
The programs included with the Ubuntu system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.
```

```
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by applicable law.
```

Eventually, the initial Subiquity installer screen appears:

```
=====
Willkommen! Bienvenue! Welcome! ????? ?????????? Welkom!
=====
Use UP, DOWN and ENTER keys to select your language.

[ English                > ]
[ Asturianu              > ]
[ Catal                 > ]
[ Hrvatski               > ]
[ Nederlands             > ]
[ Suomi                 > ]
[ Fran  ais              > ]
[ Deutsch                > ]
```

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```
[ Magyar                > ]
[ Latviešu              > ]
[ Norsk bokmål          > ]
[ Polski                > ]
[ Español               > ]
```

From this point, you can follow the normal Subiquity installation. For more details, refer to the [Subiquity installer documentation](https://canonical-subiquity.readthedocs-hosted.com/en/latest/index.html)⁷⁹.

```
=====
Keyboard configuration
=====
Please select your keyboard layout below, or select "Identify keyboard" to
detect your layout automatically.

Layout:  [ English (US)                v ]

Variant: [ English (US)                v ]

[ Identify keyboard ]

[ Done      ]
[ Back      ]
```

```
=====
Zdev setup
=====
ID                                ONLINE  NAMES                                ^
generic-ccw
0.0.0009                          >
0.0.000c                          >
0.0.000d                          >
0.0.000e                          >

dasd-eckd
0.0.0190                          >
0.0.0191                          >
0.0.019d                          >
0.0.019e                          >
0.0.0200                          >
0.0.0300                          >
0.0.0400                          >
0.0.0592                          >
v
```

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⁷⁹ <https://canonical-subiquity.readthedocs-hosted.com/en/latest/index.html>

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```
[ Continue ]
[ Back    ]
```

- If the list is long, hit the End key that will automatically scroll you down to the bottom of the Z devices list and screen.

```
=====
Zdev setup
=====
ID                ONLINE  NAMES                ^
generic-ccw
0.0.0009          >
0.0.000c          >
0.0.000d          >
0.0.000e          >

dasd-eckd
0.0.0190          >
0.0.0191          >
0.0.019d          >
0.0.019e          >
0.0.0200          > < (close)
0.0.0300          > Enable
0.0.0400          > Disable
0.0.0592          >
                                     v

[ Continue ]
[ Back    ]
```

```
=====
Zdev setup
=====
ID                ONLINE  NAMES                ^
generic-ccw
0.0.0009          >
0.0.000c          >
0.0.000d          >
0.0.000e          >

dasd-eckd
0.0.0190          >
0.0.0191          >
0.0.019d          >
0.0.019e          >
0.0.0200          online  dasda >
0.0.0300          >
0.0.0400          >
0.0.0592          >
                                     v

[ Continue ]
[ Back    ]
```

```
=====
Zdev setup
=====

dasd-eckd
0.0.0190                >
0.0.0191                >
0.0.019d                >
0.0.019e                >
0.0.0200                >
0.0.0300                >
0.0.0400                >
0.0.0592                >

qeth
0.0.0600:0.0.0601:0.0.0602      enc600 >
0.0.0603:0.0.0604:0.0.0605      >

dasd-eckd
0.0.1607                >

                                [ Continue ]
                                [ Back      ]
```

```
=====
Network connections
=====

Configure at least one interface this server can use to talk to other
machines, and which preferably provides sufficient access for updates.

NAME    TYPE  NOTES
[ enc600 eth  -          > ]
static  10.222.111.24/24
02:28:0a:00:00:39 / Unknown Vendor / Unknown Model

[ Create bond > ]

                                [ Done      ]
                                [ Back      ]
```

```
=====
Configure proxy
=====

If this system requires a proxy to connect to the internet, enter its
details here.

Proxy address:
                If you need to use a HTTP proxy to access the outside world,
                                                         (continues on next page)
```

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enter the proxy information here. Otherwise, leave this blank.

The proxy information should be given in the standard form of "http://[[user][:pass]@host[:port]]/".

[Done]
[Back]

```
=====
Configure Ubuntu archive mirror
```

If you use an alternative mirror for Ubuntu, enter its details here.

```
Mirror address: http://ports.ubuntu.com/ubuntu-ports
You may provide an archive mirror that will be used instead
of the default.
```

[Done]
[Back]

```
=====
Guided storage configuration
```

Configure a guided storage layout, or create a custom one:

(X) Use an entire disk

```
[ 0X0200      local disk 6.876G      v ]
```

```
[ ] Set up this disk as an LVM group
```

```
[ ] Encrypt the LVM group with LUKS
```

Passphrase:

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Confirm passphrase:

() Custom storage layout

[Done]
[Back]

Storage configuration

FILE SYSTEM SUMMARY

MOUNT POINT	SIZE	TYPE	DEVICE TYPE
[/	6.875G	new ext4	new partition of local disk >]

AVAILABLE DEVICES

No available devices

[Create software RAID (md) >]
[Create volume group (LVM) >]

USED DEVICES

[Done]
[Reset]
[Back]

Storage configuration

FILE SYSTEM SUMMARY

Confirm destructive action

Selecting Continue below will begin the installation process and result in the loss of data on the disks selected to be formatted.

You will not be able to return to this or a previous screen once the installation has started.

Are you sure you want to continue?

[No]
[Continue]

[Reset]
[Back]

```
=====
Profile setup
=====
Enter the username and password you will use to log in to the system. You
can configure SSH access on the next screen but a password is still needed
for sudo.

Your name: Ed Example

Your server's name: 10.222.111.24
                   The name it uses when it talks to other computers.

Pick a username: ubuntu

Choose a password: *******

Confirm your password: *******

[ Done ]
```

```
=====
SSH Setup
=====
You can choose to install the OpenSSH server package to enable secure remote
access to your server.

[ ] Install OpenSSH server

Import SSH identity: [ No v ]
                   You can import your SSH keys from Github or Launchpad.

Import Username:

[X] Allow password authentication over SSH

[ Done ]
[ Back ]
```

- It's a nice and convenient new feature to add the user's SSH keys during the installation to the system, since that makes the system login password-less on the initial login!

```
=====
SSH Setup
=====
You can choose to install the OpenSSH server package to enable secure remote
access to your server.
```

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[X] Install OpenSSH server

Import SSH identity: [from Launchpad v]
You can import your SSH keys from Github or Launchpad.

Launchpad Username: user
Enter your Launchpad username.

[X] Allow password authentication over SSH

[Done]
[Back]

SSH Setup

You can choose to install the OpenSSH server package to enable secure remote access to your server.

Confirm SSH keys

Keys with the following fingerprints were fetched. Do you want to use them?

2048 SHA256:joGsdw7NbJRkg17sRyXaegoR0iZEdDWdR9HpbC2KIw user@W520 |
(RSA)
521 SHA256:T3JzxvB6K1GzXJpP5NFgX4yXvk0jhHgVbw01F7/fZ2c
frank.heimes@canonical.com (ECDSA)

[Yes]
[No]

[Done]
[Back]

Featured Server Snaps

These are popular snaps in server environments. Select or deselect with SPACE, press ENTER to see more details of the package, publisher and versions available.

[]	kata-containers	Lightweight virtual machines that seamlessly plug into >
[]	docker	Docker container runtime >
[]	mosquitto	Eclipse Mosquitto MQTT broker >
[]	etcd	Resilient key-value store by CoreOS >
[]	stress-ng	A tool to load, stress test and benchmark a computer s >

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```
[ ] sabnzbd      SABnzbd                      >
[ ] wormhole     get things from one computer to another, safely >
[ ] slcli        Python based SoftLayer API Tool.           >
[ ] doctl        DigitalOcean command line tool            >
[ ] keepalived   High availability VRRP/BFD and load-balancing for Linu >
[ ] juju         Simple, secure and stable devops. Juju keeps complexit >
```

```
[ Done      ]
[ Back      ]
```

```
=====
Install complete!
=====
```

```

      configuring raid (mdadm) service      ^
      installing kernel
      setting up swap
      apply networking config
      writing etc/fstab
      configuring multipath
      updating packages on target system
      configuring pollinate user-agent on target
      updating initramfs configuration
finalizing installation
      running 'curtin hook'
      curtin command hook
      executing late commands
final system configuration
      configuring cloud-init
      installing openssh-server |          v

```

[View full log]

```
=====
Installation complete!
=====
```

```

      Finished install!
      apply networking config      ^
      writing etc/fstab
      configuring multipath
      updating packages on target system
      configuring pollinate user-agent on target
      updating initramfs configuration
      finalizing installation
      running 'curtin hook'
      curtin command hook
      executing late commands
final system configuration
      configuring cloud-init
      installing openssh-server
      restoring apt configuration
      downloading and installing security updates

```

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```
[ View full log ]
[ Reboot        ]
```

Installation complete!

```
=====
----- Finished install! -----
  apply networking config
  writing etc/fstab
  configuring multipath
  updating packages on target system
  configuring pollinate user-agent on target
  updating initramfs configuration
  finalizing installation
  running 'curtin hook'
  curtin command hook
  executing late commands
  final system configuration
  configuring cloud-init
  installing openssh-server
  restoring apt configuration
  downloading and installing security updates
[ Connection to 10.222.111.24 closed by remote host.
[ Rebooting... ]
Connection to 10.222.111.24 closed.
user@workstation:~$
```

Type reset to clear the screen and to revert it back to the defaults.

Now remove the old host key, since the system got a new one during the installation:

```
user@workstation:~$ ssh-keygen -f "/home/user/.ssh/known_hosts" -R
"10.222.111.24"

# Host 10.222.111.24 found: line 159
/home/user/.ssh/known_hosts updated.
Original contents retained as /home/user/.ssh/known_hosts.old
```

And finally login to the newly installed z/VM guest:

```
user@workstation:~$ ssh ubuntu@10.222.111.24
```

```
Warning: Permanently added the ECDSA host key for IP address
'10.222.111.24' to the list of known hosts.
Welcome to Ubuntu 20.04.5 LTS (GNU/Linux 5.4.0-42-generic s390x)
```

```
* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro
```

```
System information as of Wed 03 Jun 2020 05:50:05 PM UTC
```

```
System load: 0.08           Memory usage: 2%   Processes:      157
Usage of /:  18.7% of 6.70GB Swap usage:   0%   Users logged in: 0
```

```
0 updates can be installed immediately.
0 of these updates are security updates.
```

```
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the individual files
in /usr/share/doc/*/copyright.
```

```
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by applicable law.
```

```
To run a command as administrator (user "root"), use `sudo <command>`.
See `man sudo_root` for details.
```

```
ubuntu@10.222.111.24:~$ uptime
```

```
17:50:09 up 1 min,  1 user,  load average: 0.08, 0.11, 0.05
```

```
ubuntu@10.222.111.24:~$ lsb_release -a
```

```
No LSB modules are available.
Distributor ID:      Ubuntu
Description: Ubuntu 20.04.5 LTS
Release:            20.04
Codename:           focal
```

```
ubuntu@10.222.111.24:~$ uname -a
```

```
Linux 10.222.111.24 5.4.0-42-generic #30-Ubuntu SMP Wed Aug 05 16:57:22 UTC 2020
s390x s390x s390x GNU/Linux
```

```
ubuntu@10.222.111.24:~$ exit
```

```
logout
Connection to 10.222.111.24 closed.
```

```
Done!
```

Non-interactive IBM z/VM autoinstall (s390x)

This non-interactive installation uses ‘autoinstall’, which can be considered the successor to the Debian installer (d-i) and preseed on Ubuntu. This is a detailed step-by-step guide, including output and logs (which are partially a bit shortened, as indicated by ‘...’, to limit the size of this document).

The example z/VM guest here uses a direct-access storage device (*DASD*) and is connected to a regular (non-VLAN) network.

For a zFCP and a VLAN network example, please see the [non-interactive IBM LPAR \(s390x\) installation using autoinstall](#) (page 142) guide.

- Start with the preparation of the (FTP) install server (if it doesn’t already exist).

```
user@local:~$ ssh admin@installserver.local
admin@installserver:~$ mkdir -p /srv/ftp/ubuntu-daily-live-server-20.04
admin@installserver:~$ wget http://cdimage.ubuntu.com/ubuntu-server/focal/daily-live/
current/focal-live-server-s390x.iso --directory-prefix=/srv/ftp/ubuntu-daily-live-
server-20.04
--2020-06-26 12:18:48-- http://cdimage.ubuntu.com/ubuntu-server/focal/daily-live/
current/focal-live-server-s390x.iso
Resolving cdimage.ubuntu.com (cdimage.ubuntu.com)... 2001:67c:1560:8001::1d,
2001:67c:1360:8001::28, 2001:67c:1360:8001::27,
...
Connecting to cdimage.ubuntu.com (cdimage.ubuntu.com)|2001:67c:1560:8001::1d|:80...
connected.
HTTP request sent, awaiting response... 200 OK
Length: 700952576 (668M) [application/x-iso9660-image]
Saving to: ‘focal-live-server-s390x.iso’

focal-live-server-s 100%[=====>] 668.48M  2.73MB/s   in 4m 54s

2020-06-26 12:23:42 (2.27 MB/s) - ‘focal-live-server-s390x.iso’ saved [700952576/
700952576]
admin@installserver:~$
```

- The ISO image needs to be extracted now. Since files in the boot folder need to be modified, loopback mount is not an option here:

```
admin@installserver:~$ cd /srv/ftp/ubuntu-daily-live-server-20.04
admin@installserver:/srv/ftp/ubuntu-daily-live-server-20.04$ mkdir iso
admin@installserver:/srv/ftp/ubuntu-daily-live-server-20.04$ sudo mount -o loop ./
focal-live-server-s390x.iso ./iso
[sudo] password for admin:
mount: /home/user/iso-test/iso: WARNING: device write-protected, mounted read-only.
admin@installserver:/srv/ftp/ubuntu-daily-live-server-20.04$ ls -l
total 684530
-rw-rw-r-- 1 user user 700952576 Jun 26 10:12 focal-live-server-s390x.iso
dr-xr-xr-x 10 root root 2048 Jun 26 10:12 iso
admin@installserver:/srv/ftp/ubuntu-daily-live-server-20.04$
```

- Now make sure an FTP server is running in the *installserver* with */srv/ftp* as ftp-server root (as used in this example).
- Next, prepare an *autoinstall* (HTTP) server. This hosts the configuration data for the non-interactive installation.

```
admin@installserver:/srv/ftp/ubuntu-daily-live-server-20.04$ mkdir -p /srv/www/
autoinstall/zvmguest
admin@installserver:/srv/ftp/ubuntu-daily-live-server-20.04$ cd /srv/www/autoinstall/
zvmguest
admin@installserver:/srv/www/autoinstall/zvmguest$
admin@installserver:/srv/www/autoinstall/zvmguest$ echo "instance-id: $(uuidgen ||
openssl rand -base64 8)" > meta-data
admin@installserver:/srv/www/autoinstall/zvmguest$ cat meta-data
instance-id: 2c2215fb-6a38-417f-b72f-376b1cc44f01
admin@installserver:/srv/www/autoinstall/zvmguest$
```

```
admin@installserver:/srv/www/autoinstall/zvmguest$ vi user-data
admin@installserver:/srv/www/autoinstall/zvmguest$ cat user-data
```

```
#cloud-config
```

```
autoinstall:
```

```
  version: 1
```

```
  refresh-installer:
```

```
    update: yes
```

```
  reporting:
```

```
    builtin:
```

```
      type: print
```

```
  apt:
```

```
    preserve_sources_list: false
```

```
    primary:
```

```
      - arches: [amd64, i386]
```

```
        uri: http://archive.ubuntu.com/ubuntu
```

```
      - arches: [default]
```

```
        uri: http://ports.ubuntu.com/ubuntu-ports
```

```
  keyboard:
```

```
    layout: en
```

```
    variant: us
```

```
  locale: en_US
```

```
  identity:
```

```
    hostname: zvmguest
```

```
    password:
```

```
"$6$ebJ1f8wxED22bTL4F46P0"
```

```
    username: ubuntu
```

```
  user-data:
```

```
    timezone: America/Boston
```

```
  users:
```

```
    - name: ubuntu
```

```
      password:
```

```
"$6$KwuxED22bTL4F46P0"
```

```
      lock_passwd: false
```

```
  early-commands:
```

```
    - touch /tmp/lets_activate_the_s390x_devices
```

```
    - chzdev dasd -e 1f00
```

```
    - touch /tmp/s390x_devices_activation_done
```

```
  network:
```

```
    version: 2
```

```
  ethernet:
```

```
    enc600:
```

```
      addresses: [10.11.12.23/24]
```

```
      gateway4: 10.11.12.1
```

```
      nameservers:
```

```
        addresses: [10.11.12.1]
```

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```
ssh:
  install-server: true
  allow-pw: true
  authorized-keys: ['ssh-rsa meQwtZ user@workstation # ssh-import-id lp:user']
admin@installserver:~$
```

- For s390x installations, the early-commands section is the interesting part:

```
early-commands:
- touch /tmp/lets_activate_the_s390x_devices
- chzdev dasd -e 1f00
- touch /tmp/s390x_devices_activation_done
```

The first and last early-commands are optional; they only frame and indicate the real s390x command activation.

In this particular example a single *DASD ECKD* disk with the address 1f00 is enabled. zFCP disk storage can be enabled via their host (*host-bus-adapters*) addresses, for example *e000* (chzdev zfc -e e000) and *e100* (chzdev zfc -e e000). These have certain Logical Unit Numbers (LUNs) assigned, which are all automatically discovered and activated by chzdev zfc-lun -e --online. Activation of a QETH device would look like this: chzdev qeth -e 0600.

- For more details about the autoinstall config options, please have a look at the [autoinstall reference](#)⁸⁰ and [autoinstall schema](#)⁸¹ page.
- Now make sure a HTTP server is running with /srv/www as web-server root (in this particular example).
- Log in to your z/VM system using your preferred 3270 client – for example x3270 or c3270.
- Transfer the installer kernel, initrd, parmfile and exec file to the z/VM system that is used for the installation. Put these files (for example) on File Mode (Fm) A (a.k.a disk A):

```
listfiles
UBUNTU  EXEC    A1
KERNEL  UBUNTU  A1
INITRD   UBUNTU  A1
PARMFILE UBUNTU  A1
```

- Now specify the necessary autoinstall parameters in the parmfile:

```
xedit PARMFILE UBUNTU A
  PARMFILE UBUNTU 01 F 80 Trunc=80 Size=3 Line=0 Col=1 Alt=0
00000 * * * Top of File * * *
00001 ip=10.11.12.23::10.11.12.1:255.255.255.0:zvmguest:enc600:none:10.11.12.1
00002 url=ftp://installserver.local:21/ubuntu-daily-live-server-20.04/focal-live-ser
ve-ser
00003 ver-s390x.iso autoinstall ds=nocloud-net;s=http://installserver.local:80
00004 /autoinstall/zvmguest/ --- quiet
00005 * * * End of File * * *
```

⁸⁰ <https://canonical-subiquity.readthedocs-hosted.com/en/latest/reference/autoinstall-reference.html>

⁸¹ <https://canonical-subiquity.readthedocs-hosted.com/en/latest/reference/autoinstall-schema.html>

Note:

In case of any issues hitting the 80-character-per-line limit of the file, you can write parameters across two lines as long as there are no unwanted white spaces. To view all 80 characters in one line, disable the prefix area on the left. “prefix off | on” will be your friend – use it in the command area.

- You can now start the z/VM installation by executing the UBUNTU REXX script with UBUNTU.
- Now monitor the initial program load (IPL) – a.k.a. the boot-up process – of the install system. This is quite crucial, because during this process a temporary installation password is generated and displayed. The line(s) look similar to this:

```
...
|37.487141| cloud-init-1873|: Set the following 'random' passwords
|37.487176| cloud-init-1873|: installer: **i7UFdP8fhiVVMme3qqH8**
...
```

This password is needed for remotely connecting to the installer via SSH in the next step.

- So, start the REXX script:

```
UBUNTU
00: 00000004 FILES PURGED
00: RDR FILE 1254 SENT FROM zvmguest  PUN WAS 1254 RECS 102K CPY
001 A NOHOLD NO
KEEP
00: RDR FILE 1258 SENT FROM zvmguest  PUN WAS 1258 RECS 0003 CPY
001 A NOHOLD NO
KEEP
00: RDR FILE 1262 SENT FROM zvmguest  PUN WAS 1262 RECS 303K CPY
001 A NOHOLD NO
KEEP
00: 00000003 FILES CHANGED
00: 00000003 FILES CHANGED
01: HCPGSP2627I The virtual machine is placed in CP mode due to a SIGP
initial C
PU reset from CPU 00.
02: HCPGSP2627I The virtual machine is placed in CP mode due to a SIGP
initial C
PU reset from CPU 00.
03: HCPGSP2627I The virtual machine is placed in CP mode due to a SIGP
initial C
PU reset from CPU 00.
- 0.403380| Initramfs unpacking failed: Decoding failed
ln: /tmp/mountroot-fail-hooks.d//scripts/init-premount/lvm2: No such file or dir
ectory
QETH device 0.0.0600:0.0.0601:0.0.0602 configured
IP-Config: enc600 hardware address 02:28:0b:00:00:51 mtu 1500
IP-Config: enc600 guessed broadcast address 10.11.12.255
IP-Config: enc600 complete:
address: 10.11.12.23 broadcast: 10.210.210.255 netmask: 255.255.255.0

gateway: 10.11.12.1 dns0 : 10.11.12.1 dns1 : 0.0.0.0
```

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```

host      : zvmguest
rootserver: 0.0.0.0 rootpath:
filename  :
Connecting to installserver.local:21 (10.11.12.2:21)
focal-live-server-s3  2% !                               ! 15.2M  0:00:48 ETA
focal-live-server-s3  16% !*****                         ! 126M  0:00:09 ETA
focal-live-server-s3  31% !*****                         ! 236M  0:00:06 ETA
focal-live-server-s3  46% !*****                         ! 347M  0:00:04 ETA
focal-live-server-s3  60% !*****                         ! 456M  0:00:03 ETA
focal-live-server-s3  74% !*****                         ! 563M  0:00:02 ETA
focal-live-server-s3  88% !*****                         ! 667M  0:00:00 ETA
focal-live-server-s3 100% !*****                         ! 752M  0:00:00 ETA
mount: mounting /cow on /root/cow failed: No such file or directory
Connecting to plymouth: Connection refused
passwd: password expiry information changed.
- 16.748137| /dev/loop3: Can't open blockdev
- 17.908156| systemd-1|: Failed unmounting /cdrom.
- -0;1;31mFAILED -0m| Failed unmounting -0;1;39m/cdrom -0m.
- -0;32m OK -0m| Listening on -0;1;39mJournal Socket -0m.
      Mounting -0;1;39mHuge Pages File System -0m...
      Mounting -0;1;39mPOSIX Message Queue File System -0m...
      Mounting -0;1;39mKernel Debug File System -0m...
      Starting -0;1;39mJournal Service -0m...
...
[ 61.190916] cloud-init[2076]: Cloud-init v. 20.1-10-g71af48df-0ubuntu5 running
'modules:final' at Fri, 26 Jun 2020 11:02:01 +0000. Up 61.09 seconds.
[ 61.191002] cloud-init[2076]: ci-info: no authorized SSH keys fingerprints fo
und for user installer.
[ 61.191071] cloud-init[2076]: Cloud-init v. 20.1-10-g71af48df-0ubuntu5 finished at
Fri, 26 Jun 2020 11:02:01 +0000.
Datasource DataSourceNoCloudNet [seed=cmdline,
/var/lib/cloud/seed/nocloud,
http://installserver.local:80/autoinstall/zvmguest/]
[dsmode=net]. Up 61.18 seconds
[ 61.191136] cloud-init[2076]: Welcome to Ubuntu Server Installer!
[ 61.191202] cloud-init[2076]: Above you will find SSH host keys and a random
password set for the `installer` user. You can use these credentials to ssh-in
and complete the installation. If you provided SSH keys in the cloud-init datasource,
they were also provisioned to the installer user.
[ 61.191273] cloud-init[2076]: If you have access to the graphical console,
like TTY1 or HMC ASCII terminal you can complete the installation there too.

It is possible to connect to the installer over the network, which
might allow the use of a more capable terminal.

To connect, SSH to installer@10.11.12.23.

The password you should use is 'i7UFdP8fhiVVMme3qqH8'.

The host key fingerprints are:

RSA      SHA256:rBXLeUke3D4gKdsruKEajHjocxc9hr3PI
ECDSA    SHA256:KZZYFswtKxFXWQPuQS9Qp0BUoS6RHswis
ED25519  SHA256:s+5tZfagx0zffc6gYRGW3t1KcBH6f+Vt0

Ubuntu 20.04 LTS ubuntu-server scli_line0

```

- At short notice, you can even log in to the system with the user 'installer' and the temporary password that was given at the end of the boot-up process (see above) of the installation system:

```
user@workstation:~$ ssh installer@zvmguest

The authenticity of host 'zvmguest (10.11.12.23)' can't be established.
ECDSA key fingerprint is SHA256:0/dU/D8jJAEGQcbqKGE9La24IRxUPLpzzs5li9F6Vvk.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'zvmguest,10.11.12.23' (ECDSA) to the list of known
hosts.
installer@zvmguest's password:
Welcome to Ubuntu 20.04 LTS (GNU/Linux 5.4.0-37-generic s390x)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri Jun 26 11:08:18 UTC 2020

System load:      1.25      Memory usage: 4%    Processes:      192
Usage of /home:  unknown   Swap usage:  0%    Users logged in: 0

0 updates can be installed immediately.
0 of these updates are security updates.

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

the installer running on /dev/tty1 will perform the autoinstall

press enter to start a shell
```

- Please note that it informs you about a currently-running autoinstall process:

```
the installer running on /dev/tty1 will perform the autoinstall
```

- Nevertheless, we can quickly check some things – although, only until the autoinstall process is finished and the post-install reboot has been triggered:

```
root@ubuntu-server:/# ls -l /tmp/lets_activate_the_s390x_devices
-rw-r--r-- 1 root root 0 Jun 26 11:08 /tmp/lets_activate_the_s390x_devices
-rw-r--r-- 1 root root 0 Jun 26 11:09 /tmp/s390x_devices_activation_done

root@ubuntu-server:/# lszdev | grep yes
dasd-eckd    0.0.1f00                                yes yes
qeth         0.0.0600:0.0.0601:0.0.0602             yes no enc600
```

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```
root@ubuntu-server:/#
```

- If you wait long enough, you'll see that the remote session gets closed:

```
root@ubuntu-server:/# Connection to zvmguest closed by remote host.
Connection to zvmguest closed.
user@workstation:~$
```

- As well as at the console:

```
ubuntu-server login:

[[0;1;31mFAILED[0m] Failed unmounting [0;1;39m/cdrom[0m.
[ 169.161139] sd-umoun[15600]: Failed to unmount /oldroot: Device or resource busy
[ 169.161550] sd-umoun[15601]: Failed to unmount /oldroot/cdrom: Device or resource
busy
[ 169.168118] shutdown[1]: Failed to finalize file systems, loop devices, ignoring
Total: 282 Selected: 0

Command:
```

- ...and that the post-install reboot got triggered:

```
Message
Mounting [0;1;39mKernel Configuration File System[0m...
Starting [0;1;39mApply Kernel Variables[0m...
[[0;32m OK [0m] Finished [0;1;39mRemount Root and Kernel File Systems[0m.
[[0;32m OK [0m] Finished [0;1;39mUncomplicated firewall[0m.
[[0;32m OK [0m] Mounted [0;1;39mFUSE Control File System[0m.
[[0;32m OK [0m] Mounted [0;1;39mKernel Configuration File System[0m.
...
[ 35.378928] cloud-init[2565]: Cloud-init v. 20.1-10-g71af48df-0ubuntu5 runnin
g 'modules:final' at Fri, 26 Jun 2020 11:10:44 +0000. Up 35.29 seconds.
[ 35.378978] cloud-init[2565]: Cloud-init v. 20.1-10-g71af48df-0ubuntu5 finish
ed at Fri, 26 Jun 2020 11:10:44 +0000. Datasource DataSourceNone. Up 35.37 seconds
[ 35.379008] cloud-init[2565]: 2020-06-26 11:10:44,359 - cc_final_message.py[W
ARNING]: Used fallback datasource
[[0;32m OK [0m] Finished [0;1;39mExecute cloud user/final scripts[0m.
[[0;32m OK [0m] Reached target [0;1;39mCloud-init target[0m.

zvmguest login:
```

- With the completion of the reboot the autoinstall is finished and the z/VM guest is ready to use:

```
user@workstation:~$ ssh-keygen -f "/home/user/.ssh/known_hosts" -R "zvmguest"
# Host zvmguest found: line 163
/home/user/.ssh/known_hosts updated.
Original contents retained as /home/user/.ssh/known_hosts.old
user@workstation:~$ ssh ubuntu@zvmguest
The authenticity of host 'zvmguest (10.11.12.23)' can't be established.
ECDSA key fingerprint is SHA256:iGtCArEg+ZnoZlgt0kvmyy0gPY8UEI+f7zoISO+f+m/0.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'zvmguest,10.11.12.23' (ECDSA) to the list of known hosts.
Welcome to Ubuntu 20.04 LTS (GNU/Linux 5.4.0-39-generic s390x)
```

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```
* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro

System information as of Fri 26 Jun 2020 11:12:23 AM UTC

System load: 0.21           Memory usage: 3%   Processes:      189
Usage of /:  28.2% of 30.88GB Swap usage:   0%   Users logged in: 0

10 updates can be installed immediately.
0 of these updates are security updates.
To see these additional updates run: apt list --upgradable

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY,
to the extent permitted by applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@zvmguest:~$ uptime
 11:12:35 up 2 min,  1 user,  load average: 0.18, 0.17, 0.08
ubuntu@zvmguest:~$ lsb_release -a
No LSB modules are available.
Distributor ID:      Ubuntu
Description:         Ubuntu 20.04 LTS
Release:             20.04
Codename:            focal
ubuntu@zvmguest:~$ uname -a
Linux zvmguest 5.4.0-39-generic #43-Ubuntu SMP Fri Jun 19 10:27:17
UTC 2020 s390x s390x s390x
GNU/Linux
ubuntu@zvmguest:~$ lsdev | grep yes
dasd-eckd      0.0.1f00                      yes  yes
qeth           0.0.0600:0.0.0601:0.0.0602   yes  yes  enc600
ubuntu@zvmguest:~$ exit
logout
Connection to zvmguest closed.
user@workstation:~$
```

Some closing notes

- It's always best to use the latest installer and autoinstall components: either make sure the installer gets updated to the latest level, or just use a current daily live-server image.
- The ISO image specified with the kernel parameters needs to fit in the boot folder. Its kernel and initrd are specified in the 'Load from Removable Media and Server' task at the [hardware management console \(HMC\)](#).
- In addition to activating disk storage resources in early-commands, other devices like OS-A/QETH can be added and activated there, too. This is not needed for the basic network

device, as specified in the kernel parameters that are used for the installation (that one is automatically handled).

- If everything is properly set up – FTP server for the image, HTTP server for the autoinstall config files – the installation can be as quick as 2 to 3 minutes (depending on the complexity of the autoinstall YAML file).
- There is a simple way of generating a sample autoinstall YAML file: one can perform an interactive Subiquity installation, grab the file `/var/log/installer/autoinstall-user-data`, and use it as an example – but beware that the `early-commands` entries to activate the s390x-specific devices need to be added manually!

Interactive live server installation on IBM Z LPAR (s390x)

Doing an interactive (manual) live installation as described here - meaning without specifying a parmfile - has been supported in Ubuntu Server since LTS 20.04.5 ('Focal').

The following guide assumes that an FTP server to host the installation files is in place, which can be used by the 'Load from Removable Media and Server' task of the *Hardware Management Console (HMC)*.

Download and mount the ISO

Download the 'focal daily live image' (later 20.04.5 image)⁸².

Now loop-back mount the ISO to extract the four files that are needed for the installation:

```
user@workstation:~$ mkdir iso
```

```
user@workstation:~$ sudo mount -o loop ubuntu-20.04.5-live-server-s390x.iso iso
```

```
user@workstation:~$ ls -l  
./iso/boot/{ubuntu.exec,parmfile.*,kernel.u*,initrd.u*}$  
./iso/boot/initrd.ubuntu > ./iso/boot/kernel.ubuntu >  
./iso/boot/parmfile.ubuntu > ./iso/boot/ubuntu.exec
```

Make the files available via your FTP server

- Open the IBM Z HMC and navigate to 'Systems Management' on your machine.
- Select the LPAR that you are going to install Ubuntu Server on. In this example we use LPAR s1lp11.
- Now select menu: 'Recovery' → 'Load from Removable Media or Server' task.
- Fill out the 'Load from Removable Media or Server' form as follows (adapt the settings to your particular installation environment):

⁸² <http://cdimage.ubuntu.com/ubuntu/releases/20.04.5/release/ubuntu-20.04.5-live-server-s390x.iso>

```
Load from Removable Media, or Server - <machine>:s1lp11
Use this task to load operating system software or utility programs
from a CD / DVD-ROM or a server that can be accessed using FTP.
Select the source of the software:
o Hardware Management Console CD / DVD-ROM
o Hardware Management Console CD / DVD-ROM and assign for operating system use
o Hardware Management Console USB flash memory drive
o Hardware Management Console USB flash memory drive and assign for operating system
use
* FTP Source
  Host computer:      install-server
  User ID:           ftpuser
  Password:          *****
  Account (optional):
  File location (optional): ubuntu-live-server-20.04.5/boot
```

You may need to adjust the file's location according to your install server environment.

- Confirm the entered data:

```
Load from Removable Media or Server - Select Software to Install -
<machine>:s1lp11
Select the software to install.
Select  Name                                     Description
*  ubuntu-live-server-20.04.5/boot/ubuntu.ins    Ubuntu for IBM Z (default kernel)
```

- Confirm again that jobs might be cancelled if proceeding:

```
Load from Removable Media or Server Task Confirmation -
<machine>:s1lp11
Load will cause jobs to be cancelled.
Do you want to continue with this task?
ACT33501
```

- And confirm a last time that it's understood that the task is disruptive:

```
Disruptive Task Confirmation : Load from Removable Media or Server -
<machine>:s1lp11

Attention: The Load from Removable Media or Server task is disruptive.

Executing the Load from Removable Media or Server task may
adversely affect the objects listed below. Review the confirmation text
for each object before continuing with the Load from Removable Media
or Server task.

Objects that will be affected by the Load from Removable Media or
Server task

System Name      Type      OS Name      Status      Confirmation Text
<machine>:s1lp11 Image      Operating    Load from Removable Media
or Server causes operations to be disrupted, since the target is
currently in use and operating normally.

Do you want to execute the Load from Removable Media or Server task?
```

- The 'Load from Removable media or Server' task is now executed:

```
Load from Removable media or Server Progress - P00B8F67:S1LPB
Turn on context sensitive help.
Function duration time:      00:55:00
Elapsed time: 00:00:04
Select      Object Name      Status
*           <machine> s1lp11  Please wait while the image is being loaded.
```

- This may take a moment, but you will soon see:

```
Load from Removable media or Server Progress - <machine>:s1lp11
Function duration time:      00:55:00
Elapsed time: 00:00:21
Select      Object Name      Status
*           <machine> s1lp11  Success
```

- Close the 'Load from Removable media or Server' task and open the console a.k.a. 'Operating System Messages' instead. If no parmfile was configured or provided, you will find the following lines in the 'Operating System Messages' task:

```
Operating System Messages - <machine>:s1lp11

Message

Unable to find a medium container a live file system
Attempt interactive netboot from a URL?
yes no (default yes):
```

- By default, you will now see the interactive network configuration menu (again, only if no parmfile was prepared with sufficient network configuration information).
- Proceed with the interactive network configuration – in this case in a VLAN environment:

```
Unable to find a medium container a live file system
Attempt interactive netboot from a URL?
yes no (default yes):
yes
Available qeth devices:
0.0.c000 0.0.c003 0.0.c006 0.0.c009 0.0.c00c 0.0.c00f
zdev to activate (comma separated, optional):
0.0.c000
QETH device 0.0.c000:0.0.c001:0.0.c002 configured
Two methods available for IP configuration:
* static: for static IP configuration
* dhcp: for automatic IP configuration
static dhcp (default 'dhcp'):
static
ip:
10.222.111.11
gateway (default 10.222.111.1):
10.222.111.1
dns (default 10.222.111.1):
10.222.111.1
vlan id (optional):
1234
http://cdimage.ubuntu.com/ubuntu/releases/20.04.5/release/ubuntu-20.04.5-live-server-
```

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```
s390x.iso (default)
url:
ftp://10.11.12.2:21/ubuntu-live-server-20.04.5/ubuntu-20.04.5-live-server-s390x.iso
http_proxy (optional):
```

- After the last interactive step here (that this is about an optional proxy configuration), the installer will complete its boot-up process:

```
Configuring networking...
IP-Config: encc000.1234 hardware address 3e:00:10:55:00:ff mtu 1500
IP-Config: encc000.1234 guessed broadcast address 10.222.111.255
IP-Config: encc000.1234 complete:
address: 10.222.111.11 broadcast: 10.222.111.255 netmask: 255.255.255.0

gateway: 10.222.111.1 dns0 : 10.222.111.1 dns1 : 0.0.0.0

rootserver: 0.0.0.0 rootpath:
filename :
Connecting to 10.11.12.2:21 (10.11.12.2:21)
focal-live-server-s 10% |***| 72.9M 0:00:08 ETA
focal-live-server-s 25% |*****| 168M 0:00:05 ETA
focal-live-server-s 42% |*****| 279M 0:00:04 ETA
focal-live-server-s 58% |*****| 390M 0:00:02 ETA
focal-live-server-s 75% |*****| 501M 0:00:01 ETA
focal-live-server-s 89% |*****| 595M 0:00:00 ETA
focal-live-server-s 99% |*****| 662M 0:00:00 ETA
focal-live-server-s 100% |*****| 663M 0:00:00 ETA
ip: RTNETLINK answers: File exists
no search or nameservers found in /run/net-encc000.1234.conf / run/net-*.conf /run/
net6-*.conf
[ 399.808930] /dev/loop3: Can't open blockdev
[[0;1;31m SKIP [0m] Ordering cycle found, skipping [0;1;39mLogin Prompts[0m
[ 401.547705] systemd[1]: multi-user.target: Job getty.target/start deleted to
break ordering cycle starting with multi-user.target/start
[ 406.241972] cloud-init[1321]: Cloud-init v. 20.2-45-g5f7825e2-0ubuntu1 running
'init-local' at Wed, 03 Jun 2020 17:07:39 +0000. Up 406.00 seconds.
[ 407.025557] cloud-init[1348]: Cloud-init v. 20.2-45-g5f7825e2-0ubuntu1 running
'init' at Wed, 03 Jun 2020 17:07:40 +0000. Up 406.87 seconds.
[ 407.025618] cloud-init[1348]: ci-info: +++Net device info+++
[ 407.025658] cloud-init[1348]: ci-info: +-----+
+-----+
[ 407.025696] cloud-init[1348]: ci-info: | Device | Up | Addr
ess | Mask | Scope | Hw-Address |
[ 407.025731] cloud-init[1348]: ci-info: +-----+
+-----+
[ 407.025766] cloud-init[1348]: ci-info: | encc000 | True | fe80::3ca7:10f
f:fea5:c69e/64 | link | 72:5d:0d:09:ea:76 |
[ 407.025802] cloud-init[1348]: ci-info: | encc000.1234 | True | 10.245.
236.11 | 255.255.255.0 | global | 72:5d:0d:09:ea:76 |
[ 407.025837] cloud-init[1348]: ci-info: | encc000.1234 | True | fe80::3ca7:10f
f:fea5:c69e/64 | link | 72:5d:0d:09:ea:76 |
[ 407.025874] cloud-init[1348]: ci-info: | lo | True | 127.0
.0.1 | 255.0.0.0 | host | . |
[ 407.025909] cloud-init[1348]: ci-info: | lo | True | ::1/
128 | . | host | . |
[ 407.025944] cloud-init[1348]: ci-info: +-----+
+-----+
```

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```

-----+-----+-----+-----+
[ 407.025982] cloud-init[1348]: ci-info: ++++++Route I
Pv4 info+++++
[ 407.026017] cloud-init[1348]: ci-info: +-----+-----+-----+
-----+-----+
[ 407.026072] cloud-init[1348]: ci-info: | Route | Destination | Gateway
| Genmask | Interface | Flags |
[ 407.026107] cloud-init[1348]: ci-info: +-----+-----+-----+
-----+-----+
[ 407.026141] cloud-init[1348]: ci-info: | 0 | 0.0.0.0 | 10.222.111.1
| 0.0.0.0 | encc000.1234 | UG |
[ 407.026176] cloud-init[1348]: ci-info: | 1 | 10.222.111.0 | 0.0.0.0
| 255.255.255.0 | encc000.1234 | U |
[ 407.026212] cloud-init[1348]: ci-info: +-----+-----+-----+
-----+-----+
[ 407.026246] cloud-init[1348]: ci-info: ++++++Route IPv6 info+++
+++++
[ 407.026280] cloud-init[1348]: ci-info: +-----+-----+-----+
-----+-----+
[ 407.026315] cloud-init[1348]: ci-info: | Route | Destination | Gateway | Int
erface | Flags |
[ 407.026355] cloud-init[1348]: ci-info: +-----+-----+-----+
-----+-----+
[ 407.026390] cloud-init[1348]: ci-info: | 1 | fe80::/64 | :: | en
cc000 | U |
[ 407.026424] cloud-init[1348]: ci-info: | 2 | fe80::/64 | :: | encc
000.1234 | U |
[ 407.026458] cloud-init[1348]: ci-info: | 4 | local | :: | en
cc000 | U |
[ 407.026495] cloud-init[1348]: ci-info: | 5 | local | :: | encc
000.1234 | U |
[ 407.026531] cloud-init[1348]: ci-info: | 6 | ff00::/8 | :: | en
cc000 | U |
[ 407.026566] cloud-init[1348]: ci-info: | 7 | ff00::/8 | :: | encc
000.1234 | U |
[ 407.026600] cloud-init[1348]: ci-info: +-----+-----+-----+
-----+-----+
[ 407.883058] cloud-init[1348]: Generating public/private rsa key pair.
[ 407.883117] cloud-init[1348]: Your identification has been saved in /etc/ssh/
ssh_host_rsa_key
[ 407.883154] cloud-init[1348]: Your public key has been saved in /etc/ssh/ssh_
host_rsa_key.pub
[ 407.883190] cloud-init[1348]: The key fingerprint is:
[ 407.883232] cloud-init[1348]: SHA256:KX5cHC4YL9dXpvhnP6eSfS+J/zmKgg9zdLEzaEb+
RTA root@ubuntu-server
[ 407.883267] cloud-init[1348]: The key's randomart image is:
[ 407.883302] cloud-init[1348]: +---[RSA 3072]-----+
[ 407.883338] cloud-init[1348]: | . E.. |
[ 407.883374] cloud-init[1348]: | o . o |
[ 407.883408] cloud-init[1348]: | . . = +o. |
[ 407.883443] cloud-init[1348]: | + =ooo++ |
[ 407.883478] cloud-init[1348]: | + S *.o. |
[ 407.883512] cloud-init[1348]: | . = o o. |
[ 407.883546] cloud-init[1348]: | .o+o ..+o. |
[ 407.883579] cloud-init[1348]: | o=.. =o+++|
[ 407.883613] cloud-init[1348]: | .... ++*0|

```

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```
[ 407.883648] cloud-init[1348]: +----[SHA256]-----+
[ 407.883682] cloud-init[1348]: Generating public/private dsa key pair.
[ 407.883716] cloud-init[1348]: Your identification has been saved in /etc/ssh/
ssh_host_dsa_key
[ 407.883750] cloud-init[1348]: Your public key has been saved in /etc/ssh/ssh_
host_dsa_key.pub
[ 407.883784] cloud-init[1348]: The key fingerprint is:
[ 407.883817] cloud-init[1348]: SHA256:xu3vLG1BReKDy3DsuMZc/lg5y/+nhzlEmLDk/qFZ
Am0 root@ubuntu-server
[ 407.883851] cloud-init[1348]: The key's randomart image is:
[ 407.883905] cloud-init[1348]: +---[DSA 1024]-----+
[ 407.883941] cloud-init[1348]: |                ..o|
[ 407.883975] cloud-init[1348]: |                o. o o |
[ 407.884008] cloud-init[1348]: |                +.o+o+ |
[ 407.884042] cloud-init[1348]: |                ...E*oo.. |
[ 407.884076] cloud-init[1348]: |                S+o =.. |
[ 407.884112] cloud-init[1348]: |                . +o+oo.o |
[ 407.884145] cloud-init[1348]: |                **+o*o |
[ 407.884179] cloud-init[1348]: |                .oo.*+oo|
[ 407.884212] cloud-init[1348]: |                .+ ===|
[ 407.884246] cloud-init[1348]: +----[SHA256]-----+
[ 407.884280] cloud-init[1348]: Generating public/private ecdsa key pair.
[ 407.884315] cloud-init[1348]: Your identification has been saved in /etc/ssh/
ssh_host_ecdsa_key
[ 407.884352] cloud-init[1348]: Your public key has been saved in /etc/ssh/ssh_
host_ecdsa_key.pub
[ 407.884388] cloud-init[1348]: The key fingerprint is:
[ 407.884422] cloud-init[1348]: SHA256:P+hBF3fj/pu6+0KaywUYii3Lyuc09Za9/a2elCDO
gdE root@ubuntu-server
[ 407.884456] cloud-init[1348]: The key's randomart image is:
[ 407.884490] cloud-init[1348]: +---[ECDSA 256]----+
[ 407.884524] cloud-init[1348]: |                |
[ 407.884558] cloud-init[1348]: |                . |
[ 407.884591] cloud-init[1348]: |                ..E . o |
[ 407.884625] cloud-init[1348]: |                o .ooo o . |
[ 407.884660] cloud-init[1348]: |                o +S.+.. |
[ 407.884694] cloud-init[1348]: |                . +..*oo.+ . |
[ 407.884728] cloud-init[1348]: |                = o+=.+.+ |
[ 407.884762] cloud-init[1348]: |                . o.....++o oo|
[ 407.884795] cloud-init[1348]: |                oo. . +.*@*+|
[ 407.884829] cloud-init[1348]: +----[SHA256]-----+
[ 407.884862] cloud-init[1348]: Generating public/private ed25519 key pair.
[ 407.884896] cloud-init[1348]: Your identification has been saved in /etc/ssh/
ssh_host_ed25519_key
[ 407.884930] cloud-init[1348]: Your public key has been saved in /etc/ssh/ssh_
host_ed25519_key.pub
[ 407.884966] cloud-init[1348]: The key fingerprint is:
[ 407.884999] cloud-init[1348]: SHA256:CbZpkR9eFHuB1sCDZwSdSdwJzy9FpsIWRIyc9ers
hZ0 root@ubuntu-server
[ 407.885033] cloud-init[1348]: The key's randomart image is:
[ 407.885066] cloud-init[1348]: +---[ED25519 256]---+
[ 407.885100] cloud-init[1348]: |                ../%X..o |
[ 407.885133] cloud-init[1348]: |                .o&*+= |
[ 407.885167] cloud-init[1348]: |                = .+*. * . |
[ 407.885200] cloud-init[1348]: |                . B = + o |
[ 407.885238] cloud-init[1348]: |                + S . . . |
```

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```
[ 407.885274] cloud-init[1348]: |      .   o o o   |
[ 407.885308] cloud-init[1348]: |      + E      |
[ 407.885345] cloud-init[1348]: |      . .      |
[ 407.885378] cloud-init[1348]: |      .        |
[ 407.885420] cloud-init[1348]: +-----[SHA256]-----+
[ 418.521933] cloud-init[2185]: Cloud-init v. 20.2-45-g5f7825e2-0ubuntu1 running
'modules:config' at Wed, 03 Jun 2020 17:07:52 +0000. Up 418.40 seconds.
[ 418.522012] cloud-init[2185]: Set the following 'random' passwords
[ 418.522053] cloud-init[2185]: installer:C7BZrW76s4mJzmpf4eUy
ci-info: no authorized SSH keys fingerprints found for user installer.
<14>Jun  3 17:07:52 ec2:
<14>Jun  3 17:07:52 ec2: #####
#####
<14>Jun  3 17:07:52 ec2: -----BEGIN SSH HOST KEY FINGERPRINTS-----
<14>Jun  3 17:07:52 ec2: 1024 SHA256:xu3vLG1BReKdy3DsuMZc/lg5y/+nhzlEmLdk/qFZAm0
root@ubuntu-server (DSA)
<14>Jun  3 17:07:52 ec2: 256 SHA256:P+hBF3fj/pu6+0KaywUYi3Lyuc09Za9/a2elCD0gdE
root@ubuntu-server (ECDSA)
<14>Jun  3 17:07:52 ec2: 256 SHA256:CbZpkR9eFHuB1sCDZwSdSdwJzy9FpsIWRIyc9ershZ0
root@ubuntu-server (ED25519)
<14>Jun  3 17:07:52 ec2: 3072 SHA256:KX5cHC4YL9dXpvhnP6eSfS+J/zmKgg9zdLEzaEb+RTA
root@ubuntu-server (RSA)
<14>Jun  3 17:07:52 ec2: -----END SSH HOST KEY FINGERPRINTS-----
<14>Jun  3 17:07:52 ec2: #####
#####
-----BEGIN SSH HOST KEY KEYS-----
ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoYTItbmlzdHAyNTYAAAAIbmlzdHAyNTYAAABBC2zp4Fq
r1+NJOIEQIISbX+EzeJ6ucXSLi2xEvurgwq8iMYT6yYOXB0Pc/XzeFa6vBCDZk3SSSW6Lq83y7VmdRQ=
root@ubuntu-server
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIJFzgips94nJNoR4QumiyqlJoSLZ48P+NVrd7zgD5k4T
root@ubuntu-server
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQChKo060715FAjd6ImK7qZbWnL/cpgQ2A2gQEgFN0+1
joF/41ygxuw5aG0IQObWFpV9jDsMF5z4qHKzX8tFCpKC0s4uR8QBxh1dDm4wcwgcgAfVLqh7S4/R9Sqa
IFnkCzxThhNeMarcRrutY0mIzspmCg/QvFE1wrXJzl+RtOJ7GiuHHqpm76fX+6ZF1BYhkA87dXQiID2R
yUubSXKCG0NtZlgSzPqD3GB+HxRHHHLT5/Xq+njPq8jIUpqSoHtkBupsyVmcD9gDbz6vng2PuBHwZP9X
17QtyOwxddxk4xIXaTup4g8bH1oF/czswqVxNdfB7XqzROFUOD9rMIB+DwBiHsmH1kRik4wwLi6IH4hu
xrykKvfb1xcZe65kR42oDI7JbBwxvxGrOKx8DrEXnBpOWozS0IDm2ZPh3ci/0uCJ4LTIItByyCfAe/gyR
5si4SkmXrIXf5BnErZRgyJnfxKXmsFaSh7wf15w6GmsgzyD9sI2jES9+4By32ZzY0ldPi0s= root@ub
untu-server
-----END SSH HOST KEY KEYS-----
[ 418.872320] cloud-init[2203]: Cloud-init v. 20.2-45-g5f7825e2-0ubuntu1 running
'modules:final' at Wed, 03 Jun 2020 17:07:52 +0000. Up 418.79 seconds.
[ 418.872385] cloud-init[2203]: ci-info: no authorized SSH keys fingerprints fo
und for user installer.
[ 418.872433] cloud-init[2203]: Cloud-init v. 20.2-45-g5f7825e2-0ubuntu1 finish
ed at Wed, 03 Jun 2020 17:07:52 +0000. DataSource DataSourceNoCloud [seed=/var/li
b/cloud/seed/nocloud][dsmode=net]. Up 418.86 seconds
[ 418.872484] cloud-init[2203]: Welcome to Ubuntu Server Installer!
[ 418.872529] cloud-init[2203]: Above you will find SSH host keys and a random
password set for the 'installer' user. You can use these credentials to ssh-in a
nd complete the installation. If you provided SSH keys in the cloud-init datasou
rce, they were also provisioned to the installer user.
[ 418.872578] cloud-init[2203]: If you have access to the graphical console, li
ke TTY1 or HMC ASCII terminal you can complete the installation there too.
```

It is possible to connect to the installer over the network, which

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might allow the use of a more capable terminal.

To connect, SSH to `installer@10.222.111.11`.

The password you should use is `"C7BZrW76s4mJzmpf4eUy"`.

The host key fingerprints are:

```
RSA      SHA256:KX5cHC4YL9dXpvhnP6eSfS+J/zmKgg9zdLEzaEb+RTA
ECDSA    SHA256:P+hBF3fj/pu6+0KaywUYii3Lyuc09Za9/a2eLCD0gdE
ED25519  SHA256:CbZpkR9eFHuB1sCDZwSdSdwJzy9FpsIWRIyc9ershZ0
```

```
Ubuntu Focal Fossa (development branch) ubuntu-server scli_line0
ubuntu-server login:
```

- At this point you can proceed with the regular installation either by using 'Recovery' → 'Integrated ASCII Console' or with a remote SSH session.
- If the 'Integrated ASCII Console' was opened (you can hit F3 to refresh the task), the initial Subiquity installation screen is presented, which looks like this:

```
=====
Willkommen! Bienvenue! Welcome! ????? ?????????? Welkom!           [ Help ]
=====
Use UP, DOWN and ENTER keys to select your language.
[ English                               > ]
[ Asturianu                             > ]
[ Catal                                 > ]
[ Hrvatski                               > ]
[ Nederlands                             > ]
[ Suomi                                  > ]
[ Fran  ais                             > ]
[ Deutsch                                > ]
[ Magyar                                  > ]
[ Latvie  u                             > ]
[ Norsk bokm  l                         > ]
[ Polski                                 > ]
[ Espa  ol                              > ]
```

- Since the user experience is nicer in a remote SSH session, we recommend using that. However, with certain network environments it's just not possible to use a remote shell, and the 'Integrated ASCII Console' will be the only option.

Note:

At the end of the installer boot-up process, **all** necessary information is provided to proceed with a remote shell.

- The command to execute locally is:

```
user@workstation:~$ ssh installer@10.222.111.11
```

- A temporary random password for the installation was created and shared as well, which you should use without the leading and trailing double quotes:

```
"C7BZrW76s4mJzmpf4eUy"
```

- Hence the remote session for the installer can be opened by:

```
user@workstation:~$ ssh installer@10.222.111.11

The authenticity of host '10.222.111.11 (10.222.111.11)' can't be established.
ECDSA key fingerprint is SHA256:P+hBF3fj/pu6+0KaywUYii3Lyuc09Za9/a2elCD0gdE.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.222.111.11' (ECDSA) to the list of known hosts.
installer@10.222.111.11's password: C7BZrW76s4mJzmpf4eUy
```

- One may swiftly see some login messages like the following ones:

```
Welcome to Ubuntu Focal Fossa (development branch) (GNU/Linux 5.4.0-42-generic s390x)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro

System information as of Wed Jun  3 17:32:10 UTC 2020

System load:   0.0      Memory usage: 2%    Processes:    146
Usage of /home: unknown  Swap usage:   0%    Users logged in: 0

0 updates can be installed immediately.
0 of these updates are security updates.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
```

Eventually you'll reach the initial Subiquity installer screen:

```
=====
Willkommen! Bienvenue! Welcome! ????? ?????????? Welkom!
=====
Use UP, DOWN and ENTER keys to select your language.

[ English                                     > ]
[ Asturianu                                 > ]
[ Catal                                       > ]
[ Hrvatski                                   > ]
[ Nederlands                               > ]
[ Suomi                                     > ]
[ Fran  ais                                  > ]
[ Deutsch                                   > ]
[ Magyar                                    > ]
[ Latvie  u                                 > ]
[ Norsk bokm  l                             > ]
[ Polski                                    > ]
[ Espa  ol                                  > ]
```

From this point, you can follow the normal Subiquity installation. For more details, refer to the [Subiquity installer documentation](https://canonical-subiquity.readthedocs-hosted.com/en/latest/index.html)⁸³.

(I'm leaving some pretty standard screenshots here just to give an example for a basic installation ...)

```
=====
Keyboard configuration
=====
Please select your keyboard layout below, or select "Identify keyboard" to
detect your layout automatically.

Layout:  [ English (US)                v ]

Variant: [ English (US)                v ]

[ Identify keyboard ]

[ Done      ]
[ Back      ]
```

```
=====
Zdev setup
=====
ID                ONLINE  NAMES                ^
|
dasd-eckd
0.0.1600          >
0.0.1601          >
0.0.1602          >
0.0.1603          >
0.0.1604          >
0.0.1605          >
0.0.1606          >
0.0.1607          >
0.0.1608          >
0.0.1609          >
0.0.160a          >
0.0.160b          >
0.0.160c          >
0.0.160d          >                v

[ Continue      ]
[ Back          ]
```

⁸³ <https://canonical-subiquity.readthedocs-hosted.com/en/latest/index.html>

```
=====
Zdev setup
=====
ID                ONLINE  NAMES                ^
|
dasd-eckd
0.0.1600          >
0.0.1601          > < (close)
0.0.1602          > Enable
0.0.1603          > Disable
0.0.1604          >
0.0.1605          >
0.0.1606          >
0.0.1607          >
0.0.1608          >
0.0.1609          >
0.0.160a          >
0.0.160b          >
0.0.160c          >
0.0.160d          > v

[ Continue ]
[ Back ]
```

```
=====
Zdev setup
=====
ID                ONLINE  NAMES                ^
|
dasd-eckd
0.0.1600          >
0.0.1601          online  dasda  >
0.0.1602          >
0.0.1603          >
0.0.1604          >
0.0.1605          >
0.0.1606          >
0.0.1607          >
0.0.1608          >
0.0.1609          >
0.0.160a          >
0.0.160b          >
0.0.160c          >
0.0.160d          > v

[ Continue ]
[ Back ]
```

- One may hit the End key here – that will automatically scroll down to the bottom of the Z devices list and screen:

```
=====
Zdev setup
=====
0.0.f1de:0.0.f1df  > ^
0.0.f1e0:0.0.f1e1  >

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```

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```

0.0.f1e2:0.0.f1e3      >
0.0.f1e4:0.0.f1e5      >
0.0.f1e6:0.0.f1e7      >
0.0.f1e8:0.0.f1e9      >
0.0.f1ea:0.0.f1eb      >
0.0.f1ec:0.0.f1ed      >
0.0.f1ee:0.0.f1ef      >
0.0.f1f0:0.0.f1f1      >
0.0.f1f2:0.0.f1f3      >
0.0.f1f4:0.0.f1f5      >
0.0.f1f6:0.0.f1f7      >
0.0.f1f8:0.0.f1f9      >
0.0.f1fa:0.0.f1fb      >
0.0.f1fc:0.0.f1fd      > |
0.0.f1fe:0.0.f1ff      > v

[ Continue ]
[ Back     ]

```

```

=====
Network connections
=====
Configure at least one interface this server can use to talk to other
machines, and which preferably provides sufficient access for updates.

NAME      TYPE  NOTES
[ encc000  eth   -                > ]
  72:00:bb:00:aa:11 / Unknown Vendor / Unknown Model

[ encc000.1234  vlan -                > ]
  static        10.222.111.11/24
  VLAN 1234 on interface encc000

[ Create bond > ]

[ Continue ]
[ Back     ]

```

- Depending on the installer version you are using you may face a little bug here. In that case the button will be named 'Continue without network', but the network is there. If you see that, just ignore it and continue ... (If you wait long enough the label will be refreshed and corrected.)

```

=====
Configure proxy
=====
If this system requires a proxy to connect to the internet, enter its
details here.

Proxy address:

```

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If you need to use a HTTP proxy to access the outside world, enter the proxy information here. Otherwise, leave this blank.

The proxy information should be given in the standard form of "http://[[user][:pass]@host[:port]]/".

[Done]
[Back]

Configure Ubuntu archive mirror

If you use an alternative mirror for Ubuntu, enter its details here.

```
Mirror address: http://ports.ubuntu.com/ubuntu-ports
You may provide an archive mirror that will be used instead
of the default.
```

[Done]

[Back]

Guided storage configuration

Configure a guided storage layout, **or** create a custom one:

(X) Use an entire disk

```
[ 0X1601      local disk 6.877G      v ]
```

- [] Set up this disk **as** an LVM group

- [] Encrypt the LVM group **with** LUKS

Passphrase:

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Confirm passphrase:

() Custom storage layout

[Done]
[Back]

Storage configuration

FILE SYSTEM SUMMARY

MOUNT POINT	SIZE	TYPE	DEVICE TYPE
[/]	6.875G	new ext4	new partition of local disk >]

AVAILABLE DEVICES

No available devices

[Create software RAID (md) >]
[Create volume group (LVM) >]

USED DEVICES

[Done]
[Reset]
[Back]

Storage configuration

FILE SYSTEM SUMMARY

Confirm destructive action

Selecting Continue below will begin the installation process and result in the loss of data on the disks selected to be formatted.

You will not be able to return to this or a previous screen once the installation has started.

Are you sure you want to continue?

[No]
[Continue]

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```
[ Reset      ]
[ Back      ]
```

```
=====
Profile setup
=====
```

Enter the username and password you will use to log in to the system. You can configure SSH access on the next screen but a password is still needed for sudo.

Your name: Ed Example

Your server's name: s1lp11
The name it uses when it talks to other computers.

Pick a username: ubuntu

Choose a password: *****

Confirm your password: *****

```
[ Done      ]
```

```
=====
SSH Setup
=====
```

You can choose to install the OpenSSH server package to enable secure remote access to your server.

☐ Install OpenSSH server

Import SSH identity: ☐ No ☒ Yes v]
You can import your SSH keys from Github or Launchpad.

Import Username:

☒ Allow password authentication over SSH

```
[ Done      ]
[ Back      ]
```

It's a nice and convenient new feature to add the user's SSH keys during the installation to the system, since that makes the system login password-less for the initial login!

```
=====
SSH Setup
=====
```

You can choose to install the OpenSSH server package to enable secure remote access to your server.

☒ Install OpenSSH server

Import SSH identity: ☐ from Launchpad v ☐
You can import your SSH keys from Github or Launchpad.

Launchpad Username: user
Enter your Launchpad username.

☒ Allow password authentication over SSH

☐ Done ☐
☐ Back ☐

```
=====
SSH Setup
=====
```

You can choose to install the OpenSSH server package to enable secure remote access to your server.

```
----- Confirm SSH keys -----
```

Keys with the following fingerprints were fetched. Do you want to use them?

2048 SHA256:joGscmiamcaoincinaäonnväineorviZEdDWdR9Hpbc2KIw user@W520
(RSA)

521 SHA256:T3JzxvB6K1Gzidvoidhoidsaoicak0jhhgwbw01F7/fZ2c
ed.example@acme.com (ECDSA)

☐ Yes ☐
☐ No ☐

☐ Done ☐
☐ Back ☐

```
=====
Featured Server Snaps
=====
```

These are popular snaps in server environments. Select or deselect with SPACE, press ENTER to see more details of the package, publisher and versions available.

☐ kata-containers Lightweight virtual machines that seamlessly plug into >

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```
[ ] docker          Docker container runtime                >
[ ] mosquitto       Eclipse Mosquitto MQTT broker          >
[ ] etcd            Resilient key-value store by CoreOS    >
[ ] stress-ng       A tool to load, stress test and benchmark a computer s >
[ ] sabnzbd         SABnzbd                                >
[ ] wormhole        get things from one computer to another, safely    >
[ ] slcli           Python based SoftLayer API Tool.          >
[ ] doctl           DigitalOcean command line tool          >
[ ] keepalived      High availability VRRP/BFD and load-balancing for Linu >
[ ] juju            Simple, secure and stable devops. Juju keeps complexit >
```

```
[ Done      ]
[ Back      ]
```

```
=====
Install complete!
=====
```

```

      configuring raid (mdadm) service                ^
      installing kernel
      setting up swap
      apply networking config
      writing etc/fstab
      configuring multipath
      updating packages on target system
      configuring pollinate user-agent on target
      updating initramfs configuration
      finalizing installation
      running 'curtin hook'
      curtin command hook
      executing late commands
| final system configuration                          ||
| configuring cloud-init                             ||
| installing openssh-server \                         v|
```

[View full log]

```
=====
Installation complete!
=====
```

```

      Finished install!
      apply networking config                        ^
      writing etc/fstab
      configuring multipath
      updating packages on target system
      configuring pollinate user-agent on target
      updating initramfs configuration
      finalizing installation
      running 'curtin hook'
      curtin command hook
      executing late commands
| final system configuration
```

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```
| configuring cloud-init |
| installing openssh-server |
| restoring apt configuration |
| downloading and installing security updates | v
```

[View full log]
[Reboot]

Installation complete!

=====

```
----- Finished install! -----
| apply networking config | ^
| writing etc/fstab |
| configuring multipath |
| updating packages on target system |
| configuring pollinate user-agent on target |
| updating initramfs configuration |
| finalizing installation |
| running 'curtin hook' |
| curtin command hook |
| executing late commands |
| final system configuration |
| configuring cloud-init |
| installing openssh-server |
| restoring apt configuration |
| downloading and installing security updates | v
```

[Connection to 10.222.111.11 closed by remote host.

[Rebooting...]

Connection to 10.222.111.11 closed.

user@workstation:~\$

- Now type reset to clear the screen and reset it to its defaults.
- Before proceeding one needs to remove the old, temporary host key of the target system, since it was only for use during the installation:

```
user@workstation:~$ ssh-keygen -f "/home/user/.ssh/known_hosts" -R
"s1lp11"

# Host s1lp11 found: line 159
/home/user/.ssh/known_hosts updated.
Original contents retained as /home/user/.ssh/known_hosts.old
```

- And assuming the post-installation reboot is done, one can now login:

```
user@workstation:~$ ssh ubuntu@s1lp11
```

```
Warning: Permanently added the ECDSA host key for IP address
'10.222.111.11' to the list of known hosts.
Welcome to Ubuntu 20.04.5 LTS (GNU/Linux 5.4.0-42-generic s390x)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed 03 Jun 2020 05:50:05 PM UTC

System load: 0.08           Memory usage: 2%   Processes:      157
Usage of /: 18.7% of 6.70GB Swap usage:   0%   Users logged in: 0

0 updates can be installed immediately.
0 of these updates are security updates.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@s1lp11:~$ uptime
17:50:09 up 1 min,  1 user,  load average: 0.08, 0.11, 0.05
ubuntu@s1lp11:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description: Ubuntu 20.04.5 LTS
Release: 20.04
Codename: focal
ubuntu@s1lp11:~$ uname -a
Linux s1lp11 5.4.0-42-generic #30-Ubuntu SMP Wed Aug 05 16:57:22 UTC 2020
s390x s390x s390x GNU/Linux
ubuntu@s1lp11:~$ exit
logout
Connection to s1lp11 closed.
user@workstation:~$
```

Done !

Non-interactive IBM Z LPAR autoinstall (s390x)

This non-interactive installation uses `autoinstall`, which can be considered the successor to the Debian installer (`d-i`) and `preseed` on Ubuntu. This is a detailed step-by-step guide, including output and logs (which are partially a bit shortened, as indicated by ..., to limit the size of this document).

The example logical partition (LPAR) here uses zFCP storage and is connected to a VLAN network. For a *DASD* and a non-VLAN network example, please see the [non-interactive IBM z/VM \(s390x\) auto-installation](#) (page 116) guide.

Start with the preparation of the (FTP) install server (if it doesn't already exist).

```
user@local:~$ ssh admin@installserver.local
```

```
admin@installserver:~$ mkdir -p /srv/ftp/ubuntu-daily-live-server-20.04
```

```
admin@installserver:~$ wget http://cdimage.ubuntu.com/ubuntu-server/focal/
daily-live/current/focal-live-server-s390x.iso
--directory-prefix=/srv/ftp/ubuntu-daily-live-server-20.04

--2020-06-26 12:18:48-- http://cdimage.ubuntu.com/ubuntu-server/focal/daily-live/
current/focal-live-server-s390x.iso
Resolving cdimage.ubuntu.com (cdimage.ubuntu.com)... 2001:67c:1560:8001::1d,
2001:67c:1360:8001::28, 2001:67c:1360:8001::27,
...
Connecting to cdimage.ubuntu.com (cdimage.ubuntu.com)|2001:67c:1560:8001::1d|:80...
connected.
HTTP request sent, awaiting response... 200 OK
Length: 700952576 (668M) [application/x-iso9660-image]
Saving to: 'focal-live-server-s390x.iso'

focal-live-server-s 100%[=====] 668.48M  2.73MB/s   in 4m 54s

2020-06-26 12:23:42 (2.27 MB/s) - 'focal-live-server-s390x.iso' saved [700952576/
700952576]
```

The ISO image needs to be extracted now. Since files in its boot folder need to be modified, loopback mount is not an option here:

```
admin@installserver:~$ cd /srv/ftp/ubuntu-daily-live-server-20.04
```

```
admin@installserver:~/srv/ftp/ubuntu-daily-live-server-20.04$ mkdir iso
```

```
admin@installserver:~/srv/ftp/ubuntu-daily-live-server-20.04$ sudo mount -o
loop ./focal-live-server-s390x.iso ./iso

[sudo] password for admin:
mount: /home/user/iso-test/iso: WARNING: device write-protected, mounted read-only.
```

```
admin@installserver:~/srv/ftp/ubuntu-daily-live-server-20.04$ ls -l

total 684530
-rw-rw-r-- 1 user user 700952576 Jun 26 10:12 focal-live-server-s390x.iso
dr-xr-xr-x 10 root root      2048 Jun 26 10:12 iso
```

```
admin@installserver:~/srv/ftp/ubuntu-daily-live-server-20.04$ rsync -rtvz
./iso/ . && sync
```



```
sending incremental file list
skipping non-regular file "ubuntu"
skipping non-regular file "ubuntu-ports"
./
README.diskdefines
boot.catalog
md5sum.txt
ubuntu.ins
skipping non-regular file "dists/stable"
skipping non-regular file "dists/unstable"
.disk/
.disk/base_installable
.disk/casper-uuid-generic
.disk/cd_type
.disk/info
boot/
boot/README.boot
boot/initrd.off
boot/initrd.siz
boot/initrd.ubuntu
boot/kernel.ubuntu
boot/parmfile.ubuntu
boot/ubuntu.exec
boot/ubuntu.ikr
boot/ubuntu.ins
casper/
...
sent 681,509,758 bytes  received 1,857 bytes  22,344,643.11 bytes/sec
total size is 700,317,941  speedup is 1.03
```

```
admin@installserver:~/srv/ftp/ubuntu-daily-live-server-20.04$ ls -l

total 684578
dr-xr-xr-x  2 user user      4096 Jun 26 10:12 boot
-r--r--r--  1 user user      2048 Jun 26 10:12 boot.catalog
dr-xr-xr-x  3 user user      4096 Jun 26 10:12 casper
dr-xr-xr-x  3 user user      4096 Jun 26 10:11 dists
-rw-rw-r--  1 user user 700952576 Jun 26 10:12 focal-live-server-s390x.iso
dr-xr-xr-x  2 user user      4096 Jun 26 10:11 install
dr-xr-xr-x 10 root  root      2048 Jun 26 10:12 iso
-r--r--r--  1 user user      4944 Jun 26 10:12 md5sum.txt
dr-xr-xr-x  2 user user      4096 Jun 26 10:11 pics
dr-xr-xr-x  3 user user      4096 Jun 26 10:11 pool
dr-xr-xr-x  2 user user      4096 Jun 26 10:11 preseed
-r--r--r--  1 user user       236 Jun 26 10:11 README.diskdefines
-r--r--r--  1 user user       185 Jun 26 10:12 ubuntu.ins
```

Now create `.ins` and `parmfiles` dedicated to the LPAR that will be installed (here `zlinlpar`), based on the default `.ins` and `parmfiles` that are shipped with the ISO image:

```
admin@installserver:~/srv/ftp/ubuntu-daily-live-server-20.04$ chmod -R +rw
./boot > cp ./boot/ubuntu.ins ./boot/ubuntu_zlinlpar.ins > cp
./boot/parmfile.ubuntu ./boot/parmfile.zlinlpar
```

```
admin@installserver:~/srv/ftp/ubuntu-daily-live-server-20.04$ vi
./boot/ubuntu_zlinlpar.ins > cat ./boot/ubuntu_zlinlpar.ins

* Ubuntu for z Series (default kernel)
kernel.ubuntu 0x00000000
initrd.off 0x0001040c
initrd.siz 0x00010414
parmfile.zlinlpar 0x00010480
initrd.ubuntu 0x01000000
```

```
admin@installserver:~/srv/ftp/ubuntu-daily-live-server-20.04$ vi
./boot/parmfile.zlinlpar > cat ./boot/parmfile.zlinlpar

ip=10.11.12.42::10.11.12.1:255.255.255.0:zlinlpar:encc000.4711:none:10.11.12.1
vlan=encc000.4711:encc000 url=http://installserver.local:80/ubuntu-daily-live-
server-20.04/focal-live-server-s390x.iso autoinstall ds=nocloud-net;s=http://
installserver.local:80/autoinstall/zlinlpar/ --- quiet
```

Now make sure an FTP server is running in the installserver with /srv/ftp as ftp-server root (as used in this example).

Now prepare an *autoinstall* (HTTP) server, which hosts the configuration data for the non-interactive installation.

```
admin@installserver:~/srv/ftp/ubuntu-daily-live-server-20.04$ mkdir -p
/srv/www/autoinstall/zlinlpar > cd /srv/www/autoinstall/zlinlpar
```

```
admin@installserver:~/srv/www/autoinstall/zlinlpar$ echo "instance-id:
$(uuidgen || openssl rand -base64 8)" > meta-data > cat meta-data

instance-id: 2c2215fb-6a38-417f-b72f-376b1cc44f01
admin@installserver:~/srv/www/autoinstall/zlinlpar$
```

```
admin@installserver:~/srv/www/autoinstall/zlinlpar$ vi user-data > cat
user-data
```

```
#cloud-config
autoinstall:
  version: 1
  refresh-installer:
    update: yes
  reporting:
    builtin:
      type: print
  apt:
    preserve_sources_list: false
    primary:
      - arches: [amd64, i386]
        uri: http://archive.ubuntu.com/ubuntu
      - arches: [default]
        uri: http://ports.ubuntu.com/ubuntu-ports
  keyboard:
    layout: en
    variant: us
  locale: en_US
  identity:
    hostname: zlinlpar
    password:
"$6$ebJ1f8wxED22bTL4F46P0"
    username: ubuntu
  user-data:
    timezone: America/Boston
    users:
      - name: ubuntu
        password:
"$6$KwuxED22bTL4F46P0"
        lock_passwd: false
  early-commands:
    - touch /tmp/lets_activate_the_s390x_devices
    - chzdev zfcp -e e000
    - chzdev zfcp -e e100
    - chzdev zfcp-lun -e --online
    - touch /tmp/s390x_devices_activation_done
  network:
    ethernet:
      encc000: {}
    version: 2
    vlans:
      encc000.4711:
        addresses: [10.11.12.42/24]
        gateway4: 10.11.12.1
        id: 4711
        link: encc000
        nameservers:
          addresses: [10.11.12.1]
  ssh:
    install-server: true
    allow-pw: true
    authorized-keys: ['ssh-rsa meQwtZ user@workstation # ssh-import-id lp:user']
```

For s390x installations the early-commands section is the interesting part:

```
early-commands:
- touch /tmp/lets_activate_the_s390x_devices
- chzdev zfcplun -e e000
- chzdev zfcplun -e e100
- chzdev zfcplun -e --online
- touch /tmp/s390x_devices_activation_done
```

The first and last early-commands are optional; they only frame and indicate the real s390x command activation.

In this particular example, two zFCP hosts (host-bus-adapters) are enabled via their addresses *e000* (`chzdev zfcplun -e e000`) and *e100* (`chzdev zfcplun -e e100`). These have certain logical unit numbers (LUNs) assigned that are all automatically discovered and activated by `chzdev zfcplun -e --online`.

Activation of a direct-access storage device (DASD) would look like this: `chzdev dasd -e 1f00`, and a QETH device activation looks like: `chzdev qeth -e c000`.

See also

For more details about the autoinstall config options, please have a look at the [autoinstall reference](#)⁸⁴ and [autoinstall schema](#)⁸⁵ pages.

- Now make sure a HTTP server is running with `/srv/www` as web-server root (in this particular example).
- Next steps need to be done at the [hardware management console \(HMC\)](#). First, connect to the HMC and proceed with the 'Load From Removable Media and Server' task.
- Then, start the 'Load from Removable Media or Server' task under 'Recovery' → 'Load from Removable Media or Server' on your specific LPAR that you are going to install, and fill out the following fields (the contents will be of course different on your system):

```
FTP Source
Host computer:      installserver.local
User ID:           ftpuser
Password:          *****
Account (optional):
File location (optional): ubuntu-daily-live-server-20.04/boot
```

- Now confirm the entered data and click 'OK'.
- At the 'Load from Removable Media or Server - Select Software to Install' screen, choose the LPAR that is going to be installed, here:

```
ubuntu-daily-live-server-20.04/boot/ubuntu_zlinlpar.ins  Ubuntu for z Series
(default kernel)
```

- Confirm again with 'OK'.
- And another confirmation about the 'Load will cause jobs to be cancelled'.
- Then, another 'Yes' – understanding that it's a disruptive task:

⁸⁴ <https://canonical-subiquity.readthedocs-hosted.com/en/latest/reference/autoinstall-reference.html>

⁸⁵ <https://canonical-subiquity.readthedocs-hosted.com/en/latest/reference/autoinstall-schema.html>

Disruptive Task Confirmation : Load from Removable Media or Server

- Now monitor the 'Load from Removable media or Server Progress' screen and confirm it once again when the status changes from 'Please wait while the image is being loaded.' to 'Success'.
- Then navigate to 'Daily' -> 'Operating System Messages' to monitor the initial program load (IPL) of the install system ...

```

Message
chzdev: Unknown device type or device ID format: c000.4711
Use 'chzdev --help' for more information
QETH device 0.0.c000:0.0.c001:0.0.c002 configured
IP-Config: encc000.4711 hardware address 1a:3c:99:55:2a:ef mtu 1500
IP-Config: encc000.4711 guessed broadcast address 10.11.12.255
IP-Config: encc000.4711 complete:
address: 10.11.12.42    broadcast: 10.11.12.255    netmask: 255.255.255.0

gateway: 10.11.12.1    dns0      : 10.11.12.1    dns1     : 0.0.0.0

host   : zlinlpar
rootserver: 0.0.0.0 rootpath:
filename :
Connecting to installserver.local:80 (installserver.local:80)
focal-live-server-s3  5% |*                               | 39.9M  0:00:15 ETA
focal-live-server-s3 22% |*****                          | 147M   0:00:07 ETA
focal-live-server-s3 38% |*****                          | 254M   0:00:04 ETA
focal-live-server-s3 53% |*****                          | 355M   0:00:03 ETA
focal-live-server-s3 67% |*****                          | 453M   0:00:02 ETA
focal-live-server-s3 81% |*****                          | 545M   0:00:01 ETA
focal-live-server-s3 94% |*****                          | 633M   0:00:00 ETA
focal-live-server-s3 100% |*****                          | 668M   0:00:00 ETA
mount: mounting /cow on /root/cow failed: No such file or directory
Connecting to plymouth: Connection refused
passwd: password expiry information changed.
Using CD-ROM mount point /cdrom/
Identifying... [5d25356068b713167814807dd678c261-2]
Scanning disc for index files...
Found 2 package indexes, 0 source indexes, 0 translation indexes and 1 signature
Found label 'Ubuntu-Server 20.04 LTS _Focal Fossa_ - Release s390x (20200616)'
This disc is called:
'Ubuntu-Server 20.04 LTS _Focal Fossa_ - Release s390x (20200616)'
...
[ 61.190916] cloud-init[2076]: Cloud-init v. 20.1-10-g71af48df-0ubuntu5 running
'modules:final' at Fri, 26 Jun 2020 11:02:01 +0000. Up 61.09 seconds.
[ 61.191002] cloud-init[2076]: ci-info: no authorized SSH keys fingerprints fo
und for user installer.
[ 61.191071] cloud-init[2076]: Cloud-init v. 20.1-10-g71af48df-0ubuntu5 finished at
Fri, 26 Jun 2020 11:02:01 +0000.
Datasource DataSourceNoCloudNet [seed=cmdline,
/var/lib/cloud/seed/nocloud,
http://installserver.local:80/autoinstall/zlinlpar/]
[dsmode=net]. Up 61.18 seconds
[ 61.191136] cloud-init[2076]: Welcome to Ubuntu Server Installer!
[ 61.191202] cloud-init[2076]: Above you will find SSH host keys and a random
password set for the `installer` user. You can use these credentials to ssh-in

```

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and **complete** the installation. If you provided SSH keys **in** the cloud-init datasource, they were also provisioned to the installer user.

[61.191273] cloud-init[2076]: If you have access to the graphical console, like TTY1 or HMC ASCII terminal you can **complete** the installation there too.

It is possible to connect to the installer over the network, which might allow the use of a more capable terminal.

To connect, SSH to installer@10.11.12.42.

The password you should use is `'i7UFdP8fhiVVMme3qqH8''`.

The host key fingerprints are:

```
RSA      SHA256:rBXLeUke3D4gKdsruKEajHjocxc9hr3PI
ECDSA    SHA256:KZZYFswtKxFXWQPuQS9Qp0BUoS6RHswis
ED25519  SHA256:s+5tZfagx0zffc6gYRGW3t1KcBH6f+Vt0
```

```
Ubuntu 20.04 LTS ubuntu-server sclp_line0
```

- At short notice, you can even log in to the system with the user 'installer' and the temporary password that was given at the end of the boot-up process (see above) of the installation system:

```
user@workstation:~$ ssh installer@zlinlpar
```

```
The authenticity of host 'zlinlpar (10.11.12.42)' can't be established.  
ECDSA key fingerprint is SHA256:0/dU/D8jJAEGQcbqKGE9La24IRxUPLpzzs5li9F6Vvk.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added 'zlinlpar,10.11.12.42' (ECDSA) to the list of known  
hosts.  
installer@zlinlpar's password:  
Welcome to Ubuntu 20.04 LTS (GNU/Linux 5.4.0-37-generic s390x)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:       https://ubuntu.com/pro  
  
System information as of Fri Jun 26 11:08:18 UTC 2020  
  
System load:      1.25      Memory usage: 4%      Processes:       192  
Usage of /home: unknown  Swap usage:   0%      Users logged in: 0  
  
0 updates can be installed immediately.  
0 of these updates are security updates.  
  
The list of available updates is more than a week old.  
To check for new updates run: sudo apt update  
  
The programs included with the Ubuntu system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by  
applicable law.  
  
the installer running on /dev/tty1 will perform the autoinstall  
  
press enter to start a shell
```

- Notice that it informs you about a currently-running autoinstall process:

```
the installer running on /dev/tty1 will perform the autoinstall
```

- Nevertheless, we can quickly check some things – though only until the autoinstall process has finished and the post-install reboot has been triggered:

```
root@ubuntu-server:/# ls -l /tmp/lets_activate_the_s390x_devices  
-rw-r--r-- 1 root root 0 Jun 26 11:08 /tmp/lets_activate_the_s390x_devices  
-rw-r--r-- 1 root root 0 Jun 26 11:09 /tmp/s390x_devices_activation_done  
  
root@ubuntu-server:/# lszdev | grep yes  
zfcplib 0.0.e000 yes yes  
zfcplib 0.0.e100 yes yes  
zfcplib 0.0.e000:0x50050763060b16b6:0x4026400200000000 yes yes sdb sg1  
zfcplib 0.0.e000:0x50050763061b16b6:0x4026400200000000 yes yes sda sg0  
zfcplib 0.0.e100:0x50050763060b16b6:0x4026400200000000 yes yes sdd sg3  
zfcplib 0.0.e100:0x50050763061b16b6:0x4026400200000000 yes yes sdc sg2  
qeth 0.0.c000:0.0.c001:0.0.c002 yes no encc000  
root@ubuntu-server:/#
```

- If you wait long enough you'll see the remote session get closed:

```
root@ubuntu-server:/# Connection to zlinlpar closed by remote host.
Connection to zlinlpar closed.
user@workstation:~$
```

- As well as at the console:

```
ubuntu-server login:

[[0;1;31mFAILED[0m] Failed unmounting [0;1;39m/cdrom[0m.
[ 169.161139] sd-umoun[15600]: Failed to unmount /oldroot: Device or resource busy
[ 169.161550] sd-umoun[15601]: Failed to unmount /oldroot/cdrom: Device or resource
busy
[ 169.168118] shutdown[1]: Failed to finalize file systems, loop devices, ignoring
Total: 282 Selected: 0

Command:
```

- ...and that the post-install reboot gets triggered:

```
Message
Mounting [0;1;39mKernel Configuration File System[0m...
Starting [0;1;39mApply Kernel Variables[0m...
[[0;32m OK [0m] Finished [0;1;39mRemount Root and Kernel File Systems[0m.
[[0;32m OK [0m] Finished [0;1;39mUncomplicated firewall[0m.
[[0;32m OK [0m] Mounted [0;1;39mFUSE Control File System[0m.
[[0;32m OK [0m] Mounted [0;1;39mKernel Configuration File System[0m.
...
[ 35.378928] cloud-init[2565]: Cloud-init v. 20.1-10-g71af48df-0ubuntu5 runnin
g 'modules:final' at Fri, 26 Jun 2020 11:10:44 +0000. Up 35.29 seconds.
[ 35.378978] cloud-init[2565]: Cloud-init v. 20.1-10-g71af48df-0ubuntu5 finish
ed at Fri, 26 Jun 2020 11:10:44 +0000. Datasource DataSourceNone. Up 35.37 seconds
[ 35.379008] cloud-init[2565]: 2020-06-26 11:10:44,359 - cc_final_message.py[W
ARNING]: Used fallback datasource
[[0;32m OK [0m] Finished [0;1;39mExecute cloud user/final scripts[0m.
[[0;32m OK [0m] Reached target [0;1;39mCloud-init target[0m.

zlinlpar login:
```

- With the completion of the reboot, the autoinstall is finished and the LPAR is ready to use:

```
user@workstation:~$ ssh-keygen -f "/home/user/.ssh/known_hosts" -R "zlinlpar"
# Host zlinlpar found: line 163
/home/user/.ssh/known_hosts updated.
Original contents retained as /home/user/.ssh/known_hosts.old
user@workstation:~$ ssh ubuntu@zlinlpar
The authenticity of host 'zlinlpar (10.11.12.42)' can't be established.
ECDSA key fingerprint is SHA256:iGtCAReg+ZnoZlgt0kvmyy0gPY8UEI+f7zoISO+f+m/0.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'zlinlpar,10.11.12.42' (ECDSA) to the list of known hosts.
Welcome to Ubuntu 20.04 LTS (GNU/Linux 5.4.0-39-generic s390x)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro
```

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```
System information as of Fri 26 Jun 2020 11:12:23 AM UTC

System load: 0.21          Memory usage: 3%   Processes:      189
Usage of /: 28.2% of 30.88GB Swap usage:  0%   Users logged in: 0

10 updates can be installed immediately.
0 of these updates are security updates.
To see these additional updates run: apt list --upgradable

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY,
to the extent permitted by applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@zlinlpar:~$ uptime
 11:12:35 up 2 min,  1 user,  load average: 0.18, 0.17, 0.08
ubuntu@zlinlpar:~$ lsb_release -a
No LSB modules are available.
Distributor ID:      Ubuntu
Description:         Ubuntu 20.04 LTS
Release:             20.04
Codename:            focal
ubuntu@zlinlpar:~$ uname -a
Linux zlinlpar 5.4.0-39-generic #43-Ubuntu SMP Fri Jun 19 10:27:17
UTC 2020 s390x s390x s390x
GNU/Linux
ubuntu@zlinlpar:~$ lsdev | grep yes
zfcplib 0.0.e000 yes yes
zfcplib 0.0.e100 yes yes
zfcplib 0.0.e000:0x50050763060b16b6:0x4026400200000000 yes yes
sdb sg1
zfcplib 0.0.e000:0x50050763061b16b6:0x4026400200000000 yes yes
sda sg0
zfcplib 0.0.e100:0x50050763060b16b6:0x4026400200000000 yes yes
sdc sg2
zfcplib 0.0.e100:0x50050763061b16b6:0x4026400200000000 yes yes
sdd sg3
qeth 0.0.c000:0.0.c001:0.0.c002 yes yes encc000
ubuntu@zlinlpar:~$ exit
logout
Connection to zlinlpar closed.
user@workstation:~$
```

Some closing notes

- It's always best to use the latest installer and autoinstall components. Be sure to update the installer to the most recent version, or just use a current daily live-server image.
- The ISO image specified with the kernel parameters needs to fit in the boot folder. Its kernel and initrd are specified in the 'Load from Removable Media and Server' task at the hardware management console (HMC).
- In addition to activating disk storage resources in `early-commands`, other devices like OSA/QETH can be added and activated there, too. This is not needed for the basic network device, as specified in the kernel parameters used for the installation (that one is automatically handled).
- If everything is properly set up – FTP server for the image, HTTP server for the autoinstall config files – the installation can be as quick as 2 to 3 minutes. Of course this depends on the complexity of the autoinstall YAML file.
- There is a simple way of generating a sample autoinstall YAML file; one can perform an interactive Subiquity installation, grab the file `/var/log/installer/autoinstall-user-data`, and use it as an example – but beware that the `early-commands` entries to activate the s390x-specific devices need to be added manually!

3.2.2. Security

[System security](#) (page 153) is a crucial topic for any Ubuntu user. In addition to general security topics such as setting up a firewall, AppArmor profiles and user/group management, you will also find how-to guides on:

- **Authentication** with Kerberos, network user authentication with SSSD and physical authentication with smart cards
- **Cryptography** with OpenSSH
- **Virtual Private Networks** OpenVPN and WireGuard VPN

Security

While a fresh Ubuntu installation is usually safe for immediate use, there are some additional steps you can take to introduce a layered approach to your system's security. If you are new to Ubuntu, you may want to refer to our [Introduction to security](#) (page 867) first for a general overview.

General configuration

- [Users and groups management](#) (page 154) for setting up user accounts, permissions and password policies
- [Firewalls](#) (page 163) are recommended for network security
- [AppArmor](#) (page 170) limits permissions and access for the software running on your system
- [Console security](#) (page 177) for an additional physical security barrier

- [TPM-backed LUKS decryption](#) (page 179) to automate decryption of disks

User management

User management is a critical part of maintaining a secure system. Ineffective user and privilege management often leads to a system being compromised. Therefore, it is important that you understand how to protect your server through simple and effective user account management techniques.

Where is root?

Ubuntu developers decided to disable the administrative root account by default in all Ubuntu installations. This does not mean that the root account has been deleted, or that it may not be accessed. Instead, it has been given a password hash that matches no possible value, and so may not log in directly by itself.

Instead, the `sudo` utility (“superuser do”) is used to carry out system administrative duties. `sudo` allows an authorized user to temporarily elevate their privileges using their own password instead of having to know the password belonging to the root account. This provides accountability for all user actions, and gives the administrator control over which actions a user can perform with said privileges.

`sudo-rs`

From Ubuntu 25.10 (Questing Quokka) onward, `sudo` is provided by the `sudo-rs` package (a rust implementation). While `sudo-rs` is not 100% compatible with `sudo.ws`, the majority of common use cases are supported and the change should be invisible to most users.

A list of key differences is [maintained as a reference](#) (page 865)

The original `sudo` utility (maintained by Todd C. Miller), `sudo.ws` remains supported in the 25.10 and the subsequent 26.04 LTS. You will find these utilities installed with `.ws` suffix (for example `sudo.ws`, `visudo.ws`, etc.). Switching the default back to `sudo.ws` is not recommended, but if you need to do so, follow these steps:

Interactive:

```
$ update-alternatives --config sudo
```

Non-interactive:

```
$ update-alternatives --set sudo /usr/bin/sudo.ws
```

You can always switch back to `sudo-rs` using:

```
$ update-alternatives --set sudo /usr/lib/cargo/bin/sudo
```

You can learn more about the motivation for this change in the blog post [Adopting sudo-rs By Default in Ubuntu 25.10](#)⁸⁶

⁸⁶ <https://discourse.ubuntu.com/t/adopting-sudo-rs-by-default-in-ubuntu-25-10/60583>

The rest of the article should work the same with both `sudo.ws` and `sudo-rs`.

Enabling the root account

If for some reason you wish to enable the root account, you will need to give it a password:

```
$ sudo passwd
```

`sudo` will prompt you for your password, and then ask you to supply a new password for root as shown below:

```
[sudo: authenticate] Password: (enter your own password)
New password: (enter a new password for root)
Retype new password: (repeat new password for root)
passwd: password updated successfully
```

Disabling the root account password

To disable the root account password, use the following `passwd` syntax:

```
$ sudo passwd -l root
```

You can learn more about `sudo` by reading the [sudo\(8\)](https://manpages.ubuntu.com/manpages/resolute/man8/sudo.8.html)⁸⁷ manual page: `man sudo`

By default, the initial user created by the Ubuntu installer is a member of the group `sudo` which is added to the file `/etc/sudoers` as an authorized `sudo` user. To give any other account full root access through `sudo`, add them to the `sudo` group.

Listing, adding, and deleting local users

Managing local users and groups differs very little from most other [GNU/Linux](#) operating systems. Ubuntu and other Debian-based distributions encourage the use of the `adduser` package for account management.

Local users and groups are those defined in the `/etc/passwd` and `/etc/group` files, respectively. Via the Name Service Switch ([NSS](#)) mechanism, however, user and group accounts can also come from other sources, like remote Active Directory or Samba servers, OpenLDAP directory services, and more. Managing accounts in those services is an entirely different matter, and not covered in this document.

⁸⁷ <https://manpages.ubuntu.com/manpages/resolute/man8/sudo.8.html>

Listing local users

The local users are defined one per line in the `/etc/passwd` text file. The `cat /etc/passwd` command can be used to list its contents (only a few lines shown below, for brevity):

```
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
...
ubuntu:x:1000:1000:Ubuntu:/home/ubuntu:/bin/bash
```

Tip:

The structure of the `/etc/passwd` file is explained in the [passwd\(5\)](https://manpages.ubuntu.com/manpages/resolute/man5/passwd.5.html)⁸⁸ manual page.

⁸⁸ <https://manpages.ubuntu.com/manpages/resolute/man5/passwd.5.html>

There are basically three types of local users:

- system pre-installed users: These are system users essential to a working Linux system, like `root`, `man`, `lp`, and others. Their `UIDs` are in the range between 0 and 99.
- system users: These users are created dynamically, as services are installed. The UID range here is between 100 and 999.
- Regular users: Finally, the remaining range is usable for “real” users, generally representing persons using the system. This UID range starts at 1000 and, by convention, ends at 59999.

Tip:

For more details about these UID ranges, including groups and other configuration options for adding users, check the [adduser.conf.5](https://manpages.ubuntu.com/manpages/resolute/man5/adduser.conf.5.html)⁸⁹ manual page.

⁸⁹ <https://manpages.ubuntu.com/manpages/resolute/man5/adduser.conf.5.html>

To list all local usernames, we can list the contents of `/etc/passwd` with some string manipulation. For example:

```
$ cat /etc/passwd | cut -d : -f 1
```

This command will cut each line at the first column, using the “:” character as the delimiter, and result in an output showing all local users:

```
root
daemon
bin
sys
...
```

If you want to only show regular users, that is, users with an UID greater than 1000, we can use a regular expression for this filtering:

```
$ cat /etc/passwd | cut -d : -f 1,3 | grep -E ':[0-9]{4,}'
```

The regular expression `:[0-9]{4,}` means all digits, repeated 4 times or more. The output will be similar to this:

```
nobody:65534
ubuntu:1000
```

Tip:

By convention, the `nobody` users has an UID of 65534. That originated back when the user range was limited to 16 bits (so, 0-65535), and is kept for compatibility reasons.

A note about remote users

There is another command that, at first glance, looks like it lists all local users as well: `getent passwd`. What this tool is doing, however, is listing all entries from the `passwd` database, and this will include any NSS modules that might be installed and configured.

On a default installation of Ubuntu, with no extra configuration, that will indeed list only the local users, but not in the general case. If you have, for example, `libnss-ldap` installed and configured, the output of `getent passwd` will list all local users, **plus all the remote users** it finds in the LDAP server that it was configured with.

Performing such an exhaustive enumeration is frowned upon, because that can incur a large amount of network traffic, and place an unreasonable burden on the remote server. In fact, most NSS services block such enumeration actions by default.

For more information on the Name Service Switch (NSS) mechanism, please consult the [nss\(5\)](https://manpages.ubuntu.com/manpages/resolute/man5/nss.5.html)⁹⁰ and [nsswitch.conf\(5\)](https://manpages.ubuntu.com/manpages/resolute/man5/nsswitch.conf.5.html)⁹¹ manual pages.

Add a user

To add a user account, use the following syntax, and follow the prompts to give the account a password and identifiable characteristics, such as a full name, phone number, etc:

```
$ sudo adduser username
```

⁹⁰ <https://manpages.ubuntu.com/manpages/resolute/man5/nss.5.html>

⁹¹ <https://manpages.ubuntu.com/manpages/resolute/man5/nsswitch.conf.5.html>

Delete a user

To delete a user account and its primary group, use the following syntax:

```
$ sudo deluser username
```

Deleting an account does not remove their respective home folder. You must decide whether or not to delete the folder manually, or to keep it according to your desired retention policies.

Remember, any user added later with the same UID/GID as the previous owner will now have access to this folder if you have not taken the necessary precautions.

You may want to change the UID/GID values to something more appropriate, such as the root account, and perhaps even relocate the folder to avoid future conflicts:

```
$ sudo chown -R root:root /home/username/
```

```
$ sudo mkdir /home/archived_users/
```

```
$ sudo mv /home/username /home/archived_users/
```

Lock or unlock a password

To temporarily lock a user password, use the following syntax:

```
$ sudo passwd -l username
```

Similarly, to unlock a user password:

```
$ sudo passwd -u username
```

Add or delete a group

To add or delete a personalised group, use the following syntax, respectively:

```
$ sudo addgroup groupname
```

```
$ sudo delgroup groupname
```

Add a user to a group

To add a user to a group, use the following syntax:

```
$ sudo adduser username groupname
```

User profile security

When a new user is created, the `adduser` utility creates a brand new home directory named `/home/username`. The default profile is modelled after the contents found in the directory of `/etc/skel`, which includes all profile basics.

If your server will be home to multiple users, you should pay close attention to the user home directory permissions to ensure confidentiality.

Ubuntu 21.10 Impish and later

Home directories are created with private permissions (`0750` or `drwxr-x---`) by default. This ensures that only the user and members of their group can access the directory.

Ubuntu 21.04 Hirsute and earlier

Home directories were created with world read/execute permissions (`0755` or `drwxr-xr-x`) by default. This meant that all users on the system could browse and access the contents of other users home directories, which may not be suitable for your environment.

To verify your current user home directory permissions, use the following syntax:

```
$ ls -ld /home/username  
drwxr-xr-x  2 username username    4096 2007-10-02 20:03 username
```

The output shows that the directory `/home/username` has world-readable permissions. You can remove the world readable-permissions using the following command:

```
$ sudo chmod 0750 /home/username
```

Note:

Some people use the recursive option (`-R`) indiscriminately, which modifies all child folders and files. However, this is not necessary and may have undesirable/unintended consequences. Modifying only the parent directory is enough to prevent unauthorised access to anything below the parent.

A more efficient approach would be to modify the `adduser` global default permissions when creating user home folders. To do this, edit the `/etc/adduser.conf` file and modify the `DIR_MODE` variable to something appropriate, so that all new home directories will receive the correct permissions.


```
DIR_MODE=0750
```

After correcting the directory permissions using any of the previously mentioned techniques, verify the results as follows:

```
$ ls -ld /home/username
drwxr-x---  2 username username  4096 2007-10-02 20:03 username
```

Password policy

A strong password policy is one of the most important aspects of your security posture. Many successful security breaches involve simple **brute force** and **dictionary** attacks against weak passwords.

If you intend to offer any form of remote access involving your local password system, make sure you address minimum password complexity requirements, maximum password lifetimes, and frequent audits of your authentication systems.

Minimum password length

By default, Ubuntu requires a minimum password length of 6 characters, as well as some basic entropy checks. These values are controlled in the file `/etc/pam.d/common-password`, which is outlined below.

```
password      [success=1 default=ignore]      pam_unix.so obscure sha512
```

To adjust the minimum length to 8 characters, change the appropriate variable to `minlen=8`. The modification is outlined below:

```
password      [success=1 default=ignore]      pam_unix.so obscure sha512 minlen=8
```

Note:

Basic password entropy checks and minimum length rules do not apply to the administrator using `sudo`-level commands to setup a new user.

Password expiration

When creating user accounts, you should make it a policy to have a minimum and maximum password age, forcing users to change their passwords when they expire.

To view the current status of a user account:

```
$ sudo chage -l username
```

The output below shows interesting facts about the user account, namely that there are no policies applied:

```
Last password change           : Jan 20, 2015
Password expires               : never
Password inactive              : never
Account expires                : never
Minimum number of days between password change : 0
Maximum number of days between password change  : 99999
Number of days of warning before password expires : 7
```

To set any of these values, use the `chage` ("change age") command, and follow the interactive prompts:

```
$ sudo chage username
```

The following is also an example of how you can manually change the explicit expiration date (-E) to 01/31/2015, minimum password age (-m) of 5 days, maximum password age (-M) of 90 days, inactivity period (-I) of 30 days after password expiration, and a warning time period (-W) of 14 days before password expiration:

```
$ sudo chage -E 01/31/2015 -m 5 -M 90 -I 30 -W 14 username
```

To verify changes, use the same syntax as mentioned previously:

```
$ sudo chage -l username
```

The output below shows the new policies that have been established for the account:

```
Last password change           : Jan 20, 2015
Password expires               : Apr 19, 2015
Password inactive              : May 19, 2015
Account expires                : Jan 31, 2015
Minimum number of days between password change : 5
Maximum number of days between password change  : 90
Number of days of warning before password expires : 14
```

Other security considerations

Many applications use alternate authentication mechanisms that can be easily overlooked by even experienced system administrators. Therefore, it is important to understand and control how users authenticate and gain access to services and applications on your server.

SSH access by disabled passwords

Disabling or locking a user password will not prevent a user from logging into your server remotely if they have previously set up SSH public key authentication. They will still be able to gain shell access to the server, without the need for any password. Remember to check the user's home directory for files that will allow for this type of authenticated SSH access, e.g. `/home/username/.ssh/authorized_keys`.

Remove or rename the directory `.ssh/` in the user's home folder to prevent further SSH authentication access.

Be sure to check for any established SSH connections by the disabled account, as it is possible they may have existing inbound or outbound connections – then `kill` any that are found.

```
$ w | grep username (to get the pts/# terminal)
```

```
$ sudo kill -f pts/#
```

Restrict SSH access to only user accounts that should have it. For example, you may create a group called `sshlogin` and add the group name as the value associated with the `AllowGroups` variable located in the file `/etc/ssh/sshd_config`:

```
AllowGroups sshlogin
```

Then add your permitted SSH users to the group `sshlogin`, and restart the SSH service.

```
$ sudo adduser username sshlogin
```

```
$ sudo systemctl restart ssh.service
```

External user database authentication

Most enterprise networks require centralised authentication and access controls for all system resources. If you have configured your server to authenticate users against external databases, be sure to disable the user accounts both externally and locally. This way you ensure that local *fallbacks* authentication is not possible.

Firewall

The Linux kernel includes the **netfilter**⁹² subsystem, which is used to manipulate or decide the fate of network traffic headed into or through your server. All modern Linux firewall solutions use this system for packet filtering.

The kernel's packet filtering system would be of little use to administrators without a userspace interface to manage it. This is the purpose of the **iptables** utility: when a packet reaches your server, it will be handed off to the netfilter subsystem for acceptance, manipulation, or rejection based on the rules supplied to it from the userspace (via iptables). Thus, iptables is all you need to manage your firewall, if you're familiar with it, but many *frontends* are available to simplify the task. We'll take a look at the default frontend used in Ubuntu here.

ufw - Uncomplicated Firewall

The default firewall configuration tool for Ubuntu is ufw. Developed to ease iptables firewall configuration, ufw provides a user-friendly way to create an IPv4 or IPv6 host-based firewall.

ufw by default is initially disabled. From the *ufw(8)*⁹³ manual page:

ufw is not intended to provide complete firewall functionality via its command interface, but instead provides an easy way to add or remove simple rules. It is currently mainly used for host-based firewalls.

Enable or disable ufw

To enable ufw, run the following command from a terminal prompt:

```
$ sudo ufw enable
```

To disable it, you can use the following command:

```
$ sudo ufw disable
```

Open or close a port

To open a port (SSH in this case):

```
$ sudo ufw allow 22
```

Similarly, to close an opened port:

```
$ sudo ufw deny 22
```

⁹² <https://netfilter.org/>

⁹³ <https://manpages.ubuntu.com/manpages/resolute/man8/ufw.8.html>

Add or remove a rule

Rules can also be added using a **numbered** format:

```
$ sudo ufw insert 1 allow 80
```

To view the numbered format:

```
$ sudo ufw status numbered
```

To remove a rule, use delete followed by the rule:

```
$ sudo ufw delete deny 22
```

Allow access from specific hosts

It is possible to allow access from specific hosts or networks to a port. The following example allows SSH access from host 192.168.0.2 to any IP address on this host:

```
$ sudo ufw allow proto tcp from 192.168.0.2 to any port 22
```

Replace 192.168.0.2 with 192.168.0.0/24 to allow SSH access from the entire subnet.

The --dry-run option

Adding the --dry-run option to a ufw command will output the resulting rules, but not apply them. For example, the following is what would be applied if opening the HTTP port:

```
$ sudo ufw --dry-run allow http
```

```
*filter
:ufw-user-input - [0:0]
:ufw-user-output - [0:0]
:ufw-user-forward - [0:0]
:ufw-user-limit - [0:0]
:ufw-user-limit-accept - [0:0]
### RULES ###

### tuple ### allow tcp 80 0.0.0.0/0 any 0.0.0.0/0
-A ufw-user-input -p tcp --dport 80 -j ACCEPT

### END RULES ###
-A ufw-user-input -j RETURN
-A ufw-user-output -j RETURN
-A ufw-user-forward -j RETURN
-A ufw-user-limit -m limit --limit 3/minute -j LOG --log-prefix "[UFW LIMIT]: "
-A ufw-user-limit -j REJECT
-A ufw-user-limit-accept -j ACCEPT
COMMIT
Rules updated
```

Check the status

To see the firewall status, enter:

```
$ sudo ufw status
```

And for more verbose status information use:

```
$ sudo ufw status verbose
```

Note:

If the port you want to open or close is defined in `/etc/services`, you can use the port name instead of the number. In the above examples, replace 22 with `ssh`.

This is a quick introduction to using `ufw`. Please refer to the [ufw\(8\)](https://manpages.ubuntu.com/manpages/resolute/man8/ufw.8.html)⁹⁴ manual page for more information.

⁹⁴ <https://manpages.ubuntu.com/manpages/resolute/man8/ufw.8.html>

ufw application integration

Applications that open ports can include a ufw profile, which details the ports needed for the application to function properly. The profiles are kept in `/etc/ufw/applications.d`, and can be edited if the default ports have been changed.

To view which applications have installed a profile, run the following in a terminal:

```
$ sudo ufw app list
```

Similar to allowing traffic to a port, using an application profile is accomplished by entering:

```
$ sudo ufw allow Samba
```

An extended syntax is available as well:

```
$ ufw allow from 192.168.0.0/24 to any app Samba
```

Replace Samba and `192.168.0.0/24` with the application profile you are using and the IP range for your network.

Note:

There is no need to specify the **protocol** for the application, because that information is detailed in the profile. Also, note that the app name replaces the port number.

To view details about which ports and protocols, and so on, are defined for an application, enter:

```
$ sudo ufw app info Samba
```

Not all applications that require opening a network port come with ufw profiles, but if you have profiled an application and want the file to be included with the package, please file a bug against the package in Launchpad.

```
$ ubuntu-bug nameofpackage
```

IP masquerading

The purpose of IP masquerading is to allow machines with private, non-routable IP addresses on your network to access the Internet through the machine doing the masquerading. Traffic from your private network destined for the Internet must be manipulated for replies to be routable back to the machine that made the request.

To do this, the kernel must modify the **source** IP address of each packet so that replies will be routed back to it, rather than to the private IP address that made the request, which is impossible over the Internet. Linux uses **Connection Tracking** (*conntrack(8)*⁹⁵) to keep track of which connections belong to which machines and reroute each return packet accordingly.

⁹⁵ <https://manpages.ubuntu.com/manpages/resolute/man8/conntrack.8.html>

Traffic leaving your private network is thus “masqueraded” as having originated from your Ubuntu gateway machine. This process is referred to in Microsoft documentation as “Internet Connection Sharing”.

IP masquerading with ufw

IP masquerading can be achieved using custom ufw rules. This is possible because the current back-end for ufw is iptables - restore with the rules files located in `/etc/ufw/*.rules`. These files are a great place to add legacy iptables rules used without ufw, and rules that are more network gateway or bridge related.

The rules are split into two different files; rules that should be executed before ufw command line rules, and rules that are executed after ufw command line rules.

Enable packet forwarding

First, packet forwarding needs to be enabled in ufw. Two configuration files will need to be adjusted, so first, in `/etc/default/ufw` change the `DEFAULT_FORWARD_POLICY` to “ACCEPT”:

```
DEFAULT_FORWARD_POLICY="ACCEPT"
```

Then, edit `/etc/ufw/sysctl.conf` and uncomment:

```
net/ipv4/ip_forward=1
```

Similarly, for IPv6 forwarding, uncomment:

```
net/ipv6/conf/default/forwarding=1
```

Add the configuration

Now add rules to the `/etc/ufw/before.rules` file. The default rules only configure the **filter** table, and to enable masquerading the **nat** table will need to be configured. Add the following to the top of the file, just after the header comments:

```
# nat Table rules
*nat
:POSTROUTING ACCEPT [0:0]

# Forward traffic from eth1 through eth0.
-A POSTROUTING -s 192.168.0.0/24 -o eth0 -j MASQUERADE

# don't delete the 'COMMIT' line or these nat table rules won't be processed
COMMIT
```

The comments are not strictly necessary, but it is considered good practice to document your configuration. Also, when modifying any of the rules files in `/etc/ufw`, make sure these lines are the last line for each table modified:

```
# don't delete the 'COMMIT' line or these rules won't be processed
COMMIT
```


For each **Table**, a corresponding COMMIT statement is required. In these examples only the nat and filter tables are shown, but you can also add rules for the raw and mangle tables.

Note:

In the above example, replace eth0, eth1, and 192.168.0.0/24 with the appropriate interfaces and IP range for your network.

Restart ufw

Finally, disable and re-enable ufw to apply the changes:

```
$ sudo ufw disable && sudo ufw enable
```

IP masquerading should now be enabled. You can also add any additional FORWARD rules to the /etc/ufw/before.rules. It is recommended that these additional rules be added to the ufw-before-forward chain.

IP masquerading with iptables

iptables can also be used to enable masquerading.

Similarly to ufw, the first step is to enable IPv4 packet forwarding by editing /etc/sysctl.conf and uncomment the following line:

```
net.ipv4.ip_forward=1
```

If you wish to enable IPv6 forwarding also uncomment:

```
net.ipv6.conf.default.forwarding=1
```

Next, run the sysctl command to enable the new settings in the configuration file:

```
$ sudo sysctl -p
```

IP masquerading can now be accomplished with a single iptables rule, which may differ slightly based on your network configuration:

```
$ sudo iptables -t nat -A POSTROUTING -s 192.168.0.0/16 -o ppp0 -j MASQUERADE
```

The above command assumes that your private address space is 192.168.0.0/16 and that your Internet-facing device is ppp0. The syntax is broken down as follows:

- -t nat – the rule is to go into the NAT table
- -A POSTROUTING – the rule is to be appended (-A) to the POSTROUTING chain
- -s 192.168.0.0/16 – the rule applies to traffic originating from the specified address space
- -o ppp0 – the rule applies to traffic scheduled to be routed through the specified network device

- -j MASQUERADE – traffic matching this rule is to “jump” (-j) to the MASQUERADE target to be manipulated as described above

Also, each chain in the filter table (the default table, and where most – or all – packet filtering occurs) has a default **policy** of ACCEPT, but if you are creating a firewall in addition to a gateway device, you may have set the policies to DROP or REJECT, in which case your masqueraded traffic needs to be allowed through the FORWARD chain for the above rule to work:

```
$ sudo iptables -A FORWARD -s 192.168.0.0/16 -o ppp0 -j ACCEPT
```

```
$ sudo iptables -A FORWARD -d 192.168.0.0/16 -m state \> --state  
ESTABLISHED,RELATED -i ppp0 -j ACCEPT
```

The above commands will allow all connections from your local network to the Internet and all traffic related to those connections to return to the machine that initiated them.

If you want masquerading to be enabled on reboot, which you probably do, edit `/etc/rc.local` and add any commands used above. For example add the first command with no filtering:

```
$ iptables -t nat -A POSTROUTING -s 192.168.0.0/16 -o ppp0 -j MASQUERADE
```

Logs

Firewall logs are essential for recognizing attacks, troubleshooting your firewall rules, and noticing unusual activity on your network. You must include logging rules in your firewall for them to be generated, though, and logging rules must come before any applicable terminating rule (a rule with a target that decides the fate of the packet, such as ACCEPT, DROP, or REJECT).

If you are using `ufw`, you can turn on logging by entering the following in a terminal:

```
$ sudo ufw logging on
```

To turn logging off in `ufw`, replace `on` with `off` in the above command:

If you are using `iptables` instead of `ufw`, run:

```
$ sudo iptables -A INPUT -m state --state NEW -p tcp --dport 80 \> -j LOG  
--log-prefix "NEW_HTTP_CONN: "
```

A request on port 80 from the local machine, then, would generate a log in *dmesg* that looks like this (single line split into 3 to fit this document):

```
[4304885.870000] NEW_HTTP_CONN: IN=lo OUT= MAC=00:00:00:00:00:00:00:00:00:00:00:00:00:00:08:00  
SRC=127.0.0.1 DST=127.0.0.1 LEN=60 TOS=0x00 PREC=0x00 TTL=64 ID=58288 DF PROTO=TCP  
SPT=53981 DPT=80 WINDOW=32767 RES=0x00 SYN URG=0
```

The above log will also appear in `/var/log/messages`, `/var/log/syslog`, and `/var/log/kern.log`. This behavior can be modified by editing `/etc/syslog.conf` appropriately or by installing and configuring `ulogd` and using the `ULOG` target instead of `LOG`. The `ulogd` daemon

is a userspace server that listens for logging instructions from the kernel – specifically for firewalls – and can log to any file you like, or even to a PostgreSQL or MySQL database. Making sense of your firewall logs can be simplified by using a log analysing tool such as logwatch, fwana1og, fwlogwatch, or lire.

Other tools

There are many tools available to help you construct a complete firewall without intimate knowledge of iptables. A command-line tool with plain-text configuration files, for example, is Shorewall⁹⁶; a powerful solution to help you configure an advanced firewall for any network.

Further reading

- The Ubuntu Firewall⁹⁷ wiki page contains information on the development of ufw
- Also, the `ufw(8)`⁹⁸ manual page contains some very useful information: `man ufw`
- See the packet-filtering how-to⁹⁹ for more information on using iptables
- The nat how-to¹⁰⁰ contains further details on masquerading
- The iptables how-to¹⁰¹ in the Ubuntu wiki is a great resource

AppArmor

AppArmor¹⁰² is an easy-to-use Linux Security Module implementation that restricts applications' capabilities and permissions with **profiles** that are set per-program. It provides mandatory access control (MAC) to supplement the more traditional UNIX model of discretionary access control (DAC).

In Ubuntu, AppArmor is installed and loaded by default – you can check this by running `aa-status`.

It uses **profiles** of an application to determine what files and permissions the application requires. Some packages will install their own profiles, and additional profiles can be found in the `apparmor-profiles` package.

⁹⁶ <https://shorewall.org/>

⁹⁷ <https://wiki.ubuntu.com/UncomplicatedFirewall>

⁹⁸ <https://manpages.ubuntu.com/manpages/resolute/man8/ufw.8.html>

⁹⁹ <https://www.netfilter.org/documentation/HOWTO/packet-filtering-HOWTO.html>

¹⁰⁰ <https://www.netfilter.org/documentation/HOWTO/NAT-HOWTO.html>

¹⁰¹ <https://help.ubuntu.com/community/IptablesHowTo>

¹⁰² <https://apparmor.net/>

Install AppArmor Profiles

To install the `apparmor-profiles` package from a terminal prompt:

```
$ sudo apt install apparmor-profiles
```

AppArmor profiles have two modes of operation:

- **Complaining/Learning:** profile violations are permitted and logged. This is useful for testing and developing new profiles.
- **Enforced/Confined:** enforces profile policy in addition to logging the violation.

Using AppArmor

The optional `apparmor-utils` package contains command-line utilities you can use to change the AppArmor operation mode, find the status of a profile, create new profiles, etc.

AppArmor profiles are located in the `/etc/apparmor.d` directory. It also stores **abstractions** that can simplify profile authoring, such as `abstractions/base` that allows many shared libraries, writing logs to the journal, many pseudo-devices, receiving signals from unconfined processes, and many more things.

Common commands

- **apparmor_status** is used to view the current status of AppArmor profiles:

```
$ sudo apparmor_status
```

- **aa-complain** places a profile into complain mode:

```
$ sudo aa-complain /path/to/bin
```

- **aa-enforce** places a profile into enforce mode:

```
$ sudo aa-enforce /path/to/bin
```

- **apparmor_parser** is used to load a profile into the kernel. It can also be used to reload a currently-loaded profile using the `-r` option after modifying it to have the changes take effect. To reload a profile:

```
$ sudo apparmor_parser -r /etc/apparmor.d/profile.name
```

- **systemctl** can be used to reload all profiles:

```
$ sudo systemctl reload apparmor.service
```

Disabling or re-enabling a profile

The `/etc/apparmor.d/disable` directory can be used along with the `apparmor_parser -R` option to disable a profile:

```
$ sudo ln -s /etc/apparmor.d/profile.name /etc/apparmor.d/disable/
```

```
$ sudo apparmor_parser -R /etc/apparmor.d/profile.name
```

To re-enable a disabled profile, remove the symbolic link to the profile in `/etc/apparmor.d/disable/`, then load the profile using the `-a` option:

```
$ sudo rm /etc/apparmor.d/disable/profile.name
```

```
$ cat /etc/apparmor.d/profile.name | sudo apparmor_parser -a
```

Note:

Replace `profile.name` with the name of the profile you want to manipulate. Also, replace `/path/to/bin/` with the actual executable file path. For example, for the `ping` command use `/bin/ping`.

Profiles

AppArmor profiles are simple text files located in `/etc/apparmor.d/`. The files are named after the full path to the executable they profile, replacing the `/` with `.`.

For example `/etc/apparmor.d/bin.ping` is the AppArmor profile for the `/bin/ping` command.

There are two main type of rules used in profiles:

- **Path entries**, detailing which files an application can access in the file system.
- **Capability entries**, which determine what privileges a confined process is allowed to use.

As an example, take a look at `/etc/apparmor.d/bin.ping`:

```
#include <tunables/global>
/bin/ping flags=(complain) {
    #include <abstractions/base>
    #include <abstractions/consoles>
    #include <abstractions/nameservice>

    capability net_raw,
    capability setuid,
    network inet raw,

    /bin/ping mixr,
    /etc/modules.conf r,
}
```

Which can be broken down as follows:

- `#include <tunables/global>`: include statements from other files. This allows statements pertaining to multiple applications to be placed in a common file.
- `/bin/ping flags=(complain)`: path to the profiled program, also setting the mode to `complain`.
- `capability net_raw,:` allows the application access to the `CAP_NET_RAW` Posix.1e capability.
- `/bin/ping mixr,:` allows the application read and execute access to the file.

Note:

After editing a profile file the profile must be reloaded.

Create a Profile

- **Design a test plan:** Try to think about how the application should be exercised. The test plan should be divided into small test cases. Each test case should have a small description and list the steps to follow.

Some standard test cases are:

- Starting the program
- Stopping the program
- Reloading the program
- Testing all the commands supported by the init script

- **Generate the new profile:** Use `aa-genprof` to generate a new profile. From a terminal:

```
$ sudo aa-genprof executable
```

For example:

```
$ sudo aa-genprof slapd
```

- To get your new profile included in the `apparmor-profiles` package, [file a bug in Launchpad against the AppArmor package](#)¹⁰³:
 - Include your test plan and test cases
 - Attach your new profile to the bug

¹⁰³ <https://bugs.launchpad.net/ubuntu/+source/apparmor/+filebug>

Updating profiles

When the program is misbehaving, audit messages are sent to the log files. The program `aa-logprof` can be used to scan log files for AppArmor audit messages, review them and update the profiles. From a terminal:

```
$ sudo aa-logprof
```

Further pre-existing profiles

The packages `appport-profiles` and `apparmor-profiles-extra` ship some experimental profiles for AppArmor security policies. Do not expect these profiles to work out-of-the-box, but they can give you a head start when trying to create a new profile by starting off with a base that already exists.

These profiles are not considered mature enough to be shipped in `enforce` mode by default. Therefore, they are shipped in `complain` mode so that users can test them, choose which are desired, and help improve them upstream if needed.

Some even more experimental profiles carried by the package are placed in `/usr/share/doc/apparmor-profiles/extras/`

Checking and debugging denies

You will see in `dmesg` (and any log that collects kernel messages) if you have hit a **deny**. It is worth knowing that this will cover any access that was denied because it was not allowed, but explicit denies will put no message in your logs at all.

Examples might look like:

```
[1521056.552037] audit: type=1400 audit(1571868402.378:24425): apparmor="DENIED"
operation="open" profile="/usr/sbin/cups-browsed" name="/var/lib/libvirt/dnsmasq/"
pid=1128 comm="cups-browsed" requested_mask="r" denied_mask="r" fsuid=0 ouid=0
[1482106.651527] audit: type=1400 audit(1571829452.330:24323): apparmor="DENIED"
operation="sendmsg" profile="snap.lxd.lxc" pid=24115 comm="lxc" laddr=10.7.0.69
lport=48796 faddr=10.7.0.231 fport=445 family="inet" sock_type="stream" protocol=6
requested_mask="send" denied_mask="send"
```

That follows a generic structure starting with a timestamp, an audit tag and the category `apparmor="DENIED"`. From the following fields you can derive what was going on and why it was failing.

In the examples above that would be:

First example:

- operation: open (program tried to open a file)
- profile: `/usr/sbin/cups-browsed` (you'll find `/etc/apparmor.d/usr.bin.cups-browsed`)
- name: `/var/lib/libvirt/dnsmasq` (what it wanted to access)
- pid/comm: the program that triggered the access

- `requested_mask/denied_mask/fsuid/oid`: parameters of that open call

Second example:

- `operation`: `sendmsg` (program tried send via network)
- `profile`: `snap.lxd.lxc` (snaps are special, you'll find `/var/lib/snapd/apparmor/profiles/snap.lxd.lxc`)
- `pid/comm`: the program that triggered the access
- `laddr/lport/faddr/fport/family/sock_type/protocol`: parameters of the `sendmsg` call

That way you know in which profile and at what action you have to start if you consider either debugging or adapting the profiles.

Profile customization

Profiles are meant to provide security and so can't be too permissive. But often, a very special setup would work with a profile if it would *just allow this one extra access*. To handle that situation, there are three options:

- Modify the profile itself:
 - Always works, but has the drawback that profiles are in `/etc` and considered config files. So after modification on a related package update you might get a config file prompt. Worst case; depending on configuration, automatic updates might even override it and your custom rule is gone.
- Use tunables:
 - These provide variables that can be used in templates, for example if you want a custom `dir` considered as it would be a home directory. You could modify `/etc/apparmor.d/tunables/home`, which defines the base path rules used for home directories.
 - By design, these variables will only influence profiles that use them.
- Modify a local override:
 - To mitigate the drawbacks of above approaches, **local includes** were introduced, adding the ability to write arbitrary rules that not run into issues during upgrades that modify the packaged rule.
 - The files can be found in `/etc/apparmor.d/local/` and exist for the packages that are known to sometimes need slight tweaks for special setups.

Disabling or re-enabling AppArmor

Starting with Ubuntu 24.04 LTS and later, the AppArmor services are baked into the Ubuntu Kernel. In earlier versions of Ubuntu, you could disable AppArmor by not loading the service. However, it now requires setting a module parameter on the kernel command line to fully disable or re-enable AppArmor.

Disable AppArmor

Warning:

Disabling AppArmor reduces the security of your system! You should only disable AppArmor if you understand the security implications of disabling the service!

To disable AppArmor, you must do the following:

- Edit `/etc/default/grub` and add `apparmor=0` to `GRUB_CMDLINE_LINUX` in `/etc/default/grub`
- Run `sudo update-grub` to refresh the boot configuration
- Reboot the system.

Once rebooted, you can check the status of AppArmor with the following commands, and should see similar output to this:

```
$ sudo aa-status

apparmor module is loaded.
apparmor filesystem is not mounted.
```

```
$ systemctl status apparmor

❑ apparmor.service - Load AppArmor profiles
   Loaded: loaded (/usr/lib/systemd/system/apparmor.service; enabled; preset:
   enabled)
   Active: inactive (dead)
     Condition: start condition unmet at Tue 2025-01-14 08:45:04 UTC; 1min 20s ago
               └─ ConditionSecurity=apparmor was not met
        Docs: man:apparmor(7)
              https://gitlab.com/apparmor/apparmor/wikis/home/

Jan 14 08:45:04 n systemd[1]: apparmor.service - Load AppArmor profiles was skipped
because of an unmet condition check (ConditionSecurity=apparmor).
```

Re-enable AppArmor

- Remove the `apparmor=0` item from `GRUB_CMDLINE_LINUX` in `/etc/default/grub`.
- Run `sudo update-grub` to update your system boot configuration.
- Reboot your system.

You can check if AppArmor is then re-enabled with the following command, and you should see output similar to this:

```
$ sudo aa-status
```

```
apparmor module is loaded.  
119 profiles are loaded.  
24 profiles are in enforce mode.  
  /usr/bin/man  
...
```

Further reading

- See the [AppArmor Administration Guide](#)¹⁰⁴ for advanced configuration options.
- For details using AppArmor with other Ubuntu releases see the [AppArmor Community Wiki](#)¹⁰⁵ page.
- The [OpenSUSE AppArmor](#)¹⁰⁶ page is another introduction to AppArmor.
- (<https://wiki.debian.org/AppArmor>) is another introduction and basic how-to for AppArmor.
- A great place to get involved with the Ubuntu Server community and to ask for AppArmor assistance is the [#server channel on Matrix](#)¹⁰⁷. The [#security channel](#)¹⁰⁸ may also be of use.

Console security

It is difficult to defend against the damage that can be caused by someone with physical access to your environment, for example, due to theft of hard drives, power or service disruption, and so on.

Console security should be addressed as one component of your overall physical security strategy. A locked “screen door” may deter a casual criminal, or at the very least slow down a determined one, so it is still advisable to perform basic precautions with regard to console security.

Disable Ctrl+Alt+Delete

Anyone with physical access to the keyboard can use the Ctrl + Alt + Delete key combination to reboot the server without having to log on. While someone could also simply unplug the power source, you should still prevent the use of this key combination on a production server. This forces an attacker to take more drastic measures to reboot the server, and will prevent accidental reboots at the same time.

To disable the reboot action taken by pressing the Ctrl + Alt + Delete key combination, run the following two commands:

¹⁰⁴ http://www.novell.com/documentation/apparmor/apparmor201_sp10_admin/?page=/documentation/apparmor/apparmor201_sp10_admin/data/book_apparmor_admin.html

¹⁰⁵ <https://help.ubuntu.com/community/AppArmor>

¹⁰⁶ https://en.opensuse.org/SDB:AppArmor_geeks

¹⁰⁷ <https://matrix.to/#/#server:ubuntu.com>

¹⁰⁸ <https://matrix.to/#/#security:ubuntu.com>

```
$ sudo systemctl mask ctrl-alt-del.target
```

```
$ sudo systemctl daemon-reload
```

Note:

To re-enable this feature, use:

```
$ sudo systemctl unmask ctrl-alt-del.target
```

Disable Magic SysRq keys

The Linux kernel's [Magic SysRq](#)¹⁰⁹ feature allows keyboard shortcuts to send low-level commands directly to the kernel, bypassing any running applications and without requiring a login. By default, Ubuntu enables a subset of these functions, including the ability to immediately reboot (Alt + SysRq + b) or power off (Alt + SysRq + o) the machine. As with Ctrl + Alt + Delete, this is worth addressing as part of your physical access hardening.

Note:

On most keyboards, SysRq is labeled as PrtScn.

You can check which functions are currently enabled:

```
$ sysctl kernel.sysrq  
  
kernel.sysrq = 176
```

The default value of 176 enables filesystem sync (16), remounting read-only (32), and reboot/poweroff (128). To disable all SysRq functions, create an override file in `/etc/sysctl.d/` with a higher sort order than the default SysRq configuration file `10-magic-sysrq.conf`:

```
$ echo "kernel.sysrq = 0" | sudo tee /etc/sysctl.d/99-disable-sysrq.conf
```

```
$ sudo sysctl --system
```

The 99- prefix ensures this file is processed after `10-magic-sysrq.conf`, and the setting persists across reboots.

¹⁰⁹ <https://www.kernel.org/doc/html/latest/admin-guide/sysrq.html>

Note:

A list of all of the available SysRq values and a brief description explaining what each value does can be found by examining `/etc/sysctl.d/10-magic-sysrq.conf`.

Beyond that, further explanation can be found by referencing the [SysRq section on the Linux Kernel documentation website](#)¹¹⁰.

¹¹⁰ <https://www.kernel.org/doc/html/latest/admin-guide/sysrq.html>

Verify the change has taken effect:

```
$ sysctl kernel.sysrq  
  
kernel.sysrq = 0
```

Note:

To re-enable the default SysRq functions, remove the override file and reload:

```
$ sudo rm /etc/sysctl.d/99-disable-sysrq.conf
```

```
$ sudo sysctl --system
```

TPM-based LUKS decryption with Clevis

[Clevis](#)¹¹¹ is a pluggable framework for automated decryption. When combined with Trusted Platform Module ([TPM](#)) and Full Disk Encryption ([FDE](#)) via Linux Unified Key Setup ([LUKS](#)), Clevis can automatically unlock encrypted drives during boot without user intervention. The Clevis tooling is currently provided via the universe archive. While functional, it serves as a community-supported fallback for TPM-bound encryption until a fully integrated server FDE solution lands in a future LTS release.

This guide targets systems that use [dracut\(8\)](#)¹¹² as the initramfs generator.

Warning:

Keep at least one well-known LUKS passphrase for recovery. TPM measurements can change after firmware updates, Secure Boot state changes, or some [bootloader](#) changes, which can prevent automatic unlock until you rebind Clevis.

¹¹¹ <https://github.com/latchset/clevis>

¹¹² <https://manpages.ubuntu.com/manpages/resolute/man8/dracut.8.html>

Prerequisites

- An Ubuntu Server installation with LVM over LUKS, with Dracut as the initramfs generator
- A system with a TPM 2.0 module
- Root or sudo privileges

Install Clevis and Dracut integration

Install the necessary packages:

```
$ sudo apt update && sudo apt install clevis clevis-tpm2 clevis-dracut clevis-luks
```

Binding a LUKS volume

Before Dracut can auto-unlock a drive, the drive must be bound to the TPM. You can use Clevis to add a new key to the LUKS header, sealed against the TPM.

First, identify the encrypted partition before binding:

```
$ lsblk -f
```

Look for the partition with FSTYPE set to `crypto_LUKS`, then use that path in the following command as `<encrypted_partition>`.

Bind against Platform Configuration Register (PCR) 7, which tracks secure boot state. If your environment requires different trust guarantees, choose PCR values that match your threat model.

```
$ sudo clevis luks bind -d <encrypted_partition> tpm2 '{"pcr_ids": "7"}'
```

You will be prompted to enter the LUKS passphrase you created during installation. Clevis will generate a new cryptographic secret and store it in a new keyslot. It will also create a corresponding token linked to this new keyslot.

Updating the initial ramdisk

The `clevis-dracut` package provides the necessary Dracut modules to include Clevis decryption hooks in the early boot environment. After binding your root disk, you must regenerate the initial ramdisk.

To force a rebuild of the initial ramdisk for the current kernel, use:

```
$ sudo dracut -f
```

You can verify that the Clevis modules were successfully included with `lsinitrd(1)`¹¹³ by inspecting the generated image:

```
$ sudo lsinitrd | grep '^clevis'

clevis
clevis-pin-null
clevis-pin-sss
clevis-pin-tang
clevis-pin-tpm2
```

You can also verify that a Clevis token exists in the LUKS metadata – it should show a TPM pin bound to the device:

```
$ sudo clevis luks list -d <encrypted_partition>

2: tpm2 '{"hash":"sha256","key":"ecc"}'
```

For a lower-level check, inspect LUKS token metadata with `cryptsetup(8)`¹¹⁴, which should show the Clevis token and the keyslot it is bound to:

```
$ sudo cryptsetup luksDump <encrypted_partition> | grep -A2 -i Tokens

Tokens:
 0: clevis
    Keyslot: 2
```

After rebooting, confirm the root volume unlocks automatically and the system reaches the login prompt without asking for the LUKS passphrase.

Unlocking secondary disks

If your server has secondary encrypted data disks, you do not need to use dracut to unlock them, as they are unlocked later in the boot process by `systemd(1)`¹¹⁵.

To setup automatic unlocking, you must first bind the secondary disk to the TPM. Replace `<secondary_encrypted_partition>` with your secondary encrypted partition.

```
$ sudo clevis luks bind -d <secondary_encrypted_partition> tpm2 '{"pcr_ids":
"7"}'
```

Then, get the UUID of the locked LUKS partition:

```
$ blkid -s UUID -o value <secondary_encrypted_partition>
```

¹¹³ <https://manpages.ubuntu.com/manpages/resolute/man1/lsinitrd.1.html>

¹¹⁴ <https://manpages.ubuntu.com/manpages/resolute/man8/cryptsetup.8.html>

¹¹⁵ <https://manpages.ubuntu.com/manpages/resolute/man1/systemd.1.html>

Add the disk to `crypttab(5)`¹¹⁶, using `none` for the password file so `systemd(1)`¹¹⁷ automatically intercepts the prompt and recognizes the Clevis binding.

```
data_vol UUID=<UUID> none luks
```

The first field in the `/etc/crypttab` entry is the name used for the mapped device. In the example above the name is `data_vol`, so when the LUKS volume is unlocked the decrypted mapper device is exposed at `/dev/mapper/data_vol`. Add that device (`/dev/mapper/data_vol`) to `fstab(5)`¹¹⁸ as usual.

Removing a TPM binding

If you need to remove a Clevis binding (for example, if you are decommissioning a server or moving the drive), you can unbind it. First, list the active Clevis tokens to find the correct LUKS slot:

```
$ sudo clevis luks list -d <encrypted_partition>
```

Then, unbind the specific slot (replacing 1 with the slot number identified above).

```
$ sudo clevis luks unbind -d <encrypted_partition> -s 1
```

Authentication

These tools are particularly useful for more advanced or complex setups.

- [Kerberos](#) (page 182) is a network authentication protocol providing identity verification for distributed systems
- [Network user authentication with SSSD](#) (page 204) handles authentication, user/group information and authorisation from disparate network sources
- [Smart card authentication](#) (page 223) provides a physical authentication method

Kerberos

This section assumes you have some familiarity with the terms and concepts used in Kerberos.

You will need a properly configured DNS server set up before you begin. See the [Domain Name Service \(DNS\)](#) (page 306) page for instructions on how to set this up.

We recommend following this set of how-to guides in the order presented for best results.

¹¹⁶ <https://manpages.ubuntu.com/manpages/resolute/man5/crypttab.5.html>

¹¹⁷ <https://manpages.ubuntu.com/manpages/resolute/man1/systemd.1.html>

¹¹⁸ <https://manpages.ubuntu.com/manpages/resolute/man5/fstab.5.html>

How to install a Kerberos server

For this discussion, we will create an MIT Kerberos domain with the following features (edit them to fit your needs):

- **Realm:** EXAMPLE.COM
- **Primary KDC:** kdc01.example.com
- **Secondary KDC:** kdc02.example.com
- **User principal:** ubuntu
- **Admin principal:** ubuntu/admin

Prerequisites

Before installing the Kerberos server, a properly configured [DNS](#) server is needed for your domain. Since the Kerberos realm (by convention) matches the domain name, this section uses the EXAMPLE.COM domain configured in the primary server section of the [DNS documentation](#) (page 306).

Also, Kerberos is a time sensitive protocol. If the local system time between a client machine and the server differs by more than five minutes (by default), the workstation will not be able to authenticate. To correct the problem all hosts should have their time synchronized using the same Network Time Protocol (NTP) server. Check out the [NTP chapter](#) (page 963) for more details.

Install the Kerberos packages

The first step in creating a Kerberos realm is to install the krb5-kdc and krb5-admin-server packages. From a terminal enter:

```
$ sudo apt install krb5-kdc krb5-admin-server
```

You will be asked at the end of the install to supply the [hostname](#) for the Kerberos and Admin servers for the realm, which may or may not be the same server. Since we are going to create the realm, and thus these servers, type in the full hostname of this server.

Note:

By default the realm name will be domain name of the Key Distribution Center (KDC) server.

Next, create the new realm with the kdb5_newrealm utility:

```
$ sudo krb5_newrealm
```

It will ask you for a database password, which is used to encrypt the local database. Chose a secure password: its strength is not verified for you.

Configure the Kerberos server

The questions asked during installation are used to configure the `/etc/krb5.conf` and `/etc/krb5kdc/kdc.conf` files. The former is used by the Kerberos 5 libraries, and the latter configures the KDC. If you need to adjust the KDC settings, edit the file and restart the `krb5-kdc` daemon. If you need to reconfigure Kerberos from scratch, perhaps to change the realm name, you can do so by typing:

```
$ sudo dpkg-reconfigure krb5-kdc
```

Note:

The manual page for `krb5.conf` is in the `krb5-doc` package.

Let's create our first principal. Since there is no principal create yet, we need to use `kadmin.local`, which uses a local UNIX socket to talk to the KDC, and requires root privileges:

```
$ sudo kadmin.local

Authenticating as principal root/admin@EXAMPLE.COM with password.
kadmin.local: addprinc ubuntu
WARNING: no policy specified for ubuntu@EXAMPLE.COM; defaulting to no policy
Enter password for principal "ubuntu@EXAMPLE.COM":
Re-enter password for principal "ubuntu@EXAMPLE.COM":
Principal "ubuntu@EXAMPLE.COM" created.
kadmin.local: quit
```

To be able to use `kadmin` remotely, we should create an **admin principal**. Convention suggests it should be an **admin instance**, as that also makes creating a generic Access Control List (ACL) easier. Let's create an admin instance for the ubuntu principal:

```
$ sudo kadmin.local

Authenticating as principal root/admin@EXAMPLE.COM with password.
kadmin.local: addprinc ubuntu/admin
WARNING: no policy specified for ubuntu/admin@EXAMPLE.COM; defaulting to no policy
Enter password for principal "ubuntu/admin@EXAMPLE.COM":
Re-enter password for principal "ubuntu/admin@EXAMPLE.COM":
Principal "ubuntu/admin@EXAMPLE.COM" created.
kadmin.local: quit
```

Next, the new admin principal needs to have the appropriate ACL permissions. The permissions are configured in the `/etc/krb5kdc/kadm5.acl` file:

```
ubuntu/admin@EXAMPLE.COM
```

*

You can also use a more generic form for this ACL:

```
*/admin@EXAMPLE.COM
```

*

The above will grant all privileges to any admin instance of a principal. See the `kadm5`.

`acl(5)`¹¹⁹ manual page for details.

Now restart the `krb5-admin-server` for the new ACL to take effect:

```
$ sudo systemctl restart krb5-admin-server.service
```

The new user principal can be tested using the `kinit` utility:

```
$ kinit ubuntu/admin  
Password for ubuntu/admin@EXAMPLE.COM:
```

After entering the password, use the `klist` utility to view information about the Ticket Granting Ticket (TGT):

```
$ klist  
  
Ticket cache: FILE:/tmp/krb5cc_1000  
Default principal: ubuntu/admin@EXAMPLE.COM  
  
Valid starting    Expires            Service principal  
04/03/20 19:16:57  04/04/20 05:16:57  krbtgt/EXAMPLE.COM@EXAMPLE.COM  
        renew until 04/04/20 19:16:55
```

Where the cache filename `krb5cc_1000` is composed of the prefix `krb5cc_` and the user id (UID), which in this case is 1000.

`kinit` will inspect `/etc/krb5.conf` to find out which KDC to contact, and the corresponding address. The KDC can also be found via DNS lookups for special TXT and SRV records. You can add these records to your `example.com` DNS zone:

```
_kerberos._udp.EXAMPLE.COM.    IN SRV 1 0 88 kdc01.example.com.  
_kerberos._tcp.EXAMPLE.COM.    IN SRV 1 0 88 kdc01.example.com.  
_kerberos._udp.EXAMPLE.COM.    IN SRV 10 0 88 kdc02.example.com.  
_kerberos._tcp.EXAMPLE.COM.    IN SRV 10 0 88 kdc02.example.com.  
_kerberos-adm._tcp.EXAMPLE.COM. IN SRV 1 0 749 kdc01.example.com.  
_kpasswd._udp.EXAMPLE.COM.     IN SRV 1 0 464 kdc01.example.com.
```

See the [DNS chapter](#) (page 306) for detailed instructions on setting up DNS.

A very quick and useful way to troubleshoot what `kinit` is doing is to set the environment variable `KRB5_TRACE` to a file, or `stderr`, and it will show extra information. The output is quite verbose:

```
$ KRB5_TRACE=/dev/stderr kinit ubuntu/admin  
  
[2898] 1585941845.278578: Getting initial credentials for ubuntu/admin@EXAMPLE.COM  
[2898] 1585941845.278580: Sending unauthenticated request  
[2898] 1585941845.278581: Sending request (189 bytes) to EXAMPLE.COM  
[2898] 1585941845.278582: Resolving hostname kdc01.example.com  
(...)
```

Your new Kerberos realm is now ready to authenticate clients.

¹¹⁹ <https://manpages.ubuntu.com/manpages/resolute/man5/kadm5.acl.5.html>

How to configure Kerberos service principals

The specific steps to enable Kerberos for a service can vary, but in general both of the following are needed:

- A principal for the service – usually `service/host@REALM`
- A keytab accessible to the service wherever it's running – usually in `/etc/krb5.keytab`

For example, let's create a principal for an LDAP service running on the `ldap-server.example.com` host:

```
ubuntu@ldap-server:~$ sudo kadmin -p ubuntu/admin

Authenticating as principal ubuntu/admin with password.
Password for ubuntu/admin@EXAMPLE.COM:
kadmin: addprinc -randkey ldap/ldap-server.example.com
No policy specified for ldap/ldap-server.example.com@EXAMPLE.COM; defaulting to no
policy
Principal "ldap/ldap-server.example.com@EXAMPLE.COM" created.
```

Let's dig a bit into what is happening here:

- The `kadmin` command is being run on the `ldap-server` machine, not on the Key Distribution Center (KDC). We are using `kadmin` remotely.
- It's being run with `sudo`. The reason for this will become clear later.
- We are logged in on the server as `ubuntu`, but specifying an `ubuntu/admin` principal. Remember the `ubuntu` principal has no special privileges.
- The name of the principal we are creating follows the pattern `service/hostname`.
- In order to select a random secret, we pass the `-randkey` parameter. Otherwise we would be asked to type in a password.

With the principal created, we need to extract the key from the KDC and store it in the `ldap-server` host, so that the `ldap` service can use it to authenticate itself with the KDC. Still in the same `kadmin` session:

```
kadmin: ktadd ldap/ldap-server.example.com
Entry for principal ldap/ldap-server.example.com with kvno 2, encryption type aes256-cts-
hmac-sha1-96 added to keytab FILE:/etc/krb5.keytab.
Entry for principal ldap/ldap-server.example.com with kvno 2, encryption type aes128-cts-
hmac-sha1-96 added to keytab FILE:/etc/krb5.keytab.
```

This is why we needed to run `kadmin` with `sudo`: so that it can write to `/etc/krb5.keytab`. This is the system keytab file, which is the default file for all keys that might be needed for services on this host, and we can list them with `klist`. Back in the shell:

```
$ sudo klist -k
```

```
Keytab name: FILE:/etc/krb5.keytab
KVNO Principal
```

```
-----
 2 ldap/ldap-server.example.com@EXAMPLE.COM
 2 ldap/ldap-server.example.com@EXAMPLE.COM
```

If you don't have the `kadmin` utility on the target host, one alternative is to extract the keys on a different host and into a different file, and then transfer this file *securely* to the target server. For example:

```
kadmin: ktadd -k /home/ubuntu/ldap.keytab ldap/ldap-server.example.com
Entry for principal ldap/ldap-server.example.com with kvno 3, encryption type aes256-cts-hmac-sha1-96 added to keytab WRFILE:/home/ubuntu/ldap.keytab.
Entry for principal ldap/ldap-server.example.com with kvno 3, encryption type aes128-cts-hmac-sha1-96 added to keytab WRFILE:/home/ubuntu/ldap.keytab.
```

Note:

Notice how the kvno changed from 2 to 3 in the example above, when using `ktadd` a second time? This is the key version, and it invalidated the previously extracted key with kvno 2. Every time a key is extracted with `ktadd`, its version is bumped and that invalidates the previous ones!

In this case, as long as the target location is writable, you don't even have to run `kadmin` with `sudo`.

Then use `scp` to transfer it to the target host:

```
$ scp /home/ubuntu/ldap.keytab ldap-server.example.com:
```

And over there copy it to `/etc/krb5.keytab`, making sure it's mode `0600` and owned by `root:root`.

Kerberos encryption types

Encryption is at the heart of Kerberos, and it supports multiple cryptographic algorithms. The default choices are good enough for most deployments, but specific situations might need to tweak these settings.

This document will explain the basic configuration parameters of Kerberos that control the selection of encryption algorithms used in a Kerberos deployment.

Server-side configuration

There are two main server-side configuration parameters that control the encryption types used on the server for its database and its collection of principals. Both exist in `/etc/krb5kdc/kdc.conf` inside the `[realms]` section and are as follows:

- `master_key_type` Specifies the key type of the key. This is used to encrypt the database, and the default is `aes256-cts-hmac-sha1-96`.
- `supported_etypes` Specifies the default key/salt combinations of principals for this realm. The default is `aes256-cts-hmac-sha1-96:normal aes128-cts-hmac-sha1-96:normal`, and the encryption types should be listed in order of preference.

Possible values for the encryption algorithms are listed in the [MIT documentation on encryption types](https://web.mit.edu/kerberos/krb5-latest/doc/admin/conf_files/kdc_conf.html#encryption-types)¹²⁰, and the salt types can be seen in the [MIT keysalt lists](https://web.mit.edu/kerberos/krb5-latest/doc/admin/conf_files/kdc_conf.html#keysalt-lists)¹²¹.

Here is an example showing the default values (other settings removed for brevity):

```
[realms]
EXAMPLE.INTERNAL = {
    (...)
    master_key_type = aes256-cts
    supported_etypes = aes256-cts-hmac-sha1-96:normal aes128-cts-hmac-sha1-96:normal
    (...)
}
```

The key is created once per realm, when the realm is bootstrapped. That is usually done with the `krb5_newrealm` tool (see [how to install a Kerberos server](#) (page 183) for details). You can check the key type with either of these commands on the KDC server:

```
$ sudo kadmin.local

kadmin.local: getprinc K/M
Principal: K/M@EXAMPLE.INTERNAL
(...)
Number of keys: 1
Key: vno 1, aes256-cts-hmac-sha1-96
(...)
```

```
$ sudo klist -ke /etc/krb5kdc/stash

Keytab name: FILE:/etc/krb5kdc/stash
KVNO Principal
-----
  1 K/M@EXAMPLE.INTERNAL (aes256-cts-hmac-sha1-96)
```

When a new Kerberos principal is created through the `kadmin` service (via the `kadmin` or `kadmin.local` utilities), the types of encryption keys it will get are controlled via the `supported_etypes` configuration parameter.

For example, let's create an `ubuntu` principal, and check the keys that were created for it (output abbreviated):

¹²⁰ https://web.mit.edu/kerberos/krb5-latest/doc/admin/conf_files/kdc_conf.html#encryption-types

¹²¹ https://web.mit.edu/kerberos/krb5-latest/doc/admin/conf_files/kdc_conf.html#keysalt-lists

```
$ sudo kadmin.local

Authenticating as principal root/admin@EXAMPLE.INTERNAL with password.
kadmin.local: addprinc ubuntu
No policy specified for ubuntu@EXAMPLE.INTERNAL; defaulting to no policy
Enter password for principal "ubuntu@EXAMPLE.INTERNAL":
Re-enter password for principal "ubuntu@EXAMPLE.INTERNAL":
Principal "ubuntu@EXAMPLE.INTERNAL" created.

kadmin.local: getprinc ubuntu
Principal: ubuntu@EXAMPLE.INTERNAL
(...)
Number of keys: 2
Key: vno 1, aes256-cts-hmac-sha1-96
Key: vno 1, aes128-cts-hmac-sha1-96
(...)
```

Two keys were created for the ubuntu principal, following the default setting of supported_ectypes in kdc.conf for this realm.

Note:

The server config supported_ectypes has the *default* list of key types that are created for a principal. This list applies to the moment when that principal is **created** by kadmin. Changing that setting after the fact won't affect the keys that the principal in question has after that event. In particular, principals can be created with specific key types regardless of the supported_ectypes setting. See the -e parameter for the `kadmin add_principal` command¹²².

¹²² <https://web.mit.edu/kerberos/krb5-latest/doc/admin/database.html>

If we had supported_ectypes set to aes256-sha2:normal aes128-sha2:normal camellia256-cts:normal in kdc.conf, then the ubuntu principal would get three key types:

```
kadmin.local: getprinc ubuntu
Principal: ubuntu@EXAMPLE.INTERNAL
(...)
Number of keys: 3
Key: vno 1, aes256-cts-hmac-sha384-192
Key: vno 1, aes128-cts-hmac-sha256-128
Key: vno 1, camellia256-cts-cmac
```

Note:

Bootstrapping a new Kerberos realm via the `krb5_newrealm` command also creates some system principals required by Kerberos, such as kadmin/admin, kadmin/changepw and others. They will all also get the same number of keys each: one per encryption type in supported_ectypes.

Client-side configuration

When we say “client-side”, we really mean “applications linked with the Kerberos libraries”. These live on the server too, so keep that in mind.

The encryption types supported by the Kerberos libraries are defined in the `/etc/krb5.conf` file, inside the `[libdefaults]` section, via the `permitted_enctypes` parameter.

Example:

```
[libdefaults]
(...)
permitted_enctypes = aes256-cts-hmac-sha1-96 aes128-cts-hmac-sha1-96
```

This parameter contains a space-separated list of encryption type names, in order of preference. Default value: `aes256-cts-hmac-sha1-96 aes128-cts-hmac-sha1-96 aes256-cts-hmac-sha384-192 aes128-cts-hmac-sha256-128 camellia256-cts-cmac camellia128-cts-cmac`.

Possible values for the encryption algorithms are listed in [the MIT documentation](https://web.mit.edu/kerberos/krb5-latest/doc/admin/conf_files/krb5_conf.html#libdefaults)¹²³ (same ones as for the KDC).

Note:

There are more encryption-related parameters in `krb5.conf`, but most take their defaults from `permitted_enctypes`. See the [MIT libdefaults documentation](https://web.mit.edu/kerberos/krb5-latest/doc/admin/conf_files/krb5_conf.html#libdefaults)¹²⁴ for more information.

¹²⁴ https://web.mit.edu/kerberos/krb5-latest/doc/admin/conf_files/krb5_conf.html#libdefaults

Putting it all together

When a client performs Kerberos authentication and requests a ticket from the KDC, the encryption type used in that ticket is decided by picking the common set of:

- The encryption types supported by the server for that principal
- The encryption types supported by the client

If there is no common algorithm between what the client accepts, and what the server has to offer for that specific principal, then `kinit` will fail.

For example, if the principal on the server has:

```
kadmin.local: getprinc ubuntu
Principal: ubuntu@EXAMPLE.INTERNAL
(...)
Number of keys: 2
Key: vno 1, aes256-cts-hmac-sha384-192
Key: vno 1, aes128-cts-hmac-sha256-128
```

And the client's `krb5.conf` has:

¹²³ https://web.mit.edu/kerberos/krb5-latest/doc/admin/conf_files/kdc_conf.html#encryption-types

```
permitted_etypes = aes256-sha1 aes128-sha1
```

Then kinit will fail, because the client only supports sha1 variants, and the server only has sha2 to offer for that particular principal the client is requesting:

```
$ kinit ubuntu
kinit: Generic error (see e-text) while getting initial credentials
```

The server log (`journalctl -u krb5-admin-server.service`) will have more details about the error:

```
Apr 19 19:31:49 j-kdc krb5kdc[8597]: AS_REQ (2 etypes {aes256-cts-hmac-sha1-96(18),
aes128-cts-hmac-sha1-96(17)}) fd42:78f4:b1c4:3964:216:3eff:feda:118c: GET_LOCAL_TGT:
ubuntu@EXAMPLE.INTERNAL for krbtgt/EXAMPLE.INTERNAL@EXAMPLE.INTERNAL, No matching key in
entry having a permitted enctype
```

This log says that there was an AS-REQ request which accepted two encryption types, but there was no matching key type on the server database for that principal.

Changing encryption types

Changing encryption types of an existing Kerberos realm is no small task. Just changing the configuration settings won't recreate existing keys, nor add new ones. The modifications have to be done in incremental steps.

MIT Kerberos has a [guide on updating encryption types](#)¹²⁵ that covers many scenarios, including deployments with multiple replicating servers:

Further reading

- [Encryption types in MIT Kerberos](#)¹²⁶
- [krb5.conf encryption related configurations options](#)¹²⁷
- [Migrating away from older encryption types](#)¹²⁸
- [kdc.conf\(5\)](#)¹²⁹ manual page
- [krb5.conf\(5\)](#)¹³⁰ manual page
- [Kerberos V5 concepts](#)¹³¹

¹²⁵ <https://web.mit.edu/kerberos/krb5-latest/doc/admin/etypes.html#migrating-away-from-older-encryption-types>

¹²⁶ <https://web.mit.edu/kerberos/krb5-latest/doc/admin/etypes.html>

¹²⁷ <https://web.mit.edu/kerberos/krb5-latest/doc/admin/etypes.html#configuration-variables>

¹²⁸ <https://web.mit.edu/kerberos/krb5-latest/doc/admin/etypes.html#migrating-away-from-older-encryption-types>

¹²⁹ <https://manpages.ubuntu.com/manpages/resolute/man5/kdc.conf.5.html>

¹³⁰ <https://manpages.ubuntu.com/manpages/resolute/man5/krb5.conf.5.html>

¹³¹ <https://web.mit.edu/kerberos/krb5-latest/doc/basic/index.html>

How to set up secondary KDC

Once you have one Key Distribution Center (KDC) on your network, it is good practice to have a secondary KDC in case the primary becomes unavailable.

Also, if you have Kerberos clients that are on different networks (possibly separated by routers using NAT), it is wise to place a secondary KDC in each of those networks.

Note:

The native replication mechanism explained here relies on a cron job; it essentially dumps the DB on the primary and loads it back up on the secondary. You may want to take a look at using the `kldap` backend, which can use the OpenLDAP replication mechanism. This is explained in [How to set up Kerberos with OpenLDAP backend](#) (page 196).

Install the required packages

First, install the packages, and when asked for the Kerberos and Admin server names enter the name of the Primary KDC:

```
$ sudo apt install krb5-kdc krb5-admin-server
```

Once you have installed the packages, create the host principals for both KDCs. From a terminal prompt, enter:

```
$ kadmin -q "addprinc -randkey host/kdc01.example.com" > kadmin -q "addprinc  
-randkey host/kdc02.example.com"
```

Note:

The `kadmin` command defaults to using a principal like `username/admin@EXAMPLE.COM`, where `username` is your current shell user. If you need to override that, use `-p <principal-you-want>`.

Extract the **key file** for the `kdc02` principal, which is the server we are on:

```
$ sudo kadmin -p ubuntu/admin -q "ktadd host/kdc02.example.com"
```

Next, there needs to be a `kpropd.acl` file on each KDC that lists all KDCs for the realm. For example, on both the **primary and secondary KDC**, create `/etc/krb5kdc/kpropd.acl`:

```
host/kdc01.example.com@EXAMPLE.COM  
host/kdc02.example.com@EXAMPLE.COM
```

Note:

It's customary to allow both KDCs because one may want to switch their roles if one goes bad. For such an eventuality, both are already listed here.

Create an empty database on the **secondary KDC**:

```
$ sudo kdb5_util create -s
```

Now install kprop daemon, which listens for connections from the kprop utility from the **primary KDC**:

```
$ sudo apt install krb5-kpropd
```

The service will be running immediately after installation.

From a terminal on the **primary KDC**, create a dump file of the principal database:

```
$ sudo kdb5_util dump /var/lib/krb5kdc/dump
```

Still on the **Primary KDC**, extract its **key**:

```
$ sudo kadmin.local -q "ktadd host/kdc01.example.com"
```

On the **primary KDC**, run the kprop utility to push the database dump made before to the secondary KDC:

```
$ sudo kprop -r EXAMPLE.COM -f /var/lib/krb5kdc/dump kdc02.example.com  
Database propagation to kdc02.example.com: SUCCEEDED
```

Note the SUCCEEDED message, which signals that the propagation worked. If there is an error message, check /var/log/syslog on the secondary KDC for more information.

You may also want to create a cron job to periodically update the database on the **secondary KDC**. For example, the following will push the database every hour:

```
# m h dom mon dow   command  
0 * * * * root /usr/sbin/kdb5_util dump /var/lib/krb5kdc/dump && /usr/sbin/kprop -r  
EXAMPLE.COM -f /var/lib/krb5kdc/dump kdc02.example.com
```

Finally, start the krb5-kdc daemon on the **secondary KDC**:

```
$ sudo systemctl start krb5-kdc.service
```

Note:

The secondary KDC does not run an admin server, since it's a read-only copy.

From now on, you can specify both KDC servers in /etc/krb5.conf for the EXAMPLE.COM realm, in any host participating in this realm (including kdc01 and kdc02), but remember that there can only be one admin server and that's the one running on kdc01:

```
[realms]  
EXAMPLE.COM = {  
    kdc = kdc01.example.com
```

(continues on next page)

```
}  
kdc = kdc02.example.com  
admin_server = kdc01.example.com
```

The **secondary KDC** should now be able to issue tickets for the realm. You can test this by stopping the `krb5-kdc` daemon on the primary KDC, then using `kinit` to request a ticket. If all goes well you should receive a ticket from the secondary KDC. Otherwise, check `/var/log/syslog` and `/var/log/auth.log` on the secondary KDC.

How to set up basic workstation authentication

In this section we'll look at configuring a Linux system as a Kerberos client. This will allow access to any "Kerberised" services once a user has successfully logged into the system.

Note that Kerberos alone is not enough for a user to exist in a Linux system. We cannot just point the system at a Kerberos server and expect all the Kerberos principals to be able to *log in* on the Linux system, simply because these users do not *exist* locally.

Kerberos only provides authentication: it doesn't know about user groups, Linux *UIDs* and *GIDs*, home directories, etc. Normally, another network source is used for this information, such as an LDAP or Windows server, and, in the old days, NIS was used for that as well.

Set up a Linux system as a Kerberos client

If you have local users matching the principals in a Kerberos realm, and just want to switch the authentication from local to remote using Kerberos, you can follow this section. This is not a very usual scenario, but serves to highlight the separation between user authentication and user information (full name, UID, GID, home directory, groups, etc). If you just want to be able to grab tickets and use them, it's enough to install `krb5-user` and run `kinit`.

We are going to use `sssd` with a trick so that it will fetch the user information from the local system files, instead of a remote source which is the more common case.

Install the required packages

To install the packages enter the following in a terminal prompt:

```
$ sudo apt install krb5-user sssd-krb5
```

You will be prompted for the addresses of your KDCs and admin servers. If you have followed our [how to install a Kerberos server](#) (page 183) and [how to set up a secondary KDC](#) (page 192) guides, the KDCs will be (space separated):

```
kdc01.example.com kdc02.example.com
```

And the admin server will be:

```
kdc01.example.com
```

Remember that kdc02 is a read-only copy of the primary KDC, so it doesn't run an admin server.

Note:

If you have added the appropriate SRV records to [DNS](#), none of those prompts will need answering.

Configure Kerberos

If you missed the questions earlier, you can reconfigure the package to fill them in again:

```
$ sudo dpkg-reconfigure krb5-config
```

You can test the Kerberos configuration by requesting a ticket using the kinit utility. For example:

```
$ kinit ubuntu
Password for ubuntu@EXAMPLE.COM:
```

Note that kinit doesn't need the principal to exist as a local user in the system. In fact, you can kinit any principal you want. If you don't specify one, then the tool will use the username of whoever is running kinit.

Configure sssd

The only remaining configuration now is for sssd. Create the file /etc/sss/sss.conf with the following content:

```
[sss]
config_file_version = 2
domains = example.com

[pam]

[domain/example.com]
id_provider = proxy
proxy_lib_name = files
auth_provider = krb5
krb5_server = kdc01.example.com,kdc02.example.com
krb5_kpasswd = kdc01.example.com
krb5_realm = EXAMPLE.COM
```

The above configuration will use Kerberos for **authentication** (auth_provider), but will use the local system users for user and group information (id_provider).

Adjust the permissions of the config file and start sssd:

```
$ sudo chown root:root /etc/sss/sss.conf
```

```
$ sudo chmod 0600 /etc/sss/sssd.conf
```

```
$ sudo systemctl start sssd
```

Just by having installed sssd and its dependencies, PAM will already have been configured to use sssd, with a *fallback* to local user authentication. To try it out, if this is a workstation, simply switch users (in the GUI), or open a login terminal (Ctrl-Alt-number), or spawn a login shell with `sudo login`, and try logging in using the name of a Kerberos principal. Remember that this user must already exist on the local system:

```
$ sudo login

focal-krb5-client login: ubuntu
Password:
Welcome to Ubuntu Focal Fossa (development branch) (GNU/Linux 5.4.0-21-generic x86_64)

(...)

Last login: Thu Apr  9 21:23:50 UTC 2020 from 10.20.20.1 on pts/0
$ klist
Ticket cache: FILE:/tmp/krb5cc_1000_NlfnSX
Default principal: ubuntu@EXAMPLE.COM

Valid starting    Expires          Service principal
04/09/20 21:36:12  04/10/20 07:36:12  krbtgt/EXAMPLE.COM@EXAMPLE.COM
    renew until 04/10/20 21:36:12
```

You will have a Kerberos ticket already right after login.

How to set up Kerberos with OpenLDAP backend

Kerberos supports a few different database backends. The default one (which we have used in our other Kerberos guides so far) is called db2. The [DB types documentation](#)¹³² shows all the options, one of which is LDAP.

Why use LDAP?

There are several reasons why one would want to have the Kerberos principals stored in LDAP as opposed to a local on-disk database. There are also cases when it is not a good idea. Each site has to evaluate the pros and cons. Here are a few:

- Pros:
 - If you already have OpenLDAP set up for other things, such as storing users and groups, adding the Kerberos attributes can be beneficial, providing an integrated story
- Cons:

¹³² <https://web.mit.edu/kerberos/krb5-latest/doc/admin/dbtypes.html>

- Setting up the LDAP backend isn't a trivial task and shouldn't be attempted by administrators without prior knowledge of OpenLDAP
- As highlighted in the [LDAP section of DB types](#)¹³³, since krb5kdc is single-threaded there may be higher latency in servicing requests when using the OpenLDAP backend

In this guide

In this section we'll configure a primary and secondary Kerberos server to use OpenLDAP for the principal database. Note that as of version 1.18, the Key Distribution Center (KDC) from MIT Kerberos [does not support](#)¹³⁴ a primary KDC using a read-only consumer (secondary) LDAP server. What we have to consider here is that a primary KDC is read-write, and it needs a read-write backend. The secondary KDCs can use both a read-write and read-only backend, because they are expected to be read-only. Therefore there are only some possible layouts we can use:

1. Simple case:

- Primary KDC connected to primary OpenLDAP
- Secondary KDC connected to both primary and secondary OpenLDAP

2. Extended simple case:

- Multiple primary KDCs connected to one primary OpenLDAP
- Multiple secondary KDCs connected to primary and secondary OpenLDAP

3. OpenLDAP with replication:

- Multiple primary KDCs connected to all primary OpenLDAP servers

We haven't covered OpenLDAP replication in this guide, so we will show the **simple case** only. The second scenario is an extension: just add another primary KDC to the mix, talking to the same primary OpenLDAP server.

Configure OpenLDAP

We are going to install the OpenLDAP server on the same host as the KDC, to simplify the communication between them. In such a setup, we can use the `ldapi:///` transport, which is via a UNIX socket, and we don't need to set up SSL certificates to secure the communication between the Kerberos services and OpenLDAP. Note, however, that SSL is still needed for the OpenLDAP replication. See [LDAP with TLS](#) (page 484) for details.

If you want to use an existing OpenLDAP server, that's also possible, but keep in mind that you should then use SSL for the communication between the KDC and this OpenLDAP server.

First, the necessary **schema** needs to be loaded on an OpenLDAP server that has network connectivity to both the **primary** and **secondary** KDCs. The rest of this section assumes that you also have LDAP replication configured between at least two servers. For information on setting up OpenLDAP see [OpenLDAP Server](#) (page 457).

¹³³ <https://web.mit.edu/kerberos/krb5-latest/doc/admin/dbtypes.html#ldap-module-ldap>

¹³⁴ <https://krbdev.mit.edu/rt/Ticket/Display.html?id=7754#>

Note:

`cn=admin,dc=example,dc=com` is a default admin user that is created during the installation of the `slapd` package (the OpenLDAP server). The domain component will change for your server, so adjust accordingly.

- Install the necessary packages (it's assumed that OpenLDAP is already installed):

```
$ sudo apt install krb5-kdc-ldap krb5-admin-server
```

- Next, extract the `kerberos.schema.gz` file:

```
$ sudo cp /usr/share/doc/krb5-kdc-ldap/kerberos.schema.gz  
/etc/ldap/schema/ > sudo gunzip /etc/ldap/schema/kerberos.schema.gz
```

- The **Kerberos schema** needs to be added to the `cn=config` tree. This schema file needs to be converted to LDIF format before it can be added. For that we will use a helper tool, called `schema2ldif`, provided by the package of the same name which is available in the Universe archive:

```
$ sudo apt install schema2ldif
```

- To import the Kerberos schema, run:

```
$ sudo ldap-schema-manager -i kerberos.schema  
  
SASL/EXTERNAL authentication started  
SASL username: gidNumber=0+uidNumber=0,cn=peercred,cn=external,cn=auth  
SASL SSF: 0  
executing 'ldapadd -Y EXTERNAL -H ldapi:/// -f /etc/ldap/schema/kerberos.ldif'  
SASL/EXTERNAL authentication started  
SASL username: gidNumber=0+uidNumber=0,cn=peercred,cn=external,cn=auth  
SASL SSF: 0  
adding new entry "cn=kerberos,cn=schema,cn=config"
```

- With the new schema loaded, let's index an attribute often used in searches:

```
$ sudo ldapmodify -Q -Y EXTERNAL -H ldapi:/// <<EOF  
  
dn: olcDatabase={1}mdb,cn=config  
add: olcDbIndex  
olcDbIndex: krbPrincipalName eq,pres,sub  
EOF  
  
modifying entry "olcDatabase={1}mdb,cn=config"
```

- Let's create LDAP entries for the Kerberos administrative entities that will contact the OpenLDAP server to perform operations. There are two:
 - **ldap_kdc_dn**: needs to have read rights on the realm container, principal container and realm sub-trees. If **disable_last_success** and **disable_lockout** are not set,

however, then `ldap_kdc_dn` needs write access to the Kerberos container just like the admin `DN` below.

- `ldap_kadmin_dn`: needs to have read and write rights on the realm container, principal container and realm sub-trees

Here is the command to create these entities:

```
$ ldapadd -x -D cn=admin,dc=example,dc=com -W <<EOF

dn: uid=kdc-service,dc=example,dc=com
uid: kdc-service
objectClass: account
objectClass: simpleSecurityObject
userPassword: {CRYPT}x
description: Account used for the Kerberos KDC

dn: uid=kadmin-service,dc=example,dc=com
uid: kadmin-service
objectClass: account
objectClass: simpleSecurityObject
userPassword: {CRYPT}x
description: Account used for the Kerberos Admin server
EOF
Enter LDAP Password:
adding new entry "uid=kdc-service,dc=example,dc=com"

adding new entry "uid=kadmin-service,dc=example,dc=com"
```

Now let's set a password for them. Note that first the tool asks for the password you want for the specified user DN, and then for the password of the `cn=admin` DN:

```
$ ldappasswd -x -D cn=admin,dc=example,dc=com -W -S
uid=kdc-service,dc=example,dc=com

New password: <-- password you want for uid-kdc-service
Re-enter new password:
Enter LDAP Password: <-- password for the DN specified with the -D option
```

Repeat for the `uid=kadmin-service` DN. These passwords will be needed later.

You can test these with `ldapwhoami`:

```
$ ldapwhoami -x -D uid=kdc-service,dc=example,dc=com -W

Enter LDAP Password:
dn:uid=kdc-service,dc=example,dc=com
```

- Finally, update the Access Control Lists ([ACL](#)). These can be tricky, as it highly depends on what you have defined already. By default, the `slapd` package configures your database with the following ACLs:

```
olcAccess: {0}to attrs=userPassword by self write by anonymous auth by * none
olcAccess: {1}to attrs=shadowLastChange by self write by * read
olcAccess: {2}to * by * read
```


We need to insert new rules before the final to * by * read one, to control access to the Kerberos related entries and attributes:

```
$ sudo ldapmodify -Q -Y EXTERNAL -H ldapi:/// <<EOF

dn: olcDatabase={1}mdb,cn=config
add: olcAccess
olcAccess: {2}to attrs=krbPrincipalKey
    by anonymous auth
    by dn.exact="uid=kdc-service,dc=example,dc=com" read
    by dn.exact="uid=kadmin-service,dc=example,dc=com" write
    by self write
    by * none
-
add: olcAccess
olcAccess: {3}to dn.subtree="cn=krbContainer,dc=example,dc=com"
    by dn.exact="uid=kdc-service,dc=example,dc=com" read
    by dn.exact="uid=kadmin-service,dc=example,dc=com" write
    by * none
EOF

modifying entry "olcDatabase={1}mdb,cn=config"
```

This will make the existing {2} rule become {4}. Check with `sudo slapcat -b cn=config` (the output below was reformatted a bit for clarity):

```
olcAccess: {0}to attrs=userPassword
    by self write
    by anonymous auth
    by * none
olcAccess: {1}to attrs=shadowLastChange
    by self write
    by * read
olcAccess: {2}to attrs=krbPrincipalKey by anonymous auth
    by dn.exact="uid=kdc-service,dc=example,dc=com" read
    by dn.exact="uid=kadmin-service,dc=example,dc=com" write
    by self write
    by * none
olcAccess: {3}to dn.subtree="cn=krbContainer,dc=example,dc=com"
    by dn.exact="uid=kdc-service,dc=example,dc=com" read
    by dn.exact="uid=kadmin-service,dc=example,dc=com" write
    by * none
olcAccess: {4}to * by * read
```

Your LDAP directory is now ready to serve as a Kerberos principal database.

Primary KDC configuration (LDAP)

With OpenLDAP configured it is time to configure the KDC. In this example we are doing it in the same OpenLDAP server to take advantage of local UNIX socket communication.

- Reconfigure the krb5-config package if needed to get a good starting point with /etc/krb5.conf:

```
$ sudo dpkg-reconfigure krb5-config
```

- Now edit /etc/krb5.conf adding the database_module option to the EXAMPLE.COM realm section:

```
[realms]
EXAMPLE.COM = {
    kdc = kdc01.example.com
    kdc = kdc02.example.com
    admin_server = kdc01.example.com
    default_domain = example.com
    database_module = openldap_ldapconf
}
```

Then also add these new sections:

```
[dbdefaults]
ldap_kerberos_container_dn = cn=krbContainer,dc=example,dc=com

[dbmodules]
openldap_ldapconf = {
    db_library = kldap

    # if either of these is false, then the ldap_kdc_dn needs to
    # have write access
    disable_last_success = true
    disable_lockout = true

    # this object needs to have read rights on
    # the realm container, principal container and realm sub-trees
    ldap_kdc_dn = "uid=kdc-service,dc=example,dc=com"

    # this object needs to have read and write rights on
    # the realm container, principal container and realm sub-trees
    ldap_kadmin_dn = "uid=kadmin-service,dc=example,dc=com"

    ldap_service_password_file = /etc/krb5kdc/service.keyfile
    ldap_servers = ldapi:///
    ldap_conns_per_server = 5
}
```

- Next, use the kdb5_ldap_util utility to create the realm:

```
$ sudo kdb5_ldap_util -D cn=admin,dc=example,dc=com create -subtrees
dc=example,dc=com -r EXAMPLE.COM -s -H ldapi:///
```

```
Password for "cn=admin,dc=example,dc=com":
Initializing database for realm 'EXAMPLE.COM'
You will be prompted for the database Master Password.
It is important that you NOT FORGET this password.
Enter KDC database master key:
Re-enter KDC database master key to verify:
```

- Create a stash of the password used to bind to the LDAP server. Run it once for each `ldap_kdc_dn` and `ldap_kadmin_dn`:

```
$ sudo kdb5_ldap_util -D cn=admin,dc=example,dc=com stashsrvpw -f
/etc/krb5kdc/service.keyfile uid=kdc-service,dc=example,dc=com > sudo
kdb5_ldap_util -D cn=admin,dc=example,dc=com stashsrvpw -f
/etc/krb5kdc/service.keyfile uid=kadmin-service,dc=example,dc=com
```

Note:

The `/etc/krb5kdc/service.keyfile` file now contains clear text versions of the passwords used by the KDC to contact the LDAP server!

- Create a `/etc/krb5kdc/kadm5.acl` file for the admin server, if you haven't already:

```
*/admin@EXAMPLE.COM *
```

- Start the Kerberos KDC and admin server:

```
$ sudo systemctl start krb5-kdc.service krb5-admin-server.service
```

You can now add Kerberos principals to the LDAP database, and they will be copied to any other LDAP servers configured for replication. To add a principal using the `kadmin.local` utility enter:

```
$ sudo kadmin.local

Authenticating as principal root/admin@EXAMPLE.COM with password.
kadmin.local: addprinc ubuntu
WARNING: no policy specified for ubuntu@EXAMPLE.COM; defaulting to no policy
Enter password for principal "ubuntu@EXAMPLE.COM":
Re-enter password for principal "ubuntu@EXAMPLE.COM":
Principal "ubuntu@EXAMPLE.COM" created.
kadmin.local:
```

The above will create an `ubuntu` principal with a DN of `krbPrincipalName=ubuntu@EXAMPLE.COM,cn=EXAMPLE.COM,cn=krbContainer,dc=example,dc=com`.

Let's say, however, that you already have a user in your directory, and it's in `uid=testuser1,ou=People,dc=example,dc=com`. How can you add the Kerberos attributes to it? You use the `-x` parameter to specify the location. For the `ldap_kadmin_dn` to be able to write to it, we first need to update the ACLs:

```
$ sudo ldapmodify -Q -Y EXTERNAL -H ldapi:/// <<EOF

dn: olcDatabase={1}mdb,cn=config
add: olcAccess
olcAccess: {4}to dn.subtree="ou=People,dc=example,dc=com"
    by dn.exact="uid=kdc-service,dc=example,dc=com" read
    by dn.exact="uid=kadmin-service,dc=example,dc=com" write
    by * break
EOF
```

And now we can specify the new location:

```
$ sudo kadmin.local

Authenticating as principal root/admin@EXAMPLE.COM with password.
kadmin.local: addprinc -x dn=uid=testuser1,ou=People,dc=example,dc=com testuser1
WARNING: no policy specified for testuser1@EXAMPLE.COM; defaulting to no policy
Enter password for principal "testuser1@EXAMPLE.COM":
Re-enter password for principal "testuser1@EXAMPLE.COM":
Principal "testuser1@EXAMPLE.COM" created.
```

Since the specified DN already exists, `kadmin.local` will just add the required Kerberos attributes to this existing entry. If it didn't exist, it would be created from scratch, with only the Kerberos attributes, just like what happened with the `ubuntu` example above, but in the specified location.

Note:

If the `ldap_kadmin_dn` DN (`uid=kadmin-service` in our example) does not have write access to the location specified by the `-x` parameter, you will get an `Insufficient access` error.

Both places are visible for `kinit`, since, when the realm was created with `kdb5_ldap_util`, the default value for the search scope and base were taken: `subtree`, and `dc=example,dc=com`.

Secondary KDC configuration (LDAP)

The setup of the secondary KDC (and its OpenLDAP replica) is very similar. Once you have the OpenLDAP replication setup, repeat these steps on the secondary:

- Install `krb5-kdc-ldap`, `ldap-utils`. Do **not** install `krb5-admin-server`.
- Load the Kerberos schema using `schema2ldif`.
- Add the index for `krbPrincipalName`.
- Add the ACLs.
- Configure `krb5.conf` in the same way, initially. If you want to, and if you configured SSL properly, you can add `ldaps://kdc01.example.com` to the `ldap_servers` list after `ldapi://`, so that the secondary KDC can have two LDAP backends at its disposal.
- **DO NOT** run `kdb5_ldap_util`. There is no need to create the database since it's being replicated from the primary.

- Copy over the following files from the primary KDC and place them in the same location on the secondary:
 - `/etc/krb5kdc/stash`
 - `/etc/krb5kdc/service.keyfile`
- Start the KDC: `sudo systemctl start krb5-kdc.service`

Further reading

- [Configuring Kerberos with OpenLDAP back-end](#)¹³⁵
- [MIT Kerberos backend types](#)¹³⁶

See also

- Explanation: [Introduction to Kerberos](#) (page 872)

Network user authentication with SSSD

These guides will show you how to set up network user authentication with SSSD with...

How to set up SSSD with Active Directory

This section describes the use of SSSD to authenticate user logins against an Active Directory via using SSSD's "ad" provider. At the end, Active Directory users will be able to log in on the host using their AD credentials. Group membership will also be maintained.

Group Policies for Ubuntu

SSSD manages user authentication and sets initial security policies.

[ADSys](#) serves as a Group Policy client for Ubuntu, streamlining the configuration of Ubuntu systems within a Microsoft Active Directory environment. If you are interested in Group Policies support for Ubuntu, detailed information can be found in the [ADSys documentation](#)¹³⁷.

¹³⁵ https://web.mit.edu/kerberos/krb5-latest/doc/admin/conf_ldap.html#conf-ldap

¹³⁶ <https://web.mit.edu/kerberos/krb5-latest/doc/admin/dbtypes.html>

¹³⁷ <https://documentation.ubuntu.com/adsys/stable/>

Prerequisites and assumptions

This guide does not explain Active Directory, how it works, how to set one up, or how to maintain it. It assumes that a working Active Directory domain is already configured and you have access to the credentials to join a machine to that domain.

- The domain controller is:
 - Acting as an authoritative *DNS* server for the domain.
 - The primary DNS resolver (check with `resolvectl status`).
- System time is correct and in sync, maintained via a service like `chrony` or `ntp`.
- The domain used in this example is `ad1.example.com`.

Install necessary software

Install the following packages:

```
$ sudo apt install sssd-ad sssd-tools realmd adcli
```

Join the domain

We will use the `realm` command, from the `realmd` package, to join the domain and create the SSSD configuration.

Let's verify the domain is discoverable via DNS:

```
$ sudo realm -v discover ad1.example.com

* Resolving: _ldap._tcp.ad1.example.com
* Performing LDAP DSE lookup on: 10.51.0.5
* Successfully discovered: ad1.example.com
ad1.example.com
type: kerberos
realm-name: AD1.EXAMPLE.COM
domain-name: ad1.example.com
configured: no
server-software: active-directory
client-software: sssd
required-package: sssd-tools
required-package: sssd
required-package: libnss-sss
required-package: libpam-sss
required-package: adcli
required-package: samba-common-bin
```

This performs several checks and determines the best software stack to use with SSSD. SSSD can install the missing packages via `packagekit`, but we already installed them in the previous step.

Now let's join the domain:

```
$ sudo realm join ad1.example.com
```

```
Password for Administrator:
```

That was quite uneventful. If you want to see what it was doing, pass the `-v` option:

```
$ sudo realm join -v ad1.example.com
```

```
* Resolving: _ldap._tcp.ad1.example.com
* Performing LDAP DSE lookup on: 10.51.0.5
* Successfully discovered: ad1.example.com
Password for Administrator:
* Unconditionally checking packages
* Resolving required packages
* LANG=C /usr/sbin/adcli join --verbose --domain ad1.example.com --domain-realm
AD1.EXAMPLE.COM --domain-controller 10.51.0.5 --login-type user --login-user
Administrator --stdin-password
* Using domain name: ad1.example.com
* Calculated computer account name from fqdn: AD-CLIENT
* Using domain realm: ad1.example.com
* Sending NetLogon ping to domain controller: 10.51.0.5
* Received NetLogon info from: SERVER1.ad1.example.com
* Wrote out krb5.conf snippet to /var/cache/realmd/adcli-krb5-hUfTUg/krb5.d/adcli-
krb5-conf-hv2kzi
* Authenticated as user: Administrator@AD1.EXAMPLE.COM
* Looked up short domain name: AD1
* Looked up domain SID: S-1-5-21-2660147319-831819607-3409034899
* Using fully qualified name: ad-client.ad1.example.com
* Using domain name: ad1.example.com
* Using computer account name: AD-CLIENT
* Using domain realm: ad1.example.com
* Calculated computer account name from fqdn: AD-CLIENT
* Generated 120 character computer password
* Using keytab: FILE:/etc/krb5.keytab
* Found computer account for AD-CLIENT$ at: CN=AD-CLIENT,CN=Computers,DC=ad1,
DC=example,DC=com
* Sending NetLogon ping to domain controller: 10.51.0.5
* Received NetLogon info from: SERVER1.ad1.example.com
* Set computer password
* Retrieved kvno '3' for computer account in directory: CN=AD-CLIENT,CN=Computers,
DC=ad1,DC=example,DC=com
* Checking RestrictedKrbHost/ad-client.ad1.example.com
*   Added RestrictedKrbHost/ad-client.ad1.example.com
* Checking RestrictedKrbHost/AD-CLIENT
*   Added RestrictedKrbHost/AD-CLIENT
* Checking host/ad-client.ad1.example.com
*   Added host/ad-client.ad1.example.com
* Checking host/AD-CLIENT
*   Added host/AD-CLIENT
* Discovered which keytab salt to use
* Added the entries to the keytab: AD-CLIENT$@AD1.EXAMPLE.COM: FILE:/etc/krb5.
keytab
* Added the entries to the keytab: host/AD-CLIENT@AD1.EXAMPLE.COM: FILE:/etc/krb5.
keytab
* Added the entries to the keytab: host/ad-client.ad1.example.com@AD1.EXAMPLE.COM:
FILE:/etc/krb5.keytab
* Added the entries to the keytab: RestrictedKrbHost/AD-CLIENT@AD1.EXAMPLE.COM:
FILE:/etc/krb5.keytab
* Added the entries to the keytab: RestrictedKrbHost/ad-client.ad1.example.com@AD1.
EXAMPLE.COM: FILE:/etc/krb5.keytab
* /usr/sbin/update-rc.d sssd enable
* /usr/sbin/service sssd restart
* Successfully enrolled machine in realm
```


By default, `realm` will use the “Administrator” account of the domain to request the join. If you need to use another account, pass it to the tool with the `-U` option.

Another popular way of joining a domain is using a One Time Password (OTP) token. For that, use the `--one-time-password` option.

SSSD configuration

The `realm` tool already took care of creating an SSSD configuration, adding the PAM and NSS modules, and starting the necessary services.

Let’s take a look at `/etc/sss/sss.conf`:

```
[sss]
domains = ad1.example.com
config_file_version = 2
services = nss, pam

[domain/ad1.example.com]
default_shell = /bin/bash
krb5_store_password_if_offline = True
cache_credentials = True
krb5_realm = AD1.EXAMPLE.COM
realmd_tags = manages-system joined-with-adcli
id_provider = ad
fallback_homedir = /home/%u@%d
ad_domain = ad1.example.com
use_fully_qualified_names = True
ldap_id_mapping = True
access_provider = ad
```

Note:

Something very important to remember is that this file must have permissions `0600` and ownership `root:root`, or else SSSD won’t start!

Note:

```
[FAILED] Failed to listen on sssd-nss.socket...
[FAILED] Failed to listen on sssd-pam.socket...
[FAILED] Dependency failed for sssd-pam-priv.socket...
```

If you get those errors on startup, just remove the line `services = nss, pam`

Let’s highlight a few things from this config file:

- **cache_credentials:** This allows logins when the AD server is unreachable
- **fallback_homedir:** The home directory. By default, `/home/<user>@<domain>`. For example, the AD user *john* will have a home directory of `/home/john@ad1.example.com`.
- **use_fully_qualified_names:** Users will be of the form `user@domain`, not just `user`. This should only be changed if you are certain no other domains will ever join the AD forest, via one of the several possible trust relationships.

Automatic home directory creation

What the `realm` tool didn't do for us is setup `pam_mkhomedir`, so that network users can get a home directory when they login. This remaining step can be done by running the following command:

```
$ sudo pam-auth-update --enable mkhomedir
```

Testing our setup

You should now be able to fetch information about AD users. In this example, John Smith is an AD user:

```
$ getent passwd john@ad1.example.com

john@ad1.example.com:*:1725801106:1725800513:John Smith:/home/john@ad1.example.com:/bin/bash
```

Let's see his groups:

```
$ groups john@ad1.example.com

john@ad1.example.com : domain users@ad1.example.com engineering@ad1.example.com
```

Note:

If you just changed the group membership of a user, it may be a while before SSSD notices due to caching.

Finally, how about we try a login:

```
$ sudo login

ad-client login: john@ad1.example.com
Password:
Welcome to Ubuntu 20.04 LTS (GNU/Linux 5.4.0-24-generic x86_64)
...
Creating directory '/home/john@ad1.example.com'.

john@ad1.example.com@ad-client:~$
```

Notice how the home directory was automatically created.

You can also use SSH, but note that the command will look a bit funny because of the multiple @ signs:

```
$ ssh john@ad1.example.com@10.51.0.11
```

```
Welcome to Ubuntu 20.04 LTS (GNU/Linux 5.4.0-24-generic x86_64)
(...)
Last login: Thu Apr 16 21:22:55 2020

john@ad1.example.com@ad-client:~$
```

Note:

In the SSH example, public key authentication was used, so no password was required. Remember that SSH password authentication is disabled by default in `/etc/ssh/sshd_config`.

Kerberos tickets

If you install `krb5-user`, your AD users will also get a Kerberos ticket upon logging in:

```
john@ad1.example.com@ad-client:~$ klist

Ticket cache: FILE:/tmp/krb5cc_1725801106_9UxVIz
Default principal: john@AD1.EXAMPLE.COM

Valid starting    Expires          Service principal
04/16/20 21:32:12 04/17/20 07:32:12 krbtgt/AD1.EXAMPLE.COM@AD1.EXAMPLE.COM
                renew until 04/17/20 21:32:12
```

Note:

realm also configured `/etc/krb5.conf` for you, so there should be no further configuration prompts when installing `krb5-user`.

Let's test with `smbclient` using Kerberos authentication to list the shares of the domain controller:

```
john@ad1.example.com@ad-client:~$ smbclient -k -L server1.ad1.example.com

Sharename      Type      Comment
-----
ADMIN$         Disk     Remote Admin
C$             Disk     Default share
IPC$           IPC       Remote IPC
NETLOGON       Disk     Logon server share
SYSVOL         Disk     Logon server share
SMB1 disabled -- no workgroup available
```

Notice how we now have a ticket for the `cifs` service, which was used for the share list above:

```
john@ad1.example.com@ad-client:~$ klist
```

```
Ticket cache: FILE:/tmp/krb5cc_1725801106_9UxVIz
Default principal: john@AD1.EXAMPLE.COM

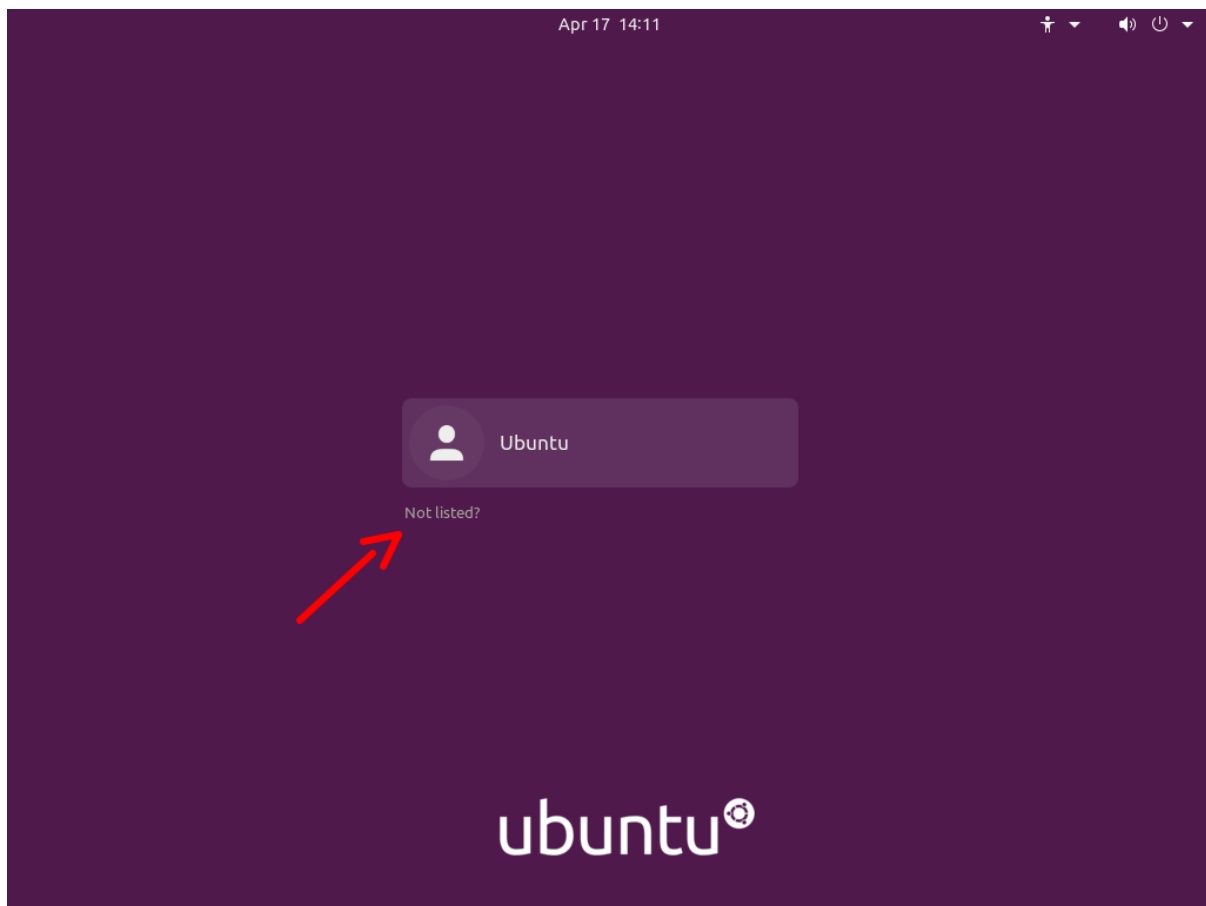
Valid starting    Expires          Service principal
04/16/20 21:32:12  04/17/20 07:32:12  krbtgt/AD1.EXAMPLE.COM@AD1.EXAMPLE.COM
        renew until 04/17/20 21:32:12
04/16/20 21:32:21  04/17/20 07:32:12  cifs/server1.ad1.example.com@AD1.EXAMPLE.COM
```

Ubuntu Desktop authentication

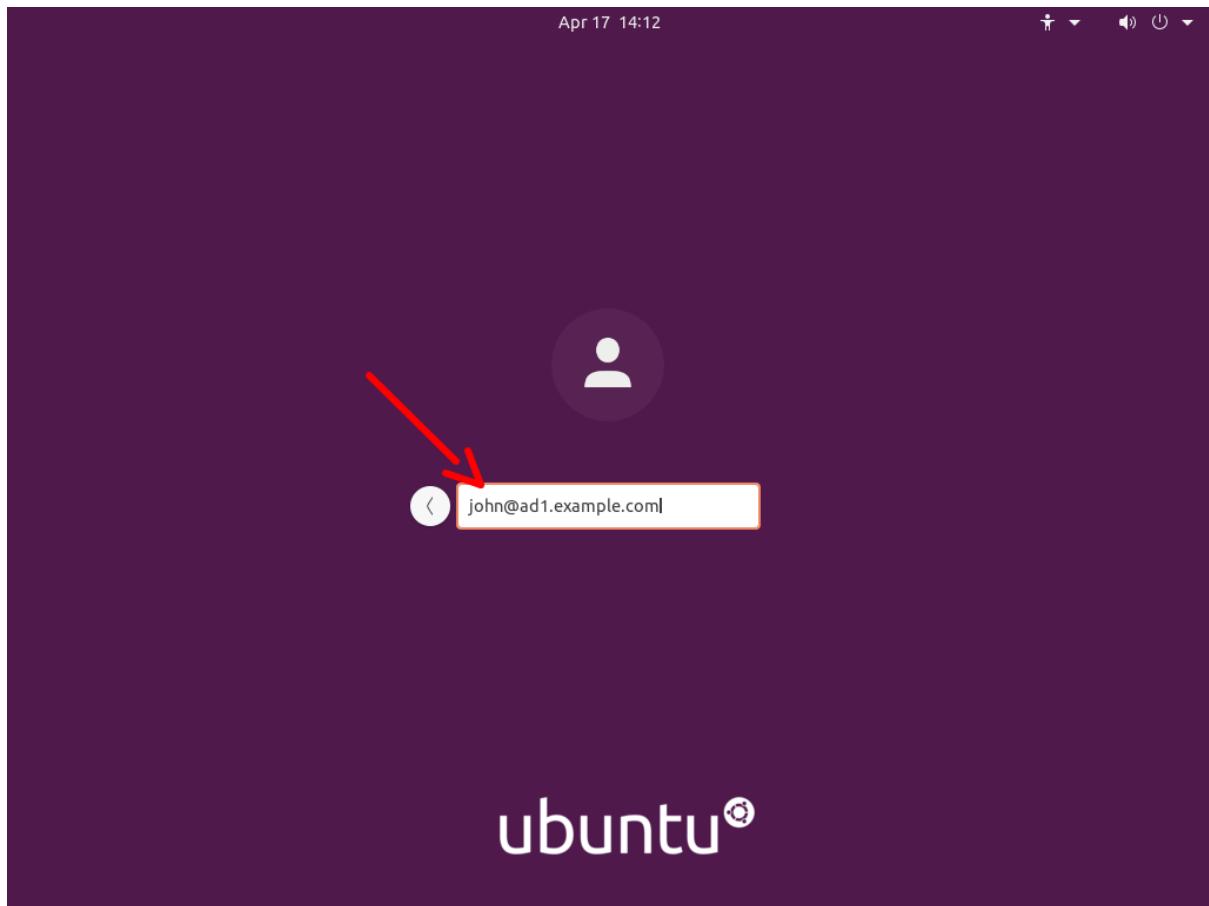
The desktop login only shows local users in the list to pick from, and that's on purpose.

To login with an Active Directory user for the first time, follow these steps:

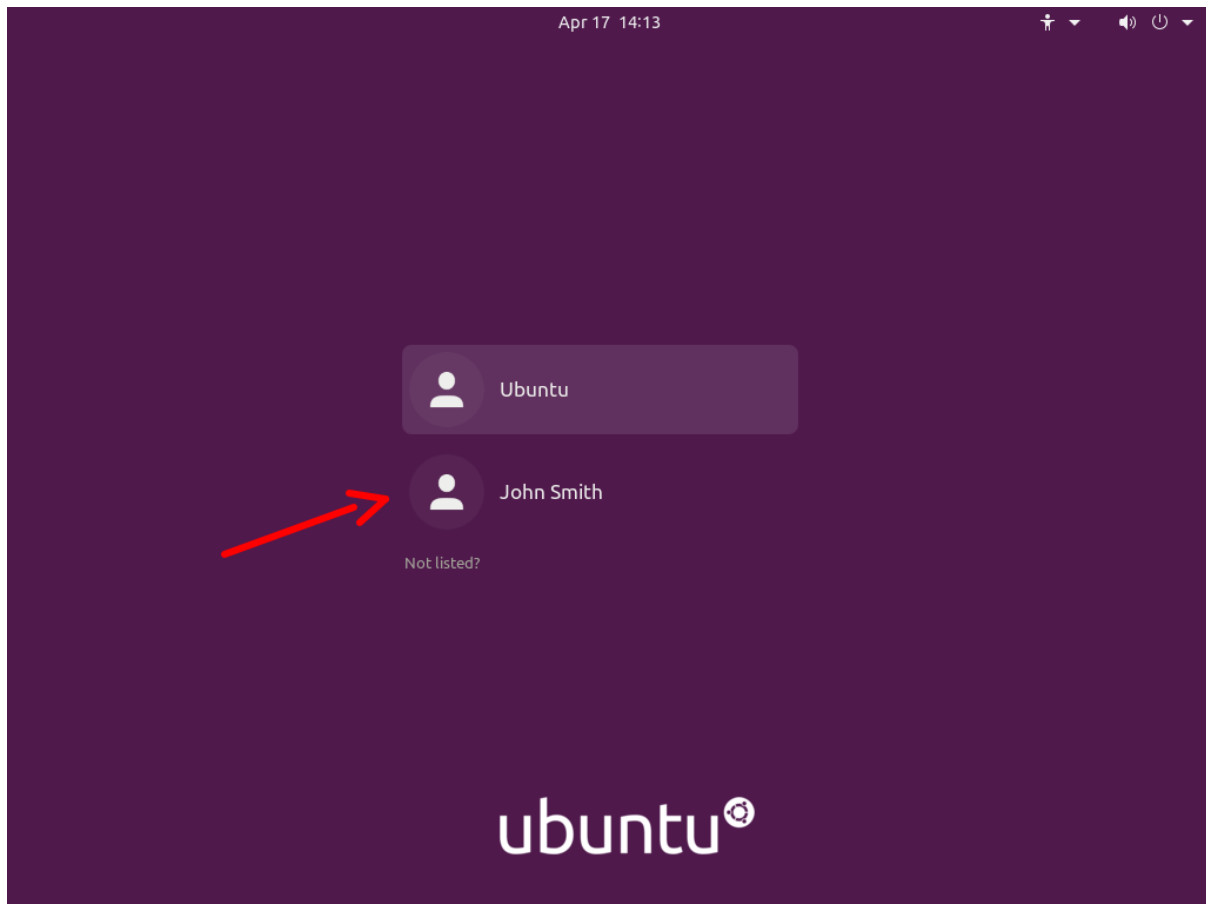
- Click on the “Not listed?” option:



- Type in the login name followed by the password:



- Next time you login, the AD user will be listed as if it was a local user:



Known issues

When logging in on a system joined with an Active Directory domain, sssd (the package responsible for this integration) will try to apply Group Policies by default. There are cases where if a specific policy is missing, the login will be denied.

This is being tracked in [LP: #1934997](https://bugs.launchpad.net/bugs/1934997)¹³⁸. Until the fix becomes available, please see comment number 5 in that bug report for existing workarounds.

Further reading

- [GitHub SSSD Project](#)¹³⁹
- [Active Directory DNS Zone Entries](#)¹⁴⁰

¹³⁸ <https://bugs.launchpad.net/bugs/1934997>

¹³⁹ <https://github.com/SSSD/sss>

¹⁴⁰ [https://learn.microsoft.com/en-us/previous-versions/windows/it-pro/windows-server-2003/cc759550\(v=ws.10\)?redirectedfrom=MSDN](https://learn.microsoft.com/en-us/previous-versions/windows/it-pro/windows-server-2003/cc759550(v=ws.10)?redirectedfrom=MSDN)

How to set up SSSD with LDAP

SSSD can also use LDAP for authentication, authorisation, and user/group information. In this section we will configure a host to authenticate users from an OpenLDAP directory.

Prerequisites and assumptions

For this setup, we need:

- An existing OpenLDAP server with SSL enabled and using the RFC2307 schema for users and groups
- A client host where we will install the necessary tools and login as a user from the LDAP server

Install necessary software

Install the following packages:

```
$ sudo apt install sssd-ldap ldap-utils
```

Configure SSSD

Create the `/etc/sss/sss.conf` configuration file, with permissions `0600` and ownership `root:root`, and add the following content:

```
[sss]
config_file_version = 2
domains = example.com

[domain/example.com]
id_provider = ldap
auth_provider = ldap
ldap_uri = ldap://ldap01.example.com
cache_credentials = True
ldap_search_base = dc=example,dc=com
```

Make sure to start the sssd service:

```
$ sudo systemctl start sssd.service
```

Note:

sss will use START_TLS by default for authentication requests against the LDAP server (the **auth_provider**), but not for the **id_provider**. If you want to also enable START_TLS for the **id_provider**, specify `ldap_id_use_start_tls = true`.

Automatic home directory creation

To enable automatic home directory creation, run the following command:

```
$ sudo pam-auth-update --enable mkhomedir
```

Check SSL setup on the client

The client must be able to use START_TLS when connecting to the LDAP server, with full certificate checking. This means:

- The client host knows and trusts the CA that signed the LDAP server certificate,
- The server certificate was issued for the correct host (ldap01.example.com in this guide),
- The time is correct on all hosts performing the TLS connection, and
- That neither certificate (CA or server's) expired.

If using a custom CA, an easy way to have a host trust it is to place it in /usr/local/share/ca-certificates/ with a .crt extension and run sudo update-ca-certificates.

Alternatively, you can edit /etc/ldap/ldap.conf and point TLS_CACERT to the CA public key file.

Note:

You may have to restart sssd after these changes: `sudo systemctl restart sssd`

Once that is all done, check that you can connect to the LDAP server using verified SSL connections:

```
$ ldapwhoami -x -ZZ -H ldap://ldap01.example.com
anonymous
```

and for ldaps (if enabled in /etc/default/slapd):

```
$ ldapwhoami -x -H ldaps://ldap01.example.com
```

The -ZZ parameter tells the tool to use START_TLS, and that it must not fail. If you have LDAP logging enabled on the server, it will show something like this:

```
slapd[779]: conn=1032 op=0 STARTTLS
slapd[779]: conn=1032 op=0 RESULT oid= err=0 text=
slapd[779]: conn=1032 fd=15 TLS established tls_ssf=256 ssf=256
slapd[779]: conn=1032 op=1 BIND dn="" method=128
slapd[779]: conn=1032 op=1 RESULT tag=97 err=0 text=
slapd[779]: conn=1032 op=2 EXT oid=1.3.6.1.4.1.4203.1.11.3
slapd[779]: conn=1032 op=2 WHOAMI
slapd[779]: conn=1032 op=2 RESULT oid= err=0 text=
```


START_TLS with err=0 and TLS established is what we want to see there, and, of course, the WHOAMI extended operation.

Final verification

In this example, the LDAP server has the following user and group entry we are going to use for testing:

```
dn: uid=john,ou=People,dc=example,dc=com
uid: john
objectClass: inetOrgPerson
objectClass: posixAccount
cn: John Smith
sn: Smith
givenName: John
mail: john@example.com
userPassword: johnsecret
uidNumber: 10001
gidNumber: 10001
loginShell: /bin/bash
homeDirectory: /home/john

dn: cn=john,ou=Group,dc=example,dc=com
cn: john
objectClass: posixGroup
gidNumber: 10001
memberUid: john

dn: cn=Engineering,ou=Group,dc=example,dc=com
cn: Engineering
objectClass: posixGroup
gidNumber: 10100
memberUid: john
```

The user john should be known to the system:

```
ubuntu@ldap-client:~$ getent passwd john
john*:10001:10001:John Smith:/home/john:/bin/bash
```

```
ubuntu@ldap-client:~$ id john
uid=10001(john) gid=10001(john) groups=10001(john),10100(Engineering)
```

And we should be able to authenticate as john:

```
ubuntu@ldap-client:~$ sudo login
```

```
ldap-client login: john
Password:
Welcome to Ubuntu Focal Fossa (development branch) (GNU/Linux 5.4.0-24-generic x86_
64)
(...)
Creating directory '/home/john'.

john@ldap-client:~$
```

How to set up SSSD with LDAP and Kerberos

With SSSD we can create a setup that is very similar to Active Directory in terms of the technologies used: using LDAP for users and groups, and Kerberos for authentication.

Prerequisites and assumptions

For this setup, we will need:

- An existing [OpenLDAP server](#) (page 457) using the RFC2307 schema for users and groups. SSL support is recommended, but not strictly necessary because authentication in this setup is being done via Kerberos, and not LDAP.
- A [Kerberos server](#) (page 183). It doesn't have to be [using the OpenLDAP backend](#) (page 196).
- A client host where we will install and configure SSSD.

Install necessary software

On the client host, install the following packages:

```
$ sudo apt install sssd-ldap sssd-krb5 ldap-utils krb5-user
```

You may be asked about the default Kerberos realm. For this guide, we are using EXAMPLE.COM.

At this point, you should already be able to obtain tickets from your Kerberos server, assuming [DNS](#) records point at it:

```
$ kinit ubuntu

Password for ubuntu@EXAMPLE.COM:

ubuntu@ldap-krb-client:~$ klist
Ticket cache: FILE:/tmp/krb5cc_1000
Default principal: ubuntu@EXAMPLE.COM

Valid starting    Expires          Service principal
04/17/20 19:51:06 04/18/20 05:51:06 krbtgt/EXAMPLE.COM@EXAMPLE.COM
        renew until 04/18/20 19:51:05
```

But we want to be able to login as an LDAP user, authenticated via Kerberos. Let's continue with the configuration.

Configure SSSD

Create the `/etc/sss/sss.conf` configuration file, with permissions `0600` and ownership `root:root`, and add the following content:

```
[sss]
config_file_version = 2
domains = example.com

[domain/example.com]
id_provider = ldap
ldap_uri = ldap://ldap01.example.com
ldap_search_base = dc=example,dc=com
auth_provider = krb5
krb5_server = kdc01.example.com,kdc02.example.com
krb5_kpasswd = kdc01.example.com
krb5_realm = EXAMPLE.COM
cache_credentials = True
```

This example uses two KDCs, which made it necessary to also specify the `krb5_kpasswd` server because the second KDC is a replica and is not running the admin server.

Start the `sss` service:

```
$ sudo systemctl start sss.service
```

Automatic home directory creation

To enable automatic home directory creation, run the following command:

```
$ sudo pam-auth-update --enable mkhomedir
```

Final verification

In this example, the LDAP server has the following user and group entry we are going to use for testing:

```
dn: uid=john,ou=People,dc=example,dc=com
uid: john
objectClass: inetOrgPerson
objectClass: posixAccount
cn: John Smith
sn: Smith
givenName: John
mail: john@example.com
uidNumber: 10001
gidNumber: 10001
```

(continues on next page)

(continued from previous page)

```
loginShell: /bin/bash
homeDirectory: /home/john

dn: cn=john,ou=Group,dc=example,dc=com
cn: john
objectClass: posixGroup
gidNumber: 10001
memberUid: john

dn: cn=Engineering,ou=Group,dc=example,dc=com
cn: Engineering
objectClass: posixGroup
gidNumber: 10100
memberUid: john
```

Note how the user john has no userPassword attribute.

The user john should be known to the system:

```
ubuntu@ldap-client:~$ getent passwd john

john*:10001:10001:John Smith:/home/john:/bin/bash
```

```
ubuntu@ldap-client:~$ id john

uid=10001(john) gid=10001(john) groups=10001(john),10100(Engineering)
```

Let's try a login as this user:

```
ubuntu@ldap-krb-client:~$ sudo login

ldap-krb-client login: john
Password:
Welcome to Ubuntu 20.04 LTS (GNU/Linux 5.4.0-24-generic x86_64)
(...)
Creating directory '/home/john'.
```

```
john@ldap-krb-client:~$ klist

Ticket cache: FILE:/tmp/krb5cc_10001_B0rxWr
Default principal: john@EXAMPLE.COM

Valid starting    Expires          Service principal
04/17/20 20:29:50 04/18/20 06:29:50  krbtgt/EXAMPLE.COM@EXAMPLE.COM
        renew until 04/18/20 20:29:50
```

We logged in using the Kerberos password, and user/group information from the LDAP server.

SSSD and KDC spoofing

When using SSSD to manage Kerberos logins on a Linux host, there is an attack scenario you should be aware of: KDC spoofing.

The objective of the attacker is to login on a workstation that is using Kerberos authentication. Let's say they know john is a valid user on that machine.

The attacker first deploys a rogue Key Distribution Center (KDC) server in the network, and creates the john principal there with a password of the attacker's choosing. What they must do now is have their rogue KDC respond to the login request from the workstation, before (or instead of) the real KDC. If the workstation isn't authenticating the KDC, it will accept the reply from the rogue server and let john in.

There is a configuration parameter that can be set to protect the workstation from this type of attack. It will have SSSD authenticate the KDC, and block the login if the KDC cannot be verified. This option is called `krb5_validate`, and it's `false` by default.

To enable it, edit `/etc/sss/sss.conf` and add this line to the domain section:

```
[sss]
config_file_version = 2
domains = example.com

[domain/example.com]
id_provider = ldap
...
krb5_validate = True
```

The second step is to create a host principal on the KDC for this workstation. This is how the KDC's authenticity is verified. It's like a "machine account", with a shared secret that the attacker cannot control and replicate in the rogue KDC. The host principal has the format `host/<fqdn>@REALM`.

After the host principal is created, its keytab needs to be stored on the workstation. This two step process can be easily done on the workstation itself via `kadmin` (not `kadmin.local`) to contact the KDC remotely:

```
$ sudo kadmin -p ubuntu/admin

kadmin: addprinc -randkey host/ldap-krb-client.example.com@EXAMPLE.COM
WARNING: no policy specified for host/ldap-krb-client.example.com@EXAMPLE.COM;
defaulting to no policy
Principal "host/ldap-krb-client.example.com@EXAMPLE.COM" created.

kadmin: ktadd -k /etc/krb5.keytab host/ldap-krb-client.example.com
Entry for principal host/ldap-krb-client.example.com with kvno 6, encryption type
aes256-cts-hmac-sha1-96 added to keytab WRFILE:/etc/krb5.keytab.
Entry for principal host/ldap-krb-client.example.com with kvno 6, encryption type
aes128-cts-hmac-sha1-96 added to keytab WRFILE:/etc/krb5.keytab.
```

Then exit the tool and make sure the permissions on the keytab file are tight:

```
$ sudo chmod 0600 /etc/krb5.keytab
```

```
$ sudo chown root:root /etc/krb5.keytab
```

You can also do it on the KDC itself using `kadmin.local`, but you will have to store the keytab temporarily in another file and securely copy it over to the workstation.

Once these steps are complete, you can restart SSSD on the workstation and perform the login. If the rogue KDC notices the attempt and replies, it will fail the host verification. With debugging we can see this happening on the workstation:

```
==> /var/log/sss/krb5_child.log <==  
(Mon Apr 20 19:43:58 2020) [[sss[krb5_child[2102]]] [validate_tgt] (0x0020): TGT  
failed verification using key for [host/ldap-krb-client.example.com@EXAMPLE.COM].  
(Mon Apr 20 19:43:58 2020) [[sss[krb5_child[2102]]] [get_and_save_tgt] (0x0020):  
1741: [-1765328377][Server host/ldap-krb-client.example.com@EXAMPLE.COM not found in  
Kerberos database]
```

And the login is denied. If the real KDC picks it up, however, the host verification succeeds:

```
==> /var/log/sss/krb5_child.log <==  
(Mon Apr 20 19:46:22 2020) [[sss[krb5_child[2268]]] [validate_tgt] (0x0400): TGT  
verified using key for [host/ldap-krb-client.example.com@EXAMPLE.COM].
```

And the login is accepted.

If you run into difficulties, refer to our troubleshooting guide.

Troubleshooting SSSD

Here are some tips to help troubleshoot SSSD.

debug_level

The debug level of SSSD can be changed on-the-fly via `sssctl`, from the `sss-tools` package:

```
$ sudo apt install sss-tools
```

```
$ sssctl debug-level <new-level>
```

Or add it to the config file and restart SSSD:

```
[sss]  
config_file_version = 2  
domains = example.com  
  
[domain/example.com]  
debug_level = 6  
...
```

Either approach will yield more logs in `/var/log/sss/*.log` and can help identify what is happening. The `sssctl` approach has the clear advantage of not having to restart the service.

Caching

Caching is useful to speed things up, but it can get in the way big time when troubleshooting. It's useful to be able to remove the cache while chasing down a problem. This can also be done with the `sssctl` tool from the `sss-tools` package.

You can either remove the whole cache:

```
$ sssctl cache-remove

Creating backup of local data...
SSSD backup of local data already exists, override? (yes/no) [no] yes
Removing cache files...
SSSD= needs to be running. Start SSSD now? (yes/no) [yes] yes
```

Or just one element:

```
$ sssctl cache-expire -u john
```

Or expire everything:

```
$ sssctl cache-expire -E
```

DNS

Kerberos is quite sensitive to [DNS](#) issues. If you suspect something related to DNS, here are two suggestions:

FQDN hostname

Make sure `hostname -f` returns a *fully qualified domain name (FQDN)*. Set it in `/etc/hostname` if necessary, and use `sudo hostnamectl set-hostname <fqdn>` to set it at runtime.

Reverse name lookup

You can try disabling a default reverse name lookup, which the `krb5` libraries do, by editing (or creating) `/etc/krb5.conf` and setting `rdns = false` in the `[libdefaults]` section:

```
[libdefaults]
rdns = false
```

See also

- Explanation: *Introduction to network user authentication with SSSD* (page 874)

Smart card authentication

One of the most popular uses for smart cards is to control access to computer systems. The owner must physically *have* the smart card, and they must know the PIN to unlock it. This provides a higher degree of security than single-factor authentication (such as just using a password). In this page, we describe how to enable smart card authentication on Ubuntu.

Note:

This guide is meant for Ubuntu Server 20.04 and newer. If you want to configure a desktop installation refer to the [desktop guide](#)¹⁴¹.

¹⁴¹ <https://ubuntu.com/tutorials/how-to-use-smart-card-authentication-in-ubuntu-desktop#2-overview>

Software requirements

The following packages must be installed to obtain a smart card configuration on Ubuntu:

- **pcscd**: contains the drivers needed to communicate with the CCID smart card readers
- **opensc-pkcs11**: (optional, depending on your smart card hardware) contains the smart card drivers, such as Personal Identify Verification (PIV) or Common Access Card (CAC)
- **sssd**: the authentication daemon that manages smart card access and certificate verification

To install these packages, run the following command in your terminal:

```
$ sudo apt install opensc-pkcs11 pcscd sssd libpam-sss
```

Hardware requirements

Any PIV or CAC smart card with the corresponding reader should be sufficient. USB smart cards like Yubikey embed the reader, and work like regular PIV cards.

Each smart card is expected to contain an X.509 certificate and the corresponding private key to be used for authentication.

Smart card PKCS#11 modules

While `opensc-pkcs11` supports a wide number of smart cards, some of them may require specific PKCS#11 modules, and you must refer to your vendor to install the proper one. From Ubuntu 20.04 LTS onward, all modules supported by `p11-kit`¹⁴² can be used.

If custom PKCS#11 modules are used, you need to ensure that `p11-kit` is properly configured¹⁴³.

In any case, `p11-kit` can be used to see all the configured modules that can be used for authentication:

```
$ p11-kit list-modules

p11-kit-trust: p11-kit-trust.so
  library-description: PKCS#11 Kit Trust Module
  library-manufacturer: PKCS#11 Kit
  library-version: 0.23
  token: System Trust
    manufacturer: PKCS#11 Kit
    model: p11-kit-trust
    serial-number: 1
    hardware-version: 0.23
    flags:
      write-protected
      token-initialized
opensc-pkcs11: opensc-pkcs11.so
  library-description: OpenSC smartcard framework
  library-manufacturer: OpenSC Project
  library-version: 0.20
  token: MARCO TREVISAN (PIN CNS0)
    manufacturer: IC: STMicroelectronics; mask:...
    model: PKCS#15 emulated
    serial-number: 6090010669298009
    flags:
      login-required
      user-pin-initialized
      token-initialized
      user-pin-locked
```

X.509 smart card certificates

The authentication is based on X.509 certificate validation and a smart card can provide one or more certificates that can be used for this purpose.

Before continuing, you may need to export or reference the certificate ID that must be used and associated to each user; such operations can be performed in one of the following three ways:

¹⁴² <https://p11-glue.github.io/p11-glue/p11-kit.html>

¹⁴³ <https://p11-glue.github.io/p11-glue/p11-kit/manual/config.html>

Using p11tool

This is a more generic implementation that just uses the PKCS#11 protocol so it should work with all modules:

```
$ sudo apt install gnutls-bin > p11tool --list-tokens
```

Alternatively, URLs can be listed via:

```
$ p11tool --list-token-urls

Token 1:
  URL: pkcs11:model=PKCS%2315%20emulated;manufacturer=IC%3A%20Infineon%3B%20mask%3A%20IDEMIA%20%280...;serial=6090033068507002;token=MARCO%20TREVISAN%20%28PIN%20CNS1%29
  Label: MARCO TREVISAN (PIN CNS1)
  Type: Hardware token
  Flags: Requires login
  Manufacturer: IC: Infineon; mask: IDEMIA (0...
  Model: PKCS#15 emulated
  Serial: 6090033068507002
  Module: opensc-pkcs11.so
```

The command above will show all the available smart cards in the system and their associated PKCS#11 URI. Copy the URI token of the selected card in the following command, which prints all certificates that can be used for authentication and their associated token URIs.

```
$ p11tool --list-all-certs 'pkcs11:token=[TOKEN-ID]'
```

So in the above example:

```
$ p11tool --list-all-certs
'pkcs11:token=MARCO%20TREVISAN%20%28PIN%20CNS1%29'

Object 0:
  URL: pkcs11:model=PKCS%2315%20emulated;manufacturer=IC%3A%20Infineon%3B%20mask%3A%20IDEMIA%20%280...;serial=6090033068507002;token=MARCO%20TREVISAN%20%28PIN%20CNS1%29;id=%02;object=CNS1;type=cert
  Type: X.509 Certificate (RSA-2048)
  Expires: ven 17 dic 2027, 00:00:00
  Label: CNS1
  ID: 02
```

Now, once the URI of the certificate that will be used for authentication is known, let's extract the **Common Name** from the certificate. In the example we are assuming that our certificate URI is `pkcs11:id=%02;type=cert`.

It can be exported as text Privacy Enhanced Mail (PEM) format using:

```
$ p11tool --export 'pkcs11:id=%02;type=cert'
```

Using opensc

```
$ sudo apt install opensc
```

Certificates can be via:

```
$ pkcs15-tool --list-certificates
```

And exported using

```
$ pkcs15-tool --read-certificate [CERTIFICATE_ID]
```

So, for example:

```
$ pkcs15-tool --list-certificates

Using reader with a card: Alcor Micro AU9560 00 00
X.509 Certificate [CNS1]
  Object Flags   : [0x00]
  Authority      : no
  Path           : 3f00140090012002
  ID             : 02
  Encoded serial : 02 10 0357B1EC0EB725BA67BD2D838DDF93D5
```

```
$ pkcs15-tool --read-certificate 2

Using reader with a card: Alcor Micro AU9560 00 00
-----BEGIN CERTIFICATE-----
MIIHXDCCBUSgAwIBAgIQA1ex7A6.....
```

Troubleshooting

The card certificate verification can be simulated using openssl:

```
$ sudo apt install openssl
```

Save the certificate, using one of the methods stated above:

```
$ pkcs15-tool --read-certificate 2 > card-cert.pem
```

```
$ p11tool --export 'pkcs11:id=%02;type=cert' > card-cert.pem
```

See the certificate contents with:

```
$ openssl x509 -text -noout -in card-cert.pem
```

Verify it is valid for the given CA, where 'Ca-Auth-CERT.pem' contains all the certificates chain:

```
$ openssl verify -verbose -CAfile CA-Auth-CERT.pem card-cert.pem
```

If only the parent CA Certificate is available, you can use `-partial_chain`:

```
$ openssl verify -verbose -partial_chain -CAfile intermediate_CA_cert.pem
```

PAM configuration

To enable smart card authentication we should rely on a module that allows PAM supported systems to use X.509 certificates to authenticate logins. The module relies on a PKCS#11 library, such as `opensc-pkcs11` to access the smart card for the credentials it will need.

When a PAM smart card module is enabled, the login process is as follows:

1. Enter login
2. Enter PIN
3. Validate the X.509 certificate
4. Map the certificate to a user
5. Verify the login and match

To enable that process we have to configure the PAM module, add the relevant certificate authorities, add the PAM module to PAM configuration and set the mapping of certificate names to logins.

Setup guide

This configuration uses SSSD as authentication mechanism, and the example shown here is showing a possible usage for local users, but more complex setups using external remote identity managers such as [FreeIPA](#), LDAP, Kerberos or others can be used.

Refer to [SSSD documentation](#)¹⁴⁴ to learn more about this.

Enable SSSD PAM service

Pam service must be enabled in SSSD configuration, it can be done by ensuring that `/etc/sss/sss.conf` contains:

```
[sssd]
services = pam

[pam]
pam_cert_auth = True
```

Further `[pam]` configuration options can be changed according to the [sss.conf\(5\)](#)¹⁴⁵ manual page.

¹⁴⁴ <https://sssd.io/docs/introduction.html>

¹⁴⁵ <https://manpages.ubuntu.com/manpages/resolute/man5/sss.conf.5.html>

Configure SSSD Certificate Authorities database

The card certificate must be allowed by a Certificate Authority, these should be part of `/etc/sss/pki/sss_auth_ca_db.pem` (or any other location configured in `[pam]` config section of `sss.conf` as `pam_cert_db_path`).

As per SSSD using openssl, we need to add the whole certificates chain to the SSSD CA certificates path (if not changed via `sss.certificate_verification`), so adding the certificates to the `pam_cert_db_path` is enough:

```
$ sudo cat Ca-Auth-CERT*.pem >> /etc/sss/pki/sss_auth_ca_db.pem
```

Certification Revocation List can be also defined in `sss.conf`, providing a CRL file path in PEM format

```
[sss]
crl_file = /etc/sss/pki/sss_auth_crl.pem
soft_crl = /etc/sss/pki/sss_auth_soft_crl.pem
```

In case that a full certificate authority chain is not available, openssl won't verify the card certificate, and so SSSD should be instructed about.

This is not suggested, but it can be done changing `/etc/sss/sss.conf` so that it contains:

```
[sss]
certificate_verification = partial_chain
```

Troubleshooting

Card certificate verification can be simulated using SSSD tools directly, by using the command SSSD's `p11_child`:

22.04 LTS onward

```
$ sudo /usr/libexec/sss/p11_child --pre -d 10 --debug-fd=2
--ca_db=/etc/sss/pki/sss_auth_ca_db.pem
```

20.04 LTS and earlier

```
$ sudo /usr/libexec/sss/p11_child --pre -d 10 --debug-fd=2
--nssdb=/etc/sss/pki/sss_auth_ca_db.pem
```

If certificate verification succeeds, the tool should output the card certificate name, its ID and the certificate itself in base64 format (other than debug data):

```
(Mon Sep 11 16:33:32:129558 2023) [p11_child[1965]] [do_card] (0x4000): Found
certificate has key id [02].
MARCO TREVISAN (PIN CNS1)
/usr/lib/x86_64-linux-gnu/pkcs11/opensc-pkcs11.so
02
CNS1
MIIHXDCBUSgAwIBAgIQ1ex7....
```

For checking if the smart card works, without doing any verification check (and so for debugging purposes the option) `--verify=no_ocsp` can also be used, while `--verify=partial_chain` can be used to do partial CA verification.

Map certificates to user names

The sss PAM module allows certificates to be used for login, though our Linux system needs to know the username associated to a certificate. SSSD provides a variety of cert mappers to do this. Each cert mapper uses specific information from the certificate to map to a user on the system. The different cert mappers may even be stacked. In other words, if the first defined mapper fails to map to a user on the system, the next one will be tried, and so on until a user is found.

For the purposes of this guide, we will use a simple local user mapping as reference.

Mapping for more complex configurations can be done following the official [SSSD documentation](https://sssd.io/design-pages/matching_and_mapping_certificates.html)¹⁴⁶ depending on [providers](https://sssd.io/design-pages/certmaps_for_LDAP_AD_file.html)¹⁴⁷. For up-to-date information on certificate mapping, please also consult the [sss-certmap\(5\)](https://manpages.ubuntu.com/manpages/resolute/man5/sss-certmap.5.html)¹⁴⁸ manual page.

Local users mapping

When using only local users, SSSD can be easily configured to define an `implicit_domain` that maps all the local users.

Certificate mapping for local users can be easily done using the certificate Subject check, in our example:

```
$ openssl x509 -noout -subject -in card-cert.pem | sed "s/, /,/g;s/ = /=/g"
subject=C=IT,O=Actalis S.p.A.,OU=REGIONE TOSCANA,SN=TREVISAN,GN=MARCO,CN=TRVMRC[...
data-removed...]/6090033068507002.UyMnHxFF3gkAeBYHhxa6V1Edazs=
```

So we can use for the user foo:

```
[sssd]
enable_files_domain = True
services = pam

[certmap/implicit_files/foo]
```

(continues on next page)

¹⁴⁶ https://sssd.io/design-pages/matching_and_mapping_certificates.html

¹⁴⁷ https://sssd.io/design-pages/certmaps_for_LDAP_AD_file.html

¹⁴⁸ <https://manpages.ubuntu.com/manpages/resolute/man5/sss-certmap.5.html>

(continued from previous page)

```
matchrule = <SUBJECT>.*CN=TRVMRC[A-Z0-9]+/6090033068507002\ .UyMnHxFF3gkAeBYHhxa6V1Edazs=.*
```

```
[pam]
```

```
pam_cert_auth = True
```

Troubleshooting

User mapping can be tested working in versions newer than Ubuntu 20.04 LTS with:

```
$ sudo dbus-send --system --print-reply \ >
--dest=org.freedesktop.sssd.infopipe \ > /org/freedesktop/sss/infopipe/Users
\ > org.freedesktop.sssd.infopipe.Users.ListByCertificate \ > string:"$(cat
card-cert.pem)" uint32:10
```

That should return the object path containing the expected user ID:

```
method return time=1605127192.698667 sender=:1.1628 -> destination=:1.1629 serial=6
reply_serial=2
  array [
    object path "/org/freedesktop/sss/infopipe/Users/implicit_5ffiles/1000"
  ]
```

Basic SSSD configuration

The SSSD configuration for accessing to the system is out of the scope of this document, however for smart card login it should contain at least such values:

```
[sssd]
```

```
# Comma separated list of domains
;domains = your-domain1, your-domain2
```

```
# comma-separated list of SSSD services
# pam might be implicitly loaded already, so the line is optional
services = pam
```

```
# You can enable debug of the SSSD daemon
# Logs will be in /var/log/sss/sss.log
;debug_level = 10
```

```
# A mapping between the SC certificate and users
;[certmap/your-domain1/<username>]
;matchrule = <SUBJECT>.*CN=<REGEX MATCHING YOUR CN>.*
```

```
[pam]
```

```
pam_cert_auth = True
```

```
# The Certificate DB to be used:
# - Needs to be an openssl CA certificates
;pam_cert_db_path = /etc/ssl/certs/ca-certificates.crt
```

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```
# You can enable debug infos for the PAM module
# Logs will be in /var/log/sss/sss_pam.log
# p11 child logs are in /var/log/sss/p11_child.log
# standard auth logs are in /var/log/auth.log
;pam_verbosity = 10
;debug_level = 10
```

In general what's in the configuration file will affect the way SSSD will call the `p11_child` tool (that is the one in charge for the actual authentication). Check `man sssd.conf` for details.

Remember that this file should be owned by root and have permission set to 600, otherwise won't be loaded and SSSD will not complain gracefully. On errors you can test running SSSD temporary with `sudo sssd -d9 -i`.

Every time the configuration is changed SSSD should be restarted (`systemctl restart sssd`).

Add `pam_sss` to PAM

The next step includes the `pam_sss` module into the PAM stack. There are various ways to do this depending on your local policy. The following example enables smart card support for general authentication.

Edit `/etc/pam.d/common-auth` to include the `pam_sss` module as follows:

For Ubuntu later than 23.10

```
$ sudo pam-auth-update
```

Then you can interactively enable SSSD profiles for smart-card-only or optional smart card access. You can also set this non-interactively.

To use smart-card only authentication:

```
$ $ sudo pam-auth-update --disable sss-smart-card-optional --enable
sss-smart-card-required
```

To use smart-card authentication with fallback:

```
$ sudo pam-auth-update --disable sss-smart-card-required --enable
sss-smart-card-optional
```


For Ubuntu 23.10 and earlier

```
$ require SSSD smart card login  
  
auth    [success=done default=die]    pam_sss.so allow_missing_name require_cert_  
auth
```

or only try to use it:

```
$ try SSSD smart card login  
  
auth    [success=ok default=ignore]    pam_sss.so allow_missing_name try_cert_auth
```

See [pam.conf\(5\)](#)¹⁴⁹, [pam_sss\(8\)](#)¹⁵⁰ for further details.

Warning:

A global configuration such as this requires a smart card for su and sudo authentication as well!

If you want to reduce the scope of this module, move it to the appropriate PAM configuration file in /etc/pam.d and ensure that it's referenced by pam_p11_allowed_services in sssd.conf.

The OS is now ready to do a smart card login for the user foo.

Troubleshooting

pamtester is your friend!

To get better debug logging, also increase the SSSD verbosity by changing /etc/sss/sss.conf so that it has:

```
[pam]  
pam_verbosity = 10  
debug_level = 10
```

You can use it to check your configuration without having to login/logout for real. First install it:

```
$ sudo apt install pamtester
```

Run the authentication service as standalone:

```
$ pamtester -v login $USER authenticate
```

Run the authentication service to get user from cert:

¹⁴⁹ <https://manpages.ubuntu.com/manpages/resolute/man5/pam.conf.5.html>

¹⁵⁰ https://manpages.ubuntu.com/manpages/resolute/man8/pam_sss.8.html

```
$ pamtester -v login "" authenticate
```

You can check what happened in the logs:

```
$ sudo less /var/log/auth.log
```

```
$ sudo less /var/log/sss/sss_pam.log
```

```
$ sudo less /var/log/sss/p11_child.log
```

Smart card authentication with SSH

One of the authentication methods supported by the SSH protocol is public key authentication. A public key is copied to the SSH server where it is stored and marked as authorized. The owner of the corresponding private key in the smart card can then SSH login to the server.

We will use `opensc-pkcs11` on the client to access the smart card drivers, and we will copy the public key from the smart card to the SSH server to make the authentication work.

The following instructions apply to Ubuntu 18.04 LTS later.

Server configuration

The SSH server and client must be configured to permit smart card authentication.

Configure the SSH server

The SSH server needs to allow public key authentication set in its configuration file and it needs the user's public key.

Ensure the server has the `PubkeyAuthentication` option set to 'yes' in its `/etc/ssh/sshd_config` file. In a default `/etc/ssh/sshd_config` in Ubuntu, the `PubkeyAuthentication` option is commented out. However, the default is 'yes'. To ensure the setting, edit the `sshd_config` file and set accordingly.

```
PubkeyAuthentication yes
```

Restart the SSH server

```
$ sudo systemctl restart sshd
```

Set the public key on the server

Extract the user's public key from the smart card on the SSH client. Use `ssh-keygen` to read the public key from the smart card and into a format consumable for SSH.

```
$ ssh-keygen -D /usr/lib/x86_64-linux-gnu/opensc-pkcs11.so > smartcard.pub
```

Copy this key to the SSH server.

```
ssh-copy-id -f -i smartcard.pub ubuntu@server-2
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "smartcard.pub"
ubuntu@server-2's password:
Number of key(s) added: 1
Now try logging into the machine, with: "ssh 'ubuntu@server-2'"
and check to make sure that only the key(s) you wanted were added.
```

Client configuration

The SSH client needs to identify its PKCS#11 provider. To do that set the `PKCS11Provider` option in the `~/.ssh/config` file of each user desiring to use SSH smart card login.

```
PKCS11Provider /usr/lib/x86_64-linux-gnu/opensc-pkcs11.so
```

Use this method to enforce SSH smart card login on a per user basis.

After this step you can SSH into the server using the smart card for authentication.

SSH authentication

See [this page on SSH authentication with smart cards](#) (page 233).

Cryptography

The Secure Shell (SSH) cryptographic protocol that provides secure channels on an unsecured network. In Ubuntu, OpenSSH is the most commonly used implementation of SSH. It provides a suite of utilities for encrypting data transfers and can also be used for remote login and authentication.

- [Obtain TLS certificates](#) (page 252) covers both Let's Encrypt (for internet-facing servers) and running your own Certificate Authority (CA) for internal networks

OpenSSH server

OpenSSH is a powerful collection of tools for remotely controlling networked computers and transferring data between them. Here we'll describe some of the configuration settings possible with the OpenSSH server application and how to change them on your Ubuntu system.

OpenSSH is a freely available version of the Secure Shell (SSH) protocol family of tools. Traditional tools, such as `telnet` or `rcp`, are insecure and transmit the user's password in cleartext when used. OpenSSH provides a server daemon and client tools to facilitate secure, encrypted, remote control and file transfer operations, effectively replacing the legacy tools.

The OpenSSH server component, `sshd`, listens continuously for client connections from any of the client tools. When a connection request occurs, `sshd` sets up the correct connection depending on the type of client tool connecting. For example, if the remote computer is connecting with the SSH client application, the OpenSSH server sets up a remote control session after authentication. If a remote user connects to an OpenSSH server with `scp`, the OpenSSH server daemon initiates a secure copy of files between the server and client after authentication.

OpenSSH can use many authentication methods, including plain password, public key cryptography, and Kerberos tickets.

Install OpenSSH

To install the OpenSSH client applications on your Ubuntu system, use this command at a terminal prompt:

```
$ sudo apt install openssh-client
```

To install the OpenSSH server application, and related support files, use this command at a terminal prompt:

```
$ sudo apt install openssh-server
```

Configure OpenSSH

To configure the default behavior of the OpenSSH server application, `sshd`, you can modify the main configuration file at `/etc/ssh/sshd_config` or add modular configuration snippets to the `/etc/ssh/sshd_config.d/` directory.

Important:

By default, Ubuntu's configuration includes the line `Include /etc/ssh/sshd_config.d/*.conf` at the very top of `/etc/ssh/sshd_config`. Because OpenSSH uses the first value set for most directives, any settings defined in files within `sshd_config.d/` will override those in the main configuration file.

Using snippets in `sshd_config.d/` makes updates easier and keeps your custom changes separate from system defaults.

For information about the configuration directives, refer to the online [sshd_config\(5\)](https://manpages.ubuntu.com/manpages/resolute/man5/sshd_config.5.html)¹⁵¹ manual page or run `man sshd_config` at a terminal prompt.

Example configuration directive

Let's take a look at an example of a configuration directive change. To make your OpenSSH server display the contents of the `/etc/issue.net` file as a pre-login banner, you can add or modify this line in the `/etc/ssh/sshd_config` or in a snippet file in `/etc/ssh/sshd_config.d/`:

```
Banner /etc/issue.net
```

Since losing an SSH server might mean losing your way to reach a server, check the configuration after changing it and before restarting the server:

```
$ sudo sshd -t
```

After making changes to the configuration, save the file. Then, restart the `sshd` server application to effect the changes using the following command:

```
$ sudo systemctl restart ssh.service
```

Disable OS information disclosure through service banner

When a client establishes a TCP connection with the OpenSSH server, the server returns a banner providing information about the host OS and OpenSSH versions.

This banner is returned before the client authenticates to the service:

```
nc -nv 127.0.0.1 22
Connection to 127.0.0.1 22 port [tcp/*] succeeded!
SSH-2.0-OpenSSH_10.2p1 Ubuntu-2ubuntu1
```

The OS related information can be suppressed from that banner through the `DebianBanner` configuration directive in the `/etc/ssh/sshd_config` or in a snippet file in `/etc/ssh/sshd_config.d/`:

```
DebianBanner no
```

As shown in the previous example, restart the `sshd` server and verify that the banner no longer contains the snippet announcing the OS related data:

```
nc -nv 127.0.0.1 22
Connection to 127.0.0.1 22 port [tcp/*] succeeded!
SSH-2.0-OpenSSH_10.2p1
```

¹⁵¹ https://manpages.ubuntu.com/manpages/resolute/man5/sshd_config.5.html

Warning:

Many other configuration directives for `sshd` are available to change the server application's behavior to fit your needs. Be advised, however, if your only method of access to a server is SSH, and you make a mistake when configuring `sshd` you may find you are locked out of the server upon restarting it. Additionally, if an incorrect configuration directive is supplied, the `sshd` server may refuse to start, so be particularly careful when editing the configuration on a remote server.

Tip:

You can use [etckeeper](#) (page 565) to track changes in your `/etc/` with `git`. Alternatively, before editing the configuration file, you should make a copy of the original `/etc/ssh/sshd_config` file and protect it from writing so you will have the original settings as a reference and to reuse as necessary. You can do this with the following commands:

```
$ sudo cp /etc/ssh/sshd_config /etc/ssh/sshd_config.original
```

```
$ sudo chmod a-w /etc/ssh/sshd_config.original
```

SSH keys

SSH allows authentication between two hosts without the need of a password, using cryptographic keys instead. [SSH-key](#) authentication uses a **private key** and a **public key**.

We recommend the **ed25519 Elliptic Curve algorithm**¹⁵² (`-t ed25519`) due to shorter key size and lower computational requirements. Alternatively, you can create a key using the **RSA Algorithm** (`-t rsa -b 4096`, for 4096 bit key size) instead.

To generate a key pair, run the following command:

```
$ ssh-keygen -t ed25519
```

During creation, you will be prompted for a key passphrase, which you would enter to use that key (every time, or instead using an [ssh-agent\(1\)](#)¹⁵³.

By default, the public key is saved in the file `~/.ssh/id_<algorithm>.pub`, while `~/.ssh/id_<algorithm>` is the private key. To allow login for a user via key, append the content of `id_ed25519.pub` (`id_rsa.pub` for RSA) to `target_machine:~/.ssh/authorized_keys` by running:

```
$ ssh-copy-id username@target_machine
```

Finally, double check the permissions on the `authorized_keys` file – only the authenticated user must have write permissions. If the permissions are not correct then change them by:

¹⁵² <https://ed25519.cr.yp.to/>

¹⁵³ <https://manpages.ubuntu.com/manpages/resolute/man1/ssh-agent.1.html>

```
$ chmod go-w .ssh/authorized_keys
```

You should now be able to SSH to the `target_machine` without being prompted for a password.

To troubleshoot this, have a look at the `target_machine`'s live logs of `ssh.service`:

```
$ sudo journalctl -fu ssh.service
```

Since you can have multiple *SSH-keys*, you can configure which to use for which target machine in `~/.ssh/config` (see *ssh_config(5)*¹⁵⁴).

Connection multiplexing

To reduce connection setup times, multiple SSH sessions to the same `target_machine` can reuse the same *TCP* connection.

Configuration to activate multiplexing in `~/.ssh/config`:

```
ControlMaster auto
ControlPath %d/.ssh/ssh_mux_%u@%l_%r@%h:%p
```

Optionally, if you want the session to still be re-usable 1 second after disconnecting, set:

```
ControlPersist 1
```

Import keys from public key servers

These days many users have already SSH keys registered with services like Launchpad or GitHub. Those can be imported with:

```
$ ssh-import-id <username-on-remote-service>
```

The prefix `lp:` is implied and means fetching from Launchpad. The alternative `gh:` will make the tool fetch from GitHub instead.

Two factor authentication

You can add an extra layer of security to the default key-based authentication using two factor authentication. You can add two factor authentication *using U2F/FIDO hardware authentication devices* (page 243). Alternatively, in cases U2F/FIDO hardware authentication devices are unavailable or impractical for your use case you can add it *using HMAC/Time based One Time Passwords (HOTP/TOTP)* (page 239).

¹⁵⁴ https://manpages.ubuntu.com/manpages/resolute/man5/ssh_config.5.html

GSSAPI/Kerberos authentication

If your network uses Kerberos, you can configure OpenSSH to authenticate users with their Kerberos tickets, and to forward those tickets to the server. See [OpenSSH with GSSAPI/Kerberos authentication](#) (page 246).

Handling unstable connections

When working on remote systems via SSH, unstable network connections or accidental disconnects can interrupt your work and terminate running processes. Terminal multiplexers solve this problem by allowing sessions to persist even after disconnection.

Using a [terminal multiplexer](#) (page 853) like `tmux` or `screen`, you can start a session on the remote machine that continues running independently of your SSH connection. If your connection drops, you can reconnect and reattach to your existing session, resuming your work exactly where you left off without losing any running processes or command output.

Further reading

- [Ubuntu Wiki SSH](#)¹⁵⁵ page.
- [OpenSSH Website](#)¹⁵⁶
- [Advanced OpenSSH Wiki Page](#)¹⁵⁷

Two factor authentication with TOTP/HOTP

For the best two factor authentication (2FA) security, we recommend using hardware authentication devices that support U2F/FIDO. See the guide on [two factor authentication with U2F/FIDO](#) (page 243) for details. However, if this is not possible or is impractical to implement in your case, TOTP/[HOTP](#) based 2FA is an improvement over no two factor at all. Smartphone apps to support this type of 2FA are common, such as Google Authenticator.

Background

The configuration presented here makes public key authentication the first factor, the TOTP/HOTP code the second factor, and makes password authentication unavailable. Apart from the usual setup steps required for public key authentication, all configuration and setup takes place on the server. No changes are required at the client end; the 2FA prompt appears in place of the password prompt.

The two supported methods are [HMAC-based One Time Password \(HOTP\)](#)¹⁵⁸ and [Time-based One Time Password \(TOTP\)](#)¹⁵⁹. Generally, TOTP is preferable if the 2FA device supports it.

¹⁵⁵ <https://help.ubuntu.com/community/SSH>

¹⁵⁶ <http://www.openssh.org/>

¹⁵⁷ <https://wiki.ubuntu.com/AdvancedOpenSSH>

¹⁵⁸ https://en.wikipedia.org/wiki/HMAC-based_one-time_password

¹⁵⁹ https://en.wikipedia.org/wiki/Time-based_one-time_password

HOTP is based on a sequence predictable only to those who share a secret. The user must take an action to cause the client to generate the next code in the sequence, and this response is sent to the server. The server also generates the next code, and if it matches the one supplied by the user, then the user has proven to the server that they share the secret. A downside of this approach is that if the user generates codes without the server following along, such as in the case of a typo, then the sequence generators can fall “out of sync”. Servers compensate by allowing a gap in the sequence and considering a few subsequent codes to also be valid; if this mechanism is used, then the server “skips ahead” to sync back up. But to remain secure, this can only go so far before the server must refuse. When HOTP falls out of sync like this, it must be reset using some out-of-band method, such as authenticating using a second backup key in order to reset the secret for the first one.

TOTP avoids this downside of HOTP by using the current timezone-independent date and time to determine the appropriate position in the sequence. However, this results in additional requirements and a different failure mode. Both devices must have the ability to tell the time, which is not practical for a USB 2FA token with no battery, for example. And both the server and client must agree on the correct time. If their clocks are skewed, then they will disagree on their current position in the sequence. Servers compensate for clock skew by allowing a few codes either side to also be valid. But like HOTP, they can only go so far before the server must refuse. One advantage of TOTP over HOTP is that correcting for this condition involves ensuring the clocks are correct at both ends; an out-of-band authentication to reset unfortunate users’ secrets is not required. When using a modern smartphone app, for example, the requirement to keep the clock correct isn’t usually a problem since this is typically done automatically at both ends by default.

Note:

It is not recommended to configure U2F/FIDO at the same time as TOTP/HOTP. This combination has not been tested, and using the configuration presented here, TOTP/HOTP would become mandatory for everyone, whether or not they are also using U2F/FIDO.

Install required software

From a terminal prompt, install the google-authenticator PAM module:

```
$ sudo apt update && sudo apt install libpam-google-authenticator
```

Note:

The libpam-google-authenticator package is in Ubuntu’s universe archive component, which receives best-effort community support only.

Configure users

Since public key authentication with TOTP/HOTP 2FA is about to be configured to be mandatory for all users, each user who wishes to continue using SSH must first set up public key authentication and then configure their 2FA keys by running the user setup tool. If this isn't done first, users will not be able to do it later over SSH, since at that point they won't have public key authentication and/or 2FA configured to authenticate with.

Configure users' key-based authentication

To set up key-based authentication, see "SSH Keys" above. Once this is done, it can be tested independently of subsequent 2FA configuration. At this stage, user authentication should work with keys only, requiring the supply of the private key passphrase only if it was configured. If configured correctly, the user should not be prompted for their password.

Configure users' TOTP/HOTP 2FA secrets

Each user needs to run the setup tool to configure 2FA. This will ask some questions, generate a key, and display a QR code for the user to import the secret into their smartphone app, such as the Google Authenticator app on Android. The tool creates the file `~/.google-authenticator`, which contains a shared secret, emergency passcodes and per-user configuration.

As a user who needs 2FA configured, from a terminal prompt run the following command:

```
$ google-authenticator
```

Follow the prompts, scanning the QR code into your 2FA app as directed.

It's important to plan for the eventuality that the 2FA device gets lost or damaged. Will this lock the user out of their account? In mitigation, it's worthwhile for each user to consider doing one or more of the following:

- Use the 2FA device's backup or cloud sync facility if it has one.
- Write down the backup codes printed by the setup tool.
- Take a photo of the QR code.
- (TOTP only) Scan the QR code on multiple 2FA devices. This only works for TOTP, since multiple HOTP 2FA devices will not be able to stay in sync.
- Ensure that the user has a different authentication path to be able to rerun the setup tool if required.

Of course, any of these backup steps also negate any benefit of 2FA should someone else get access to the backup, so the steps taken to protect any backup should be considered carefully.

Configure the SSH server

Once all users are configured, configure `sshd` itself by creating a new configuration file in `/etc/ssh/sshd_config.d/` with the following directives:

```
KbdInteractiveAuthentication yes
PasswordAuthentication no
AuthenticationMethods publickey,keyboard-interactive
```

Depending on your installation, some of these settings may be configured already, but not necessarily with the values required for this configuration. Check for and adjust existing occurrences of these configuration directives, or add new ones, as required.

Note:

On Ubuntu 20.04 LTS (Focal Fossa) and earlier, use `ChallengeResponseAuthentication yes` instead of `KbdInteractiveAuthentication yes`.

Restart the `ssh` service to pick up configuration changes:

```
$ sudo systemctl try-reload-or-restart ssh
```

Edit `/etc/pam.d/sshd` and replace the line:

```
@include common-auth
```

with:

```
auth required pam_google_authenticator.so
```

Changes to PAM configuration have immediate effect, and no separate reloading command is required.

Log in using 2FA

Now when you log in using SSH, in addition to the normal public key authentication, you will be prompted for your TOTP or HOTP code:

```
$ ssh ubuntu.server

Enter passphrase for key 'id_rsa':
(ubuntu@ubuntu.server) Verification code:
Welcome to Ubuntu (...)
ubuntu@ubuntu.server:~$
```

Special cases

On Ubuntu, the following settings are default in `/etc/ssh/sshd_config`, but if you have overridden them, note that they are required for this configuration to work correctly and must be restored as follows:

```
UsePAM yes
PubkeyAuthentication yes
```

Remember to run `sudo systemctl try-reload-or-restart ssh` for any changes made to `sshd` configuration to take effect.

Further reading

- [Wikipedia on TOTP](#)¹⁶⁰
- [Wikipedia on HOTP](#)¹⁶¹

Two factor authentication with U2F/FIDO

OpenSSH 8.2 has added [support for U2F/FIDO hardware authentication devices](#)¹⁶². These devices are used to provide an extra layer of security on top of the existing key-based authentication, as the hardware token needs to be present to finish the authentication.

It's very simple to use and setup. The only extra step is to generate a new keypair that can be used with the hardware device. For that, there are two key types that can be used: `ecdsa-sk` and `ed25519-sk`. The former has broader hardware support, while the latter might need a more recent device.

Once the keypair is generated, it can be used as you would normally use any other type of key in OpenSSH. The only requirement is that in order to use the private key, the U2F device has to be present on the host.

Example with U2F

For example, plug the U2F device in and generate a keypair to use with it:

```
$ ssh-keygen -t ecdsa-sk
```

¹⁶⁰ https://en.wikipedia.org/wiki/Time-based_one-time_password

¹⁶¹ https://en.wikipedia.org/wiki/HMAC-based_one-time_password

¹⁶² <https://www.openssh.org/txt/release-8.2>

```
Generating public/private ecdsa-sk key pair.
You may need to touch your authenticator to authorize key generation. <-- touch
device
Enter file in which to save the key (/home/ubuntu/.ssh/id_ecdsa_sk):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/ubuntu/.ssh/id_ecdsa_sk
Your public key has been saved in /home/ubuntu/.ssh/id_ecdsa_sk.pub
The key fingerprint is:
SHA256:V9PQ1MqaU8FODXdHqDiH9Mxb8XK3o5aVYDQLVL9IFRo ubuntu@focal
```

Now transfer the public part to the server to `~/.ssh/authorized_keys` and you are ready to go:

```
$ ssh -i .ssh/id_ecdsa_sk ubuntu@focal.server

Confirm user presence for key ECDSA-SK
SHA256:V9PQ1MqaU8FODXdHqDiH9Mxb8XK3o5aVYDQLVL9IFRo <-- touch device
Welcome to Ubuntu Focal Fossa (GNU/Linux 5.4.0-21-generic x86_64)
(...)
ubuntu@focal.server:~$
```

FIDO2 resident keys

FIDO2 private keys consist of two parts: a **key handle** part, stored in the private key file on disk, and a **per-device key**, which is unique to each FIDO2 token and cannot be exported from the token hardware. These are combined by the hardware at authentication time to derive the real key, which is used to sign authentication challenges.

For tokens that are required to move between computers, it can be cumbersome to have to move the private key file first. To avoid this, tokens implementing the newer FIDO2 standard support **resident keys**, where it is possible to retrieve the key handle part of the key from the hardware.

Using resident keys increases the likelihood of an attacker being able to use a stolen token device. For this reason, tokens normally enforce PIN authentication before allowing the download of keys, and users should set a PIN on their tokens before creating any resident keys. This is done via the hardware token management software.

OpenSSH allows resident keys to be generated using the `ssh-keygen` flag `-O resident` at key generation time:

```
$ ssh-keygen -t ecdsa-sk -O resident -O application=ssh:mykeyname
```

```
Generating public/private ecdsa-sk key pair.  
You may need to touch your authenticator to authorize key generation.  
Enter PIN for authenticator:  
Enter file in which to save the key (/home/ubuntu/.ssh/id_ecdsa_sk): mytoken  
Enter passphrase (empty for no passphrase):  
Enter same passphrase again:  
Your identification has been saved in mytoken  
(...)
```

This will produce a public/private key pair as usual, but it will be possible to retrieve the private key part (the key handle) from the token later. This is done by running:

```
$ ssh-keygen -K  
  
Enter PIN for authenticator:  
You may need to touch your authenticator to authorize key download.  
Enter passphrase (empty for no passphrase):  
Enter same passphrase again:  
Saved ECDSA-SK key ssh:mytoken to id_ecdsa_sk_rk_mytoken
```

It will use the part after ssh: from the application parameter from before as part of the key filenames:

```
$ ls id_ecdsa_sk_rk_mytoken*  
  
-rw----- 1 ubuntu ubuntu 598 out  4 18:49 id_ecdsa_sk_rk_mytoken  
-rw-r--r-- 1 ubuntu ubuntu 228 out  4 18:49 id_ecdsa_sk_rk_mytoken.pub
```

If you set a passphrase when extracting the keys from the hardware token, and later use these keys, you will be prompted for both the key passphrase *and* the hardware key PIN. You will also have to touch the token:

```
$ ssh -i ./id_ecdsa_sk_rk_mytoken ubuntu@focal.server  
  
Enter passphrase for key './id_ecdsa_sk_rk_mytoken':  
Confirm user presence for key ECDSA-SK  
SHA256:t+l26IgTXeURY6e36wtrq7wVYJtDVZr0+iuobs1CvVQ  
User presence confirmed  
(...)
```

It is also possible to download and add resident keys directly to ssh-agent by running

```
$ ssh-add -K
```

In this case, no file is written and the public key can be printed by running `ssh-add -L`.

Note:

If you used the `-O verify-required` option when generating the keys, or if that option is set on the SSH server via the `/etc/ssh/sshd_config` setting `PubkeyAuthOptions verify-required`, then using the agent won't work (in Ubuntu 22.04 LTS).

Further reading

- [OpenSSH 8.2 release notes](#)¹⁶³
- [Yubikey documentation for OpenSSH FIDO/FIDO2 usage](#)¹⁶⁴

OpenSSH with GSSAPI/Kerberos authentication

OpenSSH can be configured to use [GSSAPI/Kerberos](#) authentication, which allows users to authenticate using their Kerberos tickets. This is particularly useful in environments where Kerberos is already in use, such as in many enterprise networks.

This guide assumes you have a working Kerberos realm, and does not cover setting one up. For background on Kerberos concepts, see [Introduction to Kerberos](#) (page 872); to deploy a realm on Ubuntu, see [the Kerberos how-to guides](#) (page 182).

Which OpenSSH package to use

The packages to install depend on the Ubuntu release in use:

Ubuntu Resolute 26.04 LTS and earlier

The GSSAPI/Kerberos support is included in the main OpenSSH packages:

- `openssh-server` on the server
- `openssh-client` on the client

Ubuntu Stonking 26.10 and later

The GSSAPI/Kerberos support is split into separate packages, so use these:

- `openssh-server-gssapi` on the server
- `openssh-client-gssapi` on the client

To connect to a remote host using GSSAPI/Kerberos authentication, both the client and the server need to support it.

¹⁶³ <https://www.openssh.org/txt/release-8.2>

¹⁶⁴ https://developers.yubico.com/SSH/Securing_SSH_with_FIDO2.html

Configure OpenSSH on the server

GSSAPI/Kerberos support is available on the server, but not enabled by default. To enable it, create a configuration file in `/etc/ssh/sshd_config.d/`, for example `/etc/ssh/sshd_config.d/gssapi.conf`, with the following line:

```
GSSAPIAuthentication yes
```

For more information about configuration snippets, see [Configure OpenSSH](#) (page 235). The [sshd_config\(5\)](#)¹⁶⁵ manual page covers other GSSAPI/Kerberos configuration options.

Restart the OpenSSH server to apply the changes:

```
$ sudo systemctl restart ssh.service
```

In order to participate in a Kerberos realm, the OpenSSH server needs to have a so-called service principal in the Kerberos database. The service principal for OpenSSH is of the form `host/<hostname>@<REALM>`, where `<hostname>` is the fully qualified domain name of the server and `<REALM>` is the Kerberos realm.

The service principal needs to be created in the Kerberos database and a keytab file needs to be generated for it. The keytab file should be placed in `/etc/krb5.keytab` on the OpenSSH server.

Note:

For more information on service principals, consult [Kerberos service principals](#) (page 186).

To create the service principal and extract its key into the keytab, use the [kadmin\(1\)](#)¹⁶⁶ tool on the OpenSSH server. You need a principal with administrative privileges in the Kerberos realm for this. The `kadmin` tool is in the `krb5-user` package:

```
$ sudo apt install krb5-user
```

During installation, it will ask you for the Kerberos realm, and the KDC (Key Distribution Center) and Admin server hostnames. This guide uses the `EXAMPLE.COM` realm, and `server.example.com` as the hostname of the OpenSSH server.

Create the service principal, and extract its key into `/etc/krb5.keytab`. Run `kadmin` with `sudo`, because writing to `/etc/krb5.keytab` requires root privileges. Each command asks for the password of the administrative principal:

```
$ sudo kadmin -p ubuntu/admin@EXAMPLE.COM -q "addprinc -randkey  
host/server.example.com@EXAMPLE.COM"
```

```
$ sudo kadmin -p ubuntu/admin@EXAMPLE.COM -q "ktadd  
host/server.example.com@EXAMPLE.COM"
```

¹⁶⁵ https://manpages.ubuntu.com/manpages/resolute/man5/sshd_config.5.html

¹⁶⁶ <https://manpages.ubuntu.com/manpages/resolute/man1/kadmin.mit.1.html>

Note:

user/admin is normally how administrative principals are named. You would have the normal user@REALM principal for everyday tasks, and the admin “instance” (denoted by /admin) is recognized as having extra privileges. See [Configure the Kerberos server](#) (page 184) for more information.

The contents of the keytab can be verified with the `klist(1)`¹⁶⁷ command:

```
$ sudo klist -ke

Keytab name: FILE:/etc/krb5.keytab
KVNO Principal
-----
 2 host/server.example.com@EXAMPLE.COM (aes256-cts-hmac-sha384-192)
 2 host/server.example.com@EXAMPLE.COM (aes256-cts-hmac-sha1-96)
 2 host/server.example.com@EXAMPLE.COM (aes128-cts-hmac-sha1-96)
```

The server is now ready to accept GSSAPI/Kerberos authentication from clients that have valid Kerberos tickets.

Configure OpenSSH on the client

In Ubuntu, the `openssh-client` (or `openssh-client-gssapi`) package already defaults to allowing GSSAPI/Kerberos authentication, so no additional configuration is needed on the client side. However, you may want to verify that the following line is present in `/etc/ssh/ssh_config`, or in a configuration file in `/etc/ssh/ssh_config.d/`:

```
GSSAPIAuthentication yes
```

This can also be configured on a per-user and per-target-host basis in `~/.ssh/config`.

Like with the server options, there are multiple configuration options related to GSSAPI/Kerberos for the client. Consult the `ssh_config(5)`¹⁶⁸ manual page for more information.

Since the client needs to have a valid Kerberos ticket in order to authenticate with GSSAPI/Kerberos, you will need to obtain a ticket with `kinit(1)`¹⁶⁹ before connecting to the server. Install the `krb5-user` package on the client, answering the same questions as on the server:

```
$ sudo apt install krb5-user
```

Note:

Some Ubuntu deployments may be configured to obtain a Kerberos ticket automatically at login, in which case you may not need to run `kinit` manually. But it’s still useful to have the `krb5-user` package installed, as it provides the `klist` command to verify that you have a valid ticket.

¹⁶⁷ <https://manpages.ubuntu.com/manpages/resolute/man1/klist.mit.1.html>

¹⁶⁸ https://manpages.ubuntu.com/manpages/resolute/man5/ssh_config.5.html

¹⁶⁹ <https://manpages.ubuntu.com/manpages/resolute/man1/kinit.mit.1.html>

First obtain the **TGT** (Ticket Granting Ticket) for your user principal, if you don't have it already. This example uses an ubuntu principal:

```
ubuntu@client:~$ kinit  
  
Password for ubuntu@EXAMPLE.COM:
```

Confirm you have the TGT:

```
ubuntu@client:~$ klist  
  
Ticket cache: FILE:/tmp/krb5cc_1000  
Default principal: ubuntu@EXAMPLE.COM  
  
Valid starting    Expires          Service principal  
09/29/26 21:29:39 09/30/26 07:29:39 krbtgt/EXAMPLE.COM@EXAMPLE.COM
```

And now ssh to the GSSAPI/Kerberos-enabled OpenSSH server:

```
ubuntu@client:~$ ssh server.example.com  
  
Welcome to Ubuntu 26.04.1 LTS  
...  
ubuntu@server:~$
```

The server administrator can check the `/var/log/auth.log` file to verify that the user authenticated with GSSAPI/Kerberos:

```
sshd-session[954]: Accepted gssapi-with-mic for ubuntu from 10.0.10.157 port 55992 ssh2:  
ubuntu@EXAMPLE.COM
```

If you log out and run `klist` on the client, you will see that you now have a service ticket for the OpenSSH server:

```
ubuntu@client:~$ klist  
  
Ticket cache: FILE:/tmp/krb5cc_1000  
Default principal: ubuntu@EXAMPLE.COM  
  
Valid starting    Expires          Service principal  
09/29/26 21:29:39 09/30/26 07:29:39 krbtgt/EXAMPLE.COM@EXAMPLE.COM  
    renew until 09/30/26 21:29:37  
09/29/26 21:29:59 09/30/26 07:29:39 host/server.example.com@  
    renew until 09/30/26 21:29:37  
Ticket server: host/server.example.com@EXAMPLE.COM
```

Forwarding Kerberos tickets

It is possible to forward a Kerberos ticket to the remote server when logging in via SSH. This allows the user to use their Kerberos credentials on the remote server, for example to access other services that require Kerberos authentication.

This is controlled by the client option called `GSSAPIDelegateCredentials`. To enable it, add the following line to your `~/.ssh/config` file, or have an administrator add it to the system-wide `/etc/ssh/ssh_config` file:

```
GSSAPIDelegateCredentials yes
```

With this in place, the next time you ssh into the server, you will notice that you also have a Kerberos ticket there:

```
ubuntu@client:~$ ssh server.example.com
```

```
Welcome to Ubuntu 26.04.1 LTS
...
```

```
ubuntu@server:~$ klist
```

```
Ticket cache: FILE:/tmp/krb5cc_1000_Qvc5QcHuit
Default principal: ubuntu@EXAMPLE.COM
```

Valid starting	Expires	Service principal
09/29/26 21:47:13	09/30/26 07:29:39	krbtgt/EXAMPLE.COM@EXAMPLE.COM
renew until 09/30/26 21:29:37		

Note:

Tickets can only be forwarded if allowed by the Kerberos server. That is the default, but it may be disallowed in your realm. You can check if your ticket is forwardable by running `klist -f` on the client. It should have the “F” flag set if it’s forwardable.

The ticket cache file on the server is again created in `/tmp` by default, but this time with an extra random suffix appended to the filename to avoid collisions if the same user forwards another ticket from another client.

Ticket cache location

The ccache type and location are normally controlled by the `default_ccache_name` option in the `[libdefaults]` section of `/etc/krb5.conf` (see [krb5.conf\(5\)](https://manpages.ubuntu.com/manpages/resolute/man5/krb5.conf.5.html)¹⁷⁰). This option is read by the Kerberos library, and is treated as a library-provided default for tools linked with it.

For example, to store tickets in the kernel keyring instead of a file in `/tmp`, add or edit the following line in `/etc/krb5.conf` on the client:

¹⁷⁰ <https://manpages.ubuntu.com/manpages/resolute/man5/krb5.conf.5.html>

```
default_ccache_name = KEYRING:persistent:%{uid}
```

Running kinit and then klist on the client confirms the change:

```
ubuntu@client:~$ kinit
```

```
Password for ubuntu@EXAMPLE.COM:
```

```
ubuntu@client:~$ klist
```

```
Ticket cache: KEYRING:persistent:1000:1000
```

```
Default principal: ubuntu@EXAMPLE.COM
```

```
Valid starting    Expires          Service principal
09/30/26 13:48:24 09/30/26 23:48:24 krbtgt/EXAMPLE.COM@EXAMPLE.COM
        renew until 10/01/26 13:48:21
```

For forwarded tickets, it's the OpenSSH server that creates the ccache, not kinit, and the `/etc/krb5.conf` that matters is the one on the server. The same setting on the server doesn't always take effect, though: it depends on the Ubuntu release the server is running.

OpenSSH in Ubuntu Resolute 26.04 LTS and earlier ignores the `default_ccache_name` setting in `/etc/krb5.conf` on the server, so the forwarded ticket will still be stored in `/tmp` with a random suffix.

But starting with Ubuntu Stonking 26.10, the OpenSSH server will follow the `default_ccache_name` setting in `/etc/krb5.conf` on the server by default for forwarded tickets. This OpenSSH behavior is controlled by the `KerberosUniqueCCache` option in `/etc/ssh/sshd_config`. If this option is set to `no` (the default), the server will use the `default_ccache_name` setting in `/etc/krb5.conf` to determine the location and type of the ticket cache for forwarded tickets. If it is changed to `yes`, the server will always use `/tmp` with a random suffix as in previous Ubuntu releases.

If there is no `default_ccache_name` setting in `/etc/krb5.conf` on the server, the behavior is unchanged from Ubuntu Resolute 26.04 LTS and earlier.

The following table summarizes where the OpenSSH server stores forwarded tickets:

Ubuntu release	<code>default_ccache_name</code> on the server	<code>KerberosUniqueCCache</code> on the server	Forwarded ticket ccache
$\leq$ 26.04 LTS	Ignored by OpenSSH	Not available	FILE:/tmp/krb5cc_<uid>_<random>
$\geq$ 26.10	Not set	no (default) or yes	FILE:/tmp/krb5cc_<uid>_<random>
$\geq$ 26.10	Set	yes	FILE:/tmp/krb5cc_<uid>_<random>
$\geq$ 26.10	Set	no (default)	As specified by <code>default_ccache_name</code>

Here is an example forwarding a Kerberos ticket to an Ubuntu Stonking 26.10 server with the `default_ccache_name` set to `KEYRING:persistent:%{uid}` in `/etc/krb5.conf` on the server:

```
ubuntu@client:~$ ssh stonking-server.example.com
```

```
Welcome to Ubuntu Stonking Stingray (...)
```

```
...
```

```
ubuntu@stonking-server:~$ klist
```

```
Ticket cache: KEYRING:persistent:1000:krb_ccache_MU8uJl4
```

```
Default principal: ubuntu@EXAMPLE.COM
```

Valid starting	Expires	Service principal
09/30/26 14:00:40	09/30/26 23:48:24	krbtgt/EXAMPLE.COM@EXAMPLE.COM
renew until 10/01/26 13:48:21		

```
ubuntu@stonking-server:~$ grep default_ccache_name /etc/krb5.conf
```

```
default_ccache_name = KEYRING:persistent:%{uid}
```

See also

- [Introduction to Kerberos](#) (page 872)
- [How to deploy Kerberos on Ubuntu](#) (page 182)
- [Configuring Kerberos service principals](#) (page 186)

Obtain TLS certificates

Transport Layer Security (TLS) certificates authenticate your server's identity and enable encrypted connections. There are two main approaches:

- **Let's Encrypt** (recommended for internet-facing servers): a free, automated Certificate Authority (CA) whose certificates are trusted by all major browsers and clients.
- **Manual CA** (for internal networks or air-gapped environments): run your own CA and sign certificates yourself; requires distributing the CA certificate to all clients.

Use Let's Encrypt

Let's Encrypt¹⁷¹ is operated by the Internet Security Research Group (ISRG). It issues certificates via the Automated Certificate Management Environment (ACME) protocol. Certificates are valid for 90 days and can be renewed automatically.

¹⁷¹ <https://letsencrypt.org/>

Prerequisites

- A registered domain name with DNS records pointing to the server.
- Port 80 open in your firewall. The default [HTTP-01 challenge](#)¹⁷² verifies domain ownership by making an HTTP request to your server on port 80, so it must be reachable from the internet during certificate issuance and renewal.

Install Certbot

[Certbot](#)¹⁷³ is the recommended ACME client for Let's Encrypt. Install it from the snap store:

```
$ sudo snap install --classic certbot
```

Obtain a certificate

Certbot supports several modes depending on whether a web server is already running.

With an nginx server running:

Certbot's [nginx plugin](#)¹⁷⁴ finds the server block matching the specified domains in your nginx configuration (typically in `/etc/nginx/sites-enabled/`) and adds the necessary TLS directives in place, then reloads nginx:

```
$ sudo certbot --nginx -d example.com -d www.example.com
```

With an Apache server running:

Certbot's [Apache plugin](#)¹⁷⁵ finds the VirtualHost block matching the specified domains in your Apache configuration (typically in `/etc/apache2/sites-enabled/`) and adds the necessary TLS directives in place, then reloads Apache:

```
$ sudo certbot --apache -d example.com -d www.example.com
```

Without a web server (standalone mode):

Certbot starts a temporary web server on port 80 to complete the ACME challenge. Stop any service already listening on port 80 before running this.

```
$ sudo certbot certonly --standalone -d example.com -d www.example.com
```

With a web server managing the document root (webroot mode):

If you want to obtain a certificate without modifying your web server configuration, specify the directory the web server serves:

¹⁷² <https://letsencrypt.org/docs/challenge-types/#http-01-challenge>

¹⁷³ <https://certbot.eff.org/>

¹⁷⁴ <https://certbot.eff.org/docs/using.html#nginx>

¹⁷⁵ <https://certbot.eff.org/docs/using.html#apache>

```
$ sudo certbot certonly --webroot -w /var/www/html -d example.com
```

Multiple domains:

Each `-d` flag adds a domain as a Subject Alternative Name (SAN), so a single certificate can cover multiple domains and subdomains. If a certificate for any of the listed domains already exists, Certbot expands it to include all specified domains. Pass the complete desired domain list — both existing and new — in a single command:

```
$ sudo certbot --nginx -d example.com -d www.example.com -d mail.example.com
```

Other challenge methods

The modes above all use the [HTTP-01 challenge](#)¹⁷⁶ and require port 80. For servers not reachable on port 80, or when you need wildcard certificates (e.g., `*.example.com`), you can use the [DNS-01 challenge](#)¹⁷⁷ instead: it verifies domain ownership via a DNS TXT record and has no inbound port requirements. Certbot supports DNS-01 via [DNS plugins](#)¹⁷⁸ for common providers (Cloudflare, Route 53, etc.).

For environments where running an ACME client on each server is not practical — such as air-gapped networks or large fleets — an [ACME proxy server](#)¹⁷⁹ (such as `step-ca`) can act as an internal CA that forwards requests to Let's Encrypt, or a centralised ACME client such as [acme.sh](#)¹⁸⁰ or [lego](#)¹⁸¹ can manage certificate issuance and distribution centrally.

Locate the issued certificate files

After a successful run, Certbot writes the certificate files to `/etc/letsencrypt/live/<domain>/`:

File	Contents
<code>fullchain.pem</code>	Server certificate plus any intermediate certificates (use this in most service configurations)
<code>privkey.pem</code>	Private key
<code>cert.pem</code>	Server certificate only
<code>chain.pem</code>	Intermediate certificates only

Point your service configuration at the files in this directory. For example in nginx:

¹⁷⁶ <https://letsencrypt.org/docs/challenge-types/#http-01-challenge>

¹⁷⁷ <https://letsencrypt.org/docs/challenge-types/#dns-01-challenge>

¹⁷⁸ <https://certbot.eff.org/docs/using.html#dns-plugins>

¹⁷⁹ <https://smallstep.com/docs/step-ca/>

¹⁸⁰ <https://acme.sh>

¹⁸¹ <https://go-acme.github.io/lego/>

```
ssl_certificate      /etc/letsencrypt/live/example.com/fullchain.pem;  
ssl_certificate_key  /etc/letsencrypt/live/example.com/privkey.pem;
```

Automatic renewal

Certbot installs a systemd timer that attempts renewal twice a day. Check that it is active:

```
$ sudo systemctl status snap.certbot.renew.timer
```

Test the renewal process without making any changes:

```
$ sudo certbot renew --dry-run
```

Services configured using the Certbot nginx or Apache plugins are reloaded automatically after a successful renewal. For other services, add a renewal hook. Create `/etc/letsencrypt/renewal-hooks/deploy/reload-service.sh`:

```
#!/bin/sh  
systemctl reload <your-service>
```

Then make it executable:

```
$ sudo chmod +x /etc/letsencrypt/renewal-hooks/deploy/reload-service.sh
```

Replace `<your-service>` with the name of the service to reload after renewal (e.g., postfix, dovecot, slapd).

Set up a manual Certificate Authority

If the services on your network require more than a few self-signed certificates it may be worth the additional effort to set up your own internal Certification Authority (CA). Using certificates signed by your own CA allows the various services using the certificates to easily trust other services using certificates issued from the same CA.

Note:

Clients that connect to your services must trust your CA. You will need to install your CA certificate in the trust store of every client, or distribute it to users. See [Install a root CA certificate in the trust store](#) (page 259) for how to add a CA certificate to Ubuntu's trust store.

Create the CA directory structure

First, create the directories to hold the CA certificate and related files:

```
$ sudo mkdir /etc/ssl/CA
```

```
$ sudo mkdir /etc/ssl/newcerts
```

The CA needs a few additional files to operate: one to keep track of the last serial number used by the CA (each certificate must have a unique serial number), and another file to record which certificates have been issued:

```
$ sudo sh -c "echo '01' > /etc/ssl/CA/serial"
```

```
$ sudo touch /etc/ssl/CA/index.txt
```

Configure OpenSSL

The third file needed is a CA configuration file. Though not strictly necessary, it is convenient when issuing multiple certificates. Edit `/etc/ssl/openssl.cnf`, and in the `[ CA_default ]` section, change:

```
dir           = /etc/ssl           # Where everything is kept
database      = $dir/CA/index.txt  # database index file.
certificate    = $dir/certs/cacert.pem # The CA certificate
serial        = $dir/CA/serial     # The current serial number
private_key   = $dir/private/cakey.pem # The private key
```

Create the self-signed root certificate

```
$ openssl req -new -x509 -extensions v3_ca -keyout cakey.pem -out cacert.pem
-days 3650
```

You will then be asked to enter the details about the certificate. Next, install the root certificate and key:

```
$ sudo mv cakey.pem /etc/ssl/private/
```

```
$ sudo mv cacert.pem /etc/ssl/certs/
```

Generate a Certificate Signing Request (CSR)

You are now ready to start signing certificates. The first item needed is a Certificate Signing Request (CSR). If the certificate will be used by service daemons, such as Apache, Postfix, Dovecot, etc., a key without a passphrase is often appropriate. Not having a passphrase allows the services to start without manual intervention, usually the preferred way to start a daemon.

Warning:

Running your secure service without a passphrase is convenient but insecure – a compromise of the key means a compromise of the server as well.

To generate the keys for the Certificate Signing Request (CSR) run the following command from a terminal prompt:

```
$ openssl genrsa -des3 -out server.key 2048

Generating RSA private key, 2048 bit long modulus
.....+++++
.....+++++
e is 65537 (0x10001)
Enter pass phrase for server.key:
```

You can now enter your passphrase. For best security, it should contain **at least** eight characters. The minimum length when specifying `-des3` is four characters. As a best practice it should also include numbers and/or punctuation and not be a word in a dictionary. Also remember that your passphrase is case-sensitive.

Re-type the passphrase to verify. Once you have re-typed it correctly, the server key is generated and stored in the `server.key` file.

Now create the insecure key, the one without a passphrase, and shuffle the key names:

```
$ openssl rsa -in server.key -out server.key.insecure > mv server.key
server.key.secure > mv server.key.insecure server.key
```

The insecure key is now named `server.key`, and you can use this file to generate the CSR without a passphrase.

To create the CSR, run the following command at a terminal prompt:

```
$ openssl req -new -key server.key -out server.csr
```

It will prompt you to enter the passphrase. If you enter the correct passphrase, it will prompt you to enter 'Company Name', 'Site Name', 'Email ID', etc. Once you enter all these details, your CSR will be created and it will be stored in the `server.csr` file.

Sign the certificate

Enter the following to generate a certificate signed by the CA:

```
$ sudo openssl ca -in server.csr -config /etc/ssl/openssl.cnf
```

After entering the password for the CA key, you will be prompted to sign the certificate, and again to commit the new certificate. You should then see a somewhat large amount of output related to the certificate creation.

There should now be a new file, `/etc/ssl/newcerts/01.pem`, containing the same output. Copy and paste everything beginning with the `-----BEGIN CERTIFICATE-----` line and continuing through to the `-----END CERTIFICATE-----` lines to a file named after the *hostname* of the server where the certificate will be installed. For example `mail.example.com.crt` is a descriptive name.

Subsequent certificates will be named `02.pem`, `03.pem`, etc.

Note:

Replace `mail.example.com.crt` with your own descriptive name.

Install the certificate

Copy the new certificate to the host that needs it, and configure the appropriate applications to use it. The default location to install certificates is `/etc/ssl/certs`. This enables multiple services to use the same certificate without overly complicated file permissions.

```
$ sudo cp server.key /etc/ssl/private/
```

```
$ sudo cp mail.example.com.crt /etc/ssl/certs/
```

For applications that can be configured to use a CA certificate, you should also copy the `/etc/ssl/certs/cacert.pem` file to the `/etc/ssl/certs/` directory on each server that needs to trust certificates issued by your CA.

Further reading

- [Let's Encrypt documentation](https://letsencrypt.org/docs/)¹⁸²
- [Certbot documentation](https://certbot.eff.org/docs/)¹⁸³
- The Wikipedia article on [Transport Layer Security](https://en.wikipedia.org/wiki/Transport_Layer_Security)¹⁸⁴ provides background on TLS and its history.
- [About certificates](#) (page 927) explains the concepts behind public-key cryptography and certificate types.

¹⁸² <https://letsencrypt.org/docs/>

¹⁸³ <https://certbot.eff.org/docs/>

¹⁸⁴ https://en.wikipedia.org/wiki/Transport_Layer_Security

Install a root CA certificate in the trust store

Enterprise environments sometimes have a local Certificate Authority (CA) that issues certificates for use within the organization. For an Ubuntu server to be functional, and to trust the hosts in this environment, this CA must be installed in Ubuntu's trust store.

Certificate formats

There are two encoding formats for certificates:

- **Privacy Enhanced Mail (PEM):** These are human-readable and in Base64-encoded **ASCII** format.
- **Distinguished Encoding Rules (DER):** These are encoded in a more compact **binary** format, and not human readable.

To install a certificate in the trust store it must be in PEM format. A PEM certificate starts with the line `-----BEGIN CERTIFICATE-----`. If you see this, you're ready to install. If not, it is probably a DER certificate and needs to be converted before you can install it in the trust store.

Install a PEM-format certificate

Assuming your PEM-formatted root CA certificate is in `local-ca.crt`, run the following commands to install it:

- Generate a random certificate

Answer the questions asked after executing the command. When asked to input a commonName (CN), it should match the hostname of the server.

```
$ openssl req -x509 -new -nodes -keyout local-ca.key -out local-ca.crt
```

- Install the CA certificate package

```
$ sudo apt-get install -y ca-certificates
```

- Copy your certificate to the local CA certificates directory

```
$ sudo cp local-ca.crt /usr/local/share/ca-certificates
```

- Add the certificate to your trust store

```
$ sudo update-ca-certificates
```

- Verify that your certificate is in PEM format

```
$ sudo ls /etc/ssl/certs/ | grep local-ca  
local-ca.pem
```

- You can also verify that your certificate is available in the trust store by selecting a few texts from your certificate and comparing them with the certificate at the end of the trust store file to see if it matches yours.

```
$ sudo cat local-ca.crt
```

```
...  
L4z0d3b41xJtYldofPve  
-----END CERTIFICATE-----
```

```
$ sudo cat /etc/ssl/certs/ca-certificates.crt | grep L4z0d3b41xJtYldofPve  
  
L4z0d3b41xJtYldofPve
```

Note:

It is important that the certificate file has the .crt extension, otherwise it will not be processed.

After this point, you can use tools like `curl` and `wget` to connect to local sites.

Note:

This procedure updates the system trust store and affects applications that rely on the host system's OpenSSL configuration, such as `curl` and `wget`. Snap applications, including snap-packaged browsers, are unlikely to automatically trust certificates installed in the system trust store due to snap confinement.

Uninstall a PEM-format certificate

- Verify that the certificate you'd like to uninstall exists.

```
$ sudo ls /usr/local/share/ca-certificates/local-ca.crt  
  
/usr/local/share/ca-certificates/local-ca.crt
```

- Delete the certificate

```
$ sudo rm /usr/local/share/ca-certificates/local-ca.crt
```

- The certificate still exists in the trust store.

```
$ sudo cat /etc/ssl/certs/ca-certificates.crt | grep  
L4z0d3b41xJtYldofPve  
  
L4z0d3b41xJtYldofPve
```

- Update the trust store

```
$ sudo update-ca-certificates --fresh
```

- Verify that the certificate has been uninstalled

```
$ sudo cat /etc/ssl/certs/ca-certificates.crt | grep  
L4z0d3b41xJtYldofPve
```

Convert from DER to PEM format

You can convert a DER-formatted certificate called `local-ca.der` to PEM form like this:

```
$ sudo openssl x509 -inform der -outform pem -in local-ca.der -out  
local-ca.crt`
```

The CA trust store location

The CA trust store (as generated by `update-ca-certificates`) is available at the following locations:

- As a single file (PEM bundle) in `/etc/ssl/certs/ca-certificates.crt`
- As an OpenSSL-compatible certificate directory in `/etc/ssl/certs`

Virtual Private Network (VPN)

VPNs are commonly used to provide encrypted, secure access to a network. Two of the most popular choices in Ubuntu are OpenVPN and WireGuard VPN.

- [OpenVPN](#) (page 261) is a well-established option that supports many platforms besides Linux
- [WireGuard VPN](#) (page 274) is a modern and performant option that removes a lot of the complexity from configuring a VPN

How to install and use OpenVPN

OpenVPN is a flexible, reliable and secure Virtual Private Networking (VPN) solution. It belongs to the family of SSL/TLS VPN stacks (different from IPsec VPNs). This chapter will show how to install and configure OpenVPN to create a VPN.

Install the server

To install OpenVPN, run the following command in your terminal:

```
$ sudo apt install openvpn easy-rsa
```

Set up the Public Key Infrastructure (PKI)

If you want more than just pre-shared keys, OpenVPN makes it easy to set up a Public Key Infrastructure (PKI) to use SSL/TLS certificates for authentication and key exchange between the VPN server and clients.

OpenVPN can be used in a routed or bridged VPN mode and can be configured to use either UDP or TCP. The port number can be configured as well, but port 1194 is the official one; this single port is used for all communication. [VPN client implementations](#) (page 936) are available for almost anything including all Linux distributions, macOS, Windows and OpenWRT-based WLAN routers.

The first step in building an OpenVPN configuration is to establish a PKI, which consists of:

- A separate certificate (also known as a public key) and private key for the server and each client
- A primary Certificate Authority (CA) certificate and key, used to sign the server and client certificates

OpenVPN supports bi-directional authentication based on certificates, meaning that the client must authenticate the server certificate and the server must authenticate the client certificate before mutual trust is established.

Both the server and the client will authenticate each other by first verifying that the presented certificate was signed by the primary Certificate Authority (CA), and then by testing information in the now-authenticated certificate header, such as the certificate common name or certificate type (client or server).

Set up the Certificate Authority

OpenVPN uses its own internal Public Key Infrastructure (PKI) where every client and server holds a certificate signed by the same CA. This is different from obtaining a public TLS certificate for a web service; the CA here is private and only used to authenticate VPN participants.

To set up your own CA, and generate certificates and keys for an OpenVPN server with multiple clients, first copy the `easy-rsa` directory to `/etc/openvpn`. This will ensure that any changes to the scripts will not be lost when the package is updated. From a terminal, run:

```
$ sudo make-cadir /etc/openvpn/easy-rsa
```

Note:

You can alternatively edit `/etc/openvpn/easy-rsa/vars` directly, adjusting it to your needs.

As a root user, change to the newly created directory `/etc/openvpn/easy-rsa` and run:

```
$ ./easyrsa init-pki && ./easyrsa build-ca
```

The PEM passphrase set when creating the CA will be asked for every time you need to encrypt the output of a command (such as a private key). The encryption here is important, to avoid printing any private key in plain text.

Create server keys and certificates

Next, we will generate a key pair for the server:

```
$ ./easyrsa gen-req myservername nopass
```

parameters must be generated for the OpenVPN server. The following command will place them in `pki/dh.pem`:

```
$ ./easyrsa gen-dh
```

And finally, create a certificate for the server:

```
$ ./easyrsa sign-req server myservername
```

All certificates and keys have been generated in subdirectories. Common practice is to copy them to `/etc/openvpn/`:

```
$ cp pki/dh.pem pki/ca.crt pki/issued/myservername.crt  
pki/private/myservername.key /etc/openvpn/
```

Simple server configuration

Included with your OpenVPN installation are these (and many more) sample configuration files:

```
/usr/share/doc/openvpn/examples/sample-config-files/client.conf  
/usr/share/doc/openvpn/examples/sample-config-files/server.conf
```

If these files under `/usr/share/doc/*` are not available (for example, the system was provisioned as a minimal install), you can fetch them directly from the Ubuntu package repositories:

- [client.conf](https://git.launchpad.net/ubuntu/+source/openvpn/tree/sample/sample-config-files/client.conf)¹⁸⁵
- [server.conf](https://git.launchpad.net/ubuntu/+source/openvpn/tree/sample/sample-config-files/server.conf)¹⁸⁶

Start by copying the example server configuration to `/etc/openvpn/myserver.conf`:

¹⁸⁵ <https://git.launchpad.net/ubuntu/+source/openvpn/tree/sample/sample-config-files/client.conf>

¹⁸⁶ <https://git.launchpad.net/ubuntu/+source/openvpn/tree/sample/sample-config-files/server.conf>


```
$ sudo cp /usr/share/doc/openvpn/examples/sample-config-files/server.conf
/etc/openvpn/myserver.conf
```

Note:

In Ubuntu 20.04 LTS or older, the file is compressed, so do this instead:

```
$ sudo cp
/usr/share/doc/openvpn/examples/sample-config-files/server.conf.gz
/etc/openvpn/myserver.conf.gz
```

```
$ sudo gzip -d /etc/openvpn/myserver.conf.gz
```

Edit `/etc/openvpn/myserver.conf` to make sure the following lines are pointing to the certificates and keys you created in the section above.

```
ca ca.crt
cert myservername.crt
key myservername.key
dh dh.pem
tls-crypt ta.key
```

Note:

The `tls-crypt` option was introduced in OpenVPN 2.4.0. If you are running an Ubuntu Server release older than 18.04 LTS (Bionic Beaver), please use `tls-auth ta.key 0` instead.

Complete this set with a TLS Authentication (TA) key in `/etc/openvpn` for `tls-crypt` like this:

```
$ sudo openvpn --genkey secret ta.key
```

Add a config file to `/etc/sysctl.conf.d` to enable IP forwarding:

```
$ echo "net.ipv4.ip_forward = 1" | sudo tee
/etc/sysctl.d/50-enable-ipv4-forwarding.conf
```

Then apply this file through `sysctl`:

```
$ sudo sysctl -p /etc/sysctl.d/50-enable-ipv4-forwarding.conf
```

This is the minimum you need to configure to get a working OpenVPN server. You can consider studying and tweaking all the default settings we got from the sample `server.conf` file. Now you can start the server.

```
$ sudo systemctl start openvpn@myserver
```

Note:

Be aware that the `systemctl start openvpn` is **not** starting the openvpn you just defined. OpenVPN uses templated systemd jobs, `openvpn@CONFIGFILENAME`. So if, for example, your configuration file is `myserver.conf` your service is called `openvpn@myserver`. You can run all kinds of service and `systemctl` commands like `start/stop/enable/disable/preset` against a templated service like `openvpn@server`.

You will find logging and error messages in the journal. For example, if you started a [templated service](#)¹⁸⁷ `openvpn@server` you can filter for this particular message source with:

```
$ sudo journalctl -u openvpn@myserver -xe
```

The same templated approach works for all of `systemctl`, which shows the current server status:

```
$ sudo systemctl status openvpn@myserver

Loaded: loaded (/usr/lib/systemd/system/openvpn@.service; disabled; preset:
enabled)
Active: active (running) since Wed 2026-01-21 10:38:01 UTC; 37s ago
Invocation: b2a3d68935e7423e8f822bc0c88db764
Docs: man:openvpn(8)
      https://community.openvpn.net/openvpn/wiki/Openvpn24ManPage
      https://community.openvpn.net/openvpn/wiki/HOWTO
Main PID: 10556 (openvpn)
Status: "Initialization Sequence Completed"
Tasks: 1 (limit: 10)
Memory: 1.7M (peak: 1.9M)
CPU: 17ms
CGroup: /system.slice/system-openvpn.slice/openvpn@myserver.service
        └─10556 /usr/sbin/openvpn --daemon ovpn-myserver --status /run/openvpn/
myserver.status 10 --cd /etc/openvpn --script-security 2 --config /etc/openvpn/
myserver.conf --writepid /r>

Jan 21 10:38:01 server ovpn-myserver[10556]: net_iface_up: set tun0 up
Jan 21 10:38:01 server ovpn-myserver[10556]: net_addr_v4_add: 10.8.0.1/24 dev tun0
Jan 21 10:38:01 server ovpn-myserver[10556]: Could not determine IPv4/IPv6 protocol.
Using AF_INET
Jan 21 10:38:01 server ovpn-myserver[10556]: Socket Buffers: R=[212992->212992]
S=[212992->212992]
Jan 21 10:38:01 server ovpn-myserver[10556]: UDPv4 link local (bound): [AF_
INET][undef]:1194
Jan 21 10:38:01 server ovpn-myserver[10556]: UDPv4 link remote: [AF_UNSPEC]
Jan 21 10:38:01 server ovpn-myserver[10556]: MULTI: multi_init called, r=256 v=256
Jan 21 10:38:01 server ovpn-myserver[10556]: IFCONFIG POOL IPv4: base=10.8.0.2
size=253
Jan 21 10:38:01 server ovpn-myserver[10556]: IFCONFIG POOL LIST
Jan 21 10:38:01 server ovpn-myserver[10556]: Initialization Sequence Completed
```

You can enable/disable various OpenVPN services on one system, but you could also let Ubuntu do it for you. There is a config for `AUTOSTART` in `/etc/default/openvpn`. Allowed values are "all", "none" or a space-separated list of names of the VPNs. If empty, "all" is assumed.

¹⁸⁷ <https://www.freedesktop.org/software/systemd/man/latest/systemd.unit.html>

The VPN name refers to the VPN configuration file name, i.e., home would be `/etc/openvpn/home.conf`.

If you're running `systemd`, changing this variable requires running `systemctl daemon-reload` followed by a restart of the `openvpn` service (if you removed entries you may have to stop those manually).

After `systemctl daemon-reload`, a restart of the "generic" OpenVPN will restart all dependent services that the generator in `/lib/systemd/system-generators/openvpn-generator` created for your conf files when you called `daemon-reload`.

Now, check if OpenVPN created a `tun0` interface:

```
$ ip addr show dev tun0
```

Due to restricting it to `dev tun0` this will report just that:

```
5: tun0: <POINTOPOINT,MULTICAST,NOARP,UP,LOWER_UP> mtu 1500 qdisc fq_codel state
UNKNOWN group default qlen 100
    link/none
    inet 10.8.0.1 peer 10.8.0.2/32 scope global tun0
        valid_lft forever preferred_lft forever
    inet6 fe80::b5ac:7829:f31e:32c5/64 scope link stable-privacy
        valid_lft forever preferred_lft forever
```

Create client certificates

The VPN client will also need a certificate to authenticate itself to the server. Usually you create a different certificate for each client.

This can be done either on the server (as with the keys and certificates above) and then securely distributed to the client, or the client can generate and submit a request that is sent and signed by the server.

To create the certificate, enter the following in a terminal as a root user:

```
$ ./easyrsa gen-req myclient1 nopass && ./easyrsa sign-req client myclient1
```

If the first command above was done on a remote system, then copy the `.req` file to the CA server. From there, you can import it via `easyrsa import-req /incoming/myclient1.req myclient1`. Then you can go on with the second `sign-req` command.

After this is done, in both cases you will need to copy the following files to the client using a secure method:

- `pki/ca.crt`
- `pki/issued/myclient1.crt`
- `pki/private/myclient1.key`
- `ta.key`

Since the client certificates and keys are only required on the client machine, you can remove them from the server.

Simple client configuration

There are various different OpenVPN client implementations – both with and without *GUIs*. You can read more about clients in [our page on OpenVPN Clients](#) (page 936). For now, we use the command-line/service-based OpenVPN client for Ubuntu, which is part of the same package as the server. So you must install the `openvpn` package again on the client machine:

```
$ sudo apt install openvpn
```

This time, copy the `client.conf` sample config file to `/etc/openvpn/`:

```
$ sudo cp /usr/share/doc/openvpn/examples/sample-config-files/client.conf  
/etc/openvpn/
```

Copy the following client keys and certificate files you created in the section above to e.g. `/etc/openvpn/` and edit `/etc/openvpn/client.conf` to make sure the following lines are pointing to those files. If you have the files in `/etc/openvpn/` you can omit the path:

```
ca ca.crt  
cert myclient1.crt  
key myclient1.key  
tls-crypt ta.key
```

Note:

The `tls-crypt` option was introduced in OpenVPN 2.4.0. If you are running an Ubuntu Server release older than 18.04 LTS (Bionic Beaver), please use `tls-auth ta.key 1` instead.

And you have to specify the OpenVPN server name or address. Make sure the keyword **client** is in the config file, since that's what enables client mode.

```
client  
remote vpnserver.example.com 1194
```

Now start the OpenVPN client with the same templated mechanism:

```
$ sudo systemctl start openvpn@client
```

You can check the status as you did on the server:

```
$ sudo systemctl status openvpn@client
```

```
□ openvpn@client.service - OpenVPN connection to client
  Loaded: loaded (/usr/lib/systemd/system/openvpn@.service; disabled; preset:
  enabled)
  Active: active (running) since Wed 2026-01-21 10:54:27 UTC; 55s ago
  Invocation: f65eae29fc8e4403bd06236309a1db2c
    Docs: man:openvpn(8)
          https://community.openvpn.net/openvpn/wiki/Openvpn24ManPage
          https://community.openvpn.net/openvpn/wiki/HOWTO
  Main PID: 11361 (openvpn)
    Status: "Initialization Sequence Completed"
    Tasks: 1 (limit: 10)
  Memory: 1.9M (peak: 2.1M)
    CPU: 22ms
  CGroup: /system.slice/system-openvpn.slice/openvpn@client.service
          └─11361 /usr/sbin/openvpn --daemon ovpn-client --status /run/openvpn/
client.status 10 --cd /etc/openvpn --script-security 2 --config /etc/openvpn/client.
conf --writepid /run/openvpn/client.pid
```

```
Jan 21 10:54:27 client systemd[1]: Starting openvpn@client.service - OpenVPN
connection to client...
Jan 21 10:54:27 client ovpn-client[11361]: Note: --cipher is not set. OpenVPN
versions before 2.5 defaulted to BF-CBC as fallback when cipher negotiation failed
in this case. If you need this fallback please add '--data-ciphers-fallback BF-CBC'
to your configuration and/or add BF-CBC to --data-ciphers.
Jan 21 10:54:27 client ovpn-client[11361]: Note: Kernel support for ovpn-dco
missing, disabling data channel offload.
Jan 21 10:54:27 client ovpn-client[11361]: OpenVPN 2.6.15 x86_64-pc-linux-gnu [SSL
(OpenSSL)] [LZO] [LZ4] [EPOLL] [PKCS11] [MH/PKTINFO] [AEAD] [DCO]
Jan 21 10:54:27 client ovpn-client[11361]: library versions: OpenSSL 3.5.3 16 Sep
2025, LZO 2.10
Jan 21 10:54:27 client systemd[1]: Started openvpn@client.service - OpenVPN
connection to client.
Jan 21 10:54:27 client ovpn-client[11361]: DCO version: N/A
Jan 21 10:54:27 client ovpn-client[11361]: TCP/UDP: Preserving recently used remote
address: [AF_INET]10.185.198.84:1194
Jan 21 10:54:27 client ovpn-client[11361]: Socket Buffers: R=[212992->212992]
S=[212992->212992]
Jan 21 10:54:27 client ovpn-client[11361]: UDPv4 link local: (not bound)
Jan 21 10:54:27 client ovpn-client[11361]: UDPv4 link remote: [AF_INET]10.185.198.
84:1194
Jan 21 10:54:27 client ovpn-client[11361]: TLS: Initial packet from [AF_INET]10.185.
198.84:1194, sid=46d7b1e7 da534d2d
Jan 21 10:54:27 client ovpn-client[11361]: VERIFY OK: depth=1, CN=server
Jan 21 10:54:27 client ovpn-client[11361]: VERIFY KU OK
Jan 21 10:54:27 client ovpn-client[11361]: Validating certificate extended key usage
Jan 21 10:54:27 client ovpn-client[11361]: ++ Certificate has EKU (str) TLS Web
Server Authentication, expects TLS Web Server Authentication
Jan 21 10:54:27 client ovpn-client[11361]: VERIFY EKU OK
Jan 21 10:54:27 client ovpn-client[11361]: VERIFY OK: depth=0, CN=server
Jan 21 10:54:27 client ovpn-client[11361]: Control Channel: TLSv1.3, cipher TLSv1.3
TLS_AES_256_GCM_SHA384, peer certificate: 2048 bits RSA, signature: RSA-SHA256
Jan 21 10:54:27 client ovpn-client[11361]: [server] Peer Connection Initiated with
[AF_INET]10.185.198.84:1194
Jan 21 10:54:27 client ovpn-client[11361]: TLS: move_session: dest=TM_ACTIVE src=TM_
INITIAL reinit_src=1
Jan 21 10:54:27 client ovpn-client[11361]: TLS: tls_multi_process: initial untrusted
session promoted to trusted
Jan 21 10:54:27 client ovpn-client[11361]: PUSH: Received control message: 'PUSH_
REPLY,route-gateway 10.8.0.1,topology subnet,ping 10,ping-restart 120,ifconfig 10.8.
0.2 255.255.255.0,peer-id 1,cipher AES-256-GCM,protocol-flags cc-exit tls-ekm dyn-
tls-crypt,tun-mtu 1500'
Jan 21 10:54:27 client ovpn-client[11361]: OPTIONS IMPORT: --ifconfig/up options
modified
Jan 21 10:54:27 client ovpn-client[11361]: OPTIONS IMPORT: extra related options
```

```
ovpn-myserver[10556]: read UDPv4 [ECONNREFUSED]: Connection refused (fd=7,code=111)
ovpn-myserver[10556]: 10.185.198.41:40732 VERIFY OK: depth=1, CN=server
ovpn-myserver[10556]: 10.185.198.41:40732 VERIFY OK: depth=0, CN=client
ovpn-myserver[10556]: 10.185.198.41:40732 peer info: IV_VER=2.6.15
ovpn-myserver[10556]: 10.185.198.41:40732 peer info: IV_PLAT=linux
ovpn-myserver[10556]: 10.185.198.41:40732 peer info: IV_TCPNL=1
ovpn-myserver[10556]: 10.185.198.41:40732 peer info: IV_MTU=1600
ovpn-myserver[10556]: 10.185.198.41:40732 peer info: IV_NCP=2
ovpn-myserver[10556]: 10.185.198.41:40732 peer info: IV_CIPHERS=AES-256-GCM:AES-128-
GCM:CHACHA20-POLY1305
ovpn-myserver[10556]: 10.185.198.41:40732 peer info: IV_PROTO=990
ovpn-myserver[10556]: 10.185.198.41:40732 peer info: IV_LZO_STUB=1
ovpn-myserver[10556]: 10.185.198.41:40732 peer info: IV_COMP_STUB=1
ovpn-myserver[10556]: 10.185.198.41:40732 peer info: IV_COMP_STUBv2=1
ovpn-myserver[10556]: 10.185.198.41:40732 TLS: move_session: dest=TM_ACTIVE src=TM_INITIAL
reinit_src=1
ovpn-myserver[10556]: 10.185.198.41:40732 TLS: tls_multi_process: initial untrusted
session promoted to trusted
ovpn-myserver[10556]: 10.185.198.41:40732 Control Channel: TLSv1.3, cipher TLSv1.3 TLS_
AES_256_GCM_SHA384, peer certificate: 2048 bits RSA, signature: RSA-SHA256, peer temporary
key: 768 bits X25519MLKEM768
ovpn-myserver[10556]: 10.185.198.41:40732 [client] Peer Connection Initiated with [AF_
INET]10.185.198.41:40732
ovpn-myserver[10556]: MULTI: new connection by client 'client' will cause previous active
sessions by this client to be dropped. Remember to use the --duplicate-cn option if you
want multiple clients using the same certificate or username to concurrently connect.
ovpn-myserver[10556]: MULTI_sva: pool returned IPv4=10.8.0.2, IPv6=(Not enabled)
ovpn-myserver[10556]: MULTI: Learn: 10.8.0.2 -> client/10.185.198.41:40732
ovpn-myserver[10556]: MULTI: primary virtual IP for client/10.185.198.41:40732: 10.8.0.2
ovpn-myserver[10556]: SENT CONTROL [client]: 'PUSH_REPLY,route-gateway 10.8.0.1,topology
subnet,ping 10,ping-restart 120,ifconfig 10.8.0.2 255.255.255.0,peer-id 0,cipher AES-256-
GCM,protocol-flags cc-exit tls-ekm dyn-tls-crypt,tun-mtu 1500' (status=1)
ovpn-myserver[10556]: client/10.185.198.41:40732 Data Channel: cipher 'AES-256-GCM', peer-
id: 0
ovpn-myserver[10556]: client/10.185.198.41:40732 Timers: ping 10, ping-restart 240
ovpn-myserver[10556]: client/10.185.198.41:40732 Protocol options: explicit-exit-notify 1,
protocol-flags cc-exit tls-ekm dyn-tls-crypt
```

```
$ ip addr show dev tun0
```

```
6: tun0: <POINTOPOINT,MULTICAST,NOARP,UP,LOWER_UP> mtu 1500 qdisc fq_codel state
UNKNOWN group default qlen 500
    link/none
    inet 10.8.0.2/24 brd 10.8.0.255 scope global tun0
        valid_lft forever preferred_lft forever
    inet6 fe80::3c08:9432:f898:d1c0/64 scope link stable-privacy proto kernel_ll
        valid_lft forever preferred_lft forever
```

```
$ ping 10.8.0.1 -c 1
```

```
PING 10.8.0.1 (10.8.0.1) 56(84) bytes of data.  
64 bytes from 10.8.0.1: icmp_seq=1 ttl=64 time=0.335 ms
```

Note:

The OpenVPN server always uses the first usable IP address in the client network and only that IP is pingable. E.g., if you configured a /24 for the client network mask, the .1 address will be used. The P-t-P address you see in the `ip addr` output above does not usually answer ping requests.

```
$ ip route show dev tun0
```

```
10.8.0.0/24 proto kernel scope link src 10.8.0.2
```

First troubleshooting

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Advanced configuration

Advanced routed VPN configuration on server

Note:

Read the OpenVPN [hardening security guide](https://openvpn.net/community-docs/security.html)¹⁸⁸ for further security advice.

¹⁸⁸ <https://openvpn.net/community-docs/security.html>

Advanced bridged VPN configuration on server

DHCP

Prepare interface config for bridging on server

```
network:
  version: 2
  renderer: networkd
  ethernet:
    enp0s31f6:
      dhcp4: no
  bridge:
    br0:
      interfaces: [enp0s31f6]
      dhcp4: no
      addresses: [10.0.1.100/24]
      routes:
        - to: default
          via: 10.0.1.1
      nameservers:
        addresses: [10.0.1.1]
```



```
$ sudo apt update && sudo apt install networkd-dispatcher
```

```
#!/bin/sh
set -e
if [ "$IFACE" = br0 ]; then
    # no networkd-dispatcher event for 'carrier' on the physical interface
    ip link set enp0s31f6 up promisc on
fi
```

```
$ sudo chmod +x /etc/networkd-dispatcher/dormant.d/promisc_bridge
```

Prepare server config for bridging

```
;dev tun
dev tap
;server 10.8.0.0 255.255.255.0
server-bridge 10.0.0.4 255.255.255.0 10.0.0.128 10.0.0.254
```

```
$ sudo systemctl restart openvpn@myserver
```

Prepare client config for bridging

```
dev tap
;dev tun
```

```
$ sudo systemctl restart openvpn@client
```

Using OpenSSL providers

Legacy provider

```
providers legacy default
```

```
$ openvpn --providers legacy default ...
```

Note:

On Ubuntu 22.04 LTS and earlier, legacy algorithms are included in OpenVPN by default.

TPM 2.0

```
providers legacy default tpm2
```

```
$ openvpn --providers legacy default tpm2 ...
```

Note:

The provider order matters here as tpm2 currently requires the legacy provider to load first.

Further reading

- [EasyRSA](#)¹⁸⁹
- [OpenVPN quick start guide](#)¹⁹⁰
- [easy-openvpn](#)¹⁹¹
- [OpenVPN Guide](#)¹⁹²

WireGuard VPN

WireGuard VPN peer-to-site

another system in the home network

on the router

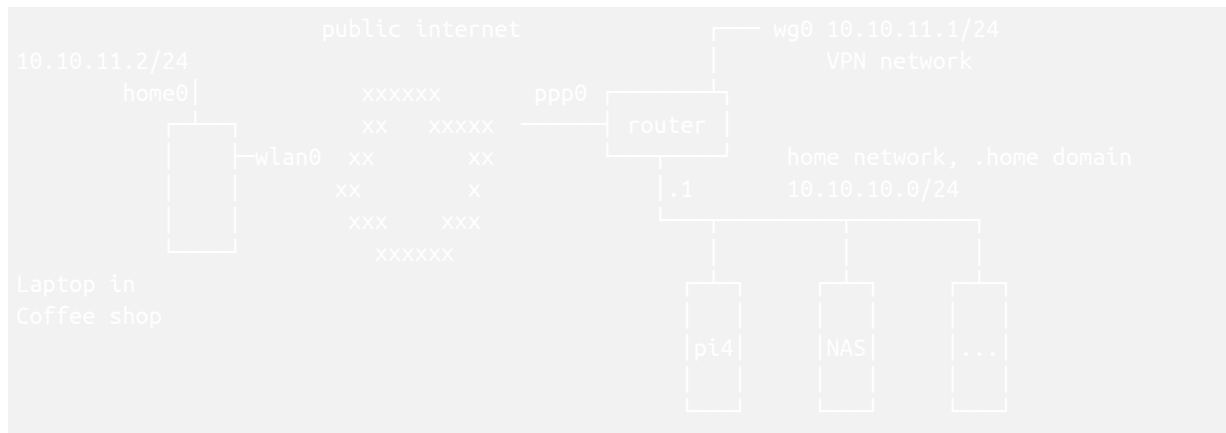
¹⁸⁹ <https://github.com/OpenVPN/easy-rsa/blob/master/README.quickstart.md>

¹⁹⁰ <https://openvpn.net/as-docs/getting-started.html>

¹⁹¹ <https://snapcraft.io/easy-openvpn-server>

¹⁹² <https://wiki.debian.org/OpenVPN>

WireGuard VPN peer-to-site (on router)



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Configuration

```
$ umask 077
```

```
$ wg genkey > laptop-private.key
```

```
$ wg pubkey < laptop-private.key > laptop-public.key
```

```
$ wg genkey > router-private.key
```

```
$ wg pubkey < router-private.key > router-public.key
```

[Interface]

```
PrivateKey = <contents-of-router-private.key>  
ListenPort = 51000  
Address = 10.10.11.1/24
```

[Peer]

```
PublicKey = <contents-of-laptop-public.key>  
AllowedIPs = 10.10.11.2
```

Important:

This configuration file contains a secret: **PrivateKey**. Make sure to adjust its permissions accordingly, as follows:

```
$ sudo chmod 0600 /etc/wireguard/wg0.conf
```

```
$ sudo chown root: /etc/wireguard/wg0.conf
```

-
-

[Interface]

```
PrivateKey = <contents-of-laptop-private.key>
ListenPort = 51000
Address = 10.10.11.2/24
```

[Peer]

```
PublicKey = <contents-of-router-public.key>
Endpoint = <home-ppp0-IP-or-hostname>:51000
AllowedIPs = 10.10.11.0/24,10.10.10.0/24
```

Important:

As before, this configuration file contains a secret: **PrivateKey**. You need to adjust its permissions accordingly, as follows:

```
$ sudo chmod 0600 /etc/wireguard/home0.conf
```

```
$ sudo chown root: /etc/wireguard/home0.conf
```

Note:

You may have noticed by now that address allocation is manual, and not via something like *DHCP*. Keep tabs on it!

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DNS

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Default Gateway

Testing

```
$ sudo wg-quick up wg0

[#] ip link add wg0 type wireguard
[#] wg setconf wg0 /dev/fd/63
[#] ip -4 address add 10.10.11.1/24 dev wg0
[#] ip link set mtu 1378 up dev wg0
```

```
$ ip a show dev wg0

9: wg0: <POINTOPOINT,NOARP,UP,LOWER_UP> mtu 1378 qdisc noqueue state UNKNOWN group
default qlen 1000
    link/none
    inet 10.10.11.1/24 scope global wg0
        valid_lft forever preferred_lft forever
```

```
$ ip route | grep wg0

10.10.11.0/24 dev wg0 proto kernel scope link src 10.10.11.1
```

```
$ sudo wg show

interface: wg0
  public key: <router public key>
  private key: (hidden)
  listening port: 51000

peer: <laptop public key>
  allowed ips: 10.10.11.2/32
```

```
$ sudo watch wg show
```

```
$ sudo wg-quick up home0
```

```
[#] ip link add home0 type wireguard  
[#] wg setconf home0 /dev/fd/63  
[#] ip -4 address add 10.10.11.2/24 dev home0  
[#] ip link set mtu 1420 up dev home0  
[#] ip -4 route add 10.10.10.0/24 dev home0
```

```
$ ip a show dev home0
```

```
24: home0: <POINTOPOINT,NOARP,UP,LOWER_UP> mtu 1420 qdisc noqueue state UNKNOWN  
group default qlen 1000  
    link/none  
    inet 10.10.11.2/24 scope global home0  
        valid_lft forever preferred_lft forever
```

```
$ ip route | grep home0
```

```
10.10.10.0/24 dev home0 scope link  
10.10.11.0/24 dev home0 proto kernel scope link src 10.10.11.2
```

```
$ ping -c 3 10.10.10.1
```

```
PING 10.10.10.1 (10.10.10.1) 56(84) bytes of data.  
64 bytes from 10.10.10.1: icmp_seq=1 ttl=64 time=603 ms  
64 bytes from 10.10.10.1: icmp_seq=2 ttl=64 time=300 ms  
64 bytes from 10.10.10.1: icmp_seq=3 ttl=64 time=304 ms
```



```
$ sudo wg show

interface: wg0
  public key: <router public key>
  private key: (hidden)
  listening port: 51000

peer: <laptop public key>
  endpoint: <laptop public IP>:51000
  allowed ips: 10.10.11.2/32
  latest handshake: 1 minute, 8 seconds ago
  transfer: 564 B received, 476 B sent
```

WireGuard on an internal system (peer-to-site)

on the home router itself

DHCP

Router changes

Network	10.10.10.0/24
Usable addresses	10.10.10.2 – 10.10.10.254 (.1 is the router)
DHCP range	10.10.10.50 – 10.10.10.254
VPN range	10.10.10.10 – 10.10.10.49

Configure the internal WireGuard endpoint

```
$ sudo apt install wireguard
```

```
$ umask 077
```

```
$ wg genkey > internal-private.key
```

```
$ wg pubkey < internal-private.key > internal-public.key
```

[Interface]

```
Address = 10.10.10.10/32
```

```
ListenPort = 51000
```

```
PrivateKey = <contents of internal-private.key>
```

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```
[Peer]
# laptop
PublicKey = <contents of laptop-public.key>
AllowedIPs = 10.10.10.11/32 # any available IP in the VPN range
```

Note:

Just like in the *peer-to-site* (page 275) scenario with WireGuard on the router, there is no Endpoint configuration here for the laptop peer, because we don't know where it will be connecting from beforehand.

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```
net.ipv4.ip_forward = 1
net.ipv4.conf.all.proxy_arp = 1
```

```
$ sudo sysctl -p /etc/sysctl.d/70-wireguard-routing.conf -w
```

```
$ sudo wg-quick up wg0
```

Configuring the peer

```
[Interface]
ListenPort = 51000
Address = 10.10.10.11/24
PrivateKey = <contents of the private key for this system>

[Peer]
PublicKey = <contents of internal-public.key>
Endpoint = <home-ppp0-IP-or-hostname>:51000
AllowedIPs = 10.10.10.0/24
```

```
$ sudo wg-quick up home0
```

Note:

There is no need to add an index number to the end of the interface name. That is a convention, but not strictly a requirement.

Testing

WireGuard VPN site-to-site

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```
ubuntu@b1:~$ mtr -n -r 10.10.10.230
```

```
Start: 2022-09-02T18:56:51+0000
```

HOST: b1	Loss%	Snt	Last	Avg	Best	Wrst	StDev
1. -- 10.10.11.1	0.0%	10	0.1	0.1	0.1	0.2	0.0
2. -- 10.10.9.0	0.0%	10	299.6	299.3	298.3	300.0	0.6
3. -- 10.10.10.230	0.0%	10	299.1	299.1	298.0	300.2	0.6

Note:

Technically, a /31 Classless Inter-Domain Routing (CIDR) network has no usable IP addresses, since the first one is the network address, and the second (and last) one is the broadcast address. RFC 3021¹⁹³ allows for it, but if you encounter routing or other networking issues, switch to a /30 CIDR and its two valid host IPs.

¹⁹³ <https://www.ietf.org/rfc/rfc3021.txt>

Configure WireGuard

```
$ sudo apt install wireguard
```

```
$ wg genkey | sudo tee /etc/wireguard/wgA.key
```

```
$ sudo cat /etc/wireguard/wgA.key | wg pubkey | sudo tee  
/etc/wireguard/wgA.pub
```

[Interface]

```
PostUp = wg set %i private-key /etc/wireguard/%i.key  
Address = 10.10.9.0/31  
ListenPort = 51000
```

[Peer]

```
# beta site  
PublicKey = <contents of /etc/wireguard/wgB.pub>  
AllowedIPs = 10.10.11.0/24,10.10.9.0/31  
Endpoint = <beta-gw-ip>:51000
```

```
$ sudo apt install wireguard
```

```
$ wg genkey | sudo tee /etc/wireguard/wgB.key
```

```
$ sudo cat /etc/wireguard/wgB.key | wg pubkey | sudo tee  
/etc/wireguard/wgB.pub
```

[Interface]

```
Address = 10.10.9.1/31  
PostUp = wg set %i private-key /etc/wireguard/%i.key
```

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```
ListenPort = 51000
```

[Peer]

```
# alpha site
```

```
PublicKey = <contents of /etc/wireguard/wgA.pub>
```

```
AllowedIPs = 10.10.10.0/24,10.10.9.0/31
```

```
Endpoint = <alpha-gw-ip>:51000
```

Important:

WireGuard is being set up on the gateways for these two networks. As such, there are no changes needed on individual hosts of each network, but keep in mind that the WireGuard tunneling and encryption is only happening between the *alpha* and *beta* gateways, and **NOT** between the hosts of each network.

Bring the interfaces up

```
$ sudo systemctl enable --now wg-quick@wgA
```

```
$ sudo systemctl enable --now wg-quick@wgB
```

Firewall and routing

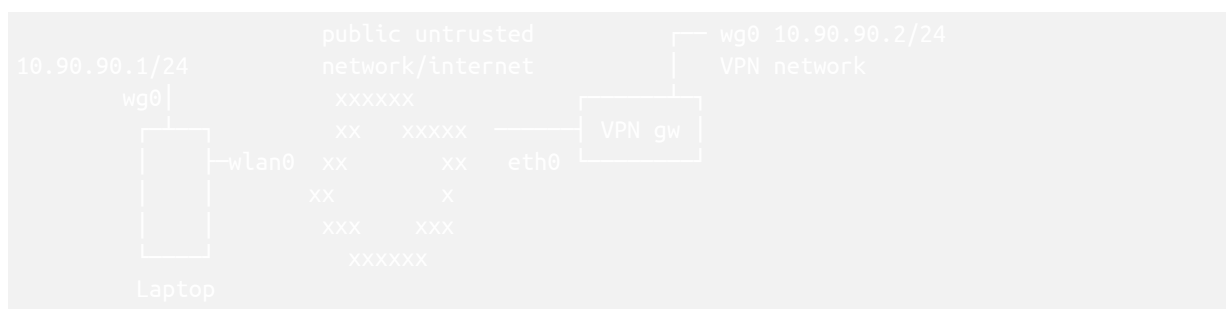
•

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```
$ sudo iptables -A FORWARD -i wgA -p tcp --dport 22 -j REJECT
```

```
$ sudo iptables -A FORWARD -i wgB -p tcp --dport 22 -j REJECT
```

Using the VPN as the default gateway



Install and configure WireGuard

```
$ sudo apt install wireguard
```

```
$ umask 077
```

```
$ wg genkey > wg0.key
```

```
$ wg pubkey < wg0.key > wg0.pub
```

```
$ sudo mv wg0.key wg0.pub /etc/wireguard
```

```
$ sudo apt install wireguard
```

```
$ umask 077
```

```
$ wg genkey > gateway0.key
```

```
$ wg pubkey < gateway0.key > gateway0.pub
```

```
$ sudo mv gateway0.key gateway0.pub /etc/wireguard
```

[Interface]

```
PostUp = wg set %i private-key /etc/wireguard/wg0.key
```

```
ListenPort = 51000
```

```
Address = 10.90.90.1/24
```

[Peer]

```
PublicKey = <contents of gateway0.pub>
```

```
Endpoint = <public IP of gateway server>
```

```
AllowedIPs = 0.0.0.0/0
```

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```
[Interface]
PostUp = wg set %i private-key /etc/wireguard/%i.key
Address = 10.90.90.2/24
ListenPort = 51000

[Peer]
PublicKey = <contents of wg0.pub>
AllowedIPs = 10.90.90.1/32
```

Routing and masquerading

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```
net.ipv4.ip_forward = 1
```

```
$ sudo sysctl -p /etc/sysctl.d/70-wireguard-routing.conf -w
```

```
$ sudo iptables -t nat -A POSTROUTING -s 10.90.90.0/24 -o eth0 -j MASQUERADE
```

```
#!/bin/sh
iptables -t nat -A POSTROUTING -s 10.90.90.0/24 -o eth0 -j MASQUERADE
```

Testing

```
$ sudo wg-quick up gateway0

[#] ip link add gateway0 type wireguard
[#] wg setconf gateway0 /dev/fd/63
[#] ip -4 address add 10.90.90.2/24 dev gateway0
[#] ip link set mtu 1378 up dev gateway0
[#] wg set gateway0 private-key /etc/wireguard/gateway0.key
```

```
$ sudo wg-quick up wg0

[#] ip link add wg0 type wireguard
[#] wg setconf wg0 /dev/fd/63
[#] ip -4 address add 10.90.90.1/24 dev wg0
[#] ip link set mtu 1420 up dev wg0
[#] wg set wg0 fwmark 51820
[#] ip -4 route add 0.0.0.0/0 dev wg0 table 51820
[#] ip -4 rule add not fwmark 51820 table 51820
[#] ip -4 rule add table main suppress_prefixlength 0
[#] sysctl -q net.ipv4.conf.all.src_valid_mark=1
[#] nft -f /dev/fd/63
[#] wg set wg0 private-key /etc/wireguard/wg0.key
```

```
$ mtr -r 1.1.1.1

Start: 2022-09-01T12:42:59+0000
HOST: laptop.lan
      Loss%  Snt  Last  Avg  Best  Wrst StDev
  1. |-- 10.90.90.2      0.0%   10 184.9 185.5 184.9 186.9   0.6
  2. |-- 10.48.128.1     0.0%   10 185.6 185.8 185.2 188.3   0.9
  (...)
  7. |-- one.one.one.one 0.0%   10 186.2 186.3 185.9 186.6   0.2
```

```
$ ip route

default via 192.168.122.1 dev enp1s0 proto dhcp src 192.168.122.160 metric 100
10.90.90.0/24 dev wg0 proto kernel scope link src 10.90.90.1
192.168.122.0/24 dev enp1s0 proto kernel scope link src 192.168.122.160 metric 100
192.168.122.1 dev enp1s0 proto dhcp scope link src 192.168.122.160 metric 100
```

DNS leaks

DNS

//dnsleaktest.com

https:

Selecting a DNS server

```
$ resolvectl status
```

```
Global
    Protocols: -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
    resolv.conf mode: stub

Link 2 (ens2)
    Current Scopes: DNS
        Protocols: +DefaultRoute +LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
    Current DNS Server: 10.48.0.5
        DNS Servers: 10.48.0.5
        DNS Domain: openstacklocal

Link 5 (gateway0)
    Current Scopes: none
        Protocols: -DefaultRoute +LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
```

[Interface]

```
PostUp = wg set %i private-key /etc/wireguard/wg0.key
PostUp = resolvectl dns %i 10.48.0.5; resolvectl domain %i \~.
ListenPort = 51000
Address = 10.90.90.1/24
```

[Peer]

```
PublicKey = <contents of gateway0.pub>
Endpoint = <public IP of gateway server>
AllowedIPs = 0.0.0.0/0
```

```
$ sudo resolvectl dns wg0 10.48.0.5; sudo resolvectl domain wg0 \~.
```

```
$ sudo wg-quick down wg0; sudo wg-quick up wg0
```

Installing your own DNS server

```
$ sudo apt install bind9
```

[Interface]

```
PostUp = wg set %i private-key /etc/wireguard/wg0.key  
PostUp = resolvectl dns %i 10.90.90.2; resolvectl domain %i |~.  
ListenPort = 51000  
Address = 10.90.90.1/24
```

[Peer]

```
PublicKey = <contents of gateway0.pub>  
Endpoint = <public IP of gateway server>  
AllowedIPs = 0.0.0.0/0
```

Common tasks in WireGuard VPN

Controlling the WireGuard interface with systemd

```
$ sudo systemctl enable wg-quick@wg0
```

```
$ sudo systemctl reload wg-quick@wg0
```

DNS resolving

DNS

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[Interface]

...

```
PostUp = resolvectl dns %i 10.10.10.1; resolvectl domain %i \~home
```

*wg-quick(8)*¹⁹⁴

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```
$ sudo wg-quick up home0
```

```
[#] ip link add home0 type wireguard
```

```
[#] wg setconf home0 /dev/fd/63
```

```
[#] ip -4 address add 10.10.11.2/24 dev home0
```

```
[#] ip link set mtu 1420 up dev home0
```

```
[#] ip -4 route add 10.10.10.0/24 dev home0
```

```
[#] resolvectl dns home0 10.10.10.1; resolvectl domain home0 \~home
```

hostname

```
$ resolvectl status home0
```

¹⁹⁴ <https://manpages.ubuntu.com/manpages/resolute/man8/wg-quick.8.html>

```
Link 26 (home0)
  Current Scopes: DNS
    Protocols: -DefaultRoute +LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
Current DNS Server: 10.10.10.1
  DNS Servers: 10.10.10.1
  DNS Domain: ~home
```

Note:

The *wg-quick(8)*¹⁹⁵ manual page documents the DNS setting of the WireGuard interface which has the same purpose, but only works if you have `resolveconf` installed. Ubuntu systems by default don't, and rely on `systemd-resolved` instead.

¹⁹⁵ <https://manpages.ubuntu.com/manpages/resolute/man8/wg-quick.8.html>

Adding another peer

- 1.
- 2.
- 3.

```
$ umask 077
```

```
$ wg genkey > ontheroad-private.key
```

```
$ wg pubkey < ontheroad-private.key > ontheroad-public.key
```

```
$ ls -la ontheroad.*
```

```
-rw----- 1 ubuntu ubuntu 45 Aug 22 20:12 ontheroad-private.key
-rw----- 1 ubuntu ubuntu 45 Aug 22 20:13 ontheroad-public.key
```

[Interface]

```
PrivateKey = <contents-of-ontheroad-private.key>
ListenPort = 51000
Address = 10.10.11.3/24
```

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[Peer]

```
PublicKey = <contents-of-router-public.key>
Endpoint = <home-ppp0-IP-or-hostname>:51000
AllowedIPs = 10.10.11.0/24,10.10.10.0/24
```

[Interface]

```
PrivateKey = <contents-of-router-private.key>
ListenPort = 51000
Address = 10.10.11.1/24
```

[Peer]

```
# laptop
PublicKey = <contents-of-laptop-public.key>
AllowedIPs = 10.10.11.2
```

[Peer]

```
# ontheroad
PublicKey = <contents-of-ontheroad-public.key>
AllowedIPs = 10.10.11.3
```

```
$ systemctl reload wg-quick@wg0
```

Note:

For this case of a “server” or “VPN gateway”, where we are just adding another peer to an existing config, the `systemctl reload` action will work well enough to insert the new peer into the WireGuard configuration. However, it won’t create new routes, or do any of the other steps that `wg-quick` does. Depending on your setup, you might need a full restart so that `wg-quick` can fully do its job.

Adding a smartphone peer

[upstream installation page](#)¹⁹⁶

```
$ sudo apt install qrencode
```

¹⁹⁶ <https://www.wireguard.com/install/>


```
$ cat phone.conf | qrencode -t ansiutf8
```

Important:

Treat this QR code as a secret, as it contains the private key for the WireGuard interface!

Security tips for WireGuard VPN

Traffic goes both ways

Using PreSharedKey

```
$ wg genpsk  
vxlX6eMMin8uhxbKEhe/i0xi8ru+q1qWzCdjESXoFZY=
```

```
[Peer]  
PublicKey = ....  
Endpoint = ....  
AllowedIPs = ....  
PresharedKey = vxlX6eMMin8uhxbKEhe/i0xi8ru+q1qWzCdjESXoFZY=
```

Note:

Both sides need to have the same PresharedKey in their respective [Peer] sections.

Preventing accidental leakage of private keys

[Interface]

```
ListenPort = 51000
Address = 10.10.11.3/24
PostUp = wg set %i private-key /etc/wireguard/%i.key
```

[Peer]

```
PublicKey = <contents-of-router-public.key>
Endpoint = 10.48.132.39:51000
AllowedIPs = 10.10.11.0/24,10.10.10.0/24
```

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Troubleshooting WireGuard VPN

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The watch wg command

```
Every 2.0s: wg                               j-wg: Fri Aug 26 17:44:37 2022

interface: wg0
  public key: +T3T3HTMeyrEDvim8FBxbYjbz+/P0eOtG3Rl9kJmM=
  private key: (hidden)
  listening port: 51000

peer: 2cJdFcNzXv4YUGyDTaht0frbsrFsCByatPnNzKts0Qo=
  endpoint: 10.172.196.106:51000
  allowed ips: 10.10.11.2/32
  latest handshake: 3 hours, 27 minutes, 35 seconds ago
  transfer: 3.06 KiB received, 2.80 KiB sent

peer: ZliZ1hlarZqvfxPMyme2ECtXdk611NB7uzLAD4McpGI=
  allowed ips: 10.10.11.3/32
```

Kernel debug messages

Note:

To be able to use the commands outlined below `kernel_lockdown(7)`¹⁹⁷ needs to be disabled. That is because the debug interfaces used are considered a way to tamper with the integrity of the kernel.

Lockdown is enabled by default in secure boot mode. Check the status via:

```
$ cat /sys/kernel/security/lockdown
none [integrity] confidentiality
```

Or that when disabled:

```
$ cat /sys/kernel/security/lockdown
[none] integrity confidentiality
```

To disable it either disable secure boot in *BIOS* or your virtualization platform.

If that is impossible, consider not using the dynamic module debug control shown below, but instead using a kernel boot parameter `wireguard.dyndbg=+p` which has the drawback that it needs a reboot to enable/disable it.

Before 20.04 [Kernel Lockdown](#)¹⁹⁸ was not yet present, therefore such systems can skip this step.

¹⁹⁷ https://manpages.ubuntu.com/manpages/resolute/man7/kernel_lockdown.7.html

¹⁹⁸ <https://documentation.ubuntu.com/security/security-features/kernel-protections/#kernel-lockdown>

```
$ echo "module wireguard +p" | sudo tee
/sys/kernel/debug/dynamic_debug/control
```

```
$ sudo dmesg -wT
```

```
$ echo "module wireguard -p" | sudo tee
/sys/kernel/debug/dynamic_debug/control
```

Destination address required

```
$ ping 10.10.11.2
```

```
PING 10.10.11.2 (10.10.11.2) 56(84) bytes of data.  
From 10.10.11.1 icmp_seq=1 Destination Host Unreachable  
ping: sendmsg: Destination address required
```

Required key not available

```
$ ping 10.10.11.1
```

```
PING 10.10.11.1 (10.10.11.1) 56(84) bytes of data.  
From 10.10.11.2 icmp_seq=1 Destination Host Unreachable  
ping: sendmsg: Required key not available
```

```
wireguard: home0: No peer has allowed IPs matching 10.10.11.1
```

See also

- [Introduction to WireGuard VPN](#)

See also

- [Introduction to security](#)
- [Security topics](#)

Our networking section

-
-

Networking

[networking](#)

[introduction to](#)

Configuration

[Configuring networks](#)

[the Netplan documentation](#)¹⁹⁹

Network tools

¹⁹⁹ <https://netplan.readthedocs.io/en/stable/>

Set up an FTP server

tation

OpenSSH documen-

-
-

vsftpd – FTP server installation

```
$ sudo apt install vsftpd
```

Anonymous FTP configuration

```
anonymous_enable=YES
```

```
$ sudo systemctl restart vsftpd
```

FTP files location

```
$ sudo mkdir -p /srv/files/ftp
```

```
$ sudo usermod -d /srv/files/ftp ftp
```

```
$ sudo systemctl restart vsftpd.service
```

User authenticated FTP configuration

```
write_enable=YES
```

```
$ sudo systemctl restart vsftpd.service
```

```
anon_upload_enable=YES
```

Warning:

Enabling anonymous FTP upload can be an extreme security risk. It is best to not enable anonymous upload on servers accessed directly from the Internet.

Securing FTP

```
chroot_local_user=YES
```

```
chroot_list_enable=YES  
chroot_list_file=/etc/vsftpd.chroot_list
```

Warning:

If `chroot_local_user` is set to YES (setting all users to be limited to just their home directory), `/etc/vsftpd.chroot_list` becomes a list of users which are NOT limited to just their home directory

```
$ sudo systemctl restart vsftpd.service
```

```
ssl_enable=YES
```

```
rsa_cert_file=/etc/ssl/certs/ssl-cert-snakeoil.pem  
rsa_private_key_file=/etc/ssl/private/ssl-cert-snakeoil.key
```

Security - Certificates

```
$ sudo systemctl restart vsftpd.service
```

```
# /etc/shells: valid login shells
/bin/csh
/bin/sh
  /usr/bin/es
  /usr/bin/ksh
  /bin/ksh
  /usr/bin/rc
  /usr/bin/tcsh
  /bin/tcsh
  /usr/bin/esh
  /bin/dash
  /bin/bash
/bin/rbash
/usr/bin/screen
/usr/sbin/nologin
```

```
auth      required      pam_shells.so
```

Further reading

- [vsftpd website](http://vsftpd.beasts.org/vsftpd_conf.html)²⁰⁰

²⁰⁰ http://vsftpd.beasts.org/vsftpd_conf.html

Domain Name Service (DNS)

domain names (FQDN)

fully qualified

Install DNS

```
$ sudo apt install bind9
```

```
$ sudo apt install dnsutils
```

DNS configuration overview

-
-
-

Configuration files

-
-
-

Set up a caching nameserver

```
options {  
    forwarders {  
        1.2.3.4;  
        5.6.7.8;  
    };  
};
```

Note:

Replace 1.2.3.4 and 5.6.7.8 with the IP addresses of actual nameservers.

```
$ sudo systemctl restart bind9.service
```

See the dig section

Set up a primary server

Forward zone file

```
zone "example.com" {  
    type master;  
    file "/etc/bind/db.example.com",  
};
```

Note:

If BIND will be receiving automatic updates to the file as with [DDNS](#), then use `/var/lib/bind/db.example.com` rather than `/etc/bind/db.example.com` both here and in the copy command below.

FQDN

```
;
; BIND data file for example.com
;
$TTL      604800
@         IN      SOA      example.com. root.example.com. (
                        2      ; Serial
                        604800 ; Refresh
                        86400  ; Retry
                        2419200 ; Expire
                        604800 ) ; Negative Cache TTL
;
@         IN      NS       ns.example.com.
@         IN      A        192.168.1.10
@         IN      AAAA     2001:db8::10
ns        IN      A        192.168.1.10
```

Common record types

Note:

Many admins like to use the “last edited” date as the Serial of a zone, such as **2020012100** which is **yyyymmddss** (where **ss** is the Serial Number)

```
$ sudo systemctl restart bind9.service
```

Reverse zone file

```
zone "1.168.192.in-addr.arpa" {  
    type master;  
    file "/etc/bind/db.192";  
};
```

Note:

Replace 1.168.192 with the first three octets of whatever network you are using. Also, name the zone file `/etc/bind/db.192` appropriately. It should match the first octet of your network.

```
;   
; BIND reverse data file for local 192.168.1.XXX net   
;   
$TTL      604800   
@         IN      SOA      ns.example.com. root.example.com. (   
                                2          ; Serial   
                                604800     ; Refresh   
                                86400      ; Retry   
                                2419200    ; Expire   
                                604800 )   ; Negative Cache TTL   
;   
@         IN      NS       ns.   
10        IN      PTR      ns.example.com.
```

```
$ sudo systemctl restart bind9.service
```

Set up a secondary server

```
zone "example.com" {
    type master;
    file "/etc/bind/db.example.com";
    allow-transfer { 192.168.1.11; };
};

zone "1.168.192.in-addr.arpa" {
    type master;
    file "/etc/bind/db.192";
    allow-transfer { 192.168.1.11; };
};
```

Note:

Replace 192.168.1.11 with the IP address of your secondary nameserver.

```
$ sudo systemctl restart bind9.service
```

```
zone "example.com" {
    type secondary;
    file "db.example.com";
    masters { 192.168.1.10; };
};

zone "1.168.192.in-addr.arpa" {
    type secondary;
    file "db.192";
    masters { 192.168.1.10; };
};
```

```
$ sudo systemctl restart bind9.service
```

```
client 192.168.1.10#39448: received notify for zone '1.168.192.in-addr.arpa'
zone 1.168.192.in-addr.arpa/IN: Transfer started.
transfer of '100.18.172.in-addr.arpa/IN' from 192.168.1.10#53:
  connected using 192.168.1.11#37531
zone 1.168.192.in-addr.arpa/IN: transferred serial 5
transfer of '100.18.172.in-addr.arpa/IN' from 192.168.1.10#53:
  Transfer completed: 1 messages,
  6 records, 212 bytes, 0.002 secs (106000 bytes/sec)
zone 1.168.192.in-addr.arpa/IN: sending notifies (serial 5)
```

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```
client 192.168.1.10#20329: received notify for zone 'example.com'
zone example.com/IN: Transfer started.
transfer of 'example.com/IN' from 192.168.1.10#53: connected using 192.168.1.11#38577
zone example.com/IN: transferred serial 5
transfer of 'example.com/IN' from 192.168.1.10#53: Transfer completed: 1 messages,
8 records, 225 bytes, 0.002 secs (112500 bytes/sec)
```

Note:

A zone is only transferred if the Serial Number on the primary is larger than the one on the secondary. If you want to have your primary DNS notify other secondary DNS servers of zone changes, you can add `also-notify { ipaddress; };` to `/etc/bind/named.conf.local` as shown in the example below:

```
zone "example.com" {
    type master;
    file "/etc/bind/db.example.com";
    allow-transfer { 192.168.1.11; };
    also-notify { 192.168.1.11; };
};

zone "1.168.192.in-addr.arpa" {
    type master;
    file "/etc/bind/db.192";
    allow-transfer { 192.168.1.11; };
    also-notify { 192.168.1.11; };
};
```

Note:

The default directory for non-authoritative zone files is `/var/cache/bind/`. This directory is also configured in AppArmor to allow the named daemon to write to it. See this page for *more information on AppArmor* (page 170).

Testing your setup

resolv.conf

DNS client configuration

```
nameserver 127.0.0.53
search example.com
```



```
$ resolvectl status
```

Note:

You should also add the IP address of the secondary nameserver to your client configuration in case the primary becomes unavailable.

dig

```
$ dig -x 127.0.0.1  
;; Query time: 1 msec  
;; SERVER: 192.168.1.10#53(192.168.1.10)
```

```
$ dig ubuntu.com  
;; Query time: 49 msec
```

```
;; Query time: 1 msec
```

ping

```
$ ping example.com
```

```
PING ns.example.com (192.168.1.10) 56(84) bytes of data:  
64 bytes from 192.168.1.10: icmp_seq=1 ttl=64 time=0.800 ms  
64 bytes from 192.168.1.10: icmp_seq=2 ttl=64 time=0.813 ms
```

named-checkzone

```
$ named-checkzone example.com /etc/bind/db.example.com  
  
zone example.com/IN: loaded serial 6  
OK
```

```
$ named-checkzone 1.168.192.in-addr.arpa /etc/bind/db.192  
  
zone 1.168.192.in-addr.arpa/IN: loaded serial 3  
OK
```

Note:

The Serial Number of your zone file will probably be different.

Quick temporary query logging

```
$ sudo rndc querylog on
```

```
$ sudo rndc querylog off
```

```
Jan 20 19 40:50 new-n1 named[816]: received control channel command 'querylog on'  
Jan 20 19 40:50 new-n1 named[816]: query logging is now on  
Jan 20 19 40:57 new-n1 named[816]: client @0x7f48ec101480 192.168.1.10#36139 (ubuntu.com):  
query: ubuntu.com IN A +E(0)K (192.168.1.10)
```

Note:

The amount of logs generated by enabling querylog could be huge!

Logging

```
logging {  
    category default { default_syslog; default_debug; };  
    category unmatched { null; };  
};
```

```
logging {  
    channel query.log {  
        file "/var/log/named/query.log";  
        severity debug 3;  
    };  
    category queries { query.log; };  
};
```

Note:

The debug option can be set from 1 to 3. If a level isn't specified, level 1 is the default.

```
$ sudo mkdir /var/log/named
```

```
$ sudo chown bind:bind /var/log/named
```

```
$ sudo systemctl restart bind9.service
```

Common record types

- *hostname*

www	IN	A	192.168.1.12
-----	----	---	--------------

-

web	IN	CNAME	www
-----	----	-------	-----

-

@	IN	MX	1	mail.example.com.
mail	IN	A	192.168.1.13	

-

@	IN	NS	ns.example.com.
@	IN	NS	ns2.example.com.
ns	IN	A	192.168.1.10
ns2	IN	A	192.168.1.11

Further reading

- *Installing DNS Security Extensions (DNSSEC)*
- Upstream BIND9 Documentation²⁰¹
- DNS and BIND²⁰²
BIND on IPv6²⁰³

DNS and

Installing DNS Security Extensions (DNSSEC)

DNS

²⁰¹ <https://bind9.readthedocs.io/en/latest/>

²⁰² <https://www.oreilly.com/library/view/dns-and-bind/0596100574/>

²⁰³ <https://www.oreilly.com/library/view/dns-and-bind/9781449308025/>

Starting point

DNS How-To

```
$ sudo apt install bind9 -y
```

```
zone "example.internal" {  
    type master;  
    file "/var/lib/bind/db.example.internal";  
};
```

```
$TTL 86400      ; 1 day  
example.internal.      IN SOA  example.internal. root.example.internal. (  
                            1          ; serial  
                           43200       ; refresh (12 hours)  
                            900        ; retry (15 minutes)  
                          1814400      ; expire (3 weeks)  
                           7200       ; minimum (2 hours)  
                           )  
example.internal.      IN NS   ns.example.internal.  
ns                      IN A    192.168.1.10  
noble                   IN A    192.168.1.11
```

```
$ sudo systemctl restart named
```

```
$ dig @127.0.0.1 +short noble.example.internal  
192.168.1.11
```

Enabling DNSSEC

```
zone "example.internal" {  
    type master;  
    file "/var/lib/bind/db.example.internal";  
    dnssec-policy default;  
    inline-signing yes;  
};
```

-
-

```
$ sudo rndc reconfig
```

```
$ sudo journalctl -u named.service -f
```

```
named[3246]: zone example.internal/IN (unsigned): loaded serial 1  
named[3246]: zone example.internal/IN (signed): loaded serial 1  
named[3246]: zone example.internal/IN (signed): receive_secure_serial: unchanged  
named[3246]: zone example.internal/IN (signed): sending notifies (serial 1)  
named[3246]: zone example.internal/IN (signed): reconfiguring zone keys  
named[3246]: keymgr: DNSKEY example.internal/ECDSAP256SHA256/44911 (CSK) created for  
policy default  
named[3246]: Fetching example.internal/ECDSAP256SHA256/44911 (CSK) from key repository.  
named[3246]: DNSKEY example.internal/ECDSAP256SHA256/44911 (CSK) is now published  
named[3246]: DNSKEY example.internal/ECDSAP256SHA256/44911 (CSK) is now active  
named[3246]: zone example.internal/IN (signed): next key event: 23-Oct-2024 22:47:12.544  
named[3246]: any newly configured zones are now loaded  
named[3246]: running  
named[3246]: resolver priming query complete: success  
named[3246]: managed-keys-zone: Key 20326 for zone . is now trusted (acceptance timer
```

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```
complete)
named[3246]: zone example.internal/IN (signed): sending notifies (serial 3)
```

-
-
-
-

```
-rw-r--r-- 1 bind bind 413 Oct 23 20:50 Kexample.internal.+013+48112.key
-rw----- 1 bind bind 215 Oct 23 20:50 Kexample.internal.+013+48112.private
-rw-r--r-- 1 bind bind 647 Oct 23 20:50 Kexample.internal.+013+48112.state
```

Verification

```
$ cp /var/cache/bind/Kexample.internal.+013+48112.key /tmp/example.key
```

```
example.internal. 3600 IN DNSKEY 257 3 13 jkmSShfyY3nSww...
```

-
-
-
-
-

```
trust-anchors {  
    example.internal. static-key 257 3 13 "jkmS5hfyY3nSww...";  
};
```

```
$ delv @127.0.0.1 -a /tmp/example.key +root=example.internal  
noble.example.internal +multiline  
  
; fully validated  
noble.example.internal. 86400 IN A 192.168.1.11  
noble.example.internal. 86400 IN RRSIG A 13 3 86400 (  
                                20241106131533 20241023195023 48112 example.  
internal.  
                                5fL4apIwCD9kt4XbzzLLxMXy3mj8Li1WZu3qzlcBpERp  
                                lXPgLODbRrWyp7L81xEFnfhecKtEYv+6Y0XaSiVRug== )
```

-
-

Connecting the dots: the parent zone

DS record format

```
$ dnssec-dsfromkey /var/cache/bind/Kexample.internal.+013+48112.key
example.internal. IN DS 48112 13 2
1212DE7DA534556F1E11898F2C7A66736D5107962A19A0BFE1C2A67D6841962A
```

DNSKEY record format

```
$ cat /var/cache/bind/Kexample.internal.+013+48112.key
; This is a key-signing key, keyid 48112, for example.internal.
; Created: 20241023205023 (Wed Oct 23 20:50:23 2024)
; Publish: 20241023205023 (Wed Oct 23 20:50:23 2024)
; Activate: 20241023205023 (Wed Oct 23 20:50:23 2024)
; SyncPublish: 20241024215523 (Thu Oct 24 21:55:23 2024)
example.internal. 3600 IN DNSKEY 257 3 13 jkmS5hfyY3nSww4tD9Fy5d+GGc3A/
zR1CFUxmN8T2TKtkgWp8dusllM 7TrIZTEg6wZxmMs754/ftoTA6jmM1g==
```

```
example.internal. 3600 IN DNSKEY 257 3 13 (jkmS5hfyY3nSww4tD9Fy5d+GGc3A/
zR1CFUxmN8T2TKtkgWp8dusllM 7TrIZTEg6wZxmMs754/ftoTA6jmM1g==);
```

Further reading

- The DNSSEC Guide from bind9²⁰⁴
- Easy-Start Guide for Signing Authoritative Zones²⁰⁵
- Creating a Custom DNSSEC Policy²⁰⁶
- Detailed DNSSEC chapter from the bind9 documentation²⁰⁷
- *delv(1)*²⁰⁸
- Working with the parent zone²⁰⁹

²⁰⁴ <https://bind9.readthedocs.io/en/stable/dnssec-guide.html>

²⁰⁵ <https://bind9.readthedocs.io/en/stable/dnssec-guide.html#signing>

²⁰⁶ <https://bind9.readthedocs.io/en/stable/dnssec-guide.html#signing-custom-policy>

²⁰⁷ <https://bind9.readthedocs.io/en/stable/chapter5.html>

²⁰⁸ <https://manpages.ubuntu.com/manpages/resolute/man1/delv.1.html>

²⁰⁹ <https://bind9.readthedocs.io/en/latest/dnssec-guide.html#working-with-the-parent-zone>

Basic DNSSEC troubleshooting

DNSSEC

DNS

Handy “bad” and “good” DNSSEC domains

- - *dnssec-failed.org*²¹⁰
 -
- - *isc.org*²¹¹
 -

Logs

BIND9 server

```
$ dig @127.0.0.1 -t A www.dnssec-failed.org

; <<>> DiG 9.20.11-Ubuntu <<>> @127.0.0.1 -t A www.dnssec-failed.org
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: SERVFAIL, id: 26260
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags;; udp: 1232
; COOKIE: 6339d7228b8587f401000000671bc2eb2fe25bdf099ef1af (good)
;; QUESTION SECTION:
;www.dnssec-failed.org.      IN      A

;; Query time: 460 msec
;; SERVER: 127.0.0.1#53(127.0.0.1) (UDP)
;; WHEN: Fri Oct 25 16:10:19 UTC 2024
;; MSG SIZE rcvd: 78
```

²¹⁰ <http://dnssec-failed.org>

²¹¹ <http://isc.org>

```
$ journalctl -u named.service -f
```

```
(...)
```

```
named[286]: validating dnssec-failed.org/DNSKEY: no valid signature found (DS)
named[286]: no valid RRSIG resolving 'dnssec-failed.org/DNSKEY/IN': 68.87.85.132#53
named[286]: validating dnssec-failed.org/DNSKEY: no valid signature found (DS)
named[286]: no valid RRSIG resolving 'dnssec-failed.org/DNSKEY/IN': 68.87.68.244#53
named[286]: validating dnssec-failed.org/DNSKEY: no valid signature found (DS)
named[286]: no valid RRSIG resolving 'dnssec-failed.org/DNSKEY/IN': 68.87.76.228#53
named[286]: validating dnssec-failed.org/DNSKEY: no valid signature found (DS)
named[286]: no valid RRSIG resolving 'dnssec-failed.org/DNSKEY/IN': 68.87.72.244#53
named[286]: validating dnssec-failed.org/DNSKEY: no valid signature found (DS)
named[286]: no valid RRSIG resolving 'dnssec-failed.org/DNSKEY/IN': 69.252.250.103
#53
named[286]: broken trust chain resolving 'www.dnssec-failed.org/A/IN': 68.87.72.244
#53
```

Client-side tooling: dig

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-

*isc.org*²¹²

Note:

We need to (temporarily) enable DNSSEC validation and potentially set DNSSEC capable upstream servers (e.g. Cloudflare's 1.1.1.1, in case your upstream DNS server cannot properly handle it) in systemd-resolved, as shown below. After our experiments, we might want to restore the default settings, using: `resolvectl revert eth0`

²¹² <http://isc.org>

```
$ resolvectl dnssec eth0 true
```

```
$ resolvectl dns eth0 1.1.1.1
```

```
$ dig @127.0.0.53 -t A +dnssec +multiline isc.org

; <<>> DiG 9.20.11-Ubuntu <<>> @127.0.0.53 -t A +dnssec +multiline isc.org
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 23376
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 5, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags: do; udp: 65494
;; QUESTION SECTION:
;isc.org.                IN A

;; ANSWER SECTION:
isc.org.                299 IN A 151.101.2.217
isc.org.                299 IN A 151.101.66.217
isc.org.                299 IN A 151.101.130.217
isc.org.                299 IN A 151.101.194.217
isc.org.                299 IN RRSIG A 13 2 300 (
                        20260130043603 20260116034316 27566 isc.org.
                        kp+MQSLXw7x0qU072l4g3/Fq815uc0v5rN77kxizdxnL
                        EMwYbgtcvI+AgDIwhm9qgLhc2G06pUUF+puJCokxNA== )

;; Query time: 26 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Thu Jan 22 13:56:58 UTC 2026
;; MSG SIZE rcvd: 203
```

-
-
-

```
$ dig @127.0.0.53 -t A +dnssec +multiline dnssec-failed.org
```

```
; <<>> DiG 9.20.11-Ubuntu <<>> @127.0.0.53 -t A +dnssec +multiline dnssec-failed.org
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: SERVFAIL, id: 2028
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags: do; udp: 65494
; EDE: 9 (DNSKEY Missing): (no SEP matching the DS found for dnssec-failed.org.)
;; QUESTION SECTION:
;dnssec-failed.org.          IN A

;; Query time: 219 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Thu Jan 22 13:57:33 UTC 2026
;; MSG SIZE rcvd: 103
```

-
-
-

```
$ journalctl -u systemd-resolved.service -f

(...)
systemd-resolved[9203]: [ ] DNSSEC validation failed for question dnssec-failed.org
IN A: no-signature
```

```
$ dig @127.0.0.53 -t A +dnssec +cd +multiline dnssec-failed.org
```

```
; <<>> DiG 9.20.11-Ubuntu <<>> @127.0.0.53 -t A +dnssec +cd +multiline dnssec-  
failed.org  
; (1 server found)  
;; global options: +cmd  
;; Got answer:  
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 32064  
;; flags: qr rd ra cd; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1  
  
;; OPT PSEUDOSECTION:  
; EDNS: version: 0, flags: do; udp: 65494  
; EDE: 9 (DNSKEY Missing): (no SEP matching the DS found for dnssec-failed.org.)  
;; QUESTION SECTION:  
;dnssec-failed.org.          IN A  
  
;; ANSWER SECTION:  
dnssec-failed.org.          299 IN A 96.99.227.255  
dnssec-failed.org.          299 IN RRSIG A 5 2 300 (  
    20260204145120 20260118144620 44973 dnssec-failed.org.  
    U8NkuyOJBdX6vA/f+o4DcLA0x5YrCV0pMCGyCnXqaTTv  
    8+QhDuwRGsAhXYtAZTx4k6q7B2NbqPFsavHKbbsEY46J  
    SLpA1MIYydHTnUPHnu+foH4Dq/QctoXbUejMySdyZwPp  
    MU5hm5IkwN8LgKjoE0YL89mr4ueu01DtESG6P+I= )  
  
;; Query time: 119 msec  
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)  
;; WHEN: Thu Jan 22 13:58:49 UTC 2026  
;; MSG SIZE rcvd: 296
```

-
-
-

Digging a bit deeper: delv

```
$ delv @127.0.0.53 -t A +dnssec +multiline dnssec-failed.org
;; resolution failed: SERVFAIL
```

```
$ delv @127.0.0.53 -t A +dnssec +cd +multiline dnssec-failed.org
;; validating dnssec-failed.org/DNSKEY: no valid signature found (DS)
;; no valid RRSIG resolving 'dnssec-failed.org/DNSKEY/IN': 127.0.0.1#53
;; broken trust chain resolving 'dnssec-failed.org/A/IN': 127.0.0.1#53
;; resolution failed: broken trust chain
```

```
$ delv @127.0.0.53 -i -t A +dnssec +cd +multiline dnssec-failed.org
; answer not validated
dnssec-failed.org.      100 IN A 96.99.227.255
```

```
$ delv @127.0.0.53 -t A +multiline +cd isc.org
; fully validated
isc.org.      299 IN A 151.101.2.217
isc.org.      299 IN A 151.101.66.217
isc.org.      299 IN A 151.101.130.217
isc.org.      299 IN A 151.101.194.217
isc.org.      299 IN RRSIG A 13 2 300 (
20250908112301 20250825104017 27566 isc.org.
9ocLTno0Ub2NmUEXdyLv0zqwPBbbUVmT3RX4aP4BQQ+h
4g839JXuCKHufXSPkWh/GJe/MveP83dDvJkrMmEIzg== )
```

Client-side tooling: resolvectl

Note:

We need to (temporarily) enable DNSSEC validation and potentially set DNSSEC capable upstream servers (e.g. Cloudflare's 1.1.1.1, in case your upstream DNS server cannot properly handle it) in systemd-resolved, as shown below. After our experiments, we might want to restore the default settings, using: `resolvectl revert eth0`

```
$ sudo resolvectl flush-caches
```

```
$ sudo resolvectl reset-server-features
```

```
$ sudo resolvectl dnssec eth0 yes
```

```
$ sudo resolvectl dns eth0 1.1.1.1
```

```
$ resolvectl dnssec
```

```
Global: no  
Link 44 (eth0): yes
```

```
$ resolvectl query --type=MX isc.org
```

```
isc.org IN MX 5 mx.pao1.isc.org          -- link: eth0  
isc.org IN MX 10 mx.ams1.isc.org         -- link: eth0
```

```
-- Information acquired via protocol DNS in 3.0ms.  
-- Data is authenticated: yes; Data was acquired via local or encrypted transport:  
no  
-- Data from: network
```

Incorrect time

```
noble.example.internal. 86400 IN RRSIG A 13 3 86400 (  
20241106131533 20241023195023 48112 example.internal.  
5fL4apIwCD9kt4XbzzLLxMXY3mj8Li1WZu3qzlcBpERp  
LXPgLODbRrWyp7L81xEFnfhcKtEYv+6Y0Xa5iVRug== )
```


-
-

```
$ date
```

```
Tue Oct 10 10:10:19 UTC 2000
```

```
$ delv @10.10.17.229 -a example.internal.key +root=example.internal  
+multiline noble.example.internal
```

```
;; validating example.internal/DNSKEY: verify failed due to bad signature  
(keyid=48112): RRSIG validity period has not begun  
;; validating example.internal/DNSKEY: no valid signature found (DS)  
;; no valid RRSIG resolving 'example.internal/DNSKEY/IN': 10.10.17.229#53  
;; broken trust chain resolving 'noble.example.internal/A/IN': 10.10.17.229#53  
;; resolution failed: broken trust chain
```

```
named[3593]: managed-keys-zone: DNSKEY set for zone '.' could not be verified with current  
keys  
named[3593]: validating ./DNSKEY: verify failed due to bad signature (keyid=20326):  
RRSIG validity period has not begun  
named[3593]: validating ./DNSKEY: no valid signature found (DS)  
named[3593]: broken trust chain resolving './NS/IN': 199.7.83.42#53  
named[3593]: resolver priming query complete: broken trust chain
```

Third-party web-based diagnostics

- <https://dnsviz.net>
- <https://dnssec-debugger.verisignlabs.com>

Further reading

- [bind9's guide to DNSSEC troubleshooting](#)²¹³
- [How to validate DNSSEC using the command line](#)²¹⁴
- [delv\(1\)](#)²¹⁵
- [dig\(1\)](#)²¹⁶

How to use Open vSwitch with DPDK

DPDK is just a library

```
$ sudo apt-get install openvswitch-switch-dpdk
```

```
$ sudo update-alternatives --set ovs-vswitchd  
/usr/lib/openvswitch-switch-dpdk/ovs-vswitchd-dpdk
```

```
$ ovs-vsctl set Open_vSwitch . "other_config:dpdk-init=true"
```

•

```
$ ovs-vsctl set Open_vSwitch . "other_config:dpdk-lcore-mask=0x1"
```

•

```
$ ovs-vsctl set Open_vSwitch . "other_config:dpdk-alloc-mem=2048"
```

•

```
$ ovs-vsctl set Open_vSwitch .  
"other_config:dpdk-extra=--pci-whitelist=0000:04:00.0"
```

```
$ sudo service openvswitch-switch restart
```

²¹³ <https://bind9.readthedocs.io/en/latest/dnssec-guide.html#basic-dnssec-troubleshooting>

²¹⁴ <https://www.cyberciti.biz/faq/unix-linux-test-and-validate-dnssec-using-dig-command-line/>

²¹⁵ <https://manpages.ubuntu.com/manpages/resolute/man1/delv.1.html>

²¹⁶ <https://manpages.ubuntu.com/manpages/resolute/man1/dig.1.html>

Note:

You need to assign devices to DPDK-compatible drivers before restarting – see the DPDK section on *unassigning the default kernel drivers* (page 956).

Open vSwitch setup²¹⁷

Attach DPDK ports to Open vSwitch

```
$ ovs-vsctl add-br ovsdpkbr0 -- set bridge ovsdpkbr0 datapath_type=netdev
```

```
$ ovs-vsctl add-port ovsdpkbr0 dpdk0 -- set Interface dpdk0 type=dpdk  
"options:dpdk-devargs=${OVSDEV_PCIID}"
```

```
$ ovs-vsctl set Interface dpdk0 "options:n_rxq=2"
```

Open vSwitch DPDK to KVM guests

```
$ ovs-vsctl add-port ovsdpkbr0 vhost-user-1 -- set Interface vhost-user-1  
type=dpdkvhostuserclient  
"options:vhost-server-path=/var/run/vhostuserclient/vhost-user-client-1"
```

²¹⁷ <https://docs.openvswitch.org/en/latest/intro/install/dpdk/#setup-ovs>

```
<interface type='vhostuser'>
<source type='unix'
path='/var/run/vhostuserclient/vhost-user-client-1'
mode='server' />
<model type='virtio' />
</interface>
```

Tuning Open vSwitch-DPDK

```
$ ovs-vsctl set Open_vSwitch . other_config:n-dpdk-rxqs=2
```

```
$ ovs-vsctl set Open_vSwitch . other_config:pmd-cpu-mask=0x6
```

See also

Check the links to [EAL Command-line Options](#) and “Open vSwitch DPDK installation” at the end of this document for more information.

Support and troubleshooting

- [DPDK Mailing Lists](#)²¹⁸
- [OpenStack Mailing Lists](#)²¹⁹
- [DPDK Launchpad Area](#)²²⁰

²¹⁸ <https://www.dpdk.org/contribute/>

²¹⁹ <http://www.openvswitch.org/mlists/>

²²⁰ <https://launchpad.net/ubuntu/+source/dpdk>

```
ovs-ctl[3560]: EAL: PCI device 0000:04:00.1 on NUMA socket 0
ovs-ctl[3560]: EAL:   probe driver: 8086:1528 rte_ixgbe_pmd
ovs-ctl[3560]: EAL:   PCI memory mapped at 0x7f2140000000
ovs-ctl[3560]: EAL:   PCI memory mapped at 0x7f2140200000
```

#Note: This log was taken with dpdk 2.2 and openvswitch 2.5 but still looks quite similar (a bit extended) these days

Captions:

CMD: that you enter

SYSLOG: (Including EAL and OVS Messages)

OVS-LOG: (Openvswitch messages)

#PREPARATION

Bind an interface to DPDK UIO drivers, make Hugepages available, enable DPDK on OVS

CMD: sudo service openvswitch-switch restart

SYSLOG:

2016-01-22T08:58:31.372Z|00003|daemon_unix(monitor)|INFO|pid 3329 died, killed (Terminated), exiting

2016-01-22T08:58:33.377Z|00002|vlog|INFO|opened log file /var/log/openvswitch/ovs-vswitchd.log

2016-01-22T08:58:33.381Z|00003|ovs_numa|INFO|Discovered 12 CPU cores on NUMA node 0

2016-01-22T08:58:33.381Z|00004|ovs_numa|INFO|Discovered 1 NUMA nodes and 12 CPU cores

2016-01-22T08:58:33.381Z|00005|reconnect|INFO|unix:/var/run/openvswitch/db.sock: connecting...

2016-01-22T08:58:33.383Z|00006|reconnect|INFO|unix:/var/run/openvswitch/db.sock: connected

2016-01-22T08:58:33.386Z|00007|bridge|INFO|ovs-vswitchd (Open vSwitch) 2.5.0

OVS-LOG:

systemd[1]: Stopping Open vSwitch...

systemd[1]: Stopped Open vSwitch.

systemd[1]: Stopping Open vSwitch Internal Unit...

ovs-ctl[3541]: * Killing ovs-vswitchd (3329)

ovs-ctl[3541]: * Killing ovssdb-server (3318)

systemd[1]: Stopped Open vSwitch Internal Unit.

systemd[1]: Starting Open vSwitch Internal Unit...

ovs-ctl[3560]: * Starting ovssdb-server

ovs-vsctl: ovs|00001|vsctl|INFO|Called as ovs-vsctl --no-wait -- init -- set Open_vSwitch . db-version=7.12.1

ovs-vsctl: ovs|00001|vsctl|INFO|Called as ovs-vsctl --no-wait set Open_vSwitch . ovs-version=2.5.0 "external-ids:system-id=\"e7c5ba80-bb14-45c1-b8eb-628f3ad03903\"" "system-type=\"Ubuntu\"" "system-version=\"16.04-xenial\""

ovs-ctl[3560]: * Configuring Open vSwitch system IDs

ovs-ctl[3560]: 2016-01-22T08:58:31Z|00001|dpdk|INFO|No -vhost_sock_dir provided - defaulting to /var/run/openvswitch

ovs-vswitchd: ovs|00001|dpdk|INFO|No -vhost_sock_dir provided - defaulting to /var/run/openvswitch

ovs-ctl[3560]: EAL: Detected lcore 0 as core 0 on socket 0

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```
ovs-ctl[3560]: EAL: Detected lcore 1 as core 1 on socket 0
ovs-ctl[3560]: EAL: Detected lcore 2 as core 2 on socket 0
ovs-ctl[3560]: EAL: Detected lcore 3 as core 3 on socket 0
ovs-ctl[3560]: EAL: Detected lcore 4 as core 4 on socket 0
ovs-ctl[3560]: EAL: Detected lcore 5 as core 5 on socket 0
ovs-ctl[3560]: EAL: Detected lcore 6 as core 0 on socket 0
ovs-ctl[3560]: EAL: Detected lcore 7 as core 1 on socket 0
ovs-ctl[3560]: EAL: Detected lcore 8 as core 2 on socket 0
ovs-ctl[3560]: EAL: Detected lcore 9 as core 3 on socket 0
ovs-ctl[3560]: EAL: Detected lcore 10 as core 4 on socket 0
ovs-ctl[3560]: EAL: Detected lcore 11 as core 5 on socket 0
ovs-ctl[3560]: EAL: Support maximum 128 logical core(s) by configuration.
ovs-ctl[3560]: EAL: Detected 12 lcore(s)
ovs-ctl[3560]: EAL: VFIO modules not all loaded, skip VFIO support...
ovs-ctl[3560]: EAL: Setting up physically contiguous memory...
ovs-ctl[3560]: EAL: Ask a virtual area of 0x100000000 bytes
ovs-ctl[3560]: EAL: Virtual area found at 0x7f2040000000 (size = 0x100000000)
ovs-ctl[3560]: EAL: Requesting 4 pages of size 1024MB from socket 0
ovs-ctl[3560]: EAL: TSC frequency is ~2397202 KHz
ovs-vswitchd[3592]: EAL: TSC frequency is ~2397202 KHz
ovs-vswitchd[3592]: EAL: Master lcore 0 is ready (tid=fc6cbb00;cpuset=[0])
ovs-vswitchd[3592]: EAL: PCI device 0000:04:00.0 on NUMA socket 0
ovs-vswitchd[3592]: EAL:   probe driver: 8086:1528 rte_ixgbe_pmd
ovs-vswitchd[3592]: EAL:   Not managed by a supported kernel driver, skipped
ovs-vswitchd[3592]: EAL: PCI device 0000:04:00.1 on NUMA socket 0
ovs-vswitchd[3592]: EAL:   probe driver: 8086:1528 rte_ixgbe_pmd
ovs-vswitchd[3592]: EAL:   PCI memory mapped at 0x7f2140000000
ovs-vswitchd[3592]: EAL:   PCI memory mapped at 0x7f2140200000
ovs-ctl[3560]: EAL: Master lcore 0 is ready (tid=fc6cbb00;cpuset=[0])
ovs-ctl[3560]: EAL: PCI device 0000:04:00.0 on NUMA socket 0
ovs-ctl[3560]: EAL:   probe driver: 8086:1528 rte_ixgbe_pmd
ovs-ctl[3560]: EAL:   Not managed by a supported kernel driver, skipped
ovs-ctl[3560]: EAL: PCI device 0000:04:00.1 on NUMA socket 0
ovs-ctl[3560]: EAL:   probe driver: 8086:1528 rte_ixgbe_pmd
ovs-ctl[3560]: EAL:   PCI memory mapped at 0x7f2140000000
ovs-ctl[3560]: EAL:   PCI memory mapped at 0x7f2140200000
ovs-vswitchd[3592]: PMD: eth_ixgbe_dev_init(): MAC: 4, PHY: 3
ovs-vswitchd[3592]: PMD: eth_ixgbe_dev_init(): port 0 vendorID=0x8086 deviceID=0x1528
ovs-ctl[3560]: PMD: eth_ixgbe_dev_init(): MAC: 4, PHY: 3
ovs-ctl[3560]: PMD: eth_ixgbe_dev_init(): port 0 vendorID=0x8086 deviceID=0x1528
ovs-ctl[3560]: Zone 0: name:<RG_MP_log_history>, phys:0x83ffffdec0, len:0x2080,
virt:0x7f213ffffdec0, socket_id:0, flags:0
ovs-ctl[3560]: Zone 1: name:<MP_log_history>, phys:0x83fd73d40, len:0x28a0c0,
virt:0x7f213fd73d40, socket_id:0, flags:0
ovs-ctl[3560]: Zone 2: name:<rte_eth_dev_data>, phys:0x83fd43380, len:0x2f700,
virt:0x7f213fd43380, socket_id:0, flags:0
ovs-ctl[3560]: * Starting ovs-vswitchd
ovs-ctl[3560]: * Enabling remote OVSDB managers
systemd[1]: Started Open vSwitch Internal Unit.
systemd[1]: Starting Open vSwitch...
systemd[1]: Started Open vSwitch.

CMD: sudo ovs-vsctl add-br ovsdpdkbr0 -- set bridge ovsdpdkbr0 datapath_type=netdev

SYSLOG:
```

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```
2016-01-22T08:58:56.344Z|00008|memory|INFO|37256 kB peak resident set size after 24.5
seconds
2016-01-22T08:58:56.346Z|00009|ofproto_dpif|INFO|netdev@ovs-netdev: Datapath supports
recirculation
2016-01-22T08:58:56.346Z|00010|ofproto_dpif|INFO|netdev@ovs-netdev: MPLS label stack
length probed as 3
2016-01-22T08:58:56.346Z|00011|ofproto_dpif|INFO|netdev@ovs-netdev: Datapath supports
unique flow ids
2016-01-22T08:58:56.346Z|00012|ofproto_dpif|INFO|netdev@ovs-netdev: Datapath does not
support ct_state
2016-01-22T08:58:56.346Z|00013|ofproto_dpif|INFO|netdev@ovs-netdev: Datapath does not
support ct_zone
2016-01-22T08:58:56.346Z|00014|ofproto_dpif|INFO|netdev@ovs-netdev: Datapath does not
support ct_mark
2016-01-22T08:58:56.346Z|00015|ofproto_dpif|INFO|netdev@ovs-netdev: Datapath does not
support ct_label
2016-01-22T08:58:56.360Z|00016|bridge|INFO|bridge ovspdkbr0: added interface ovspdkbr0
on port 65534
2016-01-22T08:58:56.361Z|00017|bridge|INFO|bridge ovspdkbr0: using datapath ID
00005a4a1ed0a14d
2016-01-22T08:58:56.361Z|00018|connmgr|INFO|ovspdkbr0: added service controller "punix:/
var/run/openvswitch/ovspdkbr0.mgmt"
```

OVS-LOG:

```
ovs-vsctl: ovs|00001|vsctl|INFO|Called as ovs-vsctl add-br ovspdkbr0 -- set bridge
ovspdkbr0 datapath_type=netdev
systemd-udevd[3607]: Could not generate persistent MAC address for ovs-netdev: No such
file or directory
kernel: [50165.886554] device ovs-netdev entered promiscuous mode
kernel: [50165.901261] device ovspdkbr0 entered promiscuous mode
```

CMD: sudo ovs-vsctl add-port ovspdkbr0 dpdk0 -- set Interface dpdk0 type=dpdk

SYSLOG:

```
2016-01-22T08:59:06.369Z|00019|memory|INFO|peak resident set size grew 155% in last 10.0
seconds, from 37256 kB to 95008 kB
2016-01-22T08:59:06.369Z|00020|memory|INFO|handlers:4 ports:1 revalidators:2 rules:5
2016-01-22T08:59:30.989Z|00021|dpdk|INFO|Port 0: 8c:dc:d4:b3:6d:e9
2016-01-22T08:59:31.520Z|00022|dpdk|INFO|Port 0: 8c:dc:d4:b3:6d:e9
2016-01-22T08:59:31.521Z|00023|dpif_netdev|INFO|Created 1 pmd threads on numa node 0
2016-01-22T08:59:31.522Z|00001|dpif_netdev(pmd16)|INFO|Core 0 processing port 'dpdk0'
2016-01-22T08:59:31.522Z|00024|bridge|INFO|bridge ovspdkbr0: added interface dpdk0 on
port 1
2016-01-22T08:59:31.522Z|00025|bridge|INFO|bridge ovspdkbr0: using datapath ID
00008cdcd4b36de9
2016-01-22T08:59:31.523Z|00002|dpif_netdev(pmd16)|INFO|Core 0 processing port 'dpdk0'
```

OVS-LOG:

```
ovs-vsctl: ovs|00001|vsctl|INFO|Called as ovs-vsctl add-port ovspdkbr0 dpdk0 -- set
Interface dpdk0 type=dpdk
ovs-vswitchd[3595]: PMD: ixgbe_dev_tx_queue_setup(): sw_ring=0x7f211a79ebc0 hw_
ring=0x7f211a7a6c00 dma_addr=0x81a7a6c00
ovs-vswitchd[3595]: PMD: ixgbe_set_tx_function(): Using simple tx code path
ovs-vswitchd[3595]: PMD: ixgbe_set_tx_function(): Vector tx enabled.
ovs-vswitchd[3595]: PMD: ixgbe_dev_rx_queue_setup(): sw_ring=0x7f211a78a6c0 sw_sc_
```

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```
ring=0x7f211a786580 hw_ring=0x7f211a78e800 dma_addr=0x81a78e800
ovs-vswitchd[3595]: PMD: ixgbe_set_rx_function(): Vector rx enabled, please make sure RX
burst size no less than 4 (port=0).
ovs-vswitchd[3595]: PMD: ixgbe_dev_tx_queue_setup(): sw_ring=0x7f211a79ebc0 hw_
ring=0x7f211a7a6c00 dma_addr=0x81a7a6c00
...

CMD: sudo ovs-vsctl add-port ovspdkbr0 vhost-user-1 -- set Interface vhost-user-1
type=dpmkvhostuser

OVS-LOG:
2016-01-22T09:00:35.145Z|00026|dpdk|INFO|Socket /var/run/openvswitch/vhost-user-1 created
for vhost-user port vhost-user-1
2016-01-22T09:00:35.145Z|00003|dpif_netdev(pmd16)|INFO|Core 0 processing port 'dpdk0'
2016-01-22T09:00:35.145Z|00004|dpif_netdev(pmd16)|INFO|Core 0 processing port 'vhost-user-
1'
2016-01-22T09:00:35.145Z|00027|bridge|INFO|bridge ovspdkbr0: added interface vhost-user-1
on port 2

SYSLOG:
ovs-vsctl: ovs|00001|vsctl|INFO|Called as ovs-vsctl add-port ovspdkbr0 vhost-user-1 --
set Interface vhost-user-1 type=dpmkvhostuser
ovs-vswitchd[3595]: VHOST_CONFIG: socket created, fd:46
ovs-vswitchd[3595]: VHOST_CONFIG: bind to /var/run/openvswitch/vhost-user-1

Eventually we can see the poll thread in top
  PID USER      PR  NI   VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
 3595 root        10  -10 4975344 103936   9916 S 100.0   0.3   33:13.56 ovs-vswitchd
```

Further reading

- [DPDK documentation](#)²²¹
- [Linux DPDK user getting started](#)²²²
- [EAL command-line options](#)²²³
- [DPDK API documentation](#)²²⁴
- [Open Vswitch DPDK installation](#)²²⁵
- [Wikipedia's definition of DPDK](#)²²⁶

²²¹ <http://dpdk.org/doc>

²²² http://dpdk.org/doc/guides/linux_gsg/index.html

²²³ http://dpdk.org/doc/guides/testpmd_app_ug/run_app.html

²²⁴ <http://dpdk.org/doc/api/>

²²⁵ <https://github.com/openvswitch/ovs/blob/branch-2.5/INSTALL.DPDK.md>

²²⁶ https://en.wikipedia.org/wiki/Data_Plane_Development_Kit

Install and maintain DOCA-OFED

Note:

DOCA-OFED²²⁷ is NVIDIA's distribution of OpenFabrics Enterprise Distribution (OFED) drivers and userspace libraries for high-performance networking. DOCA-OFED is available for Ubuntu 26.04 LTS (Resolute Raccoon).

²²⁷ <https://developer.nvidia.com/networking/doca>

Available metapackages

Metapackage	Kernel flavor
doca-ofed-aws	aws
doca-ofed-aws-64k	aws-64k
doca-ofed-azure	azure
doca-ofed-azure-fde	azure-fde
doca-ofed-gcp	gcp
doca-ofed-gcp-64k	gcp-64k
doca-ofed-generic	generic
doca-ofed-generic-64k	generic-64k
doca-ofed-generic-64k-hwe-26.04	generic-64k-hwe-26.04
doca-ofed-generic-64k-hwe-26.04-edge	generic-64k-hwe-26.04-edge
doca-ofed-generic-hwe-26.04	generic-hwe-26.04
doca-ofed-generic-hwe-26.04-edge	generic-hwe-26.04-edge
doca-ofed-oracle	oracle
doca-ofed-oracle-64k	oracle-64k

Install DOCA-OFED

*uname(1)*²²⁸

```
$ uname -r
```

*add-apt-repository(1)*²²⁹

```
$ sudo add-apt-repository -y ppa:canonical-nvidia/doca-stable
```

```
$ sudo apt update
```

*apt(8)*²³⁰

```
$ sudo apt install -y doca-ofed-<your_kernel_flavor>
```

```
$ sudo apt install -y doca-ofed-generic
```

Maintain and upgrade DOCA-OFED

```
$ apt list --upgradable
```

```
$ sudo apt update && sudo apt full-upgrade
```

Warning:

Use `sudo apt full-upgrade` for DOCA-OFED updates. Do not use only `sudo apt upgrade` for this workflow. (DOCA-OFED updates from our PPA require uninstalling the old version's package names and installing the new version's package names (ex: `doca-ofed-userspace-3.3` is uninstalled, and `doca-ofed-userspace-3.4` is installed on update). `sudo apt upgrade` will hold such updates back.)

²²⁸ <https://manpages.ubuntu.com/manpages/resolute/man1/uname.1.html>

²²⁹ <https://manpages.ubuntu.com/manpages/resolute/man1/add-apt-repository.1.html>

²³⁰ <https://manpages.ubuntu.com/manpages/resolute/man8/apt.8.html>

Collect a sosreport for support

```
$ apt list --installed | grep doca-ofed-userspace
```

```
$ sudo apt install doca-sosreport-<doca_version>
```

DHCP

How to install and configure isc-kea

Kea²³¹

DHCP

refer to this guide instead

Install isc-kea

```
$ sudo apt install kea
```

-
-
-
-

DNS

LP: #2007312²³²

²³¹ <https://www.isc.org/kea/>

²³² <https://bugs.launchpad.net/bugs/2007312>

-
-
-

Configure kea-dhcp4

```
{
  "Dhcp4": {
    "interfaces-config": {
      "interfaces": ["eth4"]
    },
    "control-socket": {
      "socket-type": "unix",
      "socket-name": "/run/kea/kea4-ctrl-socket"
    },
    "lease-database": {
      "type": "memfile",
      "lfc-interval": 3600
    },
    "valid-lifetime": 600
    "max-valid-lifetime": 7200
    "subnet4": [{
      "id": 1,
      "subnet": "192.168.1.0/24",
      "pools": [{
        "pool": "192.168.1.150 - 192.168.1.200"
      }],
      "option-data": [{
        "name": "routers",
        "data": "192.168.1.254"
      },
      {
        "name": "domain-name-servers",
        "data": "192.168.1.1, 192.168.1.2"
      },
      {
        "name": "domain-name",
        "data": "mydomain.example"
      }
    ]
  }
}]
```

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```
}  
}
```

```
$ kea-shell --host 127.0.0.1 --port 8000 --auth-user kea-api --auth-password  
$(cat /etc/kea/kea-api-password) --service dhcp4 config-reload
```

```
[ { "result": 0, "text": "Configuration successful." } ]
```

```
DHCP4_DYNAMIC_RECONFIGURATION_SUCCESS dynamic server reconfiguration succeeded with file:  
/etc/kea/kea-dhcp4.conf
```

```
$ journalctl -u kea-dhcp4-server
```

```
$ systemctl restart kea-dhcp4-server
```

Further reading

- [ISC Kea Documentation](#)²³³

²³³ <https://kb.isc.org/docs/kea-administrator-reference-manual>

How to install and configure isc-dhcp-server

Warning:

The `isc-dhcp-server` package is deprecated and unsupported since Ubuntu 24.04 LTS. It is no longer supported by its vendor. Please use [Kea](https://www.isc.org/kea/)²³⁴ or [dnsmasq](https://thekelleys.org.uk/dnsmasq/doc.html)²³⁵ instead.

²³⁴ <https://www.isc.org/kea/>

²³⁵ <https://thekelleys.org.uk/dnsmasq/doc.html>

guide instead

DHCPD

refer to this

Install isc-dhcp-server

```
$ sudo apt install isc-dhcp-server
```

Note:

You can find diagnostic messages from `dhcpcd` in `syslog`.

Configure isc-dhcp-server

```
# minimal sample /etc/dhcp/dhcpd.conf
default-lease-time 600;
max-lease-time 7200;

subnet 192.168.1.0 netmask 255.255.255.0 {
    range 192.168.1.150 192.168.1.200;
    option routers 192.168.1.254;
    option domain-name-servers 192.168.1.1, 192.168.1.2;
    option domain-name "nydomain.example";
}
```

DNS

```
INTERFACESv4="eth4"
```

```
$ sudo systemctl restart isc-dhcp-server.service
```

Further reading

- *dhcpcd.conf(5)*²³⁶
- ISC dhcp-server²³⁷

Time synchronization

Synchronize time using chrony

synchronize time using systemd

Check status of chrony client

```
Local time: Mo 2025-06-16 15:21:46 CEST
Universal time: Mo 2025-06-16 13:21:46 UTC
RTC time: Mo 2025-06-16 13:21:46
Time zone: Europe/Berlin (CEST, +0200)
System clock synchronized: yes
NTP service: active
RTC in local TZ: no
```

²³⁶ <https://manpages.ubuntu.com/manpages/resolute/man5/dhcpd.conf.5.html>

²³⁷ <https://www.isc.org/dhcp/>

```
Reference ID      : B97DBE7B (2.ntp.ubuntu.com)
Stratum          : 3
Ref time (UTC)   : Mon Jun 16 13:06:04 2025
System time      : 0.000000004 seconds slow of NTP time
Last offset      : +0.001758954 seconds
RMS offset       : 0.017604901 seconds
Frequency        : 3.889 ppm slow
Residual freq    : +0.202 ppm
Skew             : 1.458 ppm
Root delay       : 0.022837413 seconds
Root dispersion  : 0.003050051 seconds
Update interval  : 1032.5 seconds
Leap status      : Normal
```

Network Time Security (NTS)

```
pool 1.ntp.ubuntu.com iburst maxsources 1 nts prefer
pool 2.ntp.ubuntu.com iburst maxsources 1 nts prefer
pool 3.ntp.ubuntu.com iburst maxsources 1 nts prefer
pool 4.ntp.ubuntu.com iburst maxsources 1 nts prefer
pool ntp-bootstrap.ubuntu.com iburst maxsources 1 nts certset 1
```

```
pool ntp-bootstrap.ubuntu.com iburst maxsources 1 nts certset 1
```

```
$ openssl s_client -connect 1.ntp.ubuntu.com:4460 -servername
1.ntp.ubuntu.com

Server certificate
...
subject=CN=ntp.ubuntu.com
issuer=C=US, O=Let's Encrypt, CN=R13
---
```



```
Server certificate
...
subject=CN=ntp-bootstrap.ubuntu.com
issuer=CN=ubuntu
```

```
ntstrustedcerts 1 /etc/chrony/nts-bootstrap-ubuntu.crt
```

```
$ sudo chronyc -N authdata
```

Name/IP address	Mode	KeyID	Type	KLen	Last	Atmp	NAK	Cook	CLen
1.ntp.ubuntu.com	NTS	6	30	128	14d	0	0	8	64
2.ntp.ubuntu.com	NTS	6	30	128	14d	0	0	8	64
3.ntp.ubuntu.com	NTS	1	30	128	27d	0	0	8	64
4.ntp.ubuntu.com	NTS	2	30	128	20d	0	0	5	64
ntp-bootstrap.ubuntu.com	NTS	3	30	128	7d	0	0	8	64

NTS related constraints

Warning:

If the network does not allow access to the Ubuntu NTS servers and required ports, with the default configuration is in place, chrony will not be able to adjust the system's clock. To revert to NTP, edit the configuration file in `/etc/chrony/sources.d/ubuntu-ntp-pools.sources` and revert to using the listed NTP servers in favor of the NTS ones.

Note:

The certificate for `ntp-bootstrap.ubuntu.com` is installed in `/etc/chrony/nts-bootstrap-ubuntu.crt` and is used specifically for the Ubuntu NTS bootstrap server, needed for when the clock is too far off. This is added to certificate set ID "1", and defined via `/etc/chrony/conf.d/ubuntu-nts.conf`.

Configure chrony

```
# Use NTS by default
# NTS uses an additional port to negotiate security: 4460/tcp
# The normal NTP port remains in use: 123/udp
pool 1.ntp.ubuntu.com iburst maxsources 1 nts prefer
pool 2.ntp.ubuntu.com iburst maxsources 1 nts prefer
pool 3.ntp.ubuntu.com iburst maxsources 1 nts prefer
pool 4.ntp.ubuntu.com iburst maxsources 1 nts prefer
# The bootstrap server is needed by systems without a hardware clock, or a very
# large initial clock offset. The specified certificate set is defined in
# /etc/chrony/conf.d/ubuntu-nts.conf.
pool ntp-bootstrap.ubuntu.com iburst maxsources 1 nts certset 1
```

Time sources provided by DHCP (option 42)

```
# Use time sources from DHCP.
# sourcedir /run/chrony-dhcp
```

Chrony time-daemon

```
$ systemctl status chrony.service

❏ chrony.service - chrony, an NTP client/server
   Loaded: loaded (/usr/lib/systemd/system/chrony.service; enabled; preset:
   enabled)
   Active: active (running) since Mon 2025-06-02 11:27:09 CEST; 2 weeks 0 days ago
     Docs: man:chronyd(8)
           man:chronyc(1)
           man:chrony.conf(5)
  Main PID: 36027 (chronyd)
    Tasks: 2 (limit: 28388)
   Memory: 5.8M (peak: 6.8M swap: 604.0K swap peak: 4.5M)
      CPU: 5.038s
   CGroup: /system.slice/chrony.service
           └─36027 /usr/sbin/chronyd -F 1
             └─36028 /usr/sbin/chronyd -F 1

Jun 02 11:27:09 questing chronyd[36027]: Using right/UTC timezone to obtain leap
second data
Jun 02 11:27:09 questing chronyd[36027]: Loaded seccomp filter (level 1)
Jun 02 11:27:09 questing chronyd[36027]: Added pool 1.ntp.ubuntu.com
Jun 02 11:27:09 questing chronyd[36027]: Added pool 2.ntp.ubuntu.com
Jun 02 11:27:09 questing chronyd[36027]: Added pool 3.ntp.ubuntu.com
Jun 02 11:27:09 questing chronyd[36027]: Added pool 4.ntp.ubuntu.com
Jun 02 11:27:09 questing chronyd[36027]: Added pool ntp-bootstrap.ubuntu.com
Jun 02 11:27:09 questing systemd[1]: Started chrony.service - chrony, an NTP
client/server.
```

*chrony.conf(5)*²³⁸

```
$ sudo systemctl restart chrony.service
```

Next steps

[how to configure your chrony setup](#)

²³⁸ <https://manpages.ubuntu.com/manpages/resolute/man5/chrony.conf.5.html>

Further reading

- *chronyc(1)*²³⁹
- *chronyd(8)*²⁴⁰
- *chrony.conf(5)*²⁴¹

Synchronize time using `timedatectl` and `timesyncd`

Network Time Security (NTS)

client *serve the Network Time Protocol*

```
$ sudo apt-mark auto chrony && apt install systemd-timesyncd
```

Note:

If `chrony` is installed, `timedatectl` steps back to let `chrony` handle timekeeping. This ensures that no two time-syncing services will be in conflict.

Check status of `timedatectl`

```
Local time: Wed 2023-06-14 12:05:11 BST
Universal time: Wed 2023-06-14 11:05:11 UTC
RTC time: Wed 2023-06-14 11:05:11
Time zone: Europe/Isle_of_Man (BST, +0100)
System clock synchronized: yes
NTP service: active
RTC in local TZ: no
```

```
[...]
systemd-timesyncd.service active: no
```

²³⁹ <https://manpages.ubuntu.com/manpages/resolute/man1/chronyc.1.html>

²⁴⁰ <https://manpages.ubuntu.com/manpages/resolute/man8/chronyd.8.html>

²⁴¹ <https://manpages.ubuntu.com/manpages/resolute/man5/chrony.conf.5.html>

Configure timedatectl

Check status of timesyncd

```
$ systemctl status systemd-timesyncd

systemd-timesyncd.service - Network Time Synchronization
   Loaded: loaded (/lib/systemd/system/systemd-timesyncd.service; enabled;
   vendor preset: enabled)
   Active: active (running) since Fri 2018-02-23 08:55:46 UTC; 10s ago
     Docs: man:systemd-timesyncd.service(8)
  Main PID: 3744 (systemd-timesyn)
    Status: "Synchronized to time server 91.189.89.198:123 (ntp.ubuntu.com)."
```

Tasks: 2 (limit: 4915)
CGroup: /system.slice/systemd-timesyncd.service
|-3744 /lib/systemd/systemd-timesyncd

```
Feb 23 08:55:46 test-host systemd[1]: Starting Network Time Synchronization...
Feb 23 08:55:46 test-host systemd[1]: Started Network Time Synchronization.
Feb 23 08:55:46 test-host systemd-timesyncd[3744]: Synchronized to time server
91.189.89.198:123 (ntp.ubuntu.com).
```

Configure timesyncd

Next steps

how to install and configure your setup

Further reading

- `timedatectl(1)`²⁴²
- `systemd-timesyncd.service(8)`²⁴³
- [Ubuntu Time wiki page](#)²⁴⁴

How to serve the Network Time Protocol with Chrony

functioning as a client

Install chronyd

```
$ sudo apt install chrony
```

-
-

²⁴² <https://manpages.ubuntu.com/manpages/resolute/man1/timedatectl.1.html>

²⁴³ <https://manpages.ubuntu.com/manpages/resolute/man8/systemd-timesyncd.service.8.html>

²⁴⁴ <https://help.ubuntu.com/community/UbuntuTime>

Enable serving the Network Time Protocol

```
allow 1.2.3.4
```

*chrony.conf(5)*²⁴⁵

View chrony status

```
MS Name/IP address          Stratum Poll Reach LastRx Last sample
=====
^+ gamma.rueckgr.at         2   8   377   135  -1048us[-1048us] +/- 29ms
^- 2b.ncomputers.org        2   8   377   204  -1141us[-1124us] +/- 50ms
^+ www.kashra.com           2   8   377   139  +3483us[+3483us] +/- 18ms
^+ stratum2-4.NTP.TechFak.U> 2   8   377   143  -2090us[-2073us] +/- 19ms
^- zepto.mcl.gg             2   7   377    9   -774us[ -774us] +/- 29ms
^- mirrorhost.pw           2   7   377   78   -660us[ -660us] +/- 53ms
^- atto.mcl.gg              2   7   377    8   -823us[ -823us] +/- 50ms
^- static.140.107.46.78.cli> 2   8   377    9  -1503us[-1503us] +/- 45ms
^- 4.53.160.75              2   8   377   137   -11ms[ -11ms] +/- 117ms
^- 37.44.185.42             3   7   377   10  -3274us[-3274us] +/- 70ms
^- bagnikita.com            2   7   377   74  +3131us[+3131us] +/- 71ms
^- europa.ellipse.net       2   8   377   204   -790us[ -773us] +/- 97ms
^- tethys.hot-chilli.net    2   8   377   141   -797us[ -797us] +/- 59ms
^- 66-232-97-8.static.hvvc.> 2   7   377   206  +1669us[+1686us] +/- 133ms
^+ 85.199.214.102           1   8   377   205  +175us[ +192us] +/- 12ms
^* 46-243-26-34.tangos.nl   1   8   377   141  -123us[ -106us] +/- 10ms
^- pugot.canonical.com      2   8   377   21    -95us[  -95us] +/- 57ms
^- alphyn.canonical.com     2   6   377   23  -1569us[-1569us] +/- 79ms
^- golem.canonical.com      2   7   377   92  -1018us[-1018us] +/- 31ms
^- chilipepper.canonical.com 2   8   377   21  -1106us[-1106us] +/- 27ms
```

²⁴⁵ <https://manpages.ubuntu.com/manpages/resolute/man5/chrony.conf.5.html>

210 Number of sources = 20

Name/IP Address	NP	NR	Span	Frequency	Freq Skew	Offset	Std Dev
gamma.rueckgr.at	25	15	32m	-0.007	0.142	-878us	106us
2b.ncomputers.org	26	16	35m	-0.132	0.283	-1169us	256us
www.kashra.com	25	15	32m	-0.092	0.259	+3426us	195us
stratum2-4.NTP.TechFak.U>	25	14	32m	-0.018	0.130	-2056us	96us
zepto.mcl.gg	13	11	21m	+0.148	0.196	-683us	66us
mirrorhost.pw	6	5	64s	+0.117	0.445	-591us	19us
atto.mcl.gg	21	13	25m	-0.069	0.199	-904us	103us
static.140.107.46.78.cli>	25	18	34m	-0.005	0.094	-1526us	78us
4.53.160.75	25	10	32m	+0.412	0.110	-11ms	84us
37.44.185.42	24	12	30m	-0.983	0.173	-3718us	122us
bagnikita.com	17	7	31m	-0.132	0.217	+3527us	139us
europa.ellipse.net	26	15	35m	+0.038	0.553	-473us	424us
tethys.hot-chilli.net	25	11	32m	-0.094	0.110	-864us	88us
66-232-97-8.static.hvvc.>	20	11	35m	-0.116	0.165	+1561us	109us
85.199.214.102	26	11	35m	-0.054	0.390	+129us	343us
46-243-26-34.tangos.nl	25	16	32m	+0.129	0.297	-307us	198us
pugot.canonical.com	25	14	34m	-0.271	0.176	-143us	135us
alphyn.canonical.com	17	11	1100	-0.087	0.360	-1749us	114us
golem.canonical.com	23	12	30m	+0.057	0.370	-988us	229us
chilipepper.canonical.com	25	18	34m	-0.084	0.224	-1116us	169us

Pulse-Per-Second (PPS) support

GPSD

jitter

Example configuration for GPSD to feed chrony

```
$ sudo apt install gpsd chrony
```

```
$ sudo apt install pps-tools gpsd-clients
```


GPSD hardware²⁴⁶*dmesg*

```
[ 52.442199] usb 1-1.1: new full-speed USB device number 3 using xhci_hcd
[ 52.546639] usb 1-1.1: New USB device found, idVendor=067b, idProduct=2303,
bcdDevice= 4.00
[ 52.546654] usb 1-1.1: New USB device strings: Mfr=1, Product=2, SerialNumber=0
[ 52.546665] usb 1-1.1: Product: USB-Serial Controller D
[ 52.546675] usb 1-1.1: Manufacturer: Prolific Technology Inc.
[ 52.602103] usbcore: registered new interface driver usbserial_generic
[ 52.602244] usbserial: USB Serial support registered for generic
[ 52.609471] usbcore: registered new interface driver pl2303
[ 52.609503] usbserial: USB Serial support registered for pl2303
[ 52.609564] pl2303 1-1.1:1.0: pl2303 converter detected
[ 52.618366] usb 1-1.1: pl2303 converter now attached to ttyUSB0
```

```
refclock SHM 0 refid GPS precision 1e-1 offset 0.9999 delay 0.2
refclock SOCK /var/run/chrony.ttyUSB0.sock refid PPS
```

```
$ sudo systemctl restart chrony
```

```
DEVICES="/dev/ttyUSB0"
```

•

—

•

—

²⁴⁶ <https://gpsd.gitlab.io/gpsd/hardware.html>

```
pps_ldisc: PPS line discipline registered
pps pps0: new PPS source usbserial0
pps pps0: source "/dev/ttyUSB0" added
```

```
pps0: name=usbserial0 path=/dev/ttyUSB0
```

```
...
| Status:          3D FIX (7 secs) ...
```

```
trying PPS source "/dev/pps0"
found PPS source "/dev/pps0"
ok, found 1 source(s), now start fetching data...
source 0 - assert 1588140739.099526246, sequence: 69 - clear 1588140739.999663721,
sequence: 70
source 0 - assert 1588140740.099661485, sequence: 70 - clear 1588140739.999663721,
sequence: 70
source 0 - assert 1588140740.099661485, sequence: 70 - clear 1588140740.999786664,
sequence: 71
source 0 - assert 1588140741.099792447, sequence: 71 - clear 1588140740.999786664,
sequence: 71
```

```
chronyc> sources
```

```
210 Number of sources = 10
```

```
MS Name/IP address          Stratum Poll Reach LastRx Last sample
```

```
=====
#? GPS                      0    4    0    -    +0ns[  +0ns] +/-    0ns
#? PPS                      0    4    0    -    +0ns[  +0ns] +/-    0ns
```

```
chronyc> sources
```

```
210 Number of sources = 10
```

```
MS Name/IP address      Stratum Poll Reach LastRx Last sample
```

```
=====
```

#x GPS	0	4	177	24	-876ms[-876ms]	+/- 200ms
#- PPS	0	4	177	21	+916us[+916us]	+/- 63us
^- chilipepper.canonical.com	2	6	37	53	+33us[+33us]	+/- 33ms

```
chronyc> sources
```

```
210 Number of sources = 10
```

```
MS Name/IP address      Stratum Poll Reach LastRx Last sample
```

```
=====
```

#x GPS	0	4	377	20	-884ms[-884ms]	+/- 200ms
#* PPS	0	4	377	18	+6677ns[+52us]	+/- 58us
^- alphyn.canonical.com	2	6	377	20	-1303us[-1258us]	+/- 114ms

```
chronyc> sources
```

```
210 Number of sources = 11
```

```
MS Name/IP address      Stratum Poll Reach LastRx Last sample
```

```
=====
```

#? GPS	0	4	0	-	+0ns[+0ns]	+/- 0ns
#+ PPS	0	4	377	22	+154us[+154us]	+/- 8561us
^* chilipepper.canonical.com	2	6	377	50	-353us[-300us]	+/- 44ms

```
chronyc> sourcestats
```

```
210 Number of sources = 10
```

```
Name/IP Address        NP  NR  Span  Frequency  Freq Skew  Offset  Std Dev
```

```
=====
```

GPS	20	9	302	+1.993	11.501	-868ms	1208us
PPS	6	3	78	+0.324	5.009	+3365ns	41us
golem.canonical.com	15	10	783	+0.859	0.509	-750us	108us

```
$ sudo ntpshmmon -o
```

```
ntpshmmon: version 3.20
```

#	Name	Offset	Clock	Real	L	Prc
sample	NTP1	0.000223854	1588265805.000223854	1588265805.000000000	0	-10
sample	NTP0	0.125691783	1588265805.125999851	1588265805.000308068	0	-20
sample	NTP1	0.000349341	1588265806.000349341	1588265806.000000000	0	-10
sample	NTP0	0.130326636	1588265806.130634945	1588265806.000308309	0	-20
sample	NTP1	0.000485216	1588265807.000485216	1588265807.000000000	0	-10

NTS Support

curity “NTS”²⁴⁷

Network Time Secu-

NTS server

```
ntsservercert /etc/chrony/fullchain.pem  
ntsserverkey /etc/chrony/privkey.pem
```

```
$ sudo chown _chrony:_chrony /etc/chrony/*.pem
```

```
$ sudo chronyc -N serverstats
```

²⁴⁷ <https://web.archive.org/web/20241130024605/http://networktimesecurity.org/>

```
NTP packets received      : 213
NTP packets dropped      : 0
Command packets received  : 117
Command packets dropped   : 0
Client log records dropped : 0
NTS-KE connections accepted: 2
NTS-KE connections dropped : 0
Authenticated NTP packets : 197
```

```
$ sudo chronyc -N clients -k
```

Hostname	NTP	Drop	Int	IntL	Last	NTS-KE	Drop	Int	Last
=====	=====	=====	=====	=====	=====	=====	=====	=====	=====
10.172.196.173	197	0	10	-	595	2	0	5	48h
...									

*chrony.conf(5)*²⁴⁸

About certificate placement

chrony, by default, is isolated via AppArmor and uses a number of protect* features of systemd. Due to that, there are not many paths chrony can access for the certificates. But /etc/chrony/* is allowed as read-only and that is enough. Check /etc/apparmor.d/usr.sbin.chrond if you want other paths or allow custom paths in /etc/apparmor.d/local/usr.sbin.chrond.

Further reading

- [chrony FAQ](#)²⁴⁹
- [ntp.org: home of the Network Time Protocol project](#)²⁵⁰
- [pool.ntp.org: project of virtual cluster of timeservers](#)²⁵¹
- [Freedesktop.org info on timedatectl](#)²⁵²
- [Freedesktop.org info on systemd-timesyncd service](#)²⁵³
- [Feeding chrony from GPSD](#)²⁵⁴
- [Ubuntu Time](#)²⁵⁵

²⁴⁸ <https://manpages.ubuntu.com/manpages/resolute/man5/chrony.conf.5.html>

²⁴⁹ <https://chrony-project.org/faq.html>

²⁵⁰ <http://www.ntp.org/>

²⁵¹ <https://www.ntppool.org/en/>

²⁵² <https://www.freedesktop.org/software/systemd/man/latest/timedatectl.html>

²⁵³ <https://www.freedesktop.org/software/systemd/man/latest/systemd-timesyncd.service.html>

²⁵⁴ https://gpsd.gitlab.io/gpsd/gpsd-time-service-howto.html#_feeding_chrony_from_gpsd

²⁵⁵ <https://help.ubuntu.com/community/UbuntuTime>

Network shares

Network File System (NFS)

-
-
-

Warning:

If you use **NFS** and **authd** at the same time, you must add a Kerberos configuration on both the client and the server. Otherwise, you will encounter permission issues due to mismatched user and group identifiers.
For details, see [Using authd with NFS](https://documentation.ubuntu.com/authd/stable-docs/howto/use-with-nfs/)²⁵⁶.

²⁵⁶ <https://documentation.ubuntu.com/authd/stable-docs/howto/use-with-nfs/>

Installation

```
$ sudo apt install nfs-kernel-server
```

```
$ sudo systemctl start nfs-kernel-server.service
```

Configuration

```
/srv      *(ro,sync,subtree_check)
/home     *.hostname.com(rw,sync,no_subtree_check)
/scratch  *(rw,async,no_subtree_check,no_root_squash)
```

```
$ sudo mkdir /scratch
```

```
$ sudo exportfs -a
```

hostname

filesystem

*exports(5)*²⁵⁷

²⁵⁷ <https://manpages.ubuntu.com/manpages/resolute/man5/exports.5.html>

NFS Client Configuration

```
$ sudo apt install nfs-common
```

```
$ sudo mkdir /opt/example
```

```
$ sudo mount example.hostname.com:/srv /opt/example
```

Warning:

The mount point directory `/opt/example` must exist. There should be no files or subdirectories in the `/opt/example` directory, else they will become inaccessible until the `nfs` filesystem is unmounted.

```
example.hostname.com:/srv /opt/example nfs
```

Note:

Details about all the NFS related options that can go into `/etc/fstab` can be found in the [*nfs\(5\)*²⁵⁸](#) manual page.

²⁵⁸ <https://manpages.ubuntu.com/manpages/resolute/man5/nfs.5.html>

```
mount: (hint) your fstab has been modified, but systemd still uses
the old version; use 'systemctl daemon-reload' to reload.
```

```
$ sudo systemctl daemon-reload
```


Advanced configuration

*nfs.conf(5)*²⁵⁹

*nfsconf(8)*²⁶⁰

Restarting NFS services

Service	nfs-utils. service	nfs-server. service	config file
nfs-blkmap	PartOf		nfs.conf
nfs-mountd		BindTo	nfs.conf
nfsdclld			
nfs-idmapd		BindTo	nfs.conf, idmapd.conf
rpc-gssd	PartOf		nfs.conf
rpc-statd	PartOf		nfs.conf
rpc-svcgssd	PartOf	BindTo	nfs.conf

²⁵⁹ <https://manpages.ubuntu.com/manpages/resolute/man5/nfs.conf.5.html>

²⁶⁰ <https://manpages.ubuntu.com/manpages/resolute/man8/nfsconf.8.html>

*nfs.systemd(7)*²⁶¹

NFS with Kerberos

NFS server (using Kerberos)

- *DNS*
-
-
-
-

```
$ sudo apt install krb5-user
```

```
$ kadmin -p ubuntu/admin -q "addprinc -randkey nfs/nfs-server.vms"
```

```
$ sudo kadmin -p ubuntu/admin -q "ktadd nfs/nfs-server.vms"
```

Authenticating as principal ubuntu/admin with password.

Password for ubuntu/admin@VMS:

Entry for principal nfs/nfs-server.vms with kvno 2, encryption type aes256-cts-hmac-sha1-96 added to keytab FILE:/etc/krb5.keytab.

Entry for principal nfs/nfs-server.vms with kvno 2, encryption type aes128-cts-hmac-sha1-96 added to keytab FILE:/etc/krb5.keytab.

²⁶¹ <https://manpages.ubuntu.com/manpages/resolute/man7/nfs.systemd.7.html>

```
$ sudo klist -k
```

```
Keytab name: FILE:/etc/krb5.keytab  
KVNO Principal
```

```
-----  
  2 nfs/nfs-server.vms@VMS  
  2 nfs/nfs-server.vms@VMS
```

```
$ sudo apt install nfs-kernel-server
```

```
$ sudo systemctl restart nfs-server
```

```
/storage *(rw,sync,no_subtree_check,sec=krb5p)
```

```
$ sudo exportfs -rav
```

```
exporting */:/storage
```

exports(5)²⁶²

-
-
-

²⁶² <https://manpages.ubuntu.com/manpages/resolute/man5/exports.5.html>

NFS client (using Kerberos)

```
$ sudo apt install krb5-user
```

```
$ kadmin -p ubuntu/admin -q "addprinc -randkey host/nfs-client.vms"
```

```
$ sudo kadmin -p ubuntu/admin -q "ktadd host/nfs-client.vms"
```

```
$ sudo apt install nfs-common
```

```
$ sudo systemctl restart nfs-client.target
```

```
$ sudo mount nfs-server:/storage /mnt
```

```
# mount nfs-server:/storage /mnt
# ls -l /mnt/*
-rw-r--r-- 1 root root 0 Apr  5 14:50 /mnt/hello-from-nfs-server.txt
# klist
klist: No credentials cache found (filename: /tmp/krb5cc_0)
```

```
# sudo -u ubuntu -i
$ ls -l /mnt/*
ls: cannot access '/mnt/*': Permission denied
```

```
$ kinit
```

```
Password for ubuntu@VMS:
```

```
$ ls -l /mnt/*
```

```
-rw-r--r-- 1 root root 0 Apr  5 14:50 /mnt/hello-from-nfs-server.txt
```

```
$ klist
```

```
Ticket cache: FILE:/tmp/krb5cc_1000
```

```
Default principal: ubuntu@VMS
```

Valid starting	Expires	Service principal
04/05/22 17:48:50	04/06/22 03:48:50	krbtgt/VMS@VMS
	renew until 04/06/22 17:48:48	
04/05/22 17:48:52	04/06/22 03:48:50	nfs/nfs-server.vms@
	renew until 04/06/22 17:48:48	
Ticket server: nfs/nfs-server.vms@VMS		

rpc.gssd(8)²⁶³

•

Warning:

Note that this prevents automatic NFS mounts via `/etc/fstab`, unless a Kerberos ticket is obtained before.

[gssd]

`use-machine-creds=0`

²⁶³ <https://manpages.ubuntu.com/manpages/resolute/man8/rpc.gssd.8.html>

Upgrading from Ubuntu 20.04 LTS and earlier

Warning:

The `NEED_*` parameters have no effect on systemd-based installations, like Ubuntu 20.04 LTS (Focal) and Ubuntu 18.04 LTS (Bionic). In those systems, to control whether a service should be running or not, use `systemctl enable` or `systemctl disable`, respectively.

-
-

file a bug for the `nfs-utils` package²⁶⁴

²⁶⁴ <https://bugs.launchpad.net/ubuntu/+source/nfs-utils/+filebug>

Samba

Provisioning a Samba Active Directory Domain Controller

See also

For more information on why the Samba AD/DC server should not be used to provide file and print services, please refer to this [list of reasons and caveats in the Samba Wiki](#)²⁶⁵

Installation

```
$ sudo apt install samba-ad-dc krb5-user bind9-dnsutils
```

```
$ sudo systemctl disable --now smbd nmbd winbind
```

```
$ sudo systemctl mask smbd nmbd winbind
```

```
$ sudo systemctl unmask samba-ad-dc
```

```
$ sudo systemctl enable samba-ad-dc
```

²⁶⁵ [https://wiki.samba.org/index.php/Setting_up_Samba_as_an_Active_Directory_Domain_Controller#Using_the_Domain_Controller_as_a_File_Server_\(Optional\)](https://wiki.samba.org/index.php/Setting_up_Samba_as_an_Active_Directory_Domain_Controller#Using_the_Domain_Controller_as_a_File_Server_(Optional))

Note:

A Samba AD/DC deployment represents a collection of services connected to each other, and needs its own specific systemd service unit.

```
$ sudo mv /etc/samba/smb.conf /etc/samba/smb.conf.orig
```

Provisioning

-
-
-

```
$ sudo samba-tool domain provision \ > --domain EXAMPLE \ >  
--realm=EXAMPLE.INTERNAL \ > --adminpass=Passw0rd \ > --server-role=dc \ >  
--use-rfc2307 \ > --dns-backend=SAMBA_INTERNAL
```

Warning:

Providing passwords in the command line is generally unsafe. Other users on the system who can see the process listing can spot the password, and it will also be saved in the shell history, unless the command starts with a blank space.

```
(...)  
INFO ... #498: Server Role:      active directory domain controller  
INFO ... #499: Hostname:        ad  
INFO ... #500: NetBIOS Domain:  EXAMPLE  
INFO ... #501: DNS Domain:      example.internal  
INFO ... #502: DOMAIN SID:      S-1-5-21-2373640847-2123283686-338028823
```



```
INFO ... #497: Admin password: sbruR-Py>];k=KdN1H58PB#
```

Post-installation steps

DNS

```
[global]  
dns forwarder = 127.0.0.53
```

```
Global  
    Protocols: -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported  
    resolv.conf mode: stub  
  
Link 2 (enp5s0)  
    Current Scopes: DNS  
        Protocols: +DefaultRoute -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported  
Current DNS Server: 10.10.17.1  
    DNS Servers: 10.10.17.1  
    DNS Domain: lxd
```

```
[global]  
dns forwarder = 10.10.17.1
```

```
$ sudo unlink /etc/resolv.conf
```

Note:

This will make sudo issue complaints about DNS from this point on, until the Samba DNS service is up and running.

```
nameserver 127.0.0.1  
search example.internal
```

```
$ sudo systemctl disable --now systemd-resolved
```

```
$ sudo cp -f /var/lib/samba/private/krb5.conf /etc/krb5.conf
```

```
$ sudo systemctl start samba-ad-dc
```

Verification

Kerberos authentication

```
$ kinit Administrator
```

```
Password for Administrator@EXAMPLE.INTERNAL:
```

See also

If you are not familiar with Kerberos, please see our [Introduction to Kerberos](#) (page 872).

```
Ticket cache: FILE:/tmp/krb5cc_1000
Default principal: Administrator@EXAMPLE.INTERNAL

Valid starting    Expires          Service principal
07/24/24 13:52:33 07/24/24 23:52:33 krbtgt/EXAMPLE.INTERNAL@EXAMPLE.INTERNAL
    renew until 07/25/24 13:52:02
```

DNS tests

name

host-

```
$ hostname
```

```
$ hostname -f
```

Server Information

```
$ samba-tool dns serverinfo $(hostname -f)

dwVersion           : 0xece0205
fBootMethod         : DNS_BOOT_METHOD_DIRECTORY
fAdminConfigured    : FALSE
fAllowUpdate        : TRUE
fDsAvailable        : TRUE
pszServerName       : AD.example.internal
pszDsContainer      : CN=MicrosoftDNS,DC=DomainDnsZones,DC=example,
DC=internal
(...)
```

```
Ticket cache: FILE:/tmp/krb5cc_1000
Default principal: Administrator@EXAMPLE.INTERNAL

Valid starting    Expires          Service principal
07/24/24 14:29:55 07/25/24 00:29:55 krbtgt/EXAMPLE.INTERNAL@EXAMPLE.INTERNAL
        renew until 07/25/24 14:29:53
07/24/24 14:29:59 07/25/24 00:29:55 host/ad.example.internal@EXAMPLE.INTERNAL
        renew until 07/25/24 14:29:53
```

DNS records

```
for srv in _ldap._tcp._kerberos._tcp._kerberos._udp._kpasswd._udp; do
    echo -n "${srv}.example.internal: "
    dig @localhost +short -t SRV ${srv}.example.internal
done
```

```
_ldap._tcp.example.internal: 0 100 389 ad.example.internal.
_kerberos._tcp.example.internal: 0 100 88 ad.example.internal.
_kerberos._udp.example.internal: 0 100 88 ad.example.internal.
_kpasswd._udp.example.internal: 0 100 464 ad.example.internal.
```

```
$ dig @localhost +short -t A $(hostname -f)
```

User creation test

```
$ samba-tool user add noble
```

Note:

samba-tool creates **Samba** users, not local Linux users! These Samba users only exist for domain joined computers and other SMB connections/shares.

```
$ sudo samba-tool domain passwordsettings show

Password information for domain 'DC=example,DC=internal'

Password complexity: on
Store plaintext passwords: off
Password history length: 24
Minimum password length: 7
Minimum password age (days): 1
Maximum password age (days): 42
Account lockout duration (mins): 30
Account lockout threshold (attempts): 0
Reset account lockout after (mins): 30
```

```
$ sudo samba-tool domain passwordsettings set --min-pwd-length=12
```

```
$ samba-tool domain passwordsettings set --help
```

Next steps

Further reading

- [Active Directory Domain Services Overview](#)²⁶⁶
- *samba-tool(8)*²⁶⁷
- - *Choosing an integration method*

²⁶⁶ <https://learn.microsoft.com/en-us/windows-server/identity/ad-ds/get-started/virtual-dc/active-directory-domain-services-overview>

²⁶⁷ <https://manpages.ubuntu.com/manpages/resolute/man8/samba-tool.8.html>

- *Joining a Member Server*
- *Joining Workstations (without Samba services)*

Member server in an Active Directory domain

SSSD

Network User Authentication with

See also

Samba itself has the necessary tooling to join an Active Directory domain. It requires a sequence of manual steps and configuration file editing, which is **thoroughly documented on the Samba wiki**²⁶⁸. It's useful to read that documentation to get an idea of the steps necessary, and the decisions you will need to make.

Use realmd to join the Active Directory domain

Install realmd

```
$ sudo apt install realmd samba
```

DNS

*FQDN
hostname*

```
$ sudo hostnamectl hostname <yourfqdn>
```

²⁶⁸ https://wiki.samba.org/index.php/Setting_up_Samba_as_a_Domain_Member

Verify the AD server

```
$ sudo realm discover internal.example.fake

internal.example.fake
  type: kerberos
  realm-name: INTERNAL.EXAMPLE.FAKE
  domain-name: internal.example.fake
  configured: no
  server-software: active-directory
  client-software: sssd
  required-package: sssd-tools
  required-package: sssd
  required-package: libnss-sss
  required-package: libpam-sss
  required-package: adcli
  required-package: samba-common-bin
```

Join the AD domain

```
$ sudo realm join -v --membership-software=samba --client-software=winbind
internal.example.fake

* Resolving: _ldap._tcp.internal.example.fake
* Performing LDAP DSE lookup on: 10.0.16.5
* Successfully discovered: internal.example.fake
Password for Administrator:
* Unconditionally checking packages
* Resolving required packages
* Installing necessary packages: libnss-winbind samba-common-bin libpam-winbind
winbind
* LANG=C LOGNAME=root /usr/bin/net --configfile /var/cache/realmd/realmd-smb-conf.
A53N01 -U Administrator --use-kerberos=required ads join internal.example.fake
Password for [INTEAMPLE\Administrator]:
Using short domain name -- INTEAMPLE
Joined 'J1' to dns domain 'internal.example.fake'
* LANG=C LOGNAME=root /usr/bin/net --configfile /var/cache/realmd/realmd-smb-conf.
A53N01 -U Administrator ads keytab create
Password for [INTEAMPLE\Administrator]:
* /usr/sbin/update-rc.d winbind enable
* /usr/sbin/service winbind restart
* Successfully enrolled machine in realm
```

Note:

This command also installed the `libpam-winbind` package, **which allows AD users to authenticate to other services on this system via PAM, like SSH or console logins**. For example, if your SSH server allows password authentication (`PasswordAuthentication yes` in `/etc/ssh/sshd_config`), then the domain users will be allowed to login remotely on this system via SSH. If you don't expect or need AD users to log into this system (unless it's via Samba or Windows), then it's safe and probably best to remove the `libpam-winbind` package.

LP: #1980246²⁶⁹

-

```
passwd:      files systemd winbind
group:       files systemd winbind
```

```
$ getent passwd INTEXAMPLE\Administrator
```

```
INTEXAMPLE administrator:*:2000500:2000513::/home/administrator@INTEXAMPLE:/
bin/bash
```

Common installation options

*smb.conf(5)*²⁷⁰

-
-
-
-

²⁶⁹ <https://bugs.launchpad.net/bugs/1980246>

²⁷⁰ <https://manpages.ubuntu.com/manpages/resolute/man5/smb.conf.5.html>


```
$ sudo pam-auth-update --enable mkhomedir
```

Export shares

```
[storage]
path = /storage
comment = Storage share
writable = yes
guest ok = no
```

```
$ sudo mkdir -m 1777 /storage
```

```
$ sudo smbcontrol smbd reload-config
```

```
$ smbclient //j1.internal.example.fake/storage -U INTEXAMPLE\\ubuntu
Enter INTEXAMPLE\\ubuntu's password:
Try "help" to get a list of possible commands.
smb: \>
```

```
$ sudo smbstatus
```

```
Samba version 4.15.5-Ubuntu
PID      Username      Group      Machine
Version  Encryption    Signing
-----
3631     INTEXAMPLE\ubuntu INTEXAMPLE\domain users 10.0.16.1 (ipv4:10.0.16.1:39534)
SMB3_11  -              partial(AES-128-CMAC)

Service  pid    Machine    Connected at      Encryption
Signing
-----
storage  3631   10.0.16.1  Wed Jun 29 17:42:54 2022 UTC  -          -

No locked files
```

```
[storage]
path = /storage
comment = Storage share
writable = yes
guest ok = no
valid users = "@INTEXAMPLE\ LTS Releases"
```

Choose an idmap backend

- *idmap_ad(8)*²⁷¹
- *idmap_autorid(8)*²⁷²
- *idmap_rid(8)*²⁷³

Choosing an idmap backend²⁷⁴

²⁷¹ https://manpages.ubuntu.com/manpages/resolute/man8/idmap_ad.8.html

²⁷² https://manpages.ubuntu.com/manpages/resolute/man8/idmap_autorid.8.html

²⁷³ https://manpages.ubuntu.com/manpages/resolute/man8/idmap_rid.8.html

²⁷⁴ https://wiki.samba.org/index.php/Setting_up_Samba_as_a_Domain_Member#Choosing_an_idmap_backend

```
idmap config * : range = 10000-999999
idmap config intexample : backend = rid
idmap config intexample : range = 2000000-2999999
idmap config * : backend = tdb
```

```
$ id INTEXAMPLE\Administrator

uid=2000500(INTEAMPLE\administrator)
gid=2000513(INTEAMPLE\domain users)
groups=2000513(INTEAMPLE\domain users),
        2000500(INTEAMPLE\administrator),
        2000572(INTEAMPLE\denied rodc password replication group),
        2000519(INTEAMPLE\enterprise admins),
        2000518(INTEAMPLE\schema admins),
        2000520(INTEAMPLE\group policy creator owners),
        2000512(INTEAMPLE\domain admins),
        10001(BUILTIN\users),
        10000(BUILTIN\administrators)
```

Further reading

- [The Samba Wiki](https://wiki.samba.org)²⁷⁵

Set up sharing services

²⁷⁵ <https://wiki.samba.org>

Set up Samba as a file server

Share Access Control

Warning:

If you use **Samba and authd** at the same time, you must specify user and group mapping. Otherwise, you will encounter permission issues due to mismatched user and group identifiers.

If you *are* using Samba with authd, you should follow the instructions in the [steps for the server](#)²⁷⁶ guide in the authd documentation instead.

²⁷⁶ <https://documentation.ubuntu.com/authd/edge-docs/howto/use-with-samba/#steps-for-the-server>

Install Samba

```
$ sudo apt install samba
```

Configure Samba as a file server

Note:

Not all available options are included in the default configuration file. See the [smb.conf man page](#)²⁷⁷ or the [Samba HOWTO Collection](#)²⁷⁸ for more details.

²⁷⁷ <https://www.samba.org/samba/docs/current/man-html/smb.conf.5.html>

²⁷⁸ <https://www.samba.org/samba/docs/old/Samba3-HOWTO/>

```
workgroup = EXAMPLE
```

[share]

```
comment = Ubuntu File Server Share
path = /srv/samba/share
browsable = yes
guest ok = yes
read only = no
create mask = 0755
```

-
-

Note:

This example uses `/srv/samba/sharename` because, according to the *Filesystem Hierarchy Standard (FHS)*, `/srv`²⁷⁹ is where site-specific data should be served. Technically, Samba shares can be placed anywhere on the *filesystem* as long as the permissions are correct, but adhering to standards is recommended.

²⁷⁹ <https://www.pathname.com/fhs/pub/fhs-2.3.html#SRVDATAFORSERVICESPROVIDEDBYSYSTEM>

-
-
-
-

Create the directory

```
$ sudo mkdir -p /srv/samba/share
```

```
$ sudo chown nobody:nogroup /srv/samba/share/
```

Enable the new configuration

```
$ sudo systemctl restart smbd.service nmbd.service
```

Warning:

Once again, the above configuration gives full access to any client on the local network. For a more secure configuration see [Share Access Control](#) (page 383).

Further reading

- [Samba how-to collection](#)²⁸⁰
- [Samba in printed format](#)²⁸¹
- [Using Samba](#)²⁸²
- [Ubuntu Wiki Samba](#)²⁸³

Set up Samba as a print server

using Samba as a file server

Share access controls

²⁸⁰ <https://www.samba.org/samba/docs/old/Samba3-HOWTO/>

²⁸¹ <https://www.amazon.com/exec/obidos/tg/detail/-/0131882228>

²⁸² <https://www.oreilly.com/library/view/using-samba-3rd/0596007698/>

²⁸³ <https://help.ubuntu.com/community/Samba>

Install and configure CUPS

our guide on CUPS

Install Samba

```
$ sudo apt install samba
```

Configure Samba

```
workgroup = EXAMPLE
```

```
browsable = yes  
guest ok = yes
```

```
$ sudo systemctl restart smbd.service nmbd.service
```

Further reading

- [Samba HOWTO Collection](#)²⁸⁴
- [Samba HOWTO Collection](#) printed format²⁸⁵
- [Using Samba](#)²⁸⁶
- [CUPS Website](#)²⁸⁷
- [Ubuntu Wiki Samba](#)²⁸⁸

²⁸⁴ <https://www.samba.org/samba/docs/old/Samba3-HOWTO/>

²⁸⁵ <http://www.amazon.com/exec/obidos/tg/detail/-/0131882228>

²⁸⁶ <http://www.oreilly.com/catalog/9780596007690/>

²⁸⁷ <http://www.cups.org/>

²⁸⁸ <https://help.ubuntu.com/community/Samba>

Access controls

Share access controls

Authenticating Samba users

See also

How to create and manage Linux users is covered in [Users and groups management](#) (page 154).

```
$ sudo smbpasswd -a melissa
```

Note:

As this is a separate user database, the password selected for the Samba user does not need to be the same as the Linux password for that user. In fact, most Samba servers setup this way will have the Linux users setup without a valid password: these users only exist so that the corresponding Samba users can be created.

```
$ sudo smbpasswd melissa
```


Groups

Read and write permissions

```
read list = @qa  
write list = @sysadmin, vincent
```

```
admin users = melissa
```

Note:

Remember that the users listed in `smb.conf` for these access controls need to exist both as Linux users, and Samba users.

```
$ sudo smbcontrol smbd reload-config
```

Filesystem permissions

filesystem

ACL

```
UUID=66bcdd2e-8861-4fb0-b7e4-e61c569fe17d /srv ext3 noatime,relatime,acl 0 1
```

```
$ sudo mount -v -o remount /srv
```

Note:

This example assumes `/srv` is on a separate partition. If `/srv`, or wherever you have configured your share path, is part of the `/` partition then a reboot may be required.

```
$ sudo chown -R melissa /srv/samba/share/
```

```
$ sudo chgrp -R sysadmin /srv/samba/share/
```

```
$ sudo setfacl -R -m g:qa:rx /srv/samba/share/
```

Note:

The `setfacl` command above gives *execute* permissions to all files in the `/srv/samba/share` directory, which you may or may not want.

*acl(5)*²⁸⁹ *setfacl(1)*²⁹⁰

²⁸⁹ <https://manpages.ubuntu.com/manpages/resolute/man5/acl.5.html>

²⁹⁰ <https://manpages.ubuntu.com/manpages/resolute/man1/setfacl.1.html>

Further reading

- [Samba HOWTO Collection](#)²⁹¹
- [Samba HOWTO Collection](#) printed format²⁹²
- [Using Samba](#)²⁹³
- [Chapter 18](#)²⁹⁴
- [Samba ACLs page](#)²⁹⁵
- [Ubuntu Wiki Samba](#)²⁹⁶

Create a Samba AppArmor profile

in this guide

Install apparmor-profiles

```
$ sudo apt install apparmor-profiles apparmor-utils
```

Note:

This package contains profiles for several other binaries.

AppArmor profile modes

²⁹¹ <https://www.samba.org/samba/docs/old/Samba3-HOWTO/>

²⁹² <http://www.amazon.com/exec/obidos/tg/detail/-/0131882228>

²⁹³ <http://www.oreilly.com/catalog/9780596007690/>

²⁹⁴ <https://www.samba.org/samba/docs/old/Samba3-HOWTO/securing-samba.html>

²⁹⁵ <https://www.samba.org/samba/docs/old/Samba3-HOWTO/AccessControls.html>

²⁹⁶ <https://help.ubuntu.com/community/Samba>

```
Jun 30 14:41:09 ubuntu kernel: [ 621.478989] audit:
type=1400 audit(1656600069.123:418):
apparmor="ALLOWED" operation="exec" profile="smbd"
name="/usr/lib/x86_64-linux-gnu/samba/samba-bgqd" pid=4122 comm="smbd"
requested_mask="x" denied_mask="x" fsuid=0 ouid=0
target="smbd//null-/usr/lib/x86_64-linux-gnu/samba/samba-bgqd"
```

```
$ sudo aa-enforce /usr/sbin/smbd
```

```
$ sudo apparmor_parser -r -W -T /etc/apparmor.d/usr.sbin.smbd
```

Further reading

- [the Debian AppArmor guide²⁹⁷](#)

How to mount CIFS shares permanently

Introduction to Samba

Prerequisites

-
-
-

²⁹⁷ <https://wiki.debian.org/AppArmor/HowToUse>

-
-
-

Install CIFS

```
$ sudo apt-get install cifs-utils
```

Mount unprotected (guest) network folders

```
$ sudo mkdir /media/windowsshare
```

```
//servername/sharename /media/windowsshare cifs guest,uid=1000 0 0
```

- *hostname*
-
-
-

```
$ sudo mount /media/windowsshare
```

Mount password-protected network folders

```
//servername/sharename /media/windowsshare cifs username=msusername,password=mspassword 0 0
```

Create a credentials file

```
$ gedit ~/.smbcredentials
```

```
username=msusername  
password=mppassword
```

```
$ chmod 600 ~/.smbcredentials
```

```
//servername/sharename /media/windowsshare cifs credentials=/home/ubuntuusername/.  
smbcredentials 0 0
```

```
$ sudo mount /media/windowsshare
```

Changing the share ownership

GID

```
//servername/sharename /media/windowsshare cifs uid=ubuntuusername,credentials=/home/  
ubuntuusername/.smbcredentials 0 0
```

Mount network folders with authd

steps for the client²⁹⁸

Mount password-protected shares using libpam-mount

Install libpam-mount

```
$ sudo apt-get install libpam-mount
```

```
$ sudo gedit /etc/security/pam_mount.conf.xml
```

```
$ gedit ~/.pam_mount.conf.xml
```

²⁹⁸ <https://documentation.ubuntu.com/authd/edge-docs/howto/use-with-samba/#steps-for-the-client>

```
<?xml version="1.0" encoding="utf-8" ?>

<pan_mount>

<volume options="uid=%(USER),gid=100,dir_mode=0700,credentials=/home/ubuntuusername/.
smbcredentials,nosuid,nodev" user="*" mountpoint="/media/windowsshare" path="sharename"
server="servername" fstype="cifs" />

</pan_mount>
```

Troubleshooting

Login errors

-
-
-
-

*mount.cifs(8)*²⁹⁹

Mount after login instead of boot

```
//servername/sharename /media/windowsshare cifs noauto,credentials=/home/ubuntuusername/.
smbpasswd 0 0
```

²⁹⁹ <https://manpages.ubuntu.com/manpages/resolute/man8/mount.cifs.8.html>

Legacy options

NT4 domain controller (legacy)

Note:

This section is flagged as *legacy* because nowadays, Samba can be deployed in full Active Directory domain controller mode, and the old-style NT4 Primary Domain Controller is deprecated.

Primary domain controller

Install Samba

```
$ sudo apt install samba libpam-winbind
```

Configure Samba

```
workgroup = EXAMPLE  
...  
security = user
```

```
domain logons = yes  
logon path = \\%N%\%U\profile  
logon drive = H:  
logon home = \\%N%\%U  
logon script = logon.cmd  
add machine script = sudo /usr/sbin/useradd -N -g machines -c Machine -d  
/var/lib/samba -s /bin/false %u
```

Note:

If you wish to not use *Roaming Profiles* leave the `logon home` and `logon path` options commented out.

-
-
-
-
-
-

Security - Users: Adding and Deleting Users

Mapping shares

[homes]

```
comment = Home Directories
browseable = no
read only = no
create mask = 0700
directory mask = 0700
valid users = %S
```

[netlogon]

```
comment = Network Logon Service
path = /srv/samba/netlogon
guest ok = yes
read only = yes
share modes = no
```

Note:

The original `netlogon` share path is `/home/samba/netlogon`, but according to the *Filesystem Hierarchy Standard (FHS)*, `/srv` is the correct location³⁰⁰ for site-specific data provided by the system.

³⁰⁰ <https://www.pathname.com/fhs/pub/fhs-2.3.html#SRVDATAFORSERVICESPROVIDEDBYSYSTEM>

```
$ sudo mkdir -p /srv/samba/netlogon
```

```
$ sudo touch /srv/samba/netlogon/logon.cmd
```

```
$ sudo systemctl restart smbd.service nmbd.service
```

Final setup tasks

```
$ sudo net groupmap add ntgroup="Domain Admins" unixgroup=sysadmin rid=512  
type=d
```

```
$ sudo smbpasswd -a sysadmin
```

```
$ net rpc rights grant -U sysadmin "EXAMPLE\Domain Admins"  
SeMachineAccountPrivilege \ > SePrintOperatorPrivilege SeAddUsersPrivilege  
SeDiskOperatorPrivilege \ > SeRemoteShutdownPrivilege
```

Backup domain controller

BDC

DAP Backend

Samba - OpenL-

```
$ sudo apt install samba libpam-winbind
```

```
workgroup = EXAMPLE  
...  
security = user
```

```
domain logons = yes  
domain master = no
```

```
$ sudo chgrp -R admin /var/lib/samba
```

```
$ sudo scp -r username@pdc:/var/lib/samba /var/lib
```

hostname

```
$ sudo systemctl restart smb.service nmbd.service
```

Further reading

- [Samba HOWTO Collection](#)³⁰¹
- [in printed format](#)³⁰²
- [Using Samba](#)³⁰³
- [Chapter 4](#)³⁰⁴
- [Chapter 5](#)³⁰⁵
- [Ubuntu Wiki Samba](#)³⁰⁶

OpenLDAP backend (legacy)

Note:

This section is flagged as *legacy* because nowadays, Samba 4 is best integrated with its own LDAP server in Active Directory mode. Integrating Samba with LDAP as described here covers the NT4 mode, which has been deprecated for many years.

Install LDAP

LDAP with Transport Layer Security

³⁰¹ <https://www.samba.org/samba/docs/old/Samba3-HOWTO/>

³⁰² <https://www.amazon.com/exec/obidos/tg/detail/-/0131882228>

³⁰³ <https://www.oreilly.com/library/view/using-samba-3rd/0596007698/>

³⁰⁴ <https://www.samba.org/samba/docs/old/Samba3-HOWTO/samba-pdc.html>

³⁰⁵ <https://www.samba.org/samba/docs/old/Samba3-HOWTO/samba-bdc.html>

³⁰⁶ <https://help.ubuntu.com/community/Samba>

Install the software

```
$ sudo apt install samba smbldap-tools
```

Configure LDAP

-
-
-

Samba schema

```
$ sudo ldapadd -Q -Y EXTERNAL -H ldapi:/// -f  
/usr/share/doc/samba/examples/LDAP/samba.ldif
```

```
$ sudo ldapsearch -Q -LLL -Y EXTERNAL -H ldapi:/// -b cn=schema,cn=config  
'cn=*samba*'
```

Samba indices

```
dn: olcDatabase={1}mdb,cn=config
changetype: modify
replace: olcDbIndex
olcDbIndex: objectClass eq
olcDbIndex: uidNumber,gidNumber eq
olcDbIndex: loginShell eq
olcDbIndex: uid,cn eq,sub
olcDbIndex: memberUid eq,sub
olcDbIndex: member,uniqueMember eq
olcDbIndex: sambaSID eq
olcDbIndex: sambaPrimaryGroupSID eq
olcDbIndex: sambaGroupType eq
olcDbIndex: sambaSIDList eq
olcDbIndex: sambaDomainName eq
olcDbIndex: default sub,eq
```

```
$ sudo ldapmodify -Q -Y EXTERNAL -H ldapi:/// -f samba_indices.ldif
```

```
$ sudo ldapsearch -Q -LLL -Y EXTERNAL -H \> ldapi:/// -b cn=config
olcDatabase={1}mdb olcDbIndex
```

Adding Samba LDAP objects

- *hostname*
-

```
$ sudo smbldap-config
```

-
-
-
-

DN

```
$ sudo smbldap-populate -g 10000 -u 10000 -r 10000
```

gid

Samba configuration

```
# passwd backend = tdbsam
workgroup = EXAMPLE

# LDAP Settings
passdb backend = ldapsam:ldap://ldap01.example.com
ldap suffix = dc=example,dc=com
ldap user suffix = ou=People
ldap group suffix = ou=Groups
ldap machine suffix = ou=Computers
ldap idmap suffix = ou=Idmap
ldap admin dn = cn=admin,dc=example,dc=com
ldap ssl = start tls
ldap passwd sync = yes
```

Note:

The `smb.conf` as shipped by the package is quite long and has many configuration examples. An easy way to visualize it without any comments is to run `testparm -s`.


```
$ sudo smbpasswd -W
```

Network User Authentication with SSSD

```
$ sudo apt install sssd-ldap
```

```
[sss]
config_file_version = 2
domains = example.com

[domain/example.com]
id_provider = ldap
auth_provider = ldap
ldap_uri = ldap://ldap01.example.com
cache_credentials = True
ldap_search_base = dc=example,dc=com
```

```
$ sudo chmod 0600 /etc/sss/sss.conf
```

```
$ sudo chown root:root /etc/sss/sss.conf
```

```
$ sudo systemctl start sss
```

```
$ sudo systemctl restart smb.service nmb.service
```

```
$ getent group Replicators
Replicators*:552:
```

Note:

The names are case sensitive!

```
$ sudo smbpasswd -a username
```

•

```
$ sudo smbldap-useradd -a -P -m username
```

```
$ getent passwd username
```

•

```
$ sudo smbldap-userdel username
```

•

```
$ sudo smbldap-groupadd -a groupname
```

•

```
$ sudo smbldap-groupmod -m username groupname
```

•

```
$ sudo smbldap-groupmod -x username groupname
```

•

```
$ sudo smbldap-useradd -t 0 -w username
```

Further reading

- [Upstream Samba documentation collection](#)³⁰⁷
- [Upstream Samba wiki](#)³⁰⁸

See also

- [Introduction to Samba](#)

Active Directory integration

AD integration explanations

Join a domain with winbind: preparation

```
$ sudo apt install winbind libnss-winbind libpam-winbind
```

DNS

DHCP

```
network:  
  version: 2  
  ethernet:  
    eth0:  
      dhcp4: true
```

```
network:  
  version: 2  
  ethernet:  
    eth0:  
      dhcp4: true
```

(continues on next page)

³⁰⁷ <https://www.samba.org/samba/docs/>

³⁰⁸ https://wiki.samba.org/index.php/Main_Page

(continued from previous page)

```
nameservers
addresses  [10.10.4.5]
search:    [example.internal]
```

```
$ sudo netplan generate
```

```
$ sudo netplan apply
```

Note:

Be careful whenever changing network parameters over an SSH connection. If there are any mistakes, you might lose remote access!

```
$ resolvectl status

Global
    Protocols: -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
    resolv.conf mode: stub

Link 281 (eth0)
    Current Scopes: DNS
        Protocols: +DefaultRoute -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
    DNS Servers: 10.10.4.5 10.10.4.1
    DNS Domain: example.internal
```

```
# /etc/nsswitch.conf
#
# Example configuration of GNU Name Service Switch functionality.
# If you have the `glibc-doc-reference' and `info' packages installed, try:
# `info libc "Name Service Switch"' for information about this file.

passwd:      files systemd winbind
group:       files systemd winbind
(...)
```

```
$ sudo pam-auth-update --enable mkhomedir
```

Join a simple domain with the rid backend

```
[global]
security = ads
realm = EXAMPLE.INTERNAL
workgroup = EXAMPLE

idmap config * : backend          = tdb
idmap config * : range           = 100000 - 199999
idmap config EXAMPLE : backend   = rid
idmap config EXAMPLE : range     = 1000000 - 1999999

# allow logins when the DC is unreachable
winbind offline logon = yes
# this can be yes if there is absolute certainty that there is only a
# single domain involved
winbind use default domain = no
# setting these enumeration options to yes has a high performance impact
# and can cause instabilities
winbind enum groups = no
winbind enum users = no
winbind refresh tickets = yes
# if domain users should be allowed to login, they will need a login shell
template shell = /bin/bash
# the home directory template for domain users
template homedir = /home/%D/%U
kerberos method = secrets and keytab
```

```
Load smb config files from /etc/samba/smb.conf
Loaded services file OK.
Weak crypto is allowed by GnuTLS (e.g. NTLM as a compatibility fallback)

Server role: ROLE_DOMAIN_MEMBER

Press enter to see a dump of your service definitions
(...)
```

hostname

DNS

FQDN

```
$ sudo hostnamectl hostname n1.example.internal
```

```
$ sudo net ads join -U Administrator  
Password for [EXAMPLE\Administrator]:  
Using short domain name -- EXAMPLE  
Joined 'N1' to dns domain 'example.internal'
```

```
$ sudo hostnamectl hostname n1
```

```
$ sudo systemctl restart winbind.service
```

Verifying the join

```
$ sudo wbinfo -t  
checking the trust secret for domain EXAMPLE via RPC calls succeeded
```

```
$ id jammy@example.internal  
uid=1001103(EXAMPLE\jammy) gid=1000513(EXAMPLE\domain users) groups=1000513(EXAMPLE\  
domain users),1001103(EXAMPLE\jammy)
```

```
$ id EXAMPLE\\jammy  
uid=1001103(EXAMPLE\jammy) gid=1000513(EXAMPLE\domain users) groups=1000513(EXAMPLE\  
domain users),1001103(EXAMPLE\jammy)
```

```
$ n1 login: jammy@example.internal

Password:
Welcome to Ubuntu 24.04 LTS (GNU/Linux 6.5.0-26-generic x86_64)
(...)
Creating directory '/home/EXAMPLE/jammy'.

EXAMPLE\jammy@n1:~$
```

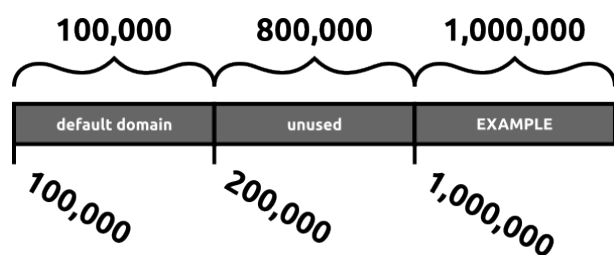
Note:

The actual login name used can have multiple formats: DOMAIN\user at the terminal login prompt, DOMAIN\\user when referred to in shell scripts (note the escaping of the '\' character), and user@domain is also accepted.

Join a forest with the rid backend

```
[global]
security = ads
realm = EXAMPLE.INTERNAL
workgroup = EXAMPLE

idmap config * : backend = tdb
# 100,000 - 199,999
idmap config * : range = 100000 - 199999
idmap config EXAMPLE : backend = rid
# 1,000,000 - 1,999,999
idmap config EXAMPLE : range = 1000000 - 1999999
```



```
$ testparm
```

```
Load smb config files from /etc/samba/smb.conf  
Loaded services file OK.  
Weak crypto is allowed by GnuTLS (e.g. NTLM as a compatibility fallback)
```

```
Server role: ROLE_DOMAIN_MEMBER
```

hostname

```
$ sudo hostnamectl hostname n3.example.internal
```

```
$ hostname
```

```
n3.example.internal
```

```
$ hostname -f
```

```
n3.example.internal
```

```
$ sudo net ads join -U Administrator
```

```
Password for [EXAMPLE\Administrator]:  
Using short domain name -- EXAMPLE  
Joined 'N3' to dns domain 'example.internal'
```

```
$ sudo hostnamectl hostname n3
```

```
$ sudo systemctl restart winbind.service
```

```
$ id jammy@example.internal
```

```
uid=1001103(EXAMPLE\jammy) gid=1000513(EXAMPLE\domain users) groups=1000513(EXAMPLE\  
domain users),1001103(EXAMPLE\jammy)
```

```
$ id noble@mydomain.internal
```

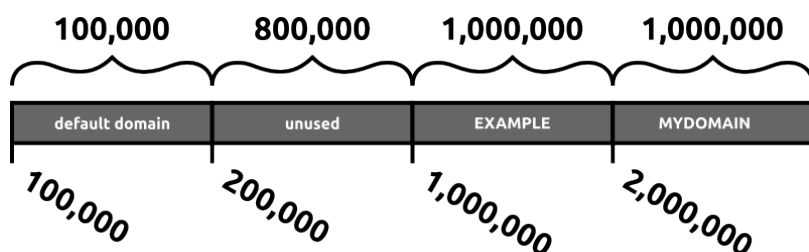
```
uid=1000000(MYDOMAIN\noble) gid=1000000(MYDOMAIN\domain users) groups=1000000(MYDOMAIN\  
domain users)
```



```
[global]
security = ads
realm = EXAMPLE.INTERNAL
workgroup = EXAMPLE

idmap config * : backend          = tdb
# 100,000 - 199,999
idmap config * : range            = 100000 - 199999
idmap config EXAMPLE : backend = rid
# 1,000,000 - 1,999,999
idmap config EXAMPLE : range     = 1000000 - 1999999

# MYDOMAIN.INTERNAL idmap configuration
idmap config MYDOMAIN : backend = rid
# 2,000,000 - 2,999,999
idmap config MYDOMAIN : range   = 2000000 - 2999999
```



```
$ id jammy@example.internal

uid=1001103(EXAMPLE\jammy) gid=1000513(EXAMPLE\domain users) groups=1000513(EXAMPLE\
domain users),1001103(EXAMPLE\jammy)
```

```
$ id noble@mydomain.internal

uid=2001104(MYDOMAIN\noble) gid=2000513(MYDOMAIN\domain users)
groups=2000513(MYDOMAIN\domain users),2001104(MYDOMAIN\noble)
```

-
-
-
-

Join a forest with the autorid backend

[global]

```
security = ads
realm = EXAMPLE.INTERNAL
workgroup = EXAMPLE

idmap config * : backend = autorid
# 1,000,000 - 19,999,999
idmap config * : range   = 1000000 - 19999999
# 1,000,000
idmap config * : rangesize = 1000000

# allow logins when the DC is unreachable
winbind offline logon = yes
# this *can* be yes if there is absolute certainty that there is only a
# single domain involved
winbind use default domain = no
# setting these enumeration options to yes has a high performance impact
# and can cause instabilities
winbind enum groups = no
winbind enum users = no
winbind refresh tickets = yes
# if domain users should be allowed to login, they will need a login shell
template shell = /bin/bash
# the home directory template for domain users
template homedir = /home/%D/%U
kerberos method = secrets and keytab
```

-
-
-

```
$ testparm

Load smb config files from /etc/samba/smb.conf
Loaded services file OK.
Weak crypto is allowed by GnuTLS (e.g. NTLM as a compatibility fallback)

Server role: ROLE_DOMAIN_MEMBER

Press enter to see a dump of your service definitions
```

hostname

DNS

```
$ sudo hostnamectl hostname n2.example.internal
```

```
$ sudo net ads join -U Administrator

Password for [EXAMPLE\Administrator]:
Using short domain name -- EXAMPLE
Joined 'N2' to dns domain 'example.internal'
```

```
$ sudo hostnamectl hostname n2
```

```
$ hostname

n2
```

```
$ hostname -f  
n2.example.internal
```

```
$ sudo systemctl restart winbind.service
```

Verifying the join

```
$ sudo wbinfo -t  
checking the trust secret for domain EXAMPLE via RPC calls succeeded
```

```
$ id jammy@example.internal  
uid=2001103(EXAMPLE\jammy) gid=2000513(EXAMPLE\domain users) groups=2000513(EXAMPLE\  
domain users),2001103(EXAMPLE\jammy)
```

```
$ id EXAMPLE\jammy  
uid=2001103(EXAMPLE\jammy) gid=2000513(EXAMPLE\domain users) groups=2000513(EXAMPLE\  
domain users),2001103(EXAMPLE\jammy)
```

```
$ n2 login: jammy@example.internal  
Password:  
Welcome to Ubuntu 24.04 LTS (GNU/Linux 6.5.0-26-generic x86_64)  
(...)  
Creating directory '/home/EXAMPLE/jammy'.  
EXAMPLE\jammy@n1:~$
```

```
$ id noble@mydomain.internal

uid=3001104(MYDOMAIN\noble) gid=3000513(MYDOMAIN\domain users)
groups=3000513(MYDOMAIN\domain users),3001104(MYDOMAIN\noble)
```

```
$ id MYDOMAIN\\noble

uid=3001104(MYDOMAIN\noble) gid=3000513(MYDOMAIN\domain users)
groups=3000513(MYDOMAIN\domain users),3001104(MYDOMAIN\noble)
```

```
$ n2 login: noble@mydomain.internal

Password:
Welcome to Ubuntu 24.04 LTS (GNU/Linux 6.8.0-31-generic x86_64)
(...)
Creating directory '/home/MYDOMAIN/noble'.

MYDOMAIN\noble@n2:~$
```

Note:

The actual login name used can have multiple formats: DOMAIN\user at the terminal login prompt, DOMAIN\\user when referred to in shell scripts (note the escaping of the '\' character), and user@domain is also accepted.

See also

- [*Introduction to Active Directory integration*](#)
- [*Samba*](#)
- [*Active Directory integration*](#)

Traffic management

Traffic shaping with tc and CAKE

What is bufferbloat?

Prerequisites

- [WAN](#)
-
-

Enable Explicit Congestion Notification (ECN)

```
# Enable explicit congestion notification for TCP (covers IPv4 and IPv6)
net.ipv4.tcp_ecn = 1
```

```
$ sudo sysctl --system
```

Create the IFB interface

```
[NetDev]
Kind=ifb
Name=wanifb
MTUBytes=1492
```

Note:

Adjust MTUBytes to match your WAN connection. For *PPPoE* connections, 1492 is typical. For standard Ethernet, use 1500. You can measure the *MTU* to be sure.

```
[Match]
Name=wanifb

[Link]
ActivationPolicy=always-up
```

```
$ sudo networkctl reload
```

```
$ ip link show wanifb
```

Create the traffic shaping script

```
#!/usr/bin/env python3
"""
Traffic shaping using CAKE qdisc.

Shapes both upload (egress) and download (ingress via IFB redirect) traffic
to reduce bufferbloat and improve latency for interactive applications.

Test results at: https://www.waveform.com/tools/bufferbloat
"""

import subprocess
import argparse
import shlex
```

(continues on next page)

```
# Configuration - adjust these values for your connection
CONFIG = {
    "wan_iface": "ext0",          # Physical WAN interface
    "ifb_iface": "wanifb",       # IFB interface for download shaping
    # set to ~90-95% of what your ISP effectively delivers (not just their promise)
    # please measure your bandwidth, and remember it can have time-of-day dependent
    # variations
    "upload_speed": "30mbit",    # you -> WAN
    "download_speed": "80mbit",  # WAN -> you
}

def run(cmd, ignore_errors=False):
    """Run a shell command."""
    print f" {cmd}"
    subprocess.run(shlex.split(cmd), check=(not ignore_errors))

def clear_all():
    """Remove existing traffic shaping rules."""
    run(f"tc qdisc del dev {CONFIG['wan_iface']} root", ignore_errors=True)
    run(f"tc qdisc del dev {CONFIG['wan_iface']} ingress", ignore_errors=True)
    run(f"tc qdisc del dev {CONFIG['ifb_iface']} root", ignore_errors=True)

def start():
    """Apply traffic shaping rules."""
    print("Clearing existing rules...")
    clear_all()

    # Ensure IFB interface is up
    run(f"ip link set dev {CONFIG['ifb_iface']} up", ignore_errors=True)

    # Shape upload (egress on WAN interface)
    print f"Setting upload limit to {CONFIG['upload_speed']}..."
    run(f"tc qdisc add dev {CONFIG['wan_iface']} root cake "
        f"bandwidth {CONFIG['upload_speed']} nat")

    # Redirect ingress traffic to IFB interface
    print "Redirecting ingress traffic to IFB interface..."
    run(f"tc qdisc add dev {CONFIG['wan_iface']} handle ffff: ingress")
    run(f"tc filter add dev {CONFIG['wan_iface']} parent ffff: protocol all "
        f"u32 match u32 0 0 action mirrored egress redirect dev {CONFIG['ifb_iface']}")

    # Shape download (egress on IFB interface)
    print f"Setting download limit to {CONFIG['download_speed']}..."
    run(f"tc qdisc add dev {CONFIG['ifb_iface']} root cake "
        f"bandwidth {CONFIG['download_speed']} wash")

    print("Traffic shaping enabled.")

def stop():
    """Remove traffic shaping rules."""
    print("Removing traffic shaping rules...")
```

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```
clear_all()
print("Traffic shaping disabled.")

def status():
    """Show current traffic shaping statistics."""
    print f"\n=== Upload ({CONFIG['wan_iface']}) ==="
    run(f"tc -s qdisc show dev {CONFIG['wan_iface']}")
    print f"\n=== Download ({CONFIG['ifb_iface']}) ==="
    run(f"tc -s qdisc show dev {CONFIG['ifb_iface']}")

def main():
    parser = argparse.ArgumentParser(
        description="Manage traffic shaping with CAKE qdisc"
    )
    parser.add_argument(
        "action",
        choices=["start", "stop", "status"],
        help="Action to perform"
    )
    args = parser.parse_args()

    actions = {"start": start, "stop": stop, "status": status}
    actions[args.action]()

if __name__ == "__main__":
    main()
```

```
$ sudo chmod +x /usr/local/bin/traffic-shaping
```

- `systemd.link(5)`³⁰⁹
-
-
-

Note:

Setting speeds slightly below your actual bandwidth leaves headroom for the shaper to work effectively. If you set the limit too high, bufferbloat can still occur in upstream devices.

³⁰⁹ <https://manpages.ubuntu.com/manpages/resolute/man5/systemd.link.5.html>

Create a systemd service

```
[Unit]
Description=Traffic shaping with CAKE
After=network-online.target
Wants=network-online.target
# Bind to the WAN and IFB interfaces
BindsTo=sys-subsystem-net-devices-ext0.device
After=sys-subsystem-net-devices-ext0.device
BindsTo=sys-subsystem-net-devices-wanifb.device
After=sys-subsystem-net-devices-wanifb.device

[Service]
Type=oneshot
ExecStart=/usr/local/bin/traffic-shaping start
ExecStop=/usr/local/bin/traffic-shaping stop
RemainAfterExit=yes

[Install]
WantedBy=multi-user.target
```

Note:

Replace `ext0` in the `BindsTo` and `After` lines with your actual WAN interface name - or rename your external interface to `ext0`. The interface name becomes part of the systemd device unit path.

```
$ sudo systemctl daemon-reload > sudo systemctl enable --now
traffic-shaping.service
```

Verify the configuration

```
$ sudo systemctl status traffic-shaping.service
```

```
$ sudo /usr/local/bin/traffic-shaping status
```

Test for bufferbloat

1. Waveform Bufferbloat Test³¹⁰
- 2.
- 3.

CAKE options explained

- - *NAT*
 - *DSCP*
 - *DSCP*
 -
 -
 -
- tc-cake(8)*³¹¹

Troubleshooting

- 1.

```
$ ip link show ext0 > ip link show wanifb
```

- 2.

```
$ sudo journalctl -eu traffic-shaping.service
```

- 3.

```
$ lsmod | grep cake
```

³¹⁰ <https://www.waveform.com/tools/bufferbloat>

³¹¹ <https://manpages.ubuntu.com/manpages/resolute/man8/tc-cake.8.html>

4.

```
$ sudo /usr/local/bin/traffic-shaping stop > sudo  
/usr/local/bin/traffic-shaping start
```

Further reading

- [Bufferbloat project](#)³¹²
- [tc\(8\)](#)³¹³
- [tc-cake\(8\)](#)³¹⁴

Printing

Install and configure a CUPS print server

Common UNIX Printing System, or CUPS³¹⁵

Install CUPS

```
$ sudo apt install cups
```

³¹² <https://www.bufferbloat.net/>

³¹³ <https://manpages.ubuntu.com/manpages/resolute/man8/tc.8.html>

³¹⁴ <https://manpages.ubuntu.com/manpages/resolute/man8/tc-cake.8.html>

³¹⁵ <https://openprinting.github.io/cups/doc/overview.html>

Configure the CUPS server

Make a copy of the configuration file

```
$ sudo cp /etc/cups/cupsd.conf /etc/cups/cupsd.conf.original
```

```
$ sudo chmod a-w /etc/cups/cupsd.conf.original
```

Configure Server administrator

```
ServerAdmin bjoy@somebigco.com
```

Configure Listen

hostname

```
Listen 127.0.0.1:631      # existing loopback Listen
Listen /var/run/cups/cups.sock # existing socket Listen
Listen 192.168.10.250:631  # Listen on the LAN interface, Port 631 (IPP)
```

```
Listen socrates:631 # Listen on all interfaces for the hostname 'socrates'
```

```
Port 631 # Listen on port 631 on all interfaces
```

```
$ man cupsd.conf
```

Configure network access

```
# Allow access to the web interface from the local network
<Location />
  Order allow,deny
  Allow localhost
  Allow 192.168.10.*
</Location>

# Allow access to the administration pages from the local network
<Location /admin>
  Order allow,deny
  Allow localhost
  Allow 192.168.10.*
</Location>

# Allow access to the admin configuration from the local network
<Location /admin/conf>
  AuthType Default
  Require user @SYSTEM
  Order allow,deny
  Allow localhost
  Allow 192.168.10.*
</Location>
```

firewall

```
$ sudo ufw allow 631/tcp
```

Post-configuration restart

```
$ sudo systemctl restart cups.service
```

Web Interface

```
$ sudo usermod -aG lpadmin username
```

Error logs

Further reading

- [CUPS Website](http://www.cups.org/)³¹⁶
- [Debian Printing Wiki](https://wiki.debian.org/Printing)³¹⁷

³¹⁶ <http://www.cups.org/>

³¹⁷ <https://wiki.debian.org/Printing>

See also

- [Networking section](#)

Managing software

-
-
-

Managing software

Install and manage packages

Update the package index

```
$ sudo apt update
```

Note:

Ubuntu releases prior to 24.04 LTS (Noble), which don't use the deb822 standard by default, have repository information defined in the `/etc/apt/sources.list` file.

Install a package

```
$ sudo apt install nmap
```


Tip:

You can install or remove multiple packages at once by separating them with spaces.

Remove a package

```
$ sudo apt remove nmap
```

Note:

While apt is a command-line tool, it is intended to be used interactively, and not to be called from non-interactive scripts. The apt-get command should be used in scripts (perhaps with the `--quiet` flag). For basic commands the syntax of the two tools is identical.

Upgrading packages

```
$ sudo apt update
```

```
$ sudo apt upgrade
```

guide on upgrading releases
APT User's

Guide³¹⁸

Aptitude

frontend

³¹⁸ <https://www.debian.org/doc/user-manuals#apt-guide>

```
$ sudo apt install aptitude
```

```
$ sudo aptitude
```

Installing packages

Remove Packages

Updating the package index

Upgrade packages

-
-
-
-
-
-
-
-

Command-line Aptitude

```
$ sudo aptitude install nmap
```

```
$ sudo aptitude remove nmap
```

*Aptitude manual pages*³¹⁹

³¹⁹ <https://manpages.ubuntu.com/manpages/resolute/man8/aptitude-curses.8.html>

dpkg

List packages

```
$ dpkg -l
```

```
$ dpkg -l | grep apache2
```

List files

```
$ dpkg -L ufw
```

```
$ dpkg -S /etc/host.conf  
base-files: /etc/host.conf
```

Note:

Many files are automatically generated during the package install process, and even though they are on the *filesystem*, `dpkg -S` may not know which package they belong to.

Installing a deb file

```
$ sudo dpkg -i zip_3.0-4_amd64.deb
```

Uninstalling packages

```
$ sudo dpkg -r zip
```

Caution:

Uninstalling packages using dpkg, is **NOT** recommended in most cases. It is better to use a package manager that handles dependencies to ensure that the system is left in a consistent state. For example, using `dpkg -r zip` will remove the zip package, but any packages that depend on it will still be installed and may no longer function correctly as a result.

dpkg(1)³²⁰

APT configuration

Automatic updates

Automatic updates

Extra repositories

using third-party repositories

³²⁰ <https://manpages.ubuntu.com/manpages/resolute/man1/dpkg.1.html>

Warning:

Be advised that packages in Universe and Multiverse are not officially supported and do not receive security patches, except through Ubuntu Pro's [Expanded Security Maintenance](#)³²¹. A subscription to [Ubuntu Pro](#)³²² is free for personal use on up to five machines. Packages in the *multiverse* repository often have licensing issues that prevent them from being distributed with a free operating system, and they may be illegal in your locality.

³²¹ <https://ubuntu.com/security/esm>

³²² <https://ubuntu.com/pro>

```
Types: deb
URIs: http://archive.ubuntu.com/ubuntu
Suites: DISTRO-SHORT-CODENAME DISTRO-SHORT-CODENAME-updates DISTRO-SHORT-CODENAME-backports
Components: main universe restricted multiverse
Signed-By: /usr/share/keyrings/ubuntu-archive-keyring.gpg
```

Logging

Further reading

- [Installing Software](#)³²³
- [APT User's Guide](#)³²⁴
- [systemd\(1\)](#)³²⁵ [systemd.timer\(5\)](#)³²⁶ [sys -](#)
- [Aptitude user's manual](#)³²⁷

³²³ <https://help.ubuntu.com/community/InstallingSoftware>

³²⁴ <https://www.debian.org/doc/user-manuals#apt-guide>

³²⁵ <https://manpages.ubuntu.com/manpages/resolute/man1/systemd.1.html>

³²⁶ <https://manpages.ubuntu.com/manpages/resolute/man5/systemd.timer.5.html>

³²⁷ <https://www.debian.org/doc/user-manuals#aptitude>

- [Adding Repositories HOWTO \(Ubuntu Wiki\)](#)³²⁸
- *Package management*

Updates

Automatic updates

-
-
-

Important:

Just adding another package repository to an Ubuntu system WILL NOT make unattended-upgrades consider it for updates! This is explained in *where to pick updates from* (page 432) later in this document.

Configuration layout

```
$ sudo apt install unattended-upgrades
```

-
-
-

Expanded Security Maintenance (ESM)³²⁹

³²⁸ <https://help.ubuntu.com/community/Repositories/Ubuntu>

³²⁹ <https://ubuntu.com/security/esm>

Enabling and disabling unattended upgrades

Upgrading

packages

```
APT::Periodic::Update-Package-Lists "1";  
APT::Periodic::Unattended-Upgrade "1";
```

```
APT::Periodic::Update-Package-Lists "0";  
APT::Periodic::Unattended-Upgrade "0";
```

```
[Timer]  
Persistent=false
```

*systemd.timer(5)*³³⁰

³³⁰ <https://manpages.ubuntu.com/manpages/resolute/man5/systemd.timer.5.html>

Where to pick updates from

Ubuntu Project documenta-

tion³³¹

```
Unattended-Upgrade::Allowed-Origins {
    "${distro_id}:${distro_codename}";
    "${distro_id}:${distro_codename}-security";
    // Extended Security Maintenance; doesn't necessarily exist for
    // every release and this system may not have it installed, but if
    // available, the policy for updates is such that unattended-upgrades
    // should also install from here by default.
    "${distro_id}ESMA:${distro_codename}-apps-security";
    "${distro_id}ESM:${distro_codename}-infra-security";
    // "${distro_id}:${distro_codename}-updates";
    // "${distro_id}:${distro_codename}-proposed";
    // "${distro_id}:${distro_codename}-backports";
};
```

Note:

The double “//” indicates a comment, so whatever follows “//” will not be evaluated.

```
Unattended-Upgrade::Allowed-Origins {
    "${distro_id}:${distro_codename}";
    "${distro_id}:${distro_codename}-security";
    // Extended Security Maintenance; doesn't necessarily exist for
    // every release and this system may not have it installed, but if
    // available, the policy for updates is such that unattended-upgrades
    // should also install from here by default.
    "${distro_id}ESMA:${distro_codename}-apps-security";
    "${distro_id}ESM:${distro_codename}-infra-security";
    "${distro_id}:${distro_codename}-updates";
    // "${distro_id}:${distro_codename}-proposed";
    // "${distro_id}:${distro_codename}-backports";
};
```

³³¹ <https://documentation.ubuntu.com/project/how-ubuntu-is-made/concepts/package-archive/#pockets>

Automatic upgrades from a PPA

Launchpad PPA³³²

<https://launchpad.net/>

`~canonical-server/+archive/ubuntu/server-backports`

```
Unattended-Upgrade::Allowed-Origins {
    "${distro_id}:${distro_codename}";
    "${distro_id}:${distro_codename}-security";
    // Extended Security Maintenance; doesn't necessarily exist for
    // every release and this system may not have it installed, but if
    // available, the policy for updates is such that unattended-upgrades
    // should also install from here by default.
    "${distro_id}ESMAppls:${distro_codename}-apps-security";
    "${distro_id}ESM:${distro_codename}-infra-security";
    "${distro_id}:${distro_codename}-updates";
    // "${distro_id}:${distro_codename}-proposed";
    // "${distro_id}:${distro_codename}-backports";
    "LP-PPA-canonical-server-server-backports:${distro_codename}";
};
```

```
2025-03-13 22:44:29,802 INFO Starting unattended upgrades script
2025-03-13 22:44:29,803 INFO Allowed origins are: o=Ubuntu,a=noble, o=Ubuntu,a=noble-
security, o=UbuntuESMAppls,a=noble-apps-security, o=UbuntuESM,a=noble-infra-security, o=LP-
PPA-canonical-server-server-backports,a=noble
2025-03-13 22:44:29,803 INFO Initial blacklist:
2025-03-13 22:44:29,803 INFO Initial whitelist (not strict):
2025-03-13 22:44:33,029 INFO Option --dry-run given, *not* performing real actions
2025-03-13 22:44:33,029 INFO Packages that will be upgraded: ibverbs-providers libibverbs1
rdma-core
2025-03-13 22:44:33,029 INFO Writing dpkg log to /var/log/unattended-upgrades/unattended-
upgrades-dpkg.log
2025-03-13 22:44:34,421 INFO All upgrades installed
2025-03-13 22:44:34,855 INFO The list of kept packages can't be calculated in dry-run
mode.
```

³³² <https://documentation.ubuntu.com/launchpad/user/Packaging/PPA/>

```
/var/lib/apt/lists/ppa.launchpadcontent.net_canonical-server_server-backports_ubuntu_
dists_noble_InRelease
```

```
-----BEGIN PGP SIGNED MESSAGE-----
Hash: SHA512

Origin: LP-PPA-canonical-server-server-backports
Label: Server Team Backports
Suite: noble
Version: 24.04
Codename: noble
Date: Tue, 03 Dec 2024 6:00:43 UTC
Architectures: amd64 arm64 armhf i386 ppc64el riscv64 s390x
Components: main
Description: Ubuntu Noble 24.04
(...)
```

How to block certain packages

Python Regular Expressions³³³

Note:

Keep in mind that blocking a package might prevent other updates from being installed if they depend on the blocked package!

```
Unattended-Upgrade::Package-Blacklist {
    "linux-";
}
```

```
Unattended-Upgrade::Package-Blacklist {
    "libc6$";
    "libc-bin$";
}
```

³³³ <https://docs.python.org/3/howto/regex.html>

Note:

The regular expressions used here behave as if the “^” character is present at the start, i.e., the `libc6$` expression will match `libc6`, but will NOT match `glibc6` for example.

```
Unattended-Upgrade::Package-Blacklist {  
    "libc(6|-bin)$";  
}
```

Notifications

-

-

-

-

-

Note:

Sending out emails like this requires the separate configuration of a package like `ssmtp(8)`³³⁴ or another minimalistic mail client that is capable of sending messages to a mail server.

³³⁴ <https://manpages.ubuntu.com/manpages/resolute/man8/ssmtp.8.html>

Notification examples

No changes applied, no errors

Subject: unattended-upgrades result for <hostname>: SUCCESS

Unattended upgrade result: No packages found that can be upgraded unattended and no pending auto-removals

Unattended-upgrades log:

Starting unattended upgrades script

Allowed origins are: o=Ubuntu,a=noble, o=Ubuntu,a=noble-security, o=UbuntuESMApps,a=noble-apps-security, o=UbuntuESM,a=noble-infra-security, o=Ubuntu,a=noble, o=Ubuntu,a=noble-security, o=UbuntuESMApps,a=noble-apps-security, o=UbuntuESM,a=noble-infra-security

Initial blacklist:

Initial whitelist (not strict):

No packages found that can be upgraded unattended and no pending auto-removals

Upgrades applied, no errors

Subject: unattended-upgrades result for nuc1: SUCCESS

Unattended upgrade result: All upgrades installed

Packages that were upgraded:

linux-firmware

Package installation log:

Log started: 2025-03-13 06:19:10

Preparing to unpack

.../linux-firmware_20240318.git3b128b60-0ubuntu2.10_amd64.deb ...

Unpacking linux-firmware (20240318.git3b128b60-0ubuntu2.10) over (20240318.git3b128b60-0ubuntu2.9) ...

Setting up linux-firmware (20240318.git3b128b60-0ubuntu2.10) ...

Processing triggers for initramfs-tools (0.142ubuntu25.5) ...

update-initramfs: Generating /boot/initrd.img-6.8.0-55-generic

Running kernel seems to be up-to-date.

The processor microcode seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.

Log ended: 2025-03-13 06:19:26

(continues on next page)

(continued from previous page)

```
Unattended-upgrades log:
Starting unattended upgrades script
Allowed origins are: o=Ubuntu,a=noble, o=Ubuntu,a=noble-security,
o=UbuntuESMApps,a=noble-apps-security,
o=UbuntuESM,a=noble-infra-security, o=Ubuntu,a=noble,
o=Ubuntu,a=noble-security, o=UbuntuESMApps,a=noble-apps-security,
o=UbuntuESM,a=noble-infra-security
Initial blacklist:
Initial whitelist (not strict):
Packages that will be upgraded: linux-firmware
Writing dpkg log to /var/log/unattended-upgrades/unattended-upgrades-dpkg.log
All upgrades installed
```

Reboots

-
-
-

*shutdown(8)*³³⁵

Note:

For more information about this time specification for the reboot, and other options like cancelling a scheduled reboot, see the *shutdown(8)*³³⁶ manual page.

³³⁶ <https://manpages.ubuntu.com/manpages/resolute/man8/shutdown.8.html>

```
2025-03-13 20:43:25,923 INFO Starting unattended upgrades script
2025-03-13 20:43:25,924 INFO Allowed origins are: o=Ubuntu,a=noble, o=Ubuntu,a=noble-
```

(continues on next page)

³³⁵ <https://manpages.ubuntu.com/manpages/resolute/man8/shutdown.8.html>

(continued from previous page)

```
security, o=UbuntuESMApps,a=noble-apps-security, o=UbuntuESM,a=noble-infra-security
2025-03-13 20:43:25,924 INFO Initial blacklist:
2025-03-13 20:43:25,924 INFO Initial whitelist (not strict):
2025-03-13 20:43:29,082 INFO Packages that will be upgraded: libc6 python3-jinja2
2025-03-13 20:43:29,082 INFO Writing dpkg log to /var/log/unattended-upgrades/unattended-
upgrades-dpkg.log
2025-03-13 20:43:39,532 INFO All upgrades installed
2025-03-13 20:43:40,201 WARNING Found /var/run/reboot-required, rebooting
2025-03-13 20:43:40,207 WARNING Shutdown msg: b"Reboot scheduled for Thu 2025-03-13
20:45:00 UTC, use 'shutdown -c' to cancel."
```

Service restarts

*needrestart(1)*³³⁷
Starting with Ubuntu 24.04 LTS³³⁸

-
-

```
$nrconf{override_rc} = {
    qr(^whatever-critical-service\.service$) => 0,
};
```

³³⁷ <https://manpages.ubuntu.com/manpages/resolute/man1/needrestart.1.html>

³³⁸ <https://discourse.ubuntu.com/t/needrestart-changes-in-ubuntu-24-04-service-restarts/44671>

When to consider disabling automatic updates

Systems which just get recreated

Manual steps required

Too much of a risk

Postponable updates

Fleet management

-
-
-

Landscape³³⁹

Postponable updates

```
Unattended-Upgrade::Postpone-For-Days "3";
```

Prompt duration

³³⁹ <https://ubuntu.com/landscape>

Who can postpone

polkit documentation³⁴⁰

Notifications in different environments

Testing and troubleshooting

*manual page*³⁴¹

-
-

```
$ sudo unattended-upgrade -v --dry-run
```

³⁴⁰ <https://www.freedesktop.org/software/polkit/docs/latest/polkit.8.html>

³⁴¹ <https://manpages.ubuntu.com/manpages/resolute/man8/unattended-upgrade.8.html>

```
Starting unattended upgrades script
Allowed origins are: o=Ubuntu,a=noble, o=Ubuntu,a=noble-security, o=UbuntuESMApps,
a=noble-apps-security, o=UbuntuESM,a=noble-infra-security, o=LP-PPA-canonical-
server-server-backports,a=noble
Initial blacklist:
Initial whitelist (not strict):
Option --dry-run given, *not* performing real actions
Packages that will be upgraded: rdma-core
Writing dpkg log to /var/log/unattended-upgrades/unattended-upgrades-dpkg.log
/usr/bin/unattended-upgrade:567: DeprecationWarning: This process (pid=1213) is
multi-threaded, use of fork() may lead to deadlocks in the child.
  pid = os.fork()
/usr/bin/dpkg --status-fd 10 --no-triggers --unpack --auto-deconfigure /var/cache/
apt/archives/rdma-core_52.0-2ubuntu1~backport24.04.202410192216~ubuntu24.04.1_amd64.
deb
/usr/bin/dpkg --status-fd 10 --configure --pending
All upgrades installed
The list of kept packages can't be calculated in dry-run mode.
```

-
-

```
$ apt-cache policy rdma-core

rdma-core:
  Installed: 50.0-2build2
  Candidate: 52.0-2ubuntu1~backport24.04.202410192216~ubuntu24.04.1
  Version table:
     52.0-2ubuntu1~backport24.04.202410192216~ubuntu24.04.1 500
        500 https://ppa.launchpadcontent.net/canonical-server/server-backports/
ubuntu noble/main amd64 Packages
    *** 50.0-2build2 500
        500 http://br.archive.ubuntu.com/ubuntu noble/main amd64 Packages
        100 /var/lib/dpkg/status
```

How to upgrade your Ubuntu release

Ubuntu Pro subscription³⁴²

release lifecycle and support period³⁴³

Understanding Upgrade Paths

Pre-upgrade checklist

-
-

Ubuntu Wiki Releases Page³⁴⁴

explanation on phased updates

1.

```
$ sudo apt update
```

```
$ sudo apt dist-upgrade -o  
APT::Get::Always-Include-Phased-Updates=true
```

2.

³⁴² <https://ubuntu.com/pro>

³⁴³ <https://ubuntu.com/about/release-cycle>

³⁴⁴ <https://wiki.ubuntu.com/Releases>

3.

-
-
-
-

Upgrade the system

```
$ sudo do-release-upgrade
```

Note:

Upgrading to a development release of Ubuntu is available using the `-d` flag. However, using the development release (or the `-d` flag) is **not recommended** for production environments.

Pre-upgrade summary

```
Do you want to start the upgrade?
```

```
5 installed packages are no longer supported by Canonical. You can  
still get support from the community.
```

```
4 packages are going to be removed. 117 new packages are going to be  
installed. 424 packages are going to be upgraded.
```

```
You have to download a total of 262 M. This download will take about  
33 minutes with a 1Mbit DSL connection and about 10 hours with a 56k  
modem.
```

```
Fetching and installing the upgrade can take several hours. Once the  
download has finished, the process cannot be canceled.
```

```
Continue [yN] Details [d]
```

Configuration changes

```
Configuration file '/etc/ssh/ssh_config'  
==> Modified (by you or by a script) since installation.  
==> Package distributor has shipped an updated version.  
What would you like to do about it ? Your options are:  
  Y or I : install the package maintainer's version  
  N or O : keep your currently-installed version  
    D    : show the differences between the versions  
    Z    : start a shell to examine the situation  
The default action is to keep your current version.  
*** ssh_config (Y/I/N/O/D/Z) [default=N] ?
```

Removing Obsolete Packages

```
Remove obsolete packages?  
  
30 packages are going to be removed.  
  
Continue [yN] Details [d]
```

Reboot

```
System upgrade is complete.  
  
Restart required  
  
To finish the upgrade, a restart is required.  
If you select 'y' the system will be restarted.  
  
Continue [yN]
```

Further reading

- [List of releases](#)³⁴⁵

How to use the Ubuntu snapshot service

-
-
-

³⁴⁵ <https://documentation.ubuntu.com/project/release-team/list-of-releases/>

Prerequisites

```
deb http://archive.ubuntu.com/ubuntu jammy main universe restricted multiverse
deb http://archive.ubuntu.com/ubuntu jammy-updates main universe restricted multiverse
deb http://archive.ubuntu.com/ubuntu jammy-backports main restricted universe multiverse
deb http://security.ubuntu.com/ubuntu jammy-security main restricted universe multiverse
```

all components in all pockets³⁴⁶

```
deb [snapshot=yes] http://archive.ubuntu.com/ubuntu jammy main universe restricted
multiverse
deb [snapshot=yes] http://archive.ubuntu.com/ubuntu jammy-updates main universe restricted
multiverse
deb [snapshot=yes] http://archive.ubuntu.com/ubuntu jammy-backports main restricted
universe multiverse
deb [snapshot=yes] http://security.ubuntu.com/ubuntu jammy-security main restricted
universe multiverse
```

Ubuntu Release	Supports Snapshot Service?	Need to configure sources?
Next Ubuntu releases	Yes	No
Ubuntu 24.04 LTS	Yes	No
Ubuntu 22.04 LTS	Yes (with apt >= 2.4.11)	Yes
Ubuntu 20.04 LTS	Yes (with apt >= 2.0.10)	Yes

³⁴⁶ <https://documentation.ubuntu.com/project/how-ubuntu-is-made/concepts/package-archive/#pockets>

Quick start

```
$ sudo apt install hello --update --snapshot 20240301T030400Z
```

Using the snapshot service

The --snapshot CLI option

```
$ sudo apt update --snapshot $SNAPSHOT_ID
```

```
$ sudo apt install $PACKAGE_NAME --snapshot $SNAPSHOT_ID
```

```
$ sudo apt install $PACKAGE_NAME --update --snapshot $SNAPSHOT_ID
```

```
$ sudo apt install $PACKAGE_NAME --update -S $SNAPSHOT_ID
```

```
$ sudo apt update --snapshot 20240501T120000Z
```

```
$ apt policy docker.io --snapshot 20240501T120000Z

docker.io:
  Installed: (none)
  Candidate: 24.0.7-0ubuntu4
  Version table:
    24.0.7-0ubuntu4 500
    500 https://snapshot.ubuntu.com/ubuntu/20240501T120000Z noble/universe amd64
Packages
```

```
$ apt policy docker.io

docker.io:
  Installed: (none)
  Candidate: 28.2.2-0ubuntu1~24.04.1
  Version table:
    28.2.2-0ubuntu1~24.04.1 500
    500 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages
    27.5.1-0ubuntu3~24.04.2 500
    500 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages
    24.0.7-0ubuntu4 500
    500 http://archive.ubuntu.com/ubuntu noble/universe amd64 Packages
```

```
$ sudo apt install docker.io --snapshot 20240501T120000Z

...
```

```
$ docker --version

Docker version 24.0.7, build 24.0.7-0ubuntu4
```

Docker package publishing history in Launchpad³⁴⁷

```
$ sudo apt install docker.io --update --snapshot 20241126T230000Z

...
```

³⁴⁷ <https://launchpad.net/ubuntu/+source/docker.io-app/+publishinghistory>

```
$ docker --version
```

```
Docker version 26.1.3, build 26.1.3-0ubuntu1~24.04.1
```

Note:

Note that, at the time of writing, the version of docker.io³⁴⁸ in the noble-security pocket is 27.5.1-0ubuntu3~24.04.2, i.e., the version used in the example above may be affected by known vulnerabilities. When using the snapshot service, do make sure to check the latest version available in your Ubuntu release's -security pocket to make sure you are not installing a vulnerable package.

³⁴⁸ <http://docker.io>

Configuring the snapshot service for specific repositories

```
Snapshot: $SNAPSHOT_ID
```

```
deb [snapshot=$SNAPSHOT_ID] http://archive.ubuntu.com/ubuntu ...
```

```
Types: deb
URIs: http://archive.ubuntu.com/ubuntu
Suites: noble noble-updates noble-backports
Components: main universe restricted multiverse
Signed-By: /usr/share/keyrings/ubuntu-archive-keyring.gpg
Snapshot: 20250530T223000Z
```

```
Types: deb
URIs: http://security.ubuntu.com/ubuntu
Suites: noble-security
Components: main universe restricted multiverse
Signed-By: /usr/share/keyrings/ubuntu-archive-keyring.gpg
Snapshot: 20250530T223000Z
```

```
$ sudo apt install --update docker.io
```

```
$ docker --version
```

```
Docker version 27.5.1, build 27.5.1-0ubuntu3~24.04.1
```

```
$ sudo apt install --update docker.io --snapshot 20251013T2300Z
```

```
...
Hit:5 https://snapshot.ubuntu.com/ubuntu/20250530T223000Z noble InRelease
Hit:6 https://snapshot.ubuntu.com/ubuntu/20250530T223000Z noble-updates InRelease
Hit:7 https://snapshot.ubuntu.com/ubuntu/20250530T223000Z noble-backports InRelease
Hit:8 https://snapshot.ubuntu.com/ubuntu/20250530T223000Z noble-security InRelease
...
docker.io is already the newest version (27.5.1-0ubuntu3~24.04.1).
0 upgraded, 0 newly installed, 0 to remove and 3 not upgraded.
```

Configuring the snapshot service globally

Warning:

If you are following these examples until here, make sure to revert the changes to the sources configuration file made in the previous section.

```
$ echo 'APT::Snapshot "20250801T111111Z";' | sudo tee
/etc/apt/apt.conf.d/50snapshot
```

```
$ sudo apt install docker.io --update
```

```
...
```

```
$ docker --version
```

```
Docker version 27.5.1, build 27.5.1-0ubuntu3~24.04.2
```

```
$ sudo apt install docker.io --update --snapshot 20251013T120000Z
```

```
$ docker --version
```

```
Docker version 28.2.2, build 28.2.2-0ubuntu1~24.04.1
```

```
$ sudo rm /etc/apt/apt.conf.d/50snapshot
```

Further reading

- The Ubuntu snapshot service official page and documentation³⁴⁹
- Securing multiple Ubuntu instances while maximising uptime³⁵⁰
- Integrating the Ubuntu Snapshot Service into systems management and update tools³⁵¹
- *Automatic updates*
- *Upgrade your release*
- *Snapshot service*
- Enable Proposed³⁵²

Troubleshooting

- *Reporting bugs*

³⁴⁹ <https://snapshot.ubuntu.com/>

³⁵⁰ <https://ubuntu.com/blog/securing-multiple-ubuntu-instances-while-maximising-uptime>

³⁵¹ <https://ubuntu.com/blog/integrating-the-ubuntu-snapshot-service-into-systems-management-and-update-tools>

³⁵² <https://ubuntu.com/project/docs/SRU/reference/enable-proposed/>

How to report a bug in Ubuntu Server

uses Launchpad³⁵³

Report bugs with `apport-cli`

Note:

The commands `apport-cli` and `ubuntu-bug` should give the same results on a command-line interface (CLI) server. The latter is actually a symlink to `apport-bug`, which is intelligent enough to know whether a desktop environment is in use, and will choose `apport-cli` if not. Since server systems tend to be CLI-only, `apport-cli` was chosen from the outset in this guide.

```
$ apport-cli PACKAGENAME
```

³⁵³ <https://launchpad.net/>

```
*** Collecting problem information
```

```
The collected information can be sent to the developers to improve the
application. This might take a few minutes.
```

```
...
```

```
*** Send problem report to the developers?
```

```
After the problem report has been sent, please fill out the form in the
automatically opened web browser.
```

```
What would you like to do? Your options are:
```

```
S: Send report (2.8 KB)
```

```
V: View report
```

```
K: Keep report file for sending later or copying to somewhere else
```

```
I: Cancel and ignore future crashes of this program version
```

```
C: Cancel
```

```
Please choose (S/V/K/I/C):
```

S: Send report

```
*** Uploading problem information
```

```
The collected information is being sent to the bug tracking system.
This might take a few minutes.
```

```
94%
```

```
*** To continue, you must visit the following URL:
```

```
https://bugs.launchpad.net/ubuntu/+source/vim/+filebug/09b2495a-e2ab-11e3-879b-
68b5996a96c8?
```

```
You can launch a browser now, or copy this URL into a browser on another computer.
```

```
Choices:
```

```
1: Launch a browser now
```

```
C: Cancel
```

```
Please choose (1/C): 1
```

*Debian alternatives system*³⁵⁴

³⁵⁴ <https://manpages.ubuntu.com/manpages/resolute/man1/update-alternatives.1.html>

V: View

K: Keep

```
What would you like to do? Your options are:  
S: Send report (2.8 KB)  
V: View report  
K: Keep report file for sending later or copying to somewhere else  
I: Cancel and ignore future crashes of this program version  
C: Cancel  
Please choose (S/V/K/I/C): k  
Problem report file: /tmp/apport.vim.1pg92p02.apport
```

```
$ apport-cli apport.vim.1pg92p02.apport
```

```
$ apport-cli vim --save apport.vim.test.apport
```

Note:

If this Internet-enabled system is non-Ubuntu/Debian, `apport-cli` is not available so the bug will need to be created manually. An apport report is also not to be included as an attachment to a bug either so it is completely useless in this scenario.

Reporting application crashes


```
-rw-r----- 1 peter    whoopsie 150K Jul 24 16:17 _usr_lib_x86_64-linux-gnu_libmenu-  
cache2_libexec_menu-cached.1000.crash
```

```
$ apport-cli
```

```
*** Send problem report to the developers?
```

After the problem report has been sent, please fill out the form in the automatically opened web browser.

What would you like to do? Your options are:

S: Send report (153.0 KB)

V: View report

K: Keep report file for sending later or copying to somewhere else

I: Cancel and ignore future crashes of this program version

C: Cancel

Please choose (S/V/K/I/C): s

```
-rw-r----- 1 peter    whoopsie 150K Jul 24 16:17 _usr_lib_x86_64-linux-gnu_libmenu-  
cache2_libexec_menu-cached.1000.crash  
-rw-rw-r-- 1 peter    whoopsie    0 Jul 24 16:37 _usr_lib_x86_64-linux-gnu_libmenu-  
cache2_libexec_menu-cached.1000.upload  
-rw----- 1 whoopsie whoopsie    0 Jul 24 16:37 _usr_lib_x86_64-linux-gnu_libmenu-  
cache2_libexec_menu-cached.1000.uploaded
```

Further reading

- [Reporting Bugs](#)³⁵⁵
- [the Apport page](#)³⁵⁶
GUI

³⁵⁵ <https://help.ubuntu.com/community/ReportingBugs>

³⁵⁶ <https://wiki.ubuntu.com/Apport>

See also

- *Managing software*

The data and storage section

-
-

Data and storage

Data management

- *OpenLDAP*
- *Databases*

OpenLDAP

Install and configure LDAP

Lightweight Directory Access Protocol³⁵⁷

Introduction to OpenLDAP

slapd (the Stand-alone LDAP Daemon)³⁵⁸

³⁵⁷ <https://ldap.com/>

³⁵⁸ <https://www.openldap.org/software/man.cgi?query=slapd>

Note:

This guide uses a database suffix of `dc=example,dc=com`. Change this to match your particular setup.

Install slapd

```
$ sudo apt install slapd ldap-utils
```

Change the instance suffix (optional)

```
$ sudo dpkg-reconfigure slapd
```

- 1.
- 2.
- 3.
- 4.

```
BASE dc=example,dc=com
URI ldap://ldap01.example.com
```

Default tree contents

Note:

Although the configuration is accessible via the LDAP protocol under the `cn=config` suffix, it is also stored on disk under `/etc/ldap/slapd.d/`. This allows the configuration to be replicated to other servers and backed up alongside the data.

-

-

DN

SASL

Default configuration tree

```
$ sudo ldapsearch -Q -LLL -Y EXTERNAL -H ldapi:/// -b cn=config dn
```

-

-

-

-

-

-

```
dn: cn=config
dn: cn=module{0},cn=config
dn: cn=schema,cn=config
dn: cn={0}core,cn=schema,cn=config
dn: cn={1}cosine,cn=schema,cn=config
dn: cn={2}nis,cn=schema,cn=config
dn: cn={3}inetorgperson,cn=schema,cn=config
dn: olcDatabase={-1}frontend,cn=config
dn: olcDatabase={0}config,cn=config
dn: olcDatabase={1}mdb,cn=config
```

-
-
-
-
-
-
-
-
-
-

Frontend

```
$ sudo ldapsearch -Q -LLL -Y EXTERNAL -H ldapi:/// -b cn=config
```

```
dn: cn=config
objectClass: olcGlobal
cn: config
olcArgsFile: /var/run/slapd/slapd.args
olcLogLevel: none
...
```

Default “data” tree

```
$ ldapsearch -x -LLL -H ldap:/// -b dc=example,dc=com dn
```

- *bindDN*
-

```
dn: dc=example,dc=com
```

Who am I?

```
$ ldapwhoami -x
```

```
anonymous
```

```
$ ldapwhoami -x -D cn=admin,dc=example,dc=com -W
```

```
Enter LDAP Password:  
dn:cn=admin,dc=example,dc=com
```

Note:

A simple bind without some sort of transport security mechanism is **clear text**, meaning the credentials are transmitted in the clear. You should [add Transport Layer Security \(TLS\) support](#) (page 484) to your OpenLDAP server as soon as possible.

```
$ ldapwhoami -Y EXTERNAL -H ldapi:/// -Q
```

```
dn:gidNumber=1000+uidNumber=1000,cn=peercred,cn=external,cn=auth
```

```
$ sudo ldapwhoami -Y EXTERNAL -H ldapi:/// -Q
```

```
dn:gidNumber=0+uidNumber=0,cn=peercred,cn=external,cn=auth
```

uid *gid*

Set up LDAP access control

Note:

OpenLDAP ACLs are explained in [Set up access control](#) (page 469)

Populate the directory

- -
- -

```
dn: ou=People,dc=example,dc=com
objectClass: organizationalUnit
ou: People

dn: ou=Groups,dc=example,dc=com
objectClass: organizationalUnit
ou: Groups

dn: cn=miners,ou=Groups,dc=example,dc=com
objectClass: posixGroup
cn: miners
gidNumber: 5000
memberUid: john

dn: uid=john,ou=People,dc=example,dc=com
objectClass: inetOrgPerson
objectClass: posixAccount
objectClass: shadowAccount
uid: john
sn: Doe
givenName: John
cn: John Doe
displayName: John Doe
uidNumber: 10000
gidNumber: 5000
userPassword: {CRYPT}x
gecos: John Doe
loginShell: /bin/bash
homeDirectory: /home/john
```

Note:

It's important that `uid` and `gid` values in your directory do not collide with local Linux users' values as defined in `/etc/passwd` and `/etc/group`. You can use higher number ranges, such as starting at 5000 or even higher.

```
$ ldapadd -x -D cn=admin,dc=example,dc=com -W -f add_content.ldif
```

```
Enter LDAP Password: *****
adding new entry "ou=People,dc=example,dc=com"

adding new entry "ou=Groups,dc=example,dc=com"

adding new entry "cn=miners,ou=Groups,dc=example,dc=com"

adding new entry "uid=john,ou=People,dc=example,dc=com"
```



```
$ ldapsearch -x -LLL -b dc=example,dc=com '(uid=john)' cn gidNumber
```

```
dn: uid=john,ou=People,dc=example,dc=com
cn: John Doe
gidNumber: 5000
```

```
$ ldapsearch -x -LLL -b dc=example,dc=com
'(&(objectClass=posixGroup)(memberUid=john))' cn gidNumber
```

```
dn: cn=miners,ou=Groups,dc=example,dc=com
cn: miners
gidNumber: 5000
```

RFC 4515³⁵⁹

```
$ ldappasswd -x -D cn=admin,dc=example,dc=com -W -S
uid=john,ou=people,dc=example,dc=com
```

```
New password:
Re-enter new password:
Enter LDAP Password:
```

```
$ ldapwhoami -x -D uid=john,ou=people,dc=example,dc=com -W
```

³⁵⁹ <https://www.rfc-editor.org/rfc/rfc4515.txt>

```
Enter LDAP Password:
dn:uid=john,ou=People,dc=example,dc=com
```

Note:

Remember that simple binds are insecure and you should [add TLS support](#) (page 484) to your server as soon as possible!

Change the configuration

Which “admin” DN to use?

Suffix	DN for making changes	Authentication mechanism
cn=config	cn=admin,cn=config	Absent
cn=config	gidNumber=0+uidNumber=0, cn=peercred,cn=external, cn=auth	SASL EXTERNAL as root via ldapi://
dc=example, dc=com	cn=admin,dc=example,dc=com	Simple bind with passwordset dur- ing install or reconfigure

Change the “admin” password

```
$ sudo ldapsearch -Q -LLL -Y EXTERNAL -H ldapi:/// -b cn=config  
'(olcSuffix=dc=example,dc=com)' olcSuffix olcRootDN olcRootPW
```

```
dn: olcDatabase={1}mdb,cn=config  
olcSuffix: dc=example,dc=com  
olcRootDN: cn=admin,dc=example,dc=com  
olcRootPW: {SSHA}Y0UjBUUmf08TC25ePVc6waI/mfvPNktk
```

Note:

Even though this `olcRootDN` is the administrative DN for the `dc=example,dc=com` suffix, it is stored under the `cn=config` suffix!

```
New password:  
Re-enter new password:  
{SSHA}VKrYMxLSKhONGRpC6rnASKNmXG2xHXFo
```

```
dn: olcDatabase={1}mdb,cn=config  
changetype: modify  
replace: olcRootPW  
olcRootPW: {SSHA}VKrYMxLSKhONGRpC6rnASKNmXG2xHXFo
```

```
$ ldapmodify -Q -Y EXTERNAL -H ldapi:/// -f changerootp.ldif
```

```
modifying entry "olcDatabase={1}mdb,cn=config"
```

Add an index

```
dn: olcDatabase={1}mdb,cn=config
add: olcDbIndex
olcDbIndex: mail eq,sub
```

```
$ sudo ldapmodify -Q -Y EXTERNAL -H ldapi:/// -f add_index.ldif
modifying entry "olcDatabase={1}mdb,cn=config"
```

```
$ sudo ldapsearch -Q -LLL -Y EXTERNAL -H ldapi:/// -b cn=config
'(olcDatabase={1}mdb)' olcDbIndex

dn: olcDatabase={1}mdb,cn=config
olcDbIndex: objectClass eq
olcDbIndex: cn,uid eq
olcDbIndex: uidNumber,gidNumber eq
olcDbIndex: member,memberUid eq
olcDbIndex: mail eq,sub
```

See also

To learn more about OpenLDAP indexes, check the upstream documentation at <https://www.openldap.org/doc/admin26/tuning.html#Indexes>

Add a schema

Note:

It is not trivial to remove a schema from the slapd-config database. Practice adding schemas on a test system.

```
$ sudo ldapadd -Q -Y EXTERNAL -H ldapi:/// -f /etc/ldap/schema/corba.ldif
adding new entry "cn=corba,cn=schema,cn=config"
```

Logging

*slapd-config(5)*³⁶⁰

Example logging with the stats level

```
dn: cn=config
changetype: modify
replace: olcLogLevel
olcLogLevel: stats
```

```
$ sudo ldapmodify -Q -Y EXTERNAL -H ldapi:/// -f logging.ldif
```

³⁶⁰ <https://manpages.ubuntu.com/manpages/resolute/man5/slapd-config.5.html>

```
$ sudo journalctl -eu slapd.service
```

Next steps

about access control

set up users and groups

Set up LDAP access control

ACL

installed the slapd package

frontend

Getting the ACLs

```
$ sudo ldapsearch -Q -LLL -Y EXTERNAL -H ldapi:/// -b cn=config  
'(olcDatabase={1}mdb)' olcAccess  
  
dn: olcDatabase={1}mdb,cn=config  
olcAccess: {0}to attrs=userPassword by self write by anonymous auth by * none  
olcAccess: {1}to attrs=shadowLastChange by self write by * read  
olcAccess: {2}to * by * read
```

```
$ sudo ldapsearch -Q -LLL -Y EXTERNAL -H ldapi:/// -b cn=config  
'(olcDatabase={-1}frontend)' olcAccess
```

```
dn: olcDatabase={-1}frontend,cn=config
olcAccess: {0}to * by dn.exact=gidNumber=0+uidNumber=0,cn=peercred,cn=external
,cn=auth manage by * break
olcAccess: {1}to dn.exact="" by * read
olcAccess: {2}to dn.base="cn=Subschema" by * read
```

Note:

The Root *DN* always has full rights to its database and does not need to be included in any ACL.

Interpreting the results

```
olcAccess: {0}to attrs=userPassword by self write by anonymous auth by * none
olcAccess: {1}to attrs=shadowLastChange by self write by * read
```

```
to attrs=userPassword
  by self write
  by anonymous auth
  by * none

to attrs=shadowLastChange
  by self write
  by * read
```

-

DIT

-

-

-

```
to *
  by * read
```

SASL identity

```
dn.exact=gidNumber=0+uidNumber=0,cn=peercred,cn=external,cn=auth
```

```
$ sudo ldapsearch -Q -LLL -Y EXTERNAL -H ldapi:/// -b cn=config  
'(olcDatabase={0}config)' olcAccess  
  
dn: olcDatabase={0}config,cn=config  
olcAccess: {0}to * by dn.exact=gidNumber=0+uidNumber=0,cn=peercred,  
cn=external,cn=auth manage by * break
```

-
-

Grant root passwordless access to the data tree

```
# Allow root user (via peercred) to manage the primary database (mdb)  
dn: olcDatabase={1}mdb,cn=config  
changetype: modify  
add: olcAccess  
# Inserted as the first ACL rule  
olcAccess: {0}to * by dn.exact="gidNumber=0+uidNumber=0,cn=peercred,cn=external,cn=auth"  
manage by * break
```

```
$ sudo ldapmodify -Q -Y EXTERNAL -H ldapi:/// -f peercred.ldif
```



```
$ sudo ldapadd -Q -Y EXTERNAL -H ldapi:/// -f add_content.ldif
```

```
$ sudo ldapsearch -Q -LLL -Y EXTERNAL -H ldapi:/// -b cn=config  
'(olcAccess=*)' olcAccess olcSuffix
```

Next steps

set up LDAP users and groups

Further reading

- [*slapd.access\(5\)*](#)³⁶¹
- [access control topic](#)³⁶²

OpenLDAP replication

ability

high avail-

4533³⁶⁴

OpenLDAP administrator's guide³⁶³

RFC

-
-

Important:

You **must** have Transport Layer Security (TLS) enabled already before proceeding with this guide. Please consult the [LDAP with TLS guide](#) (page 484) for details of how to set this up.

³⁶¹ <https://manpages.ubuntu.com/manpages/resolute/man5/slapd.access.5.html>

³⁶² <https://openldap.org/doc/admin26/guide.html#Access%20Control>

³⁶³ <https://openldap.org/doc/admin26/guide.html#LDAP%20Sync%20Replication>

³⁶⁴ <https://www.rfc-editor.org/rfc/rfc4533.txt>

Provider configuration - replication user

ACL

```
dn: cn=replicator,dc=example,dc=com
objectClass: simpleSecurityObject
objectClass: organizationalRole
cn: replicator
description: Replication user
userPassword: {CRYPT}x
```

passwordless access for root

```
$ sudo ldapadd -Q -Y EXTERNAL -H ldapi:/// -f replicator.ldif
```

```
$ ldapadd -x -ZZ -H ldap://ldap01.example.com -D cn=admin,dc=example,dc=com
-W -f replicator.ldif
```

```
$ ldappasswd -x -ZZ -H ldap://ldap01.example.com -D
cn=admin,dc=example,dc=com -W -S cn=replicator,dc=example,dc=com
```

Note:

Adjust the server URI in the -H parameter if needed to match your deployment.

-
-

```
$ sudo ldapsearch -Q -Y EXTERNAL -H ldapi:/// -LLL -b cn=config
'(olcSuffix=dc=example,dc=com)' olcAccess

dn: olcDatabase={1}mdb,cn=config
olcAccess: {0}to attrs=userPassword by self write by anonymous auth by * none
olcAccess: {1}to attrs=shadowLastChange by self write by * read
olcAccess: {2}to * by * read
```

```
dn: olcDatabase={1}mdb,cn=config
changetype: modify
add: olcAccess
olcAccess: {0}to *
    by dn.exact="cn=replicator,dc=example,dc=com" read
    by * break
-
add: olcLimits
olcLimits: dn.exact="cn=replicator,dc=example,dc=com"
    time.soft=unlimited time.hard=unlimited
    size.soft=unlimited size.hard=unlimited
```

```
$ sudo ldapmodify -Q -Y EXTERNAL -H ldapi:/// -f replicator-acl-limits.ldif
```

Provider configuration - standard replication

```
# Add indexes to the frontend db.
dn: olcDatabase={1}mdb,cn=config
changetype: modify
add: olcDbIndex
olcDbIndex: entryCSN eq
-
add: olcDbIndex
olcDbIndex: entryUUID eq

#Load the syncprov module.
dn: cn=module{0},cn=config
changetype: modify
add: olcModuleLoad
olcModuleLoad: syncprov

# syncrepl Provider for primary db
dn: olcOverlay=syncprov,olcDatabase={1}mdb,cn=config
changetype: add
objectClass: olcOverlayConfig
objectClass: olcSyncProvConfig
olcOverlay: syncprov
olcSpCheckpoint: 100 10
olcSpSessionLog: 100
```

Warning:

The LDIF above has some parameters that you should review before deploying in production on your directory. In particular – `olcSpCheckpoint` and `olcSpSessionLog`. Please see the [slapo-syncprov\(5\)](https://manpages.ubuntu.com/manpages/resolute/man5/slapo-syncprov.5.html)³⁶⁵ manual page. In general, `olcSpSessionLog` should be equal to (or preferably larger than) the number of entries in your directory. Also see [ITS #8125](https://bugs.openldap.org/show_bug.cgi?id=8125)³⁶⁶ for details on an existing bug.

³⁶⁵ <https://manpages.ubuntu.com/manpages/resolute/man5/slapo-syncprov.5.html>

³⁶⁶ https://bugs.openldap.org/show_bug.cgi?id=8125

```
$ sudo ldapadd -Q -Y EXTERNAL -H ldapi:/// -f provider_simple_sync.ldif
```

Consumer configuration - standard replication

*the installation steps
enable TLS*

```
dn: cn=module{0},cn=config
changetype: modify
add: olcModuleLoad
olcModuleLoad: syncprov

dn: olcDatabase={1}mdb,cn=config
changetype: modify
add: olcDbIndex
olcDbIndex: entryUUID eq
-
add: olcSyncrepl
olcSyncrepl: rid=0
provider=ldap://ldap01.example.com
bindmethod=simple
binddn="cn=replicator,dc=example,dc=com" credentials=<secret>
searchbase="dc=example,dc=com"
schemachecking=on
type=refreshAndPersist retry="60 +"
starttls=critical tls_reqcert=demand
-
add: olcUpdateRef
olcUpdateRef: ldap://ldap01.example.com
```

- *hostname*
- **binddn** *DN*
-

-
-
-

Note:

A successful encrypted connection via START_TLS is being enforced in this configuration, to avoid sending the credentials in the clear across the network. See [LDAP with TLS](#) (page 484) for details on how to set up OpenLDAP with trusted SSL certificates.

```
$ sudo ldapadd -Q -Y EXTERNAL -H ldapi:/// -f consumer_simple_sync.ldif
```

Provider configuration - delta replication

-
-
-

Add syncprov and accesslog overlays and DBs

```
# Add indexes to the frontend db.
dn: olcDatabase={1}mdb,cn=config
changetype: modify
add: olcDbIndex
olcDbIndex: entryCSN eq
-
add: olcDbIndex
olcDbIndex: entryUUID eq

#Load the syncprov and accesslog modules.
dn: cn=module{0},cn=config
changetype: modify
add: olcModuleLoad
olcModuleLoad: syncprov
-
add: olcModuleLoad
olcModuleLoad: accesslog

# Accesslog database definitions
```

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```
dn: olcDatabase={2}mdb,cn=config
objectClass: olcDatabaseConfig
objectClass: olcMdbConfig
olcDatabase: {2}mdb
olcDbDirectory: /var/lib/ldap/accesslog
olcSuffix: cn=accesslog
olcRootDN: cn=admin,dc=example,dc=com
olcDbIndex: default eq
olcDbIndex: entryCSN,objectClass,reqEnd,reqResult,reqStart
olcAccess: {0}to * by dn.exact="cn=replicator,dc=example,dc=com" read by * break
olcLimits: dn.exact="cn=replicator,dc=example,dc=com"
    time.soft=unlimited time.hard=unlimited
    size.soft=unlimited size.hard=unlimited

# Accesslog db syncprov.
dn: olcOverlay=syncprov,olcDatabase={2}mdb,cn=config
changetype: add
objectClass: olcOverlayConfig
objectClass: olcSyncProvConfig
olcOverlay: syncprov
olcSpNoPresent: TRUE
olcSpReloadHint: TRUE

# syncrepl Provider for primary db
dn: olcOverlay=syncprov,olcDatabase={1}mdb,cn=config
changetype: add
objectClass: olcOverlayConfig
objectClass: olcSyncProvConfig
olcOverlay: syncprov
olcSpCheckpoint: 100 10
olcSpSessionLog: 100

# accesslog overlay definitions for primary db
dn: olcOverlay=accesslog,olcDatabase={1}mdb,cn=config
objectClass: olcOverlayConfig
objectClass: olcAccessLogConfig
olcOverlay: accesslog
olcAccessLogDB: cn=accesslog
olcAccessLogOps: writes
olcAccessLogSuccess: TRUE
# scan the accesslog DB every day, and purge entries older than 7 days
olcAccessLogPurge: 07+00:00 01+00:00
```

Warning:

The LDIF above has some parameters that you should review before deploying in production on your directory. In particular – `olcSpCheckpoint`, `olcSpSessionLog`. Please see the [slapo-syncprov\(5\)](https://manpages.ubuntu.com/manpages/resolute/man5/slapo-syncprov.5.html)³⁶⁷ manual page. In general, `olcSpSessionLog` should be equal to (or preferably larger than) the number of entries in your directory. Also see [ITS #8125](https://bugs.openldap.org/show_bug.cgi?id=8125)³⁶⁸ for details on an existing bug. For `olcAccessLogPurge`, please check the [slapo-accesslog\(5\)](https://manpages.ubuntu.com/manpages/resolute/man5/slapo-accesslog.5.html)³⁶⁹ manual page.

³⁶⁷ <https://manpages.ubuntu.com/manpages/resolute/man5/slapo-syncprov.5.html>

³⁶⁸ https://bugs.openldap.org/show_bug.cgi?id=8125

³⁶⁹ <https://manpages.ubuntu.com/manpages/resolute/man5/slapo-accesslog.5.html>

```
$ sudo -u openldap mkdir /var/lib/ldap/accesslog
```

```
$ sudo ldapadd -Q -Y EXTERNAL -H ldapi:/// -f provider_sync.ldif
```

Consumer configuration

*the installation steps
enable TLS*

```
dn: cn=module{0},cn=config
changetype: modify
add: olcModuleLoad
olcModuleLoad: syncprov

dn: olcDatabase={1}mdb,cn=config
changetype: modify
add: olcDbIndex
olcDbIndex: entryUUID eq
-
add: olcSyncrepl
olcSyncrepl: rid=0
provider=ldap://ldap01.example.com
bindmethod=simple
binddn="cn=replicator,dc=example,dc=com" credentials=<secret>
searchbase="dc=example,dc=com"
logbase="cn=accesslog"
logfilter="(&(objectClass=auditWriteObject)(reqResult=0))"
schemachecking=on
type=refreshAndPersist retry="60 +"
syncdata=accesslog
starttls=critical tls_reqcert=demand
-
```

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```
add: olcUpdateRef  
olcUpdateRef: ldap://ldap01.example.com
```

-
-
-
-
-
-

Note:

Note that a successful encrypted connection via START_TLS is being enforced in this configuration, to avoid sending the credentials in the clear across the network. See [LDAP with TLS](#) (page 484) for details on how to set up OpenLDAP with trusted SSL certificates.

```
$ sudo ldapadd -Q -Y EXTERNAL -H ldapi:/// -f consumer_sync.ldif
```

Testing

```
$ ldapsearch -z1 -LLL -x -s base -b dc=example,dc=com contextCSN  
dn: dc=example,dc=com  
contextCSN: 20200423222317.722667Z#000000#000#000000
```



```
$ ldapwhoami -x -ZZ -H ldap://ldap01.example.com -D  
cn=replicator,dc=example,dc=com -W
```

```
$ ldapsearch -x -LLL -H ldap://ldap02.example.com -b dc=example,dc=com  
'(uid=john)' uid  
  
dn: uid=john,ou=People,dc=example,dc=com  
uid: john
```

Further reading

- Replication types, OpenLDAP Administrator's Guide³⁷⁰
- LDAP Sync Replication - OpenLDAP Administrator's Guide³⁷¹
- RFC 4533³⁷²

How to set up LDAP users and groups

have a working LDAP server

nsld

SSSD

User and group management - ldapscripts

Install ldapscripts

```
$ sudo apt install ldapscripts
```

³⁷⁰ <https://openldap.org/doc/admin26/guide.html#Configuring%20the%20different%20replication%20types>

³⁷¹ <https://openldap.org/doc/admin26/guide.html#LDAP%20Sync%20Replication>

³⁷² <https://www.rfc-editor.org/rfc/rfc4533.txt>

```
SERVER=ldap://ldap01.example.com
LDAPBINOPTS="-ZZ"
BINDDN='cn=admin,dc=example,dc=com'
BINDPWDFILE="/etc/ldapscripts/ldapscripts.passwd"
SUFFIX='dc=example,dc=com'
GSUFFIX='ou=Groups'
USUFFIX='ou=People'
MSUFFIX='ou=Computers'
```

Note:

Adjust **SERVER** and related **SUFFIX** options to suit your directory structure. Here, we are forcing use of **START_TLS** (-ZZ parameter). Refer to [LDAP with TLS](#) (page 484) to learn how to set up the server with TLS support.

```
$ echo -n 'password' | sudo tee /etc/ldapscripts/ldapscripts.passwd
```

```
$ sudo chmod 400 /etc/ldapscripts/ldapscripts.passwd
```

Note:

The password file must contain exactly and only the password characters, no end-of-line or anything else. The echo command above with the -n parameter achieves that by suppressing the *EOL* character \n. And in order to prevent the password from appearing in the shell history, the *echo* command line is prefixed by a space.

Manage users and groups with ldapscripts

Create a new user

```
$ sudo ldapaddgroup george
```

```
$ sudo ldapadduser george george
```

GID

Change a user's password

```
$ sudo ldapsetpasswd george

Changing password for user uid=george,ou=People,dc=example,dc=com
New Password:
Retype New Password:
Successfully set password for user uid=george,ou=People,dc=example,dc=com
```

Delete a user

```
$ sudo ldapdeleteuser george
```

Add a group

```
$ sudo ldapaddgroup qa
```

Delete a group

```
$ sudo ldapdeletegroup qa
```

Add a user to a group

```
$ sudo ldapaddusertogroup george qa
```

Remove a user from a group

```
$ sudo ldapdeleteuserfromgroup george qa
```

Manage user attributes with `ldapmodifyuser`

```
sudo ldapmodifyuser george
# About to modify the following entry :
dn: uid=george,ou=People,dc=example,dc=com
objectClass: account
objectClass: posixAccount
cn: george
uid: george
uidNumber: 10001
gidNumber: 10001
homeDirectory: /home/george
loginShell: /bin/bash
gecos: george
description: User account
userPassword:: e1NTSEF9eXF5TFcyWlhWkF1eGUybVdFWHZKRzJVMjFTSG9vcHk=

# Enter your modifications here, end with CTRL-D.
dn: uid=george,ou=People,dc=example,dc=com
replace: geocos
gecos: George Carlin
```

ldapscripts templates

```
UTEMPLATE="/etc/ldapscripts/ldapadduser.template"
```

```
$ sudo cp /usr/share/doc/ldapscripts/examples/ldapadduser.template.sample \
> /etc/ldapscripts/ldapadduser.template
```

```
dn: uid=<user>,<usuffix>,<suffix>
objectClass: inetOrgPerson
objectClass: posixAccount
cn: <user>
sn: <ask>
uid: <user>
uidNumber: <uid>
gidNumber: <gid>
homeDirectory: <home>
```

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```
loginShell: <shell>
gecos: <user>
description: User account
title: Employee
```

```
$ dpkg -L ldapscripts | grep /usr/sbin
```

Next steps

setting up Transport Layer Security (TLS)

LDAP and Transport Layer Security (TLS)

Note:

For general information on managing certificates in Ubuntu, see [About certificates](#) (page 927). For installing a custom root CA, see [Install a root CA certificate in the trust store](#) (page 259).

```
$ sudo apt install gnutls-bin ssl-cert
```

```
$ sudo certtool --generate-privkey --bits 4096 --outfile
/etc/ssl/private/mycakey.pem
```

```
cn = Example Company
ca
cert_signing_key
expiration_days = 3650
```

```
$ sudo certtool --generate-self-signed \ > --load-privkey  
/etc/ssl/private/mycakey.pem \ > --template /etc/ssl/ca.info \ > --outfile  
/usr/local/share/ca-certificates/mycacert.crt
```

Note:

The `--outfile` path is correct. We are writing the CA certificate to `/usr/local/share/ca-certificates`. This is where `update-ca-certificates` picks up trusted local CAs from. To selectively enable CAs from `/usr/share/ca-certificates`, you can run `dpkg-reconfigure ca-certificates`.

```
$ sudo update-ca-certificates  
  
Updating certificates in /etc/ssl/certs...  
1 added, 0 removed; done.  
Running hooks in /etc/ca-certificates/update.d...  
done.
```

```
$ sudo certtool --generate-privkey \ > --bits 2048 \ > --outfile  
/etc/ldap/ldap01_slapd_key.pem
```

Note:

Replace `ldap01` in the filename with your server's *hostname*. Naming the certificate and key for the host and service that will be using them will help keep things clear.

```
organization = Example Company  
cn = ldap01.example.com  
tls_www_server  
encryption_key  
signing_key  
expiration_days = 365
```

```
$ sudo certtool --generate-certificate \ > --load-privkey  
/etc/ldap/ldap01_slapd_key.pem \ > --load-ca-certificate  
/etc/ssl/certs/mycacert.pem \ > --load-ca-privkey  
/etc/ssl/private/mycakey.pem \ > --template /etc/ssl/ldap01.info \ >  
--outfile /etc/ldap/ldap01_slapd_cert.pem
```

```
$ sudo chgrp openldap /etc/ldap/ldap01_slapd_key.pem
```

```
$ sudo chmod 0640 /etc/ldap/ldap01_slapd_key.pem
```

```
dn: cn=config  
add: olcTLSCACertificateFile  
olcTLSCACertificateFile: /etc/ssl/certs/mycacert.pem  
-  
add: olcTLSCertificateFile  
olcTLSCertificateFile: /etc/ldap/ldap01_slapd_cert.pem  
-  
add: olcTLSCertificateKeyFile  
olcTLSCertificateKeyFile: /etc/ldap/ldap01_slapd_key.pem
```

```
$ sudo ldapmodify -Y EXTERNAL -H ldapi:/// -f certinfo.ldif
```

```
SLAPD_SERVICES="ldap:/// ldapi:/// ldaps://"
```

```
$ sudo systemctl restart slapd
```

```
$ ldapwhoami -x -ZZ -H ldap://ldap01.example.com  
  
anonymous
```

```
$ ldapwhoami -x -H ldaps://ldap01.example.com  
  
anonymous
```

```
$ mkdir ldap02-ssl > cd ldap02-ssl > certtool --generate-privkey \ > --bits  
2048 \ > --outfile ldap02_slapd_key.pem
```

```
organization = Example Company  
cn = ldap02.example.com  
tls_www_server  
encryption_key  
signing_key  
expiration_days = 365
```

```
$ sudo certtool --generate-certificate \ > --load-privkey  
ldap02_slapd_key.pem \ > --load-ca-certificate /etc/ssl/certs/mycacert.pem \  
> --load-ca-privkey /etc/ssl/private/mycakey.pem \ > --template ldap02.info \  
> --outfile ldap02_slapd_cert.pem
```

Note:

We had to use `sudo` to get access to the CA's private key. This means the generated certificate file is owned by root. You should change that ownership back to your regular user before copying these files over to the Consumer.

```
$ cp /etc/ssl/certs/mycacert.pem .
```

```
$ cd .. > scp -r ldap02-ssl user@consumer:
```

```
$ sudo cp ldap02_slapd_cert.pem ldap02_slapd_key.pem /etc/ldap > sudo chgrp  
openldap /etc/ldap/ldap02_slapd_key.pem > sudo chmod 0640  
/etc/ldap/ldap02_slapd_key.pem > sudo cp mycacert.pem  
/usr/local/share/ca-certificates/mycacert.crt > sudo update-ca-certificates
```



```
dn: cn=config
add: olcTLSCACertificateFile
olcTLSCACertificateFile: /etc/ssl/certs/mycacert.pem
-
add: olcTLSCertificateFile
olcTLSCertificateFile: /etc/ldap/ldap02_slapd_cert.pem
-
add: olcTLSCertificateKeyFile
olcTLSCertificateKeyFile: /etc/ldap/ldap02_slapd_key.pem
```

```
$ sudo ldapmodify -Y EXTERNAL -H ldapi:/// -f certinfo.ldif
```

```
$ ldapwhoami -x -ZZ -H ldap://ldap02.example.com
anonymous
```

```
$ ldapwhoami -x -H ldaps://ldap02.example.com
anonymous
```

Backup and restore OpenLDAP

DIT

Backup script

```
#!/bin/bash
set -euo pipefail

BACKUP_PATH=/export/backup
CONFIG_BACKUP=f${BACKUP_PATH}/config.ldif"
DATA_BACKUP="${BACKUP_PATH}/example.com.ldif"
```

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```
# create and secure backup files
touch "$CONFIG_BACKUP" "$DATA_BACKUP"
chmod 600 "$CONFIG_BACKUP" "$DATA_BACKUP"

# Backup server config
nice slapcat -b cn=config > "$CONFIG_BACKUP"
# Backup directory tree
nice slapcat -b dc=example,dc=com > "$DATA_BACKUP"

# Optionally, use a backup tool like borgbackup to store the backups off-site
```

```
$ sudo chmod +x /usr/local/bin/ldapbackup
```

Note:

These files are uncompressed text files containing everything in your directory including the tree layout, usernames, and every password. Consider making /export/backup an encrypted partition and even having the script encrypt files as it creates them.

Schedule backups with systemd

```
[Unit]
Description=LDAP backup

[Service]
Type=oneshot
ExecStart=/usr/local/bin/ldapbackup
```

```
[Unit]
Description=Run LDAP backup daily

[Timer]
OnCalendar=*-*-* 22:45:00
Persistent=true

[Install]
WantedBy=timers.target
```

```
$ sudo systemctl daemon-reload
```

```
$ sudo systemctl enable --now ldapbackup.timer
```

```
$ systemctl list-timers ldapbackup.timer
```

Restore script

```
#!/bin/bash
set -euo pipefail

BACKUP_PATH=/export/backup

if [ -n "$(ls -l /var/lib/ldap/* 2>/dev/null)" ] || [ -n "$(ls -l /etc/ldap/slapd.d/* 2>/dev/null)" ]; then
    echo "Existing database found. Run the following to remove it:"
    echo "  sudo systemctl stop slapd.service"
    echo "  sudo rm -rf /etc/ldap/slapd.d/* /var/lib/ldap/*"
    exit 1
fi

sudo systemctl stop slapd.service || :
sudo slapadd -F /etc/ldap/slapd.d -b cn=config -l ${BACKUP_PATH}/config.ldif
sudo slapadd -F /etc/ldap/slapd.d -b dc=example,dc=com -l ${BACKUP_PATH}/example.com.ldif
sudo chown -R openldap:openldap /etc/ldap/slapd.d/
sudo chown -R openldap:openldap /var/lib/ldap/
sudo systemctl start slapd.service
```

How to configure OpenLDAP with passthrough authentication

Background

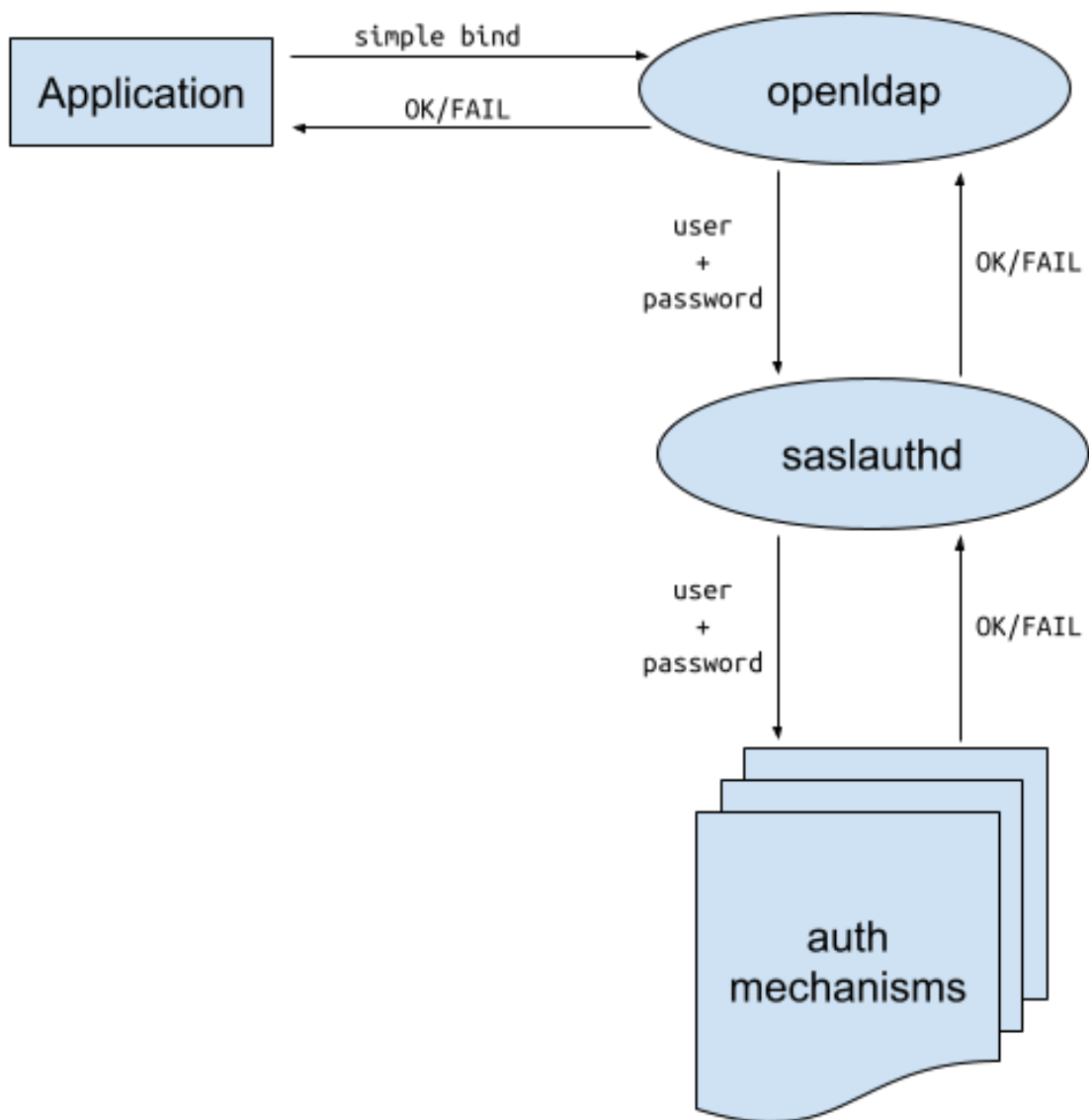
Generic Security Services Application Programming Interface³⁷³

setting up Transport Layer

Security (TLS)

³⁷³ <https://www.openldap.org/doc/admin26/sasl.html#GSSAPI>

What is passthrough authentication



Saslauthd authentication providers

-
-
-
-
-
-
-

Further reading

- [OpenLDAP.org reference](#)³⁷⁴

How to configure OpenLDAP with passthrough SASL authentication using Kerberos

Before you begin

- [How to install a Kerberos server](#)
and configure OpenLDAP *Install*
- [How to install a Kerberos server](#)
LDAP and Transport Layer Security (TLS)
- [How to install a Kerberos server](#)
service principals *How to configure Kerberos*

³⁷⁴ <https://www.openldap.org/doc/admin26/sasl.html>

Note:

This process is not the same as using [Generic Security Services Application Programming Interface³⁷⁵](https://www.openldap.org/doc/admin26/sasl.html#GSSAPI) (GSSAPI) to log into the LDAP server.

Rather it is using [simple authentication³⁷⁶](https://www.openldap.org/doc/admin26/security.html#%22simple%22%20method) with the OpenLDAP server so this should be over a [Transport Layer Security³⁷⁷](https://datatracker.ietf.org/wg/tls/documents/) (TLS) connection.

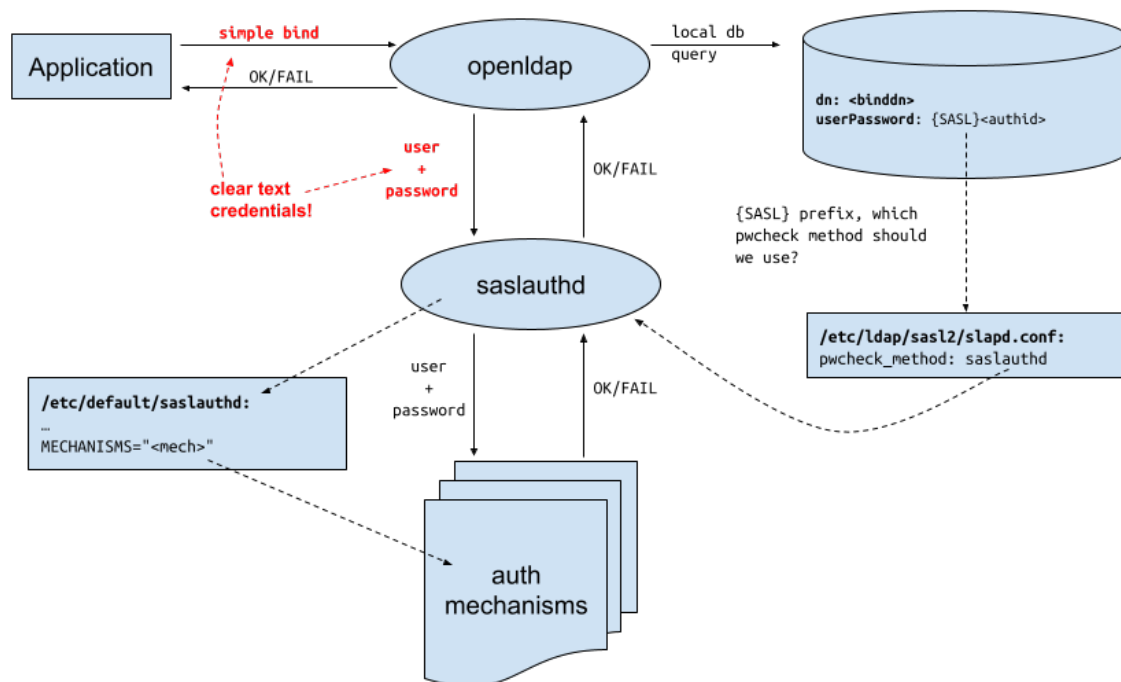
The test user we will be using is ubuntu@EXAMPLE.COM, which must exist in the Kerberos database

³⁷⁵ <https://www.openldap.org/doc/admin26/sasl.html#GSSAPI>

³⁷⁶ <https://www.openldap.org/doc/admin26/security.html#%22simple%22%20method>

³⁷⁷ <https://datatracker.ietf.org/wg/tls/documents/>

How the passthrough authentication will work



- *TLS* *LDAP and Transport Layer Security (TLS)*
-
-

Package installation

```
$ sudo apt install sasl2-bin
```

Check the hostname

```
$ hostname -f  
ldap-server.example.com
```

```
$ nslookup 10.10.17.91  
91.17.10.10.in-addr.arpa      name = ldap-server.example.com.
```

Fully Qualified Domain Name³⁷⁸

```
$ sudo vi /etc/hosts
```

```
10.10.17.91 ldap-server.example.com ldap-server
```

Create the saslauthd principal

How to configure Kerberos service principals

³⁷⁸ https://en.wikipedia.org/wiki/Fully_qualified_domain_name

```
$ sudo kadmin -p ubuntu/admin
```

```
Authenticating as principal ubuntu/admin with password.  
Password for ubuntu/admin@EXAMPLE.COM  
kadmin: addprinc -randkey host/ldap-server.example.com  
No policy specified for host/ldap-server.example.com@EXAMPLE.COM; defaulting to no policy  
Principal "host/ldap-server.example.com@EXAMPLE.COM" created.  
kadmin: ktadd host/ldap-server.example.com  
Entry for principal host/ldap-server.example.com with kvno 2, encryption type aes256-cts-hmac-sha1-96 added to keytab FILE:/etc/krb5.keytab.  
Entry for principal host/ldap-server.example.com with kvno 2, encryption type aes128-cts-hmac-sha1-96 added to keytab FILE:/etc/krb5.keytab.
```

```
$ kadmin -p ubuntu/admin -q "addprinc -randkey host/ldap-server.example.com"
```

Note:

sudo is not needed to remotely create a new principal.

```
$ sudo kadmin -p ubuntu/admin -q "ktadd host/ldap-server.example.com"
```

```
$ sudo klist -k
```

```
Keytab name: FILE:/etc/krb5.keytab  
KVNO Principal  
-----  
  2 host/ldap-server.example.com@EXAMPLE.COM  
  2 host/ldap-server.example.com@EXAMPLE.COM
```

Configure saslauthd

```
$ sudo vi /etc/default/saslauthd
```



```
...
# Which authentication mechanisms should saslauthd use? (default: pam)
#
# Available options in this Debian package:
# getpwent -- use the getpwent() library function
# kerberos5 -- use Kerberos 5
# pam      -- use PAM
# rimap    -- use a remote IMAP server
# shadow   -- use the local shadow password file
# sasldb   -- use the local sasldb database file
# ldap     -- use LDAP (configuration is in /etc/saslauthd.conf)
#
# Only one option may be used at a time. See the saslauthd man page
# for more information.
#
# Example: MECHANISMS="pam"
MECHANISMS="kerberos5"
...
```

Important:

For Ubuntu version 22.04 and earlier “START=yes” must also be added to the default config file for saslauthd to start.

Enable and start saslauthd

```
$ sudo systemctl enable --now saslauthd
```

Test saslauthd configuration

```
$ testsaslauthd -u ubuntu -p ubuntusecret
0: OK "Success."
```

```
$ testsaslauthd -u ubuntu -p ubuntusecretwrong
0: NO "authentication failed"
```

Note:

In Ubuntu 22.04 LTS and earlier, the `/run/saslauthd` directory is restricted to members of the `sasl` group, so the `testsaslauthd` commands above need to be run as root (via `sudo`) or as a user who is in the `sasl` group.

Configure OpenLDAP

```
pwcheck_method: saslauthd
```

```
$ sudo gpasswd -a openldap sasl
```

```
$ sudo systemctl restart slapd.service
```

Change the userPassword attribute

```
dn: uid=ubuntu,dc=example,dc=com
uid: ubuntu
objectClass: account
objectClass: simpleSecurityObject
userPassword: {SSHA}S+uImGneLDFeCwErKnY4mJngnVJMZAM5
```

bindDN

```
userPassword: {SASL}ubuntu@EXAMPLE.COM
```

Important:

Note how the username present in the userPassword attribute is independent of the bindDN used in the simple bind! If the userPassword attribute contained, say, {SASL}anotheruser@EXAMPLE.COM, OpenLDAP would ask saslauthd to authenticate anotheruser@EXAMPLE.COM, and not the user from the bindDN! Therefore, it's important to use OpenLDAP ACLs to prevent users from changing the userPassword attribute when using passthrough authentication!

```
$ ldapadd -x -D cn=admin,dc=example,dc=com -W -ZZ <<LDIF > dn:
uid=ubuntu,dc=example,dc=com > uid: ubuntu > objectClass: account >
objectClass: simpleSecurityObject > userPassword: {SASL}ubuntu@EXAMPLE.COM >
LDIF
```

Note:

Note how we don't need to add the posix attributes like user id, home directory, group, etc. All we really need is a directory entry that contains a userPassword attribute.

Test the authentication

```
$ ldapwhoami -D uid=ubuntu,dc=example,dc=com -W -x -ZZ
Enter LDAP Password:
```

```
dn:uid=ubuntu,dc=example,dc=com
```

```
ldap_bind: Invalid credentials (49)
```

Troubleshooting

```
$ sudo systemctl stop saslauthd
```

```
$ sudo /usr/sbin/saslauthd -a kerberos5 -d -m /var/run/saslauthd -n 1
```

Advanced options

-
-

```
krb5_keytab: /etc/saslauthd.keytab  
krb5_verify_principal: saslauthd
```

Further reading

- [OpenLDAP Passthrough Authentication](#)³⁷⁹
- [Cyrus SASL Password Verification](#)³⁸⁰
- [Cyrus SASL slapd Configuration File](#)³⁸¹
- [Kerberos Client Hostname Requirements](#)³⁸²

³⁷⁹ <https://www.openldap.org/doc/admin26/security.html#Pass-Through%20authentication>

³⁸⁰ <https://www.cyrusimap.org/sasl/sasl/components.html#password-verification-services>

³⁸¹ <https://www.cyrusimap.org/sasl/sasl/faqs/openldap-sasl-gssapi.html>

³⁸² https://web.mit.edu/kerberos/krb5-1.12/doc/admin/princ_dns.html

Set up an LDAP client

Lightweight Directory Access

Protocol (LDAP)³⁸³

Pluggable Authentication Modules (PAM)³⁸⁴

NSS

Note:

Ubuntu also supports [SSSD](https://sssd.io/)³⁸⁵ as an alternative LDAP client. SSSD provides credential caching and offline authentication. If you need those features, see [How to set up SSSD with LDAP](#) (page 214) instead.

³⁸⁵ <https://sssd.io/>

Prerequisites

- [Install and configure LDAP](#)
- [DN](#)
- [LDAP and Transport Layer Security \(TLS\)](#)
- [users/groups in LDAP](#)

Install the LDAP client daemon

```
$ sudo apt install nslcd libpam-ldapd libnss-ldapd
```

-
-

³⁸³ https://en.wikipedia.org/wiki/Lightweight_Directory_Access_Protocol

³⁸⁴ https://en.wikipedia.org/wiki/Pluggable_Authentication_Module

Configure nslcd

```
# /etc/nslcd.conf
uid nslcd
gid nslcd

uri ldaps://ldap.example.com

base dc=example,dc=com

tls_reqcert demand
tls_cacertfile /etc/ssl/certs/ca-certificates.crt
```

*nslcd.conf(5)*³⁸⁶

Set up PAM

```
$ sudo pam-auth-update
```

```
$ sudo systemctl restart nslcd
```

Test the configuration

```
$ id <username>
```

NSS

```
$ getent passwd
```

```
$ getent group
```

NSS

³⁸⁶ <https://manpages.ubuntu.com/manpages/resolute/man5/nslcd.conf.5.html>

```
$ sudo systemctl stop nslcd.service
```

```
$ sudo nslcd -n -d
```

```
$ sudo systemctl start nslcd.service
```

Configure home directories

*passwd(5)*³⁸⁷

```
map passwd homeDirectory /home/$uid
```

*nslcd.conf(5)*³⁸⁸

Restrict login access

```
# Allow members of specific LDAP groups
+:(some-admin-group) (some-user-group):ALL
# Allow the local root user and members of the local "login" group (group name is
customizable)
+:root (login):ALL
# Deny everyone else
-:ALL:ALL
```

³⁸⁷ <https://manpages.ubuntu.com/manpages/resolute/man5/passwd.5.html>

³⁸⁸ <https://manpages.ubuntu.com/manpages/resolute/man5/nslcd.conf.5.html>

```
Name: LDAP group-based login access
Default: yes
Priority: 128
Auth-Type: Additional
Auth:
    required pam_access.so nodefgroup
Account-Type: Primary
Account:
    required pam_access.so nodefgroup
Session-Type: Additional
Session:
    required pam_access.so nodefgroup
```

```
$ sudo pam-auth-update --enable ldap-access --enable ldap --enable mkhomedir
```

Note:

Local users not in any listed LDAP group can be permitted to log in by adding them to a local login group. The login group is allowed to log in through `/etc/security/access.conf`, see above. Create the group and add users as needed:

```
$ sudo groupadd -r login
```

```
$ sudo usermod -aG login <username>
```

Grant sudo access to an LDAP group

*visudo(8)*³⁸⁹

```
%your-admin-group    ALL=(ALL) ALL
```

*visudo(8)*³⁹⁰

```
%your-admin-group    ALL=(ALL) NOPASSWD: ALL
```

³⁸⁹ <https://manpages.ubuntu.com/manpages/resolute/man8/visudo.8.html>

³⁹⁰ <https://manpages.ubuntu.com/manpages/resolute/man8/visudo.8.html>

Retrieve SSH public keys from LDAP

```
AuthorizedKeysCommand /usr/local/bin/ssh-ldap-key %u
AuthorizedKeysCommandUser nobody
```

```
$ sudo apt install python3-ldap3
```

```
#!/usr/bin/env python3
"""Fetch SSH public keys for a user from an LDAP directory."""

import argparse
import logging
import ldap3

def main():
    cli = argparse.ArgumentParser(description="SSH key fetcher from LDAP")
    cli.add_argument("username")
    cli.add_argument("--ldap-server", default="ldaps://ldap.example.com")
    cli.add_argument("--base-dn", default="dc=example,dc=com")
    cli.add_argument("--verbose", "-v", action="count", default=0)
    args = cli.parse_args()

    log_level = (logging.WARNING, logging.INFO, logging.DEBUG)[min(args.verbose, 2)]
    logging.basicConfig(level=log_level, format="%(asctime)s %(message)s")

    for key in fetch_keys(args.username, args.ldap_server, args.base_dn):
        print(key)

def fetch_keys(user, server, basedn):
    conn = ldap3.Connection(server)
    if not conn.bind():
        raise RuntimeError(f"anonymous bind to {server!r} failed")

    uid = ldap3.utils.conv.escape_filter_chars(user)
    ok = conn.search(
        basedn,
        f"(&(objectClass=inetOrgPerson)(uid={uid}))",
        attributes=["sshPublicKey"],
    )
    if not ok:
        raise RuntimeError(f"LDAP search failed: {conn.result}")
    if not conn.entries:
        return []
    if len(conn.entries) > 1:
        raise RuntimeError(f"ambiguous username: {len(conn.entries)} entries returned")
```

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```
return [
    ldap3.utils.conv.to_unicode(key)
    for key in conn.entries[0].sshPublicKey
]

if __name__ == "__main__":
    main()
```

```
$ sudo install -o root -g root -m 0755 ssh-ldap-key
/usr/local/bin/ssh-ldap-key
```

```
$ sudo systemctl restart ssh
```

Further reading

- *Install and configure LDAP*
- *How to set up SSSD with LDAP*
- *LDAP and Transport Layer Security (TLS)*
- *nsld(8)*³⁹¹ *nsld.conf(5)*³⁹² *pam_access(8)*³⁹³ *visudo(8)*³⁹⁴

See also

- *How to set up SSSD with LDAP*
- *Introduction to OpenLDAP*

Databases

³⁹¹ <https://manpages.ubuntu.com/manpages/resolute/man8/nsld.8.html>

³⁹² <https://manpages.ubuntu.com/manpages/resolute/man5/nsld.conf.5.html>

³⁹³ https://manpages.ubuntu.com/manpages/resolute/man8/pam_access.8.html

³⁹⁴ <https://manpages.ubuntu.com/manpages/resolute/man8/visudo.8.html>

Install and configure a MySQL server

MySQL³⁹⁵

Install MySQL

```
$ sudo apt install mysql-server
```

```
$ sudo service mysql status
mysql.service - MySQL Community Server
  Loaded: loaded (/lib/systemd/system/mysql.service; enabled; vendor preset:
  enabled)
  Active: active (running) since Tue 2019-10-08 14:37:38 PDT; 2 weeks 5 days ago
  Main PID: 2028 (mysqld)
  Tasks: 28 (limit: 4915)
  CGroup: /system.slice/mysql.service
          └─2028 /usr/sbin/mysqld --daemonize --pid-file=/run/mysqld/mysqld.pid
  Oct 08 14:37:36 db.example.org systemd[1]: Starting MySQL Community Server...
  Oct 08 14:37:38 db.example.org systemd[1]: Started MySQL Community Server.
```

```
$ sudo ss -tap | grep mysql

LISTEN  0          151             127.0.0.1:mysql      0.0.0.0:*
users:((("mysqld",pid=149190,fd=29))
LISTEN  0          70              *:33060              *:.*
users:((("mysqld",pid=149190,fd=32))
```

```
$ sudo service mysql restart
```

```
$ sudo journalctl -u mysql
```

³⁹⁵ <https://www.mysql.com/>

Configure MySQL

```
bind-address = 192.168.0.5
```

Note:

Replace 192.168.0.5 with the appropriate address, which can be determined via the `ip address show` command.

```
$ sudo systemctl restart mysql.service
```

User setup

```
$ sudo mysql -u root
```

`auth_socket`³⁹⁶

Create a new user

```
CREATE USER 'username'@'localhost' IDENTIFIED WITH caching_sha2_password BY 'password';
```

```
CREATE USER 'username'@'localhost' IDENTIFIED WITH caching_sha2_password BY RANDOM  
PASSWORD;
```

MySQL's upstream documentation³⁹⁷

³⁹⁶ <https://dev.mysql.com/doc/mysql-secure-deployment-guide/8.0/en/secure-deployment-configure-authentication.html>

³⁹⁷ <https://dev.mysql.com/doc/mysql-secure-deployment-guide/8.0/en/secure-deployment-user-accounts.html>

```
CREATE USER 'username'@'localhost' IDENTIFIED WITH auth_socket;
```

```
CREATE USER 'username'@'localhost' IDENTIFIED WITH auth_socket AS 'system-user-username';
```

Adding user permissions

```
GRANT SELECT on *.* TO 'username'@'localhost';
```

Database engines

InnoDB³⁹⁸ MyISAM³⁹⁹

InnoDB

ACID compliant⁴⁰⁰

³⁹⁸ <https://dev.mysql.com/doc/refman/8.0/en/innodb-storage-engine.html>

³⁹⁹ <https://dev.mysql.com/doc/refman/8.0/en/myisam-storage-engine.html>

⁴⁰⁰ <https://en.wikipedia.org/wiki/ACID>

MyISAM

FULLTEXT

database⁴⁰¹

MyISAM on a production

Backups

mysqldump

MySQL Shell's Dump Utility

Percona Xtrabackup

mysqldump

logical backups⁴⁰²

```
$ mysqldump [database name] > dump.sql
```

```
$ mysqldump -u root [database name] > dump.sql
```

⁴⁰¹ <https://www.percona.com/blog/using-myisam-in-production/>

⁴⁰² https://dev.mysql.com/doc/refman/8.4/en/glossary.html#glos_logical_backup

```
$ mysql -u root [database name] < dump.sql
```

[upstream documentation](#)⁴⁰³

MySQL Shell Dump Utility

```
$ sudo apt install mysql-shell
```

```
$ mysqlsh --socket=/var/run/mysql/mysql.sock --no-password --python
```

```
util.dump_instance("/tmp/worlddump")
```

```
util.dump_tables("database name", ["table 1", "table 2"], "/tmp/taledump")
```

[dump loading utility](#)⁴⁰⁴

```
util.load_dump("/tmp/worlddump")
```

Note:

To restore data from a local file, `local_infile` needs to be enabled on the MySQL server. Activate this by accessing the server with the `mysql` command and entering `SET GLOBAL local_infile=1;`

[MySQL Shell dump documentation](#)⁴⁰⁵

⁴⁰³ <https://dev.mysql.com/doc/refman/8.4/en/mysqldump.html>

⁴⁰⁴ <https://dev.mysql.com/doc/mysql-shell/8.0/en/mysql-shell-utilities-load-dump.html>

⁴⁰⁵ <https://dev.mysql.com/doc/mysql-shell/8.4/en/mysql-shell-utilities-dump-instance-schema.html>

Percona Xtrabackup

ical backups⁴⁰⁶

MySQL Enterprise Backup⁴⁰⁷ phys-

```
$ sudo apt install percona-xtrabackup
```

```
$ xtrabackup --backup --target-dir=/tmp/worlddump
```

```
$ sudo systemctl stop mysql > xtrabackup --prepare  
--target-dir=/tmp/worlddump > sudo rm -rf /var/lib/mysql > sudo xtrabackup  
--copy-back --target-dir=/tmp/worlddump --datadir=/var/lib/mysql > sudo chown  
-R mysql:mysql /var/lib/mysql > sudo systemctl start mysql
```

Percona's upstream documentation⁴⁰⁸

Advanced configuration

Creating a tuned configuration

```
$ mysqldump --all-databases --routines -u root -p > ~/fulldump.sql
```

```
$ sudo service mysql stop
```

⁴⁰⁶ https://dev.mysql.com/doc/refman/8.4/en/glossary.html#glos_physical_backup

⁴⁰⁷ <https://www.mysql.com/products/enterprise/backup.html>

⁴⁰⁸ <https://docs.percona.com/percona-xtrabackup/8.0/>


```
$ sudo rsync -avz /etc/mysql /root/mysql-backup
```

```
$ sudo rm -rf /var/lib/mysql/* > sudo mysqld --initialize > sudo chown -R  
mysql: /var/lib/mysql > sudo service mysql start
```

```
$ cat ~/fulldump.sql | mysql
```

```
$ sudo apt install pv
```

```
$ pv ~/fulldump.sql | mysql
```

Note:

This is not necessary for all `my.cnf` changes. Most of the variables you can change to improve performance are adjustable even whilst the server is running. As with anything, make sure to have a good backup copy of your config files and data before making changes.

MySQL Tuner

MySQL Tuner⁴⁰⁹

```
$ sudo apt install mysqltuner
```

official MySQL documentation⁴¹⁰

⁴⁰⁹ <https://github.com/major/MySQLTuner-perl>

⁴¹⁰ <https://dev.mysql.com/doc/>

```
----- Recommendations -----
General recommendations:
  Run OPTIMIZE TABLE to defragment tables for better performance
  Increase table_cache gradually to avoid file descriptor limits
Variables to adjust:
  key_buffer_size (> 1.4G)
  query_cache_size (> 32M)
  table_cache (> 64)
  innodb_buffer_pool_size (>= 22G)
```

Further reading

- [MySQL Developers portal](#)⁴¹¹
 - [Getting Started with SQL: A Hands-On Approach for Beginners](#)⁴¹²
- [MySQL Developers portal](#)⁴¹¹
- [Getting Started with SQL: A Hands-On Approach for Beginners](#)⁴¹²

Install and configure PostgreSQL

PostgreSQL⁴¹⁴

Install PostgreSQL

```
$ sudo apt install postgresql
```

⁴¹¹ <https://dev.mysql.com/doc/>

⁴¹² <https://www.oreilly.com/library/view/getting-started-with/9781491938607/>

⁴¹³ <https://www.oreilly.com/library/view/sql-in-a/9780596155322/>

⁴¹⁴ <https://www.postgresql.org/>

Configure PostgreSQL

Administrator's Guide⁴¹⁵

PostgreSQL

Tip:

To configure *IDENT* authentication, add entries to the `/etc/postgresql/*/main/pg_ident.conf` file. There are detailed comments in the file to guide you.

```
listen_addresses = '*'
```

Note:

To listen on all IPv4 interfaces, set `listen_addresses` to `'0.0.0.0'`, while `:::` will listen on all IPv6 interfaces. `'*'` will cause PostgreSQL to listen on all available network interfaces, both IPv4 and IPv6.

mentation⁴¹⁶

PostgreSQL docu-

```
$ sudo -u postgres psql template1
```

```
ALTER USER postgres with encrypted password 'your_password';
```

hostssl template1	postgres	192.168.1.1/24	scram-sha-256
-------------------	----------	----------------	---------------

⁴¹⁵ <https://www.postgresql.org/docs/current/admin.html>

⁴¹⁶ <https://www.postgresql.org/docs/>

Note:

The config statement `hostssl` used here will reject TCP connections that would not use SSL. PostgreSQL in Ubuntu has the SSL feature built in and configured by default, so it works right away. On your PostgreSQL server this uses the certificate created by the `ssl-cert` package which is suitable for testing, but for production use you should replace it with a certificate from a trusted Certificate Authority (CA). See [Obtain TLS certificates](#) (page 252) for how to get a certificate from Let's Encrypt or set up your own CA.

```
$ sudo systemctl restart postgresql.service
```

Warning:

The above configuration is not complete by any means. Please refer to the [PostgreSQL Administrator's Guide](#)⁴¹⁷ to configure more parameters.

⁴¹⁷ <https://www.postgresql.org/docs/current/admin.html>

```
$ sudo apt install postgresql-client > psql --host your-servers-dns-or-ip  
--username postgres --password --dbname template1
```

Streaming replication

(WAL) XLOG⁴¹⁸

```
$ sudo -u postgres createuser --replication -P -e replicator
```

```
listen_addresses = '*'  
wal_level = replica
```

⁴¹⁸ <https://www.postgresql.org/docs/current/wal.html>

```
host replication replicator <IP address of the standby> scram-sha-256
```

```
$ sudo systemctl restart postgresql
```

```
$ sudo systemctl stop postgresql
```

```
hot_standby = on
```

```
$ sudo su - postgres > # backup the current content of the standby server
(update the version of your postgres accordingly) > cp -R
/var/lib/postgresql/14/main /var/lib/postgresql/14/main_bak > # remove all
the files in the data directory > rm -rf /var/lib/postgresql/14/main/* >
pg_basebackup -h <IP address of the main server> -D
/var/lib/postgresql/14/main -U replicator -P -v -R
```

- *hostname*
-
-
-
-
-

```
$ sudo systemctl start postgresql
```

```
$ sudo -u postgres psql -c "select * from pg_stat_replication;"
```

⁴¹⁹ <https://www.postgresql.org/docs/current/runtime-config-replication.html>

⁴²⁰ <https://www.postgresql.org/docs/current/high-availability.html>

```
$ sudo -u postgres createdb test # on the main server
```

```
$ sudo -u postgres psql -c "\l" # on the standby server
```

Backups

tor's Guide⁴²¹

PostgreSQL Administra-

Further reading

- PostgreSQL Administrator's Guide⁴²²

```
$ sudo apt install postgresql-doc
```

dblink

- On Approach for Beginners⁴²³ Getting Started with SQL: A Hands-
SQL in a Nutshell⁴²⁴

See also

- *Introduction to databases*

⁴²¹ <https://www.postgresql.org/docs/current/backup.html>

⁴²² <https://www.postgresql.org/docs/current/admin.html>

⁴²³ <https://www.oreilly.com/library/view/getting-started-with/9781491938607/>

⁴²⁴ <https://www.oreilly.com/library/view/sql-in-a/9780596155322/>

Storage and backups

- *Storage*
- *Backups and version control*

Storage

- *Logical Volume Management (LVM)*
- *iSCSI initiator and an iSCSI target.*

How to manage logical volumes

explanation of logical volume management (LVM)

Create the physical volume

```
$ sudo pvcreate /dev/sda1
```

Create the volume group

```
$ sudo vgcreate foo /dev/sda1
```

Create a logical volume

```
$ sudo lvcreate --name bar --size 5g foo
```

Resize a partition

```
$ sudo lvextend --resizefs --size +5g foo/bar
```

filesystem

Move a partition

```
$ sudo pvmove --name bar /dev/sda1
```


Create a snapshot

```
$ sudo lvcreate --snapshot --name lv_snapshot --size 5g foo/bar
```

```
$ sudo lvconvert --merge foo/lv_snapshot
```

iSCSI initiator (or client)

Wikipedia iSCSI Definition⁴²⁵:

iSCSI an acronym for **Internet Small Computer Systems Interface** , an [Internet Protocol](#)⁴²⁶ (IP)-based storage networking standard for linking data storage facilities. It provides [block-level access](#)⁴²⁷ to [storage devices](#)⁴²⁸ by carrying [SCSI](#)⁴²⁹ commands over a [TCP/IP](#)⁴³⁰ network.

iSCSI is used to facilitate data transfers over [intranets](#)⁴³¹ and to manage storage over long distances. It can be used to transmit data over [local area networks](#)⁴³² (LANs), [wide area networks](#)⁴³³ (WANs), or the [Internet](#)⁴³⁴ and can enable location-independent data storage and retrieval.

The [protocol](#)⁴³⁵ allows clients (called *initiators*) to send SCSI commands ([CDBs](#)⁴³⁶) to storage devices (*targets*) on remote servers. It is a [storage area network](#)⁴³⁷ (SAN) protocol, allowing organizations to consolidate storage into [storage arrays](#)⁴³⁸ while providing clients (such as database and web servers) with the illusion of locally attached SCSI disks.

It mainly competes with [Fibre Channel](#)⁴³⁹ , but unlike traditional *Fibre Channel*, which usually requires dedicated cabling, iSCSI can be run over long distances using existing network infrastructure.

⁴²⁵ <https://en.wikipedia.org/wiki/ISCSI>

⁴²⁶ https://en.wikipedia.org/wiki/Internet_Protocol

⁴²⁷ https://en.wikipedia.org/wiki/Block-level_storage

⁴²⁸ https://en.wikipedia.org/wiki/Computer_data_storage

⁴²⁹ <https://en.wikipedia.org/wiki/SCSI>

⁴³⁰ <https://en.wikipedia.org/wiki/TCP/IP>

⁴³¹ <https://en.wikipedia.org/wiki/Intranet>

⁴³² https://en.wikipedia.org/wiki/Local_area_network

⁴³³ https://en.wikipedia.org/wiki/Wide_area_network

⁴³⁴ <https://en.wikipedia.org/wiki/Internet>

⁴³⁵ [https://en.wikipedia.org/wiki/Protocol_\(computing\)](https://en.wikipedia.org/wiki/Protocol_(computing))

⁴³⁶ https://en.wikipedia.org/wiki/SCSI_CDB

⁴³⁷ https://en.wikipedia.org/wiki/Storage_area_network

⁴³⁸ https://en.wikipedia.org/wiki/Storage_array

⁴³⁹ https://en.wikipedia.org/wiki/Fibre_Channel

Note:

It is assumed that **you already have an iSCSI target on your local network** and have the appropriate rights to connect to it. The instructions for setting up a target vary greatly between hardware providers, so consult your vendor documentation to configure your specific iSCSI target.

Network Interfaces Configuration

netplan.io⁴⁴⁰

```
/etc/cloud/cloud.cfg.d/99-disable-network-config.cfg
```

```
{config: disabled}
```

```
/etc/netplan/50-cloud-init.yaml
```

```
network:
  ethernets:
    enp5s0:
      match:
        macaddress: 00:16:3e:af:c4:d6
      set-name: eth0
      dhcp4: true
      dhcp-identifier: mac
    enp6s0:
      match:
        macaddress: 00:16:3e:50:11:9c
      set-name: iscsi01
      dhcp4: true
      dhcp-identifier: mac
      dhcp4-overrides:
        route-metric: 300
    enp7s0:
      match:
        macaddress: 00:16:3e:b3:cc:50
      set-name: iscsi02
      dhcp4: true
      dhcp-identifier: mac
      dhcp4-overrides:
        route-metric: 300
  version: 2
  renderer: networkd
```

If you have only a single interface for the iSCSI network, make sure to follow the same instructions, but only consider the **iscsi01** interface command line examples.

⁴⁴⁰ <https://netplan.readthedocs.io/en/latest/examples/>

iSCSI Initiator Install

```
$ sudo apt install open-iscsi
```

-
-

iSCSI Initiator Configuration

/etc/iscsi/iscsid.conf

```
### startup settings

## will be controlled by systemd, leave as is
iscsid.startup = /usr/bin/iscsidnode.startup = manual

### chap settings

# node.session.auth.authmethod = CHAP

## authentication of initiator by target (session)
# node.session.auth.username = username
# node.session.auth.password = password

# discovery.sendtargets.auth.authmethod = CHAP

## authentication of initiator by target (discovery)
# discovery.sendtargets.auth.username = username
# discovery.sendtargets.auth.password = password

### timeouts

## control how much time iscsi takes to propagate an error to the
## upper layer. if using multipath, having 0 here is desirable
## so multipath can handle path errors as quickly as possible
## (and decide to queue or not if missing all paths)
node.session.timeo.replacement_timeout = 0

node.conn[0].timeo.login_timeout = 15
node.conn[0].timeo.logout_timeout = 15

## interval for a NOP-Out request (a ping to the target)
node.conn[0].timeo.noop_out_interval = 5

## and how much time to wait before declaring a timeout
node.conn[0].timeo.noop_out_timeout = 5
```

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```
## default timeouts for error recovery logics (lu & tgt resets)
node.session.err_timeo.abort_timeout = 15
node.session.err_timeo.lu_reset_timeout = 30
node.session.err_timeo.tgt_reset_timeout = 30

### retry

node.session.initial_login_retry_max = 8

### session and device queue depth

node.session.cmds_max = 128
node.session.queue_depth = 32

### performance

node.session.xmit_thread_priority = -20
```

```
$ systemctl restart iscsid.service
```

```
/etc/iscsi/initiatorname.iscsi
```

```
InitiatorName=iqn.2004-10.com.ubuntu:01 60 73517884c3
```

iSCSI Network Configuration

•

```
$ iscsi01_ip=$(ip -4 -o addr show iscsi01 | sed -r 's:.*
([0-9]{1,3}\.){3}[0-9]{1,3})/.*:\1:')

```

```
$ iscsi02_ip=$(ip -4 -o addr show iscsi02 | sed -r 's:.*
([0-9]{1,3}\.){3}[0-9]{1,3})/.*:\1:')

```

```
$ iscsi01_mac=$(ip -o link show iscsi01 | sed -r 's:.*\s+link/ether  
([0-f]{2}(\:|))\{6\}.*:\1:g')
```

```
$ iscsi02_mac=$(ip -o link show iscsi02 | sed -r 's:.*\s+link/ether  
([0-f]{2}(\:|))\{6\}.*:\1:g')
```

•

```
$ sudo iscsiadm -m iface -I iscsi01 --op=new  
New interface iscsi01 added
```

```
$ sudo iscsiadm -m iface -I iscsi01 --op=update -n iface.hwaddress -v  
$iscsi01_mac  
iscsi01 updated.
```

```
$ sudo iscsiadm -m iface -I iscsi01 --op=update -n iface.ipaddress -v  
$iscsi01_ip  
iscsi01 updated.
```

•

```
$ sudo iscsiadm -m iface -I iscsi02 --op=new  
New interface iscsi02 added
```

```
$ sudo iscsiadm -m iface -I iscsi02 --op=update -n iface.hwaddress -v  
$iscsi02_mac  
iscsi02 updated.
```

```
$ sudo iscsiadm -m iface -I iscsi02 --op=update -n iface.ipaddress -v  
$iscsi02_ip  
iscsi02 updated.
```

•

```
$ sudo iscsiadm -m discovery -I iscsi01 --op=new --op=del --type  
sendtargets --portal storage.iscsi01  
10.250.94.99:3260,1 iqn.2003-01.org.linux-iscsi.storage.x8664:sn.2c084c8320ca
```

```
$ sudo iscsiadm -m discovery -I iscsi02 --op=new --op=del --type  
sendtargets --portal storage.iscsi02  
  
10.250.93.99:3260,1 iqn.2003-01.org.linux-iscsi.storage.x8664:sn.2c084c8320ca
```

```
$ sudo iscsiadm -m node --op=update -n node.conn[0].startup -v automatic
```

```
$ sudo iscsiadm -m node --op=update -n node.startup -v automatic
```

```
$ systemctl enable open-iscsi
```

```
Synchronizing state of open-iscsi.service with SysV service script with /lib/  
systemd/systemd-sysv-install.  
Executing: /lib/systemd/systemd-sysv-install enable open-iscsi  
Created symlink /etc/systemd/system/iscsi.service → /lib/systemd/system/open-  
iscsi.service.  
Created symlink /etc/systemd/system/sysinit.target.wants/open-iscsi.service →  
/lib/systemd/system/open-iscsi.service.
```

```
$ systemctl enable iscsid
```

```
Synchronizing state of iscsid.service with SysV service script with /lib/  
systemd/systemd-sysv-install.  
Executing: /lib/systemd/systemd-sysv-install enable iscsid  
Created symlink /etc/systemd/system/sysinit.target.wants/iscsid.service → /  
lib/systemd/system/iscsid.service.
```

```
$ systemctl restart iscsid.service
```

```
$ sudo iscsiadm -m node --loginall=automatic
```

```
Logging in to [iface: iscsi02, target: iqn.2003-01.org.linux-iscsi.storage.  
x8664:sn.2c084c8320ca, portal: 10.250.93.99,3260] (multiple)  
Logging in to [iface: iscsi01, target: iqn.2003-01.org.linux-iscsi.storage.  
x8664:sn.2c084c8320ca, portal: 10.250.94.99,3260] (multiple)  
Login to [iface: iscsi02, target: iqn.2003-01.org.linux-iscsi.storage.  
x8664:sn.2c084c8320ca, portal: 10.250.93.99,3260] successful.  
Login to [iface: iscsi01, target: iqn.2003-01.org.linux-iscsi.storage.  
x8664:sn.2c084c8320ca, portal: 10.250.94.99,3260] successful.
```

Accessing the Logical Units (or LUNs)

dmesg

dmesg

```
[ 166.840694] scsi 7 0 0:4: Direct-Access      LIO-ORG   TCMU device >      0002 PQ: 0
ANSI: 5
[ 166.840892] scsi 8 0 0:4: Direct-Access      LIO-ORG   TCMU device >      0002 PQ: 0
ANSI: 5
[ 166.841741] sd 7 0 0 4: Attached scsi generic sg2 type 0
[ 166.841808] sd 8 0 0 4: Attached scsi generic sg3 type 0
[ 166.842278] scsi 7 0 0:3: Direct-Access      LIO-ORG   TCMU device >      0002 PQ: 0
ANSI: 5
[ 166.842571] scsi 8 0 0:3: Direct-Access      LIO-ORG   TCMU device >      0002 PQ: 0
ANSI: 5
[ 166.843482] sd 8 0 0 3: Attached scsi generic sg4 type 0
[ 166.843681] sd 7 0 0 3: Attached scsi generic sg5 type 0
[ 166.843706] sd 8 0 0 4: [sdd] 2097152 512-byte logical blocks: > (1.07 GB/1.00 GiB)
[ 166.843884] scsi 8 0 0:2: Direct-Access      LIO-ORG   TCMU device >      0002 PQ: 0
ANSI: 5
[ 166.843971] sd 8 0 0 4: [sdd] Write Protect is off
[ 166.843972] sd 8 0 0 4: [sdd] Mode Sense: 2f 00 00 00
[ 166.844127] scsi 7 0 0:2: Direct-Access      LIO-ORG   TCMU device >      0002 PQ: 0
ANSI: 5
[ 166.844232] sd 7 0 0 4: [sdc] 2097152 512-byte logical blocks: > (1.07 GB/1.00 GiB)
[ 166.844421] sd 8 0 0 4: [sdd] Write cache: enabled, read cache: > enabled, doesn't
support DPO or FUA
[ 166.844566] sd 7 0 0 4: [sdc] Write Protect is off
[ 166.844568] sd 7 0 0 4: [sdc] Mode Sense: 2f 00 00 00
[ 166.844846] sd 8 0 0 2: Attached scsi generic sg6 type 0
[ 166.845147] sd 7 0 0 4: [sdc] Write cache: enabled, read cache: > enabled, doesn't
support DPO or FUA
[ 166.845188] sd 8 0 0 4: [sdd] Optimal transfer size 65536 bytes
[ 166.845527] sd 7 0 0 2: Attached scsi generic sg7 type 0
[ 166.845678] sd 8 0 0 3: [sde] 2097152 512-byte logical blocks: > (1.07 GB/1.00 GiB)
[ 166.845785] scsi 8 0 0:1: Direct-Access      LIO-ORG   TCMU device >      0002 PQ: 0
ANSI: 5
[ 166.845799] sd 7 0 0 4: [sdc] Optimal transfer size 65536 bytes
[ 166.845931] sd 8 0 0 3: [sde] Write Protect is off
[ 166.845933] sd 8 0 0 3: [sde] Mode Sense: 2f 00 00 00
[ 166.846424] scsi 7 0 0:1: Direct-Access      LIO-ORG   TCMU device >      0002 PQ: 0
ANSI: 5
[ 166.846552] sd 8 0 0 3: [sde] Write cache: enabled, read cache: > enabled, doesn't
support DPO or FUA
[ 166.846708] sd 7 0 0 3: [sdf] 2097152 512-byte logical blocks: > (1.07 GB/1.00 GiB)
[ 166.847024] sd 8 0 0 1: Attached scsi generic sg8 type 0
[ 166.847029] sd 7 0 0 3: [sdf] Write Protect is off
[ 166.847031] sd 7 0 0 3: [sdf] Mode Sense: 2f 00 00 00
[ 166.847043] sd 8 0 0 3: [sde] Optimal transfer size 65536 bytes
[ 166.847133] sd 8 0 0 2: [sdg] 2097152 512-byte logical blocks: > (1.07 GB/1.00 GiB)
[ 166.849212] sd 8 0 0 2: [sdg] Write Protect is off
[ 166.849214] sd 8 0 0 2: [sdg] Mode Sense: 2f 00 00 00
[ 166.849711] sd 7 0 0 3: [sdf] Write cache: enabled, read cache: > enabled, doesn't
support DPO or FUA
[ 166.849718] sd 7 0 0 1: Attached scsi generic sg9 type 0
```

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```
[ 166.849721] sd 7 0 0 2: [sdh] 2097152 512-byte logical blocks: > (1.07 GB/1.00 GiB)
[ 166.853296] sd 8 0 0 2: [sdg] Write cache: enabled, read cache: > enabled, doesn't
support DPO or FUA
[ 166.853721] sd 8 0 0 2: [sdg] Optimal transfer size 65536 bytes
[ 166.853810] sd 7 0 0 2: [sdh] Write Protect is off
[ 166.853812] sd 7 0 0 2: [sdh] Mode Sense: 2f 00 00 00
[ 166.854026] sd 7 0 0 3: [sdf] Optimal transfer size 65536 bytes
[ 166.854431] sd 7 0 0 2: [sdh] Write cache: enabled, read cache: > enabled, doesn't
support DPO or FUA
[ 166.854625] sd 8 0 0 1: [sdi] 2097152 512-byte logical blocks: > (1.07 GB/1.00 GiB)
[ 166.854898] sd 8 0 0 1: [sdi] Write Protect is off
[ 166.854900] sd 8 0 0 1: [sdi] Mode Sense: 2f 00 00 00
[ 166.855022] sd 7 0 0 2: [sdh] Optimal transfer size 65536 bytes
[ 166.855465] sd 8 0 0 1: [sdi] Write cache: enabled, read cache: > enabled, doesn't
support DPO or FUA
[ 166.855578] sd 7 0 0 1: [sdj] 2097152 512-byte logical blocks: > (1.07 GB/1.00 GiB)
[ 166.855845] sd 7 0 0 1: [sdj] Write Protect is off
[ 166.855847] sd 7 0 0 1: [sdj] Mode Sense: 2f 00 00 00
[ 166.855978] sd 8 0 0 1: [sdi] Optimal transfer size 65536 bytes
[ 166.856305] sd 7 0 0 1: [sdj] Write cache: enabled, read cache: > enabled, doesn't
support DPO or FUA
[ 166.856701] sd 7 0 0 1: [sdj] Optimal transfer size 65536 bytes
[ 166.859624] sd 8 0 0 4: [sdd] Attached SCSI disk
[ 166.861304] sd 7 0 0 4: [sdc] Attached SCSI disk
[ 166.864409] sd 8 0 0 3: [sde] Attached SCSI disk
[ 166.864833] sd 7 0 0 3: [sdf] Attached SCSI disk
[ 166.867906] sd 8 0 0 2: [sdg] Attached SCSI disk
[ 166.868446] sd 8 0 0 1: [sdi] Attached SCSI disk
[ 166.871588] sd 7 0 0 1: [sdj] Attached SCSI disk
[ 166.871773] sd 7 0 0 2: [sdh] Attached SCSI disk
```

Introduction to device mapper multipathing

("multipath")

```
$ apt-get install multipath-tools
```

```
$ sudo multipath -r
```

```
$ sudo multipath -ll
```

```

mpathd (360014051a042fb7c41c4249af9f2cfbc) dm-3 LIO-ORG,TCMU device
size=1.0G features='0' hwhandler='0' wp=rw
|+- policy='service-time 0' prio=1 status=active
|  '- 7:0:0:4 sde 8:64 active ready running
`-+ policy='service-time 0' prio=1 status=enabled
   '- 8:0:0:4 sdc 8:32 active ready running
mpathc (360014050d6871110232471d8bcd155a3) dm-2 LIO-ORG,TCMU device
size=1.0G features='0' hwhandler='0' wp=rw
|+- policy='service-time 0' prio=1 status=active
|  '- 7:0:0:3 sdf 8:80 active ready running
`-+ policy='service-time 0' prio=1 status=enabled
   '- 8:0:0:3 sdd 8:48 active ready running
mpathb (360014051f65c6cb11b74541b703ce1d4) dm-1 LIO-ORG,TCMU device
size=1.0G features='0' hwhandler='0' wp=rw
|+- policy='service-time 0' prio=1 status=active
|  '- 7:0:0:2 sdh 8:112 active ready running
`-+ policy='service-time 0' prio=1 status=enabled
   '- 8:0:0:2 sdg 8:96 active ready running
mpatha (36001405b816e24fcab64fb88332a3fc9) dm-0 LIO-ORG,TCMU device
size=1.0G features='0' hwhandler='0' wp=rw
|+- policy='service-time 0' prio=1 status=active
|  '- 7:0:0:1 sdj 8:144 active ready running
`-+ policy='service-time 0' prio=1 status=enabled
   '- 8:0:0:1 sdi 8:128 active ready running

```

```

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```

Warning:

Do not use this in production without checking appropriate multipath configuration options in the **Device Mapper Multipathing** section. The *default multipath configuration* is sub-optimal for regular usage.

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```
$ sudo fdisk /dev/mapper/mpatha

Welcome to fdisk (util-linux 2.34).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Device does not contain a recognized partition table.
Created a new DOS disklabel with disk identifier 0x92c0322a.

Command (m for help): p
Disk /dev/mapper/mpatha: 1 GiB, 1073741824 bytes, 2097152 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 65536 bytes
Disklabel type: dos
Disk identifier: 0x92c0322a

Command (m for help): n
Partition type
   p   primary (0 primary, 0 extended, 4 free)
   e   extended (container for logical partitions)
Select (default p): p
Partition number (1-4, default 1):
First sector (2048-2097151, default 2048):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (2048-2097151, default 2097151):

Created a new partition 1 of type 'Linux' and of size 1023 MiB.

Command (m for help): w
The partition table has been altered.
```

-

filesystem

```
$ sudo mkfs.ext4 /dev/mapper/mpatha-part1
```

```
mke2fs 1.45.5 (07-Jan-2020)
Creating filesystem with 261888 4k blocks and 65536 inodes
Filesystem UUID: cdb70b1e-c47c-47fd-9c4a-03db6f038988
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376

Allocating group tables: done
Writing inode tables: done
Creating journal (4096 blocks): done
Writing superblocks and filesystem accounting information: done
```

•

```
$ sudo mount /dev/mapper/mpatha-part1 /mnt
```

•

```
$ ls /mnt

lost+found
```

Further reading

- [iscsid](#)⁴⁴¹
- [iscsi.conf](#)⁴⁴²
- [iscsid.conf](#)⁴⁴³
- [iscsi.service](#)⁴⁴⁴
- [iscsid.service](#)⁴⁴⁵
- [Open-iSCSI](#)⁴⁴⁶
- [Debian Open-iSCSI](#)⁴⁴⁷

⁴⁴¹ <https://linux.die.net/man/8/iscsid>

⁴⁴² <https://linux.die.net/man/5/iscsi.conf>

⁴⁴³ <https://github.com/open-iscsi/open-iscsi/blob/master/etc/iscsid.conf>

⁴⁴⁴ <https://github.com/open-iscsi/open-iscsi/blob/master/etc/systemd/iscsi.service.template>

⁴⁴⁵ <https://github.com/open-iscsi/open-iscsi/blob/master/etc/systemd/iscsid.service.template>

⁴⁴⁶ <http://www.open-iscsi.com/>

⁴⁴⁷ <https://wiki.debian.org/SAN/iSCSI/open-iscsi>

See also

- [About Logical Volume Management \(LVM\)](#)

Backups and version control

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Backup utilities

Bacula⁴⁴⁸ rsnapshot⁴⁴⁹

Tool	Back up	Backup Method	Install and configure
Bacula	Multiple systems over a network	Incremental backups	how to install and configure Bacula (page 533)
rsnapshot	Single system	Periodic "snapshots" of files locally or remotely with SSH	how to install and configure rsnapshot (page 555)

⁴⁴⁸ <https://www.bacula.org/>

⁴⁴⁹ <https://rsnapshot.org/>

How to install and configure Bacula

Bacula⁴⁵⁰

Bacula components

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Installing the Server Components

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⁴⁵⁰ <https://www.bacula.org/>

-

```
$ sudo apt install postgresql
```

Note:

Refer to *MySQL databases* (page 506) and *PostgreSQL databases* (page 513) for more details on these powerful databases.

```
$ sudo apt install bacula
```

Configuring the Server

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Director

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The Director resource

```
Director {
  Name = bacula-server-dir
  DIRport = 9101
  QueryFile = "/etc/bacula/scripts/query.sql"
  WorkingDirectory = "/var/lib/bacula"
  PidDirectory = "/run/bacula"
  Maximum Concurrent Jobs = 20
  Password = "<randomly generated>"
  Messages = Daemon
  #DirAddress = 127.0.0.1
}
```

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Tip:

For more details about all the options of the Director resource, please check the upstream [Director Resource](#)⁴⁵¹ documentation.

⁴⁵¹ https://www.bacula.org/15.0.x-manuals/en/main/Configuring_Director.html#SECTION002320000000000000000000

The FileSet resource

```
FileSet {
  Name = "Home Set"
  Include {
    Options {
      signature = SHA256
      aclsupport = yes
      xattrsupport = yes
    }
    Options {
      wilddir = "/home/*/Downloads"
      wildfile = "*.iso"
      exclude = yes
    }
    File = /home
  }
}
```

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Tip:

For more details about all the options of the FileSet resource, please check the upstream [FileSet Resource](https://www.bacula.org/15.0.x-manuals/en/main/Configuring_Director.html#SECTION00237000000000000000)⁴⁵² documentation.

⁴⁵² https://www.bacula.org/15.0.x-manuals/en/main/Configuring_Director.html#SECTION00237000000000000000

The Client resource

```
# Client (File Services) to backup
Client {
    Name = bacula-server-fd
    Address = bacula-server.lxd # use the real hostname instead of "localhost"
    FDPort = 9102
    Catalog = MyCatalog
    Password = "<randomly generated>"
    File Retention = 60 days
    Job Retention = 6 months
    AutoPrune = yes
}
```

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Tip:

For more details about all the options of the Client resource, please check the upstream [Client Resource⁴⁵³](https://www.bacula.org/15.0.x-manuals/en/main/Configuring_Director.html#SECTION00231300000000000000000000) documentation.

⁴⁵³ https://www.bacula.org/15.0.x-manuals/en/main/Configuring_Director.html#SECTION00231300000000000000000000

The Pool resource

```
Pool {  
  Name = File  
  Pool Type = Backup  
  Recycle = yes           # Bacula can automatically recycle Volumes  
  AutoPrune = yes        # Prune expired volumes  
  Volume Retention = 365 days # one year  
  Maximum Volume Bytes = 50G # Limit Volume size  
  Maximum Volumes = 100     # Limit number of Volumes in Pool  
  Label Format = "Vol-"     # Auto label  
}
```

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Tip:

For more details about all the options of the Pool resource, please check the upstream [Pool Resource](#)⁴⁵⁴ documentation.

⁴⁵⁴ https://www.bacula.org/15.0.x-manuals/en/main/Configuring_Director.html#SECTION002316000000000000000000

The Storage resource

```
Storage {
  Name = FileBackup
  Address = bacula-server.lxd
  SDPort = 9103
  # For this password, use:
  # sudo grep ^SDPASSWD /etc/bacula/common_default_passwords
  Password = "<SDPASSWD value>"
  Device = FileBackup
  Media Type = File
}
```

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Tip:

For more details about all the options of the `Pool` resource, please check the upstream [Storage Resource](https://www.bacula.org/15.0.x-manuals/en/main/Configuring_Director.html#blb:StorageResource)⁴⁵⁵ documentation.

⁴⁵⁵ https://www.bacula.org/15.0.x-manuals/en/main/Configuring_Director.html#blb:StorageResource

```
Storage {
  Name = bacula-server-sd
  SDPort = 9103
  WorkingDirectory = "/var/lib/bacula"
  Pid Directory = "/run/bacula"
  Plugin Directory = "/usr/lib/bacula"
  Maximum Concurrent Jobs = 20
  Encryption Command = "/etc/bacula/scripts/key-manager.py getkey"
  #SDAddress = 127.0.0.1
}
```

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```
Device {  
  Name = FileBackup  
  Media Type = File  
  Archive Device = /storage/backups  
  Random Access = yes  
  Automatic Mount = yes  
  Removable Media = no  
  Always Open = no  
  Label Media = yes  
}
```

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```
Director {  
  Name = bacula-server-dir  
  Password = "<randomly generated>"  
}
```

-

-

```
$ sudo systemctl restart bacula-sd.service
```

The Job resource

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```
JobDefs {  
  Name = "DefaultJob"  
  Type = Backup  
  Level = Incremental  
  Client = bacula-server-fd  
  FileSet = "Home Set"  
  Schedule = "WeeklyCycle"  
  Storage = FileBackup  
  Messages = Standard  
  Pool = File  
  SpoolAttributes = yes  
  Priority = 10  
  Write Bootstrap = "/var/lib/bacula/%c.bsr"  
}
```

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```
Job {  
  Name = "DirectorHomeBackup"  
  JobDefs = "DefaultJob"  
}
```

Tip:

A job has many attributes that can only be specified once. This means that jobs are pretty much specific to what is being backed up, and from where, among other things. It therefore helps to come up with a naming convention.

The upstream documentation has a section about [Naming Resources](https://docs.baculasystems.com/BEPlanningAndPreparation/BackupPolicy/NamingResources/index.html)⁴⁵⁶ with some suggestions.

⁴⁵⁶ <https://docs.baculasystems.com/BEPlanningAndPreparation/BackupPolicy/NamingResources/index.html>

```
Job {  
  Name = "RestoreFiles"  
  Type = Restore  
  Client = bacula-server-fd  
  Storage = FileBackup  
  # The FileSet and Pool directives are not used by Restore Jobs  
  # but must not be removed  
  FileSet = "Home Set"  
  Pool = File  
  Messages = Standard  
  Where = /storage/restore  
}
```

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Tip:

For more details about all the options of the Job resource, please check the upstream [Job Resource](https://www.bacula.org/15.0.x-manuals/en/main/Configuring_Director.html#blb:JobResource)⁴⁵⁷ documentation.

⁴⁵⁷ https://www.bacula.org/15.0.x-manuals/en/main/Configuring_Director.html#blb:JobResource

Storage daemon

```
$ sudo mkdir -m 0700 /storage /storage/backups /storage/restore
```

```
$ sudo chown bacula: -R /storage
```

Tip:

For more details about the Storage daemon configuration options, please check the upstream [Storage Daemon](https://www.bacula.org/15.0.x-manuals/en/main/Storage_Daemon_Configuratio.html)⁴⁵⁸ documentation.

⁴⁵⁸ https://www.bacula.org/15.0.x-manuals/en/main/Storage_Daemon_Configuratio.html

File daemon

```
FileDaemon {  
    Name = bacula-server-fd  
    FDport = 9102  
    WorkingDirectory = /var/lib/bacula  
    Pid Directory = /run/bacula  
    Maximum Concurrent Jobs = 20  
    Plugin Directory = /usr/lib/bacula  
    #FDAddress = 127.0.0.1  
}
```

```
$ sudo systemctl restart bacula-fd.service
```


Tip:

For more details about the File daemon configuration, please check the upstream [File Daemon⁴⁵⁹](#) documentation.

⁴⁵⁹ https://www.bacula.org/15.0.x-manuals/en/main/Client_File_daemon_Configur.html

Console

tion⁴⁶⁰

Console Configura-

```
$ sudo adduser <username> bacula
```

Warning:

Be mindful of who is added to the bacula group: members of this group are able to read all the data that is being backed up!

Cleaning up

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⁴⁶⁰ https://www.bacula.org/15.0.x-manuals/en/main/Console_Configuration.html

```
$ sudo systemctl restart bacula-dir.service bacula-fd.service
bacula-sd.service
```

Running a backup job

```
$ sudo bconsole
```

```
Connecting to Director localhost:9101
1000 OK: 10002 bacula-server-dir Version: 15.0.3 (25 March 2025)
Enter a period to cancel a command.
*
```

Command	Description
=====	=====
run	Run a job

Arguments:

```
job=<job-name> client=<client-name>
fileset=<FileSet-name> level=<level-keyword>
storage=<storage-name> where=<directory-prefix>
when=<universal-time-specification> pool=<pool-name>
nextpool=<next-pool-name> comment=<text> accurate=<bool> spooldata=<bool> yes
```

When at a prompt, entering a period cancels the command.

```
*run
Using Catalog "MyCatalog"
A job name must be specified.
The defined Job resources are:
    1: HomeBackup
    2: BackupCatalog
    3: RestoreFiles
Select Job resource (1-3): 1
Run Backup job
JobName: HomeBackup
Level: Incremental
Client: bacula-server-fd
FileSet: Home Set
Pool: File (From Job resource)
Storage: FileBackup (From Job resource)
When: 2025-10-20 20:21:03
Priority: 10
OK to run? (Yes/mod/no): yes
Job queued. JobId=7
```

-
-
-
-
-
-
-
-

```
+-----+-----+-----+-----+-----+-----+-----+
-----+
| jobid | name      | starttime          | type | level | jobfiles | jobbytes |
jobstatus |
+-----+-----+-----+-----+-----+-----+-----+
-----+
|      7 | HomeBackup | 2025-10-20 20:21:11 | B    | I      |          0 |          0 | T
|
+-----+-----+-----+-----+-----+-----+-----+
-----+
```

Tip:

For a list of status and error codes, check the upstream [Job status and Error codes tables](https://www.bacula.org/15.0.x-manuals/en/main/Job_status_Error_codes.html#blb:director:job:status)⁴⁶¹.

⁴⁶¹ https://www.bacula.org/15.0.x-manuals/en/main/Job_status_Error_codes.html#blb:director:job:status

```
+-----+
| logtext
|
+-----+
| bacula-server-dir JobId 7: Start Backup JobId 7, Job=HomeBackup.2025-10-20_20.21.08_03
| bacula-server-dir JobId 7: Connected to Storage "FileBackup" at bacula-server.
|xd:9103 with TLS
| bacula-server-dir JobId 7: Using Device "FileBackup" to write.
|
...
Build OS:          x86_64-pc-linux-gnu ubuntu 25.10
JobId:             7
Job:               HomeBackup.2025-10-20_20.21.08_03
Backup Level:      Incremental, since=2025-10-20 18:11:00
Client:            "bacula-server-fd" 15.0.3 (25Mar25) x86_64-pc-linux-gnu,
ubuntu,25.10
FileSet:           "Home Set" 2025-10-20 16:27:31
Pool:              "File" (From Job resource)
Catalog:           "MyCatalog" (From Client resource)
Storage:           "FileBackup" (From Job resource)
...
Non-fatal FD errors: 0
SD Errors:          0
FD termination status: OK
SD termination status: OK
Termination:        Backup OK
|
...
```

```
-rw-r----- 1 bacula tape 345K Oct 20 20:21 /storage/backups/Vol-0001
```

Restoring a backup

First you select one or more JobIds that contain files to be restored. You will be presented several methods of specifying the JobIds. Then you will be allowed to select which files from those JobIds are to be restored.

To select the JobIds, you have the following choices:

1: List last 20 Jobs run

...

5: Select the most recent backup for a client

...

Select item: (1-14): 5

Automatically selected Client: bacula-server-fd

Automatically selected FileSet: Home Set

jobid	level	jobfiles	jobbytes	starttime	volumename
4	F	266	312,756	2025-10-20 17:34:43	Vol-0001
5	I	3	32	2025-10-20 18:11:00	Vol-0001

You have selected the following JobIds: 4,5

...

You are now entering file selection mode where you add (mark) and remove (unmark) files to be restored. No files are initially added, unless you used the "all" keyword on the command line.

Enter "done" to leave this mode.

cwd is: /

\$

```
$ dir
```

```
drwxr-xr-x  1 root    root          12 2025-10-20 14:03:44 /home/
```

```
$ cd home/ubuntu
```

```
cwd is: /home/ubuntu/
```

```
$ dir
```

```
-rw----- 1 ubuntu ubuntu          32 2025-10-20 18:10:51 /home/ubuntu/.
bash_history
-rw-r--r-- 1 ubuntu ubuntu        220 2025-10-20 14:03:50 /home/ubuntu/.
bash_logout
-rw-r--r-- 1 ubuntu ubuntu       3830 2025-10-20 14:03:50 /home/ubuntu/.
bashrc
-rw-r--r-- 1 ubuntu ubuntu        807 2025-10-20 14:03:44 /home/ubuntu/.
profile
drwx----- 1 ubuntu ubuntu         30 2025-10-20 14:03:45 /home/ubuntu/.
ssh/
-rw-rw-r-- 1 ubuntu ubuntu         0 2025-10-20 18:10:50 /home/ubuntu/
this-is-on-the-server.txt
```

```
$ mark .tmux.conf
```

```
1 file marked.
```

```
$ done
```

```
Bootstrap records written to /var/lib/bacula/bacula-server-dir.restore.1.bsr
```

```
The Job will require the following (*=>InChanger):
```

Volume(s)	Storage(s)	SD Device(s)
=====		
Vol-0001	FileBackup	FileBackup

```
Volumes marked with "*" are in the Autochanger.
```

```
1 file selected to be restored.
```

```
Run Restore job
```

```
JobName:      RestoreFiles
Bootstrap:    /var/lib/bacula/bacula-server-dir.restore.1.bsr
Where:        /storage/restore
Replace:      Always
FileSet:      Home Set
Backup Client: bacula-server-fd
Restore Client: bacula-server-fd
Storage:      FileBackup
When:         2025-10-20 21:02:37
Catalog:      MyCatalog
Priority:      10
Plugin Options: *None*
OK to run? (Yes/mod/no):
```

```
OK to run? (Yes/mod/no): yes
Job queued. JobId=8
*
```

```
-rw-r--r-- 1 ubuntu ubuntu 2.4K Oct 20 14:03 /storage/restore/home/ubuntu/.tmux.conf
```

```
OK to run? (Yes/mod/no): mod
Parameters to modify:
  1: Level
  2: Storage
  3: Job
  4: FileSet
  5: Restore Client
  6: When
  7: Priority
  8: Bootstrap
  9: Where
 10: File Relocation
 11: Replace
 12: JobId
 13: Plugin Options
Select parameter to modify (1-13): 9
Please enter the full path prefix for restore (/ for none): /
Run Restore job
JobName:      RestoreFiles
Bootstrap:    /var/lib/bacula/bacula-server-dir.restore.2.bsr
Where:
Replace:      Always
FileSet:      Home Set
Backup Client: bacula-server-fd
Restore Client: bacula-server-fd
Storage:      FileBackup
When:         2025-10-20 21:11:55
Catalog:      MyCatalog
Priority:      10
Plugin Options: *None*
OK to run? (Yes/mod/no): yes
Job queued. JobId=9
```


Adding a client

```
$ sudo apt install bacula-fd
```

```
Director {  
    Name = bacula-server-dir # same as Director's Name on the Director server  
    Password = "<randomly generated>"  
}
```

-
-

```
FileDaemon {  
    Name = workstation1-fd  
    FDport = 9102          # where we listen for the director  
    WorkingDirectory = /var/lib/bacula  
    Pid Directory = /run/bacula  
    Maximum Concurrent Jobs = 20  
    Plugin Directory = /usr/lib/bacula  
    #FDAddress = 127.0.0.1 # default is to listen on all interfaces  
}
```

```
Messages {  
    Name = Standard  
    director = bacula-server-dir = all, !skipped, !restored, !verified, !saved  
}
```

```
$ sudo systemctl restart bacula-fd.service
```

```
Client {
    Name = workstation1-fd
    Address = workstation1.lxd
    FdPort = 9102
    Catalog = MyCatalog
    Password = "<to be filled in>" # password from bacula-fd.conf on workstation1-fd
    File Retention = 60 days
    Job Retention = 6 months
    AutoPrune = yes
}
```

-
-
-

```
Job {
    Name = "BackupWorkstation"
    JobDefs = "DefaultJob"
    Client = workstation1-fd
}
```

```
$ sudo systemctl restart bacula-dir.service
```

```
*list clients
Automatically selected Catalog: MyCatalog
Using Catalog "MyCatalog"
+-----+-----+-----+-----+
| clientid | name           | fileretention | jobretention |
+-----+-----+-----+-----+
|      1  | bacula-server-fd |    5,184,000 | 15,552,000 |
|      2  | workstation1-fd  |    5,184,000 | 15,552,000 |
+-----+-----+-----+-----+
```

```
*run
Using Catalog "MyCatalog"
A job name must be specified.
The defined Job resources are:
    1: HomeBackup
    2: BackupWorkstation
    3: BackupCatalog
    4: RestoreFiles
Select Job resource (1-4): 2
Run Backup job
JobName: BackupWorkstation
Level: Incremental
Client: workstation1-fd
FileSet: Home Set
Pool: File (From Job resource)
Storage: FileBackup (From Job resource)
When: 2025-10-21 21:02:48
Priority: 10
OK to run? (Yes/mod/no): yes
Job queued. JobId=14
You have messages.
```

```
*restore
...
    5: Select the most recent backup for a client
...
Select item: (1-14): 5
Defined Clients:
    1: bacula-server-fd
    2: workstation1-fd
Select the Client (1-2): 2
...
$ cd /home/ubuntu
cwd is: /home/ubuntu/
$ dir this*
-rw-rw-r-- 1 ubuntu ubuntu      0 2025-10-21 21:02:08 /home/ubuntu/
this-is-workstation1.txt
```

Further reading

- [Bacula documentation](#)⁴⁶²
- [Bacula home page](#)⁴⁶³

⁴⁶² <https://www.bacula.org/documentation/documentation/>

⁴⁶³ <https://www.bacula.org/>

How to install and configure rsnapshot

rsnapshot⁴⁶⁴ *filesystem*

Install rsnapshot

```
$ sudo apt-get install rsnapshot
```

OpenSSH documentation

Configure rsnapshot

```
snapshot_root    /var/cache/rsnapshot/
```

Number of backups to keep

```
retain    daily    6
retain    weekly    7
retain    monthly   4
```

⁴⁶⁴ <https://rsnapshot.org/>

Remote machine access

```
ssh_args      -p 22222
```

What to backup

```
backup /home/      localhost/  
backup /etc/       localhost/  
backup /usr/local/ localhost/
```

```
backup root@example.com:/home/ example.com/ +rsync_long_args=-b,limit=16,  
exclude=core  
backup root@example.com:/etc/   example.com/ exclude=mtab,exclude=core
```

*rsnapshot(1)*⁴⁶⁵

Test configuration

```
$ sudo rsnapshot configtest
```

```
$ sudo rsnapshot -t daily
```

```
$ sudo rsnapshot daily
```

⁴⁶⁵ <https://manpages.ubuntu.com/manpages/resolute/man1/rsnapshot.1.html>

Scheduling backups

```
0 4 * * *      root    /usr/bin/rsnapshot daily
0 3 * * 1      root    /usr/bin/rsnapshot weekly
0 2 1 * *      root    /usr/bin/rsnapshot monthly
```

-
-
-

Backups - Shell Scripts

Further reading

- [rsnapshot official web page](#)⁴⁶⁶
- [*rsnapshot\(1\)*](#)⁴⁶⁷
- [*rsync\(1\)*](#)⁴⁶⁸

Shell scripts

- [*how to use shell scripts for backups*](#)
- [*An example of archive rotation with shell scripts*](#)

⁴⁶⁶ <https://rsnapshot.org/>

⁴⁶⁷ <https://manpages.ubuntu.com/manpages/resolute/man1/rsnapshot.1.html>

⁴⁶⁸ <https://manpages.ubuntu.com/manpages/resolute/man1/rsync.1.html>


```
#!/bin/bash
#####
#
# Backup to NFS mount script.
#
#####

# What to backup.
backup_files="/home /var/spool/mail /etc /root /boot /opt"

# Where to backup to.
dest="/mnt/backup"

# Create archive filename.
day=$(date +%A)
hostname=$(hostname -s)
archive_file="$hostname-$day.tgz"

# Print start status message.
echo "Backing up $backup_files to $dest/$archive_file"
date
echo

# Backup the files using tar.
tar czf $dest/$archive_file $backup_files

# Print end status message.
echo
echo "Backup finished"
date

# Long listing of files in $dest to check file sizes.
ls -lh $dest
```

Running the script

Run from a terminal

```
$ chmod u+x backup.sh
```

```
$ sudo ./backup.sh
```


Run with cron

```
# m h dom mon dow  command
```

-
-
-
-
-
-

```
$ sudo crontab -e
```

Note:

Using `sudo` with the `crontab -e` command edits the `root` user's crontab. This is necessary if you are backing up directories only the root user has access to.

```
# m h dom mon dow  command
0 0 * * * bash /usr/local/bin/backup.sh
```

Note:

The `backup.sh` script will need to be copied to the `/usr/local/bin/` directory in order for this entry to run properly. The script can reside anywhere on the file system, simply change the script path appropriately.

Restoring from the archive

-

```
$ tar -tzvf /mnt/backup/host-Monday.tgz
```

-

```
$ tar -xzf /mnt/backup/host-Monday.tgz -C /tmp etc/hosts
```

-

```
$ cd / > sudo tar -xzf /mnt/backup/host-Monday.tgz
```

Note:

This will overwrite the files currently on the file system.

Archive rotation shell script

simple backup shell script

Rotating NFS archives

-
-
-

```
#!/bin/bash
#####
#
# Backup to NFS mount script with
# grandparent-parent-child rotation.
#
#####

# What to backup.
backup_files="/home /var/spool/mail /etc /root /boot /opt"

# Where to backup to.
dest="/mnt/backup"

# Setup variables for the archive filename.
day=$(date +%A)
hostname=$(hostname -s)

# Find which week of the month 1-4 it is.
day_num=$(date +%d)
if (( $day_num <= 7 )) then
    week_file="$hostname-week1.tgz"
elif (( $day_num > 7 && $day_num <= 14 )) then
    week_file="$hostname-week2.tgz"
elif (( $day_num > 14 && $day_num <= 21 )) then
    week_file="$hostname-week3.tgz"
elif (( $day_num > 21 && $day_num < 32 )) then
    week_file="$hostname-week4.tgz"
fi

# Find if the Month is odd or even.
month_num=$(date +%m)
month=$(expr $month_num % 2)
if [ $month -eq 0 ]; then
    month_file="$hostname-month2.tgz"
else
    month_file="$hostname-month1.tgz"
fi

# Create archive filename.
if [ $day_num == 1 ]; then
    archive_file=$month_file
elif [ $day != "Saturday" ]; then
    archive_file="$hostname-$day.tgz"
else
    archive_file=$week_file
fi

# Print start status message.
echo "Backing up $backup_files to $dest/$archive_file"
date
echo

# Backup the files using tar.
tar czf $dest/$archive_file $backup_files

# Print end status message.
```

(continues on next page)

(continued from previous page)

```
echo
echo "Backup finished"
date

# Long listing of files in $dest to check file sizes.
ls -lh $dest/
```

Running the script

Backing up to tape drives

```
#!/bin/bash
#####
#
# Backup to tape drive script.
#
#####

# What to backup.
backup_files="/home /var/spool/mail /etc /root /boot /opt"

# Where to backup to.
dest="/dev/st0"

# Print start status message.
echo "Backing up $backup_files to $dest"
date
echo

# Make sure the tape is rewound.
mt -f $dest rewind

# Backup the files using tar.
tar czf $dest $backup_files

# Rewind and eject the tape.
mt -f $dest rewoffl

# Print end status message.
echo
echo "Backup finished"
date
```

Note:

The default device name for a SCSI tape drive is `/dev/st0`. Use the appropriate device path for your system.

```
$ mt -f /dev/st0 rewind
```

```
$ tar -xzf /dev/st0 -C /tmp etc/hosts
```

Further reading

- [Advanced Bash-Scripting Guide](#)⁴⁷⁰
- [cron how-to wiki page](#)⁴⁷¹
- [GNU tar manual](#)⁴⁷²
- [Backup Rotation Scheme](#)⁴⁷³
- - [cpio](#)⁴⁷⁴
 - [dd](#)⁴⁷⁵
 - [rsnapshot](#)⁴⁷⁶
Tools - rsnapshot
 - [rsync\(1\)](#)⁴⁷⁷

⁴⁷⁰ <https://tldp.org/LDP/abs/html/>

⁴⁷¹ <https://help.ubuntu.com/community/CronHowto>

⁴⁷² <http://www.gnu.org/software/tar/manual/index.html>

⁴⁷³ http://en.wikipedia.org/wiki/Backup_rotation_scheme

⁴⁷⁴ <http://www.gnu.org/software/cpio/>

⁴⁷⁵ <http://www.gnu.org/software/coreutils/>

⁴⁷⁶ <https://rsnapshot.org/>

⁴⁷⁷ <https://manpages.ubuntu.com/manpages/resolute/man1/rsync.1.html>

Version control

- *etckeeper*
- *Install gitolite*

etckeeper

etckeeper⁴⁷⁸

Install etckeeper

```
$ sudo apt install etckeeper
```

Initialize etckeeper

Note:

It is possible to undo this by entering the following command:

```
$ sudo etckeeper uninit
```

⁴⁷⁸ <https://etckeeper.branchable.com/>

Configure autocommit frequency

```
$ sudo etckeeper commit "Reason for configuration change"
```

```
$ sudo etckeeper vcs log /etc/passwd
```

```
$ sudo apt install postfix
```

```
[master 5a16a0d] committing changes in /etc made by "apt install postfix"
Author: Your Name <xyz@example.com>
36 files changed, 2987 insertions(+), 4 deletions(-)
create mode 100755 init.d/postfix
create mode 100644 insserv.conf.d/postfix
create mode 100755 network/if-down.d/postfix
create mode 100755 network/if-up.d/postfix
create mode 100644 postfix/dynamicmaps.cf
create mode 100644 postfix/main.cf
create mode 100644 postfix/main.cf.proto
create mode 120000 postfix/makedefs.out
create mode 100644 postfix/master.cf
create mode 100644 postfix/master.cf.proto
create mode 100755 postfix/post-install
create mode 100644 postfix/postfix-files
create mode 100755 postfix/postfix-script
create mode 100755 ppp/ip-down.d/postfix
create mode 100755 ppp/ip-up.d/postfix
create mode 120000 rc0.d/K01postfix
create mode 120000 rc1.d/K01postfix
create mode 120000 rc2.d/S01postfix
create mode 120000 rc3.d/S01postfix
create mode 120000 rc4.d/S01postfix
create mode 120000 rc5.d/S01postfix
create mode 120000 rc6.d/K01postfix
create mode 100755 resolvconf/update-libc.d/postfix
create mode 100644 rsyslog.d/postfix.conf
create mode 120000 systemd/system/multi-user.target.wants/postfix.service
create mode 100644 ufw/applications.d/postfix
```

```
$ sudo etckeeper vcs status
```

```
$ sudo etckeeper vcs diff
```

```
$ sudo etckeeper commit "added new host"
```

Further reading

- [etckeeper](https://etckeeper.branchable.com/)⁴⁷⁹
- [the Git website](https://git-scm.com/)⁴⁸⁰

How to set up Gitolite

Gitolite

*"bare git"*⁴⁸¹

SSH-key

Install a Gitolite server

```
$ sudo apt install gitolite3
```

⁴⁷⁹ <https://etckeeper.branchable.com/>

⁴⁸⁰ <https://git-scm.com/>

⁴⁸¹ <https://git-scm.com/docs/gitglossary#Documentation/gitglossary.txt-aiddefbarerepositoryabarerepository>

Configure Gitolite

```
$ sudo useradd --system --home /home/git --create-home git
```

SSH-key
SSH keys in our OpenSSH

guide

SSH-key

```
$ cp ~/.ssh/id_ed25519.pub /tmp/admin.pub
```

```
$ sudo -i -u git gitolite setup -pk /tmp/admin.pub
```

•

•

•

—

•

•

```
$ ssh git@$yourserver
```

```
hello admin, this is git@your-gitolite-server running gitolite3
```

```
R W      gitolite-admin
```

```
R W      testing
```

Managing Gitolite users and repositories

```
$ git clone git@$yourserver:gitolite-admin.git
```

```
$ git commit
```

```
$ git push
```

-
-

Adding a Gitolite user

nsswitch

Deleting a Gitolite user

Managing repositories

Example configuration

```
# Gitolite config
# Users are created by having their public key in keydir/$username.pub

# Group creation
@bestproject      = name1 name2
@projectwatchers  = name3 @bestproject

# This repo itself
repo    gitolite-admin
RW+     =   admin
R       =   alice

# A repo with access to anybody
repo    testing
RW+     = @all

# A repo with special privileges, to tags and branches
repo    some/awesome/project
RW      =   alice @bestproject
RW+     =   bob
RW+    dev/      =   @bestproject
R       =   @projectwatchers carol
# bestproject members and alice can push code (but not force-push)
# bestproject members can force-push branches starting with dev/
# bob can forcepush anything
# projectwatchers and carol have readonly access
```

page 1⁴⁸²

page 2⁴⁸³

Using your server

```
$ git clone git@$server:some/awesome/project.git
```

```
$ git remote add $somename git@$server:some/awesome/project.git
```

⁴⁸² <https://gitolite.com/gitolite/conf.html>

⁴⁸³ <https://gitolite.com/gitolite/conf-2.html>

Further reading

- [Gitolite's code repository](#)⁴⁸⁴
- [Gitolite's documentation](#)⁴⁸⁵
- [Gitolite Essentials](#)⁴⁸⁶
- [Git homepage](#)⁴⁸⁷

See also

- [Backup and restore OpenLDAP](#)

See also

- [Data and storage](#)
- [Mail services](#)
-
-
-

Mail services

- 1.
- 2.
- 3.
- 4.

⁴⁸⁴ <https://github.com/sitaramc/gitolite>

⁴⁸⁵ <https://gitolite.com/gitolite/>

⁴⁸⁶ <https://www.packtpub.com/en-us/product/gitolite-essentials-9781783282388>

⁴⁸⁷ <https://git-scm.com/>

Mail User Agent

Thunderbird⁴⁸⁸

this short guide⁴⁸⁹

Mail Transfer Agent

Postfix⁴⁹⁰

sendmail⁴⁹¹

- *Install Postfix*

Exim4⁴⁹²

- *Install Exim4*

Install and configure Postfix

Note:

This guide does not cover setting up Postfix *Virtual Domains*. For information on Virtual Domains and other advanced configurations see the references list at the end of this page.

Install Postfix

Postfix⁴⁹³

```
$ sudo apt install postfix
```

Warning:

The mail-stack-delivery metapackage has been deprecated in Focal. The package still exists for compatibility reasons, but won't setup a working email system.

⁴⁸⁸ <https://www.thunderbird.net/>

⁴⁸⁹ <https://snapcraft.io/install/thunderbird/ubuntu>

⁴⁹⁰ <https://www.postfix.org/>

⁴⁹¹ <https://www.authsmtp.com/sendmail/index.html>

⁴⁹² <https://www.exim.org/>

⁴⁹³ <https://www.postfix.org/>

Configure Postfix

-
-
-
-

```
$ sudo dpkg-reconfigure postfix
```

-
-
-
-
-
-
-
-
-

Configure mailbox format

```
$ sudo postconf -e 'home_mailbox = Maildir/'
```

SMTP authentication

Configure SMTP authentication

```
$ sudo postconf -e 'smtpd_sasl_type = dovecot' > sudo postconf -e  
'smtpd_sasl_path = private/auth' > sudo postconf -e 'smtpd_sasl_local_domain  
=' > sudo postconf -e 'smtpd_sasl_security_options = noanonymous' > sudo  
postconf -e 'broken_sasl_auth_clients = yes' > sudo postconf -e  
'smtpd_sasl_auth_enable = yes' > sudo postconf -e  
'smtpd_recipient_restrictions = \ >  
permit_sasl_authenticated,permit_mynetworks,reject_unauth_destination'
```

Note:

The `smtpd_sasl_path` config parameter is a path relative to the Postfix queue directory.

Configure TLS

Obtain TLS certificates

```
$ sudo postconf -e 'smtp_tls_security_level = may' > sudo postconf -e  
'smtpd_tls_security_level = may' > sudo postconf -e  
'smtp_tls_note_starttls_offer = yes' > sudo postconf -e  
'smtpd_tls_chain_files =  
/etc/ssl/private/server.key,/etc/ssl/certs/server.crt' > sudo postconf -e
```

```
'smtpd_tls_loglevel = 1' > sudo postconf -e 'smtpd_tls_received_header = yes'  
> sudo postconf -e 'myhostname = mail.example.com'
```

```
$ sudo postconf -e 'smtpd_tls_CAfile = /etc/ssl/certs/cacert.pem'
```

Obtain TLS certificates

Outcome of initial configuration

```
# See /usr/share/postfix/main.cf.dist for a commented, more complete  
# version  
  
smtpd_banner = $myhostname ESMTP $mail_name (Ubuntu)  
biff = no  
  
# appending .domain is the MUA's job.  
append_dot_mydomain = no  
  
# Uncomment the next line to generate "delayed mail" warnings  
#delay_warning_time = 4h  
  
myhostname = server1.example.com  
alias_maps = hash:/etc/aliases  
alias_database = hash:/etc/aliases  
myorigin = /etc/mailname  
mydestination = server1.example.com, localhost.example.com, localhost  
relayhost =  
mynetworks = 127.0.0.0/8  
mailbox_command = procmail -a "$EXTENSION"  
mailbox_size_limit = 0  
recipient_delimiter = +  
inet_interfaces = all  
smtpd_sasl_local_domain =  
smtpd_sasl_auth_enable = yes  
smtpd_sasl_security_options = noanonymous  
broken_sasl_auth_clients = yes  
smtpd_recipient_restrictions =  
permit_sasl_authenticated,permit_mynetworks,reject_unauth_destination  
smtpd_tls_auth_only = no  
smtpd_tls_security_level = may  
smtpd_tls_security_level = may  
smtpd_tls_note_starttls_offer = yes  
smtpd_tls_key_file = /etc/ssl/private/smtpd.key  
smtpd_tls_cert_file = /etc/ssl/certs/smtpd.crt  
smtpd_tls_CAfile = /etc/ssl/certs/cacert.pem  
smtpd_tls_loglevel = 1  
smtpd_tls_received_header = yes
```

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```
smtpd_tls_session_cache_timeout = 3600s
tls_random_source = dev:/dev/urandom
```

```
$ sudo systemctl restart postfix.service
```

SASL

RFC2554⁴⁹⁴

SASL⁴⁹⁵

```
mynetworks = 127.0.0.0/8, [::1]/128
```

Configure SASL

```
$ sudo apt install dovecot-core
```

```
service auth {
    # auth_socket_path points to this userdb socket by default. It's typically
    # used by dovecot-lda, doveadm, possibly imap process, etc. Its default
    # permissions make it readable only by root, but you may need to relax these
    # permissions. Users that have access to this socket are able to get a list
    # of all usernames and get results of everyone's userdb lookups.
    unix_listener auth-userdb {
        #mode = 0600
        #user =
        #group =
    }

    # Postfix smtp-auth
    unix_listener /var/spool/postfix/private/auth {
        mode = 0660
        user = postfix
        group = postfix
    }
}
```

⁴⁹⁴ <https://www.ietf.org/rfc/rfc2554.txt>

⁴⁹⁵ <https://www.ietf.org/rfc/rfc2222.txt>

```
auth_mechanisms = plain
```

```
auth_mechanisms = plain login
```

```
$ sudo systemctl restart dovecot.service
```

Test your setup

```
$ telnet mail.example.com 25
```

```
$ ehlo mail.example.com
```

```
250-STARTTLS
250-AUTH LOGIN PLAIN
250-AUTH=LOGIN PLAIN
250 8BITMIME
```

Troubleshooting

Escaping chroot

```
smtp      inet  n       -       -       -       smtpd
```

```
smtp      inet  n       -       n       -       smtpd
```

```
$ sudo service postfix restart
```

SMTPS

```
smtps      inet  n       -       -       -       smtpd
-o smtpd_tls_wrappermode=yes
-o smtpd_sasl_auth_enable=yes
-o smtpd_client_restrictions=permit_sasl_authenticated,reject
-o milter_macro_daemon_name=ORIGINATING
```

Log viewing

```
$ tail -f /var/log/mail.err
```

Increase logging detail

```
$ sudo postconf -e 'smtpd_tls_loglevel = 4'
```

```
$ sudo systemctl reload postfix.service
```

Logging mail delivery

```
$ sudo postconf -e 'debug_peer_list = problem.domain'
```

```
$ sudo systemctl reload postfix.service
```

Increase daemon verbosity

```
smtp      unix  -      -      -      -      -      smtp -v
```

```
$ sudo systemctl reload postfix.service
```

Log SASL debug info

```
auth_debug=yes  
auth_debug_passwords=yes
```

```
$ sudo systemctl reload dovecot.service
```

Note:

Some of the options above can drastically increase the amount of information sent to the log files. Remember to return the log level back to normal after you have corrected the problem – then reload the appropriate daemon for the new configuration to take effect.

Further reading

- [Postfix website](#)⁴⁹⁶
- [Postfix: The Definitive Guide](#)⁴⁹⁷
- [Ubuntu Wiki Postfix](#)⁴⁹⁸
- [Debian Wiki Postfix](#)⁴⁹⁹
[Postfix Tutorials](#)⁵⁰⁰
- [set up mailman3 with postfix](#)⁵⁰¹

⁴⁹⁶ <http://www.postfix.org/documentation.html>

⁴⁹⁷ <https://www.oreilly.com/library/view/postfix-the-definitive/0596002122/>

⁴⁹⁸ <https://help.ubuntu.com/community/Postfix>

⁴⁹⁹ <https://wiki.debian.org/Postfix>

⁵⁰⁰ <https://wiki.debian.org/Postfix/Tutorials>

⁵⁰¹ <https://docs.mailman3.org/projects/mailman/en/latest/src/mailman/docs/mta.html>

Install and configure Exim4

Install Exim4

Exim4⁵⁰²

```
$ sudo apt install exim4
```

Configure Exim4

```
$ sudo dpkg-reconfigure exim4-config
```

Note:

The default configuration layout for Exim4 is the single configuration file one.

```
/etc/exim4/  
├─ conf.d  
│  ├── acl  
│  ├── auth  
│  ├── main  
│  ├── retry  
│  ├── rewrite  
│  ├── router  
│  └─ transport
```

```
$ sudo update-exim4.conf
```

⁵⁰² <https://www.exim.org/>

Warning:

You should never manually edit the configuration file `/var/lib/exim4/config.autogenerated`, because it is updated automatically every time you run `update-exim4.conf`. Any changes you make to it will eventually be lost.

```
$ sudo systemctl restart exim4
```

Start the Exim4 daemon

```
$ sudo service exim4 start
```

SMTP authentication

-
-

Warning:

All configuration steps shown from now on will assume a split-configuration mode for Exim4. If you have selected the non-split mode, then all commands that edit a configuration file under `/etc/exim4/conf.d` in the sections below should be replaced with editing the single file `/etc/exim4/exim4.conf.template`.

Enabling TLS

```
$ sudo /usr/share/doc/exim4-base/examples/exim-gencert
```

Warning:

This will install a self-signed certificate. If deploying this system in production, obtain a trusted certificate instead. See [Obtain TLS certificates](#) (page 252) for how to get a certificate, or how to set up your own Certificate Authority (CA) for internal deployments.

```
MAIN_TLS_ENABLE = yes
```

Authenticating existing Linux users

Note:

To manage local Linux users, please refer to [User management](#) (page 154).

```
plain_saslauthd_server:
    driver = plaintext
    public_name = PLAIN
    server_condition = ${if saslauthd{${auth2}{${auth3}}}{1}{0}}
    server_set_id = $auth2
    server_prompts = :
    .ifndef AUTH_SERVER_ALLOW_NOTLS_PASSWORDS
    server_advertise_condition = ${if eq{${tls_cipher}{}{}}{*}}
    .endif

login_saslauthd_server:
    driver = plaintext
    public_name = LOGIN
    server_prompts = "Username:: : Password::"
    # don't send system passwords over unencrypted connections
    server_condition = ${if saslauthd{${auth1}{${auth2}}}{1}{0}}
```

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```
server_set_id = $auth1
.ifdef AUTH_SERVER_ALLOW_NOTLS_PASSWORDS
server_advertise_condition = ${if eq{$tls_cipher}{*}}{*}}
.endif
```

```
$ sudo gpasswd -a Debian-exim sasl
```

```
$ sudo update-exim4.conf
```

```
$ sudo systemctl restart exim4
```

```
$ sudo apt install sasl2-bin
```

```
MECHANISMS="pam"
```

Note:

In Ubuntu 22.04 LTS (Jammy) and earlier, we also need to add `START="yes"` to `/etc/default/saslauthd`.

```
$ sudo systemctl enable saslauthd
```

```
$ sudo systemctl start saslauthd
```


Authenticating arbitrary users

```
plain_server:
    driver = plaintext
    public_name = PLAIN
    server_condition = "${if crypteq[$auth3]{$extract[1]:}{$lookup[$auth2]lsearch
[CONFFDIR/passwd]{$value}{*:*)}}}{1}{0}}"
    server_set_id = $auth2
    server_prompts = :
    .ifndef AUTH_SERVER_ALLOW_NOTLS_PASSWORDS
    server_advertise_condition = ${if eq{$tls_in_cipher}{}}{*}}
    .endif

login_server:
    driver = plaintext
    public_name = LOGIN
    server_prompts = "Username:: : Password::"
    server_condition = "${if crypteq[$auth2]{$extract[1]:}{$lookup[$auth1]lsearch
[CONFFDIR/passwd]{$value}{*:*)}}}{1}{0}}"
    server_set_id = $auth1
    .ifndef AUTH_SERVER_ALLOW_NOTLS_PASSWORDS
    server_advertise_condition = ${if eq{$tls_in_cipher}{}}{*}}
    .endif
```

Warning:

DO NOT enable both these and the `_saslauthd_server` variants (from “Authenticating existing Linux users” above) at the same time!

```
username:crypted-password:cleartext-password
```

```
$ sudo /usr/share/doc/exim4-base/examples/exim-adduser
```

```
User: ubuntu
Password: ubuntusecret
```

```
ubuntu:$1$ZvPA$HTddFobmJD1vURtJHBmbw/:ubuntusecret
```

```
$ sudo chown root:Debian-exim /etc/exim4/passwd
```

```
$ sudo chmod 0640 /etc/exim4/passwd
```

-
-
-

Warning:

The `/usr/share/doc/exim4-base/examples/exim-adduser` serves mostly as an example and is not able to handle many scenarios. For example, it won't check if the username you are providing already exists in the `passwd` file, which can lead to multiple entries for the same user, with unpredictable results.

```
$ sudo update-exim4.conf
```

```
$ sudo systemctl restart exim4
```

Note:

There is no need to restart Exim4 after making changes to the `/etc/exim4/passwd` file.

Troubleshooting

```
$ testsaslauthd -u ubuntu -p ubuntusecret
```

```
0: OK "Success."
```

```
$ sudo apt install cyrus-clients --no-install-recommends
```

*cyrus-smtpdtest(1)*⁵⁰³

```
/usr/lib/cyrus/bin/smtpdtest -t "" -a ubuntu -w ubuntusecret n-exim.lxd
```

-
-
-
-

⁵⁰³ <https://manpages.ubuntu.com/manpages/resolute/man1/cyrus-smtpdtest.1.html>

```
S: 220 n-exim ESMTTP Exim 4.97 Ubuntu Mon, 23 Jun 2025 21:11:59 +0000
C: EHLO smtptest
S: 250-n-exim Hello n-exim.lxd [10.10.17.9]
S: 250-SIZE 52428800
S: 250-8BITMIME
S: 250-PIPELINING
S: 250-PIPECONNECT
S: 250-CHUNKING
S: 250-STARTTLS
S: 250-PRDR
S: 250 HELP
C: STARTTLS
S: 220 TLS go ahead
verify error:num=18:self-signed certificate
TLS connection established: TLSv1.3 with cipher TLS_AES_256_GCM_SHA384 (256/256
bits)
C: EHLO smtptest
S: 250-n-exim Hello n-exim.lxd [10.10.17.9]
S: 250-SIZE 52428800
S: 250-8BITMIME
S: 250-PIPELINING
S: 250-PIPECONNECT
S: 250-AUTH PLAIN LOGIN
S: 250-CHUNKING
S: 250-PRDR
S: 250 HELP
C: AUTH LOGIN
S: 334 VXNlcm5hbWU6
C: dWJ1bnR1
S: 334 UGFzc3dvcmQ6
C: dWJ1bnR1c2VjcmV0
S: 235 Authentication succeeded
Authenticated.
Security strength factor: 256
```

-
-
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-

Tip:

Want to obtain the original username and password back from the base64 encoded values? Feed those values to the base64 -d tool. Example, using the value from the session above:

```
$ echo -n dWJ1bnR1c2VjcmV0 | base64 -d; echo
```

```
ubuntusecret
```

To test the PLAIN mechanism, add the -m plain command-line option:

```
/usr/lib/cyrus/bin/smtptest -t "" -a ubuntu -w ubuntusecret -m plain n-exim.lxd
```

In the new output, PLAIN was selected:

```
S: 220 n-exim ESMTP Exim 4.97 Ubuntu Mon, 23 Jun 2025 21:15:39 +0000
C: EHLO smtptest
S: 250-n-exim Hello n-exim.lxd [10.10.17.9]
S: 250-SIZE 52428800
S: 250-8BITMIME
S: 250-PIPELINING
S: 250-PIPECONNECT
S: 250-CHUNKING
S: 250-STARTTLS
S: 250-PRDR
S: 250 HELP
C: STARTTLS
S: 220 TLS go ahead
verify error:num=18:self-signed certificate
TLS connection established: TLSv1.3 with cipher TLS_AES_256_GCM_SHA384 (256/256
bits)
C: EHLO smtptest
S: 250-n-exim Hello n-exim.lxd [10.10.17.9]
S: 250-SIZE 52428800
S: 250-8BITMIME
S: 250-PIPELINING
S: 250-PIPECONNECT
S: 250-AUTH PLAIN LOGIN
S: 250-CHUNKING
S: 250-PRDR
S: 250 HELP
C: AUTH PLAIN AHVidW50dQB1YnVudHVzZWNYZXQ=
S: 235 Authentication succeeded
Authenticated.
Security strength factor: 256
```

Troubleshooting tips

Here are some troubleshooting tips.

Permissions

- If using saslauthd: Can the Debian-exim user read and write to the saslauthd socket in /run/saslauthd/mux socket?
- If using /etc/exim4/passwd: Can the Debian-exim user read this file?

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Config

Mail Delivery Agent

Dovecot⁵⁰⁷
Maildir⁵⁰⁹

mbox⁵⁰⁸

- *Install Dovecot*

Install and configure Dovecot

Install Dovecot

```
$ sudo apt install dovecot-imapd dovecot-pop3d
```

Configure Dovecot

```
!include_try /usr/share/dovecot/protocols.d/*.protocol
```

web site⁵¹⁰

the Dovecot

⁵⁰⁷ <https://dovecot.org/>

⁵⁰⁸ <https://en.wikipedia.org/wiki/Mbox>

⁵⁰⁹ <https://en.wikipedia.org/wiki/Maildir>

⁵¹⁰ <https://doc.dovecot.org/main/>

Restart the Dovecot daemon

```
$ sudo service dovecot restart
```

```
bhuvan@rainbow:~$ telnet localhost pop3
```

```
Trying 127.0.0.1...
Connected to localhost.localdomain.
Escape character is '^]'.
+OK Dovecot ready.
```

Dovecot SSL configuration

```
$ sudo openssl req -new -x509 -days 1000 -nodes -out
"/etc/dovecot/dovecot.pem" \ > -keyout "/etc/dovecot/private/dovecot.pem"
```

```
ssl_cert = </etc/dovecot/private/dovecot.pem
ssl_key = </etc/dovecot/private/dovecot.key
```

See also

See [Obtain TLS certificates](#) (page 252) for how to get a trusted certificate or set up your own CA.

Configure a firewall for an email server

-
-
-
-

Further reading

- [Dovecot website](#)⁵¹¹
- [Dovecot manual](#)⁵¹²
- [Dovecot Ubuntu Wiki](#)⁵¹³

Web services section

-
-
-

Web services

Introduction to web services

Proxy servers

⁵¹¹ <https://dovecot.org/>

⁵¹² <https://doc.dovecot.org/latest/>

⁵¹³ <https://help.ubuntu.com/community/Dovecot>

How to install a Squid server

refer to this guide

Install Squid

```
$ sudo apt install squid
```

Configure Squid

Protect the original config file

```
$ sudo cp /etc/squid/squid.conf /etc/squid/squid.conf.original
```

```
$ sudo chmod a-w /etc/squid/squid.conf.original
```

Change TCP port

```
http_port 8888
```

Set the hostname

hostname

```
visible_hostname weezie
```

Configure the memory cache

```
cache_mem 512 MB  
memory_replacement_policy lru
```

Configure on-disk cache

```
cache_dir <Type> <Directory-Name> <Size-in-MB> <L1-Dirs> <L2-Dirs> [options]
```

```
cache_dir ufs /var/spool/squid 10000 16 256
```

-
-
-
-

Configure cached objects size limits

```
maximum_object_size 512 MB
minimum_object_size 0 KB
maximum_object_size_in_memory 512 KB
```

Configure cached objects lifetime

```
refresh_pattern regex min percent max [options]
```

```
refresh_pattern -i \.(gif|jpg|png|css|js)$ 10080 90% 43200
refresh_pattern . 0 20% 4320
```

Caching HTTPS content

HTTPS
HTTPS

Squid HTTPS documentation⁵¹⁴

Other caching configuration options

Squid configuration guide⁵¹⁵

⁵¹⁴ <https://wiki.squid-cache.org/Features/HTTPS>

⁵¹⁵ <https://www.squid-cache.org/Doc/config/>

Access control

- *ACL*

```
acl fortytwo_network src 192.168.42.0/24
```

-

```
http_access allow fortytwo_network
```

-

```
acl biz_network src 10.1.42.0/24  
acl biz_hours time M T W T F 9 00-17 00
```

-

```
http_access allow biz_network biz_hours
```

Restart the Squid server

```
$ sudo squid -k parse
```

```
$ sudo systemctl restart squid.service
```

Note:

If a formerly customized squid3 was used to set up the spool at `/var/log/squid3` to be a mount point, but otherwise kept the default configuration, the upgrade will fail. The upgrade tries to rename/move files as needed, but it can't do so for an active mount point. In that case you will need to adapt either the mount point or the config in `/etc/squid/squid.conf` so that they match. The same applies if the **include** config statement was used to pull in more files from the old path at `/etc/squid3/`. In those cases you should move and adapt your configuration accordingly.

Troubleshooting

```
$ sudo squid -v
```

```
$ sudo systemctl status squid
```

```
$ sudo cat /var/log/squid/cache.log
```

```
$ sudo cat /var/log/squid/access.log
```

```
$ curl http://localhost:8888/squid-internal-mgr/info
```

-
-
-
-
-

Further reading

- [The Squid Website](#)⁵¹⁶
- [Ubuntu Wiki page on Squid](#)⁵¹⁷

Web servers

Apache2

How to install Apache2

The Apache HTTP Server⁵¹⁸

Install apache2

```
$ sudo apt install apache2
```

Configure apache2

Files

-

⁵¹⁶ <http://www.squid-cache.org/>

⁵¹⁷ <https://help.ubuntu.com/community/Squid>

⁵¹⁸ <https://httpd.apache.org/>

Note:

Historically, the main Apache2 configuration file was `httpd.conf`, named after the “httpd” daemon. In other distributions (or older versions of Ubuntu), the file might be present. In modern releases of Ubuntu, all configuration options have been moved to `apache2.conf` and the below referenced directories and `httpd.conf` no longer exists.

-
-
-

Directories

-
-
-
-
-
-

Detailed configuration

- [*Apache2 configuration settings*](#)
- [*how to extend Apache2*](#)

Further reading

- [Apache2 Documentation⁵¹⁹](#)
- [Apache Cookbook⁵²⁰](#)

How to configure Apache2 settings

installed Apache2

Basic directives

Note:

The directives set for a virtual host only apply to that particular virtual host. If a directive is set server-wide and not defined in the virtual host settings, the default setting is used. For example, you can define a Webmaster email address and not define individual email addresses for each virtual host.

```
$ sudo cp /etc/apache2/sites-available/000-default.conf
/etc/apache2/sites-available/mynewsite.conf
```

⁵¹⁹ <https://httpd.apache.org/docs/2.4/>

⁵²⁰ <https://www.oreilly.com/library/view/apache-cookbook-2nd/9780596529949/>

The ServerAdmin directive

Found in `/etc/apache2/sites-available`

The Listen directive

Found in `/etc/apache2/ports.conf`

-
-
-

The ServerName directive (optional)

FQDN

The ServerAlias directive

```
ServerAlias *.mynewsite.com
```

The DocumentRoot directive

```
$ sudo a2ensite mynewsite
```

```
$ sudo systemctl restart apache2.service
```

Note:

Be sure to replace `mynewsite` with a more descriptive name for the VirtualHost. One method is to name the file after the **ServerName** directive of the VirtualHost.

```
$ sudo a2dissite mynewsite
```

```
$ sudo systemctl restart apache2.service
```

Apache2 server default settings

The DirectoryIndex

HTML

The ErrorDocument

CustomLog and ErrorLog

The Options directive

```
<Directory /var/www/html/mynewsite>  
...  
</Directory>
```

-

Caution:

Most files should not be run as CGI scripts. This would be very dangerous. CGI scripts should be kept in a directory separate from and outside your **DocumentRoot**, and only this directory should have the ExecCGI option set. This is the default, and the default location for CGI scripts is `/usr/lib/cgi-bin`.

-

Apache SSI documentation (Ubuntu community)⁵²¹ *HTML*

-

⁵²¹ <https://help.ubuntu.com/community/ServerSideIncludes>

-

Caution:

For security reasons, this should usually not be set, and certainly should not be set on your DocumentRoot directory. Enable this option carefully on a per-directory basis **only** if you are certain you want users to see the entire contents of the directory.

-

[Apache2 documentation on this option](#)⁵²²

-

Apache2 daemon settings

-

-

-

Warning:

Unless you know exactly what you are doing, do not set the User directive to root. Using root as the User will create large security holes for your Web server.

-

⁵²² https://httpd.apache.org/docs/2.4/mod/mod_negotiation.html#multiviews

Extending Apache2

how to extend

Apache2

Further reading

- [Apache2 Documentation](#)⁵²³
- [Apache Cookbook](#)⁵²⁴

How to use Apache2 modules

Installing and handling modules

```
$ sudo apt install libapache2-mod-wsgi-py3
```

```
$ sudo a2dismod wsgi
```

```
$ sudo systemctl restart apache2.service
```

```
$ sudo a2enmod wsgi
```

⁵²³ <https://httpd.apache.org/docs/2.4/>

⁵²⁴ <https://www.oreilly.com/library/view/apache-cookbook-2nd/9780596529949/>

```
$ sudo systemctl restart apache2.service
```

Configure Apache2 for HTTPS

```
$ sudo a2enmod ssl
```

Note:

For more information on generating a key and obtaining a certificate see [Certificates](#) (page 927).

```
$ sudo a2ensite default-ssl
```

Note:

The directories `/etc/ssl/certs` and `/etc/ssl/private` are the default locations. If you install the certificate and key in another directory make sure to change `SSLCertificateFile` and `SSLCertificateKeyFile` appropriately.

```
$ sudo systemctl restart apache2.service
```

Sharing write permission

```
$ sudo chgrp -R webmasters /var/www/html
```

```
$ sudo chmod -R g=rwX /var/www/html/
```

Warning:

The apache2 daemon will run as the www-data user, which has a corresponding www-data group. These **should not** be granted write access to the document root, as this would mean that vulnerabilities in Apache or the applications it is serving would allow attackers to overwrite the served content.

Further reading

- [Apache2 Documentation](#)⁵²⁵
- [Apache Cookbook](#)⁵²⁶
- *Install Apache2*
- *Configure Apache2*
- *Extend Apache2 with modules*

Nginx

How to install nginx

Apache

⁵²⁵ <https://httpd.apache.org/docs/2.4/>

⁵²⁶ <http://shop.oreilly.com/product/9780596529949.do>

Install nginx

```
$ sudo apt update && sudo apt install nginx
```

Verify nginx is running

```
$ sudo systemctl status nginx

■ nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/lib/systemd/system/nginx.service; enabled; vendor preset:
   enabled)
   Active: active (running) since Sun 2023-08-20 01:04:22 UTC; 53s ago
     Docs: man:nginx(8)
    Process: 28210 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process
   on; (code=exited, status=0/SUCCESS)
    Process: 28211 ExecStart=/usr/sbin/nginx -g daemon on; master_process on;
   (code=exited, status=0/SUCCESS)
   Main PID: 28312 (nginx)
    Tasks: 13 (limit: 76969)
    Memory: 13.1M
      CPU: 105ms
    CGroup: /system.slice/nginx.service
            └─28312 "nginx: master process /usr/sbin/nginx -g daemon on; master_
process on;"
            └─28314 "nginx: worker process" "" "" "" "" "" "" "" "" "" "" "" ""
" "" "" "" ""
...
```


Restarting nginx

```
$ sudo systemctl restart nginx
```

Enable/disable nginx manually

```
$ sudo systemctl disable nginx
```

```
$ sudo systemctl enable nginx
```

The default nginx homepage

```
Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further
configuration is
required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.
```

Setting up nginx

- [How to configure nginx](#)
- [How to use nginx modules](#)

Further reading

- [nginx documentation](https://nginx.org/en/docs/)⁵²⁷
-

How to configure nginx

installed nginx

Server blocks

```
server {
    listen 80 default_server;
    listen [::]:80 default_server;

    root /var/www/html;

    # Add index.php to the list if you are using PHP
    index index.html index.htm index.nginx-debian.html;

    server_name _;

    location / {
        # First attempt to serve request as file, then
        # as directory, then fall back to displaying a 404.
        try_files $uri $uri/ =404;
    }
}
```

filesystem

```
server {
    listen      80;
    root        /srv/my-site/html;
    index        index.html;
    server_name  my-site.org www.my-site.org;

    location / {
```

(continues on next page)

⁵²⁷ <https://nginx.org/en/docs/>

(continued from previous page)

```
        try_files $uri $uri/ =404;  
    }  
}
```

```
$ sudo mkdir -p /srv/my-site/html
```

```
$ sudo chmod -R 755 /srv/my-site/html
```

```
$ echo "<html><body><h1>My Site!</h1></body></html>" >  
/srv/my-site/html/index.html
```

```
$ sudo systemctl reload nginx
```

```
$ www-browser www.my-site.org
```

Multi-site hosting

```
$ sudo ln -s /etc/nginx/sites-available/<your-domain>  
/etc/nginx/sites-enabled/
```

```
$ sudo rm /etc/nginx/sites-enabled/default
```

SSL and HTTPS

```
server {  
    listen            80;  
    server_name       our-site.org www.our-site.org;  
    return            301 https://$host$request_uri;  
}  
  
server {  
    listen            443 ssl;  
  
    root              /srv/our-site/html;  
    index             index.html;  
  
    server_name       our-site.org www.our-site.org;  
  
    ssl_certificate    our-site.org.crt;  
    ssl_certificate_key our-site.org.key;  
    ssl_protocols      TLSv1 TLSv1.1 TLSv1.2 TLSv1.3;  
    ssl_ciphers         HIGH:!aNULL:!MD5;  
    ssl_session_timeout 15m;  
  
    location / {  
        try_files $uri $uri/ =404;  
    }  
}
```

Obtain TLS certificates

OpenSSL

GnuTLS

Setting up nginx

- [*How to use nginx modules*](#)

Further reading

- [nginx's beginner's guide](#)⁵²⁸ *FastCGI*
- [nginx documentation](#)⁵²⁹

How to use nginx modules

Available modules

```
$ nginx -V 2>&1 | tr -- - '\n' | grep _module
http_ssl_module
http_stub_status_module
http_realip_module
...
http_image_filter_module=dynamic
http_perl_module=dynamic
http_xslt_module=dynamic
stream_geoip_module=dynamic
```

```
$ apt policy libnginx-mod-http-image-filter
```

⁵²⁸ https://nginx.org/en/docs/beginners_guide.html

⁵²⁹ https://nginx.org/en/docs/http/configuring_https_servers.html

```
libnginx-mod-http-image-filter:
  Installed: (none)
  Candidate: 1.24.0-1ubuntu1
  Version table:
    1.24.0-1ubuntu1 500
    500 http://archive.ubuntu.com/ubuntu mantic/main amd64 Packages
```

```
$ sudo apt install libnginx-mod-http-image-filter

...
The following NEW packages will be installed:
  libnginx-mod-http-image-filter
0 upgraded, 1 newly installed, 0 to remove and 34 not upgraded.
...
Triggering nginx reload
...
```

Enabling and disabling dynamic modules

```
$ ls /etc/nginx/modules-*

/etc/nginx/modules-available:

/etc/nginx/modules-enabled:
50-mod-http-image-filter.conf
```

```
$ sudo mv /etc/nginx/modules-enabled/50-mod-http-image-filter.conf
/etc/nginx/modules-available/
```

```
$ service nginx restart
```

Configuring modules

```
## /etc/nginx/modules-enabled/50-mod-http-image-filter.conf
load_module modules/nginx_http_image_filter_module.so;
```

```
location /img/ {
    image_filter resize 240 360;
    image_filter rotate 180;
    image_filter_buffer 16M;
    error_page 415 = /415.html;
}
```

Further reading

- [nginx documentation](https://nginx.org/en/docs/)⁵³⁰
-
- *Install nginx*
- *Configure nginx*
- *Extend nginx with modules*

Web programming

⁵³⁰ <https://nginx.org/en/docs/>

How to install and configure PHP

PHP⁵³¹

HTML

Prerequisites

- [our Apache guide](#)
- [our MySQL guide](#)

Install PHP

```
$ sudo apt install php libapache2-mod-php
```

Install optional packages

-

```
$ sudo apt install php-cli
```

-

```
$ sudo apt install php-cgi
```

-

```
$ sudo apt install php-mysql
```

-

```
$ sudo apt install php-pgsql
```

⁵³¹ <https://www.php.net/>

Configure PHP

```
$ sudo systemctl restart apache2.service
```

Test your setup

```
<?php  
phpinfo();  
?>
```

Docu-

mentRoot

Further reading

- [the php.net documentation](https://www.php.net/docs.php)⁵³²
-
- [Apache MySQL PHP Ubuntu Wiki](https://help.ubuntu.com/community/ApacheMySQLPHP)⁵³³

How to install and configure Ruby on Rails

[Ruby on Rails](https://rubyonrails.org/)⁵³⁴

⁵³² <https://www.php.net/docs.php>

⁵³³ <https://help.ubuntu.com/community/ApacheMySQLPHP>

⁵³⁴ <https://rubyonrails.org/>

Prerequisites

- [our Apache guide](#)
- [our MySQL guide](#)

Install rails

```
$ sudo apt install rails ruby-dev build-essential
```

*bundler(1)*⁵³⁵

```
$ sudo apt install --no-install-recommends rails ruby-dev build-essential
```

```
$ sudo apt install libmysqlclient-dev
```

Create a new application (optional)

```
$ rails new blog --database=mysql --skip-bundle
```

```
$ cd blog > bundle config set --local path vendor/bundle > bundle install
```

⁵³⁵ <https://manpages.ubuntu.com/manpages/resolute/man1/bundler.1.html>

Configure the web server

DocumentRoot

```
DocumentRoot /path/to/rails/application/public
```

```
<Directory "/path/to/rails/application/public">
  Options Indexes FollowSymLinks MultiViews ExecCGI
  AllowOverride All
  Order allow,deny
  allow from all
  AddHandler cgi-script .cgi
</Directory>
```

```
$ sudo a2enmod rewrite
```

```
$ sudo chown -R www-data:www-data /path/to/rails/application/public
```

```
$ sudo chown -R www-data:www-data /path/to/rails/application/tmp
```

```
$ RAILS_ENV=production bundle exec rake assets:precompile
```

Configure the database

```
# MySQL
production
  adapter: mysql2
  username: user
  password: password
  host: 127.0.0.1
  database: app
```

our MySQL guide

```
$ RAILS_ENV=production bundle exec rake db:create > RAILS_ENV=production  
bundle exec rake db:migrate
```

daemonize

Further reading

- [Ruby on Rails](#)⁵³⁶
- [Agile Development with Rails](#)⁵³⁷

See also

- [Web services](#)

Graphics

Graphics

On-system GPU

NVIDIA drivers installation

Warning:

NVIDIA drivers installed from sources outside of those listed in this guide could potentially overwrite those provided by ubuntu-drivers and may break secure boot.

⁵³⁶ <https://rubyonrails.org/>

⁵³⁷ <https://pragprog.com/titles/rails7/agile-web-development-with-rails-7/>

NVIDIA drivers releases

1. [NVIDIA drivers releases](#) on the NVIDIA website⁵³⁸
2. [NVIDIA drivers releases](#) *ERD* in the NVIDIA documentation⁵³⁹

Check driver versions

```
$ cat /proc/driver/nvidia/version
```

The recommended way (ubuntu-drivers tool)

Ensure your system and kernel is up-to-date

```
$ sudo apt update && sudo apt upgrade
```

⁵³⁸ <https://www.nvidia.com/en-us/drivers/unix/>

⁵³⁹ <https://docs.nvidia.com/datacenter/tesla/index.html>

Check the available drivers for your hardware

```
$ sudo ubuntu-drivers list
```

```
$ sudo ubuntu-drivers list --gpgpu
```

```
nvidia-driver-470  
nvidia-driver-470-server  
nvidia-driver-535  
nvidia-driver-535-open  
nvidia-driver-535-server  
nvidia-driver-535-server-open  
nvidia-driver-550  
nvidia-driver-550-open  
nvidia-driver-550-server  
nvidia-driver-550-server-open
```

Installing the drivers for generic use (e.g. desktop and gaming)

```
$ sudo ubuntu-drivers install
```

```
$ sudo ubuntu-drivers install nvidia:535
```

Installing the drivers on servers and/or for computing purposes

```
$ sudo ubuntu-drivers install --gpgpu
```

```
$ sudo ubuntu-drivers install --gpgpu nvidia:535-server
```

```
$ sudo apt install nvidia-utils-535-server
```

Optional step

```
$ sudo apt install nvidia-fabricmanager-535 libnvidia-nscq-535
```

Note:

While `nvidia-fabricmanager` and `libnvidia-nscq` do not have the same `-server` label in their name, they are really meant to match the `-server` drivers in the Ubuntu archive. For example, `nvidia-fabricmanager-535` will match the `nvidia-driver-535-server` package version (not the `nvidia-driver-535` package).

Manual driver installation (using APT)

Installing the kernel modules

Installing the pre-compiled NVIDIA modules for your kernel

```
$ sudo apt install  
linux-modules-nvidia-${DRIVER_BRANCH}${SERVER}-${LINUX_FLAVOUR}
```

ABI

```
$ sudo apt-cache policy  
linux-modules-nvidia-${DRIVER_BRANCH}${SERVER}-${(uname -r)}
```

```
$ sudo apt install linux-modules-nvidia-${DRIVER_BRANCH}${SERVER}-${uname -r}
```

Building your own kernel modules using the NVIDIA DKMS package

DKMS

```
$ sudo apt install linux-headers-${LINUX_FLAVOUR}
```

```
$ sudo apt-cache policy linux-headers-${uname -r}
```

```
$ sudo apt install linux-headers-${uname -r}
```

```
$ sudo apt install nvidia-dkms-${DRIVER_BRANCH}${SERVER}
```

```
$ sudo ubuntu-drivers install --include-dkms
```


Installing the user-space drivers and the driver libraries

```
$ sudo apt install nvidia-driver-${DRIVER_BRANCH}${SERVER}
```

(Optional) Installing Fabric Manager and the NSCQ library

```
$ sudo apt install nvidia-fabricmanager-${DRIVER_BRANCH}  
libnvidia-nscq-${DRIVER_BRANCH}
```

Note:

While `nvidia-fabricmanager` and `libnvidia-nscq` do not have the same `-server` label in their name, they are really meant to match the `-server` drivers in the Ubuntu archive. For example, `nvidia-fabricmanager-535` will match the `nvidia-driver-535-server` package version (not the `nvidia-driver-535` package).

Switching between pre-compiled and DKMS modules

- 1.
- 2.

Uninstalling the NVIDIA drivers

```
$ sudo apt --purge remove '*nvidia*${DRIVER_BRANCH}*
```

```
$ apt-mark showmanual | grep nvidia`.
```

```
$ sudo apt autoremove
```

Transitional packages to new driver branches

current support matrix⁵⁴⁰

Troubleshooting

Driver/library version mismatch error

`nvidia-smi`⁵⁴¹

```
Failed to initialize NVML: Driver/library version mismatch
```

```
$ sudo dmesg
```

```
NVRM: API mismatch: the client has the version 570.172.08, but
NVRM: this kernel module has the version 570.158.01. Please
NVRM: make sure that this kernel module and all NVIDIA driver
NVRM: components have the same version.
```

No devices were found error

`nvidia-smi`⁵⁴²

```
No devices were found
```

`nouveau`⁵⁴³

⁵⁴⁰ <https://docs.nvidia.com/datacenter/tesla/drivers/index.html>

⁵⁴¹ <https://developer.nvidia.com/system-management-interface>

⁵⁴² <https://developer.nvidia.com/system-management-interface>

⁵⁴³ <https://nouveau.freedesktop.org/>

```
$ lsmod | grep nouveau
```

```
$ echo "blacklist nouveau" | sudo tee /etc/modprobe.d/disable-nouveau.conf >
echo "options nouveau modeset=0" | sudo tee -a
/etc/modprobe.d/disable-nouveau.conf > sudo rmmod nouveau || true > sudo
update-initramfs -u
```

Virtual GPU

GPU virtualisation with QEMU/KVM

Graphics

frontend

-

-

libvirt section

ease-of-use and capability⁵⁴⁴

-

GL

virgil3d⁵⁴⁵
Kernels >= 4.4⁵⁴⁶

⁵⁴⁴ <https://cpaelzer.github.io/blogs/006-mediated-device-to-pass-parts-of-your-gpu-to-a-guest/>

⁵⁴⁵ <https://virgil3d.github.io/>

⁵⁴⁶ <https://www.kraxel.org/blog/2016/09/using-virtio-gpu-with-libvirt-and-spice/>

set up for device passthrough⁵⁴⁷

-

GPUs

IOMMU

```
-device vfio-pci,host=05:00.0,bus=1,addr=00.0,multifunction=on,x-vga=on -device vfio-pci,host=05:00.1,bus=1,addr=00.1
```

-

```
-display gtk,gl=on -device vfio-pci,syfsdev=/sys/bus/pci/devices/0000:00:02.0/4dd511f6-ec08-11e8-b839-2f163ddee3b3,display=on,rombar=0
```

about vGPU at ⁵⁴⁸

Ubuntu GPU mdev evaluation⁵⁴⁹

Intel⁵⁵⁰

Nvidia⁵⁵¹

AMD

libvirt section
BIOS

Preparing the input-output memory management unit (IOMMU)

ture⁵⁵³

options you can enable/configure⁵⁵²

IOMMU fea-

⁵⁴⁷ <https://www.kernel.org/doc/Documentation/vfio-mediated-device.txt>

⁵⁴⁸ <https://www.kraxel.org/blog/2018/04/vgpu-display-support-finally-merged-upstream/>

⁵⁴⁹ <https://cpaelzer.github.io/blogs/006-mediated-device-to-pass-parts-of-your-gpu-to-a-guest/>

⁵⁵⁰ https://github.com/intel/gvt-linux/wiki/GVTg_Setup_Guide

⁵⁵¹ <https://docs.nvidia.com/vgpu/latest/grid-vgpu-user-guide/index.html>

⁵⁵² <https://www.kernel.org/doc/html/latest/admin-guide/kernel-parameters.html?highlight=iommu>

⁵⁵³ <https://www.kernel.org/doc/html/latest/arch/x86/iommu.html>

Note:

The card used in all examples below e.g. when filtering for or assigning PCI IDs, is an NVIDIA V100 on PCI ID 41.00.0

```
$ lspci | grep 3D
```

```
41:00.0 3D controller: NVIDIA Corporation GV100GL [Tesla V100 PCIe 16GB] (rev a1)
```

DMAR

```
$ dmesg | grep -i -e DMAR -e IOMMU
[ 3.509232] iommu: Default domain type: Translated
...
[ 4.516995] pci 0000:00:01.0: Adding to iommu group 0
...
[ 4.702729] perf/amd_iommu: Detected AMD IOMMU #0 (2 banks, 4 counters/bank).
```

```
$ dmesg | grep -i -e DMAR -e IOMMU | grep $(lspci | awk '/ 3D / {print $1}' )
[ 4.598150] pci 0000:41:00.0: Adding to iommu group 66
```

```
$ find /sys/kernel/iommu_groups/ -name "*$(lspci | awk '/ 3D / {print $1})*"
/sys/kernel/iommu_groups/66/devices/0000:41:00.0
```

```
$ ll /sys/kernel/iommu_groups/66/devices/
lrwxrwxrwx 1 root root 0 Jan  3 06:57 0000:41:00.0 -> ../../../../devices/
pci0000:40/0000:40:03.1/0000:41:00.0/
```

Note:

Some older or non-server boards tend to group devices in one IOMMU group, which isn't very useful as it means you'll need to pass "all or none of them" to the same guest.

Preparations for PCI and mediated devices pass-through – block host drivers

```
$ echo "blacklist nouveau" | sudo tee /etc/modprobe.d/blacklist-nouveau.conf
> echo "options nouveau modeset=0" | sudo tee -a
/etc/modprobe.d/blacklist-nouveau.conf > sudo update-initramfs -u > sudo
reboot ```
```

You can check which kernel modules are loaded and available via `lspci -v`:`

```
```{terminal}
:copy:
:user:
:host:
:dir:
lspci -v | grep -A 10 " 3D "

41:00.0 3D controller: NVIDIA Corporation GV100GL [Tesla V100 PCIe 16GB] (rev a1)
...
Kernel modules: nvidiafb, nouveau
```

Kernel driver in use: nouveau

**Preparations for mediated devices pass-through - driver**

supported and available from Nvidia<sup>554</sup>

ration of vGPU on Ubuntu<sup>555</sup>

installation and configu-

<sup>554</sup> <https://docs.nvidia.com/vgpu/latest/product-support-matrix/ubuntu.html>

<sup>555</sup> <https://docs.nvidia.com/vgpu/latest/grid-vgpu-user-guide/index.html#ubuntu-install-configure-vgpu>

```
$ lsmod | grep nvidia
```

```
nvidia_vgpu_vfio 53248 38
nvidia 35282944 586 nvidia_vgpu_vfio
mdev 24576 2 vfio_mdev,nvidia_vgpu_vfio
drm 491520 6 drm_kms_helper,drm_vram_helper,nvidia
```

### Note:

While it works without a vGPU manager, to get the full capabilities you'll need to configure the **vGPU manager (that came with above package)**<sup>556</sup> and a license server so that each guest can get a license for the vGPU provided to it. Please see **Nvidia's documentation for the license server**<sup>557</sup>. While not officially supported on Linux (as of Q1 2022), it's worthwhile to note that it runs fine on Ubuntu with `sudo apt install unzip default-jre tomcat9 liblog4j2-java libslf4j-java` using `/var/lib/tomcat9` as the server path in the license server installer.

It's also worth mentioning that the Nvidia license server went **EOL on 31 July 2023**<sup>558</sup>. At that time, it was replaced by the **NVIDIA License System**<sup>559</sup>.

<sup>556</sup> <https://docs.nvidia.com/vgpu/latest/grid-vgpu-user-guide/index.html#install-vgpu-package-ubuntu>

<sup>557</sup> <https://docs.nvidia.com/vgpu/latest/grid-license-server-user-guide/index.html>

<sup>558</sup> <https://docs.nvidia.com/vgpu/news/vgpu-software-license-server-eol-notice/index.html>

<sup>559</sup> <https://docs.nvidia.com/license-system/latest/nvidia-license-system-quick-start-guide/index.html>

```
general status
$ systemctl status nvidia-vgpu-mgr
 Loaded: loaded (/lib/systemd/system/nvidia-vgpu-mgr.service; enabled; vendor preset:
 enabled)
 Active: active (running) since Tue 2021-09-14 07:30:19 UTC; 3min 58s ago
 Process: 1559 ExecStart=/usr/bin/nvidia-vgpu-mgr (code=exited, status=0/SUCCESS)
 Main PID: 1564 (nvidia-vgpu-mgr)
 Tasks: 1 (limit: 309020)
 Memory: 1.1M
 CGroup: /system.slice/nvidia-vgpu-mgr.service
 └─1564 /usr/bin/nvidia-vgpu-mgr

Sep 14 07:30:19 node-watt systemd[1]: Starting NVIDIA vGPU Manager Daemon...
Sep 14 07:30:19 node-watt systemd[1]: Started NVIDIA vGPU Manager Daemon.
Sep 14 07:30:20 node-watt nvidia-vgpu-mgr[1564]: notice: vmiop_env_log: nvidia-vgpu-mgr
daemon started

Entries when a guest gets a vGPU passed
Sep 14 08:29:50 node-watt nvidia-vgpu-mgr[2866]: notice: vmiop_log: (0x0): gpu-pci-id :
0x4100
Sep 14 08:29:50 node-watt nvidia-vgpu-mgr[2866]: notice: vmiop_log: (0x0): vgpu_type :
Quadro
Sep 14 08:29:50 node-watt nvidia-vgpu-mgr[2866]: notice: vmiop_log: (0x0): Framebuffer:
0x1dc000000
```

(continues on next page)

(continued from previous page)

```
Sep 14 08:29:50 node-watt nvidia-vgpu-mgr[2866]: notice: vmiop_log: (0x0): Virtual Device
Id: 0x1db4:0x1252
Sep 14 08:29:50 node-watt nvidia-vgpu-mgr[2866]: notice: vmiop_log: (0x0): FRL Value: 60
FPS
Sep 14 08:29:50 node-watt nvidia-vgpu-mgr[2866]: notice: vmiop_log: ##### vGPU Manager
Information: #####
Sep 14 08:29:50 node-watt nvidia-vgpu-mgr[2866]: notice: vmiop_log: Driver Version: 470.68
Sep 14 08:29:50 node-watt nvidia-vgpu-mgr[2866]: notice: vmiop_log: (0x0): vGPU supported
range: (0x70001, 0xb0001)
Sep 14 08:29:50 node-watt nvidia-vgpu-mgr[2866]: notice: vmiop_log: (0x0): Init frame copy
engine: syncing...
Sep 14 08:29:50 node-watt nvidia-vgpu-mgr[2866]: notice: vmiop_log: (0x0): vGPU migration
enabled
Sep 14 08:29:50 node-watt nvidia-vgpu-mgr[2866]: notice: vmiop_log: display_init inst: 0
successful

Entries when a guest grabs a license
Sep 15 06:55:50 node-watt nvidia-vgpu-mgr[4260]: notice: vmiop_log: (0x0): vGPU license
state: Unlicensed (Unrestricted)
Sep 15 06:55:52 node-watt nvidia-vgpu-mgr[4260]: notice: vmiop_log: (0x0): vGPU license
state: Licensed

In the guest the card is then fully recognized and enabled
$ nvidia-smi -a | grep -A 2 "Licensed Product"
 vGPU Software Licensed Product
 Product Name : NVIDIA RTX Virtual Workstation
 License Status : Licensed
```

mediated device<sup>560</sup>

*firmware*

NVIDIA documentation<sup>561</sup>

```
$ sudo mdevctl types
```

```
...
nvidia-300
 Available instances: 0
 Device API: vfio-pci
 Name: GRID V100-8C
 Description: num_heads=1, frl_config=60, framebuffer=8192M, max_resolution=4096x2160,
max_instance=2
```

<sup>560</sup> <https://www.kernel.org/doc/html/latest/driver-api/vfio-mediated-device.html>

<sup>561</sup> <https://docs.nvidia.com/vgpu/latest/grid-vgpu-user-guide/index.html#ubuntu-install-configure-vgpu>



```
$ $ sudo mdevctl define --parent 0000:41:00.0 --type nvidia-300
bc127e23-aaaa-4d06-a7aa-88db2dd538e0
```

```
$ $ sudo mdevctl define --parent 0000:41:00.0 --type nvidia-300
1360ce4b-2ed2-4f63-abb6-8cdb92100085
```

```
$ $ sudo mdevctl start --parent 0000:41:00.0 --uuid
bc127e23-aaaa-4d06-a7aa-88db2dd538e0
```

```
$ $ sudo mdevctl start --parent 0000:41:00.0 --uuid
1360ce4b-2ed2-4f63-abb6-8cdb92100085
```

```
$ sudo mdevctl list -d
bc127e23-aaaa-4d06-a7aa-88db2dd538e0 0000:41:00.0 nvidia-108 manual (active)
1360ce4b-2ed2-4f63-abb6-8cdb92100085 0000:41:00.0 nvidia-108 manual (active)
```

## Passing through PCI or mediated devices

```
<hostdev mode='subsystem' type='pci' managed='yes'>
 <source>
 <address domain='0x0000' bus='0x41' slot='0x00' function='0x0' />
 </source>
</hostdev>
```

```
<hostdev mode='subsystem' type='mdev' managed='no' model='vfio-pci' display='on'>
 <source>
 <address uuid='634fc146-50a3-4960-ac30-f09e5cedc674' />
 </source>
</hostdev>
```

part of the guest definition<sup>562</sup>

<sup>562</sup> <https://libvirt.org/formatdomain.html#usb-pci-scsi-devices>

## Note:

This works great on Focal, but `type='none'` as well as `display='off'` weren't available on Bionic. If this level of control is required one would need to consider using the [Ubuntu Cloud Archive](https://wiki.ubuntu.com/OpenStack/CloudArchive)<sup>563</sup> or [Server-Backports](https://launchpad.net/~canonical-server/+archive/ubuntu/server-backports)<sup>564</sup> for a newer stack of the virtualisation components.

<sup>563</sup> <https://wiki.ubuntu.com/OpenStack/CloudArchive>

<sup>564</sup> <https://launchpad.net/~canonical-server/+archive/ubuntu/server-backports>

s390x vfio-ccw<sup>565</sup>

- *Virtual GPU (vGPU) with QEMU/KVM*

*Virtualisation*

## Virtualisation

*Introduction to virtualization*

## Virtual machines

## LXD containers and virtual machines

LXD<sup>566</sup>

interface<sup>567</sup>

REST APIs<sup>568</sup>

built-in graphical user

to a cluster<sup>569</sup>

such as a MicroCloud<sup>570</sup>

<sup>565</sup> <https://docs.kernel.org/arch/s390/vfio-ccw.html>

<sup>566</sup> <https://canonical.com/lxd>

<sup>567</sup> [https://canonical.com/lxd/docs/latest/howto/access\\_ui/](https://canonical.com/lxd/docs/latest/howto/access_ui/)

<sup>568</sup> [https://canonical.com/lxd/docs/latest/restapi\\_landing/](https://canonical.com/lxd/docs/latest/restapi_landing/)

<sup>569</sup> <https://canonical.com/lxd/docs/latest/explanation/clusters/>

<sup>570</sup> <https://canonical.com/microcloud>

## Online resources

[the official LXD documentation](#)<sup>571</sup>  
[Ubuntu Discourse forum](#)<sup>572</sup>

[YouTube channel](#)<sup>573</sup>

## Installation

```
$ sudo snap install lxd
```

## Configuration

```
$ lxd init
```

- [Btrfs](#)<sup>574</sup> [ZFS](#)<sup>575</sup>
- 
- 

```
$ gpasswd -a joe lxd
```

[How to initialize LXD](#)<sup>576</sup>

---

<sup>571</sup> <https://canonical.com/lxd/docs/default/>

<sup>572</sup> <https://discourse.ubuntu.com/c/lxd/126>

<sup>573</sup> <https://www.youtube.com/c/LXDvideos>

<sup>574</sup> <https://btrfs.readthedocs.io/>

<sup>575</sup> [https://openzfs.org/wiki/Main\\_Page](https://openzfs.org/wiki/Main_Page)

<sup>576</sup> <https://canonical.com/lxd/docs/latest/howto/initialize/>

## Creating your first container

### Creating a container

- 
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```
$ lxc launch remote:image containername
```

```
$ lxc launch --vm remote:image vmname
```

```
$ lxc init remote:image containername
```

```
$ lxc image list ubuntu:
```

```
$ lxc image info ubuntu:noble
```

```
$ lxc image info ubuntu:lts/arm64
```

```
$ lxc launch ubuntu:noble n1
```

```
$ lxc list
```

```
$ lxc info n1
```

```
$ lxc exec n1 -- sudo -i -u ubuntu
```

```
$ lxc shell n1
```

## LXD server configuration

```
$ lxc list
```

```
$ lxc config set core.https_address :8443
```

## Authentication

```
$ lxc remote add r1 r1.example.com:8443
```

```
$ lxc config trust add r1 certfile.crt
```

## Backing store

*filesystem*

Storage configuration<sup>577</sup>

## Container configuration

```
$ lxc config device add c1 opt-mount disk source=/opt path=/opt
```

```
$ lxc help config
```

```
$ lxc config edit c1
```

---

<sup>577</sup> <https://canonical.com/lxd/docs/latest/explanation/storage/>

## Profiles

```
$ lxc config device add c1 eth1 none
```

## Nesting

```
$ lxc config set container1 security.nesting true
```

## Limits

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the configuration documentation<sup>578</sup>

---

<sup>578</sup> [https://canonical.com/lxd/docs/latest/reference/instance\\_options/](https://canonical.com/lxd/docs/latest/reference/instance_options/)

## UID mappings and privileged containers

*GIDs*  
*subuid(5)*<sup>579</sup>

```
$ lxc config set c1 security.privileged true
```

## AppArmor

## Seccomp

## Raw LXC configuration

*lxc.*

*container.conf(5)*<sup>580</sup>

---

<sup>579</sup> <https://manpages.ubuntu.com/manpages/resolute/man5/subuid.5.html>

<sup>580</sup> <https://manpages.ubuntu.com/manpages/resolute/man5/lxc.container.conf.5.html>



## Snapshots

```
$ lxc move c1 final-beta
```

```
$ lxc snapshot c1 snapshot-name
```

```
$ lxc restore c1 snapshot-name
```

```
$ lxc copy c1/snapshot-name new-container
```

## Publishing images

```
$ lxc publish c1/snapshot-name --alias foo-2.0
```

```
$ lxc publish c1/snapshot-name --alias foo-2.0 public=true
```

```
$ lxc image edit foo-2.0
```

## Image export and import

```
$ lxc image export foo-2.0 foo-2.0.tar.gz
```

```
$ lxc image import foo-2.0.tar.gz --alias foo-2.0 --public
```

## Troubleshooting

```
$ journalctl -u lxd
```

```
$ lxc info c1 --show-log
```

## How to create a VM with Multipass

Multipass<sup>581</sup>

```
$ sudo snap install multipass
```

```
$ sudo apt update && sudo apt install snapd
```

## Find available images

---

<sup>581</sup> <https://canonical.com/multipass>

Image	Aliases	Version	Description
core	core16	20200818	Ubuntu Core 16
core18		20211124	Ubuntu Core 18
core20		20230119	Ubuntu Core 20
core22		20230717	Ubuntu Core 22
22.04	jammy	20260515	Ubuntu 22.04 LTS
24.04	noble	20260518	Ubuntu 24.04 LTS
25.10	questing	20260520	Ubuntu 25.10
26.04	resolute,lts	20260520	Ubuntu 26.04 LTS
charm-dev		latest	A development and
testing environment for charmers			
docker		latest	A Docker environment
with Portainer and related tools			
minikube		latest	minikube is local
Kubernetes			

## Launch a fresh instance of Ubuntu 26.04 LTS

```
$ multipass launch 26.04
Launched: cleansing-guanaco
```

## Check out the running instances

```
$ multipass list
```

Name	State	IPv4	Image
cleansing-guanaco	Running	10.140.26.17	Ubuntu 26.04 LTS

## Learn more about the VM instance you just launched

```
$ multipass info cleansing-guanaco
```

```
Name: cleansing-guanaco
State: Running
Snapshots: 0
IPv4: 10.140.26.17
Release: Ubuntu 26.04 LTS
Image hash: dc5b5a43c267 (Ubuntu 26.04 LTS)
CPU(s): 1
Load: 0.74 0.35 0.13
Disk usage: 2.2GiB out of 4.8GiB
Memory usage: 347.4MiB out of 950.8MiB
Mounts: --
```

### Connect to a running instance

```
$ multipass shell cleansing-guanaco

Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-15-generic x86_64)
(...)

ubuntu@cleansing-guanaco:~$
```

### Disconnect from the instance

### Run commands inside an instance from outside

```
$ multipass exec cleansing-guanaco -- lsb_release -a

No LSB modules are available.
Distributor ID: Ubuntu
Description: Ubuntu 26.04 LTS
Release: 26.04
Codename: resolute
```

## Stop or start an instance

```
$ multipass stop cleansing-guanaco
```

```
$ multipass start cleansing-guanaco
```

## Delete the instance

```
$ multipass delete cleansing-guanaco
```

```
$ multipass list
```

Name	State	IPv4	Image
cleansing-guanaco	Deleted	--	Not Available

```
$ multipass purge
```

```
$ multipass list
```

```
No instances found.
```

## Integrate with the rest of your virtualisation

*ESXi*  
*uvtool*

*Libvirt*

*QEMU KVM*  
*virt-manager*<sup>582</sup>

*guide*<sup>583</sup>

---

<sup>582</sup> <https://virt-manager.org/>

<sup>583</sup> <https://documentation.ubuntu.com/multipass/latest/how-to-guides/customise-multipass/set-up-a-graphical-interface/>

```
$ sudo multipass set local.driver=libvirt
```

#### Note:

If you are having issues interacting with Multipass after switching to the libvirt driver, check if there is a restriction by [AppArmor](#), for example. AppArmor may have a default policy which restricts the Multipass service from interacting with the [libvirt service](#)<sup>584</sup>. So you need to add an explicit permission that allows it.

<sup>584</sup> <https://libvirt.org/daemons.html>

[virt-manager](#)<sup>585</sup>

```
$ multipass launch lts
```

```
Launched: engaged-amberjack
```

```
$ virsh list
```

Id	Name	State
-----	-----	-----
15	engaged-amberjack	running

Mul-

[tipass drivers documentation](#)<sup>586</sup>

## Get help

```
$ multipass help
```

```
$ multipass help <command>
```

```
$ multipass help --all
```

[Multipass documentation](#)<sup>587</sup>

<sup>585</sup> <https://virt-manager.org/>

<sup>586</sup> <https://documentation.ubuntu.com/multipass/latest/how-to-guides/customise-multipass/set-up-the-driver/>

<sup>587</sup> <https://documentation.ubuntu.com/multipass/latest/>

## Create cloud image VMs with UVtool

### Creating virtual machines using uvtool

#### Install uvtool packages

- 
- 

```
$ sudo apt -y install uvtool uvtool-libvirt
```

#### Note:

You might need to log out and log in again to use libvirt without root privilege. If not, you might encounter a Permission denied error in the later steps.

#### Get the Ubuntu cloud image with uvt-simplestreams-libvirt

```
$ uvt-simplestreams-libvirt --verbose sync arch=amd64
```

```
$ uvt-simplestreams-libvirt query
```

```
release=bionic arch=amd64 label=daily (20191107)
release=focal arch=amd64 label=daily (20191029)
...
```

```
$ uvt-simplestreams-libvirt sync release=DISTRO-SHORT-CODENAME arch=amd64
```

```
$ uvt-simplestreams-libvirt sync --source
http://cloud-images.ubuntu.com/daily [... further options]
```

### Create a valid SSH key

```
$ ssh-keygen

Generating public/private rsa key pair.
Enter file in which to save the key (/home/ubuntu/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/ubuntu/.ssh/id_rsa.
Your public key has been saved in /home/ubuntu/.ssh/id_rsa.pub.
The key fingerprint is:
4d:ba:5d:57:c9:49:ef:b5:ab:71:14:56:6e:2b:ad:9b ubuntu@DISTRO-SHORT-CODENAMES
The key's randomart image is:
+--[RSA 2048]-----+
| ..|
| o.=|
| **|
| + o+=|
| S=.|
| o . .+ .|
| . . o o |
| * |
| E |
+-----+

```



## Create the VM using uvt-kvm

```
$ uvt-kvm create firsttest
```

```
$ uvt-kvm create secondtest release=DISTRO-SHORT-CODENAME
```

```
$ uvt-kvm wait secondtest
```

## Connect to the running VM

```
$ uvt-kvm ssh secondtest
```

```
$ uvt-kvm ip secondtest
192.168.122.199
```

```
$ ssh -i ~/.ssh/id_rsa ubuntu@192.168.122.199
[...]
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@secondtest:~$
```

## Get the list of running VMs

## Destroy your VM

```
$ uvt-kvm destroy secondtest
```

### Note:

Unlike libvirt's `destroy` or `undefine` actions, this will (by default) also remove the associated virtual storage files.

## More `uv`-`kvm` options

- 
- 
- 
- 
- 
- 

*uv*-*kvm*(1)<sup>588</sup>

## QEMU

### Note:

Please bear in mind that invoking QEMU manually may sometimes require your user to be part of the `kvm` group.

<sup>588</sup> <https://manpages.ubuntu.com/manpages/resolute/man1/uv-kvm.1.html>

## Virtualisation with QEMU

QEMU<sup>589</sup>

KVM<sup>590</sup>

command line interface<sup>591</sup>

monitor<sup>592</sup>

*libvirt*

## Install QEMU/KVM

```
$ kvm-ok
```

### Note:

The command `kvm-ok` is part of the `cpu-checker` package. Install it with `sudo apt install cpu-checker` if it is not already present.

```
$ sudo apt-get install qemu-system
```

## Boot a VM

```
$ qemu-system-x86_64 -m 4G -enable-kvm -cdrom
http://releases.ubuntu.com/noble/ubuntu-24.04.3-live-server-amd64.iso
```

### Caution:

This example is just for illustration purposes - it is not generally recommended without verifying the checksums. [Multipass](#) (page 641) and [UVTool](#) (page 646) are much better ways to get actual guests easily.

<sup>589</sup> [https://wiki.qemu.org/Main\\_Page](https://wiki.qemu.org/Main_Page)

<sup>590</sup> [https://www.linux-kvm.org/page/Main\\_Page](https://www.linux-kvm.org/page/Main_Page)

<sup>591</sup> <https://qemu-project.gitlab.io/qemu/system/invoke.html>

<sup>592</sup> <https://qemu-project.gitlab.io/qemu/system/monitor.html>

## VNC

### Create a virtual disk

```
$ qemu-img create -f qcow2 disk.qcow 5G
```

[virt-manager](#)<sup>593</sup>

[libvirt](#)<sup>594</sup>

```
$ qemu-system-x86_64 options image[s]
```

[QEMU manpage](#)<sup>595</sup> [qemu-img\(1\)](#)<sup>596</sup>

[QEMU documentation](#)<sup>597</sup>

### Create QEMU VMs with up to 1024 vCPUs

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<sup>593</sup> <https://virt-manager.org/>

<sup>594</sup> <https://libvirt.org/>

<sup>595</sup> <https://manpages.ubuntu.com/manpages/resolute/man1/qemu-system.1.html>

<sup>596</sup> <https://manpages.ubuntu.com/manpages/resolute/man1/qemu-img.1.html>

<sup>597</sup> <https://www.qemu.org/documentation/>

## Requirement

QEMU version	Ubuntu release	Supported machine types
QEMU 8.2.1+	Ubuntu 24.04 LTS (Noble)	ubuntu (Native support, also supports Jammy and Mantic machine types)
QEMU 8.0.4	Ubuntu 23.10 (Mantic)	pc-q35-mantic-maxcpus, pc-i440fx-mantic-maxcpus (Also supports Jammy machine types)
QEMU 6.2	Ubuntu 22.04 LTS (Jammy)	pc-q35-jammy-maxcpus, pc-i440fx-jammy-maxcpus

## Configuration by Ubuntu release

### 24.04 Noble

```
$ qemu-system-x86_64 -M ubuntu,accel=kvm,kernel-irqchip=split -device intel-iommu,intremap=on -smp cpus=300,maxcpus=300 ...
```

### 23.10 Mantic

```
$ qemu-system-x86_64 -M pc-q35-mantic-maxcpus,accel=kvm,kernel-irqchip=split -device intel-iommu,intremap=on -smp cpus=300,maxcpus=300 ...
```

## 22.04 Jammy

```
$ qemu-system-x86_64 -M pc-q35-jammy-maxcpus,accel=kvm,kernel-irqchip=split
-device intel-iommu,intremap=on -smp cpus=300,maxcpus=300 ...
```

## Boot ARM64 virtual machines on QEMU

### Note:

This requires Ubuntu 20.04 LTS or later.

## Install QEMU to run ARM64 virtual machines

```
$ sudo apt install qemu-system-arm
```

## Create necessary support files

*EFI*

```
$ truncate -s 64m varstore.img
```

```
$ truncate -s 64m efi.img
```

```
$ dd if=/usr/share/qemu-efi-aarch64/QEMU_EFI.fd of=efi.img conv=notrunc
```

## Fetch the Ubuntu cloud image

Ubuntu cloud image<sup>598</sup>

## Run QEMU natively on an ARM64 host

```
$ qemu-system-aarch64 \ > -enable-kvm \ > -m 1024 \ > -cpu host \ > -M virt \ > -nographic \ > -drive if=pflash,format=raw,file=efi.img,readonly=on \ > -drive if=pflash,format=raw,file=varstore.img \ > -drive if=none,file=jammy-server-cloudimg-arm64.img,id=hd0 \ > -device virtio-blk-device,drive=hd0 -netdev type=tap,id=net0 \ > -device virtio-net-device,netdev=net0
```

## Run an emulated ARM64 VM on x86

```
$ qemu-system-aarch64 \ > -m 2048 \ > -cpu max \ > -M virt \ > -nographic \ > -drive if=pflash,format=raw,file=efi.img,readonly=on \ > -drive if=pflash,format=raw,file=varstore.img \ > -drive if=none,file=jammy-server-cloudimg-arm64.img,id=hd0 \ > -device virtio-blk-device,drive=hd0 \ > -netdev type=tap,id=net0 \ > -device virtio-net-device,netdev=net0
```

<sup>598</sup> <https://cloud-images.ubuntu.com/>

## Troubleshooting

### No output and no response

## Further reading

- *Virtualizing graphics using QEMU/KVM*
- *Using QEMU to create a microvm*

## Confidential Computing with AMD

- *AMD-SME*
- *AMD-SEV*
- *AMD-SEV-ES*
- *AMD-SEV-SNP*

*AMD-SEV-SNP*



## Host configuration

AMD's documentation<sup>599</sup>

BMC

```
$ cat /sys/module/kvm_amd/parameters/sev
```

```
Y
```

```
$ cpuid -1 -l 0x8000001f
```

CPU:

AMD Secure Encryption (0x8000001f):

```
SME: secure memory encryption support = true
SEV: secure encrypted virtualize support = true
VM page flush MSR support = false
SEV-ES: SEV encrypted state support = true
SEV-SNP: SEV secure nested paging = true
```

```
$ qemu-system-x86_64 \ > -enable-kvm \ > -nographic \ > -machine q35 -smp 6
-m 6G \ > -drive "if=virtio,format=qcow2,file=disk.img" \ > -net
nic,model=e1000 -net user,hostfwd=tcp::2222-:22 \ > -cpu EPYC-v4 \ > -machine
memory-encryption=sev0,vmport=off \ > -object
memory-backend-memfd,id=ram1,size=6G,share=true,prealloc=false \ > -machine
memory-backend=ram1 \ > -object sev-snp-guest,id=sev0,cbitpos=47,reduced-
phys-bits=1,kernel-hashes=on,policy=0x30000 \ > -kernel ./vmlinuz \ > -append
"root=/dev/vda1 console=ttyS0" \ > -bios /usr/share/ovmf/OVMF.amdsev.fd
```

- 

EDK II<sup>600</sup>

```
-object sev-snp-guest,id=sev0,cbitpos=47,reduced-phys-bits=1,kernel-hashes=on \
```

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<sup>599</sup> <https://docs.amd.com/v/u/en-US/58207-using-sev-with-amd-epyc-processors>

<sup>600</sup> [https://en.wikipedia.org/wiki/TianoCore\\_EDK\\_II](https://en.wikipedia.org/wiki/TianoCore_EDK_II)

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QEMU documentation<sup>601</sup>

tion<sup>602</sup> AMD SEV<sup>603</sup>

invoca-

## Guest configuration

- <https://cloud-images.ubuntu.com/releases/noble/release/ubuntu-24.04-server-cloudimg-amd64.img>
- <https://cloud-images.ubuntu.com/releases/noble/release/unpacked/>

snpguest<sup>604</sup>

```
$ sudo ./snpguest report --random attestation-report.bin request-file.txt
```

## Confidential Computing with Intel Trust Domain Extensions (TDX)

Intel TDX<sup>605</sup>

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<sup>601</sup> <https://www.qemu.org/docs/master/system/i386/amd-memory-encryption.html>

<sup>602</sup> <https://www.qemu.org/docs/master/system/invocation.html>

<sup>603</sup> <https://www.qemu.org/docs/master/system/i386/amd-memory-encryption.html#sevapi>

<sup>604</sup> <https://github.com/virtee/snpguest>

<sup>605</sup> <https://www.intel.com/content/www/us/en/developer/tools/trust-domain-extensions/overview.html>

Supported hardware

Processor	Code Name	TDX Version	Module
4th Gen Intel Xeon Scalable Processors (select SKUs with Intel TDX)	Sapphire Rapids	1.5.x	
5th Gen Intel Xeon Scalable Processors	Emerald Rapids	1.5.x	
Intel Xeon 6 Processors with E-Cores	Sierra Forest	1.5.x	
Intel Xeon 6 Processors with P-Cores	Granite Rapids	2.0.x	

Configure host

BIOS

Note:

The following is a sample BIOS configuration. The necessary BIOS settings or menus may differ based on the platform used. Reach out to your OEM/ODM or independent BIOS vendor for platform-specific instructions.

- 
- 
- 
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- 
- 
- 
-

```
$ cat /proc/cmdline
```

```
BOOT_IMAGE=/boot/vmlinuz-6.19.0-3-generic root=UUID=f6ce4201-d6d1-4505-bdf5-019ee4fe842e ro nohibernate kvm_intel.tdx=1
```

```
$ sudo dmesg | grep -i tdx
```

```
[1.824045] virt/tdx: BIOS enabled: private KeyID range [32, 64)
[1.824049] virt/tdx: Disable ACPI S3. Turn off Intel TDX in the BIOS to use ACPI S3.
[24.195089] virt/tdx: 1050644 KB allocated for PAMT
[24.195098] virt/tdx: module initialized
```

## Install virtualization stack

```
$ sudo apt update
```

```
$ sudo apt install \ > qemu-system-x86 \ > ovmf-inteltdx \ >
libvirt-daemon-system \ > libvirt-clients
```

- 
- 
- 
- 

```
$ ls -l /usr/share/ovmf/OVMF.inteltdx.ms.fd
```

```
-rw-r--r-- 1 root root 4194304 Jan 15 12:00 /usr/share/ovmf/OVMF.inteltdx.ms.fd
```

## Create a guest image

```
$ wget https://cloud-images.ubuntu.com/resolute/current/resolute-server-cloudimg-amd64.img
```

```
$ sudo apt install cloud-image-utils
```

```
$ cat > user-data.yaml << 'EOF' > #cloud-config > chpasswd: > list: | >
root:root > expire: False > ssh_pwauth: True > disable_root: false > EOF
```

```
$ cloud-localds user-data.img user-data.yaml
```

### Note:

For production use, set a strong password and configure SSH key-based authentication instead of password authentication.

## Launch a TD guest using QEMU

```
$ qemu-system-x86_64 \ > -accel kvm \ > -smp 32 \ > -m 16G \ > -cpu host \ >
-object
'{"qom-type":"tdx-guest","id":"tdx","quote-generation-socket":{"type":
"vsock", "cid":"2","port":"4050"}}' \ > -object
memory-backend-ram,id=mem0,size=16G \ > -machine
q35,kernel_irqchip=split,confidential-guest-support=tdx,memory-backend=mem0 \
> -bios /usr/share/ovmf/OVMF.inteltdx.ms.fd \ > -nographic \ > -nodefaults \
> -vga none \ > -drive
file=resolute-server-cloudimg-amd64.img,if=none,id=virtio-disk0 \ > -device
virtio-blk-pci,drive=virtio-disk0 \ > -drive
file=user-data.img,if=none,id=cloud-init,format=raw \ > -device
virtio-blk-pci,drive=cloud-init \ > -serial stdio
```

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QEMU documentation<sup>606</sup>

## Verify Intel TDX enablement in the guest

```
$ sudo dmesg | grep tdx
[0.000000] tdx: Guest detected
[11.162378] systemd[1]: Detected confidential virtualization tdx.
```

```
$ ls -l /dev/tdx_guest
crw----- 1 root root 10, 125 Jan 1 00:00 /dev/tdx_guest
```

## Launch a TDX VM with libvirt

```
$ cat > tdx-vm.xml << 'EOF' > <domain type='kvm'
xmlns:qemu='http://libvirt.org/schemas/domain/qemu/1.0'> >
<name>tdx-guest</name> > <memory unit='GiB'>16</memory> > <memoryBacking> >
<source type="anonymous"/> > <access mode="private"/> > </memoryBacking> >
<vcpu placement="static">16</vcpu> > <os> <type arch='x86_64'
machine='q35'>hvm</type> > <loader type='rom'
readonly='yes'>/usr/share/ovmf/OVMF.inteltdx.ms.fd</loader> > <boot
dev='hd'> > </os> > <features> > <acpi/> > <apic/> > <ioapic driver='qemu'> >
```

<sup>606</sup> <https://www.qemu.org/docs/master/system/i386/tdx.html>

```
> </features> > <clock offset='utc'> > <timer name='hpet' present='no' /> >
</clock> > <cpu mode='host-passthrough'> > <topology sockets='1' cores='16'
threads='1' /> > </cpu> > <devices> >
<emulator>/usr/bin/qemu-system-x86_64</emulator> > <disk type="file"
device="disk"> > <driver name="qemu" type="qcow2"/> > <source
file="/var/lib/libvirt/images/resolute-server-cloudimg-amd64.img"/> > <target
dev="vda" bus="virtio"/> > </disk> > <console type='pty'> > <target
type='virtio' port='1' /> > </console> > </devices> > <launchSecurity
type='tdx'> > <policy>0x10000000</policy> > <quoteGenerationService> >
<SocketAddress type='vsock' cid='2' port='4050' /> > </quoteGenerationService>
> </launchSecurity> > </domain> > EOF
```

```
$ sudo virsh define tdx-vm.xml
```

```
$ sudo virsh start tdx-guest
```

```
$ sudo virsh console tdx-guest
```

- 
- 
- 
- 
- 

## Manage the VM

```
$ sudo virsh list --all
```

```
$ sudo virsh shutdown tdx-guest
```

```
$ sudo virsh destroy tdx-guest
```

```
$ sudo virsh undefine tdx-guest
```

## Troubleshooting Intel TDX

- 

```
$ sudo apt install msr-tools
```

- 

```
$ sudo rdmsr 0x982 -f 1:1 1
```

- 

```
$ sudo rdmsr 0x1401 -f 11:11 1
```

- 

```
$ echo $((0x$(sudo rdmsr 0x87 -f 63:32)))
```

- 

```
$ echo $((0x$(sudo rdmsr 0x87 -f 63:32)))
```

- 

```
$ sudo rdmsr 0xa0
```



updating to the latest Intel TDX

Module<sup>607</sup>

#### Note:

If you chose to “Update Intel TDX Module via Binary Deployment”, make sure you’re using the correct Intel TDX Module version for your hardware. See the [Supported hardware](#) (page 658) table.

## Virtualization Hardware Enablement Stack

Ubuntu Resolute 26.04<sup>608</sup>

- **qemu-hwe**<sup>609</sup>
- **libvirt-hwe**<sup>610</sup>
- **edk2-hwe**<sup>611</sup>
- **seabios-hwe**<sup>612</sup> *BIOS*

## Helper package

```
root@enticed-cichlid:~# apt install ubuntu-helper-virt-hwe
```

```
root@enticed-cichlid:~# ubuntu_virt_helper
Installed variant: none
```

```
1 packages:
- ubuntu-helper-virt-hwe (1:10.2.1+ds-1ubuntu4, src:qemu-hwe)
```

<sup>607</sup> [https://cc-enabling.trustedservices.intel.com/intel-tdx-enabling-guide/04/hardware\\_setup/#deploy-specific-intel-tdx-module-version](https://cc-enabling.trustedservices.intel.com/intel-tdx-enabling-guide/04/hardware_setup/#deploy-specific-intel-tdx-module-version)

<sup>608</sup> <https://releases.ubuntu.com/resolute/>

<sup>609</sup> <https://launchpad.net/ubuntu/+source/qemu-hwe>

<sup>610</sup> <https://launchpad.net/ubuntu/+source/libvirt-hwe>

<sup>611</sup> <https://launchpad.net/ubuntu/+source/edk2-hwe>

<sup>612</sup> <https://launchpad.net/ubuntu/+source/seabios-hwe>

## Impact on current usage

```
root@enticed-cichlid:~# apt install qemu-system-x86
```

```
root@enticed-cichlid:~# ubuntu_virt_helper --verbose
Installed variant: base
```

15 packages:

- ovmf (2025.11-3ubuntu7, src:edk2, auto)
- ovmf-amdsev (2025.11-3ubuntu7, src:edk2, auto)
- ovmf-generic (2025.11-3ubuntu7, src:edk2, auto)
- ovmf-inteltdx (2025.11-3ubuntu7, src:edk2, auto)
- qemu-block-extra (1:10.2.1+ds-1ubuntu3, src:qemu, auto)
- qemu-system-common (1:10.2.1+ds-1ubuntu3, src:qemu, auto)
- qemu-system-data (1:10.2.1+ds-1ubuntu3, src:qemu, auto)
- qemu-system-gui (1:10.2.1+ds-1ubuntu3, src:qemu, auto)
- qemu-system-modules-opengl (1:10.2.1+ds-1ubuntu3, src:qemu, auto)
- qemu-system-modules-spice (1:10.2.1+ds-1ubuntu3, src:qemu, auto)
- qemu-system-x86 (1:10.2.1+ds-1ubuntu3, src:qemu, manual)
- qemu-utils (1:10.2.1+ds-1ubuntu3, src:qemu, auto)
- seabios (1.17.0-1ubuntu1, src:seabios, auto)
- ubuntu-helper-virt-hwe (1:10.2.1+ds-1ubuntu4, src:qemu-hwe, manual)
- ubuntu-virt (1:10.2.1+ds-1ubuntu3, src:qemu, auto)

## Uncontrolled variant switches and dependency management implications

```
root@enticed-cichlid:~# apt install qemu-utils-hwe
The following packages were automatically installed and are no longer required:
 adwaita-icon-theme libflac14
 ...
 x11-common
Use 'sudo apt autoremove' to remove them.
```

```
Installing:
 qemu-utils-hwe
```

```
Installing dependencies:
 qemu-block-extra-hwe ubuntu-virt-hwe
```

(continues on next page)

(continued from previous page)

```
REMOVING:
 ovmf ovmf-inteltdx qemu-system-data qemu-system-modules-spice
seabios
 ovmf-amdsew qemu-block-extra qemu-system-gui qemu-system-x86
ubuntu-virt
 ovmf-generic qemu-system-common qemu-system-modules-opengl qemu-utils

Summary:
 Upgrading: 0, Installing: 3, Removing: 14, Not Upgrading: 9
 Download size: 2496 kB
 Freed space: 118 MB

Continue? [Y/n]
```

```
Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 ubuntu-virt-hwe all 1:10.2.
1+ds-1ubuntu4 [11.6 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute/main amd64 qemu-utils-hwe amd64 1:10.2.
1+ds-1ubuntu4 [2389 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute/main amd64 qemu-block-extra-hwe amd64
1:10.2.1+ds-1ubuntu4 [95.4 kB]
Fetched 2496 kB in 1s (3358 kB/s)
Virt: switch of the ubuntu-virt[-hwe] stack is detected and there are orphaned removals
(no counterpart being installed):
- ovmf
- ovmf-amdsew
- ovmf-generic
- ovmf-inteltdx
- qemu-system-x86
- qemu-system-modules-spice
- qemu-system-common
- qemu-system-data
- qemu-system-gui
- qemu-system-modules-opengl
- seabios
Virt: Please install them back and use the ubuntu-virt-helper script to switch between
ubuntu-virt[-hwe] packages.
(Reading database ... 96412 files and directories currently installed.)
Removing ovmf (2025.11-3ubuntu7) ...
Removing ovmf-amdsew (2025.11-3ubuntu7) ...
Removing ovmf-generic (2025.11-3ubuntu7) ...
```

```
root@enticed-cichlid:~# ubuntu_virt_helper --verbose
Installed variant: hwe
```

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```
4 packages:
- qemu-block-extra-hwe (1:10.2.1+ds-1ubuntu4, src:qemu-hwe, auto)
- qemu-utils-hwe (1:10.2.1+ds-1ubuntu4, src:qemu-hwe, manual)
- ubuntu-helper-virt-hwe (1:10.2.1+ds-1ubuntu4, src:qemu-hwe, manual)
- ubuntu-virt-hwe (1:10.2.1+ds-1ubuntu4, src:qemu-hwe, auto)
```

## Proper variant switch using the helper script

```
root@enticed-cichlid:~# ubuntu_virt_helper switch
Switching from base to hwe variant...
Installing:
 ovmf-amdsev-hwe ovmf-inteltdx-hwe qemu-system-data-hwe qemu-system-
modules-spice-hwe seabios-hwe
 ovmf-generic-hwe qemu-block-extra-hwe qemu-system-gui-hwe qemu-system-
x86-hwe ubuntu-virt-hwe
 ovmf-hwe qemu-system-common-hwe qemu-system-modules-opengl-hwe qemu-utils-hwe

Suggested packages:
 samba vde2 passt

REMOVING:
 ovmf ovmf-generic qemu-block-extra qemu-system-data qemu-system-modules-
opengl qemu-system-x86 seabios
 ovmf-amdsev ovmf-inteltdx qemu-system-common qemu-system-gui qemu-system-modules-
spice qemu-utils ubuntu-virt

Summary:
 Upgrading: 0, Installing: 14, Removing: 14, Not Upgrading: 9
 Download size: 25.5 MB
 Freed space: 7168 B

Continue? [Y/n]
```

```
root@enticed-cichlid:~# ubuntu_virt_helper --verbose
Installed variant: hwe

15 packages:
- ovmf-amdsev-hwe (2025.11-3ubuntu8, src:edk2-hwe, auto)
```

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```
- ovmf-generic-hwe (2025.11-3ubuntu8, src:edk2-hwe, auto)
- ovmf-hwe (2025.11-3ubuntu8, src:edk2-hwe, auto)
- ovmf-inteltdx-hwe (2025.11-3ubuntu8, src:edk2-hwe, auto)
- qemu-block-extra-hwe (1:10.2.1+ds-1ubuntu4, src:qemu-hwe, auto)
- qemu-system-common-hwe (1:10.2.1+ds-1ubuntu4, src:qemu-hwe, auto)
- qemu-system-data-hwe (1:10.2.1+ds-1ubuntu4, src:qemu-hwe, auto)
- qemu-system-gui-hwe (1:10.2.1+ds-1ubuntu4, src:qemu-hwe, auto)
- qemu-system-modules-opengl-hwe (1:10.2.1+ds-1ubuntu4, src:qemu-hwe, auto)
- qemu-system-modules-spice-hwe (1:10.2.1+ds-1ubuntu4, src:qemu-hwe, auto)
- qemu-system-x86-hwe (1:10.2.1+ds-1ubuntu4, src:qemu-hwe, manual)
- qemu-utils-hwe (1:10.2.1+ds-1ubuntu4, src:qemu-hwe, auto)
- seabios-hwe (1:17.0-1ubuntu2, src:seabios-hwe, auto)
- ubuntu-helper-virt-hwe (1:10.2.1+ds-1ubuntu4, src:qemu-hwe, manual)
- ubuntu-virt-hwe (1:10.2.1+ds-1ubuntu4, src:qemu-hwe, auto)
```

- *LXD containers and virtual machines*
- *Create VMs with Multipass*
- *Create cloud image VMs with UVtool*
- *QEMU*
- *Confidential Computing with AMD*
- *Confidential Computing with Intel*
- *Hardware Enablement Virtualization Stack*

## VM tooling

### Libvirt

libvirt library<sup>613</sup>

Kernel-based Virtual Machine (KVM)<sup>614</sup>

```
$ kvm-ok
```

#### Note:

This command is part of the package `cpu-checker`, you might need to install it because it is not installed by default.

<sup>613</sup> <https://libvirt.org/>

<sup>614</sup> [https://www.linux-kvm.org/page/Main\\_Page](https://www.linux-kvm.org/page/Main_Page)

**Note:**

On many computers with processors supporting hardware-assisted virtualisation, you must activate an option in the *BIOS* to enable it.

## Virtual networking

SLiRP<sup>615</sup>

how to configure a bridge<sup>616</sup>  
netplan.io documentation<sup>617</sup>

## Install libvirt

```
$ sudo apt update
```

```
$ sudo apt install qemu-kvm libvirt-daemon-system
```

```
$ sudo adduser $USER libvirt
```

**Note:**

If you add the current user, log out and back in for the new group membership to take effect.

<sup>615</sup> <https://en.wikipedia.org/wiki/Slirp>

<sup>616</sup> <https://netplan.readthedocs.io/en/latest/netplan-yaml/#properties-for-device-type-bridges>

<sup>617</sup> <https://netplan.readthedocs.io/en/latest/>

- 
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- 

*UVTool*

*Multipass*

*Graphical User Interface (GUI)*

*Virtual*

*Machine Manager / Viewer*

## Virtual machine management

- *Multipass*
- *UVTool*
- *virt-\* tools*
- *OpenStack*<sup>618</sup>

## Manage VMs with virsh

- 

```
$ virsh list
```

- 

```
$ virsh start <guestname>
```

- 

```
$ virsh autostart <guestname>
```

- 

```
$ virsh reboot <guestname>
```

- 

---

<sup>618</sup> <https://ubuntu.com/openstack>

```
$ virsh save <guestname> save-my.state
```

- 

```
$ virsh restore save-my.state
```

- 

```
$ virsh shutdown <guestname>
```

- 

```
$ virsh attach-disk <guestname> /dev/cdrom /media/cdrom
```

- 

```
$ virsh edit <guestname>
```

XML representation that defines the guest<sup>619</sup>

### *Virtual Machine Manager / Viewer*

#### Note:

If `virsh` (or other `vir*` tools) connects to something other than the default `qemu-kvm/system` hypervisor, you can find alternatives for the `--connect` option using `man virsh` or the [libvirt docs](https://libvirt.org/docs)<sup>620</sup>.

---

<sup>620</sup> <https://libvirt.org/uri.html>

### system and session scope

```
$ virsh --connect qemu:///session list
```

---

<sup>619</sup> <https://libvirt.org/formatdomain.html>



```
$ export LIBVIRT_DEFAULT_URI=qemu:///session
```

## How each mode runs

SLiRP<sup>621</sup>

```
root 11351 0.0 4.3 752264 39292 ? Ssl 15:15 0:00 /usr/sbin/
libvirtd --timeout 120
ubuntu 13104 0.0 4.3 748272 39360 ? Sl 23:01 0:00 /usr/sbin/
libvirtd --timeout=120
debian 14364 2.4 4.1 747848 37660 ? Sl 23:18 0:00 /usr/sbin/
libvirtd --timeout=120
```

- 
- 

## AppArmor differences

---

<sup>621</sup> <https://en.wikipedia.org/wiki/Slirp>

```
user@host:~$... > 7 processes are in enforce mode. > ... >
/usr/bin/qemu-system-x86_64 (13660)
libvirt-03cf1350-1de5-4400-96e3-16dce6a9a921 > /usr/sbin/libvirtd (11351)
libvirtd
```

```
user@host:~$ virsh --connect qemu:///system domuuid tcg-minimal >
03cf1350-1de5-4400-96e3-16dce6a9a921
```

## When to use session mode

Aspect	qemu:///system	qemu:///session
Runs as	root (via a system-wide daemon)	the invoking user
Guest process owner	the libvirt-qemu user by default	the invoking user
Configuration directory	/etc/libvirt/	~/.config/libvirt/
Default storage pool	/var/lib/libvirt/images/	~/.local/share/libvirt/images/
Default networking	NAT-ed default network (virbr0)	user-mode (SLiRP <sup>622</sup> ) only
Bridged networking	managed by libvirt	requires the setuid qemu-bridge-helper

<sup>622</sup> <https://en.wikipedia.org/wiki/Slirp>

- 
- 
- 
- 

#### ➔ See also

There is more information about this topic in the [libvirt FAQ](#)<sup>623</sup> and this [blog post](#)<sup>624</sup> about the topic.

## Migration

- [Offline migration](#)<sup>625</sup>
- [Live migration](#)<sup>626</sup>
- [Postcopy migration](#)<sup>627</sup>

```
$ virsh migrate --help
```

*Live migration with libvirt/QEMU*

## CPU model and topology

- 
- 
- 
- 

<sup>623</sup> <https://wiki.libvirt.org/FAQ.html#what-is-the-difference-between-qemu-system-and-qemu-session-which-one-should-i-use>

<sup>624</sup> <https://blog.wikichoon.com/2016/01/qemusystem-vs-qemusession.html>

<sup>625</sup> <https://libvirt.org/migration.html#offline-migration>

<sup>626</sup> <https://libvirt.org/migration.html>

<sup>627</sup> <https://wiki.qemu.org/Features/PostCopyLiveMigration>

```
example snippet from qemu command line generated by libvirt
... -cpu Haswell-noTSX-IBRS,vmx=on,pdcm=off,...,vmx-entry-load-efer=on,vmx-eptp-switching=on ...
```

```
<capabilities>
 <host>
 <uuid>30303837-3831-584d-5135-323430354a30</uuid>
 <cpu>
 <arch>x86_64</arch>
 <model>Haswell-noTSX-IBRS</model>
 <vendor>Intel</vendor>
 <microcode version='73' />
 <signature family='6' model='63' stepping='2' />
 <counter name='tsc' frequency='2397195000' scaling='no' />
 <topology sockets='1' dies='1' cores='6' threads='2' />
 <maxphysaddr mode='emulate' bits='46' />
 <feature name='vme' />
 <feature name='ds' />
 <feature name='acpi' />
 <feature name='ss' />
```

upstream CPU model and features documentation<sup>628</sup>

## Host CPU capabilities

### Device passthrough/hotplug

- *hotplug*
- 

---

<sup>628</sup> <https://libvirt.org/formatcaps.html#host-cpu-model-and-features>

## guest description<sup>629</sup>

```
<hostdev mode='subsystem' type='usb' managed='yes'>
 <source>
 <vendor id='0x0b6d' />
 <product id='0x3880' />
 </source>
</hostdev>
```

```
<hostdev mode='subsystem' type='pci' managed='yes'>
 <source>
 <address domain='0x0000' bus='0x04' slot='0x10' function='0x0' />
 </source>
</hostdev>
```

### Note:

Getting the virtual function is device-dependent and cannot be fully covered here. In general, it involves setting up an *IOMMU*, registering via *VFIO*<sup>630</sup>, and sometimes requesting a number of VFs.

<sup>630</sup> <https://www.kernel.org/doc/Documentation/vfio.txt>

## ppc64el<sup>631</sup>

```
$ sudo modprobe vfio-pci
identify device
$ lspci -n -s 0005:01:01.3
0005:01:01.3 0200 10df:e228 (rev 10)
register and request VFs
$ echo 10df e228 | sudo tee /sys/bus/pci/drivers/vfio-pci/new_id
$ echo 4 | sudo tee /sys/bus/pci/devices/0005\:01\:00 0/sriov_numvfs
```

```
virsh attach-device <guestname> <device-xml>
Use the Device in the Guest
virsh detach-device <guestname> <device-xml>
```

<sup>629</sup> <https://libvirt.org/formatdomain.html>

<sup>631</sup> <https://wiki.debian.org/ppc64el>

## Access QEMU Monitor via libvirt

QEMU Monitor<sup>632</sup>

```
$ virsh qemu-monitor-command --hmp focal-test-log 'drive_add 0
if=none,file=/var/lib/libvirt/images/test.img,format=raw,id=disk1'
```

```
$ virsh qemu-monitor-command --hmp y-ipns 'info registers'

RAX=00ffffc000000000 RBX=ffff8f0f5d5c7e48 RCX=0000000000000000 RDX=ffffea00007571c0
RSI=0000000000000000 RDI=ffff8f0fdd5c7e48 RBP=ffff8f0f5d5c7e18 RSP=ffff8f0f5d5c7df8
[...]
```

## Huge pages

*huge pages*

64-bit CPUs<sup>633</sup>

32-bit CPUs<sup>634</sup>

x86\_64<sup>635</sup>

1 byte each<sup>636</sup>

Memory Management Unit (MMU)<sup>637</sup>

<sup>632</sup> <https://en.wikibooks.org/wiki/QEMU/Monitor>

<sup>633</sup> [https://en.wikipedia.org/wiki/64-bit\\_computing](https://en.wikipedia.org/wiki/64-bit_computing)

<sup>634</sup> [https://en.wikipedia.org/wiki/32-bit\\_computing](https://en.wikipedia.org/wiki/32-bit_computing)

<sup>635</sup> <https://en.wikipedia.org/wiki/X86-64>

<sup>636</sup> [https://en.wikipedia.org/wiki/Memory\\_address](https://en.wikipedia.org/wiki/Memory_address)

<sup>637</sup> [https://en.wikipedia.org/wiki/Memory\\_management\\_unit](https://en.wikipedia.org/wiki/Memory_management_unit)

## Translation Lookaside Buffer (TLB)

*memory to the host*      *pre-allocated*      *map the VM's*

transparent huge pages<sup>638</sup>

### Tip:

When using device passthrough on very large guests, there is an extra benefit of using huge pages, as it is faster to do the initial memory clear on the VFIO *DMA* pin.

## Huge page allocation

```
$ echo 256 | sudo tee /sys/kernel/mm/hugepages/hugepages-2048kB/nr_hugepages
```

```
$ cat /sys/kernel/mm/hugepages/hugepages-2048kB/nr_hugepages
256
```

`default_hugepagesz`<sup>639</sup>

<sup>638</sup> <https://www.kernel.org/doc/html/next/admin-guide/mm/transhuge.html>

<sup>639</sup> <https://www.kernel.org/doc/html/v5.4/admin-guide/kernel-parameters.html>

```
$ grep Hugesize /proc/meminfo
```

```
Hugesize: 2048 kB
```

```
$ tail /sys/kernel/mm/hugepages/hugepages-*/nr_hugepages`
```

```
==> /sys/kernel/mm/hugepages/hugepages-1048576kB/nr_hugepages <==
0
```

```
==> /sys/kernel/mm/hugepages/hugepages-2048kB/nr_hugepages <==
2
```

Numa node<sup>640</sup>

boot or runtime<sup>641</sup>

kernel documentation<sup>642</sup>

hugeadm<sup>643</sup>

```
$ apt install libhugetlbfs-bin
```

```
$ hugeadm --list-all-mounts
```

Mount Point	Options
/dev/hugepages	rw,relatime,pagesize=2M

```
$ hugeadm --explain
```

## Huge page usage in libvirt

```
<memoryBacking>
 <hugepages/>
</memoryBacking>
```

<sup>640</sup> <https://www.kernel.org/doc/html/v5.4/vm/numa.html>

<sup>641</sup> <https://www.kernel.org/doc/Documentation/vm/hugetlbpage.txt>

<sup>642</sup> <https://www.kernel.org/doc/Documentation/vm/hugetlbpage.txt>

<sup>643</sup> <https://linux.die.net/man/8/hugeadm>



libvirt memory backing documentation<sup>645</sup> Numa nodes<sup>644</sup>

## Controlling addressing bits

### -hpb machine types

provided special machine-types<sup>646</sup>

*HPB*

```
$ cat /proc/cpuinfo | grep '^address sizes'

...
an older server with a E5-2620
address sizes : 46 bits physical, 48 bits virtual
a laptop with an i7-8550U
address sizes : 39 bits physical, 48 bits virtual
```

<sup>644</sup> <https://www.kernel.org/doc/html/v5.4/vm/numa.html>

<sup>645</sup> <https://libvirt.org/formatdomain.html#memory-backing>

<sup>646</sup> <https://bugs.launchpad.net/bugs/1776189>

## maxphysaddr guest configuration

CPU

model and topology section<sup>647</sup>

```
<maxphysaddr mode='passthrough' />
```

```
<maxphysaddr mode='emulate' bits='42' />
```

```
<maxphysaddr mode='passthrough' limit='41' />
```

## AppArmor isolation

AppArmor rules for a guest<sup>648</sup>

- 
- 

- 

- 

- 

deny

rules<sup>649</sup>

<sup>647</sup> <https://libvirt.org/formatdomain.html#cpu-model-and-topology>

<sup>648</sup> <https://gitlab.com/apparmor/apparmor/-/wikis/Libvirt#implementation-overview>

<sup>649</sup> <https://gitlab.com/apparmor/apparmor/-/wikis/FAQ#what-is-default-deny-allow-listing>

- 

guest XML<sup>650</sup>

## Sharing files between Host<->Guest

```
<memoryBacking>
 <access mode='shared' />
</memoryBacking>
```

```
<memoryBacking>
 <hugepages>
 <page size='2048' unit='KiB' />
 </hugepages>
 <access mode='shared' />
</memoryBacking>
```

```
<filesystem type='mount' accessmode='passthrough'>
 <driver type='virtiofs' />
 <source dir='/var/guests/h-virtiofs' />
 <target dir='myfs' />
</filesystem>
```

```
$ sudo mount -t virtiofs myfs /mnt/
```

libvirt domain/filesystem documentation<sup>651</sup>

---

<sup>650</sup> <https://libvirt.org/formatdomain.html>

<sup>651</sup> <https://libvirt.org/formatdomain.html#filesystems>

### Note:

While virtiofs works with  $\geq 20.10$  (Groovy),  $\geq 21.04$  (Hirsute) made it more convenient, especially in small environments (no hard requirement to specify guest Numa topology or use huge pages). If you need to set up on 20.10 or want more details, the [libvirt knowledge-base about virtiofs](https://libvirt.org/kbase/virtiofs.html)<sup>652</sup> provides additional information.

<sup>652</sup> <https://libvirt.org/kbase/virtiofs.html>

## Resources

- [KVM home page](#)<sup>653</sup>
- [libvirt home page](#)<sup>654</sup>
  - [domains](#)<sup>655</sup> [storage](#)<sup>656</sup>
- [Ubuntu Wiki KVM](#)<sup>657</sup>
- [linux-kvm](#)<sup>658</sup>
- [Introduction to Memory Management in Linux](#)<sup>659</sup>

## Virtual Machine Manager

[Virtual Machine Manager](#)<sup>660</sup>

## Install virt-manager

```
$ sudo apt install virt-manager
```

*Graphical User Interface (GUI)*

<sup>653</sup> [https://linux-kvm.org/page/Main\\_Page](https://linux-kvm.org/page/Main_Page)

<sup>654</sup> <https://libvirt.org/>

<sup>655</sup> <https://libvirt.org/formatdomain.html>

<sup>656</sup> <https://libvirt.org/formatstorage.html>

<sup>657</sup> <https://help.ubuntu.com/community/KVM>

<sup>658</sup> [https://www.linux-kvm.org/page/How\\_to\\_assign\\_devices\\_with\\_VT-d\\_in\\_KVM#Assigning\\_the\\_device](https://www.linux-kvm.org/page/How_to_assign_devices_with_VT-d_in_KVM#Assigning_the_device)

<sup>659</sup> <https://www.youtube.com/watch?v=7aONIVSXiJ8>

<sup>660</sup> <https://virt-manager.org/>

```
$ virt-manager
```

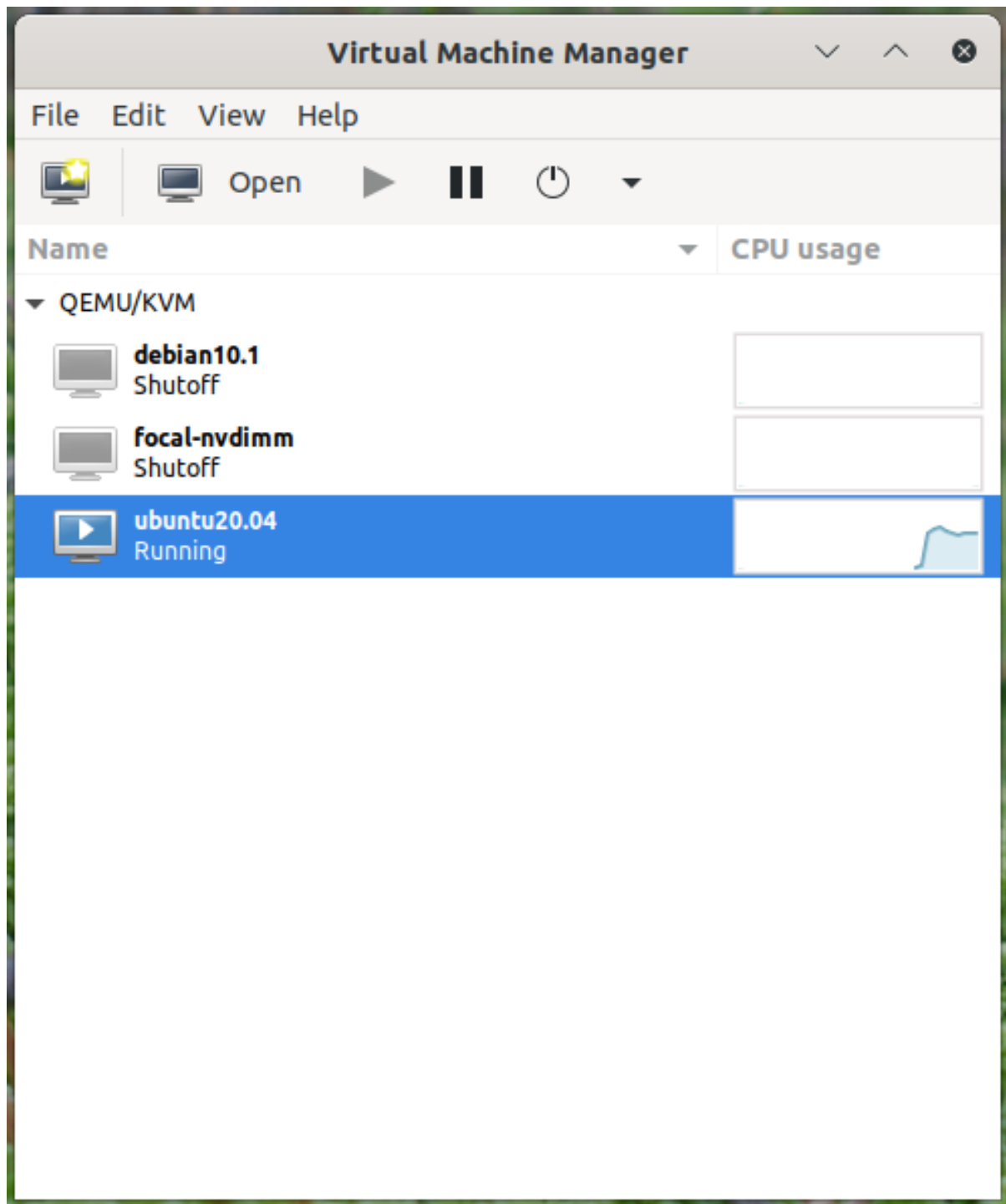
```
$ virt-manager -c qemu+ssh://virtnode1.mydomain.com/system
```

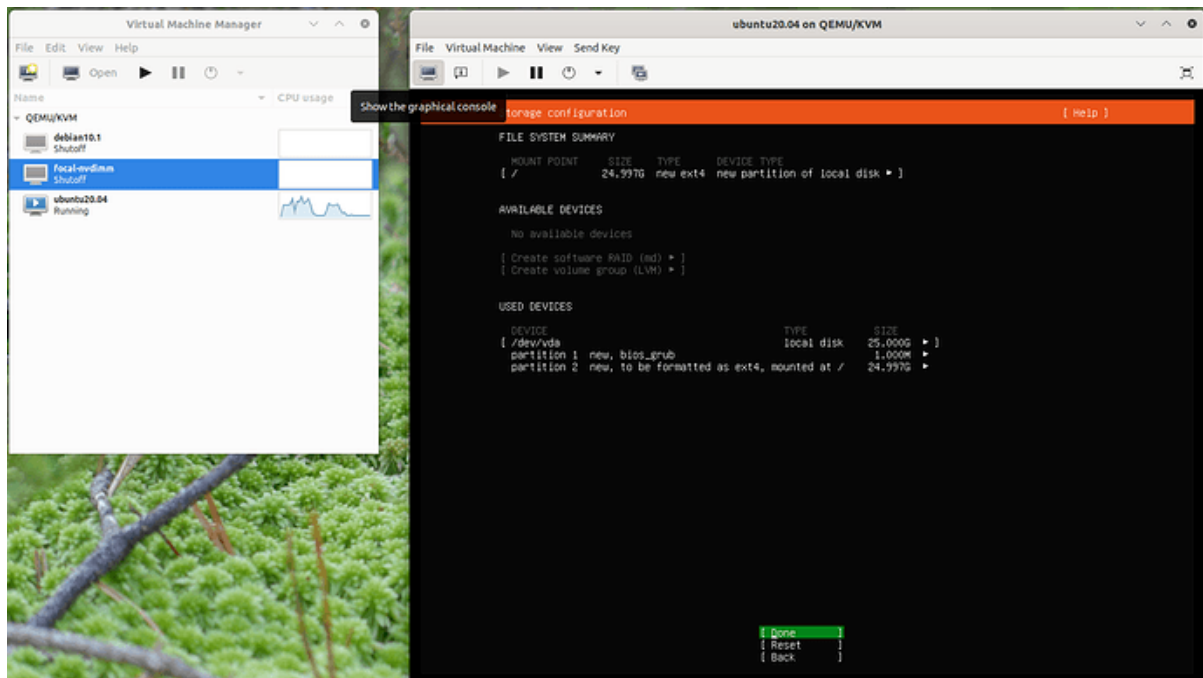
#### Note:

The above example assumes that SSH connectivity between the management system and the target system has already been configured, and uses **SSH keys** for authentication. SSH keys are needed because libvirt sends the password prompt to another process. See our guide on OpenSSH for details on [how to set up SSH keys](#) (page 237).

## Use virt-manager to manage guests

### Guest lifecycle

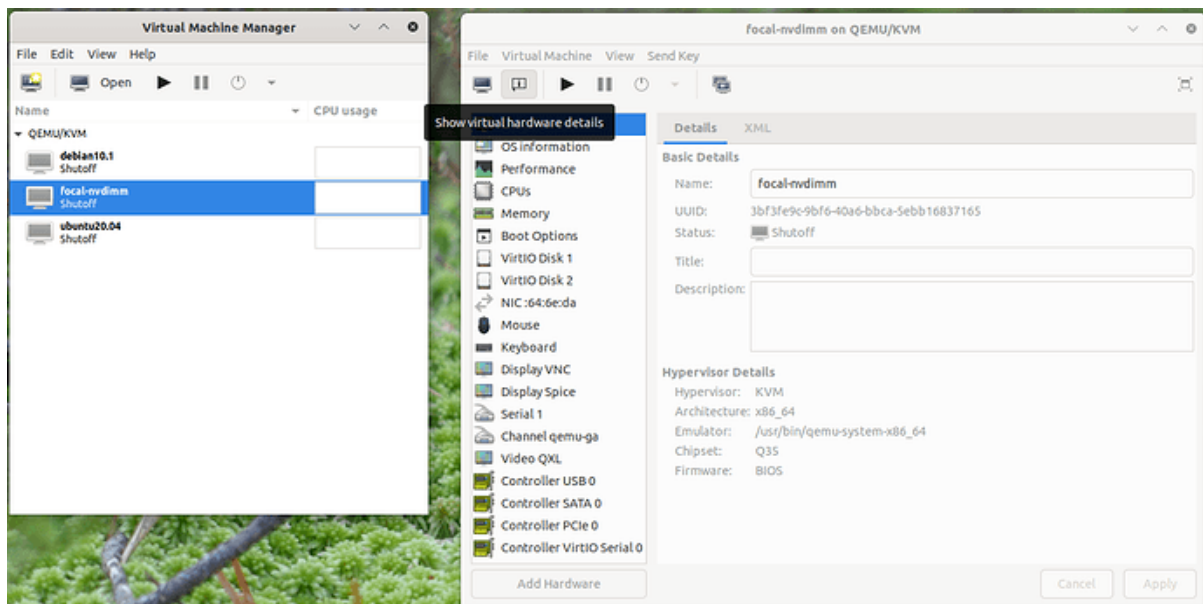




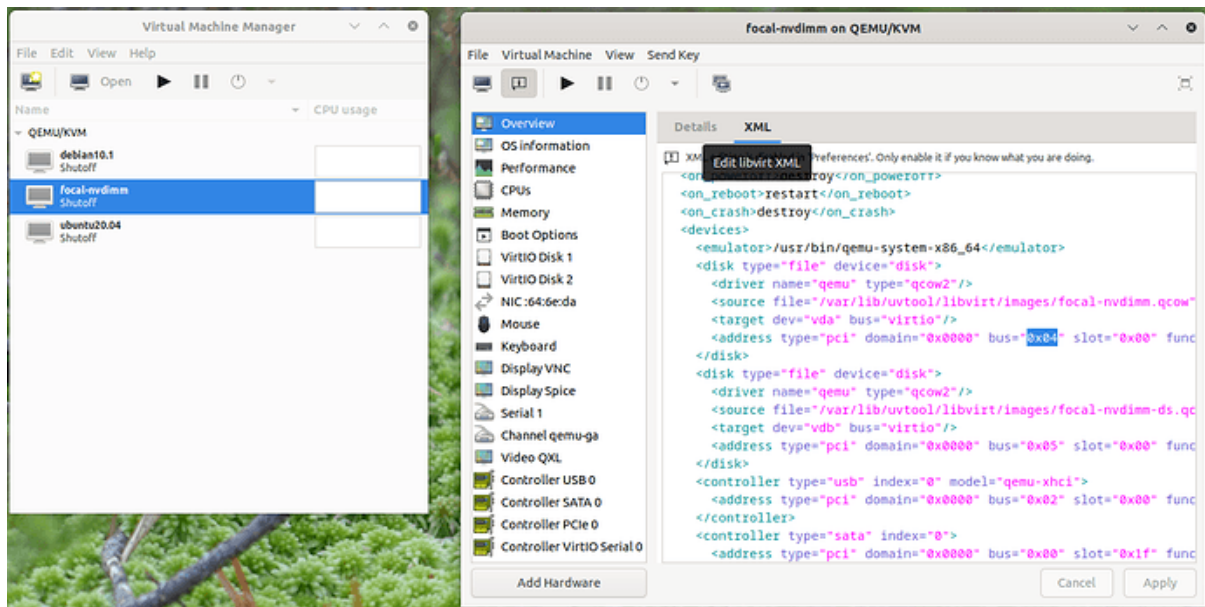
## Guest modification

tion<sup>661</sup>

guest representa-



<sup>661</sup> <https://libvirt.org/formatdomain.html>



*libvirt-client*

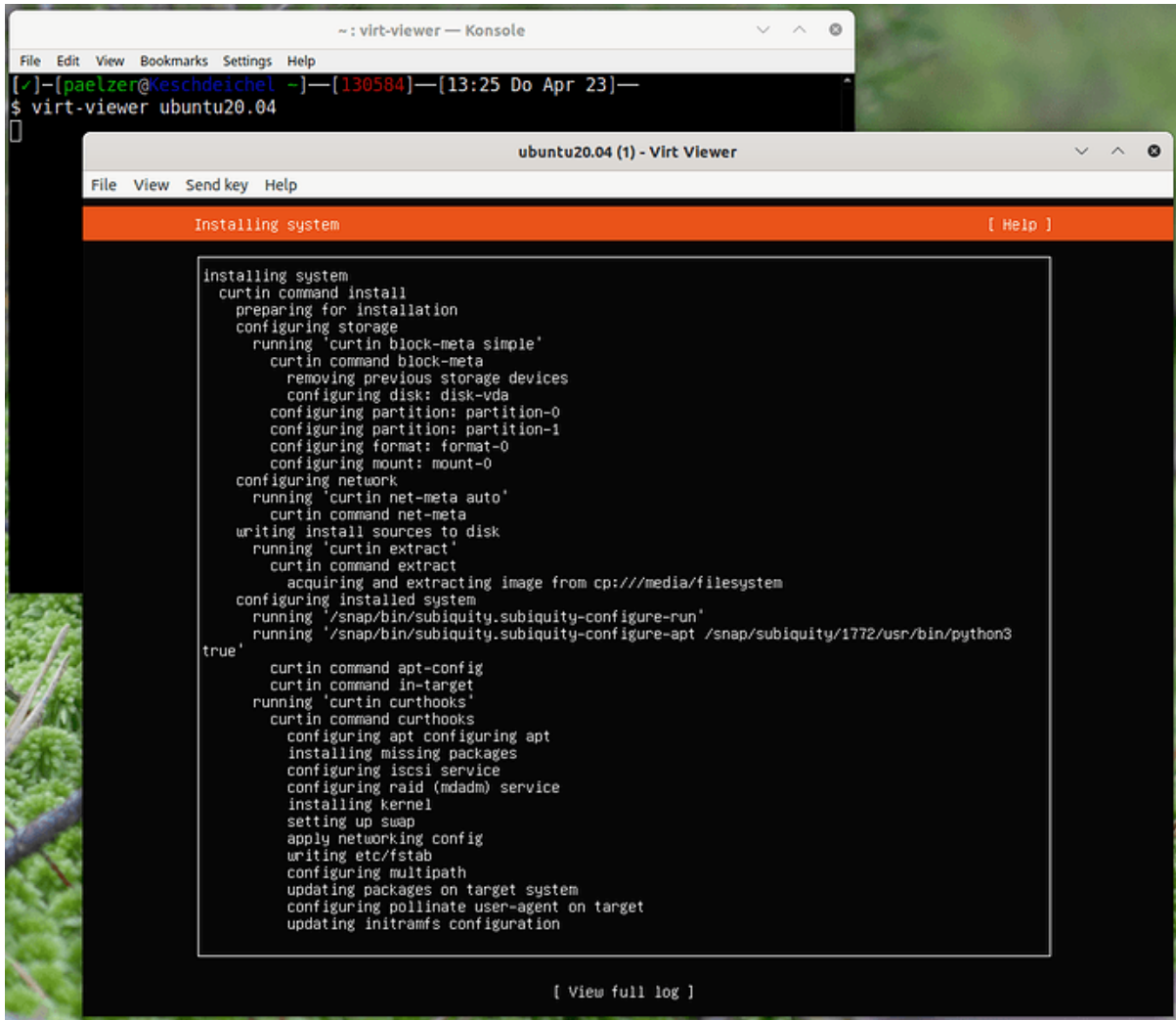
## Virtual Machine Viewer (virt-viewer)

### Install virt-viewer

```
$ sudo apt install virt-viewer
```

```
$ virt-viewer <guestname>
```





```

~: virt-viewer — Konsole
File Edit View Bookmarks Settings Help
[✓]—[paelzer@keschdeichel ~]—[130584]—[13:25 Do Apr 23]—
$ virt-viewer ubuntu20.04

ubuntu20.04 (1) - Virt Viewer
File View Send key Help

Installing system [Help]

installing system
 curtin command install
 preparing for installation
 configuring storage
 running 'curtin block-meta simple'
 curtin command block-meta
 removing previous storage devices
 configuring disk: disk-vda
 configuring partition: partition-0
 configuring partition: partition-1
 configuring format: format-0
 configuring mount: mount-0
 configuring network
 running 'curtin net-meta auto'
 curtin command net-meta
 writing install sources to disk
 running 'curtin extract'
 curtin command extract
 acquiring and extracting image from cp:///media/filesystem
 configuring installed system
 running '/snap/bin/subiquity.subiquity-configure-run'
 running '/snap/bin/subiquity.subiquity-configure-apt /snap/subiquity/1772/usr/bin/python3
true'
 curtin command apt-config
 curtin command in-target
 running 'curtin curthooks'
 curtin command curthooks
 configuring apt configuring apt
 installing missing packages
 configuring iscsi service
 configuring raid (mdadm) service
 installing kernel
 setting up swap
 apply networking config
 writing etc/fstab
 configuring multipath
 updating packages on target system
 configuring pollinate user-agent on target
 updating initramfs configuration

[View full log]

```

```
$ virt-viewer -c qemu+ssh://virtnode1.mydomain.com/system <guestname>
```

## virt-install

### Install virt-install

```
$ sudo apt install virtinst
```

```
$ virt-install \ > --name web_devel \ > --ram 8192 \ > --disk
path=/home/doug/vm/web_devel.img,bus=virtio,size=50 \ > --cdrom
focal-desktop-amd64.iso \ > --network network=default,model=virtio \ >
--graphics vnc,listen=0.0.0.0 \ > --noautoconsole \ > --hvm \ > --vcpus=4
```

*virt-install(1)*<sup>662</sup>

- 
- 
- 
- 
- 
- 
- 
- 
- 

---

<sup>662</sup> <https://manpages.ubuntu.com/manpages/resolute/man1/virt-install.1.html>

## virt-clone

```
$ virt-clone --auto-clone --original focal
```

- 
- 

### Caution:

Please be aware that this is a full clone, therefore any secrets, keys, and for example, `/etc/machine-id`, will be shared. This will cause issues with security and anything that needs to identify the machine like [DHCP](#). You most likely want to edit those afterward and de-duplicate them as needed.

## Resources

- [KVM](#)<sup>663</sup>
- [libvirt home page](#)<sup>664</sup>
- [Virtual Machine Manager](#)<sup>665</sup>

## How to enable nested virtualization

### Important:

Nested virtualization is enabled by default on Ubuntu. If you are using Ubuntu, it's unlikely that you will need to manually enable the feature. If you check (using the steps below) and discover that nested virtualization is enabled, then you will not need to do anything further.

<sup>663</sup> [https://linux-kvm.org/page/Main\\_Page](https://linux-kvm.org/page/Main_Page)

<sup>664</sup> <https://libvirt.org/>

<sup>665</sup> <https://virt-manager.org/>

## Check if nested virtualization is enabled

*AMD*

```
$ lsmod | grep -i kvm

kvm_intel 204800 0
kvm 1347584 1 kvm_intel
```

## If the module is loaded

```
$ cat /sys/module/kvm_intel/parameters/nested
```

```
$ cat /sys/module/kvm_amd/parameters/nested
```

## If the module is not loaded

```
$ modprobe kvm-intel nested=1
```

```
$ modprobe kvm-amd nested=1
```

## Enable nested virtualization

```
$ sudo modprobe -r <module>
```

```
$ sudo modprobe <module>
```

### Note:

Make sure to stop all running VMs that might use the kernel module and prevent its reload.

```
$ sudo modprobe -r kvm-intel
```

```
$ sudo modprobe kvm-intel
```

```
$ cat /sys/module/kvm_intel/parameters/nested
```

## Check and enable nested virtualization inside a VM

```
$ egrep "svm|vmx" /proc/cpuinfo
```

- 
-

- 

on the OpenStack wiki<sup>666</sup>

```
<cpu mode='host-model' check='partial' />
```

- 

## Limitations of nested virtualization

- *How to use the libvirt library with virsh*
- *How to use virt-manager and other virt\* tools*
- *How to enable nested virtualisation*

## Ubuntu in other virtual environments

### How to set up Ubuntu on Hyper-V

type 1 hypervisor<sup>667</sup> *ESXi*

- 

- 

- 

for Linux<sup>668</sup>

Windows Subsystem

## Hyper-V system requirements

system requirements for Hyper-V<sup>669</sup>

- 

- 

*BIOS*

- 

<sup>666</sup> <https://wiki.openstack.org/wiki/LibvirtXMLCPUModel>

<sup>667</sup> <https://en.wikipedia.org/wiki/Hypervisor#Classification>

<sup>668</sup> <https://learn.microsoft.com/windows/wsl/about>

<sup>669</sup> <https://learn.microsoft.com/en-us/windows-server/virtualization/hyper-v/system-requirements-for-hyper-v-on-windows>

- 

## Install Hyper-V

graded<sup>670</sup>

would need to be up-

## Install Hyper-V on Windows client

### Install Hyper-V graphically

- 1.
- 2.
- 3.
- 4.
- 5.

### Install Hyper-V using PowerShell

- 1.
- 2.

```
Enable-WindowsOptionalFeature -Online -FeatureName Microsoft-Hyper-V -All
```

- 3.

## Install Hyper-V on Windows Server

```
Install-WindowsFeature -Name Hyper-V -IncludeManagementTools -Restart
```

installing the Hyper-V role on Windows Server<sup>671</sup>

<sup>670</sup> <https://support.microsoft.com/en-us/windows/upgrade-windows-home-to-windows-pro-ef34d520-e73f-3198-c525-d1a218>

<sup>671</sup> <https://learn.microsoft.com/en-us/windows-server/virtualization/hyper-v/get-started/install-the-hyper-v-role-on-windows-server>

## Install Ubuntu on Hyper-V

### Using Quick Create

#### Note:

Quick Create is only available on Windows 10 and Windows 11 **client** editions. It is not available on Windows Server. If you are running Windows Server, use the [New Virtual Machine Wizard](#) (page 696) method instead.

1.

2.

- 

- 

3.

4.

- 

- 

5.

6.

7.

8.



## Using the New Virtual Machine Wizard

1. [official Ubuntu source](#)<sup>672</sup>
- 2.
- 3.
- 4.
5.
  - 
  - 
  - 
  - 
  - 
  -
- 6.
- 7.
- 8.
- 9.

## Further reading

[creating a virtual machine in Hyper-V](#)<sup>673</sup>

- [Setting up Ubuntu on Hyper-V](#)

---

<sup>672</sup> <https://ubuntu.com/download/server>

<sup>673</sup> <https://learn.microsoft.com/en-us/windows-server/virtualization/hyper-v/get-started/create-a-virtual-machine-in-hyper-v>

## See also

- [Virtualisation and containers](#)

## Containers

- 
- 

## Containers

- [LXD](#)
- [Docker for sysadmins](#)
- [How to run rocks on your server](#)
- [Skopeo for sysadmins](#)

## Docker for system admins

## Installation

```
$ sudo apt-get install -y docker.io docker-compose
```

### Note:

This guide uses the Docker packages provided by Ubuntu (`docker.io`), which are supported by Canonical. The instructions and examples below are written for these packages. Canonical also provides a [Docker snap](#)<sup>674</sup>. This guide does not cover the snap installation path.

---

<sup>674</sup> <https://snapcraft.io/docker>

## Configuring storage

### How to configure volumes

- 

```
$ docker volume create my-vol

my-vol
```

- 

```
$ docker volume ls

DRIVER VOLUME NAME
local my-vol
```

- 

```
$ docker volume inspect my-vol

[
 {
 "CreatedAt": "2023-10-25T00:53:24Z",
 "Driver": "local",
 "Labels": null,
 "Mountpoint": "/var/lib/docker/volumes/my-vol/_data",
 "Name": "my-vol",
 "Options": null,
 "Scope": "local"
 }
]
```

- 

```
$ docker run --name web-server -d \ > --mount source=my-vol,target=/app
 \ > ubuntu/apache2

0709c1b632801fddd767deddda0d273289ba423e9228cc1d77b2194989e0a882
```

- 

```
$ docker inspect web-server --format '{{ json .Mounts }}' | jq .
```

```
[
 {
 "Type": "volume",
 "Name": "my-vol",
 "Source": "/var/lib/docker/volumes/my-vol/_data",
 "Destination": "/app",
 "Driver": "local",
 "Mode": "z",
 "RW": true,
 "Propagation": ""
 }
]
```

- 

```
$ docker stop web-server
```

```
$ docker rm web-server
```

```
$ docker volume rm my-vol
```

## How to configure bind mounts

- 

```
$ docker run -d \ > --name web-server \ > --mount
type=bind,source="$(pwd)",target=/app \ > ubuntu/apache2
6f5378e34d6c6811702e16d047a5a80f18adbd9d8a14b11050ae3c3353bf8d2a
```

- 

```
$ docker inspect web-server --format '{{ json .Mounts }}' | jq .
[
 {
 "Type": "bind",
 "Source": "/root",
 "Destination": "/app",
 "Mode": "",
 "RW": true,
 "Propagation": "rprivate"
 }
]
```

-

```
$ docker stop web-server
```

```
$ docker rm web-server
```

## How to configure tmpfs

- 

```
$ docker run --name web-server -d \ > --mount type=tmpfs,target=/app \ > ubuntu/apache2
```

```
03483cc28166fc5c56317e4ee71904941ec5942071e7c936524f74d732b6a24c
```

- 

```
$ docker inspect web-server --format '{{ json .Mounts }}' | jq .
```

```
[
 {
 "Type": "tmpfs",
 "Source": "",
 "Destination": "/app",
 "Mode": "",
 "RW": true,
 "Propagation": ""
 }
]
```

## Choosing the right containerd snapshotter

-

**copy**

**user**

**host**

**dir**

export MOUNT\_POINT="/var/lib/containerd/io.containerd.snapshotter.v1"

Install ZFS:

```
```{terminal}
:copy:
:user:
:host:
:dir:
apt install zfsutils-linux -y
```

Create a 5GB file to use as a ZFS pool:

```
$ fallocate -l 5G /zfs-pool.img
```

Create the pool:

```
$ sudo zpool create mypool /zfs-pool.img
```

Create the ZFS dataset and mount it to your chosen mount point:

```
$ sudo zfs create -o
mountpoint=$MOUNT_POINT mypool/docker
```

Verify that it mounted successfully:

```
$ sudo zfs list
```

NAME	USED	AVAIL	REFER
MOUNTPOINT			
mypool	162K	4.36G	24K /mypool
mypool/docker	39K	4.36G	39K /var/lib/
containerd/io.containerd.snapshotter.v1.zfs			

```
$ docker info | grep "Storage Driver"
```

```
Storage Driver: overlayfs
```

```
$ docker info | grep "driver-type"
driver-type: io.containerd.snapshotter.v1
```

-

- 1.

```
$ systemctl stop docker
```

- 2.

```
[plugins."io.containerd.snapshotter.v1.zfs"]
root_path = "/my/cool/new/mount/point.zfs"
```

- 3.

```
{
  "storage-driver": "zfs"
}
```

- 4.

```
$ systemctl restart docker
```

-

```
$ docker info | grep "Storage Driver" > Storage Driver: zfs
```

```
$ docker info | grep "driver-type"
driver-type: io.containerd.snapshotter.v1
```

Configuring networking

-

```
$ docker network create --driver bridge my-net
D84efaca11d6f643394de31ad8789391e3ddf29d46faecf0661849f5ead239f7
```

-

```
$ docker network ls
```

NETWORK ID	NAME	DRIVER	SCOPE
1f55a8891c4a	bridge	bridge	local
9ca94be2c1a0	host	host	local
d84efaca11d6	my-net	bridge	local
5d300e6a07b1	none	null	local

```
$ docker network inspect my-net
```

```
[
  {
    "Name": "my-net",
    "Id": "d84efaca11d6f643394de31ad8789391e3ddf29d46faecf0661849f5ead239f7",
    "Created": "2023-10-25T22:18:52.972569338Z",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": {},
      "Config": [
        {
          "Subnet": "172.18.0.0/16",
          "Gateway": "172.18.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Containers": {},
    "Options": {},
    "Labels": {}
  }
]
```


Connecting a new container to an existing network

```
$ docker run -d --name c1 --network my-net ubuntu/apache2
C7aa78f45ce3474a276ca3e64023177d5984b3df921aadf97e221da8a29a891e
```

•

```
$ docker inspect c1 --format '{{ json .NetworkSettings }}' | jq .
{
  "Bridge": "",
  "SandboxID":
"ee1cc10093fd5d4a30c056cef47abbfa564e770272e1e5f681525fdd85555",
  "HairpinMode": false,
  "LinkLocalIPv6Address": "",
  "LinkLocalIPv6PrefixLen": 0,
  "Ports": {
    "80/tcp": null
  },
  "SandboxKey": "/var/run/docker/netns/ee1cc10093fd",
  "SecondaryIPAddresses": null,
  "SecondaryIPv6Addresses": null,
  "EndpointID": "",
  "Gateway": "",
  "GlobalIPv6Address": "",
  "GlobalIPv6PrefixLen": 0,
  "IPAddress": "",
  "IPPrefixLen": 0,
  "IPv6Gateway": "",
  "MacAddress": "",
  "Networks": {
    "my-net": {
      "IPAMConfig": null,
      "Links": null,
      "Aliases": ["c7aa78f45ce3"],
      "NetworkID":
"d84efaca11d6f643394de31ad8789391e3ddf29d46faecf0661849f5ead239f7",
      "EndpointID":
"1cb76d44a484d302137bb4b042c8142db8e931e0c63f44175a1aa75ae8af9cb5",
      "Gateway": "172.18.0.1",
      "IPAddress": "172.18.0.2",
      "IPPrefixLen": 16,
      "IPv6Gateway": "",
      "GlobalIPv6Address": "",
      "GlobalIPv6PrefixLen": 0,
      "MacAddress": "02:42:ac:12:00:02",
      "DriverOpts": null
    }
  }
}
```

Connecting a running container to an existing network

-

- 1.

```
$ docker run -d --name c2 ubuntu/nginx  
Fea22fbb6e3685eae28815f3ad8c8a655340ebcd6a0c13f3aad0b45d71a20935
```

- 2.

```
$ docker network connect my-net c2
```

```
$ docker inspect c2 --format '{{ json .NetworkSettings }}' | jq .
```

```
{
  "Bridge": "",
  "SandboxID":
"82a7ea6efd679dffcc3e4392e0e5da61a8cccf33dd78eb5381c9792a4c01f366",
  "HairpinMode": false,
  "LinkLocalIPv6Address": "",
  "LinkLocalIPv6PrefixLen": 0,
  "Ports": {
    "80/tcp": null
  },
  "SandboxKey": "/var/run/docker/netns/82a7ea6efd67",
  "SecondaryIPAddresses": null,
  "SecondaryIPv6Addresses": null,
  "EndpointID":
"490c15cf3bcb149dd8649e3ac96f71addd13f660b4ec826dc39e266184b3f65b",
  "Gateway": "172.17.0.1",
  "GlobalIPv6Address": "",
  "GlobalIPv6PrefixLen": 0,
  "IPAddress": "172.17.0.3",
  "IPPrefixLen": 16,
  "IPv6Gateway": "",
  "MacAddress": "02:42:ac:11:00:03",
  "Networks": {
    "bridge": {
      "IPAMConfig": null,
      "Links": null,
      "Aliases": null,
      "NetworkID":
"1f55a8891c4a523a288aca8881dae0061f9586d5d91c69b3a74e1ef3ad1bfcf4",
      "EndpointID":
"490c15cf3bcb149dd8649e3ac96f71addd13f660b4ec826dc39e266184b3f65b",
      "Gateway": "172.17.0.1",
      "IPAddress": "172.17.0.3",
      "IPPrefixLen": 16,
      "IPv6Gateway": "",
      "GlobalIPv6Address": "",
      "GlobalIPv6PrefixLen": 0,
      "MacAddress": "02:42:ac:11:00:03",
      "DriverOpts": null
    },
    "my-net": {
      "IPAMConfig": {},
      "Links": null,
      "Aliases": ["fea22fbb6e36"],
      "NetworkID":
"d84efaca11d6f643394de31ad8789391e3ddf29d46faecf0661849f5ead239f7",
      "EndpointID":
"17856b7f6902db39ff6ab418f127d75d8da597fdb8af0a6798f35a94be0cb805",
      "Gateway": "172.18.0.1",
      "IPAddress": "172.18.0.3",
      "IPPrefixLen": 16,
      "IPv6Gateway": "",
      "GlobalIPv6Address": "",
      "GlobalIPv6PrefixLen": 0,
      "MacAddress": "02:42:ac:12:00:03",
      "DriverOpts": {}
    }
  }
}
```

Modifying the default network “bridge”

-

```
$ vim /etc/docker/daemon.json

{
  "bip": "192.168.1.1/24",
  "fixed-cidr": "192.168.1.0/25",
  "fixed-cidr-v6": "2001:db8::/64",
  "mtu": 1500,
  "default-gateway": "192.168.1.254",
  "default-gateway-v6": "2001:db8:abcd::89",
  "dns": ["10.20.1.2", "10.20.1.3"]
}
```

-

```
$ systemctl restart docker
```

-

```
$ docker network inspect bridge
```

Exposing a container port to the host

-

-

-

-

```
$ docker run -d --name web-server --publish 8080:80 ubuntu/nginx
f451aa1990db7d2c9b065c6158e2315997a56a764b36a846a19b1b96ce1f3910
```

-

```
$ docker inspect web-server --format '{{ json .NetworkSettings.Ports }}'
| jq .
{
  "80/tcp": [
    {
      "HostIp": "0.0.0.0",
      "HostPort": "8080"
    },
    {
      "HostIp": ":::",
      "HostPort": "8080"
    }
  ]
}
```

hostname

DNS

Managing logs

Modifying the logging driver via the docker daemon file

-

```
$ vim /etc/docker/daemon.json

{
  "log-driver": "journald"
}
```

Modifying the logging driver when creating a container

-

```
$ docker run -d --name web-server --log-driver=journald ubuntu/nginx
1c08b667f32d8b834f0d9d6320721e07de5f22168cfc8a024d6e388daf486dfa
```

-

```
$ docker inspect web-server --format '{{ json .HostConfig.LogConfig }}'
| jq .
{
  "Type": "journald",
  "Config": {}
}
```

-

```
$ docker logs web-server

/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to
perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-
templates.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
```

```
{
  "log-driver": "syslog",
  "log-opts": {
    "cache-disabled": "true",
    "syslog-address": "udp://1.2.3.4:1111"
  }
}
```

```
$ docker logs web-server
```

```
Error response from daemon: configured logging driver does not support reading
```

Rootless Docker

Prerequisites

```
$ sudo apt install rootlesskit uidmap slirp4netns fuse-overlayfs
```

-
-

⁶⁷⁵ <https://docs.docker.com/engine/logging/configure/>

-
-

```
$ grep ^$(whoami): /etc/subuid
```

```
$ grep ^$(whoami): /etc/subgid
```

```
yourusername:XXXXXX:65536
```

Installation

Note:

If you are already running the rootful Docker daemon, disable it before proceeding:

```
$ sudo systemctl disable --now docker.service docker.socket
```

```
$ sudo rm /var/run/docker.sock
```

```
$ /usr/share/docker.io/contrib/dockerd-rootless-setuptool.sh install
```

```
$ docker info | grep Context
```

```
Context: rootless
```


Persisting the rootless daemon

```
$ systemctl --user enable docker
```

```
$ loginctl enable-linger $(whoami)
```

Expose Docker API socket

Over SSH (recommended)

```
$ ssh -l <REMOTEUSER> <REMOTEHOST> 'echo $DOCKER_HOST' >  
unix:///run/user/1001/docker.sock
```

```
$ docker -H ssh://<REMOTEUSER>@<REMOTEHOST> run ...
```

```
$ docker context create remote --docker  
"host=ssh://<REMOTEUSER>@<REMOTEHOST>"
```

```
$ docker context use remote
```

Over TCP

```
$ DOCKERD_ROOTLESS_ROOTLESSKIT_FLAGS="-p 0.0.0.0:2376:2376/tcp"  
/usr/share/docker.io/contrib/docker-rootless.sh \ > -H tcp://0.0.0.0:2376 \ >  
--tlsverify --tlscacert=ca.pem --tlscert=cert.pem --tlskey=key.pem
```

Expose privileged ports

```
$ sudo setcap cap_net_bind_service=ep /usr/bin/rootlesskit
```

```
$ systemctl --user restart docker
```

Further reading

- *Docker storage, networking, and logging*

How to run rocks on your server

Deploying rocks with Docker

rocks⁶⁷⁶

Docker

```
$ sudo apt-get install -y docker.io docker-compose
```

possible parameters⁶⁷⁷

“Usage” section⁶⁷⁸

```
$ sudo docker run -d --name grafana-container -e TZ=UTC -p 3000:3000
ubuntu/grafana:10.3.3-22.04_stable

Unable to find image 'ubuntu/grafana:10.3.3-22.04_stable' locally
10.3.3-22.04_stable: Pulling from ubuntu/grafana
bccd10f490ab: Already exists
549078d9d057: Pull complete
6ef870aa8500: Pull complete
2b475da7ccbd: Pull complete
Digest: sha256:df566ef90ecb14267a459081949ee7b6693fa573b97a7134a9a6722207275caa
Status: Downloaded newer image for ubuntu/grafana:10.3.3-22.04_stable
356e623ef2c16bc7d810bddccad8d7980f9c633aefc3a88bc8761eac4e1b1c50
```

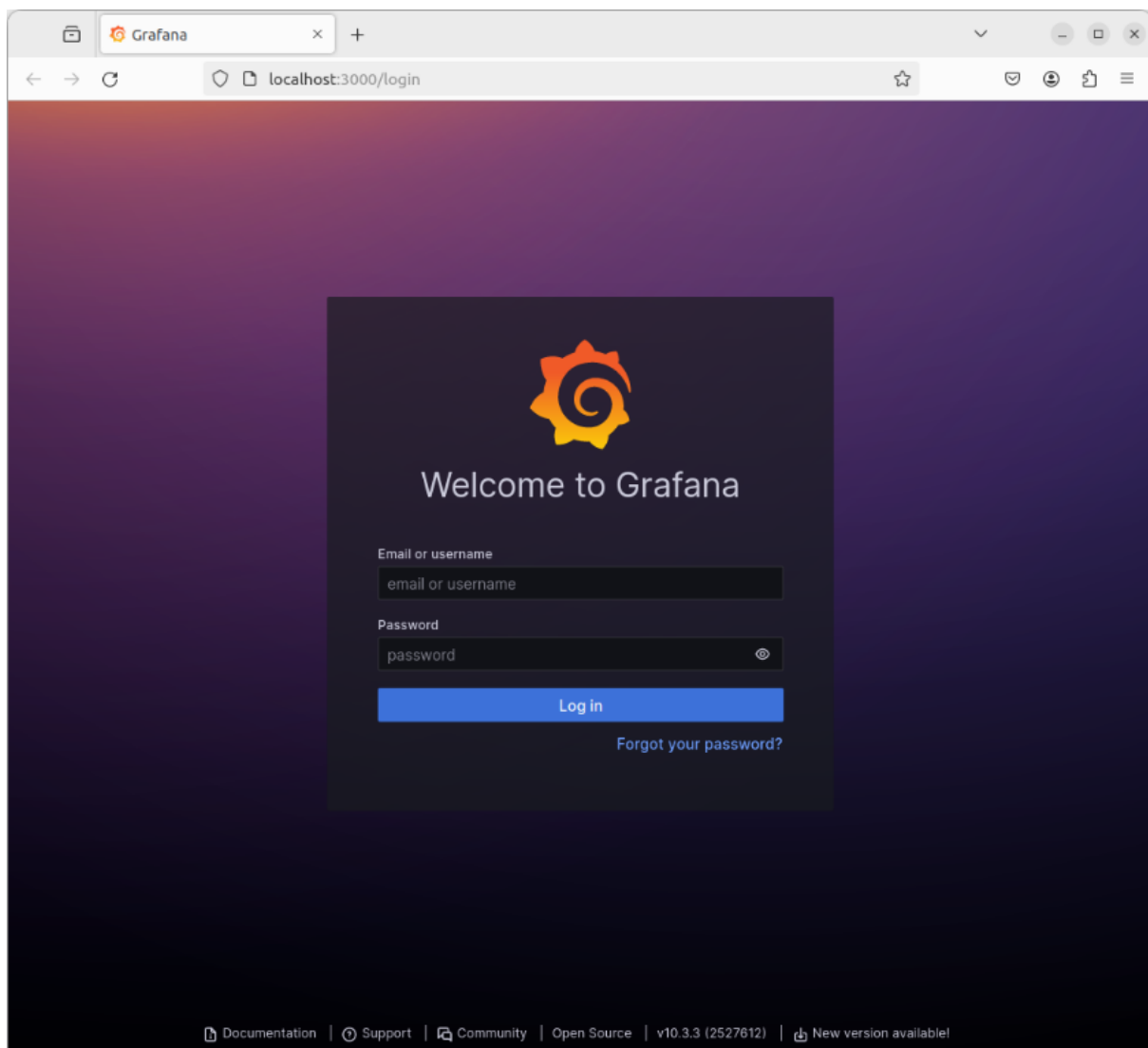
⁶⁷⁶ <https://documentation.ubuntu.com/rockcraft/latest/explanation/rocks/>

⁶⁷⁷ <https://docs.docker.com/reference/cli/docker/container/run/>

⁶⁷⁸ <https://hub.docker.com/r/ubuntu/grafana>

-
-
-
-
-

snap "channels"⁶⁷⁹



```
$ curl -s http://localhost:3000/login | grep "<title>"
```

⁶⁷⁹ <https://snapcraft.io/docs/channels>

```
<title>Grafana</title>
```

```
$ sudo docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED
STATUS	PORTS	NAMES	
356e623ef2c1	ubuntu/grafana:10.3.3-22.04_stable	"/bin/pebble enter -..."	17
minutes ago	Up 17 minutes	0.0.0.0:3000->3000/tcp, :::3000->3000/tcp	grafana-container

```
$ sudo docker stop grafana-container
```

```
$ sudo docker rm grafana-container
```

[Grafana rock's documentation](#)⁶⁸⁰

Multi-container deployment

[Grafana](#)⁶⁸²

[Prometheus](#)⁶⁸¹

```
global:
  scrape_interval: 1m

scrape_configs:
  job_name: 'prometheus'
  scrape_interval: 1m
  static_configs:
    - targets: ['localhost:9090']
```

⁶⁸⁰ <https://hub.docker.com/r/ubuntu/grafana>

⁶⁸¹ <https://hub.docker.com/r/ubuntu/prometheus>

⁶⁸² <https://hub.docker.com/r/ubuntu/grafana>

```
services:
  grafana:
    image: ubuntu/grafana:10.3.3-22.04_stable
    container_name: grafana-container
    environment:
      TZ: UTC
    ports:
      - "3000:3000"
  prometheus:
    image: ubuntu/prometheus:2.49.1-22.04_stable
    container_name: prometheus-container
    environment:
      TZ: UTC
    ports:
      - "9090:9090"
    volumes:
      - ./prometheus.yml:/etc/prometheus/prometheus.yml
```

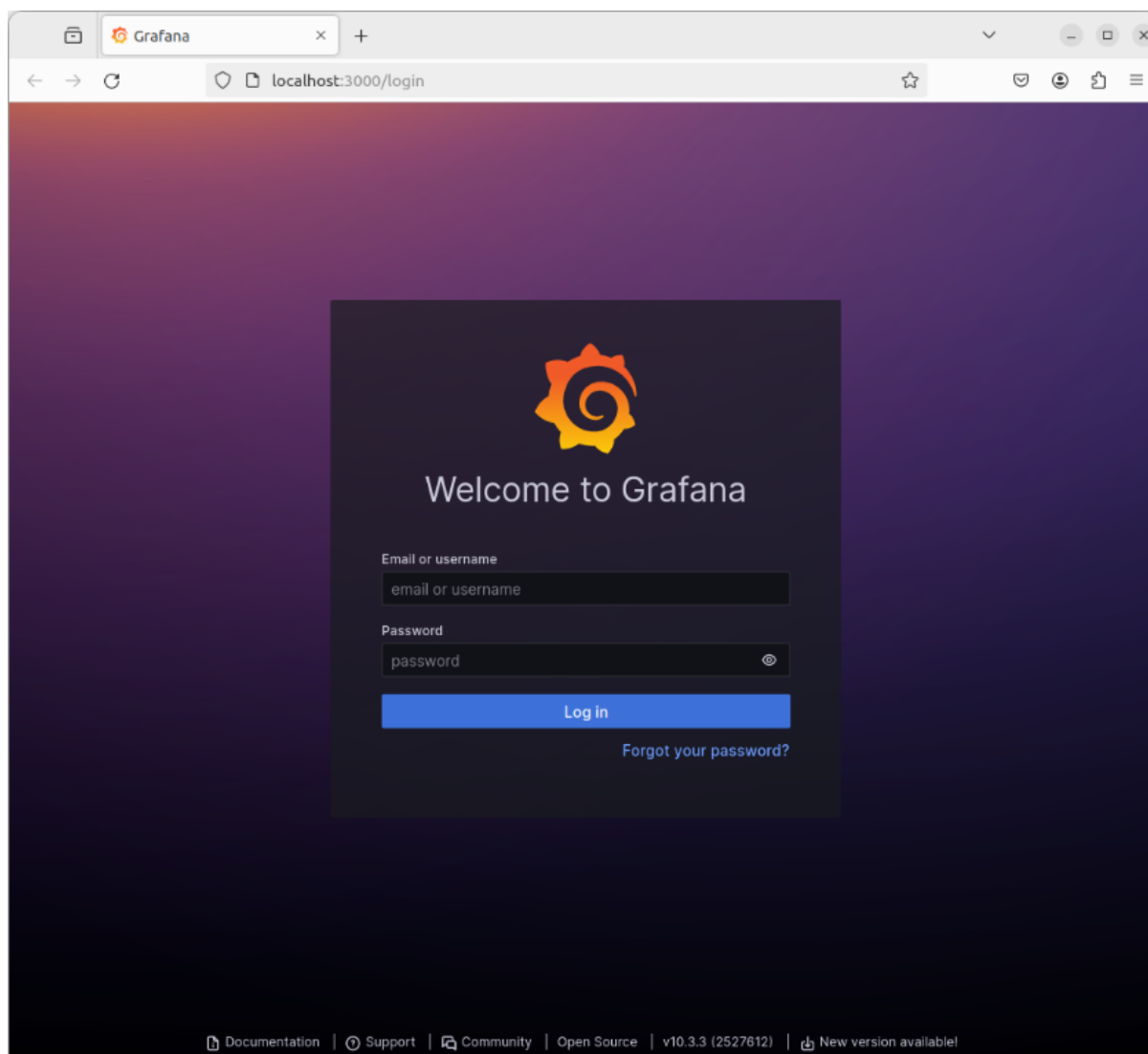
```
$ sudo docker compose up -d

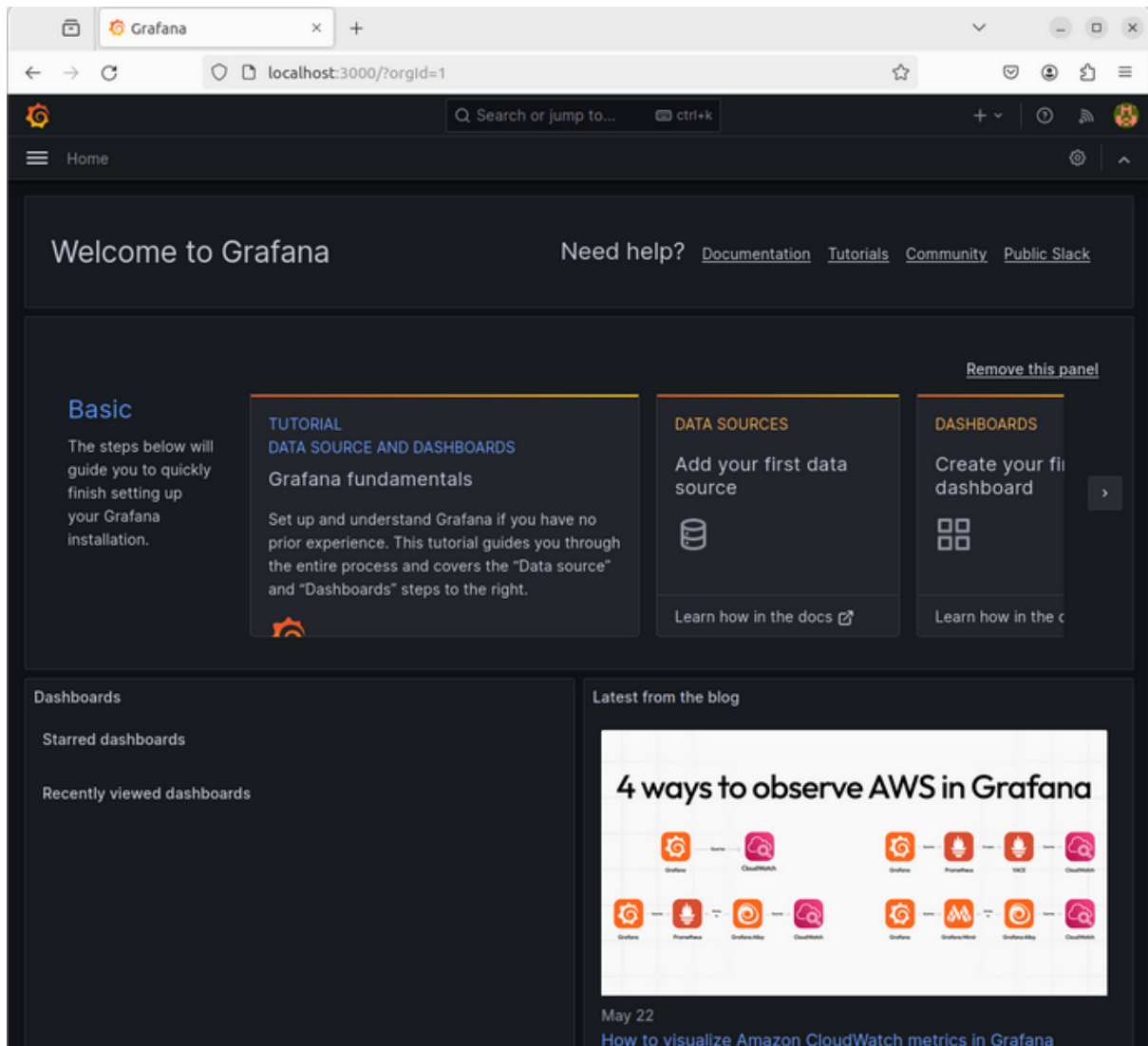
[+] Running 10/10
 ✓ grafana Pulled
   ✓ bccd10f490ab Already exists
   ✓ 549078d9d057 Pull complete
   ✓ 6ef870aa8500 Pull complete
   ✓ 2b475da7ccbd Pull complete
 ✓ prometheus Pulled
   ✓ a8b1c5f80c2d Already exists
   ✓ f021062473aa Pull complete
   ✓ 9c6122d12d1d Pull complete
   ✓ 274b56f68abe Pull complete
[+] Running 3/3
 ✓ Network compose_default      Created
 ✓ Container prometheus-container Started
 ✓ Container grafana-container  Started
```

```
$ sudo docker compose ps
```

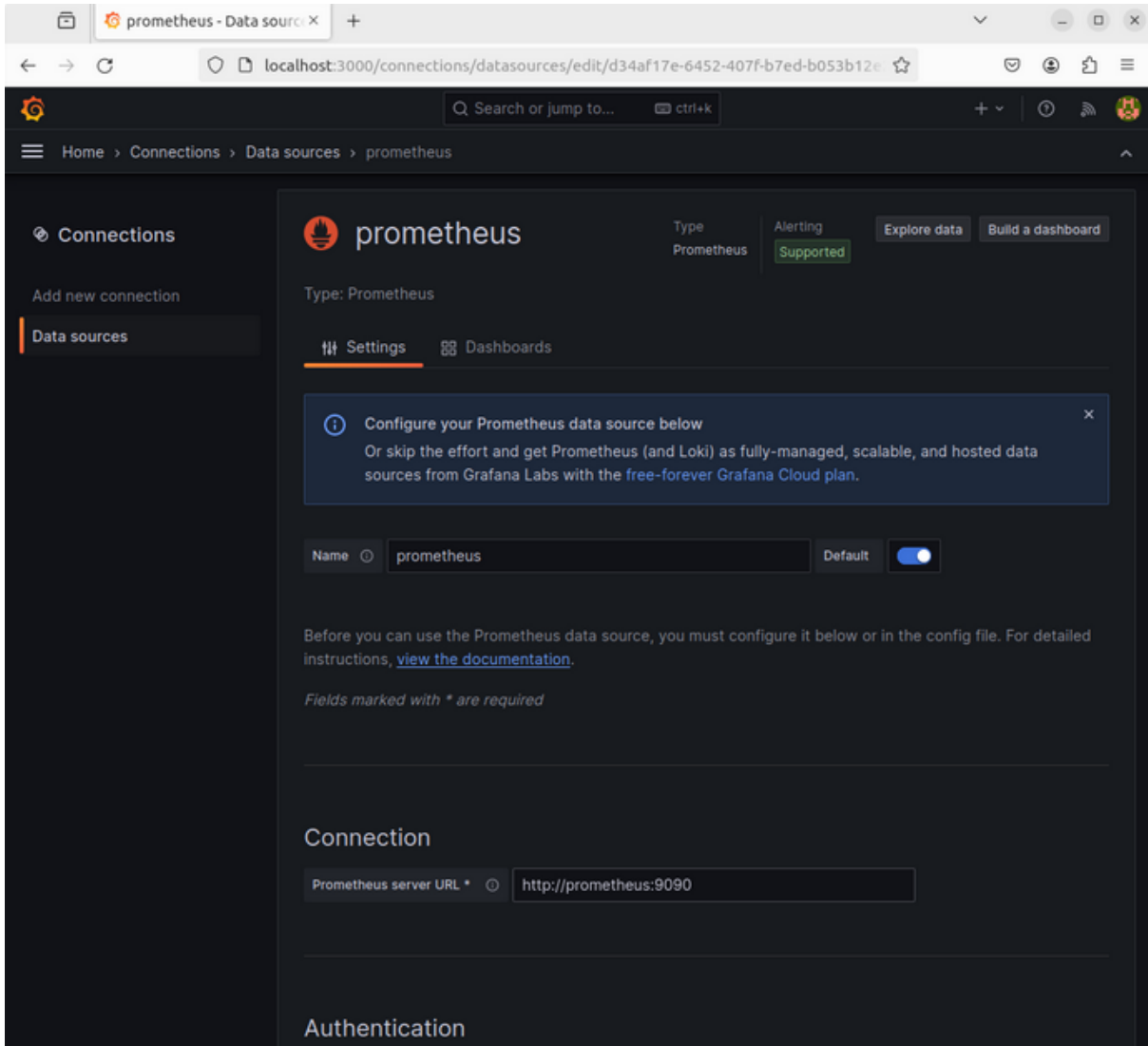
NAME	SERVICE	IMAGE	CREATED	STATUS	PORTS	COMMAND
grafana-container	grafana	ubuntu/grafana:10.3.3-22.04_stable	3 seconds ago	Up 3 seconds	0.0.0.0:3000->3000/tcp, :::3000->3000/tcp	"/bin/pebble enter -...
prometheus-container	prometheus	ubuntu/prometheus:2.49.1-22.04_stable	3 seconds ago	Up 3 seconds	0.0.0.0:9090->9090/tcp, :::9090->9090/tcp	"/bin/pebble enter -...

<http://localhost:3000>





<http://prometheus:9090>



prometheus - Data source

localhost:3000/connections/datasources/edit/d34af17e-6452-407f-b7ed-b053b12e

Search or jump to... ctrl+k

Home > Connections > Data sources > prometheus

Connections

Add new connection

Data sources

prometheus

Type: Prometheus

Alerting: Supported

Explore data Build a dashboard

Type: Prometheus

Settings Dashboards

Configure your Prometheus data source below

Or skip the effort and get Prometheus (and Loki) as fully-managed, scalable, and hosted data sources from Grafana Labs with the [free-forever Grafana Cloud plan](#).

Name prometheus Default ☒

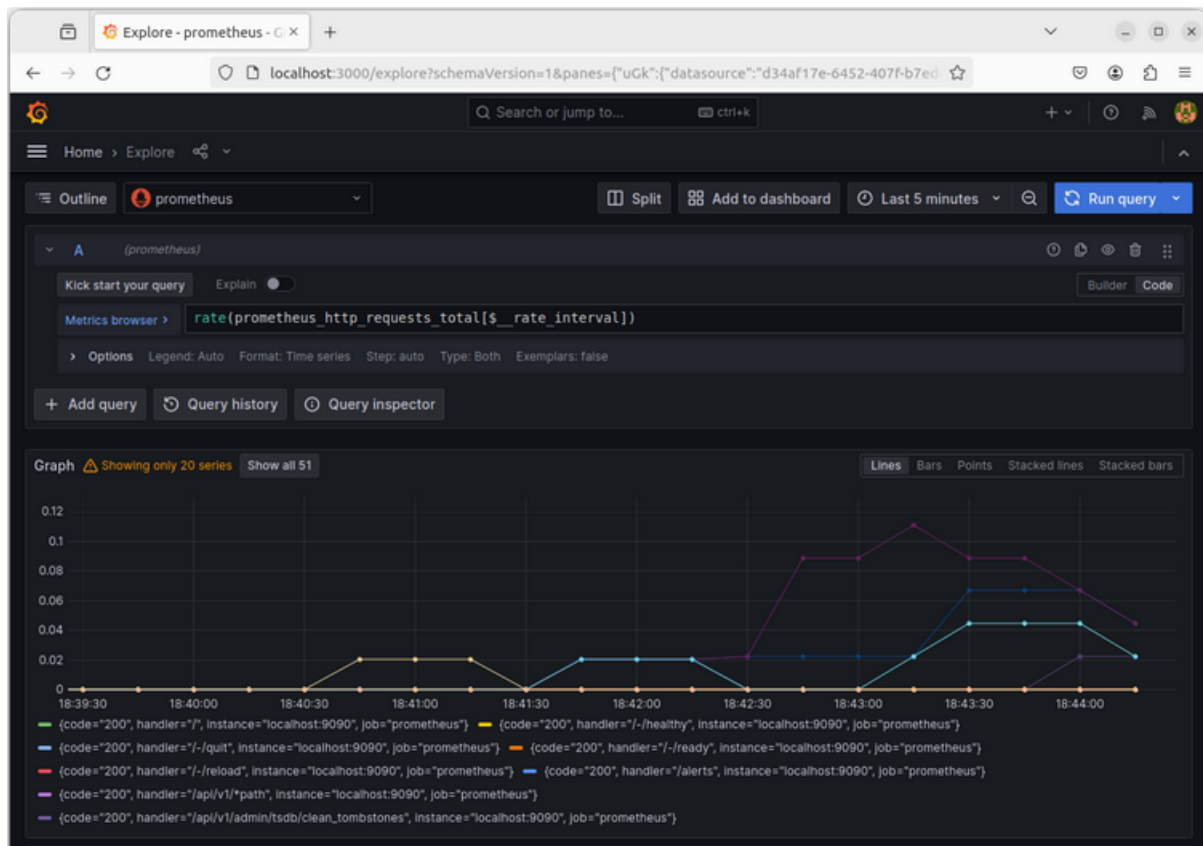
Before you can use the Prometheus data source, you must configure it below or in the config file. For detailed instructions, [view the documentation](#).

Fields marked with * are required

Connection

Prometheus server URL * http://prometheus:9090

Authentication



Next steps

MicroK8s⁶⁸³

⁶⁸³ <https://canonical.com/microk8s/docs>

Skopeo for system admins

Skopeo⁶⁸⁴

Install Skopeo

*skopeo(1)*⁶⁸⁵

```
$ sudo apt install skopeo
```

Image names

docker://

docker-daemon:

containers-storage:

Podman⁶⁸⁶

Buildah⁶⁸⁷

dir:

oci: and oci-archive:

OCI

*containers-transport(5)*⁶⁸⁸

Inspect a remote image

*skopeo-inspect(1)*⁶⁸⁹

```
$ skopeo inspect docker://docker.io/ubuntu:26.04
```

⁶⁸⁴ <https://github.com/containers/skopeo>

⁶⁸⁵ <https://manpages.ubuntu.com/manpages/resolute/man1/skopeo.1.html>

⁶⁸⁶ <https://podman.io/>

⁶⁸⁷ <https://buildah.io/>

⁶⁸⁸ <https://manpages.ubuntu.com/manpages/resolute/man5/containers-transport.5.html>

⁶⁸⁹ <https://manpages.ubuntu.com/manpages/resolute/man1/skopeo-inspect.1.html>

```
{
  "Name": "docker.io/library/ubuntu",
  "Digest":
    "sha256:2260313b31c8c011cd2eebe728008efac1b3982be73eb71348ea2648d2c0e09b",
  "RepoTags": [
    "24.04",
    "25.10",
    "26.04",
    "latest"
  ],
  "Created": "2026-08-17T09:00:47.315779976Z",
  "DockerVersion": "",
  "Labels": {
    "org.opencontainers.image.created": "2026-08-17T09:02:45.677319+00:00",
    "org.opencontainers.image.title": "ubuntu",
    "org.opencontainers.image.version": "26.04"
  },
  "Architecture": "amd64",
  "Os": "linux",
  "Layers": [
    "sha256:06e9d71331fb2b620a4f6c8064e0f84b284bb69a42c7c57b1c962bd4a4cdee76",
    "sha256:f3db1cd940786339b09d8a60e47c66fea9502d788e6fab5bec91a4a77d4ced1c"
  ],
  "LayersData": [
    {
      "MIMEType": "application/vnd.oci.image.layer.v1.tar+gzip",
      "Digest":
        "sha256:06e9d71331fb2b620a4f6c8064e0f84b284bb69a42c7c57b1c962bd4a4cdee76",
      "Size": 41569203,
      "Annotations": {
        "ci.umo.uncompressed_blob_size": "111523840"
      }
    },
    {
      "MIMEType": "application/vnd.oci.image.layer.v1.tar+gzip",
      "Digest":
        "sha256:f3db1cd940786339b09d8a60e47c66fea9502d788e6fab5bec91a4a77d4ced1c",
      "Size": 393,
      "Annotations": {
        "ci.umo.uncompressed_blob_size": "10240"
      }
    }
  ],
  "Env": [
    "PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin"
  ]
}
```

*skopeo-list-tags(1)*⁶⁹⁰

```
$ skopeo list-tags docker://docker.io/ubuntu
```

⁶⁹⁰ <https://manpages.ubuntu.com/manpages/resolute/man1/skopeo-list-tags.1.html>

```
{
  "Repository": "docker.io/library/ubuntu",
  "Tags": [
    "22.04",
    "24.04",
    "26.04",
    "latest"
  ]
}
```

Authenticate to a registry

*login(1)*⁶⁹¹

skopeo-

*containers-auth.json(5)*⁶⁹²

```
$ skopeo login registry.example.com

Username: myuser
Password:
Login Succeeded!
```

Copy images

Between registries

*skopeo-copy(1)*⁶⁹³

```
$ skopeo copy \> docker://docker.io/ubuntu:26.04 \>
docker://registry.example.com/mirror/ubuntu:26.04

Getting image source signatures
Copying blob 06e9d71331fb done
Copying blob f3db1cd94078 done
Copying config af52039db3 done
Writing manifest to image destination
```

⁶⁹¹ <https://manpages.ubuntu.com/manpages/resolute/man1/skopeo-login.1.html>

⁶⁹² <https://manpages.ubuntu.com/manpages/resolute/man5/containers-auth.json.5.html>

⁶⁹³ <https://manpages.ubuntu.com/manpages/resolute/man1/skopeo-copy.1.html>

*json(5)*⁶⁹⁴ *containers-registries.d(5)*⁶⁹⁵

containers-policy.

To and from local storage

```
$ skopeo copy docker://docker.io/ubuntu:26.04 docker-daemon:ubuntu:26.04
```

```
$ skopeo copy docker://docker.io/ubuntu:26.04  
oci-archive:ubuntu-26.04.tar:ubuntu:26.04
```

Mirror multiple images

*skopeo-sync(1)*⁶⁹⁶

```
$ skopeo sync --src docker --dest docker docker.io/ubuntu:26.04  
registry.example.com/mirror  
  
INFO[0000] Tag presence check                               imagename="docker.io/  
ubuntu:26.04" tagged=true  
INFO[0000] Copying image ref 1/1                             from="docker://ubuntu:26.04  
" to="docker://registry.example.com/mirror/ubuntu:26.04"  
Getting image source signatures  
Copying blob 06e9d71331fb done  
Copying blob f3db1cd94078 done  
Copying config af52039db3 done  
Writing manifest to image destination  
INFO[0005] Synced 1 images from 1 sources
```

```
$ skopeo sync --src docker --dest dir --scoped docker.io/ubuntu:26.04  
/media/usb
```

⁶⁹⁴ <https://manpages.ubuntu.com/manpages/resolute/man5/containers-policy.json.5.html>

⁶⁹⁵ <https://manpages.ubuntu.com/manpages/resolute/man5/containers-registries.d.5.html>

⁶⁹⁶ <https://manpages.ubuntu.com/manpages/resolute/man1/skopeo-sync.1.html>

Syncing from a YAML file

YAML

```
docker.io:  
  images:  
    ubuntu:  
      - "24.04"  
      - "26.04"  
  images-by-tag-regex:  
    nginx: ^1\.2[0-9]-alpine$
```

```
$ skopeo sync --src yaml --dest docker --dry-run sync.yaml  
registry.example.com/mirror
```

```
INFO[0000] Processing repo registry=docker.io
repo=ubuntu
INFO[0000] Processing repo registry=docker.io
repo=nginx
INFO[0000] Querying registry for image tags registry=docker.io
repo=nginx
INFO[0000] Getting tags image=docker.io/library/
nginx
WARN[0001] Running in dry-run mode
INFO[0001] Would have copied image ref 1/2 from="docker://ubuntu:24.04"
" to="docker://registry.example.com/mirror/ubuntu:24.04"
INFO[0001] Would have copied image ref 2/2 from="docker://ubuntu:26.04"
" to="docker://registry.example.com/mirror/ubuntu:26.04"
INFO[0001] Would have copied image ref 1/10 from="docker://nginx:1.20-
alpine" to="docker://registry.example.com/mirror/nginx:1.20-alpine"
INFO[0001] Would have copied image ref 2/10 from="docker://nginx:1.21-
alpine" to="docker://registry.example.com/mirror/nginx:1.21-alpine"
INFO[0001] Would have copied image ref 3/10 from="docker://nginx:1.22-
alpine" to="docker://registry.example.com/mirror/nginx:1.22-alpine"
INFO[0001] Would have copied image ref 4/10 from="docker://nginx:1.23-
alpine" to="docker://registry.example.com/mirror/nginx:1.23-alpine"
INFO[0001] Would have copied image ref 5/10 from="docker://nginx:1.24-
alpine" to="docker://registry.example.com/mirror/nginx:1.24-alpine"
INFO[0001] Would have copied image ref 6/10 from="docker://nginx:1.25-
alpine" to="docker://registry.example.com/mirror/nginx:1.25-alpine"
INFO[0001] Would have copied image ref 7/10 from="docker://nginx:1.26-
alpine" to="docker://registry.example.com/mirror/nginx:1.26-alpine"
INFO[0001] Would have copied image ref 8/10 from="docker://nginx:1.27-
alpine" to="docker://registry.example.com/mirror/nginx:1.27-alpine"
INFO[0001] Would have copied image ref 9/10 from="docker://nginx:1.28-
alpine" to="docker://registry.example.com/mirror/nginx:1.28-alpine"
INFO[0001] Would have copied image ref 10/10 from="docker://nginx:1.29-
alpine" to="docker://registry.example.com/mirror/nginx:1.29-alpine"
INFO[0001] Would have synced 12 images from 2 sources
```

Delete an image from a registry

*skopeo-delete(1)*⁶⁹⁷

```
$ skopeo delete docker://registry.example.com/mirror/ubuntu:26.04
```

⁶⁹⁷ <https://manpages.ubuntu.com/manpages/resolute/man1/skopeo-delete.1.html>

Further reading

- [Skopeo upstream documentation](#)⁶⁹⁸
- *Container tools in the Ubuntu space*

See also

- *Virtualisation and containers*

High Availability

High availability

- *Install a Distributed Replicated Block Device*

Distributed Replicated Block Device (DRBD)

Install DRBD

```
$ sudo apt install drbd-utils
```

Note:

If you are using the **virtual kernel** as part of a virtual machine you will need to manually compile the drbd module. It may be easier to install the `linux-modules-extra-$(uname -r)` package inside the virtual machine.

⁶⁹⁸ <https://github.com/containers/skopeo>

Configure DRBD

filesystem

DNS

guide to DNS

```
global { usage-count no; }
common { syncer { rate 100M; } }
resource r0 {
    protocol C;
    startup {
        wfc-timeout 15;
        degr-wfc-timeout 60;
    }
    net {
        cram-hmac-alg sha1;
        shared-secret "secret";
    }
    on drbd01 {
        device /dev/drbd0;
        disk /dev/sdb1;
        address 192.168.0.1 7788;
        meta-disk internal;
    }
    on drbd02 {
        device /dev/drbd0;
        disk /dev/sdb1;
        address 192.168.0.2 7788;
        meta-disk internal;
    }
}
```

Note:

There are many other options in `/etc/drbd.conf`, but for this example the default values are enough.

```
$ scp /etc/drbd.conf drbd02:~
```

```
$ sudo mv drbd.conf /etc/
```

```
$ sudo drbdadm create-md r0
```

```
$ sudo systemctl start drbd.service
```

```
$ sudo drbdadm -- --overwrite-data-of-peer primary all
```

```
$ watch -n1 cat /proc/drbd
```

```
$ sudo mkfs.ext3 /dev/drbd0
```

```
$ sudo mount /dev/drbd0 /srv
```

Testing

```
$ sudo cp -r /etc/default /srv
```

```
$ sudo umount /srv
```

```
$ sudo drbdadm secondary r0
```

```
$ sudo drbdadm primary r0
```

```
$ sudo mount /dev/drbd0 /srv
```

Further reading

- [DRBD web site](#)⁶⁹⁹
- [`drbd.conf\(5\)`](#)⁷⁰⁰
- [`drbdadm\(8\)`](#)⁷⁰¹

See also

- [High Availability](#)
- [High availability](#)

Observability

[Canonical Observability Stack](#)⁷⁰²

Observability

[Canonical Observability Stack](#)⁷⁰³

- [How to migrate from LMA to COS Lite](#)⁷⁰⁴
- [Integrate COS Lite with un-charmed applications](#)⁷⁰⁵
- [The OpenTelemetry Collector snap](#)⁷⁰⁶

Classical options

⁶⁹⁹ <https://linbit.com/linbit-software-download-page-for-linstor-and-drbd-linux-driver/>

⁷⁰⁰ <https://manpages.ubuntu.com/manpages/resolute/man5/drbd.conf.5.html>

⁷⁰¹ <https://manpages.ubuntu.com/manpages/resolute/man8/drbdadm.8.html>

⁷⁰² <https://charmhub.io/topics/canonical-observability-stack>

⁷⁰³ <https://documentation.ubuntu.com/observability/track-2/>

⁷⁰⁴ <https://documentation.ubuntu.com/observability/track-2/how-to/migrate-lma-to-cos-lite/>

⁷⁰⁵ <https://documentation.ubuntu.com/observability/track-2/how-to/integrating-cos-lite-with-uncharmed-applications/>

⁷⁰⁶ <https://snapcraft.io/opentelemetry-collector>

Set up your LMA stack

Note:

LMA to COS

The LMA stack is being succeeded by the Canonical Observability Stack (COS). While the current LMA still works, most users are recommended to consider COS instead. For more information, refer to [this COS topic](#)⁷⁰⁷. In environments with more limited resources, there is also [COS lite](#)⁷⁰⁸.

⁷⁰⁷ <https://charmhub.io/topics/canonical-observability-stack/>

⁷⁰⁸ <https://charmhub.io/topics/canonical-observability-stack/editions/lite>

Architectural overview

Telegraf⁷⁰⁹

Prometheus⁷¹⁰

Grafana⁷¹¹

Prometheus Alertmanager⁷¹²

Alertmanager

Getting started

Note:

For clarity, we'll refer to these two hosts as named: `workload` and `monitor`. If you use other *hostnames*, substitute your preferred names as we go through this guide.

⁷⁰⁹ <https://docs.influxdata.com/telegraf/v1/>

⁷¹⁰ <https://prometheus.io/docs/introduction/overview/>

⁷¹¹ <https://grafana.com/docs/>

⁷¹² <https://prometheus.io/docs/alerting/latest/alertmanager/>

Ports

Prometheus	monitor:9090
Alertmanager	monitor:9093
Grafana	monitor:3000
Telegraf	workload:9273

Set up the Workload node

```
$ lxc launch ubuntu:20.10 workload
```

```
Creating workload  
Starting workload
```

```
$ lxc exec workload -- bash
```

```
workload:~# apt update && apt install telegraf
```

```
[[inputs.cpu]]  
  percpu = true  
  totalcpu = true  
  collect_cpu_time = false  
  report_active = false
```

```
[[outputs.prometheus_client]]
  listen = "workload:9273"
  metric_version = 2

#[[outputs.influxdb]]
```

```
workload:~# systemctl restart telegraf
workload:~# systemctl status telegraf
■ telegraf.service - The plugin-driven server agent for reporting metrics into InfluxDB
   Loaded: loaded (/lib/systemd/system/telegraf.service; enabled; vendor preset:
   enabled)
   Active: active (running) since Sat 2020-10-31 02:17:57 UTC; 6s ago
     Docs: https://github.com/influxdata/telegraf
   Main PID: 2562 (telegraf)
      Tasks: 17 (limit: 77021)
     Memory: 42.2M
    CGroup: /system.slice/telegraf.service
            └─2562 /usr/bin/telegraf -config /etc/telegraf/telegraf.conf -
config-directory /etc/telegraf/telegraf.d

...I! Loaded inputs: swap system cpu disk diskio kernel mem processes
...I! Loaded outputs: http prometheus_client
...I! [agent] Config: Interval:10s, Quiet:false, Hostname:"workload", Flush Interval:10s
...I! [outputs.prometheus_client] Listening on http://127.0.0.1:9273/metrics
```

```
workload:~# wget -O- http://workload:9273/metrics

# HELP cpu_usage_guest Telegraf collected metric
# TYPE cpu_usage_guest gauge
cpu_usage_gues{cpu="cpu-total",host="workload"} 0
cpu_usage_gues{cpu="cpu0",host="workload"} 0
cpu_usage_gues{cpu="cpu1",host="workload"} 0
cpu_usage_gues{cpu="cpu10",host="workload"} 0
...
cpu_usage_idle{cpu="cpu-total",host="workload"} 92.74914376428686
cpu_usage_idle{cpu="cpu0",host="workload"} 86.72897196325539
cpu_usage_idle{cpu="cpu1",host="workload"} 90.11857707405758
cpu_usage_idle{cpu="cpu10",host="workload"} 95.95141700494543
```

Set up the Monitor node

```
$ lxc launch ubuntu:20.10 monitor
```

```
Creating monitor  
Starting monitor
```

```
$ lxc exec monitor -- bash
```

```
monitor:~# ip addr | grep 'inet .* global'  
inet 10.69.244.104/24 brd 10.69.244.255 scope global dynamic eth0
```

```
monitor:~# wget -O- http://workload:9273/metrics
```

```
monitor:~# apt install snapd
```

Install Prometheus

```
monitor:~# snap install prometheus  
monitor:~# snap install prometheus-alertmanager
```

```
monitor:~# snap services  
Service          Startup Current Notes  
lxd.activate      enabled inactive -  
lxd.daemon        enabled inactive socket-activated  
prometheus.prometheus enabled active -  
prometheus-alertmanager.alertmanager enabled active -
```

```
visualizer:~# ss -tulpn | grep prometheus  
tcp    LISTEN  0      128          *:9090      *:.*      users:(("prometheus",pid=618,fd=8))
```

```
monitor:~# journalctl | grep "prometheus.*config"
...
...msg="Completed loading of configuration file" filename=/var/snap/prometheus/32/
prometheus.yml
```

```
scrape_configs:
  # The job name is added as a label `job=<job_name>` to any timeseries scraped from this
  # config.
  - job_name: 'prometheus'

    # metrics_path defaults to '/metrics'
    # scheme defaults to 'http'.

    static_configs:
      - targets: ['localhost:9090']

  - job_name: 'telegraf'
    static_configs:
      - targets: ['workload:9273']
```

```
monitor:~# snap restart prometheus
```


Prometheus Time Series Collection and Processing Server - Mozilla Firefox

Prometheus Time Series x +

10.69.244.104:9090/graph

Prometheus Alerts Graph Status Help

☐ Enable query history [Try experimental React UI](#)

Expression (press Shift+Enter for newlines)

Execute - insert metric at cursor -

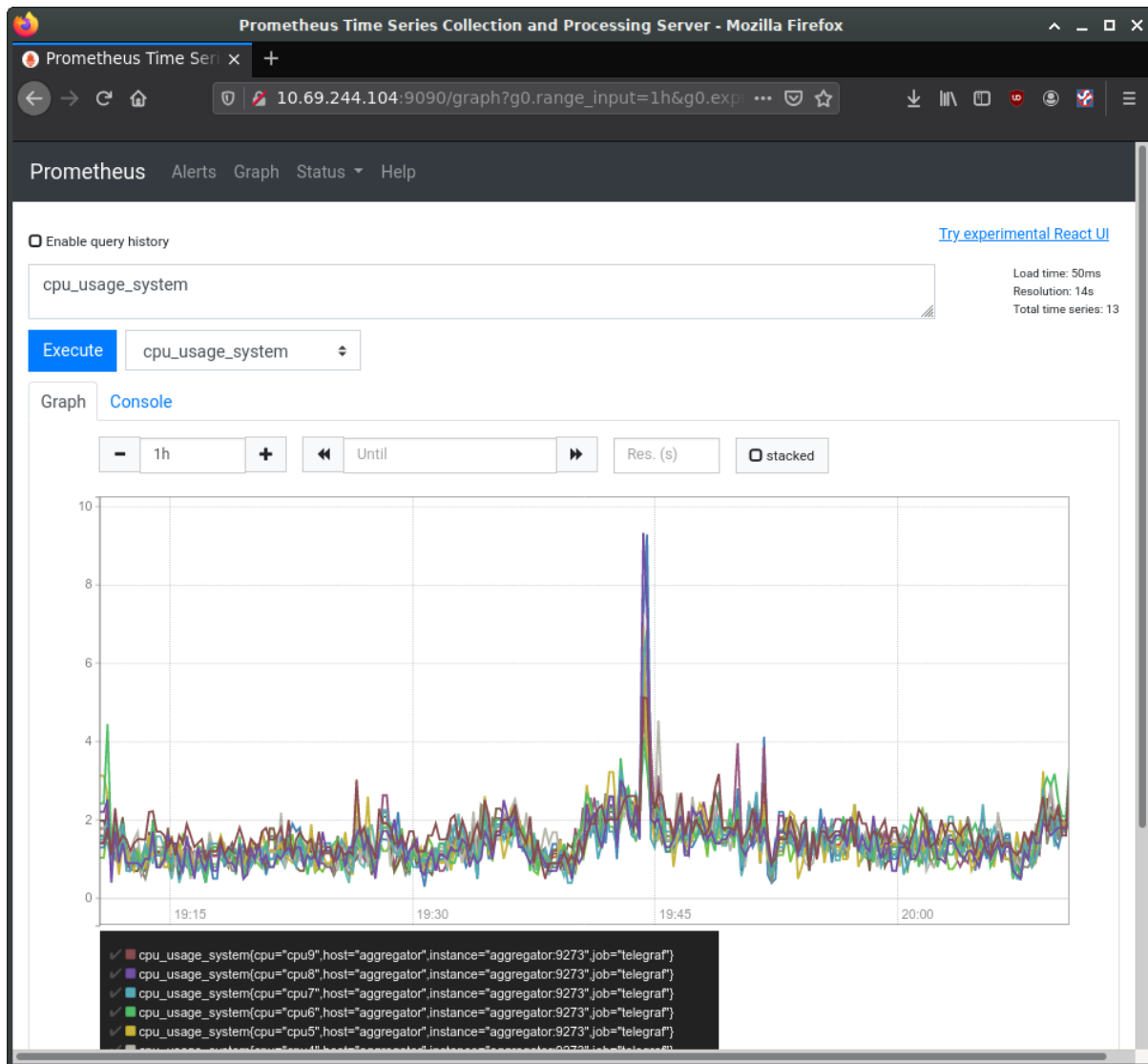
Graph Console

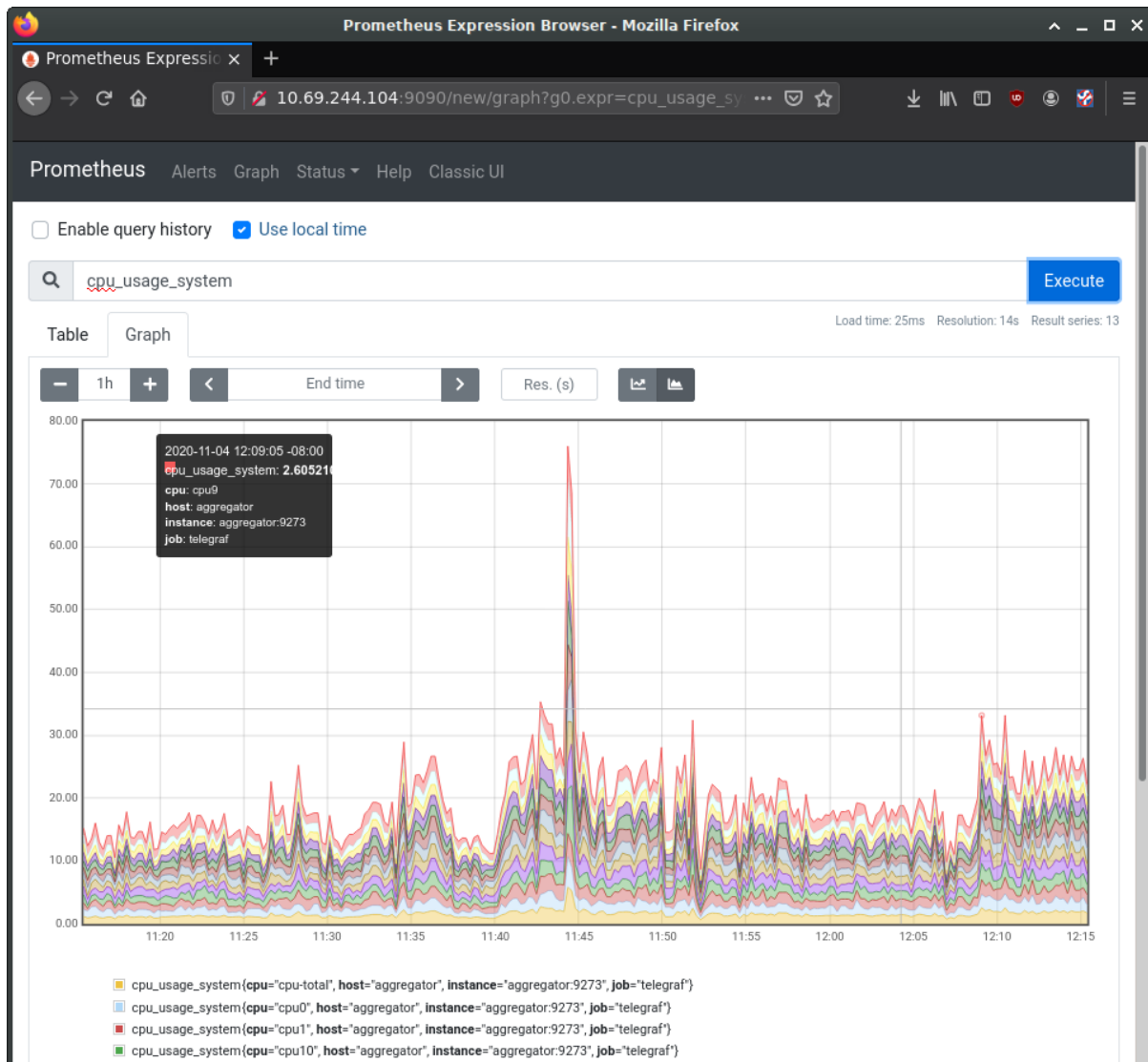
◀ Moment ▶

Element	Value
no data	

[Remove Graph](#)

Add Graph





Configure Alertmanager

```
## /var/snap/prometheus/current/prometheus.yml
#...
# Alertmanager configuration
alerting:
  alertmanagers:
    - static_configs:
        - targets:
            - 127.0.0.1:9093
rule_files:
  - 'alerts.yml'
```

```
## /var/snap/prometheus/current/alerts.yml
```

(continues on next page)

(continued from previous page)

```
groups:
- name: demo-alerts
  rules:
    - alert: HighLoad
      expr: node_load1 > 2.0
      for: 60m
      labels:
        severity: normal
      annotations:
        description: '{{ $labels.instance }} of job {{ $labels.job }} is under high
load.'
        summary: Instance {{ $labels.instance }} under high load.
        value: '{{ $value }}'

    - alert: InstanceDown
      expr: up == 0
      for: 5m
      labels:
        severity: major
      annotations:
        summary: "Instance {{ $labels.instance }} down"
        description: "{{ $labels.instance }} of job {{ $labels.job }} has been down for
more than 5 minutes."
```

```
## /var/snap/prometheus-alertmanager/current/alertmanager.yml
global:
  resolve_timeout: 5m

route:
  group_by: ['alertname']
  group_wait: 10s
  group_interval: 10s
  repeat_interval: 1h

inhibit_rules:
- source_match:
    severity: 'critical'
  target_match:
    severity: 'warning'
  equal: ['alertname', 'dev', 'instance']
```

```
workload:~# snap restart prometheus-alertmanager
```

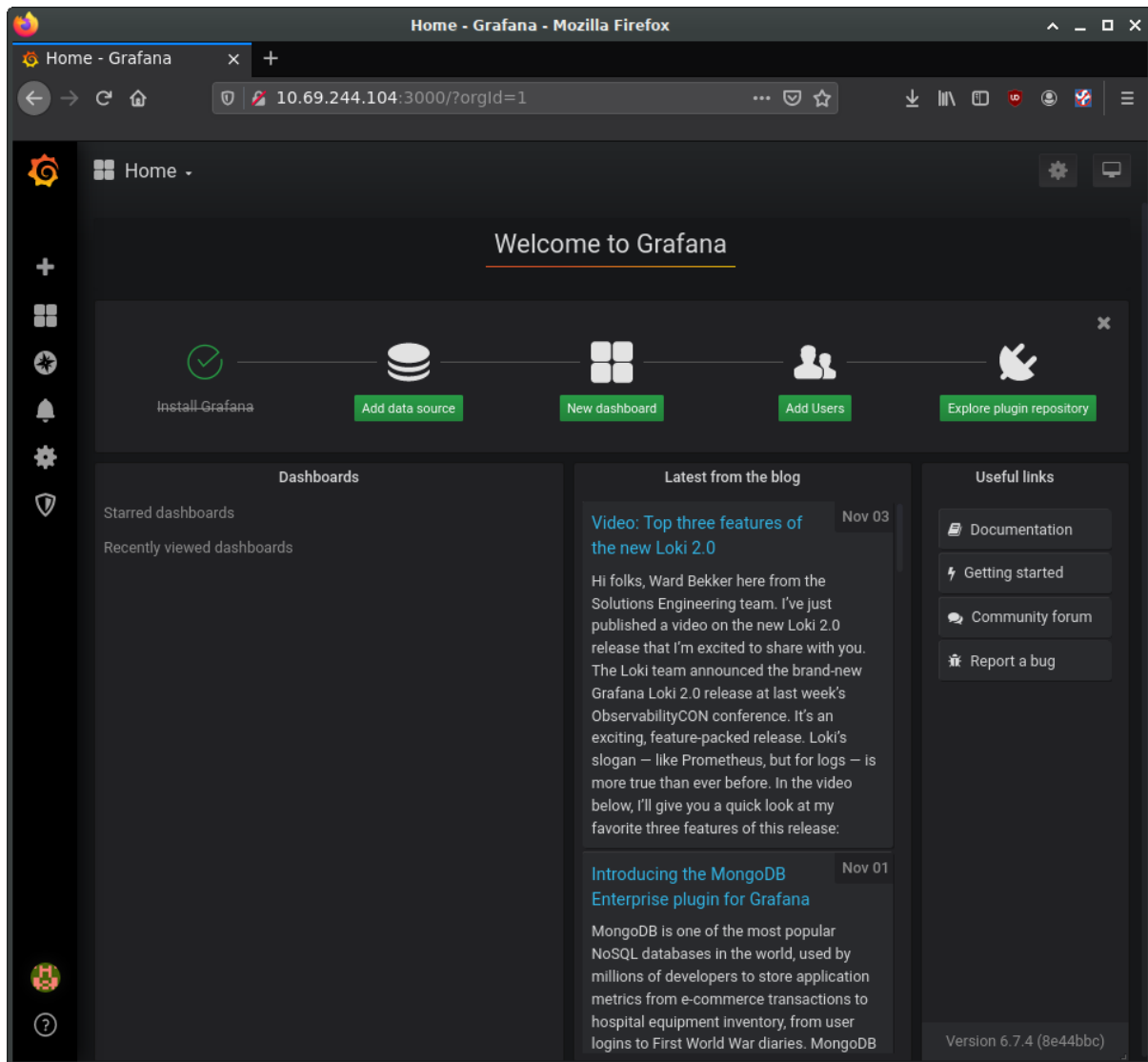
Install Grafana

```
monitor:~# snap install grafana
grafana 6.7.4 from Alvaro Uria (aluria) installed
```

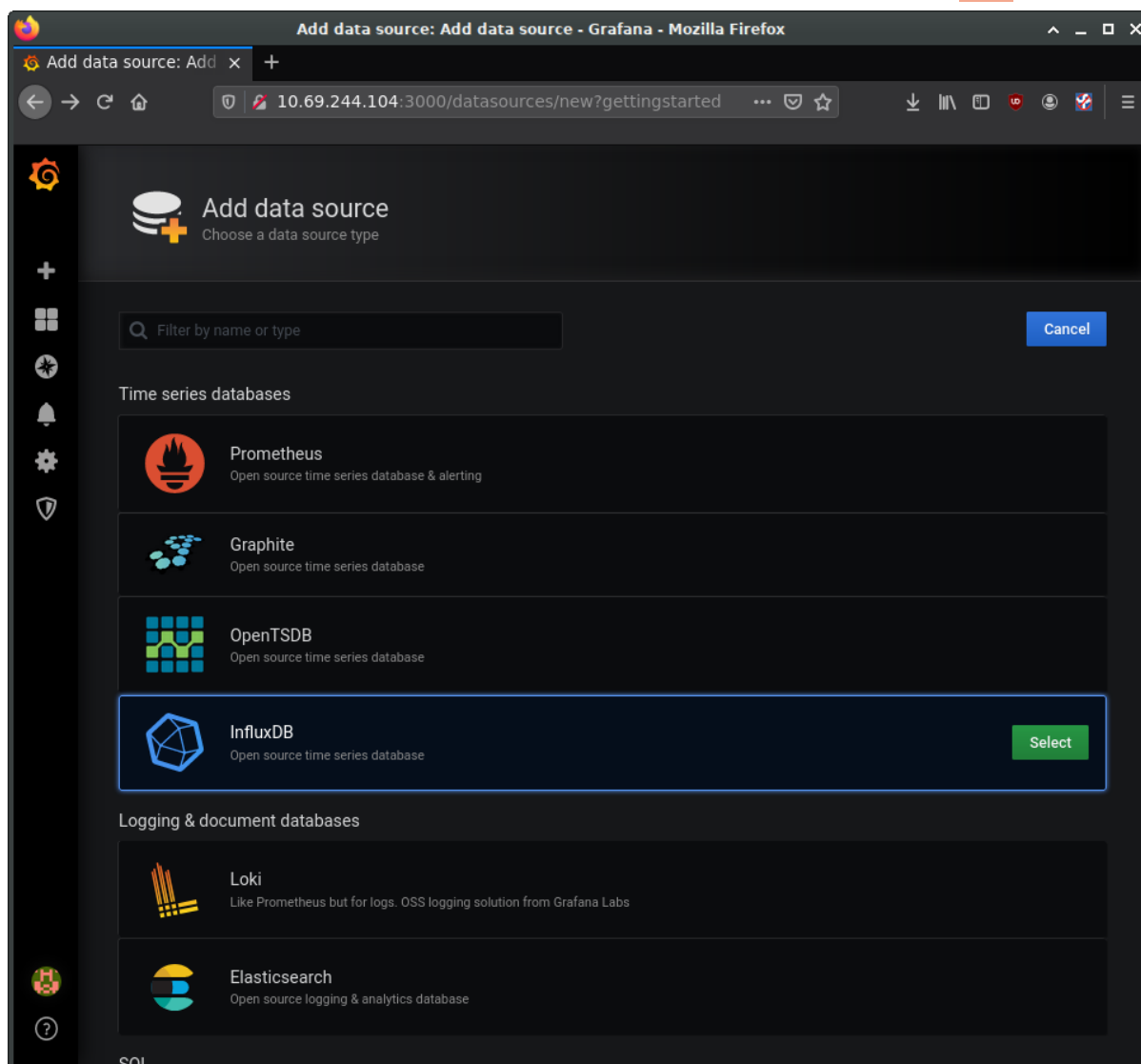
```
# ss -tulpn | grep grafana
tcp    LISTEN  0      128          *:3000      *:.*        users:((
"grafana-server",pid=1449,fd=10))
```

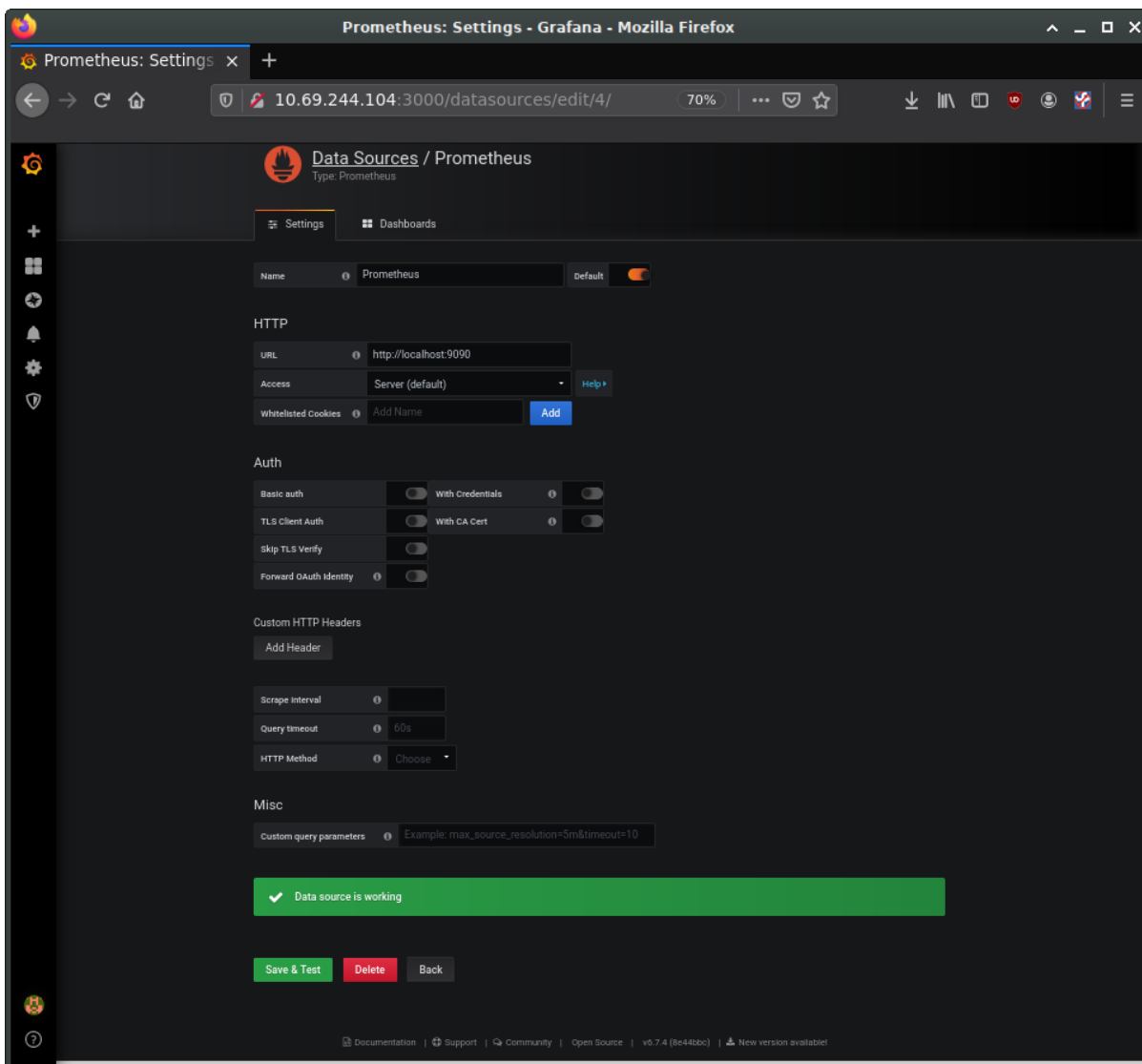
```
monitor:~# journalctl | grep "grafana.*conf"
... msg="Config loaded from" logger=settings file=/snap/grafana/36/conf/defaults.ini
... msg="Config overridden from Environment variable" logger=settings var="GF_PATHS_
PROVISIONING=/var/snap/grafana/common/conf/provisioning"
... error="open /var/snap/grafana/common/conf/provisioning/datasources: no such file or
directory"
...
```

```
$ firefox http://10.69.244.104:3000
```



-
-
-





The screenshot shows the Prometheus Settings page in Grafana, accessed via Mozilla Firefox. The browser address bar shows the URL `10.69.244.104:3000/datasources/edit/4/`. The page title is "Prometheus: Settings - Grafana - Mozilla Firefox".

The main content area is titled "Data Sources / Prometheus" with the subtitle "Type: Prometheus". Below this, there are tabs for "Settings" and "Dashboards".

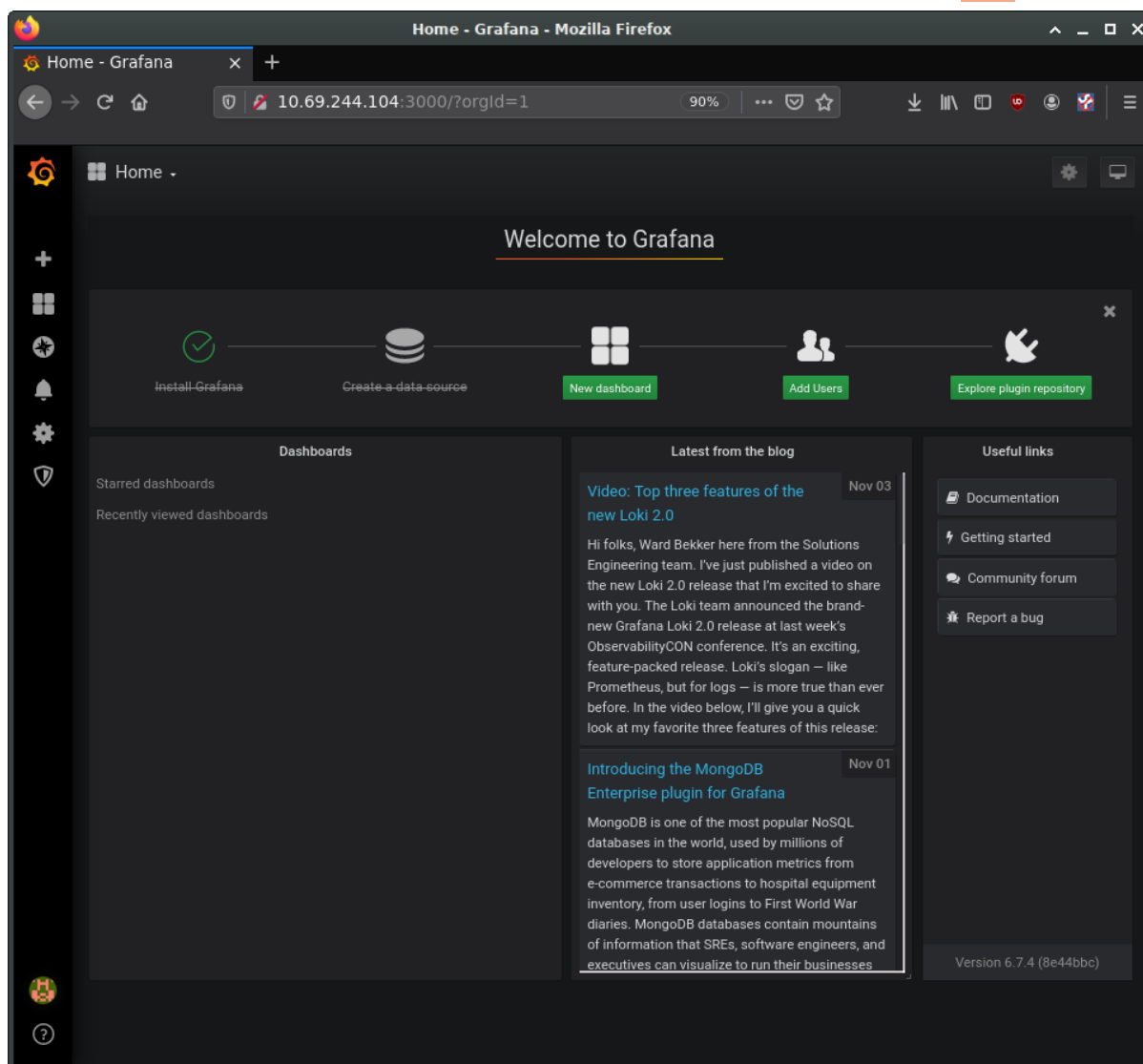
The "Settings" tab is active, showing the following configuration options:

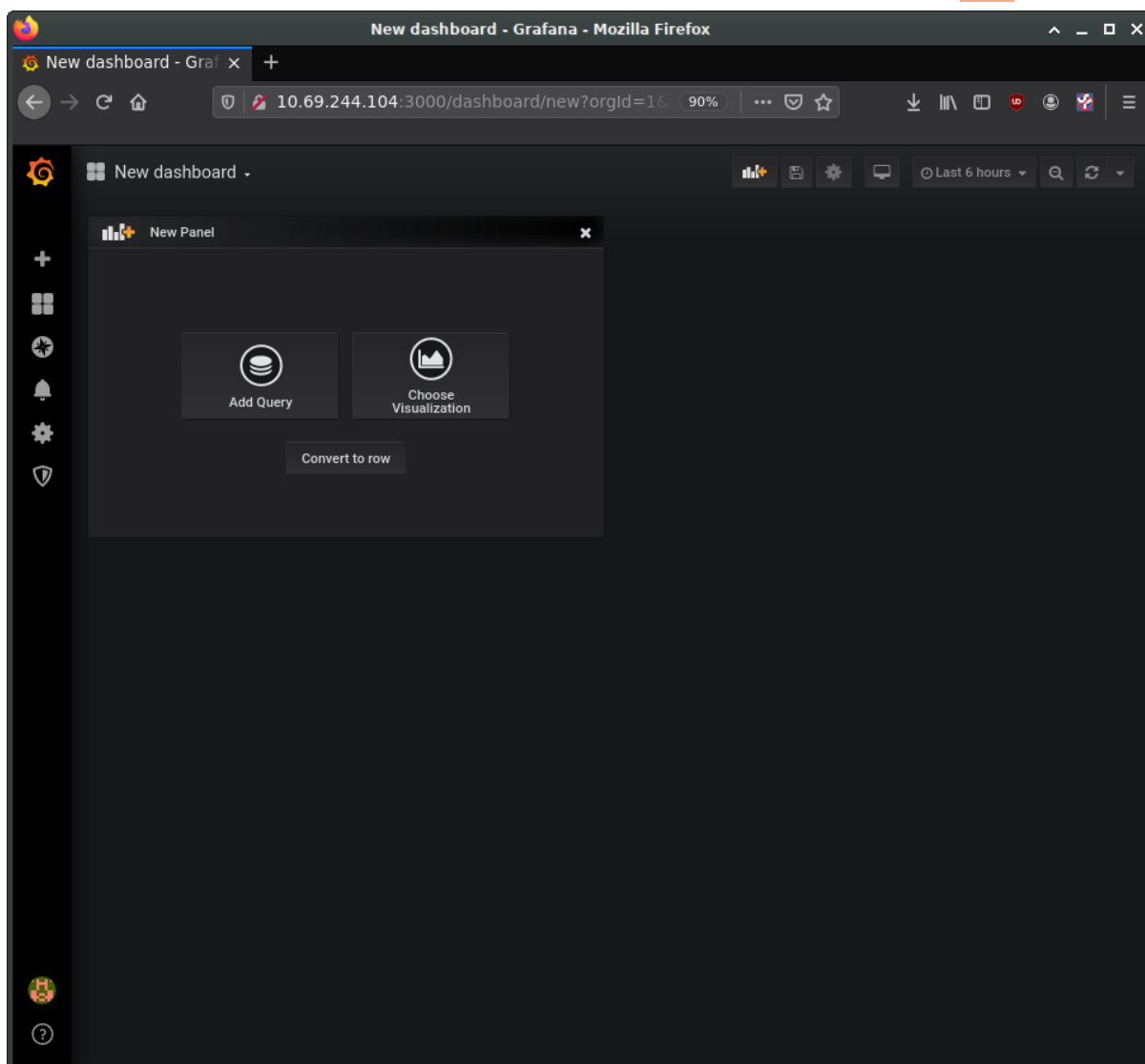
- Name:** Prometheus (Default is selected)
- HTTP:**
 - URL:** http://localhost:9090
 - Access:** Server (default) (Help icon)
 - Whitelisted Cookies:** Add Name (Add button)
- Auth:**
 - Basic auth:** With Credentials (toggle off)
 - TLS Client Auth:** With CA Cert (toggle off)
 - Skip TLS Verify:** (toggle off)
 - Forward OAuth Identity:** (toggle off)
- Custom HTTP Headers:** Add Header
- Scrape interval:** (input field)
- Query timeout:** 60s
- HTTP Method:** Choose
- Misc:** Custom query parameters (Example: max_source_resolution=5m&timeout=10)

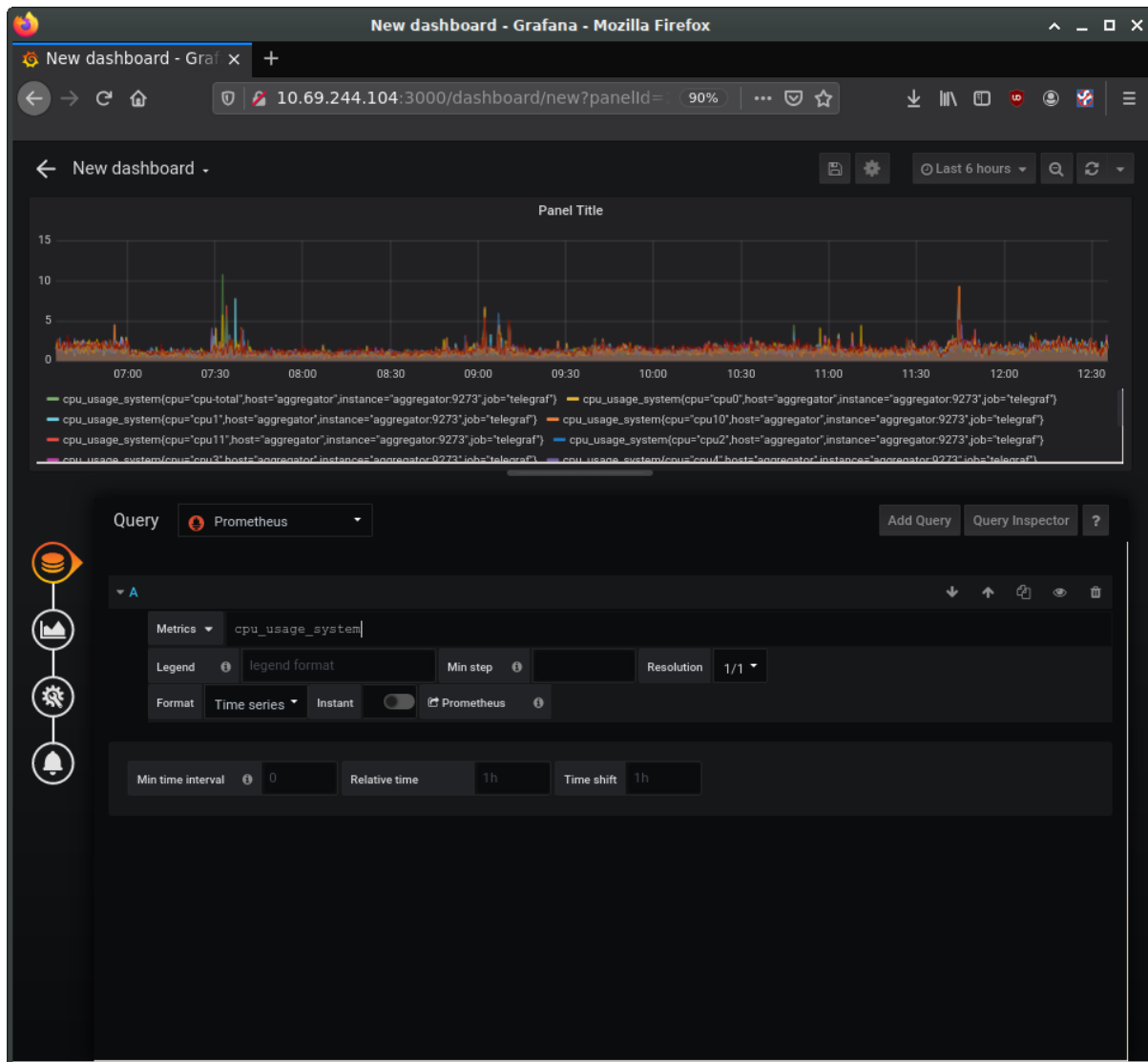
A green status bar at the bottom of the settings area indicates "Data source is working".

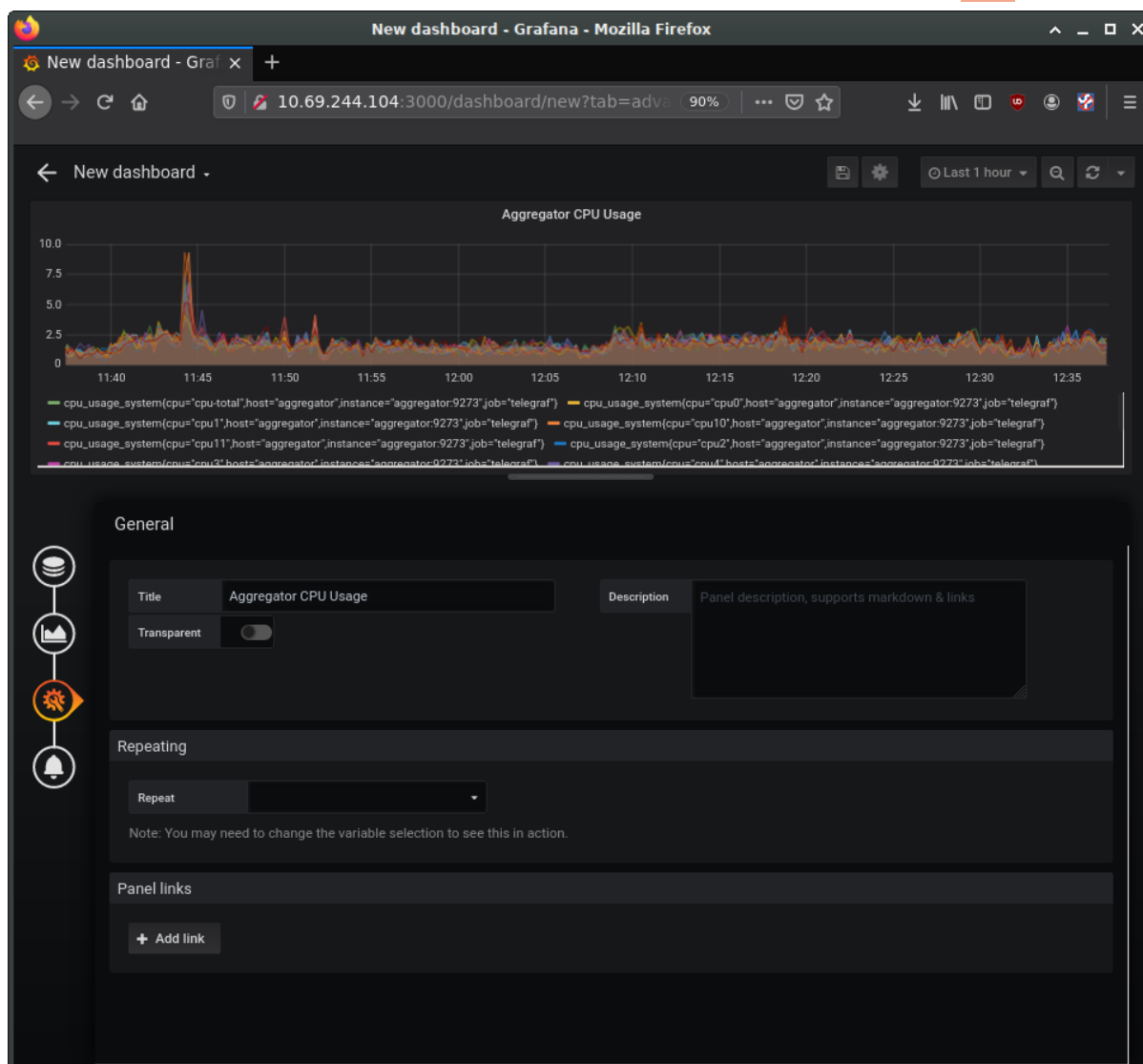
At the bottom of the page, there are buttons for "Save & Test", "Delete", and "Back".

The footer of the page contains links for Documentation, Support, Community, Open Source, and version information (v6.7.4 (8e44bdc)). A note indicates "New version available!".









New dashboard - Grafana - Mozilla Firefox

10.69.244.104:3000/dashboard/new?tab=adv

New dashboard -

Aggregator CPU Usage

Save dashboard as...

Dashboard name
Aggregator

Folder
General

Copy tags
☐

Save Cancel

General

Title
Aggregator CPU Usage

Description
Panel description, supports markdown & links

Transparent
☐

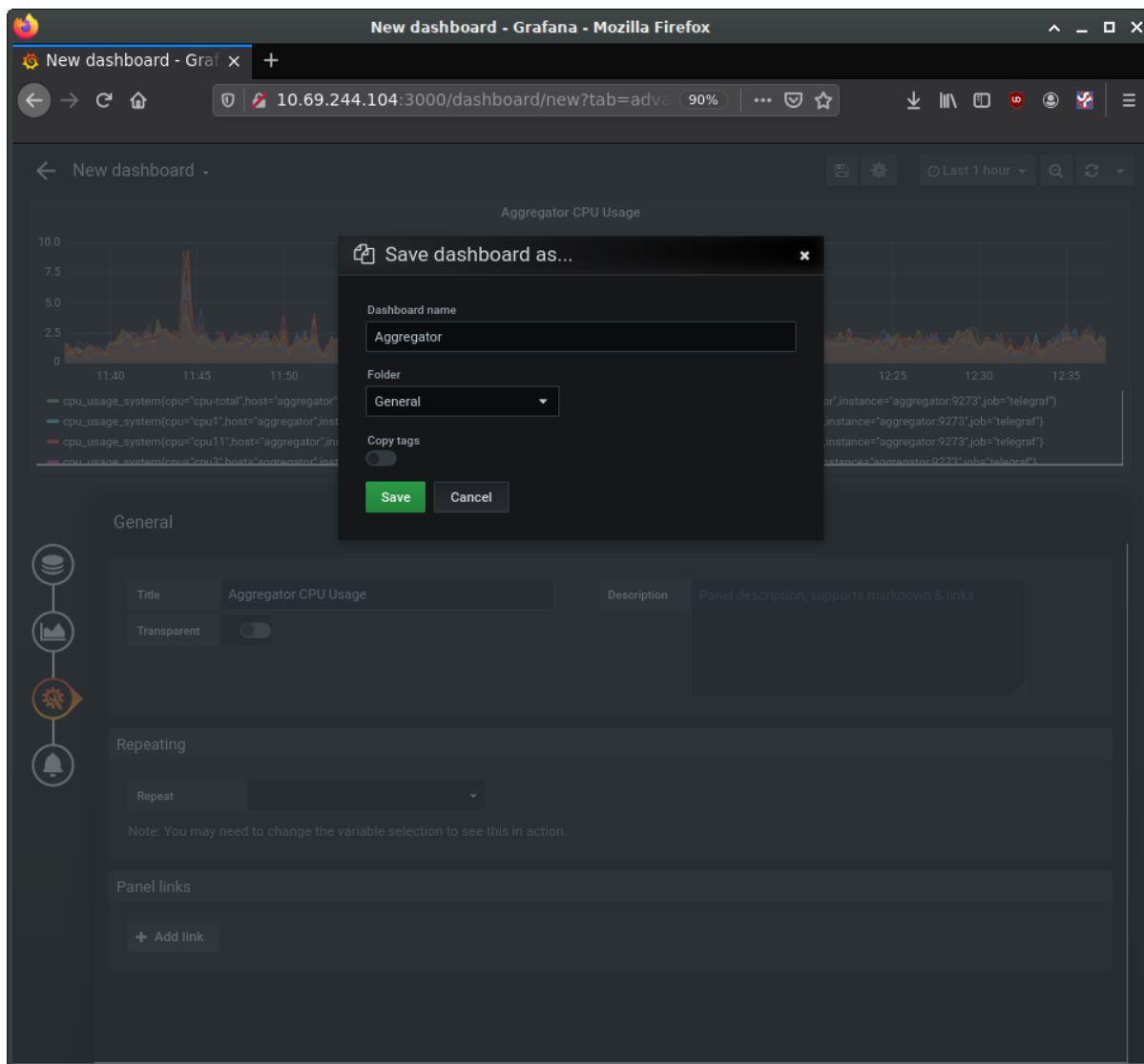
Repeating

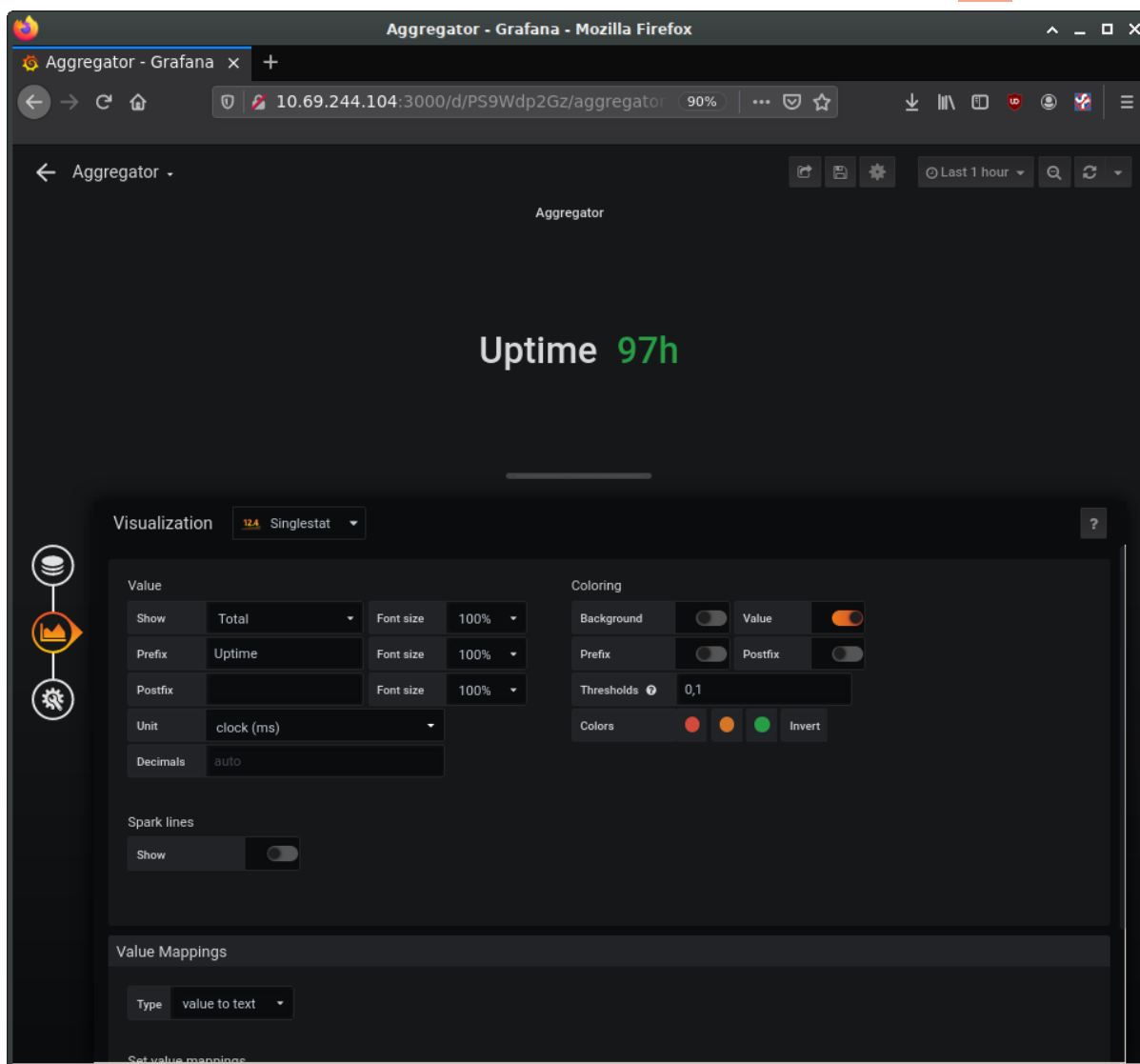
Repeat
[Dropdown]

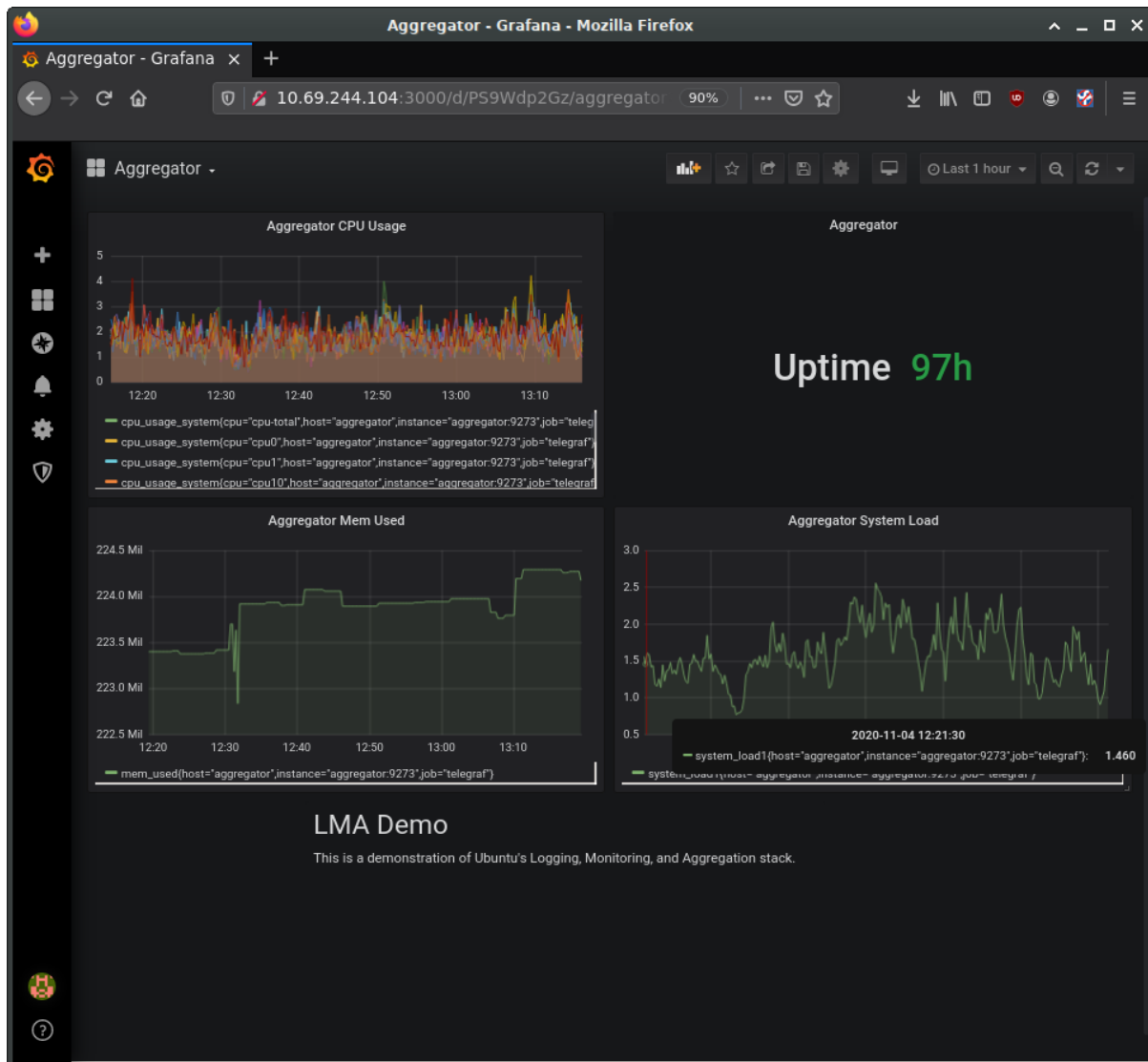
Note: You may need to change the variable selection to see this in action.

Panel links

+ Add link







How to install and configure Logwatch

Logwatch⁷¹³

Install Logwatch

```
$ sudo apt install logwatch
```

⁷¹³ <https://sourceforge.net/projects/logwatch/>

Configure Logwatch

```
$ sudo cp /usr/share/logwatch/default.conf/logwatch.conf /etc/logwatch/conf/
```

```
Output = mail
MailTo = me@mydomain.org
MailFrom = logwatch@host1.mydomain.org
Detail = Low
Service = All
```

```
Service = "-http"
Service = "-eximstats"
```

```
$ sudo logwatch --detail Low --range today
```



```
##### Logwatch 7.4.3 (12/07/16) #####
Processing Initiated: Fri Apr 24 16:58:14 2020
Date Range Processed: today
                      ( 2020-Apr-24 )
                      Period is day.

Detail Level of Output: 0
Type of Output/Format: stdout / text
Logfiles for Host: `host1.mydomain.org`
#####

----- pam_unix Begin -----

sudo:
Sessions Opened:
  bryce -> root: 1 Time(s)

----- pam_unix End -----

----- rsnapshot Begin -----

ERRORS:
/usr/bin/rsnapshot hourly: completed, but with some errors: 5 Time(s)
/usr/bin/rsync returned 127 while processing root@host2:/etc/: 5 Time(s)
/usr/bin/rsync returned 127 while processing root@host2:/home/: 5 Time(s)
/usr/bin/rsync returned 127 while processing root@host2:/proc/uptime: 5 Time(s)
/usr/bin/rsync returned 127 while processing root@host3:/etc/: 5 Time(s)
/usr/bin/rsync returned 127 while processing root@host3:/home/: 5 Time(s)
/usr/bin/rsync returned 127 while processing root@host3:/proc/uptime: 5 Time(s)

----- rsnapshot End -----

----- SSHD Begin -----

Users logging in through sshd:
  bryce:
    192.168.1.123 (`host4.mydomain.org`): 1 time

----- SSHD End -----

----- Sudo (secure-log) Begin -----

bryce => root
\-----
/bin/bash - 1 Time(s).

----- Sudo (secure-log) End -----

----- Disk Space Begin -----
```

Filesystem	Size	Used	Avail	Use%	Mounted on
/dev/sdc1	220G	19G	190G	9%	/
/dev/loop1	157M	157M	0	100%	/snap/gnome-3-28-1804/110
/dev/loop11	1.0M	1.0M	0	100%	/snap/gnome-logs/81
/dev/md5	9.1T	7.3T	1.8T	81%	/srv/Products
/dev/md6	9.1T	5.6T	3.5T	62%	/srv/Archives
/dev/loop14	2.0M	2.0M	0	100%	/snap/gnome-system-monitor/127

Further reading

- `logwatch(8)`⁷¹⁴

How to install and configure Munin

Munin⁷¹⁵
`hostnames`

Prerequisites

Apache2 will need to be installed

Install munin and munin-node

```
$ sudo apt install munin
```

```
$ sudo apt install munin-node
```

Configure munin

```
## First our "normal" host.  
[server02]  
    address 172.18.100.101
```

Note:

Replace `server02` and `172.18.100.101` with the actual hostname and IP address for your server.

⁷¹⁴ <https://manpages.ubuntu.com/manpages/resolute/man8/logwatch.8.html>

⁷¹⁵ <https://munin-monitoring.org/>

```
Require ip 172.18.100.1
```

Note:

Replace 172.18.100.1 with the actual IP address for the host you want to allow.

Configure `munin-node`

```
allow ^172\.18\.100\.100$
```

Note:

Replace `^172\.18\.100\.100$` with the IP address for your `munin` server.

```
$ sudo systemctl restart munin-node.service
```

Test the setup**Additional Plugins**

DNS DHCP

```
$ sudo apt install munin-plugins-extra
```

Note:

The `munin` package recommends `munin-node`, which recommends `munin-plugins-extra`. This means that in the most common use cases, you will already have `munin-plugins-extra` installed after following the instructions in this page.

Further reading

- [Munin](#)⁷¹⁶
- [Munin Documentation](#)⁷¹⁷

How to install and configure Nagios Core 4

Installation

hostnames

-
-

Install Nagios Core 4 on server01

```
user@server01:~$ sudo apt install nagios4 nagios-nrpe-plugin
```

Install and configure Postfix

Domain Name

Service (DNS)

⁷¹⁶ <https://munin-monitoring.org/>

⁷¹⁷ <https://munin.readthedocs.io/en/latest/>

Create a user for accessing the Nagios web interface

```
user@server01:~$ sudo htdigest -c /etc/nagios4/htdigest.users "Nagios4"  
nagiosadmin
```

Nagios CGI security documentation⁷¹⁸

Install nagios-nrpe-server on server02

```
user@server02:~$ sudo apt install nagios-nrpe-server
```

Note:

NRPE allows you to execute local checks on remote hosts. There are other ways of accomplishing this through other Nagios plugins, as well as other checks.

Configuring the Nagios server

/etc/nagios4

/etc/nagios-plugins/config

/etc/nagios

⁷¹⁸ <https://assets.nagios.com/downloads/nagioscore/docs/nagioscore/4/en/cgisecurity.html>

Enable authentication

Apache configuration for Nagios

```
user@server01:~$ sudo cp /etc/nagios4/apache2.conf  
/etc/nagios4/apache2.conf.orig
```

```
#Require ip ::1/128 fc00::/7 fe80::/10 10.0.0.0/8 169.254.0.0/16 172.16.0.0/12 192.168.0.  
0/16
```

```
<Files "cmd.cgi">  
    Options ExecCGI  
    AuthDigestDomain "Nagios4"  
    AuthDigestProvider file  
    AuthUserFile "/etc/nagios4/htdigest.users"  
    AuthGroupFile "/etc/group"  
    AuthName "Nagios4"  
    AuthType Digest  
    #Require all granted  
    Require valid-user  
</Files>
```

Force all users to authenticate

```
#<Files "cgi.cmd">
```

```
#</Files>
```

Note:

The Alias `/nagios4` in directive `Alias /nagios4 /usr/share/nagios4/htdocs` can be changed to be anything you need. This is how you would reach the Nagios interface from the browser. In this setup, you would enter into the browser: `server01/nagios4`.

For more information on the topic of authentication and authorization, see both [Apache's documentation](https://httpd.apache.org/docs/2.2/howto/auth.html)⁷¹⁹ and [Nagios's documentation](https://assets.nagios.com/downloads/nagioscore/docs/nagioscore/4/en/cgisecurity.html)⁷²⁰. If you need granular access for users, use the `/etc/nagios4/cgi.cfg` file to define the access the user is authorized for.

⁷¹⁹ <https://httpd.apache.org/docs/2.2/howto/auth.html>

⁷²⁰ <https://assets.nagios.com/downloads/nagioscore/docs/nagioscore/4/en/cgisecurity.html>

Nagios web interface

```
user@server01:~$ sudo cp /etc/nagios4/apache2.conf  
/etc/apache2/sites-available/nagios4.conf
```

```
user@server01:~$ sudo a2ensite nagios4.conf
```

```
user@server01:~$ sudo systemctl reload apache2
```

```
user@server01:~$ sudo systemctl reload nagios4
```

```
user@server01:~$ sudo systemctl status apache2
```

```
user@server01:~$ sudo systemctl status nagios4
```

how-to configure Apache2

Modifying the Nagios Apache site

```
Alias /nagios4 /usr/share/nagios4/htdocs
```

```
Alias /nagios /usr/share/nagios4/htdocs
```

```
user@server01:~$ sudo a2dissite nagios4
```

```
user@server01:~$ sudo cp /etc/nagios4/apache2.conf  
/etc/apache2/sites-available/nagios4.conf
```

```
user@server01:~$ sudo a2ensite nagios4.conf
```

```
user@server01:~$ sudo systemctl reload apache2
```


Working with Nagios users

```
user@server01:~$ sudo htdigest -c /etc/nagios4/htdigest.users "Nagios4"  
nagiosuser
```

```
authorized_for_system_information=nagiosadmin
```

```
authorized_for_system_information=nagiosadmin, nagiosuser
```

```
authorized_for_all_services=nagiosadmin  
authorized_for_hosts=nagiosadmin
```

```
authorized_for_all_services=nagiosadmin, nagiosuser  
authorized_for_all_hosts=nagiosadmin, nagiosuser
```

```
user@server01:~$ sudo systemctl reload nagios4
```

Note:

When using the `htdigest` utility, use the `-c` option only when first creating a user. Omit the `-c` to change password afterwards. The `/etc/nagios4/cgi.cfg` has many options. Review the file for usage.

Working with resource objects

Creating host config file for server02

```
user@server01:~$ sudo cp /etc/nagios4/objects/localhost.cfg  
/etc/nagios4/conf.d/server02.cfg
```

```
define host {  
    use             generic-host      ;Name of host template to use  
    host_name       server02          ;This host definition will inherit all variables  
that are defined  
    alias           server 02         ;in (or inherited by) the linux-server host  
template definition.  
    address         172.10.100.2  
}
```

```
# Define a DNS check service on server02  
define service {  
    use             local-service  
    host_name       server02  
    service_description DNS  
    check_command   check_dns  
    notification_enabled 0  
}
```

Testing the Nagios configuration and troubleshooting

```
user@server01:~$ sudo nagios4 -v /etc/nagios4/nagios.cfg
```

```
user@server01:~$ sudo systemctl restart nagios4
```

Hostgroup definitions

```
user@server02:~$ sudo apt install mysql-server
```

See also

- *Install and configure a MySQL server* (page 506)
- *MYSQL tutorial by tutorials24x7.com*⁷²¹
- *MySQL official documentation*⁷²²

```
#check MySQL servers
define service {
    hostgroup_name      mysql-servers
    service_description MySQL
    check_command        check_mysql_cmdlinecred!nagios!secret!$HOSTADDRESS
    use                  generic-service
    notification_interval 0; set > 0 to be re-notified
}
```

```
#MySQL group definition
define hostgroup {
```

(continues on next page)

⁷²¹ <https://www.tutorials24x7.com/mysql/how-to-install-mysql-8-on-ubuntu-2004-lts>

⁷²² <https://www.mysql.com>

(continued from previous page)

```
}
    hostgroup_name    mysql-servers
    alias             MySQL_servers
    members           localhost, server02
}
```

Nagios user authentication to MySQL

```
user@server02:~$ mysql -u root -p -e "create user nagios identified by
'secret';"
```

```
user@server02:~$ sudo mysql -u root -p > mysql> CREATE USER 'nagios'
IDENTIFIED BY 'secret'; > mysql> flush privileges; > mysql> exit
```

Note:

The Nagios user will need to be added to all hosts in the mysql-servers hostgroup.

```
user@server01:~$ sudo systemctl restart nagios4
```

NRPE configuration

```
#NRPE disk check
define service {
    use                generic-service
    host_name          server02
    server_description nrpe-disk-check
    check_command       check_nrpe_1arg!check_all_disks!172.10.100.2
}
```

```
allowed_hosts=172.10.100.1
```

```
command[check_all_disks]=/usr/lib/nagios/plugins/check_disk -w 20% -c 10% -e
```

```
user@server02:~$ sudo systemctl restart nagios-nrpe-server.service
```

```
user@server01:~$ sudo systemctl restart nagios4
```

Further reading

- [Nagios website](#)⁷²³
- [Nagios Core Documentation](#)⁷²⁴ [Nagios Core 4 Documentation](#)⁷²⁵
- [list of books](#)⁷²⁶

How to use Nagios with Munin

Note:

Nagios Core 3 has been deprecated and is now replaced by Nagios Core 4. The nagios3 package was last supported in Bionic, so subsequent releases should use nagios4 instead.

⁷²³ <https://www.nagios.com>

⁷²⁴ <https://assets.nagios.com/downloads/nagioscore/docs/nagioscore/4/en/whatsnew.html>

⁷²⁵ <https://assets.nagios.com/downloads/nagioscore/docs/nagioscore/4/en/>

⁷²⁶ <https://www.nagios.org/about/propaganda/books-2/>

Install Nagios 3

On server01

```
$ sudo apt install nagios3 nagios-nrpe-plugin
```

```
$ sudo htpasswd /etc/nagios3/htpasswd.users nagiosadmin
```

```
$ sudo htpasswd /etc/nagios3/htpasswd.users steve
```

On server02

```
$ sudo apt install nagios-nrpe-server
```

Note:

NRPE allows you to run local checks on remote hosts. There are other ways of accomplishing this, including through other Nagios plugins.

Configuration overview

-
-
-
-

*DNS**host group***Note:**

See these additional guides for details on setting up *Apache* (page 597), *Domain Name Service (DNS)* (page 306), and *MySQL* (page 506).

-
-
-
-
-

Configure Nagios

```
$ sudo cp /etc/nagios3/conf.d/localhost_nagios2.cfg \ >
/etc/nagios3/conf.d/server02.cfg
```

Note:

In the above and following command examples, replace “server01”, “server02”, 172.18.100.100, and 172.18.100.101 with the host names and IP addresses of your servers.

```
define host{
    use                generic-host ; Name of host template to use
    host_name          server02
    alias              Server 02
    address             172.18.100.101
}
```

(continues on next page)

(continued from previous page)

```
# check DNS service.
define service {
    use                generic-service
    host_name          server02
    service_description DNS
    check_command       check_dns!172.18.100.101
}
```

```
$ sudo systemctl restart nagios3.service
```

```
# check MySQL servers.
define service {
    hostgroup_name      mysql-servers
    service_description MySQL
    check_command        check_mysql_cmdlinecred!nagios!secret!$HOSTADDRESS
    use                 generic-service
    notification_interval 0 ; set > 0 if you want to be renotified
}
```

```
# MySQL hostgroup.
define hostgroup {
    hostgroup_name      mysql-servers
    alias               MySQL servers
    members              localhost, server02
}
```

```
$ mysql -u root -p -e "create user nagios identified by 'secret';"
```

Note:

The nagios user will need to be added all hosts in the mysql-servers host group.

```
$ sudo systemctl restart nagios3.service
```

```
# NRPE disk check.
define service {
```

(continues on next page)

(continued from previous page)

```
use generic-service
host_name server02
service_description nrpe-disk
check_command check_nrpe_1arg!check_all_disks!172.18.100.101
}
```

```
allowed_hosts=172.18.100.100
```

```
command[check_all_disks]=/usr/lib/nagios/plugins/check_disk -w 20% -c 10% -e
```

```
$ sudo systemctl restart nagios-nrpe-server.service
```

```
$ sudo systemctl restart nagios3.service
```

Install Munin

setting up Apache

On server01

```
$ sudo apt install munin
```

On server02

```
$ sudo apt install munin-node
```

Configure Munin on server01

```
## First our "normal" host.  
[server02]  
    address 172.18.100.101
```

Note:

Replace server02 and 172.18.100.101 with the actual hostname and IP address of your server.

Configure munin-node on server02

```
allow ^172\.18\.100\.100$
```

Note:

Replace ^172\.18\.100\.100\$ with IP address for your Munin server.

```
$ sudo systemctl restart munin-node.service
```

Note:

Since this is a new install it may take some time for the graphs to display anything useful.

Additional plugins

DHCP

```
$ sudo apt install munin-plugins-extra
```

Further reading

- [Munin](#)⁷²⁷ [Nagios](#)⁷²⁸
- [Munin Documentation](#)⁷²⁹
- [Nagios Online Documentation](#)⁷³⁰
- [list of books](#)⁷³¹
- [Nagios Ubuntu Wiki](#)⁷³²

These debugging pages

Debugging

- *About debuginfod*
- *Debug symbol packages*
- *Introduction to eBPF*
- *Kernel crash dump*

About debuginfod

`debuginfod`⁷³³

manually install

HTTPS

⁷²⁷ <https://munin-monitoring.org/>

⁷²⁸ <https://www.nagios.org/>

⁷²⁹ <https://munin.readthedocs.io/en/latest/>

⁷³⁰ <https://www.nagios.org/documentation/>

⁷³¹ <https://www.nagios.org/about/propaganda/books-2/>

⁷³² <https://help.ubuntu.com/community/Nagios3>

⁷³³ <https://sourceware.org/elfutils/Debuginfod.html>

Using the service

supported Ubuntu releases⁷³⁴

GNU Debugger (GDB)⁷³⁵

```
export DEBUGINFOD_URLS="https://debuginfod.ubuntu.com"
```

GDB

```
set debuginfod enabled on
```

Example session with GDB

```
$ gdb -q program
```

```
Reading symbols from program...
```

```
This GDB supports auto-downloading debuginfo from the following URLs:
```

```
https://debuginfod.ubuntu.com
```

```
Enable debuginfod for this session? (y or [n])
```

```
Enable debuginfod for this session? (y or [n]) y
```

```
Debuginfod has been enabled.
```

```
To make this setting permanent, add 'set debuginfod enabled on' to .gdbinit.
```

```
Downloading 0.20 MB separate debug info for /home/ubuntu/program
```

```
Reading symbols from /home/ubuntu/.cache/debuginfod_client/
```

```
c0fbda15a807f880e9d0b2dcc635eeeb1f0f728e/debuginfo...
```

```
(gdb)
```

⁷³⁴ <https://releases.ubuntu.com/>

⁷³⁵ <https://www.sourceware.org/gdb/>

Opting out of the service

```
unset DEBUGINFOD_URLS
```

```
set debuginfod enabled off
```

How does debuginfod find the debug symbols for the binary I am debugging?

```
$ readelf -j .note.gnu.build-id /bin/bash

Displaying notes found in: .note.gnu.build-id
Owner          Data size      Description
GNU            0x00000014     NT_GNU_BUILD_ID (unique build ID bitstring)
Build ID: 7426cc87c256cb386b52891bbe4300f2f5bab8c6
```

HTTPS

Can ddeb packages co-exist with debuginfod?

Can I use debuginfod with my own binary that links against system libraries?

Bug reports

[Launchpad for ubuntu-debuginfod](#)⁷³⁶

Debug symbol packages

[Valgrind](#)⁷³⁸

[backtrace](#)⁷³⁷

Debuginfod

[\(GDB\)](#)⁷³⁹

[GNU Debugger](#)

to our Debuginfod guide

Getting -dbgsym.ddeb packages

Import the signing key

[signing key](#)⁷⁴⁰

```
$ sudo apt install ubuntu-dbgsym-keyring
```

⁷³⁶ <https://bugs.launchpad.net/ubuntu-debuginfod/+filebug>

⁷³⁷ <https://wiki.ubuntu.com/Backtrace>

⁷³⁸ <https://wiki.ubuntu.com/Valgrind>

⁷³⁹ <https://www.sourceware.org/gdb/>

⁷⁴⁰ https://help.ubuntu.com/community/Repositories/Ubuntu#Authentication_Tab

Create a ddebs.list file

```
$ echo "Types: deb > URIs: http://ddebs.ubuntu.com/ > Suites: $(lsb_release
-cs) $(lsb_release -cs)-updates $(lsb_release -cs)-proposed > Components:
main restricted universe multiverse > Signed-by:
/usr/share/keyrings/ubuntu-dbgsym-keyring.gpg" | \ > sudo tee -a
/etc/apt/sources.list.d/ddebs.sources
```

ries⁷⁴²

this article⁷⁴¹

adding other reposito-

```
$ deb http://ddebs.ubuntu.com focal main restricted universe multiverse
```

Note:

Make sure you replace focal with the Ubuntu release name you're using.

Update package list

```
$ sudo apt-get update
```

Manual install of debug packages

```
$ sudo apt-get install PACKAGE-dbgsym
```

```
$ sudo apt-get install xserver-xorg-core-dbgsym
```

```
$ sudo apt-get install libc6-dbg
```

⁷⁴¹ <https://help.ubuntu.com/community/Repositories/Ubuntu>

⁷⁴² https://help.ubuntu.com/community/Repositories/Ubuntu#Adding_Other_Repositories

The debian-goodies tool

```
$ sudo apt install debian-goodies
```

```
$ find-dbgsym-packages [core_path|running_pid|binary_path]
```

“debs” versus “ddebs”

their own separate repository⁷⁴³

⁷⁴³ <http://ddebs.ubuntu.com>

Kernel crash dump

-
-
-
-
-

Kernel crash dump mechanism

Installation

```
$ sudo apt install kdump-tools
```

```
-----| Configuring kdump-tools |-----  
  
If you choose this option, the kdump-tools mechanism will be enabled. A  
reboot is still required in order to enable the crashkernel kernel  
parameter.  
  
Should kdump-tools be enabled by default?  
  
                <Yes>                <No>  
-----|
```

kdump is enabled by default where applicable

- -
 -
 -
- -

Disabling kdump

```
$ sudo apt remove kdump-tools
```

```
USE_KDUMP=0
```

Activating config changes

Verify if kdump is ready

```
$ kdump-config show

DUMP_MODE:          kdump
USE_KDUMP:          1
KDUMP_COREDIR:      /var/crash
crashkernel addr: 0x60000000
    /var/lib/kdump/vmlinuz: symbolic link to /boot/vmlinuz-6.18.0-8-generic
kdump initrd:
    /var/lib/kdump/initrd.img: symbolic link to /var/lib/kdump/initrd.img-6.18.0-8-generic
current state:      ready to kdump

crashkernel suggested size: 417M

kexec command:
    /sbin/kexec -p -s --command-line="BOOT_IMAGE=/vmlinuz-6.18.0-8-generic
root=UUID=0a86b691-f733-4cb0-9c5c-b88e0ef9e212 ro console=tty1 console=ttyS0 reset_
devices systemd.unit=kdump-tools-dump.service nr_cpus=1 irqpoll usbcore.nousb" --
initrd=/var/lib/kdump/initrd.img /var/lib/kdump/vmlinuz
```

Reserving Memory

```
$ cat /proc/cmdline

BOOT_IMAGE=/vmlinuz-6.18.0-8-generic
root=UUID=0a86b691-f733-4cb0-9c5c-b88e0ef9e212 ro console=tty1 console=ttyS0
crashkernel=2G-4G:320M,4G-32G:512M,32G-64G:1024M,64G-128G:2048M,128G-:4096M
```

```
crashkernel=<range1>:<size1>[,<range2>:<size2> ...]@offset|
range=start-[end] 'start' is inclusive and 'end' is exclusive.
```

```
crashkernel=2G-4G:320M,4G-32G:512M,32G-64G:1024M,64G-128G:2048M,128G-:4096M
```

-
-
-
-

```
$ dmesg | grep -i crash
```

```
[ 0.004623] crashkernel reserved: 0x0000000060000000 - 0x0000000074000000 (320 MB)
```

Configuration

Local kernel crash dumps

Remote kernel crash dumps using the SSH protocol

```
# -----  
# Remote dump facilities:  
# SSH - username and hostname of the remote server that will receive the dump  
#       and dmesg files.  
# SSH_KEY - Full path of the ssh private key to be used to login to the remote  
#            server. use kdump-config propagate to send the public key to the  
#            remote server  
# HOSTTAG - Select if hostname of IP address will be used as a prefix to the  
#            timestamped directory when sending files to the remote server.  
#            'ip' is the default.  
SSH="ubuntu@kdump-netcrash"
```

hostname

```
$ sudo kdump-config propagate
```

```
Need to generate a new ssh key...  
The authenticity of host 'kdump-netcrash (192.168.1.74)' can't be established.  
ECDSA key fingerprint is SHA256:IMp+5Y28qhbd+tevFCWrEXyKdD4dI3yN40Vlu3CBBQ4.  
Are you sure you want to continue connecting (yes/no)? yes  
ubuntu@kdump-netcrash's password:  
propagated ssh key /root/.ssh/kdump_id_rsa to server ubuntu@kdump-netcrash
```

```
$ kdump-config show
```

```
DUMP_MODE:      kdump
USE_KDUMP:      1
KDUMP_SYSCTL:   kernel.panic_on_oops=1
KDUMP_COREDIR:  /var/crash
crashkernel addr: 0x2c000000
               /var/lib/kdump/vmlinuz: symbolic link to /boot/vmlinuz-4.4.0-10-generic
kdump initrd:
               /var/lib/kdump/initrd.img: symbolic link to /var/lib/kdump/initrd.img-4.4.0-10-generic
SSH:            ubuntu@kdump-netcrash
SSH_KEY:        /root/.ssh/kdump_id_rsa
HOSTTAG:        ip
current state:  ready to kdump
```

Remote kernel crash dumps using the NFS protocol

```
# NFS -      Hostname and mount point of the NFS server configured to receive
#           the crash dump. The syntax must be {HOSTNAME}:{MOUNTPOINT}
#           (e.g. remote:/var/crash)
#
NFS="kdump-netcrash:/var/crash"
```

```
$ kdump-config show

DUMP_MODE:      kdump
USE_KDUMP:      1
KDUMP_SYSCTL:   kernel.panic_on_oops=1
KDUMP_COREDIR:  /var/crash
crashkernel addr: 0x2c000000
               /var/lib/kdump/vmlinuz: symbolic link to /boot/vmlinuz-4.4.0-10-generic
kdump initrd:
               /var/lib/kdump/initrd.img: symbolic link to /var/lib/kdump/initrd.img-4.4.0-10-generic
NFS:            kdump-netcrash:/var/crash
HOSTTAG:        hostname
current state:  ready to kdump
```

Troubleshooting

```
$ kdump-config show

DUMP_MODE:          kdump
USE_KDUMP:          1
KDUMP_COREDIR:      /var/crash
crashkernel addr:
  /var/lib/kdump/vmlinuz: symbolic link to /boot/vmlinuz-6.18.0-8-generic
kdump initrd:
  /var/lib/kdump/initrd.img: symbolic link to /var/lib/kdump/initrd.img-6.18.0-8-generic
current state:      Not ready to kdump

crashkernel suggested size: 329M

kexec command:
  no kexec command recorded
```

```
$ systemctl status kdump-tools

...
... kdump-tools[5348]: Memory for crashkernel is not reserved
... kdump-tools[5348]: Please reserve memory by passing "crashkernel=Y@X" parameter
to kernel
... kdump-tools[5348]: Then try to loading kdump kernel
```

Testing the crash dump mechanism

Warning:

Testing the crash dump mechanism **will cause a system reboot**. In certain situations, this can cause data loss if the system is under heavy load. If you want to test the mechanism, make sure that the system is idle or under very light load.

```
$ cat /proc/sys/kernel/sysrq
```

Note:

On 24.10 and earlier the location of this file was at `/etc/sysctl.d/10-magic-sysrq.conf`

```
$ sudo sysctl -w kernel.sysrq=1
```

```
[ 977.208267] sysrq: Trigger a crash
[ 977.209313] Kernel panic - not syncing: sysrq triggered crash
[ 977.210684] CPU: 0 UID: 0 PID: 1567 Comm: bash Kdump: loaded Not tainted 6.18.0-8-
generic #8-Ubuntu PREEMPT(voluntary)
[ 977.215248] Hardware name: QEMU Standard PC (Q35 + ICH9, 2009) /LXD, BIOS unknown 2/2/
2022
[ 977.216319] Call Trace:
...
```

```
[ 5.171504] kdump-tools[740]: Starting kdump-tools:
[ 5.174084] kdump-tools[747]: * running makedumpfile --dump-dmesg /proc/vmcore /var/
crash/202601210724/dmesg.202601210724
[ 5.185243] kdump-tools[765]: The kernel version is not supported.
[ 5.188411] kdump-tools[765]: The makedumpfile operation may be incomplete.
[ 5.191528] kdump-tools[765]: The dmesg log is saved to /var/crash/202601210724/dmesg.
202601210724.
[ 5.196300] kdump-tools[765]: makedumpfile Completed.
[ 5.199056] kdump-tools[747]: * kdump-tools: saved dmesg content in /var/crash/
202601210724
[ 5.204580] kdump-tools[747]: * running makedumpfile -F -c -d 31 /proc/vmcore |
compress > /var/crash/202601210724/dump-incomplete
[ 5.210935] kdump-tools[769]: The kernel version is not supported.
[ 5.214499] kdump-tools[769]: The makedumpfile operation may be incomplete.
Copying data : [100.0 %] \ eta: 0s
[ 7.000839] kdump-tools[769]: The dumpfile is saved to STDOUT.
[ 7.003347] kdump-tools[769]: nakedumpfile Completed.
[ 7.006824] kdump-tools[747]: * kdump-tools: saved vmcore in /var/crash/202601210724
```



```
$ ll /var/crash
drwxr-xr-x  2 root root  4096 Jan 21 07:31 202601210724/
-rw-r--r--  1 root root 28838 Jan 21 07:24 linux-image-6.18.0-8-generic-
202601210724.crash
```

```
$ ll /var/crash/202601210724
-rw-----  1 root root   82264 Jan 21 07:24 dmesg.202601210724
-rw-r--r--  1 root root 51753823 Jan 21 07:24 dump.202601210724
```

Analyzing a crash dump

Getting -dbgsym.ddeb

packages

Note:

This does not strictly need to happen on the system that had the crash, it can be copied away for analysis. But the examples are simplified to use e.g. `uname -r` to automatically select the right kernel - when running on a different system these might need to be adapted.

```
$ apt-get install linux-image-$(uname -r)-dbgsym
```

```
$ crash /usr/lib/debug/boot/vmlinux-6.18.0-8-generic
/var/crash/202601210724/dump.202601210724
```

```
...
  KERNEL: /usr/lib/debug/boot/vmlinux-6.18.0-8-generic
  DUMPFILE: /var/crash/202601210724/dump.202601210724 [PARTIAL DUMP]
  CPUS: 1
  DATE: Thu Jan  1 00:00:00 UTC 1970
  UPTIME: 00:16:16
  LOAD AVERAGE: 0.00, 0.00, 0.00
  TASKS: 157
  NODENAME: r-vm
  RELEASE: 6.18.0-8-generic
  VERSION: #8-Ubuntu SMP PREEMPT_DYNAMIC Wed Dec 17 16:14:24 UTC 2025
  MACHINE: x86_64 (3792 Mhz)
  MEMORY: 3 GB
  PANIC: "Kernel panic - not syncing: sysrq triggered crash"
  PID: 1567
  COMMAND: "bash"
  TASK: ffff89cd07908000 [THREAD_INFO: ffff89cd07908000]
  CPU: 0
  STATE: TASK_RUNNING (PANIC)

crash>
```

PANIC: "Kernel panic - not syncing: sysrq triggered crash"

Further reading

- [kdump kernel documentation](#)⁷⁴⁴
- [Analyzing Linux Kernel Crash](#)⁷⁴⁵

Introduction to eBPF

eBPF⁷⁴⁶

⁷⁴⁴ <https://www.kernel.org/doc/Documentation/kdump/kdump.txt>

⁷⁴⁵ <https://www.dedoimedo.com/computers/crash-analyze.html>

⁷⁴⁶ <https://ebpf.io/>

How eBPF works

-
-
-

eBPF in Ubuntu Server

application developer and sysadmin experience in Ubuntu⁷⁴⁷

enhance the

```
$ dpkg -L bpftrace bpfcc-tools | grep -E '/s?bin/.*$' | xargs -n1 basename
```

⁷⁴⁷ <https://discourse.ubuntu.com/t/spec-include-performance-tooling-in-ubuntu/43134>

```
# bashreadline.bt
```

Example - Determine what commands are executed

-
-

```
$ sudo execsnoop-bpfcc -Uu root -T
```

```
$ sudo apt install --reinstall vim-nox
```

TIME	UID	PCOMM	PID	PPID	RET	ARGS
10:58:07	1000	sudo	1323101	1322857	0	/usr/bin/sudo apt install --reinstall vim-nox
10:58:10	0	apt	1323107	1323106	0	/usr/bin/apt install --reinstall vim-nox
10:58:10	0	dpkg	1323108	1323107	0	/usr/bin/dpkg --print-foreign-architectures

(continues on next page)

(continued from previous page)

```
...
10:58:12 0      sh      1323134 1323107 0 /bin/sh -c /usr/sbin/dpkg-preconfigure
--apt || true
...
10:58:13 0      tar      1323155 1323152 0 /usr/bin/tar -x -f --warning=no-
timestamp
10:58:14 0      vim-nox.prerm 1323157 1323150 0 /var/lib/dpkg/info/vim-nox.prerm
upgrade 2:9.1.0016-1ubuntu7.3
10:58:14 0      dpkg-deb 1323158 1323150 0 /usr/bin/dpkg-deb --fsys-tarfile /var/
cache/apt/archives/vim-nox_2%3a9.1.0016-1ubuntu7.3_amd64.deb
...
10:58:14 0      update-alternat 1323171 1323163 0 /usr/bin/update-alternatives --install
/usr/bin/vimdiff vimdiff /usr/bin/vim.nox 40
...
10:58:17 0      snap     1323218 1323217 0 /usr/bin/snap advise-snap --from-apt
10:58:17 1000  git      1323224 1323223 0 /usr/bin/git rev-parse --abbrev-ref
HEAD
10:58:17 1000  git      1323226 1323225 0 /usr/bin/git status --porcelain
10:58:17 1000  vte-urlencode-c 1323227 1322857 0 /usr/libexec/vte-urlencode-cwd
```

eBPF can be modified to your needs

Example - Find out which files QEMU is loading

-
-

```
$ sudo /usr/sbin/opensnoop-bpfcc --full-path --name qemu-system-x86
```

```
$ lxc launch ubuntu-daily:n n-vm-test --ephemeral --vm
```

```
1308728 qemu-system-x86 -1 2 PID COMM FD ERR PATH
/snap/lxd/current/zfs-2.2/lib/glibc-hwcaps/x86-64-v3/libpixman-1.so.0
1308728 qemu-system-x86 -1 2 /snap/lxd/current/zfs-2.2/lib/glibc-hwcaps/x86-64-v2/
libpixman-1.so.0
1308728 qemu-system-x86 -1 2 /snap/lxd/current/zfs-2.2/lib/tls/haswell/x86_64/
libpixman-1.so.0
...
1313104 qemu-system-x86 58 0 /sys/dev/block/230:16/queue/zoned
1313104 qemu-system-x86 20 0 /dev/fd/4
```

Focusing on a specific file type

```
$ sudo cp /usr/sbin/opensnoop-bpfcc /usr/sbin/opensnoop-bpfcc.new
```

```
$ $ sudo vim /usr/sbin/opensnoop-bpfcc.new
```

```
...
```

```
$ $ diff -Naur /usr/sbin/opensnoop-bpfcc /usr/sbin/opensnoop-bpfcc.new
```

```
--- /usr/sbin/opensnoop-bpfcc          2025-10-24 12:17:00.000000000 +0000
+++ /usr/sbin/opensnoop-bpfcc.new      2026-01-19 10:45:13.620367441 +0000
@@ -40,6 +40,7 @@
     ./opensnoop -u 1000                # only trace UID 1000
     ./opensnoop -d 10                  # trace for 10 seconds only
     ./opensnoop -n main                # only print process names containing "main"
+   ./opensnoop -c path                # only print paths containing "fname"
     ./opensnoop -e                    # show extended fields
     ./opensnoop -f O_WRONLY -f O_RDWR # only print calls for writing
     ./opensnoop -F                    # show full path for an open file with
relative path
@@ -71,6 +72,9 @@
     parser.add_argument("-n", "--name",
        type=ArgString,
        help="only print process names containing this name")
+parser.add_argument("-c", "--contains",
+   type=ArgString,
+   help="only print paths containing this string (implies --full-path)")
     parser.add_argument("--ebpf", action="store_true",
        help=argparse.SUPPRESS)
     parser.add_argument("-e", "--extended_fields", action="store_true",
@@ -83,6 +87,8 @@
        help="size of the perf ring buffer "
        "(must be a power of two number of pages and defaults to 64)")
     args = parser.parse_args()
+if args.contains is not None:
+   args.full_path = True
     debug = 0
     if args.duration:
        args.duration = timedelta(seconds=int(args.duration))
@@ -478,6 +484,12 @@
        if args.name and bytes(args.name) not in event.comm:
            skip = True

+   paths = entries[event.id]
+   paths.reverse()
+   entire_path = os.path.join(*paths)
+   if args.contains and bytes(args.contains) not in entire_path:
+       skip = True
+
     if not skip:
        if args.timestamp:
            delta = event.ts - initial_ts
@@ -502,9 +514,7 @@
        if not args.full_path:
            printb(b"%s" % event.name)
        else:
-           paths = entries[event.id]
-           paths.reverse()
-           printb(b"%s" % os.path.join(*paths))
+           printb(b"%s" % entire_path)

     if args.full_path:
        try:
```

```
$ sudo /usr/sbin/opensnoop-bpffcc.new --contains '.bin' --name  
qemu-system-x86
```

PID	COMM	FD	ERR	PATH
1316661	qemu-system-x86	21	0	/snap/lxd/current/share/qemu//kvmvapic.bin
1316661	qemu-system-x86	39	0	/snap/lxd/current/share/qemu//vgabios-virtio.bin
1316661	qemu-system-x86	39	0	/snap/lxd/current/share/qemu//vgabios-virtio.bin

Use for other purposes, the limit is your imagination

```
$ sudo /usr/sbin/opensnoop-bpffcc.new --name 'apache2' --contains '/etc'
```

PID	COMM	FD	ERR	PATH
1319357	apache2	3	0	/etc/apache2/apache2.conf
1319357	apache2	4	0	/etc/apache2/mods-enabled
1319357	apache2	4	0	/etc/apache2/mods-enabled/access_compat.load
...				
1319357	apache2	4	0	/etc/apache2/ports.conf
1319357	apache2	4	0	/etc/apache2/conf-enabled
1319357	apache2	4	0	/etc/apache2/conf-enabled/charset.conf
1319357	apache2	4	0	/etc/apache2/conf-enabled/localized-error-pages.conf
...				
1319357	apache2	4	0	/etc/apache2/sites-enabled/000-default.conf

eBPF's limitations

Possibly lost 84 samples

same discussion⁷⁴⁸

⁷⁴⁸ <https://github.com/iovisor/bcc/issues/1033>

Conclusion

Further reading

- [Introduction to eBPF video, given at the Ubuntu summit 2024](#)⁷⁴⁹
- [what is eBPF](#)⁷⁵⁰
- [upstream documentation](#)⁷⁵¹
- [list of eBPF based solutions](#)⁷⁵²

Further reading

- [The Ubuntu Project documentation](#)⁷⁵³
- [Glossary](#)
- [Command-line cheat sheet](#)

[contributions page](#)

A	B	C	D	E	F	
G	H	I	J	K	L	M
	N	O	P	Q	R	S
	T	U	V	W	X	Y
	Z					

⁷⁴⁹ <https://www.youtube.com/live/byPpJW5l6pg?t=30314s>

⁷⁵⁰ <https://ebpf.io/what-is-ebpf/>

⁷⁵¹ <https://docs.kernel.org/bpf/>

⁷⁵² <https://ebpf.io/applications/>

⁷⁵³ <https://documentation.ubuntu.com/project/contributors/debugging/>

A

ABI Application Binary Interface

API

ACL Access Control List

- *the ACL manual page*⁷⁵⁴
- *OpenLDAP* *Kerberos*

AD Active Directory

ADSys

AD
SSSD

- *the ADSys documentation*⁷⁵⁵
- *Group Policy Object* *SSSD*

AES Advanced Encryption Standard

•

Alertmanager

⁷⁵⁴ <https://manpages.ubuntu.com/manpages/resolute/man5/acl.5.html>

⁷⁵⁵ <https://documentation.ubuntu.com/adsys/stable/>

- the Alertmanager documentation⁷⁵⁶

- *Prometheus*

ALUA

Asymmetric Logical Unit Access

- *MPIO SCSI SAN*

AMD

Advanced Micro Devices

-
-
-

-

AMD-SME

AMD Secure Memory Encryption

-

AMD-SEV

AMD Secure Encrypted Virtualization

-

AMD-SEV-ES

AMD Secure Encrypted Virtualization - Encrypted State

AMD-SEV

⁷⁵⁶ <https://prometheus.io/docs/alerting/latest/alertmanager/>

-

AMD-SEV-SNP

AMD Secure Encrypted Virtualization - Secure Nested Paging

AMD-SEV-ES

-

Ansible

- The Ansible website⁷⁵⁷

-

Apache2

HTTP

HTTP

SSL TLS

- The Apache project documentation⁷⁵⁸

-

API

Application Programming Interface

⁷⁵⁷ <https://docs.ansible.com/>

⁷⁵⁸ <https://httpd.apache.org/docs/2.4/>

AppArmor

Mandatory Access Control (MAC)

- The AppArmor website⁷⁵⁹

-

Apport

- The Apport Wiki Page⁷⁶⁰

-

APT

Advanced Package Tool

- *Install and manage packages*

armhf

ARM hard-float

-

ARP

Address Resolution Protocol

MAC ad-

dress

⁷⁵⁹ <https://apparmor.net/>

⁷⁶⁰ <https://wiki.ubuntu.com/Apport>

-

ASCII
American Standard Code for Information Interchange

async
asynchronous

-

attestation

-

-

Authenticator

bind

-

autocommit

- [autocommit behavior in Python's sqlite3 module](#)⁷⁶¹

-

autodetect

-

autoinstall

- [The Autoinstall documentation](#)⁷⁶²

-

autorid

UID *GID*

- [The autorid Samba Wiki](#)⁷⁶³

-

AWS

Amazon Web Services

⁷⁶¹ <https://docs.python.org/3/library/sqlite3.html#transaction-control>

⁷⁶² <https://canonical-subiquity.readthedocs-hosted.com/en/latest/intro-to-autoinstall.html>

⁷⁶³ https://wiki.samba.org/index.php/Idmap_config_autorid

- The AWS documentation⁷⁶⁴

-

B

backend

Backports

Backtrace

BDC

Backup Domain Controller

Samba server *PDC*

bind

bindDN

DN

bind

BIOS

Basic Input/Output System

UEFI

BMC

Baseboard Management Controller

bootloader

BIOS *UEFI*

⁷⁶⁴ <https://docs.aws.amazon.com/>

BOOTP
Bootstrap Protocol

bootstrap

btrfs
B-tree File System

C

CA
Certificate Authority

CAC
Common Access Card

CARP
Cache Array Routing Protocol

CCID
Chip Card Interface Device

CDB
Command Descriptor Block

CGNAT
Carrier-Grade Network Address Translation

CGI
Common Gateway Interface

checksums

chrony

chroot

CIDR
Classless Inter-Domain Routing

CIFS
Common Internet File System

CIS
Center for Internet Security

CLVM
Clustered Logical Volume Manager

CMS
Configuration Management System

CN
Common Name

colocation

conffile

config

connectionless

containerization

CPU
Central Processing Unit

CRL
Certificate Revocation List

crypto
cryptographic

CSR
Certificate Signing Request

CVE
Common Vulnerabilities and Exposures

D

DAC

Discretionary access control

-

daemonize

DARPA

Defense Advanced Research Projects Agency

DAS

Direct Attached Storage

- *NAS*

- *SAN*

DASD

Direct Access Storage Device

-

Datagram

-

dblink

Database link

-

DC
Domain Component

DDNS
Dynamic Domain Name System

-

deb822

- [RFC 822](#)⁷⁶⁵
- [deb822\(5\)](#)⁷⁶⁶

debconf
configuration management system

deduplication

denylist

-

DER
Distinguished Encoding Rules

DGC
Distributed Garbage Collection

DHCP
Dynamic Host Configuration Protocol

⁷⁶⁵ <https://datatracker.ietf.org/doc/html/rfc822>

⁷⁶⁶ <https://manpages.ubuntu.com/manpages/resolute/man5/deb822.5.html>

DHCPD
Dynamic Host Configuration Protocol Daemon

DIT
Directory Information Tree

•

DKMS
Dynamic Kernel Module Support

•

DMA
Direct Memory Access

DMAR
Direct Memory Access Remapping

Direct Memory Access

virtual machines (VMs)

dmesg

DN
Distinguished Name

OU DC

CN

DNS
Domain Name System

•

dnsmasq

DNS DHCP

DNSSEC Domain Name System Security Extensions

DNS

-

Docker

-

DocumentRoot

DoT DNS over TLS

Transport Layer Security (TLS)

- [RFC 7858](#)⁷⁶⁷

-

-

DoH DNS over HTTPS

HTTPS

- [RFC 8484](#)⁷⁶⁸

-

-

dpkg

⁷⁶⁷ <https://datatracker.ietf.org/doc/html/rfc7858>

⁷⁶⁸ <https://datatracker.ietf.org/doc/html/rfc8484>

DRBD
Distributed Replicated Block Device

•

DSCP
Differentiated Services Code Point

•

DTLS
Datagram Transport Layer Security

TLS

UDP

E

EAL
Environment Abstraction Layer

ECKD
Extended Count Key Data

EFI
Extensible Firmware Interface

UEFI

BIOS

ELinks

GUI

Engenio

SAN

NAS

EOL
End of life

ERD
Enterprise Ready Drivers

ESM
Expanded Security Maintenance

LTS

ESXi

F

failover
Storage Area Network (SAN)

fallbacks

FastCGI
Fast Common Gateway Interface
CGI

FC
Fiber Channel

Storage Area Network (SAN)

FDE
Full-Disk Encryption

FHS
Filesystem Hierarchy Standard

Fileset

-

filesystem

-

FTP
File Transfer Protocol

- *SFTP*
- *rsync*
- *SCP*

FIPS
Federal Information Processing Standard

- *NIST*⁷⁶⁹

Fluentd

-

FQDN
Fully Qualified Domain Name

DNS

-

FreeIPA
Free Identity, Policy, and Audit

LDAP

CA

SSSD

Kerberos

-

⁷⁶⁹ https://en.wikipedia.org/wiki/National_Institute_of_Standards_and_Technology

Freemove

Internet Relay Chat (IRC)

frontend

-

-

OS

GPU

backend

-

slapd

-

fsck

File System Check

filesystem

-

FULLTEXT

- Full-Text Search Functions⁷⁷⁰

-

FW

Firmware

-

RAM

-

PCIe

GPUs

⁷⁷⁰ <https://dev.mysql.com/doc/refman/8.4/en/fulltext-search.html>

G

gcplogs

•

gcrypt

•

GDB
GNU Debugger

•

GELF
Graylog Extended Log Format

JSON

Graylog Logstash Fluentd

•

GFS2

filesystem

•

GB

GID
Group ID

•

gitolite
Gitolite

- **GKE**
Google Kubernetes Engine

- **GL**
Graphics Library
API

- **GNU**
GNU's Not Unix

- **GnuTLS**
GNU's Not Unix Transport Layer Security

- *SSL TLS DTLS*

- **GPG**
GNU Privacy Guard

- **GPS**
Global Positioning System

- *PPS*

- **GPSD**
GPS daemon

- *Chrony*

GPU
Graphics Processing Unit

•

Graylog

gelf

•

GPO
Group Policy Object

•

GRUB
GRand Unified Bootloader

GSSAPI
Generic Security Services Application Program Interface
API

•

GTK
GIMP Toolkit

graphical user interfaces (GUIs)

•

GUI
Graphical User Interface

GZIP
GNU Zip

GNU

-
- *fileset*

-

-

H

HA
High Availability

-

HBA
Host Bus Adapter

-

HMAC
Hash-based Message Authentication Code
Message Authentication Code

-

HMC
Hardware Management Console

OS

hostgroup

-

hostname

IP address

HOTP
HMAC-based One-Time Password

OTP *HMAC*

hotplug

HPB
Host Physical Bits

CPU

HPC
High Performance Computing

HSG
High-availability Storage Group

HSV
Highly-available Storage Virtualization

HTCP
Hyper Text Caching Protocol

RFC2756⁷⁷¹ *HTTP*

HTML
HyperText Markup Language

HTML Living Standard⁷⁷²

HTTP
HyperText Transfer Protocol

HyperText Markup Language (HTML) *IETF*
Request for Comment (RFC) *HTTP Documentation⁷⁷³*

HTTDP
HyperText Transfer Protocol Daemon

HTML
HTTP

HTTPS
HyperText Transfer Protocol Secure

HTTP

⁷⁷¹ <https://www.rfc-editor.org/info/rfc2756/>

⁷⁷² <https://html.spec.whatwg.org/multipage/>

⁷⁷³ <https://httpwg.org/specs/>

TLS

hugepage

Translation Lookaside Buffer (TLB)

HWE
Hardware Enablement

I

ICMP
Internet Control Message Protocol

ICP
Internet Cache Protocol

IDENT
Identification Protocol

IMAP
Internet Messages Access Protocol

init
initialization

Intel-TDX
Intel Trust Domain Extensions

.

I/O
Input/Output

IOMMU
Input-Output Memory Management Unit

IoT
Internet of Things

IP **Internet Protocol**

datagram

- *Networking key concepts*

•

IP address

- *Networking key concepts*

•

IPC **Inter-Process Communication**

IPL **Initial Program Load**

IPMI **Intelligent Platform Management Interface**

IPP **Internet Printing Protocol**

IPSec **Internet Protocol Security**

IPVS **IP Virtual Server**

IQN **iSCSI Qualified Name**

IRC **Internet Relay Chat**

IRQ **Interrupt Request**

ISC
Internet Systems Consortium

iSCSI
Internet Small Computer System Interface
SCSI

- *iSCSI initiator (or client)*

-
-

ISO
International Organization for Standardization

ISP
Internet Service Provider

J

jitter

-

journal

temd

sys-

- *journal*.conf(5)⁷⁷⁴
- Docker journal documentation⁷⁷⁵

⁷⁷⁴ <https://manpages.ubuntu.com/manpages/resolute/man5/journal.conf.5.html>

⁷⁷⁵ <https://docs.docker.com/engine/logging/drivers/journal/>

-

JSON
JavaScript Object Notation

- [The JSON website](https://www.json.org/json-en.html)⁷⁷⁶

K

KDC
Key Distribution Center

keepalive

Kerberos

keypair

keyring

keysalt

keyservers

keytab

keytool

KVM
Kernel-based Virtual Machine

⁷⁷⁶ <https://www.json.org/json-en.html>

L

LAN

Local Area Network

LDAP

Lightweight Directory Access Protocol

LDIF

LDAP Data Interchange Format

lightervisor

LinuxONE

Load-balancing

localhost

hostname

IP address

•

Log files

Logstash

Logwatch

LPAR

Logical Partition

LSI

Logic Systems Incorporated

LTS

Long-Term Support

- [Ubuntu Releases](#)⁷⁷⁷

LU
Logical Unit

LUA
Lua Scripting Language

LUKS
Linux Unified Key Setup

LUN
Logical Unit Number

LV
Logical Volume

LVM
Logical Volume Manager

LXC
Linux Containers

LXD
Linux Container Daemon

M

MAAS
Metal as a Service

MAC
Message Authentication Code

MAC address

manual page

⁷⁷⁷ <https://documentation.ubuntu.com/project/release-team/ubuntu-releases/>

MCE
Machine Check Exception

MD5
Message Digest Algorithm 5

MDA
Mail Delivery Agent

mdev
Minimal Device Manager

metapackage

METAR
Meteorological Aerodrome Report

microservices

microVMs

MOTD
Message of the Day

mount points

MPIO
Multipath Input/Output

MSA
Modular Smart Array

MTA
Mail Transfer Agent

MTU
Maximum Transmission Unit

MUA
Mail User Agent

Multipass

Multipath

Multiview

MySQL

N

nameserver

namespace

NAS

Network Attached Storage

- *DAS*
- *SAN*

NAT

Network Address Translation

Netboot

Netfilter

Netplan

NFS

Network File System

NFV

Network Functions Virtualization

nginx

NIC

Network Interface Card

NIS
Network Information Service

NMI
Non-Maskable Interrupt

NRPE
Nagios Remote Plugin Executor

NSCQ
Network System Configuration Queue

NSS
Name Service Switch

NTP
Network Time Protocol

NTS
Network Time Security

NUMA
Non-Uniform Memory Access

CPU affinity, NUMA placement, and isolation

Nvidia

NVMe
Non-Volatile Memory Express

NVRAM
Non-Volatile Random Access Memory

NVSwitch
NVIDIA NVLink Switch

O

OCF
Open Cluster Framework

OCFS
Oracle Cluster File System

OCI
Open Container Initiative

OCS
Online Certificate Status Protocol

OpenLDAP

OpenSC
Open Smart Card

OpenSSH

OpenSSL

OpenStack

OpenSUSE

OpenVPN

OpenVZ

OpenWRT

OS
Operating System

OSA
Open Systems Adapter

OSI
Open Systems Interconnection

OSPF
Open Shortest Path First

•

OTE
Operational Test and Evaluation

OTP
One-Time Password

OU
Organizational Unit

OverlayFS

OVS
Open vSwitch

P

PAM
Pluggable Authentication Module

page fault
page faults

[Linux kernel memory management documentation](https://docs.kernel.org/admin-guide/mm/index.html)⁷⁷⁸

passthrough

PB

TB

PCI
Peripheral Component Interconnect

⁷⁷⁸ <https://docs.kernel.org/admin-guide/mm/index.html>

PCIe
Peripheral Component Interconnect Express

PCS
Pacemaker/Corosync Stack

PDC
Primary Domain Controller

PEM
Privacy Enhanced Mail

Petitboot

PgSQL

PHP
PHP: HyperText Preprocessor

PID
Process Identifier

pingable

PIV
Personal Identity Verification

PKCS
Public-Key Cryptography Standards

PKI
Public Key Infrastructure

pluggable

PMD
Poll Mode Driver

pockets

- [Ubuntu Project: pockets](#)⁷⁷⁹

⁷⁷⁹ <https://documentation.ubuntu.com/project/how-ubuntu-is-made/concepts/package-archive/#pockets>

POSIX
Portable Operating System Interface (for UNIX)

Postcopy

Postfix

PostgreSQL

PostScript

Power 8
Power 9

PowerShell

PPA
Personal Package Archive

ppc
PowerPC

PPD
PostScript Printer Description

PPPoE
Point-to-Point Protocol over Ethernet

.
PPS
Pulse Per Second

Preboot

preseed

Prometheus

proxy

PTP
Precision Time Protocol

PTR
Pointer Record

PXE
Preboot Execution Environment

PXELINUX
PXE Linux Loader

Q

QA
Quality Assurance

QDevice
Quorum Device

QEMU
Quick Emulator

QETH
QDIO Ethernet

quickstart

R

RAM
Random Access Memory

RangeSize

RDAC
Redundant Disk Array Controller

RDBMS
Relational Database Management System

RDN
Relative Distinguished Name

renderer

REXX
Restructured Extended Executor

RFC
Request For Comments

rid
Relative Identifier

RIP
Router Information Protocol

•

RISC-V
Reduced Instruction Set Computing - Version Five

Rocks

ROM
Read-Only Memory

rootDN
Root Distinguished Name

rootfs
Root File System

routable

RSA
Rivest-Shamir-Adleman

rsync
Remote Sync

- *FTP*
- *SCP*
- *SFTP*

RTC **Real-Time Clock**

runtime

S

Samba

- *Introduction to Samba*

SAN **Storage Area Network**

- *NAS*

sandboxed

- -
 -
 -
 -
-
-

SANLOCK
Storage Area Network Locking Daemon

SAN

SASL
Simple Authentication and Security Layer

SBD
Storage-Based Death

sbin
System Binaries

schema

SCP
Secure Copy Protocol

SSH

- *FTP*
- *SFTP*
- *rsync*

Scrollback

•

SCSI
Small Computer System Interface

SDN
Software-Defined Networking

seccomp
Secure computing mode

SFTP
SSH File Transfer Protocol

SSH

- *FTP*
- *SCP*
- *rsync*

SGI
Silicon Graphics Inc.

SHA
Secure Hash Algorithm

sharding

Shell

SHM
Shared Memory

Shorewall
Shoreline Firewall

SIDs
Security Identifiers

SIMD
Single Instruction, Multiple Data

slapd
Standalone LDAP Daemon

SLAT
Second Level Address Translation

smart card

SMB
Server Message Block

SMS
Short Message Service

SMTP
Simple Mail Transfer Protocol

SMTPS
SMTP Secure

Snap

snapt

snapshot

Snap store

SNMP
Simple Network Management Protocol

SOA
Start of Authority

Solaris

SPC
SCSI Primary Commands

Splunk

SRU
Stable Release Update

SSD
Solid State Drive

SSH
Secure Shell

SSH-key
Secure Shell Key

SSH

OpenSSH

SSH

SSI
Server-Side Includes

SSL
Secure Sockets Layer

SSO
Single Sign-On

SSSD
System Security Services Daemon

stateful

STDIN
Standard Input

STDOUT
Standard Output

STDERR
Standard Error

STK
StorageTek

storage

subcommand

Subiquity

subnet
subnetwork

substring

subvolume

sudo
superuser do

superblock

symlink

syslog

systemctl

Systemd

T

tasksel
Task selector

TB

GB

TCP
Transmission Control Protocol

TCP

TDX
Trust domain extensions

•

•
TEE
Trusted Execution Environment

TFTP
Trivial File Transfer Protocol

UDP

TCP

TGS
Ticket Granting Service

TGT
Ticket Granting Ticket

TLB
Translation Lookaside Buffer

CPU

TLS
Transport Layer Security

HTTPS

tmpfs
Temporary Filesystem

tmux
Terminal Multiplexer

topologies

TOTP
Time-based One-Time Password

TPM
Trusted Platform Module

Traceback

Traceroute

TTYS
Teletype Terminals

-
-
-

TXT
Trusted Execution Technology

-
-
-

U

UDA
Unified Data Architecture

UDP
User Datagram Protocol

UEFI
Unified Extensible Firmware Interface

UFW
Uncomplicated Firewall

UID
User Identifier

UI
User Interface

unicast

unmount

untrusted

- 1.
- 2.
- 3.

uptime

URI
Uniform Resource Identifier

URL
Uniform Resource Locator

userspace

USN
Update Sequence Number

usr

UUIDs
Universally Unique Identifiers

V

vCPU
Virtual Central Processing Unit

VM

CPU

VCS
Version Control System

veth
Virtual Ethernet

VFIO
Virtual Function I/O

VM

Input-Output Memory Management Unit

VFS
Virtual File System

VFs
Virtual Functions

VG
Volume Group

Logical Volume Manager

vGPU
Virtual Graphics Processing Unit

GPU

VM

virsh
Virtual Shell

VirtIO
Virtualization I/O

virtual

virtualization

virtual

VLAN
Virtual Local Area Network

VM
Virtual Machine

CPU RAM

VNC
Virtual Network Computing

VPN
Virtual Private Network

IP

VRRP
Virtual Router Redundancy Protocol

vsftpd
Very Secure FTP Daemon

W

WAL
Write-Ahead Log

WAN
Wide Area Network

WCCP
Web Cache Communication Protocol

Web server

HTTP HTTPS

winbind
Windows Bind

AD

Samba

WireGuard

VPN

WLAN
Wireless Local Area Network

WSGI
Web Server Gateway Interface

WWID
World Wide Identifier

LUN

X

X.509

Certificate Authority

xhtml
Extensible HyperText Markup Language

HTML

HTML

HTML

XML
Extensible Markup Language

Y

YAML
YAML Ain't Markup Language

Yubikey

Z

zFCP
zSeries Fibre Channel Protocol

SCSI

ZFS
Zettabyte File System

z/VM

CLI basics

the terminal

Welcome to

Navigation

Command	Description	Online manual page
<code>pwd</code>	Print the current working directory	<i>pwd(1)</i> ⁷⁸⁰
<code>ls</code>	List all files and directories in the current directory	<i>ls(1)</i> ⁷⁸¹
<code>ls -l</code>	Long list, with permissions, owner, size, and date	
<code>ls -a</code>	List all, including hidden files (file names starting with <code>.</code>)	
<code>ls -la</code> or <code>ll</code>	Long listing including hidden files	
<code>ls -ld <dir></code>	Show the directory entry itself, not its contents, for directory <code><dir></code>	
<code>ls -l <file></code>	Show file type, permissions, owner, and size for <code><file></code>	
<code>cd <path></code>	Change to the specified directory	<i>cd(1posix)</i> ⁷⁸²
<code>cd ..</code>	Move up to the parent directory	
<code>cd ~</code>	Return to the home directory	
<code>cd -</code>	Return to the previous directory	
<code>cd /</code>	Move to the root directory	
<code>cd ../../</code>	Move up two levels at once	

⁷⁸⁰ <https://manpages.ubuntu.com/manpages/resolute/man1/pwd.1.html>

⁷⁸¹ <https://manpages.ubuntu.com/manpages/resolute/man1/ls.1.html>

⁷⁸² <https://manpages.ubuntu.com/manpages/resolute/man1/cd.1posix.html>

Working with the filesystem

Command	Description	Online manual page
<code>mkdir <name></code>	Create a new directory	<i>mkdir(1)</i> ⁷⁸³
<code>rmdir <name></code>	Remove an empty directory	<i>rmdir(1)</i> ⁷⁸⁴
<code>tree</code>	Display directory structure as a tree	<i>tree(1)</i> ⁷⁸⁵
<code>tree -L <n> <path></code>	Display tree to depth n	

Working with files

Command	Description	Online manual page
<code>touch <file></code>	Create a new (empty) file, or update the timestamp of an existing file	<i>touch(1)</i> ⁷⁸⁶
<code>cp <src> <dest></code>	Copy a file from a source (src) to a new destination (dest)	<i>cp(1)</i> ⁷⁸⁷
<code>cp -a <src> <dest></code>	Copy a directory recursively, preserving permissions	
<code>mv <src> <dest></code>	Rename a file or directory	<i>mv(1)</i> ⁷⁸⁸
<code>mv <src>/file <dest>/file</code>	Move a file without renaming it	
<code>rm <file></code>	Remove a file	<i>rm(1)</i> ⁷⁸⁹
<code>rm -r <dir></code>	Remove a non-empty directory and all its contents	
<code>cat <file></code>	Print the contents of <file> to the screen	<i>cat(1)</i> ⁷⁹⁰
<code>less <file></code>	View file contents one page at a time	<i>less(1)</i> ⁷⁹¹
<code>head -n 20 <file></code>	Print the first 20 lines of a file (default with no flag is 10)	<i>head(1)</i> ⁷⁹²
<code>tail -n 20 <file></code>	Print the last 20 lines of a file (default with no flag is 10)	<i>tail(1)</i> ⁷⁹³
<code>wc -l <file></code>	Count the number of lines in a file	<i>wc(1)</i> ⁷⁹⁴

⁷⁸³ <https://manpages.ubuntu.com/manpages/resolute/man1/mkdir.1.html>

⁷⁸⁴ <https://manpages.ubuntu.com/manpages/resolute/man1/rmdir.1.html>

⁷⁸⁵ <https://manpages.ubuntu.com/manpages/resolute/man1/tree.1.html>

Users, groups, and permissions

Command	Description	Online manual page
<code>chmod <mode> <file></code>	Change file permissions (symbolic or numeric notation)	<i>chmod(1)</i> ⁷⁹⁵
<code>chown <user>[:<group>] <file></code>	Change file owner and group	<i>chown(1)</i> ⁷⁹⁶
<code>umask</code>	Show or set the default permissions mask	<i>umask(2)</i> ⁷⁹⁷
<code>sudo <command></code>	Run a command with administrator privileges	<i>sudo(8)</i> ⁷⁹⁸

Searching

Command	Description	Online manual page
<code>grep <pattern> <file></code>	Search for lines matching a pattern in a file	<i>grep(1)</i> ⁷⁹⁹
<code>grep <name> /etc/passwd</code>	Look up a user account	
<code>grep <name> /etc/group</code>	Look up group membership	
<code>find <path> -name <name> -o -name <pattern></code>	Search for files matching a name pattern	<i>find(1)</i> ⁸⁰⁰
<code>which <command></code>	Show the full path to an executable command	<i>which(1)</i> ⁸⁰¹
<code>file <name></code>	Determine the type of a file	<i>file(1)</i> ⁸⁰²

⁷⁸⁶ <https://manpages.ubuntu.com/manpages/resolute/man1/touch.1.html>

⁷⁸⁷ <https://manpages.ubuntu.com/manpages/resolute/man1/cp.1.html>

⁷⁸⁸ <https://manpages.ubuntu.com/manpages/resolute/man1/mv.1.html>

⁷⁸⁹ <https://manpages.ubuntu.com/manpages/resolute/man1/rm.1.html>

⁷⁹⁰ <https://manpages.ubuntu.com/manpages/resolute/man1/cat.1.html>

⁷⁹¹ <https://manpages.ubuntu.com/manpages/resolute/man1/less.1.html>

⁷⁹² <https://manpages.ubuntu.com/manpages/resolute/man1/head.1.html>

⁷⁹³ <https://manpages.ubuntu.com/manpages/resolute/man1/tail.1.html>

⁷⁹⁴ <https://manpages.ubuntu.com/manpages/resolute/man1/wc.1.html>

⁷⁹⁵ <https://manpages.ubuntu.com/manpages/resolute/man1/chmod.1.html>

⁷⁹⁶ <https://manpages.ubuntu.com/manpages/resolute/man1/chown.1.html>

⁷⁹⁷ <https://manpages.ubuntu.com/manpages/resolute/man2/umask.2.html>

⁷⁹⁸ <https://manpages.ubuntu.com/manpages/resolute/man8/sudo-rs.8.html>

CLI in depth

The command line in depth

Redirecting input and output

Command	Description
<code>cmd > file</code>	Redirect stdout (1) to a file (overwrites existing content)
<code>cmd >> file</code>	Append stdout (1) to the end of <file> (does not overwrite existing contents)
<code>cmd < file</code>	Pass a file's contents as stdin (0) to a command (cmd)
<code>cmd > /dev/null</code>	Discard stdout (1) entirely
<code>cmd 2> file</code>	Redirect stderr (2) to a file
<code>cmd >> file 2>&1</code>	Redirect both stdout (1) and stderr (2) to a file
<code>cmd << EOF</code>	Here-document: feed multi-line text as stdin, ended by a line containing EOF

Pipes and command chaining

Command	Description
<code>cmd1 cmd2</code>	Pipe stdout (1) of cmd1 to stdin (0) of cmd2
<code>cmd1; cmd2</code>	Run commands in sequence regardless of exit status
<code>cmd1 && cmd2</code>	Run cmd2 only if cmd1 succeeds (exit code 0)
<code>cmd1 cmd2</code>	Run cmd2 only if cmd1 fails (non-zero exit code)
<code>cmd &</code>	Run a command in the background
<code>{ cmd1; cmd2; } > file</code>	Group commands and redirect their combined output
<code>(cmd1; cmd2)</code>	Run commands in a subshell (side effects don't affect the current shell)
<code>cmd1 \</code>	Continue a long command on the next line

⁷⁹⁹ <https://manpages.ubuntu.com/manpages/resolute/man1/grep.1.html>

⁸⁰⁰ <https://manpages.ubuntu.com/manpages/resolute/man1/find.1.html>

⁸⁰¹ <https://manpages.ubuntu.com/manpages/resolute/man1/which.debianutils.1.html>

⁸⁰² <https://manpages.ubuntu.com/manpages/resolute/man1/file.1.html>

Variables and environment

Command	Description	Online manual page
<code>env</code>	List all environment variables	<i>env(1)</i> ⁸⁰³
<code>echo \$VAR</code>	Print the value of variable VAR	See note below this table
<code>VAR=value</code>	Set a shell variable (available in current shell only)	
<code>export VAR=value</code>	Set a variable and export it to child processes	
<code>unset VAR</code>	Remove a variable	
<code>echo \$?</code>	Print the exit status of the last command (0 = success)	
<code>echo \$\$</code>	Print the PID of the current shell	

Note:

If you are using the bash, dash or zsh shell, `echo` is a builtin command documented under the shell manual page. For example, the [bash builtin shell section](#)⁸⁰⁴. If your shell doesn't have an `echo` builtin command, use the [*echo\(1\)*](#)⁸⁰⁵ manual page instead.

⁸⁰⁴ <https://manpages.ubuntu.com/manpages/resolute/man1/bash.1.html#shell-builtin-commands>

⁸⁰⁵ <https://manpages.ubuntu.com/manpages/resolute/man1/echo.1.html>

⁸⁰³ <https://manpages.ubuntu.com/manpages/resolute/man1/env.1.html>

Shell expansion

Command	Description
~	Expands to the home directory (\$HOME)
\$VAR, \${VAR}	Variable expansion: replaced by the value of VAR
\$(command)	Command substitution: replaced by the output of command
\$((expr))	Arithmetic expansion: evaluates integer arithmetic, e.g. <code>\$((5 + 3))</code> evaluates to 8
{a,b,c}	Brace expansion: generates separate words, e.g. {a,b,c} → a b c
{1..10}	Range expansion: generates a sequence, e.g. {1..5} → 1 2 3 4 5
*	Glob: matches any sequence of characters in a filename
?	Glob: matches any single character in a filename
[abc]	Glob: matches any one of the listed characters in a filename

Getting help

Command	Description	Online manual page
man <command>	Open the manual page for a command	
man man	Open the manual page for man itself	<i>man(1)</i> ⁸⁰⁶
man -k <keyword>	Search man page descriptions for a keyword	
<command> --help	Print a brief help summary for a command	

⁸⁰⁶ <https://manpages.ubuntu.com/manpages/resolute/man1/man.1.html>

System requirements

Architecture

- *AMD*
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-
-
-
-

Ubuntu certified servers⁸⁰⁷

Memory

Storage

⁸⁰⁷ <https://ubuntu.com/certified/servers>

High availability

corresponding commands

Migrate from crmsh to pcs

Action	crmsh
Show configuration (raw XML)	crm configure show xml
Show configuration (human-friendly)	crm configure show
Show cluster status	crm status
Put a node in standby mode	crm node standby NODE
Remove a node from standby mode	crm node online NODE
Set cluster property	crm configure property PROPERTY=VALUE
List resource agent classes	crm ra classes
List available resource agents by standard	crm ra list ocf
List available resource agents by OCF provider	crm ra list ocf pacemaker
List available resource agent parameters	crm ra info AGENT
Show available fence agent parameters	crm ra info stonith:AGENT
Create a resource	crm configure primitive NAME AGENT params PA
Show configuration of all resources	crm configure show
Show configuration of one resource	crm configure show RESOURCE
Show configuration of fencing resources	crm resource status
Start a resource	crm resource start RESOURCE

Action	crmsd
Stop a resource	<code>crm resource stop RESOURCE</code>
Remove a resource	<code>crm configure delete RESOURCE</code>
Modify a resource's instance parameters	<code>crm resource param RESOURCE set PARAMETER=VALUE</code>
Delete a resource's instance parameters	<code>crm resource param RESOURCE delete PARAMETER</code>
List current resource defaults	<code>crm configure show type:rsc_defaults</code>
Set resource defaults	<code>crm configure rsc_defaults OPTION=VALUE</code>
List current operation defaults	<code>crm configure show type:op_defaults</code>
Set operation defaults	<code>crm configure op_defaults OPTION=VALUE</code>
Clear fail counts for a resource	<code>crm resource cleanup RESOURCE</code>
Create a colocation constraint	<code>crm configure colocation NAME INFINITY: RESOURCE</code>
Create an ordering constraint	<code>crm configure order NAME mandatory: RESOURCE</code>
Create a location constraint	<code>crm configure location NAME RESOURCE 50: NODE</code>
Move a resource to a specific node	<code>crm resource move RESOURCE NODE</code>
Move a resource away from its current node	<code>crm resource ban RESOURCE NODE</code>
Remove any constraints created by moving a resource	<code>crm resource unmove RESOURCE</code>

See also

- [Set up a distributed replicated block device \(DRBD\)](#)
- [High Availability](#)

Other tools

- [Terminal Multiplexers](#)
- [pam_motd](#)
- [sudo-rs](#)

Terminal multiplexers

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tmux

*tmux(1)*⁸⁰⁸

Control key and keybindings

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⁸⁰⁸ <https://manpages.ubuntu.com/manpages/resolute/man1/tmux.1.html>

Session management

```
$ tmux new-session -s myproject
```

```
$ tmux list-sessions
```

```
$ tmux ls
```

```
$ tmux attach-session -t myproject
```

```
$ tmux a -t myproject
```

```
$ tmux attach-session -t myproject -d
```

```
$ tmux attach
```

```
$ tmux kill-session -t myproject
```

Window management

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Note:

When you exit the shell in a window (by typing `exit` or pressing `Ctrl+d`), that window closes. If it was the last window in a session, the entire `tmux` session terminates.

Pane management

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Console-based screen sharing

```
$ tmux new-session -s shared
```

```
$ tmux attach-session -t shared
```

```
$ tmux attach-session -t shared -r
```

```
$ tmux -S /tmp/shared-session new-session -s collaboration > chmod 777  
/tmp/shared-session
```

```
$ tmux -S /tmp/shared-session attach-session -t collaboration
```

Note:

For multi-user sharing, ensure both users have appropriate permissions on the socket file. Consider security implications when sharing sessions between different users.

Configuration

```
$ set -g mouse on
```

```
$ set-option -g history-limit 10000
```

```
$ tmux source-file ~/.tmux.conf
```

Scripting and automation

*tmux(1)*⁸⁰⁹

⁸⁰⁹ <https://manpages.ubuntu.com/manpages/resolute/man1/tmux.1.html>

tmux plugins

```
$ sudo apt install tmux-plugin-manager
```

Note:

While tmux plugins can be powerful, be aware that they introduce third-party code and configuration into your system. The same considerations discussed in the [Third party repository usage](#) (page 976) documentation apply here - evaluate trustworthiness, security implications, and maintenance status before using plugins, especially in production environments.

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Other options

Mosh (mobile shell)

```
$ sudo apt install mosh
```

```
$ mosh user@remote-server
```

GNU Screen

Byobu

```
$ sudo apt install byobu
```

```
$ byobu
```

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-
-

`pam_motd`

Packages adding to MOTD

Landscape-sysinfo

-

Landscape⁸¹⁰

```
System load:          0.0
Usage of /:           25.7% of 8.55GB
Memory usage:         31%
Swap usage:           0%
Processes:            125
Users logged in:      0
IPv4 address for enp5s0: 10.185.198.41
IPv6 address for enp5s0: fd42:50cb:3a48:1f2c:216:3eff:fe27:c18a
```

Note:

You can run `landscape-sysinfo` manually at any time.

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filesystem checks (fsck)

⁸¹⁰ <https://ubuntu.com/landscape>

Other default MOTD entries

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- ubuntu-pro-client⁸¹¹
-
-
-
-
-
-

Customization

```
#!/bin/bash

# 1. Get timestamp and calculate nearest sysadmin day
NOW=$(date +%s)
THIS_YEAR=$(date +%Y)
TARGET_PREV=$(date -d "$((THIS_YEAR-1))-08-01 last friday" +%s)
TARGET_CURR=$(date -d "$THIS_YEAR-08-01 last friday" +%s)
TARGET_NEXT=$(date -d "$((THIS_YEAR+1))-08-01 last friday" +%s)

# 2. Loop to find the shortest distance
MIN_DIFF=999999999
CHOSEN_DAYS=0
TIMING=""
for TARGET in $TARGET_PREV $TARGET_CURR $TARGET_NEXT; do
    # Calculate difference
    DIFF=$(( (TARGET - NOW) / 86400 ))
```

(continues on next page)

⁸¹¹ https://documentation.ubuntu.com/pro-client/en/latest/explanations/motd_messages/

(continued from previous page)

```
# Get absolute value for comparison
if [ $DIFF -lt 0 ] then
    ABS_DIFF=$((DIFF * -1))
    CURRENT_TIMING="ago"
else
    ABS_DIFF=$DIFF
    CURRENT_TIMING="in"
fi

# Update if this is the closest date found so far
if [ $ABS_DIFF -lt $MIN_DIFF ] then
    MIN_DIFF=$ABS_DIFF
    CHOSEN_DAYS=$ABS_DIFF
    TIMING=$CURRENT_TIMING
fi
done

# 3. Report to gain some admin-love
if [ "$CHOSEN_DAYS" -eq 0 ] then
    echo -e "System Administrator Appreciation Day is Today! Gifts welcome."
elif [ "$TIMING" == "ago" ] then
    echo -e "System Administrator Appreciation Day is ${CHOSEN_DAYS} days ago but we'd
    appreciate to be treated well every day."
else
    echo -e "System Administrator Appreciation Day is in ${CHOSEN_DAYS} days but we'd
    appreciate to be treated well every day."
fi
```

```
$ chmod +x /etc/update-motd.d/98-admin-reminder
```

```
$ ssh ubuntu@yoursystem
```

```
...
System Administrator Appreciation Day is 181 days ago but we'd appreciate to be
treated well every day.
```

Note:

Be careful as this will be executed on every login. The usual pattern to prevent that from slowing down logins is to do anything even slightly complex asynchronously and only cat pre-generated content in the actual MOTD handling. Furthermore consider using stamp files to only update at regular intervals instead of at every login. This example was meant to be kept simple, but the existing serious MOTD entries placed there by the packages make use of such, please look at those for writing your own.

Further reading

- [update-motd\(5\)](#)⁸¹²
- [weather](#)⁸¹³

System utility replacements

[Ubuntu Discourse](#)⁸¹⁴

- [GNU Coreutils](#)⁸¹⁵ [utils Coreutils](#)⁸¹⁶
- [sudo.ws](#)⁸¹⁷ [sudo-rs](#)⁸¹⁸

rust-coreutils

[utils Coreutils Documentation](#)⁸¹⁹

⁸¹² <https://manpages.ubuntu.com/manpages/resolute/man5/update-motd.5.html>

⁸¹³ <https://debaday.debian.net/2007/10/04/weather-check-weather-conditions-and-forecasts-on-the-command-line/>

⁸¹⁴ <https://discourse.ubuntu.com/t/carefully-but-purposefully-oxidising-ubuntu/56995>

⁸¹⁵ <https://www.gnu.org/software/coreutils/manual>

⁸¹⁶ <https://utils.org/coreutils/docs>

⁸¹⁷ <https://www.sudo.ws/docs/man/sudo.man/>

⁸¹⁸ <https://github.com/trifectatechfoundation/sudo-rs>

⁸¹⁹ <https://utils.org/coreutils/docs/index.html>

Release-specific exceptions

Release	Utilities configured to use GNU Coreutils
Ubuntu 24.04 LTS and earlier	All
Ubuntu 25.10	chmod, chown, cp, df, mv, rm, true
Ubuntu 26.04 LTS	cp, df, mv, rm, true
Ubuntu 26.10 and later	None

Note:

Although the `rust-coreutils` package is available in Ubuntu 24.04 LTS, it cannot be configured as the default coreutils. The `coreutils-from-<provider>` packages required to switch the provider are only available starting with Ubuntu 25.10.

Switching the coreutils provider (Ubuntu 25.10 and later)

Note:

`update-alternatives` is not currently suitable for switching between GNU Coreutils and `utils` Coreutils, as `alternatives` are not safe for Essential packages. Instead, use the `coreutils-from-<provider>` packages to switch providers.

```
$ sudo apt install coreutils-from-gnu coreutils-from-utils-  
--allow-remove-essential
```

```
$ sudo apt install coreutils-from-utils coreutils-from-gnu-  
--allow-remove-essential
```

build-essential dependency

Note:

As a result, these packages may currently conflict with `coreutils-from-gnu` when switching the `coreutils` provider.

sudo-rs

Differences

1.
 - [differences-from-original-sudo](#)⁸²⁰
 - [aim-of-the-project](#)⁸²¹
- 2.

⁸²⁰ <https://github.com/trifectatechfoundation/sudo-rs#differences-from-original-sudo>

⁸²¹ <https://github.com/trifectatechfoundation/sudo-rs?tab=readme-ov-file#aim-of-the-project>

See more information⁸²²

3.

4.

LDAP authentication via PAM⁸²³

5.

CLI flags parity⁸²⁴

Our security explanations

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Security

⁸²² <https://github.com/trifectatechfoundation/sudo-rs/issues/1242>

⁸²³ <https://github.com/trifectatechfoundation/sudo-rs/issues/445>

⁸²⁴ <https://github.com/trifectatechfoundation/sudo-rs/issues/129>

General configuration

- *Introduction to security*
- *Security suggestions*

Introduction to security

Server security guidance

security suggestions

security

advanced

Ubuntu Security documentation⁸²⁵

Ubuntu Pro

Ubuntu Pro⁸²⁶

Client⁸²⁷

Ubuntu Pro

⁸²⁵ <https://documentation.ubuntu.com/security/>

⁸²⁶ <https://ubuntu.com/pro>

⁸²⁷ <https://documentation.ubuntu.com/pro-client/en/latest/>

Vulnerability management

itory⁸²⁸

Main repos-

Expanded Security Maintenance⁸³⁰
vulnerabilities⁸²⁹

Kernel application hardening

Livepatch here⁸³¹

about

Security Compliance and Certification

CIS⁸³³

FIPS⁸³²

Reporting vulnerabilities

disclosure policy⁸³⁴

Security suggestions

⁸²⁸ <https://documentation.ubuntu.com/project/how-ubuntu-is-made/concepts/package-archive/#components>

⁸²⁹ <https://ubuntu.com/security/cves/about>

⁸³⁰ <https://documentation.ubuntu.com/security/security-updates/esm/>

⁸³¹ <https://ubuntu.com/security/livepatch/docs>

⁸³² <https://documentation.ubuntu.com/security/compliance/fips/>

⁸³³ <https://documentation.ubuntu.com/security/compliance/usg/>

⁸³⁴ <https://ubuntu.com/security/disclosure-policy>

For any Ubuntu system

Keep your system up-to-date

1.

```
$ sudo apt update && sudo apt upgrade
```

```
$ sudo apt install unattended-upgrades
```

*unattended-upgrades(8)*⁸³⁵

2.

- *Package management*
- *understand the risks and how to minimize them.*

3.

how to upgrade

4. **Ubuntu Pro**⁸³⁶

- Expanded Security Maintenance (ESM)⁸³⁷
- Livepatch⁸³⁸

*official documentation*⁸³⁹

⁸³⁵ <https://manpages.ubuntu.com/manpages/resolute/man8/unattended-upgrades.8.html>

⁸³⁶ <https://ubuntu.com/pro>

⁸³⁷ <https://documentation.ubuntu.com/security/security-updates/esm/>

⁸³⁸ <https://ubuntu.com/security/livepatch/docs>

⁸³⁹ <https://documentation.ubuntu.com/pro-client/en/latest/>

Access Control

1. [principle of least privilege](#)⁸⁴⁰

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-

guide on user management

Network security

- 1.

Firewall

- 2.

OpenSSH

guide to

Physical security

- *Smart card authentication*
- *Smart card authentication with SSH*
- *Console security*

Suggestions for complex setups

[Ubuntu Security](#)⁸⁴¹

Advanced Access Control

- 1.

introduction to OpenLDAP
on how to set up OpenLDAP

- 2.

⁸⁴⁰ https://en.wikipedia.org/wiki/Principle_of_least_privilege

⁸⁴¹ <https://documentation.ubuntu.com/security/>

introduction to Kerberos

set up and use Kerberos

3.

introduction to SSSD
how-to section

4.

Active Directory integration
ADSys
*tation*⁸⁴²

learn how to set it up

ADSys documen-

5.

to set up AppArmor

how

Security of communications

1.

Introduction to WireGuard VPN
Guard VPN

set up Wire-

2.

about the available clients

how to install OpenVPN

⁸⁴² <https://documentation.ubuntu.com/adsys/latest/>

Cryptography

about crypto

libraries

Introduction to cryptographic libraries

About certificates

Install root CA certificate in

the trust store

Compliance and auditing

Ubuntu Security documentation⁸⁴³

Authentication

- *Introduction to Kerberos*
- *Introduction to SSSD*
- *About DNSSEC*

Introduction to Kerberos

Overview

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⁸⁴³ <https://documentation.ubuntu.com/security/compliance/>

- *hostname*
- *ACL*
- *DNS*
-
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Further reading

- MIT Kerberos⁸⁴⁴
- Libera.Chat⁸⁴⁶ Ubuntu Server⁸⁴⁵
- Another guide for installing Kerberos on Debian, includes PKINIT⁸⁴⁷

⁸⁴⁴ <http://web.mit.edu/Kerberos/>

⁸⁴⁵ <https://matrix.to/#/#server:ubuntu.com>

⁸⁴⁶ <https://libera.chat/>

⁸⁴⁷ <http://techpubs.spinlocksolutions.com/dklar/kerberos.html>

Introduction to network user authentication with SSSD

System Security Services Daemon (SSSD)⁸⁴⁸

Common deployment scenarios

- *Active Directory*
- *LDAP*
- *LDAP and Kerberos*

Integration with PAM and NSS

Troubleshooting

SSSD troubleshooting guide

DNS Security Extensions (DNSSEC)

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⁸⁴⁸ <https://sssd.io/>

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What is it?

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Simplified 12-step DNSSEC validation process⁸⁴⁹

⁸⁴⁹ <https://bind9.readthedocs.io/en/latest/dnssec-guide.html#the-12-step-dnssec-validation-process-simplified>

DNS daemons

Daemon	Type	DNSSEC support
systemd-resolv	Stub solver (local)	Yes. Disabled by default. Controlled via DNSSEC=... setting in /etc/systemd/resolved.conf.d.
dnsmasq	Stub solver	Yes. Disabled by default. Controlled via dnssec and conf-file=.../trust-anchors.conf settings.
bind9	Recursive Resolver	Yes. Enabled by default via dnssec-validation auto; setting.

-
-

```
$ sudo mkdir -p /etc/systemd/resolved.conf.d
```

```
$ sudo tee /etc/systemd/resolved.conf.d/10-dnssec.conf >/dev/null <<EOF
[Resolve]
DNSSEC=yes
EOF
```

```
$ sudo systemctl reload systemd-resolved.service
```

```
$ dig @127.0.0.53 isc.org +dnssec
[...]
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 5, AUTHORITY: 0, ADDITIONAL: 1
```

LP: #2119652⁸⁵¹ ref2⁸⁵² ref3⁸⁵³ LP: #2121483⁸⁵⁴**Warning:**

Be aware that enforcing DNSSEC in strict mode can lead to errors, especially for local, unsigned domains and you would only be able to reach such services by accessing them through their IP address directly. For example this could manifest itself by errors like `DNS_PROBE_FINISHED_NXDOMAIN` in your browser, when trying to access services in the local network.

- <https://internet.nl/test-connection/>
- <https://wander.science/projects/dns/dnssec-resolver-test/>

Note:

In case of issues with Domain Name resolution, make sure to remove relevant drop-in configs in `/etc/resolved.conf.d` and execute `systemctl reload systemd-resolved.service`. This will reset any DNSSEC configuration to the default `DNSSEC=no` and can be confirmed by executing `resolvectl dnssec`.

New Resource Records (RRs)

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-
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⁸⁵⁰ <https://datatracker.ietf.org/doc/html/rfc3225>

⁸⁵¹ <https://bugs.launchpad.net/bugs/2119652>

⁸⁵² <https://lists.fedoraproject.org/archives/list/devel@lists.fedoraproject.org/message/AFHNUEHKC5KJVGBGSJBH2BMESUAGDF4H/>

⁸⁵³ <https://lists.fedoraproject.org/archives/list/devel@lists.fedoraproject.org/message/P63RI3VBQ7NGL3AKMTR7PCVHVSCPYPCLF/>

⁸⁵⁴ <https://bugs.launchpad.net/bugs/2121483>

```
$ dig @1.1.1.1 +dnssec -t MX isc.org

; <<>> DiG 9.18.28-0ubuntu0.24.04.1-Ubuntu <<>> @1.1.1.1 +dnssec -t MX isc.org
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 51256
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 3, AUTHORITY: 0, ADDITIONAL: 1

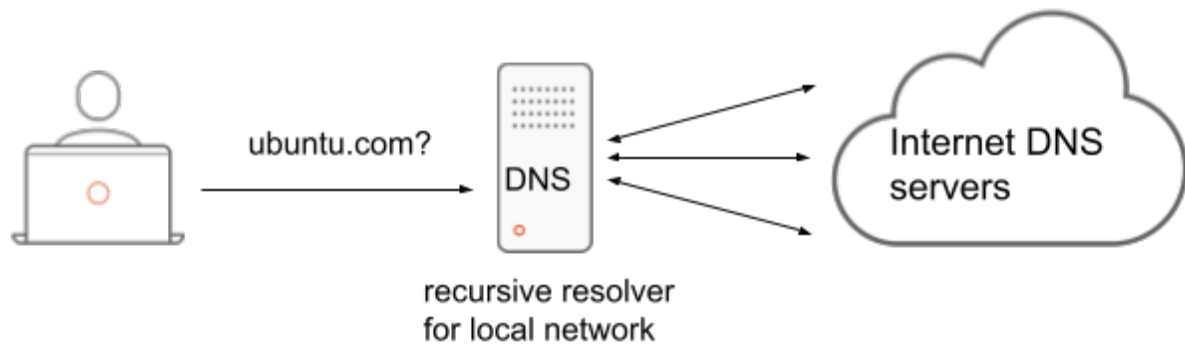
;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags: do; udp: 1232
;; QUESTION SECTION:
;isc.org.                IN      MX

;; ANSWER SECTION:
isc.org.                300     IN      MX      5 mx.pao1.isc.org.
isc.org.                300     IN      MX      10 mx.ams1.isc.org.
isc.org.                300     IN      RRSIG   MX 13 2 300 20241029080338
20241015071431 27566 isc.org. LG/
cvFmZ8jLz+CM14foaCtwsyCTwKXfVBZV2jcl2UV8zV79QRLs0YXJ3 sjag1vYCqc+Q5AwUi2DB8L/
wZR6EJQ==

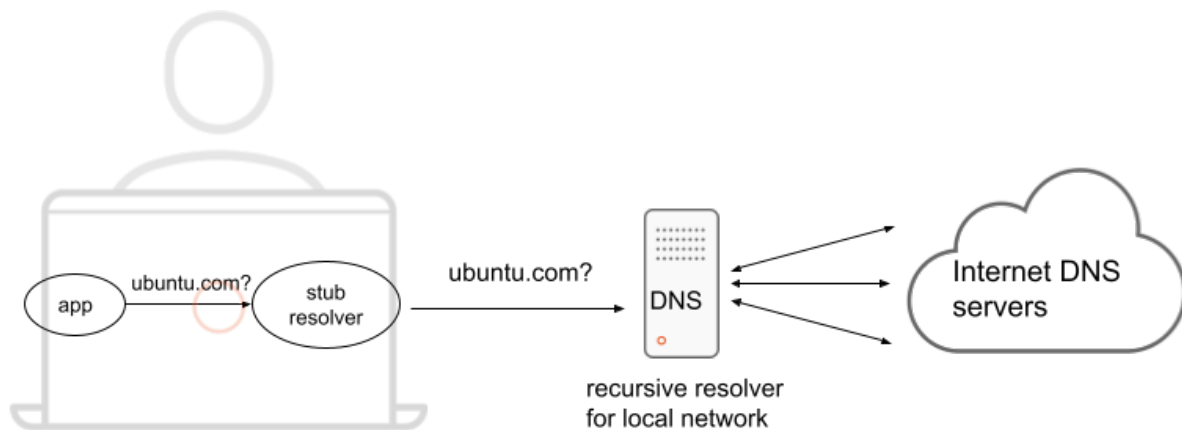
;; Query time: 199 msec
;; SERVER: 1.1.1.1#53(1.1.1.1) (UDP)
;; WHEN: Tue Oct 22 16:44:33 UTC 2024
;; MSG SIZE rcvd: 187
```

Other uses

Where does the DNSSEC validation happen?



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hostname

```
nameserver 127.0.0.53
options edns0 trust-ad
```



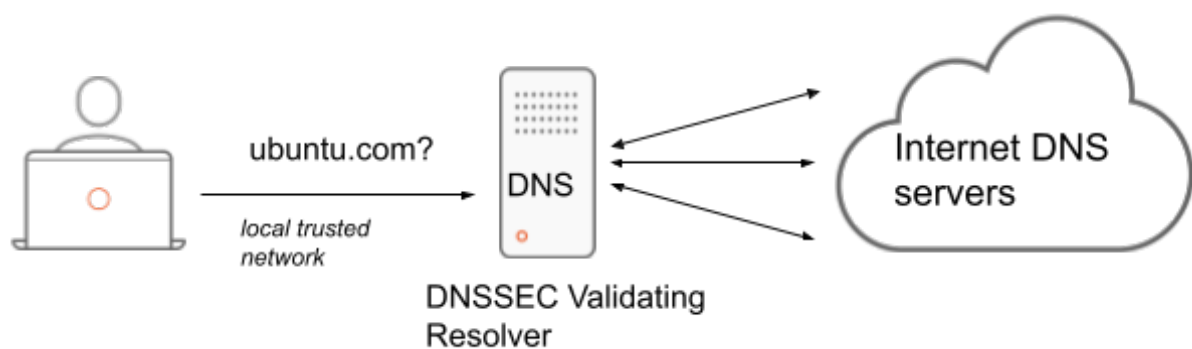
```
Global
    Protocols: -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
    resolv.conf mode: stub

Link 12 (eth0)
    Current Scopes: DNS
    Protocols: +DefaultRoute -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
Current DNS Server: 10.10.17.1
    DNS Servers: 10.10.17.1
    DNS Domain: lxd
```

DHCP

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Validating Resolver



Note:

Up to Ubuntu 24.04 LTS, version 1:9.18.34-1, the default was explicitly stated as `dnssec-validation auto;` in `/etc/bind/named.conf.options`.

```
nameserver 127.0.0.53
options edns0 trust-ad
```

*resolv.conf(5)*⁸⁵⁵

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```
nameserver 10.10.17.11
options edns0 trust-ad
```

-

⁸⁵⁵ <https://manpages.ubuntu.com/manpages/resolute/man5/resolv.conf.5.html>

•

Local DNSSEC validation

DNS-over-TLS (DoT) DNS-over-HTTPS (DoH)

```
$ resolvectl dnssec
```

```
Global: no
```

```
Link 44 (eth0): no
```

```
$ resolvectl query --type=MX isc.org
```

```
isc.org IN MX 10 mx.ams1.isc.org
```

```
-- link: eth0
```

```
isc.org IN MX 5 mx.pao1.isc.org
```

```
-- link: eth0
```

```
-- Information acquired via protocol DNS in 37.2ms.
```

```
-- Data is authenticated: no; Data was acquired via local or encrypted transport: no
```

```
-- Data from: network
```

```
$ sudo resolvectl dnssec eth0 true
```

```
$ resolvectl query --type=MX isc.org

isc.org IN MX 5 mx.pao1.isc.org      -- link: eth0
isc.org IN MX 10 mx.ams1.isc.org     -- link: eth0

-- Information acquired via protocol DNS in 3.0ms.
-- Data is authenticated: yes; Data was acquired via local or encrypted transport:
no
-- Data from: network
```

```
-- Data is authenticated: yes
```

What happens when DNSSEC validation fails

```
$ resolvectl query www.dnssec-failed.org

www.dnssec-failed.org: resolve call failed: Could not resolve 'www.dnssec-failed.org',
server or network returned error SERVFAIL
```

```
Oct 22 17 14:50 n-dns named[285]: validating dnssec-failed.org/DNSKEY: no valid signature
found (DS)
Oct 22 17 14:50 n-dns named[285]: no valid RRSIG resolving 'dnssec-failed.org/DNSKEY/IN':
68.87.68.244#53
...
Oct 22 17 14:52 n-dns named[285]: broken trust chain resolving 'www.dnssec-failed.org/
AAAA/IN': 68.87.72.244#53
```

```
$ sudo resolvectl dnssec eth0 true
```

```
$ resolvectl query www.dnssec-failed.org

www.dnssec-failed.org: resolve call failed: DNSSEC validation failed: no-signature
```

```
$ host www.dnssec-failed.org  
Host www.dnssec-failed.org not found: 2(SERVFAIL)
```

Further reading

- [DNSSEC - What Is It and Why Is It Important](#)⁸⁵⁶
- [Tool to visualize the DNSSEC chain of trust of a domain](#)⁸⁵⁷
- [DANE](#)⁸⁵⁸
- [RFC 4255](#)⁸⁵⁹
- *Basic DNSSEC troubleshooting*

Cryptography

- *Our cryptography section*
- *Certificates*
- *How to obtain certificates*

Cryptography

Introduction to cryptographic libraries

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⁸⁵⁶ <https://www.icann.org/resources/pages/dnssec-what-is-it-why-important-2019-03-05-en>

⁸⁵⁷ <https://dnsviz.net/>

⁸⁵⁸ https://en.wikipedia.org/wiki/DNS-based_Authentication_of_Named_Entities

⁸⁵⁹ <https://datatracker.ietf.org/doc/html/rfc4255>

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- *GnuTLS*
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-
- *gcrypt*

Determining which libraries are being used by an application

Follow dynamic links

Check static linking

Look at the plugins

Examine the execution of external binaries

Indirect use of libraries or executables

Check documentation more carefully

Detailed examples of how to locate crypto libraries

Use dpkg to check package dependencies

```
$ dpkg -S /usr/bin/lynx
```

```
lynx: /usr/bin/lynx
```

```
$ dpkg -s lynx | grep -E "^(Depends|Recommends)"
```

```
Depends: libbsd0 (>= 0.0), libbz2-1.0, libc6 (>= 2.34), libgnutls30 (>= 3.7.0),  
libidn2-0 (>= 2.0.0), libncursesw6 (>= 6), libtinfo6 (>= 6), zlib1g (>= 1:1.1.4),  
lynx-common  
Recommends: mime-support
```

Using ldd to list dynamic libraries

```
$ ldd /usr/bin/lynx  
  
linux-vdso.so.1 (0x00007ffffd2df000)  
libz.so.1 => /lib/x86_64-linux-gnu/libz.so.1 (0x00007feb69d77000)  
libbz2.so.1.0 => /lib/x86_64-linux-gnu/libbz2.so.1.0 (0x00007feb69d64000)  
libidn2.so.0 => /lib/x86_64-linux-gnu/libidn2.so.0 (0x00007feb69d43000)  
libncursesw.so.6 => /lib/x86_64-linux-gnu/libncursesw.so.6 (0x00007feb69d07000)  
libtinfo.so.6 => /lib/x86_64-linux-gnu/libtinfo.so.6 (0x00007feb69cd5000)  
libgnutls.so.30 => /lib/x86_64-linux-gnu/libgnutls.so.30 (0x00007feb69aea000)  
libbsd.so.0 => /lib/x86_64-linux-gnu/libbsd.so.0 (0x00007feb69ad0000)  
libc.so.6 => /lib/x86_64-linux-gnu/libc.so.6 (0x00007feb698a8000)  
libunistring.so.2 => /lib/x86_64-linux-gnu/libunistring.so.2  
(0x00007feb696fe000)  
libp11-kit.so.0 => /lib/x86_64-linux-gnu/libp11-kit.so.0 (0x00007feb695c3000)  
libtasn1.so.6 => /lib/x86_64-linux-gnu/libtasn1.so.6 (0x00007feb695ab000)  
libnettle.so.8 => /lib/x86_64-linux-gnu/libnettle.so.8 (0x00007feb69565000)  
libhogweed.so.6 => /lib/x86_64-linux-gnu/libhogweed.so.6 (0x00007feb6951b000)  
libgmp.so.10 => /lib/x86_64-linux-gnu/libgmp.so.10 (0x00007feb69499000)  
/lib64/ld-linux-x86-64.so.2 (0x00007feb69fe6000)  
libmd.so.0 => /lib/x86_64-linux-gnu/libmd.so.0 (0x00007feb6948c000)  
libffi.so.8 => /lib/x86_64-linux-gnu/libffi.so.8 (0x00007feb6947f000)
```

```
$ objdump -x /usr/bin/lynx|grep NEEDED
```



```
NEEDED          libz.so.1
NEEDED          libbz2.so.1.0
NEEDED          libidn2.so.0
NEEDED          libncursesw.so.6
NEEDED          libtinfo.so.6
NEEDED          libgnutls.so.30
NEEDED          libbsd.so.0
NEEDED          libc.so.6
```

```
$ lddtree /usr/bin/lynx

lynx => /usr/bin/lynx (interpreter => /lib64/ld-linux-x86-64.so.2)
libz.so.1 => /lib/x86_64-linux-gnu/libz.so.1
libbz2.so.1.0 => /lib/x86_64-linux-gnu/libbz2.so.1.0
libidn2.so.0 => /lib/x86_64-linux-gnu/libidn2.so.0
libunistring.so.2 => /lib/x86_64-linux-gnu/libunistring.so.2
libncursesw.so.6 => /lib/x86_64-linux-gnu/libncursesw.so.6
libtinfo.so.6 => /lib/x86_64-linux-gnu/libtinfo.so.6
libgnutls.so.30 => /lib/x86_64-linux-gnu/libgnutls.so.30
libp11-kit.so.0 => /lib/x86_64-linux-gnu/libp11-kit.so.0
libffi.so.8 => /lib/x86_64-linux-gnu/libffi.so.8
libtasn1.so.6 => /lib/x86_64-linux-gnu/libtasn1.so.6
libnettle.so.8 => /lib/x86_64-linux-gnu/libnettle.so.8
libhogweed.so.6 => /lib/x86_64-linux-gnu/libhogweed.so.6
libgmp.so.10 => /lib/x86_64-linux-gnu/libgmp.so.10
ld-linux-x86-64.so.2 => /lib64/ld-linux-x86-64.so.2
libbsd.so.0 => /lib/x86_64-linux-gnu/libbsd.so.0
libmd.so.0 => /lib/x86_64-linux-gnu/libmd.so.0
libc.so.6 => /lib/x86_64-linux-gnu/libc.so.6
```

Check package headers for static linking

-
-

```
$ dpkg -s rclone | grep -E "^(Depends|Recommends)"

Depends: libc6 (>= 2.34)
```

```
$ dpkg -s rclone | grep Built-Using
```

```
Built-Using: go-md2man-v2 (= 2.0.1+ds1-1), golang-1.18 (= 1.18-1ubuntu1), golang-bazil-fuse (= 0.0~git20160811.0.371fbbd-3), ...
```

<https://git.launchpad.net/ubuntu/+source/rclone/tree/debian/control?h=ubuntu/jammy-devel#n7>

```
$ apt-cache show golang-golang-x-crypto-dev | grep ^Source:
```

```
Source: golang-go.crypto
```

Note:

If there is no Source: line, then it means the name of the source package is the same as the binary package that was queried.

Next steps

- [OpenSSL guide](#)
- [GnuTLS guide](#)
- [Troubleshooting TLS/SSL](#)

Common crypto libraries

OpenSSL

Structure of the config file

SSL config manual page⁸⁶⁰

```
openssl_conf = <name-of-conf-section>

[<name-of-conf-section>]
ssl_conf = <name-of-ssl-section>

[<name-of-ssl-section>]
server = <name of section>
client = <name of section>
system_default = <name of section>
```

```
openssl_conf = openssl_init

[openssl_init]
ssl_conf = ssl_sect

[ssl_sect]
system_default = system_default_sect

[system_default_sect]
CipherString = DEFAULT:@SECLEVEL=2
```

SSL_CTX_set_security_level(3)⁸⁶¹

⁸⁶⁰ <https://manpages.ubuntu.com/manpages/resolute/man5ssl/config.5ssl.html>

⁸⁶¹ https://manpages.ubuntu.com/manpages/resolute/man3/SSL_CTX_set_security_level.3.html

Note:

In Ubuntu Jammy, TLS versions below 1.2 are **disabled** in OpenSSL's SECLEVEL=2 due to [this patch](#)⁸⁶².

⁸⁶² <https://git.launchpad.net/ubuntu/+source/openssl/tree/debian/patches/tls1.2-min-seclevel2.patch?h=ubuntu/jammy-devel>

CLEVEL=2⁸⁶³

set to SE-

openssl ciphers⁸⁶⁴

```
$ openssl ciphers -s -v DEFAULT:@SECLEVEL=2

TLS_AES_256_GCM_SHA384      TLSv1.3 Kx=any      Au=any  Enc=AESGCM(256)
Mac=AEAD
TLS_CHACHA20_POLY1305_SHA256 TLSv1.3 Kx=any      Au=any  Enc=CHACHA20/
POLY1305(256) Mac=AEAD
TLS_AES_128_GCM_SHA256      TLSv1.3 Kx=any      Au=any  Enc=AESGCM(128)
Mac=AEAD
ECDHE-ECDSA-AES256-GCM-SHA384 TLSv1.2 Kx=ECDH      Au=ECDSA Enc=AESGCM(256)
Mac=AEAD
(...)
```

Note:

The `openssl ciphers` command will output even ciphers that are not allowed, unless the `-s` switch is given. That option tells the command to list only **supported** ciphers.

CONF_cmd(3)⁸⁶⁵

SSL_

Cipher strings, cipher suites, cipher lists

phers⁸⁶⁶

openssl ci-

⁸⁶³ <https://git.launchpad.net/ubuntu/+source/openssl/tree/debian/rules?h=ubuntu/jammy-devel#n15>

⁸⁶⁴ <https://docs.openssl.org/3.0/man1/openssl-ciphers/>

⁸⁶⁵ https://manpages.ubuntu.com/manpages/resolute/man3/SSL_CONF_cmd.3.html

⁸⁶⁶ <https://docs.openssl.org/3.0/man1/openssl-ciphers/>

- upstream documentation⁸⁶⁷
-

```
[system_default_sect]
CipherString = DEFAULT:@SECLEVEL=2
CipherSuites = TLS_AES_256_GCM_SHA384:TLS_CHACHA20_POLY1305_SHA256
```

openssl ciphers examples

```
$ openssl ciphers -s -v
```

⁸⁶⁷ <https://docs.openssl.org/3.0/man1/openssl-ciphers/>

TLS_AES_256_GCM_SHA384 Mac=AEAD	TLSv1.3 Kx=any	Au=any	Enc=AESGCM(256)
TLS_CHACHA20_POLY1305_SHA256 POLY1305(256) Mac=AEAD	TLSv1.3 Kx=any	Au=any	Enc=CHACHA20/
TLS_AES_128_GCM_SHA256 Mac=AEAD	TLSv1.3 Kx=any	Au=any	Enc=AESGCM(128)
ECDHE - ECDSA - AES256 - GCM - SHA384 Mac=AEAD	TLSv1.2 Kx=ECDH	Au=ECDSA	Enc=AESGCM(256)
ECDHE - RSA - AES256 - GCM - SHA384 Mac=AEAD	TLSv1.2 Kx=ECDH	Au=RSA	Enc=AESGCM(256)
DHE - RSA - AES256 - GCM - SHA384 Mac=AEAD	TLSv1.2 Kx=DH	Au=RSA	Enc=AESGCM(256)
ECDHE - ECDSA - CHACHA20 - POLY1305 POLY1305(256) Mac=AEAD	TLSv1.2 Kx=ECDH	Au=ECDSA	Enc=CHACHA20/
ECDHE - RSA - CHACHA20 - POLY1305 POLY1305(256) Mac=AEAD	TLSv1.2 Kx=ECDH	Au=RSA	Enc=CHACHA20/
DHE - RSA - CHACHA20 - POLY1305 POLY1305(256) Mac=AEAD	TLSv1.2 Kx=DH	Au=RSA	Enc=CHACHA20/
ECDHE - ECDSA - AES128 - GCM - SHA256 Mac=AEAD	TLSv1.2 Kx=ECDH	Au=ECDSA	Enc=AESGCM(128)
ECDHE - RSA - AES128 - GCM - SHA256 Mac=AEAD	TLSv1.2 Kx=ECDH	Au=RSA	Enc=AESGCM(128)
DHE - RSA - AES128 - GCM - SHA256 Mac=AEAD	TLSv1.2 Kx=DH	Au=RSA	Enc=AESGCM(128)
ECDHE - ECDSA - AES256 - SHA384 Mac=SHA384	TLSv1.2 Kx=ECDH	Au=ECDSA	Enc=AES(256)
ECDHE - RSA - AES256 - SHA384 Mac=SHA384	TLSv1.2 Kx=ECDH	Au=RSA	Enc=AES(256)
DHE - RSA - AES256 - SHA256 Mac=SHA256	TLSv1.2 Kx=DH	Au=RSA	Enc=AES(256)
ECDHE - ECDSA - AES128 - SHA256 Mac=SHA256	TLSv1.2 Kx=ECDH	Au=ECDSA	Enc=AES(128)
ECDHE - RSA - AES128 - SHA256 Mac=SHA256	TLSv1.2 Kx=ECDH	Au=RSA	Enc=AES(128)
DHE - RSA - AES128 - SHA256 Mac=SHA256	TLSv1.2 Kx=DH	Au=RSA	Enc=AES(128)
ECDHE - ECDSA - AES256 - SHA Mac=SHA1	TLSv1 Kx=ECDH	Au=ECDSA	Enc=AES(256)
ECDHE - RSA - AES256 - SHA Mac=SHA1	TLSv1 Kx=ECDH	Au=RSA	Enc=AES(256)
DHE - RSA - AES256 - SHA Mac=SHA1	SSLv3 Kx=DH	Au=RSA	Enc=AES(256)
ECDHE - ECDSA - AES128 - SHA Mac=SHA1	TLSv1 Kx=ECDH	Au=ECDSA	Enc=AES(128)
ECDHE - RSA - AES128 - SHA Mac=SHA1	TLSv1 Kx=ECDH	Au=RSA	Enc=AES(128)
DHE - RSA - AES128 - SHA Mac=SHA1	SSLv3 Kx=DH	Au=RSA	Enc=AES(128)
AES256 - GCM - SHA384 Mac=AEAD	TLSv1.2 Kx=RSA	Au=RSA	Enc=AESGCM(256)
AES128 - GCM - SHA256 Mac=AEAD	TLSv1.2 Kx=RSA	Au=RSA	Enc=AESGCM(128)
AES256 - SHA256 Mac=SHA256	TLSv1.2 Kx=RSA	Au=RSA	Enc=AES(256)
AES128 - SHA256 Mac=SHA256	TLSv1.2 Kx=RSA	Au=RSA	Enc=AES(128)
AES256 - SHA Mac=SHA1	SSLv3 Kx=RSA	Au=RSA	Enc=AES(256)
AES128 - SHA Mac=SHA1	SSLv3 Kx=RSA	Au=RSA	Enc=AES(128)

```
$ openssl ciphers -s -v 'DEFAULTS:-AES128:-SHA1'
```

TLS_AES_256_GCM_SHA384 Mac=AEAD	TLSv1.3 Kx=any	Au=any	Enc=AESGCM(256)
TLS_CHACHA20_POLY1305_SHA256 POLY1305(256) Mac=AEAD	TLSv1.3 Kx=any	Au=any	Enc=CHACHA20/
TLS_AES_128_GCM_SHA256 Mac=AEAD	TLSv1.3 Kx=any	Au=any	Enc=AESGCM(128)
ECDHE-ECDSA-AES256-GCM-SHA384 Mac=AEAD	TLSv1.2 Kx=ECDH	Au=ECDSA	Enc=AESGCM(256)
ECDHE-RSA-AES256-GCM-SHA384 Mac=AEAD	TLSv1.2 Kx=ECDH	Au=RSA	Enc=AESGCM(256)
DHE-RSA-AES256-GCM-SHA384 Mac=AEAD	TLSv1.2 Kx=DH	Au=RSA	Enc=AESGCM(256)
ECDHE-ECDSA-CHACHA20-POLY1305 POLY1305(256) Mac=AEAD	TLSv1.2 Kx=ECDH	Au=ECDSA	Enc=CHACHA20/
ECDHE-RSA-CHACHA20-POLY1305 POLY1305(256) Mac=AEAD	TLSv1.2 Kx=ECDH	Au=RSA	Enc=CHACHA20/
DHE-RSA-CHACHA20-POLY1305 POLY1305(256) Mac=AEAD	TLSv1.2 Kx=DH	Au=RSA	Enc=CHACHA20/
ECDHE-ECDSA-AES256-SHA384 Mac=SHA384	TLSv1.2 Kx=ECDH	Au=ECDSA	Enc=AES(256)
ECDHE-RSA-AES256-SHA384 Mac=SHA384	TLSv1.2 Kx=ECDH	Au=RSA	Enc=AES(256)
DHE-RSA-AES256-SHA256 Mac=SHA256	TLSv1.2 Kx=DH	Au=RSA	Enc=AES(256)
AES256-GCM-SHA384 Mac=AEAD	TLSv1.2 Kx=RSA	Au=RSA	Enc=AESGCM(256)
AES256-SHA256 Mac=SHA256	TLSv1.2 Kx=RSA	Au=RSA	Enc=AES(256)

openssl ciphers manual page⁸⁶⁸

```
$ openssl ciphers -s -v -ciphersuites
```

```
TLS_AES_256_GCM_SHA384:TLS_CHACHA20_POLY1305_SHA256 'DEFAULTS:-AES128:-SHA1'
```

⁸⁶⁸ <https://docs.openssl.org/3.0/man1/openssl-ciphers/>

TLS_AES_256_GCM_SHA384 Mac=AEAD	TLSv1.3 Kx=any	Au=any	Enc=AESGCM(256)
TLS_CHACHA20_POLY1305_SHA256 POLY1305(256) Mac=AEAD	TLSv1.3 Kx=any	Au=any	Enc=CHACHA20/
ECDHE-ECDSA-AES256-GCM-SHA384 Mac=AEAD	TLSv1.2 Kx=ECDH	Au=ECDSA	Enc=AESGCM(256)
ECDHE-RSA-AES256-GCM-SHA384 Mac=AEAD	TLSv1.2 Kx=ECDH	Au=RSA	Enc=AESGCM(256)
DHE-RSA-AES256-GCM-SHA384 Mac=AEAD	TLSv1.2 Kx=DH	Au=RSA	Enc=AESGCM(256)
ECDHE-ECDSA-CHACHA20-POLY1305 POLY1305(256) Mac=AEAD	TLSv1.2 Kx=ECDH	Au=ECDSA	Enc=CHACHA20/
ECDHE-RSA-CHACHA20-POLY1305 POLY1305(256) Mac=AEAD	TLSv1.2 Kx=ECDH	Au=RSA	Enc=CHACHA20/
DHE-RSA-CHACHA20-POLY1305 POLY1305(256) Mac=AEAD	TLSv1.2 Kx=DH	Au=RSA	Enc=CHACHA20/
ECDHE-ECDSA-AES256-SHA384 Mac=SHA384	TLSv1.2 Kx=ECDH	Au=ECDSA	Enc=AES(256)
ECDHE-RSA-AES256-SHA384 Mac=SHA384	TLSv1.2 Kx=ECDH	Au=RSA	Enc=AES(256)
DHE-RSA-AES256-SHA256 Mac=SHA256	TLSv1.2 Kx=DH	Au=RSA	Enc=AES(256)
AES256-GCM-SHA384 Mac=AEAD	TLSv1.2 Kx=RSA	Au=RSA	Enc=AESGCM(256)
AES256-SHA256 Mac=SHA256	TLSv1.2 Kx=RSA	Au=RSA	Enc=AES(256)

Config file examples

Only use TLSv1.3

```
[system_default_sect]
CipherString = DEFAULT:@SECLEVEL=2
MinProtocol = TLSv1.3
```

```
$ curl https://j-server.lxd/stats
curl: (35) error:0A00042E:SSL routines::tlsv1 alert protocol version
```



```
$ wget https://j-server.lxd/stats

--2023-01-06 13:41:50-- https://j-server.lxd/stats
Resolving j-server.lxd (j-server.lxd)... 10.0.100.87
Connecting to j-server.lxd (j-server.lxd)|10.0.100.87|:443... connected.
OpenSSL: error:0A00042E:SSL routines::tlsv1 alert protocol version
Unable to establish SSL connection.
```

Use only AES256 with TLSv1.3

```
[system_default_sect]
CipherString = DEFAULT:@SECLEVEL=2
CipherSuites = TLS_AES_256_GCM_SHA384
MinProtocol = TLSv1.3
```

the Troubleshooting section

```
$ sudo openssl s_server -cert j-server.pem -key j-server.key -port 443 -www
```

Note:

Be sure to use another system for this server, or else it will be subject to the same `/etc/ssl/openssl.cnf` constraints you are testing on the client, and this can lead to very confusing results.

```
$ wget https://j-server.lxd/stats -O /dev/stdout -q | grep Cipher -w

New, TLSv1.3, Cipher is TLS_AES_256_GCM_SHA384
Cipher      : TLS_AES_256_GCM_SHA384
```

```
$ sudo openssl s_server -cert j-server.pem -key j-server.key -port 443 -www
-ciphersuites TLS_CHACHA20_POLY1305_SHA256
```

```
$ wget https://j-server.lxd/stats -O /dev/stdout
```

```
--2023-01-06 14:20:55-- https://j-server.lxd/stats
Resolving j-server.lxd (j-server.lxd)... 10.0.100.87
Connecting to j-server.lxd (j-server.lxd)|10.0.100.87|:443... connected.
OpenSSL: error:0A000410:SSL routines::ssl3 alert handshake failure
Unable to establish SSL connection.
```

Drop AES128 entirely

```
[system_default_sect]
CipherString = DEFAULT:@SECLEVEL=2:!AES128
CipherSuites = TLS_AES_256_GCM_SHA384:TLS_CHACHA20_POLY1305_SHA256
MinProtocol = TLSv1.2
```

```
$ sudo openssl s_server -cert j-server.pem -key j-server.key -port 443 -www
-tls1_2
```

```
$ wget https://j-server.lxd/stats -O /dev/stdout -q | grep Cipher -w

New, TLSv1.2, Cipher is ECDHE-RSA-AES256-GCM-SHA384
Cipher      : ECDHE-RSA-AES256-GCM-SHA384
```

```
$ sudo openssl s_server -cert j-server.pem -key j-server.key -port 443 -www
-tls1_2 -cipher 'DEFAULT:!AES256:!CHACHA20'
```

```
$ wget https://j-server.lxd/stats -O /dev/stdout -q | grep Cipher -w

New, TLSv1.2, Cipher is ECDHE-RSA-AES128-GCM-SHA256
Cipher      : ECDHE-RSA-AES128-GCM-SHA256
```

```
$ echo | openssl s_client -connect j-server.lxd:443 | grep -w -i cipher  
  
4007F4F9D47F0000:error:0A000410:SSL routines:ssl3_read_bytes:sslv3 alert handshake  
failure:../ssl/record/rec_layer_s3.c:1584:SSL alert number 40  
New, (NONE), Cipher is (NONE)
```

```
$ echo | openssl s_client -connect j-server.lxd:443 --cipher DEFAULT:AES128  
2>&1| grep -w -i cipher  
  
New, TLSv1.2, Cipher is ECDHE-RSA-AES128-GCM-SHA256  
Cipher      : ECDHE-RSA-AES128-GCM-SHA256
```

Further reading

- [OpenSSL home page](https://www.openssl.org/)⁸⁶⁹
- - https://docs.openssl.org/3.0/man3/SSL_CTX_set_security_level/
 - <https://www.feistyduck.com/library/openssl-cookbook/online/openssl-command-line/understanding-security-levels.html>
- [SSL_CONF_cmd\(3\)](https://manpages.ubuntu.com/manpages/resolute/man3/SSL_CONF_cmd.3.html)⁸⁷⁰

GnuTLS

GnuTLS

⁸⁶⁹ <https://www.openssl.org/>

⁸⁷⁰ https://manpages.ubuntu.com/manpages/resolute/man3/SSL_CONF_cmd.3.html

Structure of the config file

```
[global]
override-mode = blocklist

[priorities]
SYSTEM = NORMAL:-MD5

[overrides]
tls-disabled-mac = sha1
```

The global section

-
-

```
[global]
override-mode = allowlist

[overrides]
secure-hash = sha256
enabled-curve = secp256r1
secure-sig = ecdsa-secp256r1-sha256
enabled-version = tls1.3
tls-enabled-cipher = aes-128-gcm
tls-enabled-mac = aead
tls-enabled-group = secp256r1
```

```
[global]
override-mode = blocklist

[overrides]
tls-disabled-cipher = aes-128-cbc
tls-disabled-cipher = aes-256-cbc
tls-disabled-mac = sha1
tls-disabled-group = group-ffdhe8192
```

disabling algorithms and protocols⁸⁷¹

Priority strings

defined in `debian/rules`⁸⁷²

```
NORMAL: -VERS-ALL:+VERS-TLS1.3:+VERS-TLS1.2:+VERS-DTLS1.2:%PROFILE_MEDIUM
```

defined in this table⁸⁷³

```
$ gnutls-cli --list --priority SECURE256
```

Cipher suites for SECURE256

TLS_AES_256_GCM_SHA384	0x13, 0x02	TLS1.3
TLS_CHACHA20_POLY1305_SHA256	0x13, 0x03	TLS1.3
TLS_ECDHE_ECDSA_AES_256_GCM_SHA384	0xc0, 0x2c	TLS1.2
TLS_ECDHE_ECDSA_CHACHA20_POLY1305	0xcc, 0xa9	TLS1.2
TLS_ECDHE_ECDSA_AES_256_CCM	0xc0, 0xad	TLS1.2
TLS_ECDHE_RSA_AES_256_GCM_SHA384	0xc0, 0x30	TLS1.2
TLS_ECDHE_RSA_CHACHA20_POLY1305	0xcc, 0xa8	TLS1.2
TLS_RSA_AES_256_GCM_SHA384	0x00, 0x9d	TLS1.2
TLS_RSA_AES_256_CCM	0xc0, 0x9d	TLS1.2
TLS_DHE_RSA_AES_256_GCM_SHA384	0x00, 0x9f	TLS1.2
TLS_DHE_RSA_CHACHA20_POLY1305	0xcc, 0xaa	TLS1.2
TLS_DHE_RSA_AES_256_CCM	0xc0, 0x9f	TLS1.2

Protocols: VERS-TLS1.3, VERS-TLS1.2, VERS-TLS1.1, VERS-TLS1.0, VERS-DTLS1.2, VERS-DTLS1.0

Ciphers: AES-256-GCM, CHACHA20-POLY1305, AES-256-CBC, AES-256-CCM

MACs: AEAD

Key Exchange Algorithms: ECDHE-ECDSA, ECDHE-RSA, RSA, DHE-RSA

Groups: GROUP-SECP384R1, GROUP-SECP521R1, GROUP-FFDHE8192

PK-signatures: SIGN-RSA-SHA384, SIGN-RSA-PSS-SHA384, SIGN-RSA-PSS-RSAE-SHA384, SIGN-ECDSA-SHA384, SIGN-ECDSA-SECP384R1-SHA384, SIGN-EdDSA-Ed448, SIGN-RSA-SHA512, SIGN-RSA-PSS-SHA512, SIGN-RSA-PSS-RSAE-SHA512, SIGN-ECDSA-SHA512, SIGN-ECDSA-SECP521R1-SHA512

⁸⁷¹ https://www.gnutls.org/manual/html_node/Disabling-algorithms-and-protocols.html

⁸⁷² <https://git.launchpad.net/ubuntu/+source/gnutls28/tree/debian/rules?h=ubuntu/jammy-devel#n38>

⁸⁷³ https://gnutls.org/manual/html_node/Priority-Strings.html#tab_003aprio_002dkeywords

```
$ gnutls-cli --list --priority SECURE256:-CHACHA20-POLY1305

Cipher suites for SECURE256:-CHACHA20-POLY1305
TLS_AES_256_GCM_SHA384          0x13, 0x02      TLS1.3
TLS_ECDHE_ECDSA_AES_256_GCM_SHA384 0xc0, 0x2c      TLS1.2
TLS_ECDHE_ECDSA_AES_256_CCM      0xc0, 0xad      TLS1.2
TLS_ECDHE_RSA_AES_256_GCM_SHA384 0xc0, 0x30      TLS1.2
TLS_RSA_AES_256_GCM_SHA384       0x00, 0x9d      TLS1.2
TLS_RSA_AES_256_CCM              0xc0, 0x9d      TLS1.2
TLS_DHE_RSA_AES_256_GCM_SHA384   0x00, 0x9f      TLS1.2
TLS_DHE_RSA_AES_256_CCM          0xc0, 0x9f      TLS1.2

Protocols: VERS-TLS1.3, VERS-TLS1.2, VERS-TLS1.1, VERS-TLS1.0, VERS-DTLS1.2, VERS-DTLS1.0
Ciphers: AES-256-GCM, AES-256-CBC, AES-256-CCM
MACs: AEAD
Key Exchange Algorithms: ECDHE-ECDSA, ECDHE-RSA, RSA, DHE-RSA
Groups: GROUP-SECP384R1, GROUP-SECP521R1, GROUP-FFDHE8192
PK-signatures: SIGN-RSA-SHA384, SIGN-RSA-PSS-SHA384, SIGN-RSA-PSS-RSAE-SHA384, SIGN-ECDSA-SHA384, SIGN-ECDSA-SECP384R1-SHA384, SIGN-EdDSA-Ed448, SIGN-RSA-SHA512, SIGN-RSA-PSS-SHA512, SIGN-RSA-PSS-RSAE-SHA512, SIGN-ECDSA-SHA512, SIGN-ECDSA-SECP521R1-SHA512
```

[priorities]

MYSET = SECURE256:-CHACHA20-POLY1305

```
$ gnutls-cli --list --priority @MYSET
```

Verification profile (overrides)

[overrides]

do not allow applications use the LOW or VERY-WEAK profiles.
min-verification-profile = legacy

key sizes and security

parameters⁸⁷⁴

⁸⁷⁴ https://gnutls.org/manual/html_node/Selecting-cryptographic-key-sizes.html#tab_003akey_002dsizes

Practical examples

Only use TLSv1.3

```
[global]
override-mode = blocklist

[overrides]
default-priority-string = NORMAL:-VERS-TLS-ALL:+VERS-TLS1.3
```

```
$ sudo openssl s_server -cert j-server.pem -key j-server.key -port 443
-no_tls1_3 -www
```

```
$ gnutls-cli j-server.lxd

Processed 125 CA certificate(s).
Resolving 'j-server.lxd:443'...
Connecting to '10.0.100.87:443'...
*** Fatal error: A TLS fatal alert has been received.
*** Received alert [70]: Error in protocol version
```

```
$ lftp -c "cat https://j-server.lxd/status"

cat: /status: Fatal error: gnutls_handshake: A TLS fatal alert has been received.
```

```
$ lftp -c "set ssl:priority NORMAL:+VERS-TLS-ALL; cat
https://j-server.lxd/status" | grep ^New

New, TLSv1.2, Cipher is ECDHE-RSA-AES256-GCM-SHA384
```

```
[global]
override-mode = blocklist
```

```
[overrides]
disabled-version = tls1.1
disabled-version = tls1.2
disabled-version = tls1.0
```

```
$ lftp -c "set ssl:priority NORMAL:+VERS-TLS-ALL; cat
https://j-server.lxd/status" | grep ^New

cat: /status: Fatal error: gnutls_handshake: A TLS fatal alert has been received.
```

Use only AES256 with TLSv1.3

AES

```
[global]
override-mode = blocklist

[overrides]
disabled-version = tls1.1
disabled-version = tls1.2
disabled-version = tls1.0
tls-disabled-cipher = AES-128-GCM
```

```
$ sudo openssl s_server -cert j-server.pem -key j-server.key -port 443
-ciphersuites TLS_AES_128_GCM_SHA256 -www
```

```
$ gnutls-cli j-server.lxd

Processed 126 CA certificate(s).
Resolving 'j-server.lxd:443'...
Connecting to '10.0.100.87:443'...
*** Fatal error: A TLS fatal alert has been received.
*** Received alert [40]: Handshake failed
```

```
$ gnutls-cli --priority="NORMAL:+AES-128-GCM" j-server.lxd
```



```
Processed 126 CA certificate(s).
Resolving 'j-server.lxd:443'...
Connecting to '10.0.100.87:443'...
*** Fatal error: A TLS fatal alert has been received.
*** Received alert [40]: Handshake failed
```

Further reading

- System-wide configuration⁸⁷⁵
- Priority strings⁸⁷⁶
- min-verification-profile values⁸⁷⁷
- Disabling algorithms and protocols⁸⁷⁸
- Invoking the `gnutls-cli` command line tool⁸⁷⁹

Network Security Services (NSS)

Config file

LP: #2016303⁸⁸⁰

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⁸⁷⁵ https://www.gnutls.org/manual/html_node/System_002dwide-configuration-of-the-library.html

⁸⁷⁶ https://gnutls.org/manual/html_node/Priority-Strings.html

⁸⁷⁷ https://gnutls.org/manual/html_node/Selecting-cryptographic-key-sizes.html#tab_003akey_002dsizes

⁸⁷⁸ https://www.gnutls.org/manual/html_node/Disabling-algorithms-and-protocols.html

⁸⁷⁹ https://gnutls.org/manual/html_node/gnutls_002dcli-Invocation.html

⁸⁸⁰ <https://bugs.launchpad.net/bugs/2016303>

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Configuring the NSS policy module

```
library=  
name=Policy  
NSS=flags=policyOnly,moduleDB  
config="disallow=<list> allow=<list> flags=<flags>"
```

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DTLS

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Practical examples

Bootstrapping the NSS database

```
$ sudo apt install libnss3-tools
```

```
$ mkdir -p ~/.pki/nssdb
```

```
$ certutil -d ~/.pki/nssdb -N
```

```
library=
name=NSS Internal PKCS #11 Module
parameters=configdir='/home/ubuntu/.pki/nssdb' certPrefix='' keyPrefix='' secmod='secmod.
db' flags= updatedir='' updateCertPrefix='' updateKeyPrefix='' updateid=''
updateTokenDescription=''
NSS=Flags=internal,critical trustOrder=75 cipherOrder=100 slotParams=(1={slotFlags=[ECC,
RSA,DSA,DH,RC2,RC4,DES,RANDOM,SHA1,MD5,MD2,SSL,TLS,AES,Camellia,SEED,SHA256,SHA512]
askpw=any timeout=30}))
```

```
library=
name=NSS Internal PKCS #11 Module
parameters=configdir='/home/ubuntu/.pki/nssdb' certPrefix='' keyPrefix='' secmod='secmod.
db' flags= updatedir='' updateCertPrefix='' updateKeyPrefix='' updateid=''
updateTokenDescription=''
NSS=Flags=internal,critical trustOrder=75 cipherOrder=100 slotParams=(1={slotFlags=[ECC,
RSA,DSA,DH,RC2,RC4,DES,RANDOM,SHA1,MD5,MD2,SSL,TLS,AES,Camellia,SEED,SHA256,SHA512]
askpw=any timeout=30}))
```

```
library=
name=Policy
NSS=flags=policyOnly,moduleDB
config="allow=tls-version-min=tlsl.3"
```

Test setup

```
$ openssl req -new -x509 -days 30 -nodes -subj "/CN=localhost" -out
localhost.pem -keyout localhost.key
```

```
$ certutil -d ~/.pki/nssdb -A -a -i localhost.pem -t TCP -n localhost
```

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certutil manpage⁸⁸¹

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-

```
$ openssl s_server -accept 4443 -cert localhost.pem -key localhost.key -www
```

The tstclnt tool

```
$ tstclnt -d ~/.pki/nssdb -h localhost -p 4443
```

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```
$ echo "GET / HTTP/1.0" | tstclnt -d ~/.pki/nssdb -h localhost -p 4443 2>&1 |  
grep ^New
```

```
New, TLSv1.3, Cipher is TLS_AES_128_GCM_SHA256  
^C
```

⁸⁸¹ <https://manpages.ubuntu.com/manpages/resolute/man1/certutil.1.html>

Only use TLSv1.3

```
library=  
name=Policy  
NSS=flags=policyOnly,moduleDB  
config="allow=tls-version-min=tls1.3"
```

```
$ openssl s_server -accept 4443 -cert localhost.pem -key localhost.key -www  
-no_tls1_3
```

```
$ echo "GET / HTTP/1.0" | tstclnt -d ~/.pki/nssdb -h localhost -p 4443 2>&1 |  
grep ^New > echo $?  
1
```

```
$ echo "GET / HTTP/1.0" | tstclnt -d ~/.pki/nssdb -h localhost -p 4443 2>&1  
tstclnt: write to SSL socket failed: SSL_ERROR_PROTOCOL_VERSION_ALERT: Peer reports  
incompatible or unsupported protocol version.
```

```
$ openssl s_server -accept 4443 -cert localhost.pem -key localhost.key -www
```

```
$ echo "GET / HTTP/1.0" | tstclnt -d ~/.pki/nssdb -h localhost -p 4443 2>&1 |  
grep ^New  
New, TLSv1.3, Cipher is TLS_AES_128_GCM_SHA256  
^C
```

Use only AES256 with TLSv1.3

AES

```
library=  
name=Policy  
NSS=flags=policyOnly,moduleDB  
config="disallow=aes128-gcm allow=tls-version-min=tls1.3"
```

```
$ echo "GET / HTTP/1.0" | tstclnt -d ~/.pki/nssdb -h localhost -p 4443 2>&1 |  
grep ^New  
  
New, TLSv1.3, Cipher is TLS_CHACHA20_POLY1305_SHA256
```

```
config="disallow=aes128-gcm:chacha20-poly1305 allow=tls-version-min=tls1.3"
```

```
$ echo "GET / HTTP/1.0" | tstclnt -d ~/.pki/nssdb -h localhost -p 4443 2>&1 |  
grep ^New  
  
New, TLSv1.3, Cipher is TLS_AES_256_GCM_SHA384
```

Further reading

- In the source code⁸⁸²
- In the tests (policy)⁸⁸³
- In the tests (SSL policy)⁸⁸⁴

⁸⁸² <https://git.launchpad.net/ubuntu/+source/nss/tree/nss/lib/pk11wrap/pk11pars.c#n144>

⁸⁸³ <https://git.launchpad.net/ubuntu/+source/nss/tree/nss/tests/policy>

⁸⁸⁴ <https://git.launchpad.net/ubuntu/+source/nss/tree/nss/tests/ssl/sslpolicy.txt>

Java cryptography configuration

GUI

```
$ sudo apt install default-jre-headless
```

Config file

```
$ java -version 2>&1 | head -n 1
```

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-

java.security configuration file⁸⁸⁵

```
$ java -Djava.security.properties=file://$HOME/java.security
```

⁸⁸⁵ <https://git.launchpad.net/ubuntu/+source/openjdk-lts/tree/src/java.base/share/conf/security/java.security/#n520>


```
$ java -Djava.security.properties==file://$HOME/java.security
```

Practical examples

Test setup

```
$ openssl req -new -x509 -days 30 -nodes -subj "/CN=localhost" -out  
localhost.pem -keyout localhost.key
```

```
$ sudo cp localhost.pem /usr/local/share/ca-certificates/localhost-test.crt
```

```
$ sudo update-ca-certificates
```

```
$ openssl s_server -accept 4443 -cert localhost.pem -key localhost.key | grep  
^CIPHER
```

```
$ keytool -J-Djava.security.properties=file://$HOME/java.security -printcert  
-sslserver localhost:4443 > /dev/null;echo $?
```

-
-

Note:

keytool is not really intended as a tool to test SSL/TLS connections, but being part of the Java packaging makes it convenient and it's enough for our purposes.

Only use TLSv1.3

```
jdk.tls.disabledAlgorithms=TLSv1, TLSv1.1, TLSv1.2, SSLv3, SSLv2
```

```
$ keytool -J-Djava.security.properties=file://$HOME/java.security -printcert  
-sslserver localhost:4443 > /dev/null;echo $?
```

```
0
```

```
$ openssl s_server -accept 4443 -key localhost.key -cert localhost.pem | grep  
^CIPHER  
  
CIPHER is TLS_AES_256_GCM_SHA384
```

```
$ keytool \ > -J-Djava.security.properties=file://$HOME/java.security \ >
-J-Djdk.tls.client.protocols=TLSv1.2 \ > -printcert -sslserver localhost:4443

keytool error: java.lang.Exception: No certificate from the SSL server
```

```
$ openssl s_server -accept 4443 -key localhost.key -cert localhost.pem
-no_tls1_3

Using default temp DH parameters
ACCEPT
ERROR
40676E75B37F0000:error:0A000102:SSL routines:tls_early_post_process_client_
hello:unsupported protocol:../ssl/statem/statem_srvr.c:1657:
shutting down SSL
CONNECTION CLOSED
```

```
$ keytool -J-Djava.security.properties=file://$HOME/java.security -printcert
-sslserver localhost:4443

keytool error: java.lang.Exception: No certificate from the SSL server
```

```
$ keytool -J-Djava.security.properties=file://$HOME/java.security -printcert
-sslserver localhost:4443 -v

keytool error: java.lang.Exception: No certificate from the SSL server
java.lang.Exception: No certificate from the SSL server
    at java.base/sun.security.tools.keytool.Main.doPrintCert(Main.java:2981)
    at java.base/sun.security.tools.keytool.Main.doCommands(Main.java:1292)
    at java.base/sun.security.tools.keytool.Main.run(Main.java:421)
    at java.base/sun.security.tools.keytool.Main.main(Main.java:414)
Caused by: javax.net.ssl.SSLHandshakeException: Received fatal alert: protocol_
version
...
```

Prevent a specific cipher

Java Security Standard Algorithm Names⁸⁸⁶

```
jdk.tls.disabledAlgorithms=TLSv1, TLSv1.1, TLSv1.2, SSLv3, SSLv2, TLS_AES_256_GCM_SHA384
```

```
$ keytool -J-Djava.security.properties=file://$HOME/java.security -printcert  
-sslserver localhost:4443 > /dev/null; echo $?  
0
```

```
$ openssl s_server -accept 4443 -key localhost.key -cert localhost.pem | grep  
^CIPHER  
CIPHER is TLS_AES_128_GCM_SHA256
```

Blocking cipher “elements”

```
jdk.tls.disabledAlgorithms=TLSv1, TLSv1.1, TLSv1.3, SSLv2, SSLv3
```

```
$ keytool -J-Djava.security.properties=file://$HOME/java.security -printcert  
-sslserver localhost:4443 > /dev/null; echo $?  
0
```

```
$ openssl s_server -accept 4443 -key localhost.key -cert localhost.pem | grep  
^CIPHER
```

⁸⁸⁶ <https://docs.oracle.com/en/java/javase/11/docs/specs/security/standard-names.html>

```
CIPHER is ECDHE-RSA-AES256-GCM-SHA384
```

```
jdk.tls.disabledAlgorithms=TLSv1, TLSv1.1, TLSv1.3, SSLv2, SSLv3, AES_256_GCM
```

```
$ openssl s_server -accept 4443 -key localhost.key -cert localhost.pem | grep  
^CIPHER  
  
CIPHER is ECDHE-RSA-CHACHA20-POLY1305
```

Further reading

- [Java's Cryptographic Algorithms settings](#)⁸⁸⁷
- [Standard Algorithm Names](#)⁸⁸⁸
- [keytool upstream documentation](#)⁸⁸⁹
- [java.security file with comments](#)⁸⁹⁰

BIND 9 DNSSEC cryptography selection

DNS

DNSSEC validation

```
$ dig @127.0.0.1 isc.org +dnssec +multiline
```

⁸⁸⁷ https://www.java.com/en/configure_crypto.html

⁸⁸⁸ <https://docs.oracle.com/en/java/javase/12/docs/specs/security/standard-names.html>

⁸⁸⁹ <https://docs.oracle.com/en/java/javase/11/tools/keytool.html>

⁸⁹⁰ <https://git.launchpad.net/ubuntu/+source/openjdk-lts/tree/src/java.base/share/conf/security/java.security?h=applied/ubuntu/jammy-devel#n520>

```
; <<>> DiG 9.18.12-0ubuntu0.22.04.1-Ubuntu <<>> @127.0.0.1 isc.org +dnssec
+multiline
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 57669
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags: do; udp: 1232
; COOKIE: 71aa6b4e4ca6bb4b01000000643fee81edf0840b48d28d44 (good)
;; QUESTION SECTION:
;isc.org.                IN A

;; ANSWER SECTION:
isc.org.                 300 IN A 149.20.1.66
isc.org.                 300 IN RRSIG A 13 2 300 (
                        20230510161711 20230410161439 27566 isc.org.
                        EUA5QPEjtVC0scPsvf1c/EIBKHRpS8ektiWiOqk6nb3t
                        JhJAt9uCr3e0KNAcc3WDU+wJzEvqDyJrLZoELqT/pQ== )

;; Query time: 740 msec
;; SERVER: 127.0.0.1#53(127.0.0.1) (UDP)
;; WHEN: Wed Apr 19 13:37:05 UTC 2023
;; MSG SIZE rcvd: 183
```

```
$ dig @127.0.0.1 www.dnssec-failed.org +dnssec +multiline

; <<>> DiG 9.18.12-0ubuntu0.22.04.1-Ubuntu <<>> @127.0.0.1 www.dnssec-failed.org
+dnssec +multiline
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: SERVFAIL, id: 56056
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags: do; udp: 1232
; COOKIE: 541f6c66a216acdb01000000643fef9ebb21307fee2ea0e3 (good)
;; QUESTION SECTION:
;www.dnssec-failed.org.  IN A

;; Query time: 1156 msec
;; SERVER: 127.0.0.1#53(127.0.0.1) (UDP)
;; WHEN: Wed Apr 19 13:41:50 UTC 2023
;; MSG SIZE rcvd: 78
```

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```
$ journalctl -u named.service -n 10
```

```
Apr 19 13:41:50 j named[3018]: validating dnssec-failed.org/DNSKEY: no valid
signature found (DS)
Apr 19 13:41:50 j named[3018]: no valid RRSIG resolving 'dnssec-failed.org/DNSKEY/IN
': 68.87.76.228#53
Apr 19 13:41:50 j named[3018]: broken trust chain resolving 'www.dnssec-failed.org/
A/IN': 68.87.85.132#53
(...)
```

```
$ dig @127.0.0.1 www.dnssec-failed.org +dnssec +multiline +cd
```

```
; <<>> DiG 9.18.12-0ubuntu0.22.04.1-Ubuntu <<>> @127.0.0.1 www.dnssec-failed.org
+dnssec +multiline +cd
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->HEADER<<- opcode: QUERY, status: NOERROR, id: 42703
;; flags: qr rd ra cd; QUERY: 1, ANSWER: 3, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags: do; udp: 1232
; COOKIE: 3d6a4f4ff0014bdc01000000643ff01c3229ed7d798c5f8d (good)
;; QUESTION SECTION:
;www.dnssec-failed.org.          IN A

;; ANSWER SECTION:
www.dnssec-failed.org.          7031 IN      A 68.87.109.242
www.dnssec-failed.org.          7031 IN      A 69.252.193.191
www.dnssec-failed.org.          7074 IN      RRSIG A 5 3 7200 (
                                20230505145108 20230418144608 44973 dnssec-failed.
org.
                                R6/u+5Gv3rH93g08uNvz3sb9ErQNuVFKu6W5rtUleXF/
                                vkqJXbNe8grMuIv6Y+CNEP6jRBu0jOBPncb5cXbfcmfo
                                CoV0jpsLySxt4D1EUL4yByWm2ZAdXRrk6A8SaldIdDv8
                                9t+FguTdQrZv9Si+afKrLyC7L/mltXMllq3stDI= )

;; Query time: 0 msec
;; SERVER: 127.0.0.1#53(127.0.0.1) (UDP)
;; WHEN: Wed Apr 19 13:43:56 UTC 2023
;; MSG SIZE rcvd: 287
```

Restricting DNSSEC algorithms

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```
disable-algorithms "." {  
    RSAMD5;  
    DSA;  
};  
disable-ds-digest "." {  
    GOST;  
};
```

DNSSEC Algorithm Numbers⁸⁹¹

```
isc.org.      300 IN RRSIG A 13 2 300 (
```

```
disable-algorithms "." {  
    RSAMD5;  
    DSA;  
    ECDSAP256SHA256;  
};
```

```
$ sudo systemctl restart bind9.service
```

```
$ dig @127.0.0.1 isc.org +dnssec +multiline
```

⁸⁹¹ <https://www.iana.org/assignments/dns-sec-alg-numbers/dns-sec-alg-numbers.xhtml>


```
; <<>> DiG 9.18.1-1ubuntu1-Ubuntu <<>> @127.0.0.1 isc.org +dnssec +multiline
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 43893
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags: do; udp: 1232
; COOKIE: 6527ce585598025d01000000643ff8fa02418ce38af13fa7 (good)
;; QUESTION SECTION:
;isc.org.                IN A

;; ANSWER SECTION:
isc.org.                 300 IN A 149.20.1.66
isc.org.                 300 IN RRSIG A 13 2 300 (
                        20230510161711 20230410161439 27566 isc.org.
                        EUA5QPEjtVC0scPsvf1c/EIBKHRpS8ektiWiOqk6nb3t
                        JhJAt9uCr3e0KNAcc3WDU+wJzEvqDyJrLZoELqT/pQ== )

;; Query time: 292 msec
;; SERVER: 127.0.0.1#53(127.0.0.1) (UDP)
;; WHEN: Wed Apr 19 14:21:46 UTC 2023
;; MSG SIZE rcvd: 183
```

Apr 19 14:23:01 j-bind9 named[2786]: validating isc.org/A: no valid signature found

Note:

Remember now to remove the disabling of ECDSAP256SHA256 from `/etc/bind/named.conf` options and restart BIND 9. This change was just a quick test!

Further reading

- [ISC DNSSEC Guide](#)⁸⁹²
- [DNSSEC troubleshooting section of the ISC DNSSEC guide](#)⁸⁹³
- [Standard algorithms used for DNSSEC](#)⁸⁹⁴

⁸⁹² <https://bind9.readthedocs.io/en/v9.18.14/dnssec-guide.html>

⁸⁹³ <https://bind9.readthedocs.io/en/v9.18.14/dnssec-guide.html#basic-dnssec-troubleshooting>

⁸⁹⁴ <https://www.iana.org/assignments/dns-sec-alg-numbers/dns-sec-alg-numbers.xhtml>

OpenSSH crypto configuration

Warning:

Be careful when restricting cryptographic algorithms in SSH, specially on the server side. You can inadvertently lock yourself out of a remote system!

Algorithm configuration general rules

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Note:

The output of the `ssh -Q <name>` command will not take into consideration the configuration changes that may have been made. It cannot therefore be used to test the crypto configuration changes.

Configuration settings

-
-
- provided
via a patch⁸⁹⁵ *Generic Security Services Application Program Interface (GSSAPI)*
-
-
-
-

```
$ sudo sshd -T | grep ciphers
```

⁸⁹⁵ <https://git.launchpad.net/ubuntu/+source/openssh/tree/debian/patches/gssapi.patch>

```
ciphers chacha20-poly1305@openssh.com,aes128-gcm@openssh.com,aes256-gcm@openssh.com,  
aes128-ctr,aes192-ctr,aes256-ctr
```

OpenSSH examples

Which cipher was used?

```
$ ssh -v <server> exit 2>&1 | grep "cipher:"  
debug1: kex: server->client cipher: chacha20-poly1305@openssh.com MAC: <implicit>  
compression: none  
debug1: kex: client->server cipher: chacha20-poly1305@openssh.com MAC: <implicit>  
compression: none
```

```
$ ssh -v -c aes128-ctr <server> exit 2>&1 | grep "cipher:"  
debug1: kex: server->client cipher: aes128-ctr MAC: umac-64-etm@openssh.com  
compression: none  
debug1: kex: client->server cipher: aes128-ctr MAC: umac-64-etm@openssh.com  
compression: none
```

Remove AES 128 from server

AES

```
$ sudo sshd -T | grep ciphers
```

```
ciphers chacha20-poly1305@openssh.com,aes128-gcm@openssh.com,aes256-gcm@openssh.com,
aes128-ctr,aes192-ctr,aes256-ctr
```

```
Ciphers -aes128*
```

```
$ sudo sshd -T | grep ciphers
```

```
ciphers chacha20-poly1305@openssh.com,aes256-gcm@openssh.com,aes192-ctr,aes256-ctr
```

```
$ sudo systemctl restart ssh.service
```

```
$ ssh -c aes128-ctr <server>
```

```
Unable to negotiate with 10.0.102.49 port 22: no matching cipher found. Their offer:
chacha20-poly1305@openssh.com,aes192-ctr,aes256-ctr,aes256-gcm@openssh.com
```

Prioritize AES 256 on the client

```
$ ssh -c ^aes256-ctr -v <server> exit 2>&1 | grep "cipher:"
```

```
debug1: kex: server->client cipher: aes256-ctr MAC: umac-64-etm@openssh.com
compression: none
debug1: kex: client->server cipher: aes256-ctr MAC: umac-64-etm@openssh.com
compression: none
```

```
$ ssh -c ^aes256-ctr -v <server> exit 2>&1 | grep "cipher:"
```

```
debug1: kex: server->client cipher: chacha20-poly1305@openssh.com MAC: <implicit>
compression: none
debug1: kex: client->server cipher: chacha20-poly1305@openssh.com MAC: <implicit>
compression: none
```

Further reading

- [OpenSSH upstream documentation index](#)⁸⁹⁶
- [sshd_config\(5\)](#)⁸⁹⁷
- [ssh_config\(5\)](#)⁸⁹⁸

Troubleshooting TLS/SSL

Separate the client and server

Tools

OpenSSL server and client apps

```
$ openssl req -new -x509 -nodes -days 30 -out myserver.pem -keyout  
myserver.key
```

hostname

```
$ openssl s_server -cert myserver.pem -key myserver.key
```

⁸⁹⁶ <https://www.openssh.org/manual.html>

⁸⁹⁷ https://manpages.ubuntu.com/manpages/resolute/man5/sshd_config.5.html

⁸⁹⁸ https://manpages.ubuntu.com/manpages/resolute/man5/ssh_config.5.html

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```
$ echo | openssl s_client -connect server:port 2>&1 | grep '^New'
```

```
$ echo | openssl s_client -connect j-server.lxd:443 2>&1 | grep '^New'
```

New, TLSv1.3, Cipher is TLS_AES_256_GCM_SHA384

```
$ echo | openssl s_client -connect j-server.lxd:443 -no_tls1_3 2>&1 | grep '^New'
```

New, TLSv1.2, Cipher is ECDHE-RSA-AES256-GCM-SHA384

```
$ echo | openssl s_client -connect j-server.lxd:443 -no_tls1_3 2>&1 -cipher  
DEFAULT:-AES256 | grep '^New'
```

New, TLSv1.2, Cipher is ECDHE-RSA-CHACHA20-POLY1305

The `sslsan` tool

```
$ sslscan j-server.lxd
```

*sslscan(1)*⁸⁹⁹

⁸⁹⁹ <https://manpages.ubuntu.com/manpages/resolute/man1/sslscan.1.html>

Further reading

- [*OpenSSL s_server*](#)⁹⁰⁰
- [*OpenSSL s_client*](#)⁹⁰¹
- [*sslscan\(1\)*](#)⁹⁰²
- <https://badssl.com>⁹⁰³

About certificates

HTTPS *HTTP*

Types of certificates

Note:

Self-signed certificates should not be used in most production environments.

HTTPS

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-

⁹⁰⁰ https://manpages.ubuntu.com/manpages/resolute/man1/openssl-s_server.1.html

⁹⁰¹ https://manpages.ubuntu.com/manpages/resolute/man1/openssl-s_client.1.html

⁹⁰² <https://manpages.ubuntu.com/manpages/resolute/man1/sslscan.1.html>

⁹⁰³ <https://badssl.com/>

- 1.
- 2.
- 3.
- 4.
- 5.

Generate a Certificate Signing Request (CSR)

Warning:

Running your secure service without a passphrase is convenient because you will not need to enter the passphrase every time you start your secure service. But it is insecure – a compromise of the key means a compromise of the server as well.

```
$ openssl genrsa -des3 -out server.key 2048
```

```
Generating RSA private key, 2048 bit long modulus
.....+++++
.....+++++
e is 65537 (0x10001)
Enter pass phrase for server.key:
```

```
$ openssl rsa -in server.key -out server.key.insecure > mv server.key
server.key.secure > mv server.key.insecure server.key
```

```
$ openssl req -new -key server.key -out server.csr
```

Creating a self-signed certificate

```
$ openssl x509 -req -days 365 -in server.csr -signkey server.key -out
server.crt
```

Warning:

If your secure server is to be used in a production environment, you probably need a CA-signed certificate. It is not recommended to use self-signed certificates in production environments.

Install the certificate

```
$ sudo cp server.crt /etc/ssl/certs > sudo cp server.key /etc/ssl/private
```

HTTPS

Certification Authority

Obtain TLS certificates

Further reading

- [HTTPS page](https://en.wikipedia.org/wiki/HTTPS)⁹⁰⁴ *HTTPS*
- [OpenSSL Home Page](https://www.openssl.org/)⁹⁰⁵
-

Virtual Private Network (VPN)

- *Introduction to WireGuard VPN*
- *OpenVPN clients*

Introduction to WireGuard VPN

⁹⁰⁴ <https://en.wikipedia.org/wiki/HTTPS>

⁹⁰⁵ <https://www.openssl.org/>

`ip-route(8)`⁹⁰⁶ `ip-address(8)`⁹⁰⁷

WireGuard concepts

`wg(8)`⁹⁰⁸ `wg-quick(8)`⁹⁰⁹

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Note:

Cryptography is not simple. When we say that, for example, a private key is used to decrypt or sign traffic, and a public key is used to encrypt or verify the authenticity of traffic, this is a simplification and is hiding a lot of important details. WireGuard has a detailed explanation of its protocols and cryptography handling [on its website](https://www.wireguard.com/protocol/)⁹¹⁰.

⁹¹⁰ <https://www.wireguard.com/protocol/>

⁹⁰⁶ <https://manpages.ubuntu.com/manpages/resolute/man8/ip-route.8.html>

⁹⁰⁷ <https://manpages.ubuntu.com/manpages/resolute/man8/ip-address.8.html>

⁹⁰⁸ <https://manpages.ubuntu.com/manpages/resolute/man8/wg.8.html>

⁹⁰⁹ <https://manpages.ubuntu.com/manpages/resolute/man8/wg-quick.8.html>

`wg(8)`⁹¹¹

`wg-quick(8)`⁹¹²

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```
$ sudo wg-quick up wg0
```

[Interface]

```
PrivateKey = eJdSgoS7BZ/uWkuSREN+vhCJPPr3M3U1B3v1Su/amWk=  
ListenPort = 51000  
Address = 10.10.11.10/24
```

[Peer]

```
# office  
PublicKey = xewmdxiLjgebpcItF1ouRo0ntrgFekquRJJZQ0+vsQVs=  
Endpoint = wg.example.com:51000 # fake endpoint, just an example  
AllowedIPs = 10.10.11.0/24, 10.10.10.0/24
```

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⁹¹¹ <https://manpages.ubuntu.com/manpages/resolute/man8/wg.8.html>

⁹¹² <https://manpages.ubuntu.com/manpages/resolute/man8/wg-quick.8.html>

-

```
$ umask 077
```

```
$ wg genkey > wg0.key
```

```
$ wg pubkey < wg0.key > wg0.pub
```

```
$ sudo wg-quick up wg0  
[#] ip link add wg0 type wireguard  
[#] wg setconf wg0 /dev/fd/63  
[#] ip -4 address add 10.10.11.10/24 dev wg0  
[#] ip link set mtu 1420 up dev wg0  
[#] ip -4 route add 10.10.10.0/24 dev wg0
```

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-
-

```
$ ip route get 10.10.10.201  
  
10.10.10.201 dev wg0 src 10.10.11.10 uid 1000  
cache
```

troubleshooting section

Peers

Putting it all together

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ACL

Peer-to-site

- *About peer-to-site*
- *Set up peer-to-site “on router”*
- *Set up peer-to-site on an internal device*

Site-to-site

- *Set up site-to-site*

Default gateway

- *Using the VPN as the default gateway*

Other common tasks, hints and tips

- *Common tasks*
- *Security tips*
- *Troubleshooting*

Note:

Throughout this guide, we will sometimes mention a VPN “connection”. This is technically false, as WireGuard uses UDP and there is no persistent connection. The term is used just to facilitate understanding, and means that the peers in the examples know each other and have completed a handshake already.

Further reading

- [WireGuard website](https://www.wireguard.com)⁹¹³
- [WireGuard Quickstart](https://www.wireguard.com/quickstart/)⁹¹⁴
- [wg\(8\)](https://manpages.ubuntu.com/manpages/resolute/man8/wg.8.html)⁹¹⁵
- [wg-quick\(8\)](https://manpages.ubuntu.com/manpages/resolute/man8/wg-quick.8.html)⁹¹⁶
- [Detailed explanation](https://www.wireguard.com/protocol/)⁹¹⁷

⁹¹³ <https://www.wireguard.com>

⁹¹⁴ <https://www.wireguard.com/quickstart/>

⁹¹⁵ <https://manpages.ubuntu.com/manpages/resolute/man8/wg.8.html>

⁹¹⁶ <https://manpages.ubuntu.com/manpages/resolute/man8/wg-quick.8.html>

⁹¹⁷ <https://www.wireguard.com/protocol/>

OpenVPN client implementations

OpenVPN server

Linux Network-Manager GUI for OpenVPN

GUI

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OpenVPN with GUI for Mac OS X

Tunnelblick⁹¹⁸

recommended by upstream⁹¹⁹

has an alternative⁹²⁰

⁹¹⁸ <https://tunnelblick.net>

⁹¹⁹ <https://openvpn.net/vpn-server-resources/connecting-to-access-server-with-macos/>

⁹²⁰ <https://openvpn.net/as-docs/tutorials/tutorial--connect-with-macos.html>

OpenVPN with GUI for Win

OpenVPN Windows Installer⁹²¹

updated guide by the upstream⁹²²

Further reading

- [OpenVPN](#)⁹²³
- [OpenVPN hardening security guide](#)⁹²⁴
- [OpenVPN: Building and Integrating Virtual Private Networks](#)⁹²⁵

See also

- [Security](#)

Our networking section

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-

Networking

⁹²¹ <https://openvpn.net/community/>

⁹²² https://community.openvpn.net/openvpn/wiki/Easy_Windows_Guide

⁹²³ <https://openvpn.net/>

⁹²⁴ <https://openvpn.net/community-docs/hardening-openvpn-security-178571.html>

⁹²⁵ <https://www.packtpub.com/en-us/product/openvpn-building-and-integrating-virtual-private-networks-9781847190680>

Introduction

Introduction to networking

Networking key concepts

Networking key concepts

Network configuration with Netplan

Netplan⁹²⁶
YAML configuration file⁹²⁷

Netplan

more about

configuring networks"

Network tools and services

DHCP

an explanatory guide

install isc-kea

DHCP server options

⁹²⁶ <https://netplan.io/>

⁹²⁷ <https://netplan.readthedocs.io/en/stable/netplan-yaml/>

Time synchronization

time synchronization

read our how-to guide

serve NTP using chrony

how to

The DPDK library

Data Plane Development Kit (DPDK)⁹²⁸
about DPDK and its use in Ubuntu

with Open vSwitch

how to use DPDK

Other networking functionality

Samba

introduction to Samba

Networking key concepts

The Transmission Control Protocol and Internet Protocol (TCP/IP)

DARPA

⁹²⁸ <https://www.dpdk.org/>

TCP/IP overview

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Datagram

-

TCP/IP configuration

DHCP

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Tip:

The IP address (in CIDR notation), gateway (default route), and nameserver addresses are typically specified via Netplan configuration files in `/etc/netplan/`. For more information, view the system manual page for `netplan`, using `man netplan.5` at the terminal prompt.

IP routing

RIP *OSPF*

About TCP and UDP

Internet Control Messaging Protocol (ICMP)

Request For Comments (RFC) #792⁹²⁹

About daemons

Further reading

- [*TCP*⁹³⁰ *IP*⁹³¹](#)
- [TCP/IP Tutorial and Technical Overview⁹³²](#)
- [*netplan\(5\)*⁹³³](#)

⁹²⁹ <https://www.rfc-editor.org/rfc/rfc792>

⁹³⁰ <https://manpages.ubuntu.com/manpages/resolute/man7/tcp.7.html>

⁹³¹ <https://manpages.ubuntu.com/manpages/resolute/man7/ip.7.html>

⁹³² <https://www.redbooks.ibm.com/abstracts/gg243376.html>

⁹³³ <https://manpages.ubuntu.com/manpages/resolute/man5/netplan.5.html>

Configuration

network configuration

About Netplan

About Netplan

Netplan⁹³⁴

YAML configuration file⁹³⁵

introduction to Netplan⁹³⁶

configuring networks

list of how-to guides⁹³⁷

Configuring networks

Netplan⁹³⁸

YAML configuration file⁹³⁹

the Netplan documentation⁹⁴⁰

⁹³⁴ <https://netplan.io/>

⁹³⁵ <https://netplan.readthedocs.io/en/stable/netplan-yaml/>

⁹³⁶ <https://netplan.readthedocs.io/en/stable/structure-id/>

⁹³⁷ <https://netplan.readthedocs.io/en/stable/howto/>

⁹³⁸ <https://netplan.io/>

⁹³⁹ <https://netplan.readthedocs.io/en/stable/netplan-yaml/>

⁹⁴⁰ <https://netplan.readthedocs.io/en/stable/>

Ethernet interfaces

Identify Ethernet interfaces

```
$ ip addr

1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default
qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s25: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group
default qlen 1000
    link/ether 00:16:3e:e2:52:42 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 10.102.66.200/24 brd 10.102.66.255 scope global dynamic eth0
        valid_lft 3257sec preferred_lft 3257sec
    inet6 fe80::216:3eff:fee2:5242/64 scope link
        valid_lft forever preferred_lft forever
```

```
$ sudo netplan status -a
```

```
Online state: online
DNS Addresses: 127.0.0.53 (stub)
DNS Search: internal

[] 1: lo ethernet UNKNOWN/UP (unmanaged)
    MAC Address: 00:00:00:00:00:00
    Addresses: 127.0.0.1/8
               ::1/128

[] 2: enp0s25 ethernet UP (networkd: enp0s25)
    MAC Address: 00:16:3e:e2:52:42 (ASIX Electronics Corp.)
    Addresses: 10.102.66.200/24 (dynamic, dhcp)
               fe80::216:3eff:fee2:5242/64 (link)
    DNS Addresses: 10.102.66.1
                  fd00::cece:1eff:fe3d:c737
    DNS Search: internal
    Routes: default via 10.102.66.1 from 10.102.66.200 metric 100 (dhcp)
            10.102.66.0/24 from 10.102.66.200 metric 100 (link)
            2001:16b8:bd99:3d00::/56 via fe80::cece:1eff:fe3d:c737 metric 100
(ra)
        default via fe80::cece:1eff:fe3d:c737 metric 100 (ra)
```

```
$ sudo lshw -class network

*-network
   description: Ethernet interface
  product: MT26448 [ConnectX EN 10GigE, PCIe 2.0 5GT/s]
  vendor: Mellanox Technologies
  physical id: 0
  bus info: pci@0004:01:00.0
  logical name: eth4
  version: b0
  serial: e4:1d:2d:67:83:56
  slot: U78CB.001.WZS09KB-P1-C6-T1
  size: 10Gbit/s
  capacity: 10Gbit/s
  width: 64 bits
  clock: 33MHz
  capabilities: pm vpd msix pciexpress bus_master cap_list ethernet physical fibre 10000bt-fd
  configuration: autonegotiation=off broadcast=yes driver=mlx4_en
  driverversion=4.0-0 duplex=full firmware=2.9.1326 ip=192.168.1.1 latency=0 link=yes
  multicast=yes port=fibre speed=10Gbit/s
  resources: iomemory:24000-23fff irq:481 memory:3fe200000000-3fe2000fffff
             memory:240000000000-240007ffffff
```

Ethernet Interface logical names

```
network:
  version: 2
  renderer: networkd
  ethernets
    eth_lan0:
      dhcp4: true
      match
        macaddress: 00:11:22:33:44:55
      set-name: eth_lan0
```

Ethernet Interface settings

```
$ sudo ethtool eth4

Settings for eth4:
    Supported ports: [ FIBRE ]
    Supported link modes:   10000baseT/Full
    Supported pause frame use: No
    Supports auto-negotiation: No
    Supported FEC modes: Not reported
    Advertised link modes:  10000baseT/Full
    Advertised pause frame use: No
    Advertised auto-negotiation: No
    Advertised FEC modes: Not reported
    Speed: 10000Mb/s
    Duplex: Full
    Port: FIBRE
    PHYAD: 0
    Transceiver: internal
    Auto-negotiation: off
    Supports Wake-on: d
    Wake-on: d
    Current message level: 0x00000014 (20)
                          link ifdown
    Link detected: yes
```

IP addressing

Temporary IP address assignment

GNU

```
$ sudo ip addr add 10.102.66.200/24 dev enp0s25
```

```
$ ip link set dev enp0s25 up
```

```
$ ip link set dev enp0s25 down
```

```
$ ip address show dev enp0s25

10: enp0s25: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group
default qlen 1000
    link/ether 00:16:3e:e2:52:42 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 10.102.66.200/24 brd 10.102.66.255 scope global dynamic eth0
        valid_lft 2857sec preferred_lft 2857sec
    inet6 fe80::216:3eff:fee2:5242/64 scope link
        valid_lft forever preferred_lft forever6
```

```
$ sudo ip route add default via 10.102.66.1
```

```
$ ip route show

default via 10.102.66.1 dev eth0 proto dhcp src 10.102.66.200 metric 100
10.102.66.0/24 dev eth0 proto kernel scope link src 10.102.66.200
10.102.66.1 dev eth0 proto dhcp scope link src 10.102.66.200 metric 100
```

DNS

```
$ sudo resolvectl dns enp0s25 8.8.8.8 8.8.4.4
```

```
$ ip addr flush eth0
```

Note:

Flushing the IP configuration using the `ip` command does not clear the contents of `systemd-resolved`. You must remove or modify those entries manually, e.g by calling `sudo resolvectl revert enp0s25`. Or re-boot, which should also cause the symlink `/etc/resolv.conf -> /run/systemd/resolve/stub-resolv.conf`, to be restored.

Dynamic IP address assignment (DHCP client)

```
network:
  version: 2
  renderer: networkd
  ethernets:
    enp3s0:
      dhcp4: true
```

```
$ sudo netplan apply
```

Static IP address assignment

```
network:
  version: 2
  renderer: networkd
  ethernets:
```

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(continued from previous page)

```
eth0:
  addresses:
    - 10.10.10.2/24
  routes:
    - to: default
      via: 10.10.10.1
  nameservers
    search [mydomain, otherdomain]
    addresses: [10.10.10.1, 1.1.1.1]
```

```
$ sudo netplan apply
```

Note:

netplan in Ubuntu Bionic 18.04 LTS doesn't understand the "to: default" syntax to specify a default route, and should use the older gateway4: 10.10.10.1 key instead of the whole routes: block.

```
$ ip address show lo

1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default
qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
```

Adding a multiple IP addresses to one interface

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Multiple static IP addresses

```
network:
  version: 2
  ethernets:
    eno1:
      addresses:
        - 192.168.2.99/24
        - 192.168.2.100/24:
            label: eno1:1
        - 192.168.2.101/24:
            label: eno1:some-label
```

Dynamic IP addresses

```
network:
  version: 2
  ethernets:
    eno1:
      addresses:
        - 192.168.2.100/24:
            label: eno1:some-label
      dhcp4: true
```

Note:

A single interface can only have one address assigned by DHCP. To have multiple dynamic addresses, configure multiple interfaces. Do not add DHCP if it is configured on the same interface in another netplan file, such as one written by `cloud-init`.

Apply and verify the IP addresses

```
$ sudo netplan apply
```

```
$ netplan status #displays the current network state
```

```
$ ip addr #shows all network interfaces
```

```
$ ip addr show <interface_name> #shows single interface
```

```
$ hostname -i #shows only the available IP addresses on the host
```

```
$ ping 192.168.0.100 #configured IP addresses
```

```
$ ping 192.168.0.101
```

Name resolution

hostnames

DNS client configuration

```
/etc/resolv.conf -> /run/systemd/resolve/stub-resolv.conf
```

```
network:
  version: 2
  renderer: networkd
  ethernets:
    enp0s25:
      addresses:
        - 192.168.0.100/24
      routes:
        - to: default
          via: 192.168.0.1
      nameservers
```

(continues on next page)

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```
search [mydomain, otherdomain]
addresses: [1.1.1.1, 8.8.8.8, 4.4.4.4]
```

```
network:
  version: 2
  renderer: networkd
  ethernet:
    enp0s25:
      addresses:
        - 192.168.0.100/24
      routes:
        - to: default
          via: 192.168.0.1
      nameservers:
        search [example.com, sales.example.com, dev.example.com]
        addresses: [1.1.1.1, 8.8.8.8, 4.4.4.4]
```

Fully Qualified Domain Name (FQDN)

- 1.
- 2.
- 3.

Static hostnames

```
127.0.0.1    localhost
127.0.1.1    ubuntu-server
10.0.0.11    server1 server1.example.com vpn
10.0.0.12    server2 server2.example.com mail
```

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```
10.0.0.13  server3 server3.example.com www
10.0.0.14  server4 server4.example.com file
```

Note:

In this example, notice that each of the servers were given aliases in addition to their proper names and FQDN's. *Server1* has been mapped to the name *vpn*, *server2* is referred to as *mail*, *server3* as *www*, and *server4* as *file*.

Name Service Switch (NSS) configuration

```
hosts:          files mdns4_minimal [NOTFOUND=return] dns mdns4
```

-
-
-
-
-

```
hosts:          files dns [NOTFOUND=return] mdns4_minimal mdns4
```

Bridging multiple interfaces

```
network:
  version: 2
```

(continues on next page)

(continued from previous page)

```
renderer: networkd
ethernets
  enp3s0
    dhcp4: no
bridges:
  br0:
    dhcp4: yes
interfaces
- enp3s0
```

```
$ sudo netplan apply
```

networkd-dispatcher for hook scripts

Netplan configuration⁹⁴¹

*dispatcher(8)*⁹⁴²

networkd-

Note:

If you are on Desktop (not Ubuntu Server) the network is driven by Network Manager - in that case you need [NM Dispatcher scripts](#)⁹⁴³ instead.

⁹⁴³ <https://networkmanager.dev/docs/api/latest/>

Netplan FAQ has a great table⁹⁴⁴

Netplan FAQ also has an example⁹⁴⁵

⁹⁴¹ <https://netplan.readthedocs.io/en/stable/netplan-yaml/>

⁹⁴² <https://manpages.ubuntu.com/manpages/resolute/man8/networkd-dispatcher.8.html>

⁹⁴⁴ <https://netplan.io/faq/>

⁹⁴⁵ <https://netplan.io/faq/>

Further reading

- [Ubuntu Wiki Network page](#)⁹⁴⁶
- [Netplan website](#)⁹⁴⁷ [examples](#)⁹⁴⁸
- [manual page](#)⁹⁴⁹
- [systemd-resolved\(8\)](#)⁹⁵⁰
- [netplan configuration reference](#)⁹⁵¹

Network tools

- [About DPDK](#)

About DPDK

EAL

Prerequisites

DPDK 17.08⁹⁵²

supported NICs⁹⁵³

⁹⁴⁶ <https://help.ubuntu.com/community/Network>

⁹⁴⁷ <https://netplan.io>

⁹⁴⁸ <https://netplan.readthedocs.io/en/stable/examples/>

⁹⁴⁹ <https://manpages.ubuntu.com/manpages/resolute/man5/netplan.5.html>

⁹⁵⁰ <https://manpages.ubuntu.com/manpages/resolute/man8/systemd-resolved.8.html>

⁹⁵¹ <https://netplan.readthedocs.io/en/stable/netplan-yaml/#properties-for-device-type-bridges>

⁹⁵² <https://git.dpdk.org/dpdk/commit/?id=f27769f796a0639368117ce22fb124b6030dbf73>

⁹⁵³ <http://core.dpdk.org/supported/>

many more “in-universe”⁹⁵⁴

Unassign the default kernel drivers

```
iommu=pt intel_iommu=on
```

AMD

```
amd_iommu=pt
```

firmware

Note:

VirtIO is special. DPDK can directly work on these devices without `vfio_pci/uio_pci_generic`. However, to avoid issues that might arise from the kernel and DPDK managing the device, you still need to unassign the kernel driver.

```
dpdk_nic_bind.py --help
```

Usage

```
dpdk-devbind.py [options] DEVICE1 DEVICE2 ....
```

where DEVICE1, DEVICE2 etc, are specified via PCI `"domain:bus:slot.func"` syntax or `"bus:slot.func"` syntax. For devices bound to Linux kernel drivers, they may also be referred to by Linux interface name e.g. `eth0`, `eth1`, `em0`, `em1`, etc.

Options:
`--help`, `--usage`:

(continues on next page)

⁹⁵⁴ <https://documentation.ubuntu.com/project/how-ubuntu-is-made/concepts/package-archive/#components>

```
Display usage information and quit

-s, --status:
    Print the current status of all known network, crypto, event
    and mempool devices.
    For each device, it displays the PCI domain, bus, slot and function
    along with a text description of the device. Depending upon whether the
    device is being used by a kernel driver, the igb_uio driver, or no
    driver, other relevant information will be displayed:
    * the Linux interface name e.g. if=eth0
    * the driver being used e.g. drv=igb_uio
    * any suitable drivers not currently using that device
      e.g. unused=igb_uio
    NOTE: if this flag is passed along with a bind/unbind option, the
    status display will always occur after the other operations have taken
    place.

--status-dev:
    Print the status of given device group. Supported device groups are:
    "net", "crypto", "event", "mempool" and "compress"

-b driver, --bind=driver:
    Select the driver to use or "none" to unbind the device

-u, --unbind:
    Unbind a device (Equivalent to "-b none")

--force:
    By default, network devices which are used by Linux - as indicated by
    having routes in the routing table - cannot be modified. Using the
    --force flag overrides this behavior, allowing active links to be
    forcibly unbound.
    WARNING: This can lead to loss of network connection and should be used
    with caution.

Examples:
-----

To display current device status:
    dpdk-devbind.py --status

To display current network device status:
    dpdk-devbind.py --status-dev net

To bind eth1 from the current driver and move to use igb_uio
    dpdk-devbind.py --bind=igb_uio eth1

To unbind 0000:01:00.0 from using any driver
    dpdk-devbind.py -u 0000:01:00.0

To bind 0000:02:00.0 and 0000:02:00.1 to the ixgbe kernel driver
    dpdk-devbind.py -b ixgbe 02:00.0 02:00.1
```

DPDK device configuration

```
# <bus>          Currently only "pci" is supported
# <id>           Device ID on the specified bus
# <driver>        Driver to bind against (vfio-pci or uio_pci_generic)
#
# Be aware that the two DPDK compatible drivers uio_pci_generic and vfio-pci are
# part of linux-image-extra-<VERSION> package.
# This package is not always installed by default - for example in cloud-images.
# So please install it in case you run into missing module issues.
#
# <bus> <id>      <driver>
pci 0000:04:00.0 uio_pci_generic
pci 0000:04:00.1 vfio-pci
```

```
Network devices using DPDK-compatible driver
=====
0000:04:00.0 'Ethernet Controller 10-Gigabit X540-AT2' drv=uio_pci_generic unused=ixgbe

Network devices using kernel driver
=====
0000:02:00.0 'NetXtreme BCM5719 Gigabit Ethernet PCIe' if=eth0 drv=tg3 unused=uio_pci_
generic *Active*
0000:02:00.1 'NetXtreme BCM5719 Gigabit Ethernet PCIe' if=eth1 drv=tg3 unused=uio_pci_
generic
0000:02:00.2 'NetXtreme BCM5719 Gigabit Ethernet PCIe' if=eth2 drv=tg3 unused=uio_pci_
generic
0000:02:00.3 'NetXtreme BCM5719 Gigabit Ethernet PCIe' if=eth3 drv=tg3 unused=uio_pci_
generic
0000:04:00.1 'Ethernet Controller 10-Gigabit X540-AT2' if=eth5 drv=ixgbe unused=uio_pci_
generic

Other network devices
=====
<none>
```

DPDK hugepage configuration

```
NR_2M_PAGES=1024  
NR_1G_PAGES=4
```

Compile DPDK applications

valid pkg-config file⁹⁵⁵

```
$ sudo apt-get install dpdk-dev libdpdk-dev
```

```
$ gcc testdpdkprog.c $(pkg-config --libs --cflags libdpdk) -o testdpdkprog
```

Open vSwitch build system⁹⁵⁶

fallbacks

⁹⁵⁵ <https://people.freedesktop.org/~dbn/pkg-config-guide.html>

⁹⁵⁶ <https://github.com/openvswitch/ovs/blob/master/acinclude.m4#L283>


```
$ sudo apt-get install build-dep dpdk
```

DPDK in KVM guests

```
<cpu mode='host-passthrough'>
```

```
<driver name="vhost" queues="4"/>
```

Use DPDK

Open vSwitch

DPDK

Further reading

- DPDK documentation⁹⁵⁷
- Release Notes matching the version packages in Ubuntu 16.04 LTS⁹⁵⁸
- Linux DPDK user getting started⁹⁵⁹
- EAL command-line options⁹⁶⁰
- DPDK API documentation⁹⁶¹
- Open vSwitch DPDK installation⁹⁶²
- Wikipedia definition of DPDK⁹⁶³

⁹⁵⁷ <http://core.dpdk.org/doc/>

⁹⁵⁸ https://doc.dpdk.org/guides/rel_notes/release_2_2.html

⁹⁵⁹ https://doc.dpdk.org/guides/linux_gsg/index.html

⁹⁶⁰ https://doc.dpdk.org/guides/testpmd_app_ug/run_app.html

⁹⁶¹ <https://doc.dpdk.org/api/>

⁹⁶² <https://github.com/openvswitch/ovs/blob/branch-2.5/INSTALL.DPDK.md>

⁹⁶³ https://en.wikipedia.org/wiki/Data_Plane_Development_Kit

DHCP

- *About DHCP*

About DHCP

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- *DNS*
- *Hostname*
-
-
-

DHCP configuration

Manual allocation (MAC address)

Dynamic allocation (address pool)

Automatic allocation

Available servers

- Kea⁹⁶⁴
how to install and configure isc-kea
- dnsmasq⁹⁶⁵

Note:

The `isc-dhcp-server` package is deprecated and unsupported since Ubuntu 24.04 LTS. Please use `isc-kea` or `dnsmasq` instead.

information on how to configure it

⁹⁶⁴ <https://www.isc.org/kea/>

⁹⁶⁵ <https://thekelleys.org.uk/dnsmasq/doc.html>

Further reading

- [isc-dhcp-server Ubuntu Wiki](https://help.ubuntu.com/community/isc-dhcp-server)⁹⁶⁶
- [dhcpd.conf\(5\)](https://manpages.ubuntu.com/manpages/resolute/man5/dhcpd.conf.5.html)⁹⁶⁷
- [ISC dhcp-server](https://www.isc.org/dhcp/)⁹⁶⁸
- [ISC Kea Documentation](https://kb.isc.org/docs/kea-administrator-reference-manual)⁹⁶⁹

Time synchronisation

- [About time synchronisation](#)

About time synchronization

synchronize time with chrony

use chrony to serve NTP

timedatectl and timesyncd

Note:

Chrony is replacing systemd-timesyncd as the default in Ubuntu 25.10 and ntpdate/ntpd, which was the default before Ubuntu 18.04 LTS.

On upgraded systems (from Ubuntu 25.04 or below) systemd-timesyncd might still be the active time-daemon and thus render the new chrony service disabled. Similarly, while the use of ntpd is no longer recommended, it might still be installed to retain any previous behavior/config on upgrades (up to Ubuntu 22.04 LTS).

⁹⁶⁶ <https://help.ubuntu.com/community/isc-dhcp-server>

⁹⁶⁷ <https://manpages.ubuntu.com/manpages/resolute/man5/dhcpd.conf.5.html>

⁹⁶⁸ <https://www.isc.org/dhcp/>

⁹⁶⁹ <https://kb.isc.org/docs/kea-administrator-reference-manual>

How time synchronization works

About chrony

About `timedatectl`

-
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Further reading

- [ntp.org](http://www.ntp.org/): home of the Network Time Protocol project⁹⁷⁰
- [pool.ntp.org](https://www.ntppool.org/en/): project of virtual cluster of timeservers⁹⁷¹
- `timedatectl(1)`⁹⁷²
- `systemd-timesyncd.service(8)`⁹⁷³
- [chrony FAQ](https://chrony-project.org/faq.html)⁹⁷⁴
- [Feeding chrony from GPSD](https://gpsd.gitlab.io/gpsd/time-service-howto.html#_feeding_chrony_from_gpsd)⁹⁷⁵
- [Ubuntu Time wiki page](https://help.ubuntu.com/community/UbuntuTime)⁹⁷⁶

⁹⁷⁰ <http://www.ntp.org/>

⁹⁷¹ <https://www.ntppool.org/en/>

⁹⁷² <https://manpages.ubuntu.com/manpages/resolute/man1/timedatectl.1.html>

⁹⁷³ <https://manpages.ubuntu.com/manpages/resolute/man8/systemd-timesyncd.service.8.html>

⁹⁷⁴ <https://chrony-project.org/faq.html>

⁹⁷⁵ https://gpsd.gitlab.io/gpsd/time-service-howto.html#_feeding_chrony_from_gpsd

⁹⁷⁶ <https://help.ubuntu.com/community/UbuntuTime>

Network shares

- *Introduction to Samba*
- *Active Directory integration*

Introduction to Samba

Samba⁹⁷⁷

Samba functionality

File and printer sharing services

- *configured as a file server*
- *configured as a print server*

⁹⁷⁷ <https://www.samba.org/>

Directory services

- [How to set up Ubuntu Server as a member of an Active Directory domain](#)
- [How to set up Ubuntu Server as a Windows NT4-style domain controller](#)
- [How to set up Ubuntu Server as an LDAP in Windows NT4 mode](#)

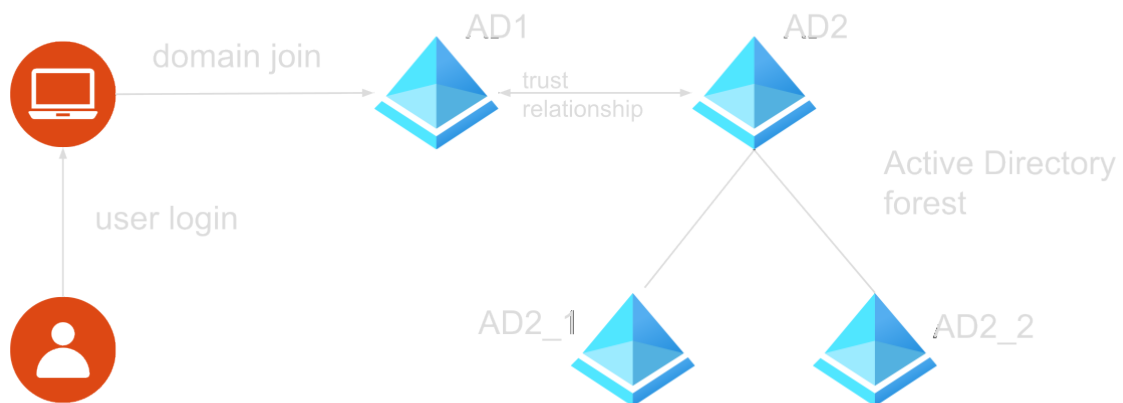
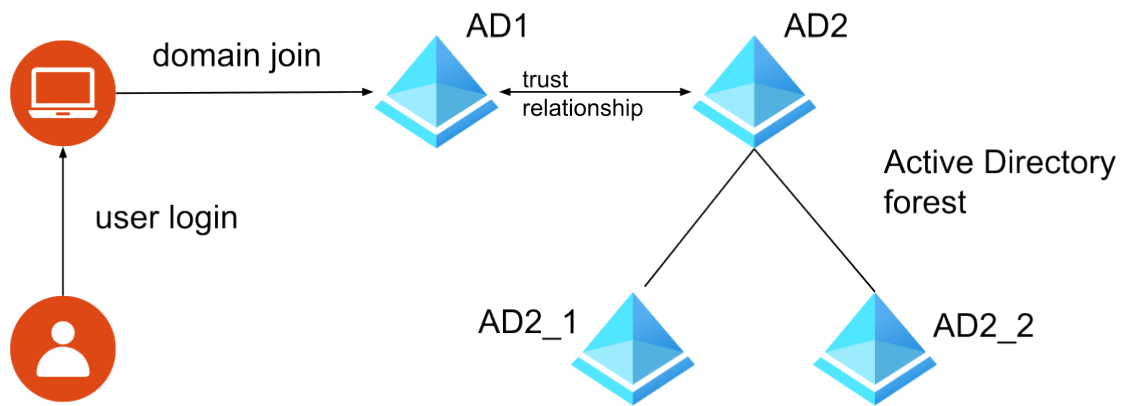
Authentication and access

- [How to set up Ubuntu Server as an LDAP directory](#) [controlling access to shared directories](#)
- [How to set up Ubuntu Server as a Samba file server](#) [set up a profile for Samba](#)
- [How to set up Ubuntu Server as an Internet File System \(CIFS\) share](#) [how to set up Common Internet File System \(CIFS\) shares](#)

Active Directory integration

Introduction

Introduction to Active Directory integration



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-
- *group policy objects*
-

- *Choosing an integration method*
- *Security identifiers*
- *Identity mapping backends*
- *The rid idmap backend*
- *The autorid idmap backend*

Further reading

- - Security Identifiers (SIDs)⁹⁷⁸
 - Active Directory Domain Services Overview⁹⁷⁹
- - Choosing an identity mapping backend⁹⁸⁰
 - rid identity mapping backend⁹⁸¹
 - autorid identity mapping backend⁹⁸²
- - *idmap_tdb(8)*⁹⁸³
 - *idmap_rid(8)*⁹⁸⁴
 - *idmap_autorid(8)*⁹⁸⁵
 - *wbinfo(1)*⁹⁸⁶

⁹⁷⁸ <https://learn.microsoft.com/en-us/windows-server/identity/ad-ds/manage/understand-security-identifiers>

⁹⁷⁹ <https://learn.microsoft.com/en-us/windows-server/identity/ad-ds/get-started/virtual-dc/active-directory-domain-services-overview>

⁹⁸⁰ https://wiki.samba.org/index.php/Setting_up_Samba_as_a_Domain_Member#Choosing_an_idmap_backend

⁹⁸¹ https://wiki.samba.org/index.php/Idmap_config_rid

⁹⁸² https://wiki.samba.org/index.php/Idmap_config_autorid

⁹⁸³ https://manpages.ubuntu.com/manpages/resolute/man8/idmap_tdb.8.html

⁹⁸⁴ https://manpages.ubuntu.com/manpages/resolute/man8/idmap_rid.8.html

⁹⁸⁵ https://manpages.ubuntu.com/manpages/resolute/man8/idmap_autorid.8.html

⁹⁸⁶ <https://manpages.ubuntu.com/manpages/resolute/man1/wbinfo.1.html>

Configuration

Choosing an integration method

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Requirement/Method	SSSD	idmap_rid	idmap_autorid
Deterministic ID	Y	Y	N
Undetermined ID	Y	Y	Y
Member server	N	Y	Y
Not a member server	Y	Y	Y
AD multi-domain (forest)	N	N(*)	Y
AD single domain	Y	Y	Y

Joining a forest with the rid backend

- *Security identifiers*
- *Identity mapping backends*
- *The rid idmap backend*
- *The autorid idmap backend*

Security identifiers (SIDs)

Note:

For a full explanation of what an SID is, and how they are structured and allocated, please refer to [Understand Security Identifiers](https://learn.microsoft.com/en-us/windows-server/identity/ad-ds/manage/understand-security-identifiers)⁹⁸⁷.

⁹⁸⁷ <https://learn.microsoft.com/en-us/windows-server/identity/ad-ds/manage/understand-security-identifiers>

S-1-5-<Domain-Identifier>--<Relative-Identifier>

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- *Identity mapping backends*
- *The rid idmap backend*
- *The autorid idmap backend*

Identity Mapping (idmap) backends

- `idmap_ad(8)`⁹⁸⁸
- `idmap_rid(8)`⁹⁸⁹
- `idmap_aurid(8)`⁹⁹⁰

`tdb(8)`⁹⁹¹

`idmap_`

user ID ranges on an Ubuntu/Debian system⁹⁹²

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⁹⁸⁸ https://manpages.ubuntu.com/manpages/resolute/man8/idmap_ad.8.html

⁹⁸⁹ https://manpages.ubuntu.com/manpages/resolute/man8/idmap_rid.8.html

⁹⁹⁰ https://manpages.ubuntu.com/manpages/resolute/man8/idmap_aurid.8.html

⁹⁹¹ https://manpages.ubuntu.com/manpages/resolute/man8/idmap_tdb.8.html

⁹⁹² <https://www.debian.org/doc/debian-policy/ch-opersys.html#uid-and-gid-classes>

- *The rid idmap backend*
- *The autorid idmap backend*

The RID IDMap backend

*idmap_rid(8)*⁹⁹³
GIDs

-
-

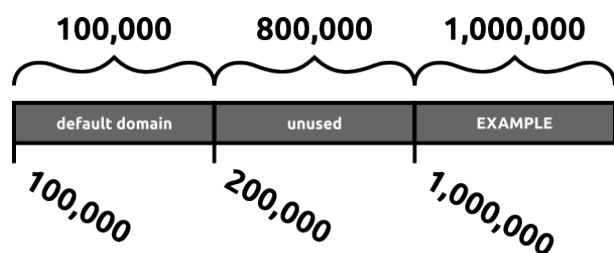
```
[global]
...
security = ads
realm = EXAMPLE.INTERNAL
workgroup = EXAMPLE
...
idmap config * : backend      = tdb
idmap config * : range       = 100000 - 199999

idmap config EXAMPLE : backend = rid
idmap config EXAMPLE : range  = 1000000 - 1999999
```

Note:

This is not yet a complete configuration file, and is just illustrating the IDMap configuration. More is needed to join an Active Directory domain.

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-



⁹⁹³ https://manpages.ubuntu.com/manpages/resolute/man8/idmap_rid.8.html

Important:

Planning a range of IDs to be used for the mapping critically important. Such a range can never be reduced, just expanded (carefully!), and it must **NEVER** overlap with another allocated range.

[global]

```
...
security = ads
realm = EXAMPLE.INTERNAL
workgroup = EXAMPLE
...
idmap config * : backend      = tdb
idmap config * : range       = 100000 - 199999

idmap config EXAMPLE : backend = rid
idmap config EXAMPLE : range  = 1000000 - 1999999

idmap config COMPANY : backend = rid
idmap config COMPANY : range  = 2000000 - 2999999
```

Pros and cons of the RID backend

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- *The autorid IDMap backend*

The AutoRID IDMap backend

*idmap_autorid(8)*⁹⁹⁴

-

-

[global]

```
...
security = ads
realm = EXAMPLE.INTERNAL
workgroup = EXAMPLE
...
idmap config * : backend = autorid
# 1,000,000 - 19,999,999
idmap config * : range    = 1000000 - 19999999
# 1,000,000
idmap config * : rangesize = 1000000
```

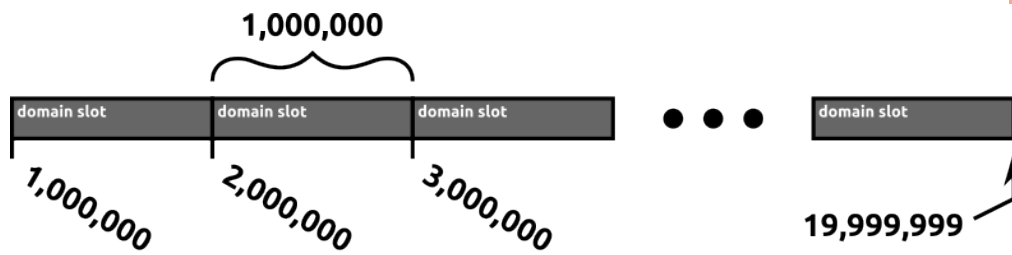
-

-

-

-

⁹⁹⁴ https://manpages.ubuntu.com/manpages/resolute/man8/idmap_autorid.8.html



Note:

The `autorid.tdb` domain mapping database file is kept in `/var/lib/samba/` and should be backed up regularly.

Pros and cons of the AutoRID backend

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See also

- [*Active Directory integration*](#)

See also

- [*Networking section*](#)

Managing software

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Managing software

Package sources

- [*Third party repository usage*](#)

Third party repository usage

Why not use third party software?

Security risk

dpkg

System integrity

Lack of official Ubuntu support

A better solution to third party APT repositories: snaps

snaps⁹⁹⁵

published in the snap store⁹⁹⁶
GPG

Mitigating the risks

Security risk mitigation

-
-

⁹⁹⁵ <https://snapcraft.io/docs/get-started>

⁹⁹⁶ <https://snapcraft.io/store>

-

System integrity mitigation

-

-

Dealing with third party APT repositories in Ubuntu

Add the repository

follow best practices⁹⁹⁷

Fetch the GPG key

-

-

```
$ sudo wget -O /usr/share/keyrings/externalrepo-archive-keyring.asc \ >  
https://thirdpartyrepo.com/ubuntu/externalrepo-archive-keyring.asc
```

⁹⁹⁷ <https://wiki.debian.org/DebianRepository/UseThirdParty>

```
$ curl -fsSL  
https://thirdpartyrepo.com/ubuntu/externalrepo-archive-keyring.asc \> | sudo  
gpg --dearmor -o /usr/share/keyrings/externalrepo-archive-keyring.gpg
```

Sources file entry

-
-

```
Types: deb  
URIs: https://thirdpartyrepo.com/ubuntu  
Suites: resolute  
Components: main  
Signed-By: /usr/share/keyrings/externalrepo-archive-keyring.asc
```

```
deb [signed-by=/usr/share/keyrings/externalrepo-archive-keyring.gpg] https://  
thirdpartyrepo.com/ubuntu/ resolute main
```

Configure pinning for the repository

```
Package: *  
Pin: origin thirdpartyrepo.com  
Pin-Priority: 100
```

Debian Reference guide⁹⁹⁸

How to remove a repository

-
-
-

```
$ apt remove --purge \ > $(grep "^Package: "  
/var/lib/apt/lists/#<SELECT_THE_FILE_FOR_YOUR_REPOSITORY>#_*_Packages \ > |  
cut -d " " -f2 | sort -u | \ > xargs dpkg-query -W  
-f='${binary:Package}\t${db:Status-Abbrev}\n' 2> /dev/null | \ > awk '/\tii  
${print $1}')
```

⁹⁹⁸ https://www.debian.org/doc/manuals/debian-reference/ch02.en.html#_tweaking_candidate_version

A special case: Ubuntu PPAs

```
$ add-apt-repository ppa:thirdparty/externalrepo
```

```
$ ppa-purge ppa:thirdparty/externalrepo
```

Configuration

- *Changing package files*
- *Configuration managers*

Changing package files

Configuration files

```
[mysqld]
#
# * Basic Settings
#
user          = mysql
# pid-file    = /var/run/mysqld/mysqld.pid
# socket      = /var/run/mysqld/mysqld.sock
port          = 3307
```

Note:

Some packages do not automatically create files for you to edit in their `.d` directories. In these cases it is often acceptable to just create an additional config file by any name there. When in doubt, check the package's documentation to confirm.

```
$ systemctl restart mysql
```

```
$ netstat -tunpevaW | grep -i 3307
```

tcp	0	0	127.0.0.1:3307	0.0.0.0:*	LISTEN	106
	416022		1730/mysqld			

Systemd files

Systemd⁹⁹⁹

⁹⁹⁹ <https://systemd.io/>


```
$ sudo systemctl edit apache2
```

```
### Editing /etc/systemd/system/apache2.service.d/override.conf
### Anything between here and the comment below will become the new contents of the file

### Lines below this comment will be discarded

### /lib/systemd/system/apache2.service
# [Unit]
# Description=The Apache HTTP Server
# After=network.target remote-fs.target nss-lookup.target
# Documentation=https://httpd.apache.org/docs/2.4/
#
# [Service]
# Type=forking
# Environment=APACHE_STARTED_BY_SYSTEMD=true
# ExecStart=/usr/sbin/apachectl start
# ExecStop=/usr/sbin/apachectl graceful-stop
# ExecReload=/usr/sbin/apachectl graceful
# KillMode=mixed
# PrivateTmp=true
# Restart=on-abort
...
```

```
### Editing /etc/systemd/system/apache2.service.d/override.conf
### Anything between here and the comment below will become the new contents of the file

[Service]
Restart=on-failure

### Lines below this comment will be discarded
...
```

Note:

Some options, such as `ExecStart` are additive. If you would like to fully override them add an extra line that clears it (e.g. `ExecStart=`) before providing new options. See [Systemd's man page](https://www.freedesktop.org/software/systemd/man/latest/systemd.service.html)¹⁰⁰⁰ for more information.

¹⁰⁰⁰ <https://www.freedesktop.org/software/systemd/man/latest/systemd.service.html>

```
$ sudo systemctl daemon-reload
```

```
$ systemctl status apache2

□ apache2.service - The Apache HTTP Server
   Loaded: loaded (/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Drop-In: /etc/systemd/system/apache2.service.d
            └─override.conf
            /run/systemd/system/service.d
            └─zzz-lxc-service.conf
   Active: active (running) since Fri 2023-02-17 16:39:22 UTC; 27min ago
     Docs: https://httpd.apache.org/docs/2.4/
  Main PID: 4735 (apache2)
    Tasks: 55 (limit: 76934)
   Memory: 6.5M
      CPU: 65ms
   CGroup: /system.slice/apache2.service
           └─4735 /usr/sbin/apache2 -k start
           └─4736 /usr/sbin/apache2 -k start
           └─4737 /usr/sbin/apache2 -k start
...

```

AppArmor

AppArmor

```
Configuration file '/etc/apparmor.d/usr.bin.swtpm'
==> Modified (by you or by a script) since installation.
==> Package distributor has shipped an updated version.
What would you like to do about it ? Your options are:
  Y or I : install the package maintainer's version
  N or O : keep your currently-installed version
  D      : show the differences between the versions
  Z      : start a shell to examine the situation
The default action is to keep your current version.
*** usr.bin.swtpm (Y/I/N/O/D/Z) [default=N] ?

```

```
/var/customtpm/** rwk,
```

Note:

Although most local profiles have the same name as the maintainer's, you can often check what file is included based on the main profile's contents. In `swtvm`'s case, `/etc/apparmor.d/usr.bin.swtvm` contains the lines:

```
# Site-specific additions and overrides. See local/README for details.  
#include <local/usr.bin.swtvm>
```

showing that the local profile is located at `/etc/apparmor.d/local/usr.bin.swtvm`

Restoring configuration files

dpkg

```
$ dpkg -S /etc/rsyslog.conf  
rsyslog: /etc/rsyslog.conf
```

```
$ rm /etc/rsyslog.conf
```

```
$ dpkg -S /etc/rsyslog.conf  
rsyslog: /etc/rsyslog.conf
```

```
$ sudo apt install --reinstall -o Dpkg::Options::="--force-confmiss" rsyslog
```

¹⁰⁰¹ <https://ubuntu.com/tutorials/beginning-apparmor-profile-development>

```
...  
Preparing to unpack .../rsyslog_8.2302.0-1ubuntu3_amd64.deb ...  
Unpacking rsyslog (8.2302.0-1ubuntu3) over (8.2302.0-1ubuntu3) ...  
Setting up rsyslog (8.2302.0-1ubuntu3) ...
```

```
Configuration file '/etc/rsyslog.conf', does not exist on system.  
Installing new config file as you requested.
```

*dpkg(1)*¹⁰⁰²

-
-
-
-
-

```
$ echo badentry >> /etc/rsyslog.conf
```

```
$ sudo apt install --reinstall -o Dpkg::Options::="--force-confask" rsyslog
```

¹⁰⁰² <https://manpages.ubuntu.com/manpages/resolute/man1/dpkg.1.html>

```
...
Preparing to unpack .../rsyslog_8.2302.0-1ubuntu3_amd64.deb ...
Unpacking rsyslog (8.2302.0-1ubuntu3) over (8.2302.0-1ubuntu3) ...
Setting up rsyslog (8.2302.0-1ubuntu3) ...
```

```
Configuration file '/etc/rsyslog.conf'
==> Modified (by you or by a script) since installation.
    Version in package is the same as at last installation.
    What would you like to do about it ? Your options are:
    Y or I : install the package maintainer's version
    N or O : keep your currently-installed version
    D      : show the differences between the versions
    Z      : start a shell to examine the situation
    The default action is to keep your current version.
*** rsyslog.conf (Y/I/N/O/D/Z) [default=N] ? D
--- /etc/rsyslog.conf 2023-04-18 07:11:50.427040350 +0000
+++ /etc/rsyslog.conf.dpkg-new 2023-02-23 16:58:03.000000000 +0000
@@ -51,4 +51,3 @@
 # Include all config files in /etc/rsyslog.d/
 #
 $IncludeConfig /etc/rsyslog.d/*.conf
-badentry
```

```
Configuration file '/etc/rsyslog.conf'
==> Modified (by you or by a script) since installation.
    Version in package is the same as at last installation.
    What would you like to do about it ? Your options are:
    Y or I : install the package maintainer's version
    N or O : keep your currently-installed version
    D      : show the differences between the versions
    Z      : start a shell to examine the situation
    The default action is to keep your current version.
*** rsyslog.conf (Y/I/N/O/D/Z) [default=N] ? y
Installing new version of config file /etc/rsyslog.conf ...
```

```
$ rm -rf /etc/vim
```

```
$ dpkg -S /etc/vim
```

```
vim-common, vim-tiny: /etc/vim
```

```
$ sudo apt install --reinstall -o Dpkg::Options::="--force-confmiss"
vim-common vim-tiny
```

```
...  
Configuration file '/etc/vim/vimrc', does not exist on system.  
Installing new config file as you requested.  
...  
Configuration file '/etc/vim/vimrc.tiny', does not exist on system.  
Installing new config file as you requested.
```

Configuration managers

Ansible

Ansible

-

Puppet

Puppet¹⁰⁰³

-

¹⁰⁰³ <https://www.puppet.com/>

Further reading

- [Wikipedia table](#)¹⁰⁰⁴ this

Updates

- [About apt upgrade and phased updates](#)
- [Advance testing of updates in best practice server deployments](#)

About apt upgrade and phased updates

```
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages have been kept back:
  (Names of <X> held back packages listed here)
0 upgraded, 0 newly installed, 0 to remove and <X> not upgraded.
```

What are phased updates?

¹⁰⁰⁴ https://en.wikipedia.org/wiki/Comparison_of_open-source_configuration_management_software

Why is Ubuntu doing this?

Note:

It should be mentioned here that security updates are *never* phased.

Can I turn off phased updates?

Can I apt upgrade the individual packages? (and should I?)

How do I turn off phased updates?

```
Update-Manager::Always-Include-Phased-Updates true;  
APT::Get::Always-Include-Phased-Updates true;
```

Why is this suddenly happening now?

implemented in APT in 21.04¹⁰⁰⁵

How does it actually work?

wiki page on phased updates¹⁰⁰⁶

How can I find out more information about currently phased packages?

```
apt policy <package>
```

¹⁰⁰⁵ <https://discourse.ubuntu.com/t/phased-updates-in-apt-in-21-04/20345>

¹⁰⁰⁶ <https://wiki.ubuntu.com/PhasedUpdates>

```
libglapi-mesa:
  Installed: 22.0.5-0ubuntu0.3
  Candidate: 22.2.5-0ubuntu0.1~22.04.1
  Version table:
    22.2.5-0ubuntu0.1~22.04.1 500 (phased 20%)
      500 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages
*** 22.0.5-0ubuntu0.3 100
      100 /var/lib/dpkg/status
    22.0.1-1ubuntu2 500
      500 http://archive.ubuntu.com/ubuntu jammy/main amd64 Packages
```

currently being phased¹⁰⁰⁷

Further reading

- [excellent post on AskUbuntu](#)¹⁰⁰⁸
CC BY-SA 4.0¹⁰⁰⁹
- [phasing Ubuntu Stable Release Updates](#)¹⁰¹⁰
- [Ubuntu wiki](#)¹⁰¹¹ [Error Tracker](#)¹⁰¹² [apt preferences](#)¹⁰¹³

Advance testing of updates in best-practice server deployments

¹⁰⁰⁷ <https://ubuntu-archive-team.ubuntu.com/phased-updates.html>

¹⁰⁰⁸ <https://askubuntu.com/questions/1431940/what-are-phased-updates-and-why-does-ubuntu-use-them>

¹⁰⁰⁹ <https://creativecommons.org/licenses/by-sa/4.0/>

¹⁰¹⁰ <https://ubuntu-archive-team.ubuntu.com/phased-updates.html>

¹⁰¹¹ <https://wiki.ubuntu.com/PhasedUpdates>

¹⁰¹² <https://wiki.ubuntu.com/ErrorTracker/PhasedUpdates>

¹⁰¹³ https://manpages.ubuntu.com/manpages/resolute/man5/apt_preferences.5.html

Information about proposed updates

Pending SRU page¹⁰¹⁴

Testing a single proposed update

Enable proposed wiki page¹⁰¹⁵

Test all proposed updates together

¹⁰¹⁴ <https://ubuntu-archive-team.ubuntu.com/pending-sru.html>

¹⁰¹⁵ <https://wiki.ubuntu.com/Testing/EnableProposed>

How to report a regression

-
-
-
-

pending SRU report¹⁰¹⁶

General information on reporting bugs¹⁰¹⁷

See also

- *Managing software*

In data and storage

-
-

Data and storage

Data management

¹⁰¹⁶ <https://ubuntu-archive-team.ubuntu.com/pending-sru.html>

¹⁰¹⁷ <https://help.ubuntu.com/community/ReportingBugs>

Introduction to OpenLDAP

Lightweight Directory Access Protocol¹⁰¹⁸

RFC 4510¹⁰¹⁹

Key concepts and terms

- *DIT*
-
-
-
- *DN*
-

Note:

The terms **object**, **container**, and **node** have certain connotations but they all essentially mean the same thing as **entry** (the technically correct term).

-
-
-

```
dn: cn=John Doe,dc=example,dc=com
cn: John Doe
givenName: John
```

(continues on next page)

¹⁰¹⁸ <https://ldap.com/>

¹⁰¹⁹ <https://datatracker.ietf.org/doc/html/rfc4510>

(continued from previous page)

```
sn: Doe
telephoneNumber: +1 888 555 6789
telephoneNumber: +1 888 555 1232
mail: john@example.com
manager: cn=Larry Smith,dc=example,dc=com
objectClass: inetOrgPerson
objectClass: organizationalPerson
objectClass: person
objectClass: top
```

RFC 2849¹⁰²⁰

Our OpenLDAP guide

- *Install and configure LDAP*
- *LDAP Access Control*
- *LDAP users and groups*
- *SSL/TLS*
- *Replication*
- *Backup and restore*

Further reading

- [OpenLDAP administrators guide](#)¹⁰²¹
- [RFC 4515: LDAP string representation of search filters](#)¹⁰²²
- [LDAP for Rocket Scientists](#)¹⁰²³
- [LDAP System Administration](#)¹⁰²⁴
- [Mastering OpenLDAP](#)¹⁰²⁵

¹⁰²⁰ <https://datatracker.ietf.org/doc/html/rfc2849>

¹⁰²¹ <https://openldap.org/doc/admin25/>

¹⁰²² <https://www.rfc-editor.org/rfc/rfc4515.txt>

¹⁰²³ <https://www.zytrax.com/books/ldap/>

¹⁰²⁴ <https://www.oreilly.com/library/view/ldap-system-administration/1565924916/>

¹⁰²⁵ <https://www.packtpub.com/en-us/product/mastering-openldap-configuring-securing-and-integrating-directory-services-978>

Introduction to databases

MySQL¹⁰²⁶

PostgreSQL¹⁰²⁷

section

how-to

- *How to install and configure MySQL*
- *How to install and configure PostgreSQL*

Storage

- *Storage*
- *Multipath*

Storage

- *About LVM*

About Logical Volume Management (LVM)

Logical Volume Management¹⁰²⁸

refer to this guide

¹⁰²⁶ <https://www.mysql.com/>

¹⁰²⁷ <https://www.postgresql.org/>

¹⁰²⁸ https://en.wikipedia.org/wiki/Logical_volume_management

Key concepts

-
-
-

filesystem

Resizing partitions

Moving Partitions

Snapshots

See also

- [*Storage*](#)

Multipath

Introduction

- [*Introduction to Device mapper multipathing*](#)

Configuration

- [*Configuration options and overview*](#)
- [*Configuration examples*](#)
- [*Common tasks and procedures*](#)

Introduction to device mapper multipathing (“multipath”)

How a multipath device works in practice

What identifier does a multipath device use?

What does multipath provide?

- *failover*
-

Configuring multipath for a storage system

Multipath components

Component	Description
dm_ multipath kernel module	Reroutes I/O and supports failover for paths and path groups.
mul- tipath command	Lists and configures multipath devices. While this command is normally started up with <code>/etc/rc.sysinit</code> , it can also be started up by a <code>udev</code> program whenever a block device is added, or it can be run by the <code>initramfs</code> file system.
multi- pathd daemon	Monitors paths; as paths fail and come back, it may initiate path group switches. Provides for interactive changes to multipath devices. This daemon must be restarted for any changes to the <code>/etc/multipath.conf</code> file to take effect.
kpartx com- mand	Creates device-mapper devices for the partitions on a device. It is necessary to use this command for DOS-based partitions with multipath. The <code>kpartx</code> is provided in its own package, but the <code>multipath-tools</code> package depends on it.

Multipath setup overview

- 1.
- 2.
- 3.
- 4.
- 5.

DM-Multipath setup

DM-Multipath configuration

Example of a multipath device in use

FC

HBAs

-
-

See also

For information on the multipath configuration defaults, including the `user_friendly_names` and `alias` configuration options, see *DM-Multipath configuration* (page 1005).

Consistent multipath device names in a cluster

use LVM

1.

2.

```
$ systemctl stop multipath-tools.service & multipath -F
```

3.

4.

```
$ systemctl start multipath-tools.service
```

Multipath device attributes

DM-

Multipath configuration

Multipath devices in logical volumes

```
$ pvcreate /dev/mapper/mpatha
```

Note:

If you try to create an LVM physical volume on a whole device on which you have configured partitions, the `pvcreate` command will fail.

fallbacks

```
filter = [ "r/block/", "r/disk/", "r/sd.*/", "a/.*/" ]
```

```
$ update-initramfs -u -k all
```

Note:

Every time either `/etc/lvm.conf` or `/etc/multipath.conf` is updated, the `initrd` should be rebuilt to reflect these changes. This is imperative when *denylists* and filters are necessary to maintain a stable storage configuration.

Multipath configuration options and overview

device mapper multipathing introduction

Configuration file overview

```
<section> {  
    <attribute> <value>  
    ...  
    <subsection> {  
        <attribute> <value>  
        ...  
    }  
}
```

-
-
-
-
-
-

Configuration file defaults

*in the manual pages*¹⁰²⁹

- [illegible]

¹⁰²⁹ <https://manpages.ubuntu.com/manpages/resolute/man5/multipath.conf.5.html>

-

-
-
-
-
-

Note:

Previously, the `multipath-tools` project provided a complete configuration file with all the most commonly used options for each of the most-used storage devices. Currently, you can see all those default options by running `sudo multipath -t`. This will dump a used configuration file including all the embedded default options.

Configuration file blacklist and exceptions

By devnode

```
blacklist {
    devnode "^(ram|zram|raw|loop|fd|md|dm-|sr|scd|st|dcssblk)[0-9]"
    devnode "^(td|hd|vd)[a-z]"
    devnode "^cciss!c[0-9]d[0-9]*"
}
```

By wwid

By device

```
device {
    vendor "LENOVO"
    product "Universal Xport"
}
```

By property

Blocklist by protocol

```
$ multipathd show paths format "%d %P"
```

Blocklist exceptions

```
blacklist_exceptions {  
    property "(SCSI_IDENT_|ID_WWN)"  
}
```

Note:

A common use is to blocklist “everything” using a catch-all regular expression, and create specific `blacklist_exceptions` entries for those devices that should be handled by `multipath-tools`.

Configuration file multipath section

-

- ## Optional attributes

multipath.conf(5)¹⁰³⁰

- ¹⁰³⁰ <https://manpages.ubuntu.com/manpages/resolute/man5/multipath.conf.5.html>

Example

```
multipaths {
    multipath {
        wwid          3600508b4000156d700012000000b0000
        alias         yellow
        path_grouping_policy multibus
        path_selector  "round-robin 0"
        failback      manual
        rr_weight      priorities
        no_path_retry  5
    }
    multipath {
        wwid          1DEC_____321816758474
        alias         red
    }
}
```

Configuration file devices section

- 1.
- 2.
- 3.
- 4.
- 5.
- 6.

- *DGC*
- *Engenio*

•

Optional attributes

•

-
-
-
-
-
-
-

Example

```
devices {
  device {
    vendor "3PARdata"
    product "VV"
    path_grouping_policy "group_by_prio"
    hardware_handler "1 alua"
    prio "alua"
    failback "immediate"
    no_path_retry 18
    fast_io_fail_tmo 10
    dev_loss_tmo "infinity"
  }
  device {
    vendor "DEC"
    product "HSG80"
    path_grouping_policy "group_by_prio"
    path_checker "hp_sw"
    hardware_handler "1 hp_sw"
    prio "hp_sw"
    no_path_retry "queue"
  }
}
```

Multipath configuration examples

1. *Introduction to device mapper multipathing*
2. *Configuration options and overview*

Basic setup

the configuration overview

```
$ sudo multipath -t
```

Example of a defaults section

```
defaults {  
    #  
    # name      : polling_interval  
    # scope     : multipathd  
    # desc      : interval between two path checks in seconds. For  
    #              properly functioning paths, the interval between checks  
    #              will gradually increase to (4 * polling_interval).  
    # values    : n > 0  
    # default   : 5  
    #  
    polling_interval 10  
  
    #  
    # name      : path_selector  
    # scope     : multipath & multipathd  
    # desc      : the default path selector algorithm to use  
    #              these algorithms are offered by the kernel multipath target  
    # values    : "round-robin 0" = Loop through every path in the path group,  
    #              sending the same amount of IO to each.  
    #              "queue-length 0" = Send the next bunch of IO down the path  
    #              with the least amount of outstanding IO.  
    #              "service-time 0" = Choose the path for the next bunch of IO  
    #              based on the amount of outstanding IO to  
    #              the path and its relative throughput.  
    # default   : "service-time 0"  
    #  
    path_selector "round-robin 0"
```

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```
#
# name      : path_grouping_policy
# scope     : multipath & multipathd
# desc      : the default path grouping policy to apply to unspecified
#             multipaths
# values    : failover          = 1 path per priority group
#             multibus          = all valid paths in 1 priority group
#             group_by_serial   = 1 priority group per detected serial
#                               number
#             group_by_prio     = 1 priority group per path priority
#                               value
#             group_by_node_name = 1 priority group per target node name
# default   : failover
#
path_grouping_policy multibus

#
# name      : uid_attribute
# scope     : multipath & multipathd
# desc      : the default udev attribute from which the path
#             identifier should be generated.
# default   : ID_SERIAL
#
uid_attribute "ID_SERIAL"

#
# name      : getuid_callout
# scope     : multipath & multipathd
# desc      : the default program and args to callout to obtain a unique
#             path identifier. This parameter is deprecated.
#             This parameter is deprecated, superseded by uid_attribute
# default   : /lib/udev/scsi_id --whitelisted --device=/dev/%n
#
getuid_callout "/lib/udev/scsi_id --whitelisted --device=/dev/%n"

#
# name      : prio
# scope     : multipath & multipathd
# desc      : the default function to call to obtain a path
#             priority value. The ALUA bits in SPC-3 provide an
#             exploitable prio value for example.
# default   : const
#
# prio "alua"

#
# name      : prio_args
# scope     : multipath & multipathd
# desc      : The arguments string passed to the prio function
#             Most prio functions do not need arguments. The
#             datacore prioritizer need one.
# default   : (null)
#
# prio_args "timeout=1000 preferredsds=foo"

#
```

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```
# name      : features
# scope     : multipath & multipathd
# desc      : The default extra features of multipath devices.
#            Syntax is "num[ feature_0 feature_1 ...]", where 'num' is the
#            number of features in the following (possibly empty) list of
#            features.
# values    : queue_if_no_path = Queue IO if no path is active; consider
#            using the 'no_path_retry' keyword instead.
#            no_partitions    = Disable automatic partitions generation via
#            kpartx.
# default   : "0"
#
features    "0"
#features   "1 queue_if_no_path"
#features   "1 no_partitions"
#features   "2 queue_if_no_path no_partitions"

#
# name      : path_checker, checker
# scope     : multipath & multipathd
# desc      : the default method used to determine the paths' state
# values    : readsector0|tur|emc_clariion|hp_sw|directio|rdac|cciss_tur
# default   : directio
#
path_checker directio

#
# name      : rr_min_io
# scope     : multipath & multipathd
# desc      : the number of IO to route to a path before switching
#            to the next in the same path group for the bio-based
#            multipath implementation. This parameter is used for
#            kernels version up to 2.6.31; newer kernel version
#            use the parameter rr_min_io_rq
# default   : 1000
#
rr_min_io 100

#
# name      : rr_min_io_rq
# scope     : multipath & multipathd
# desc      : the number of IO to route to a path before switching
#            to the next in the same path group for the request-based
#            multipath implementation. This parameter is used for
#            kernels versions later than 2.6.31.
# default   : 1
#
rr_min_io_rq 1

#
# name      : flush_on_last_del
# scope     : multipathd
# desc      : If set to "yes", multipathd will disable queueing when the
#            last path to a device has been deleted.
# values    : yes|no
# default   : no
```

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```
#
flush_on_last_del yes

#
# name      : max_fds
# scope     : multipathd
# desc      : Sets the maximum number of open file descriptors for the
#             multipathd process.
# values    : max|n > 0
# default   : None
#
max_fds 8192

#
# name      : rr_weight
# scope     : multipath & multipathd
# desc      : if set to priorities the multipath configurator will assign
#             path weights as "path prio * rr_min_io"
# values    : priorities|uniform
# default   : uniform
#
rr_weight priorities

#
# name      : failback
# scope     : multipathd
# desc      : tell the daemon to manage path group failback, or not to.
#             0 means immediate failback, values >0 means deferred
#             failback expressed in seconds.
# values    : manual|immediate|n > 0
# default   : manual
#
failback immediate

#
# name      : no_path_retry
# scope     : multipath & multipathd
# desc      : tell the number of retries until disable queueing, or
#             "fail" means immediate failure (no queueing),
#             "queue" means never stop queueing
# values    : queue|fail|n (>0)
# default   : (null)
#
no_path_retry fail

#
# name      : queue_without_daemon
# scope     : multipathd
# desc      : If set to "no", multipathd will disable queueing for all
#             devices when it is shut down.
# values    : yes|no
# default   : yes
queue_without_daemon no

#
# name      : user_friendly_names
```

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```
# scope : multipath & multipathd
# desc : If set to "yes", using the bindings file
#         /etc/multipath/bindings to assign a persistent and
#         unique alias to the multipath, in the form of mpath<n>.
#         If set to "no" use the WWID as the alias. In either case
#         this be will be overridden by any specific aliases in this
#         file.
# values : yes|no
# default : no
user_friendly_names yes

#
# name : mode
# scope : multipath & multipathd
# desc : The mode to use for the multipath device nodes, in octal.
# values : 0000 - 0777
# default : determined by the process
mode 0644

#
# name : uid
# scope : multipath & multipathd
# desc : The user id to use for the multipath device nodes. You
#         may use either the numeric or symbolic uid
# values : <user_id>
# default : determined by the process
uid 0

#
# name : gid
# scope : multipath & multipathd
# desc : The group id to user for the multipath device nodes. You
#         may use either the numeric or symbolic gid
# values : <group_id>
# default : determined by the process
gid disk

#
# name : checker_timeout
# scope : multipath & multipathd
# desc : The timeout to use for path checkers and prioritizers
#         that issue scsi commands with an explicit timeout, in
#         seconds.
# values : n > 0
# default : taken from /sys/block/sd<x>/device/timeout
checker_timeout 60

#
# name : fast_io_fail_tmo
# scope : multipath & multipathd
# desc : The number of seconds the scsi layer will wait after a
#         problem has been detected on a FC remote port before failing
#         IO to devices on that remote port.
# values : off | n >= 0 (smaller than dev_loss_tmo)
# default : determined by the OS
fast_io_fail_tmo 5
```

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```
#
# name      : dev_loss_tmo
# scope     : multipath & multipathd
# desc      : The number of seconds the scsi layer will wait after a
#              problem has been detected on a FC remote port before
#              removing it from the system.
# values    : infinity | n > 0
# default   : determined by the OS
dev_loss_tmo 120

#
# name      : bindings_file
# scope     : multipath
# desc      : The location of the bindings file that is used with
#              the user_friendly_names option.
# values    : <full_pathname>
# default   : "/var/lib/multipath/bindings"
# bindings_file "/etc/multipath/bindings"

#
# name      : wwids_file
# scope     : multipath
# desc      : The location of the wwids file multipath uses to
#              keep track of the created multipath devices.
# values    : <full_pathname>
# default   : "/var/lib/multipath/wwids"
# wwids_file "/etc/multipath/wwids"

#
# name      : reservation_key
# scope     : multipath
# desc      : Service action reservation key used by mpathpersist.
# values    : <key>
# default   : (null)
# reservation_key "mpathkey"

#
# name      : force_sync
# scope     : multipathd
# desc      : If set to yes, multipath will run all of the checkers in
#              sync mode, even if the checker has an async mode.
# values    : yes|no
# default   : no
force_sync yes

#
# name      : config_dir
# scope     : multipath & multipathd
# desc      : If not set to an empty string, multipath will search
#              this directory alphabetically for files ending in ".conf"
#              and it will read configuration information from these
#              files, just as if it was in /etc/multipath.conf
# values    : "" or a fully qualified pathname
# default   : "/etc/multipath/conf.d"
```

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```
#
# name      : delay_watch_checks
# scope     : multipathd
# desc      : If set to a value greater than 0, multipathd will watch
#             paths that have recently become valid for this many
#             checks.  If they fail again while they are being watched,
#             when they next become valid, they will not be used until
#             they have stayed up for delay_wait_checks checks.
# values    : no|<n> > 0
# default   : no
delay_watch_checks 12

#
# name      : delay_wait_checks
# scope     : multipathd
# desc      : If set to a value greater than 0, when a device that has
#             recently come back online fails again within
#             delay_watch_checks checks, the next time it comes back
#             online, it will marked and delayed, and not used until
#             it has passed delay_wait_checks checks.
# values    : no|<n> > 0
# default   : no
delay_wait_checks 12
}
```

Example of a multipaths section

Note:

You can obtain the WWIDs for your LUNs by running: `multipath -ll` after the service `multipath-tools.service` has been restarted.

```
multipaths {
    multipath {
        wwid 36000000000000000e00000000030001
        alias yellow
    }
    multipath {
        wwid 36000000000000000e00000000020001
        alias blue
    }
    multipath {
        wwid 36000000000000000e00000000010001
        alias red
    }
    multipath {
        wwid 36000000000000000e00000000040001
        alias green
    }
    multipath {
        wwid 36000000000000000e00000000050001
        alias purple
    }
}
```

(continues on next page)

(continued from previous page)

```
}  
}
```

Example of a devices section

```
# devices {  
#     device {  
#         vendor "IBM"  
#         product "2107900"  
#         path_grouping_policy group_by_serial  
#     }  
# }  
#
```

Example of a blacklist section

```
# name      : blacklist  
# scope     : multipath & multipathd  
# desc      : list of device names to discard as not multipath candidates  
#  
# Devices can be identified by their device node name "devnode",  
# their WWID "wwid", or their vendor and product strings "device"  
# default : fd, hd, md, dm, sr, scd, st, ram, raw, loop, dcscblk  
#  
# blacklist {  
#     wwid 26353900f02796769  
#     devnode "^((ram|raw|loop|fd|md|dm-|sr|scd|st)[0-9])\\*$"  
#     devnode "^hd[a-z]"  
#     devnode "^dcscblk[0-9]\\*$"  
#     device {  
#         vendor DEC.*  
#         product MSA[15]00  
#     }  
# }
```

Example of a blacklist exception section

```
# name      : blacklist_exceptions  
# scope     : multipath & multipathd  
# desc      : list of device names to be treated as multipath candidates  
#             even if they are on the blacklist.  
#  
# Note: blacklist exceptions are only valid in the same class.  
# It is not possible to blacklist devices using the devnode keyword  
# and to exclude some devices of them using the wwid keyword.  
# default : -  
#  
# blacklist_exceptions {  
#     devnode "^dasd[c-d]+[0-9]\\*$"
```

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```
#      wwid      "IBM.75000000092461.4d00.34"  
#      wwid      "IBM.75000000092461.4d00.35"  
#      wwid      "IBM.75000000092461.4d00.36"  
# }
```

Common tasks and procedures

1. *Introduction to device mapper multipathing*
2. *Configuration options and overview*
3. *Configuration examples*

Resizing online multipath devices

```
$ sudo multipath -ll  
  
mpathb (360014056eee8ec6e1164fcb959086482) dm-0 LIO-ORG,lun01  
size=1.0G features='0' hwhandler='1 alua' wp=rw  
|+- policy='service-time 0' prio=50 status=active  
|  '- 7:0:0:1 sde 8:64 active ready running  
`+- policy='service-time 0' prio=50 status=enabled  
  '- 8:0:0:1 sdf 8:80 active ready running  
mpatha (36001405e3c2841430ee4bf3871b1998b) dm-1 LIO-ORG,lun02  
size=1.0G features='0' hwhandler='1 alua' wp=rw  
|+- policy='service-time 0' prio=50 status=active  
|  '- 7:0:0:2 sdc 8:32 active ready running  
`+- policy='service-time 0' prio=50 status=enabled  
  '- 8:0:0:2 sdd 8:48 active ready running
```

```
$ echo 1 | sudo tee /sys/block/sde/device/rescan
```

```
1
```

```
$ echo 1 | sudo tee /sys/block/sdf/device/rescan
```

```
1
```

```
$ sudo multipath -ll

mpathb (360014056eee8ec6e1164fcb959086482) dm-0 LIO-ORG,lun01
size=1.0G features='0' hwhandler='1 alua' wp=rw
|+- policy='service-time 0' prio=50 status=active
|  '- 7:0:0:1 sde 8:64 active ready running
`-+- policy='service-time 0' prio=50 status=enabled
   '- 8:0:0:1 sdf 8:80 active ready running
mpatha (36001405e3c2841430ee4bf3871b1998b) dm-1 LIO-ORG,lun02
size=1.0G features='0' hwhandler='1 alua' wp=rw
|+- policy='service-time 0' prio=50 status=active
|  '- 7:0:0:2 sdc 8:32 active ready running
`-+- policy='service-time 0' prio=50 status=enabled
   '- 8:0:0:2 sdd 8:48 active ready running
```

```
$ sudo multipathd resize map mpathb
```

```
ok
```

```
$ sudo multipath -ll

mpathb (360014056eee8ec6e1164fcb959086482) dm-0 LIO-ORG,lun01
size=2.0G features='0' hwhandler='1 alua' wp=rw
|+- policy='service-time 0' prio=50 status=active
|  '- 7:0:0:1 sde 8:64 active ready running
.-+- policy='service-time 0' prio=50 status=enabled
   '- 8:0:0:1 sdf 8:80 active ready running
mpatha (36001405e3c2841430ee4bf3871b1998b) dm-1 LIO-ORG,lun02
size=1.0G features='0' hwhandler='1 alua' wp=rw
|+- policy='service-time 0' prio=50 status=active
|  '- 7:0:0:2 sdc 8:32 active ready running
.-+- policy='service-time 0' prio=50 status=enabled
   '- 8:0:0:2 sdd 8:48 active ready running
```

filesystem

Move root file system from a single path device to a multipath device

Note:

Whenever `multipath.conf` is updated, `initrd` should be updated as well by running: `update-initramfs -u -k all`. The reason for this is that `multipath.conf` is copied to the RAM disk, and is integral to determining the available devices to map via its `denylist` and `devices` sections.

The multipathd daemon

the example configuration page

Multipath command output

```
action_if_any: alias (wwid_if_different_from_alias) dm_device_name_if_known vendor,product
size=size features='features' hwhandler='hardware_handler' wp=write_permission_if_known
```

```
-+- policy='scheduling_policy' prio=prio_if_known
status=path_group_status_if_known
```

```
`- host:channel:id:lun devnode major:minor dm_status_if_known path_status
online_status
```

```
mpathb (360014056eee8ec6e1164fcb959086482) dm-0 LIO-ORG,lun01
size=2.0G features='0' hwhandler='1 alua' wp=rw
|-+- policy='service-time 0' prio=50 status=active
|  ` 7:0:0:1 sde 8:64 active ready running
`-+- policy='service-time 0' prio=50 status=enabled
   ` 8:0:0:1 sdf 8:80 active ready running
```

Multipath queries with the multipath command

-
-
-
-
-

Note:

The default verbosity level of multipath is 2 and can be globally modified by defining the verbosity attribute in the defaults section of `multipath.conf`

```
mpathb (360014056eee8ec6e1164fcb959086482) dm-0 LIO-ORG,lun01
size=2.0G features='0' hwhandler='1 alua' wp=rw
|+- policy='service-time 0' prio=0 status=active
|  '- 7:0:0:1 sde 8:64 active undef running
`+- policy='service-time 0' prio=0 status=enabled
   '- 8:0:0:1 sdf 8:80 active undef running
mpatha (36001405e3c2841430ee4bf3871b1998b) dm-1 LIO-ORG,lun02
size=1.0G features='0' hwhandler='1 alua' wp=rw
|+- policy='service-time 0' prio=0 status=active
|  '- 7:0:0:2 sdc 8:32 active undef running
`+- policy='service-time 0' prio=0 status=enabled
   '- 8:0:0:2 sdd 8:48 active undef running
```

Determining device mapper entries with dmsetup

```
$ sudo dmsetup ls
```

```
mpathb (253:0)  
mpatha (253:1)
```

```
$ ls -lahd /dev/dm*
```

```
brw-rw---- 1 root disk 253, 0 Apr 27 14:49 /dev/dm-0  
brw-rw---- 1 root disk 253, 1 Apr 27 14:47 /dev/dm-1
```

Troubleshooting with the multipathd interactive console

```
$ sudo multipathd -k
```

```
> show config  
> CTRL-D
```

```
$ sudo multipathd -k
```

```
> reconfigure  
> CTRL-D
```

```
$ sudo multipathd -k
```

```
> show paths  
> CTRL-D
```

```
$ echo 'show config' | sudo multipathd -k
```

See also

- [Data and storage](#)
- [Backup section](#)

[Web services](#)

Web services

Introduction

Introduction to web services

[on the key concepts](#)

Squid proxy server

[about Squid proxy servers](#)
[install a Squid server](#)

[how to](#)

Web servers

- [How to install and configure Apache2](#)
- [How to configure Apache2 for your needs](#)
- [How to extend Apache2's functionality with modules](#)

- *How to install Nginx*
- *How to configure Nginx*
- *How to use Nginx modules*

Database server

how to install MySQL

PostgreSQL

Scripting languages

install Ruby on Rails

how to install PHP

how to

Key concepts

About web servers

HTTP

HTML

URLs

Qualified Domain Name (FQDN)
*Ubuntu Web site*¹⁰³¹

Fully

`www.ubuntu.com`

`community`¹⁰³²

`www.ubuntu.com/community`

¹⁰³¹ <https://ubuntu.com/>

¹⁰³² <https://ubuntu.com/community>

Transfer protocols

HTTP

HTTPS

HTTP status codes

HTTP

HTTP

-
-
-
-
-

check the RFC 2616¹⁰³³

Implementation

-
-

HTML

About Squid proxy servers

Squid¹⁰³⁴

HTTP

¹⁰³³ <https://www.w3.org/Protocols/rfc2616/rfc2616-sec6.html#sec6.1.1>

¹⁰³⁴ <http://www.squid-cache.org/>

Caching

HTTP

DNS

HTCP

Filtering

HTTP

HTTP

Installing and configuring

our

installation guide

Further reading

- The Squid website¹⁰³⁵
- Ubuntu Wiki page on Squid¹⁰³⁶

¹⁰³⁵ <http://www.squid-cache.org/>

¹⁰³⁶ <https://help.ubuntu.com/community/Squid>

See also

- *Web services*

Virtualisation and containers

Virtualisation and containers

- *Introduction to virtualization*

Introduction to virtualization

Virtual machines

and technologies

the different VM tools

Containers

System containers

Application containers

overview

take a look at this

Virtualization

- *VM tools overview*
- *QEMU microvm*
- *QEMU/libvirt live migration*
- *QEMU machine types*

VM tools in the Ubuntu space

KVM

Kernel-Based Virtual Machine (KVM)¹⁰³⁷

AMD

¹⁰³⁷ https://www.linux-kvm.org/page/Main_Page

QEMU

Quick Emulator (QEMU)¹⁰³⁸

command line interface¹⁰³⁹ a monitor¹⁰⁴⁰

how to set up QEMU

libvirt

libvirt¹⁰⁴¹

to use libvirt

how

Multipass and UVtool

Multipass¹⁰⁴² UVtool¹⁰⁴³

¹⁰³⁸ <https://www.qemu.org/>

¹⁰³⁹ <https://qemu-project.gitlab.io/qemu/system/invoke.html>

¹⁰⁴⁰ <https://qemu-project.gitlab.io/qemu/system/monitor.html>

¹⁰⁴¹ <https://libvirt.org/>

¹⁰⁴² <https://canonical.com/multipass>

¹⁰⁴³ <https://launchpad.net/uvtool>

Multipass

how to use UVtool

virt-manager

Virt-manager¹⁰⁴⁴

graphical user interface (GUI)

this how-to guide

QEMU microvm

1. `microvm` machine type¹⁰⁴⁵
2. *firmware (FW)* called `qboot`¹⁰⁴⁶
- 3.

¹⁰⁴⁴ <https://virt-manager.org/>

¹⁰⁴⁵ <https://www.qemu.org/docs/master/system/i386/microvm.html>

¹⁰⁴⁶ <https://github.com/bonzini/qboot>

Why a special workload?

- *bootloaders*
- *filesystem*
- BusyBox¹⁰⁴⁷
-
- *kata containers*¹⁰⁴⁸
- *Firecracker*¹⁰⁴⁹
- *LXD*
- *LXD*
- *slim* from dockerfiles¹⁰⁵⁰
- *krunvm* from OCI images¹⁰⁵¹

¹⁰⁴⁷ <https://github.com/SecureVirtualization/busybox>

¹⁰⁴⁸ <https://katacontainers.io/>

¹⁰⁴⁹ <https://firecracker-microvm.github.io/>

¹⁰⁵⁰ <https://github.com/ottomatica/slim>

¹⁰⁵¹ <https://github.com/containers/krunvm>

Create the example workload artifact

sliced rock¹⁰⁵²

OCI-compatible¹⁰⁵³

```
$ sudo rockcraft.skopeo --insecure-policy copy  
oci-archive:chiselled-hello_latest_amd64.rock oci:chiselled-hello.oci:latest
```

```
$ sudo apt install oci-image-tool
```

```
$ oci-image-tool unpack --ref name=latest chiselled-hello.oci  
/tmp/chiselled-hello
```

```
$ mkdir /tmp/chiselled-hello/{dev,proc,sys,run,var}
```

```
$ sudo apt install guestfs-tools
```

```
$ sudo virt-make-fs --format=qcow2 --size=50M /tmp/chiselled-hello  
chiselled-hello.qcow2
```

Run the stripped workload in QEMU

```
$ sudo qemu-system-x86_64 -m 128M -machine accel=kvm \ > -kernel  
/boot/vmlinuz-$(uname -r) \ > -append 'console=ttyS0 root=/dev/vda  
fsck.mode=skip init=/usr/bin/hello' \ > -nodefaults -no-user-config \ >  
-display none -serial mon:stdio \ > -drive  
file=chiselled-hello.qcow2,index=0,format=qcow2,media=disk,if=none,id=virtio1  
\ > -device virtio-blk-pci,drive=virtio1
```

¹⁰⁵² <https://documentation.ubuntu.com/rockcraft/stable/tutorial/chisel/>

¹⁰⁵³ <https://github.com/opencontainers/image-spec/blob/main/spec.md>

```
...
[ 2.116207] Run /usr/bin/hello as init process
Hello, world!
```

command-line element	Explanation
sudo	sudo is a simple way for this example to work, but not recommended. Scenarios outside of an example should use separate kernel images and a user that is a member of the kvm group to access /dev/kvm.
qemu-system-x86_64	Call the usual binary of QEMU used for system virtualization.
-m 128M	Allocate 128 megabytes of RAM for the guest.
-machine accel=kvm	Enable KVM.
-kernel /boot/vmlinuz-\$(uname -r)	Load the currently running kernel for the guest.
-append '...'	This passes four arguments to the kernel explained one by one in the following rows.
con-sole=ttyS0	Tells the kernel which serial console it should send its output to.
root=/dev/vda	Informs it where to expect the root device matching the virtio-block device we provide.
fsck.mode=skip	Instructs the kernel to skip filesystem checks, which saves time.
init=/usr/bin/hello	Tell the kernel to directly start our test workload.
-nodefaults	Do not create the default set of devices.
-no-user-config	Do not load any user provided config files.
-display none	Disable video output (due to -nodefaults and -display none we do not also need -nographic).
-serial mon:stdio	Map the virtual serial port and the monitor (for debugging) to stdio.
-drive ... -device ...	Provide our test image as virtio-based block device.

Using qboot and microvm

SeaBIOS¹⁰⁵⁵

UEFI¹⁰⁵⁶

qboot¹⁰⁵⁴

QEMU microvm¹⁰⁵⁷

Firecracker¹⁰⁵⁸

microvm¹⁰⁵⁹

```
$ sudo qemu-system-x86_64-microvm -m 128M -machine accel=kvm,rtc=on \ > -bios
/usr/share/qemu/qboot.rom \ > -kernel /boot/vmlinuz-$(uname -r) \ > -append
'console=ttyS0 root=/dev/vda fsck.mode=skip init=/usr/bin/hello' \ >
-nodefaults -no-user-config \ > -display none -serial mon:stdio \ > -drive
file=chiselled-hello.qcow2,index=0,format=qcow2,media=disk,if=none,id=virtio1
-device virtio-blk-device,drive=virtio1
```

¹⁰⁵⁴ <https://github.com/bonzini/qboot>

¹⁰⁵⁵ <https://www.seabios.org>

¹⁰⁵⁶ <https://github.com/tianocore/edk2>

¹⁰⁵⁷ <https://github.com/qemu/qemu/blob/master/docs/system/i386/microvm.rst>

¹⁰⁵⁸ <https://firecracker-microvm.github.io/>

¹⁰⁵⁹ <https://github.com/qemu/qemu/blob/master/docs/system/i386/microvm.rst#running-a-microvm-based-vm>

command- line ele- ment	Explanation
qemu- system- x86_64- microvm	Call the lighter, feature-reduced QEMU binary.
-bios /usr/ share/ qemu/ qboot. rom	Running QEMU as <code>qemu-system-x86_64-microvm</code> will auto-select <code>/usr/share/seabios/bios-microvm.bin</code> which is a simplified SeaBIOS for this purpose. But, for the example shown here we want the even simpler <code>qboot</code> , so in addition we set <code>-bios /usr/share/qemu/qboot.rom</code> .
<i>info</i>	QEMU will auto-select the <code>microvm</code> machine type, equivalent to <code>-M microvm</code> which therefore doesn't need to be explicitly included here.
... virtio- blk- device ...	This feature-reduced QEMU only supports <code>virtio-bus</code> , so we need to switch the type <code>virtio-blk-pci</code> to <code>virtio-blk-device</code> .

Sadly, polluting this nice showcase there is currently an issue with the RTC initialisation not working in this mode - which makes the guest kernel wait $\sim 1.3 + \sim 1.4$ seconds. See this [QEMU bug¹⁰⁶⁰](https://bugs.launchpad.net/qemu/bugs/2074073) if you are curious about that.

But these changes were not about making the guest faster once it runs, instead it mostly is about the initialisation time (and kernel init by having less virtual hardware). And that we can check despite this issue.

¹⁰⁶⁰ <https://bugs.launchpad.net/qemu/bugs/2074073>

Alternative - using virtiofs

```
$ sudo rockcraft.skopeo --insecure-policy copy  
oci-archive:chiselled-hello_latest_amd64.rock oci:chiselled-hello.oci:latest
```

```
$ sudo apt install oci-image-tool
```

```
$ oci-image-tool unpack --ref name=latest chiselled-hello.oci  
/tmp/chiselled-hello
```

```
$ sudo apt install virtiofsd > /usr/libexec/virtiofsd  
--socket-path=/tmp/vfsd.sock --shared-dir /tmp/chiselled-hello  
...  
[INFO virtiofsd] Waiting for vhost-user socket connection...
```

```
...  
-object memory-backend-memfd,id=mem,share=on,size=128M \  
-numa node,memdev=mem -chardev socket,id=char0,path=/tmp/vfsd.sock \  
-device vhost-user-fs-pci,queue-size=1024,chardev=char0,tag=myfs  
...
```

Live migration with libvirt/QEMU

•

-

-

-

Versioned machine types

QEMU Machine types

CPU baseline and named features

```
-cpu EPYC-Turin,x2apic=on,tsc-deadline=on,hypervisor=on,tsc-adjust=on,  
spec-ctrl=on,stibp=on,flush-l1d=on,ssbd=on,cmp-legacy=on,virt-ssbd=on,  
tsa-sq-no=on,tsa-l1-no=on,pcid=off,la57=off
```

```
...  
<cpu mode='host-model' check='partial' />  
...
```

```
...  
<cpu mode='custom' match='exact' check='full'>  
  <model fallback='forbid'>EPYC-Turin</model>  
  <vendor>AMD</vendor>  
  <feature policy='require' name='x2apic' />  
  <feature policy='require' name='tsc-deadline' />  
  ...  
  <feature policy='require' name='tsa-l1-no' />  
  <feature policy='disable' name='pcid' />  
  <feature policy='disable' name='la57' />  
...
```

Determining the lowest common CPU denominator

*virsh(1)*¹⁰⁶¹

```
virsh capabilities > host1-caps.xml
```

¹⁰⁶¹ <https://manpages.ubuntu.com/manpages/resolute/man1/virsh.1.html>

```
virsh hypervisor-cpu-baseline --migratable host1-caps.xml host2-caps.xml
```

```
<cpu mode='custom' match='exact'>
  <model fallback='forbid'>EPYC-Turin</model>
  <vendor>AMD</vendor>
  <feature policy='require' name='x2apic' />
  <feature policy='require' name='tsc-deadline' />
  ...
</cpu>
```

Further reading

- QEMU documentation on migration¹⁰⁶²
- libvirt documentation on migration¹⁰⁶³

QEMU Machine types

```
...
pc-q35-8.0      Standard PC (Q35 + ICH9, 2009)
pc-q35-8.1      Standard PC (Q35 + ICH9, 2009)
pc-q35-8.2      Standard PC (Q35 + ICH9, 2009)
pc-q35-9.0      Standard PC (Q35 + ICH9, 2009)
pc-q35-9.1      Standard PC (Q35 + ICH9, 2009)
pc-q35-9.2      Standard PC (Q35 + ICH9, 2009)
pc-q35-10.0     Standard PC (Q35 + ICH9, 2009)
pc-q35-10.1     Standard PC (Q35 + ICH9, 2009)
pc-q35-10.2     Standard PC (Q35 + ICH9, 2009)
```

¹⁰⁶² <https://www.qemu.org/docs/master/devel/migration/main.html>

¹⁰⁶³ <https://libvirt.org/migration.html>

Ubuntu-specific machine types

Launchpad bug¹⁰⁶⁴

```
$ root@resolute:~# qemu-system-x86_64 -machine help | grep resolute > ubuntu
Ubuntu 26.04 PC (i440FX + PIIX, 1996) (alias of pc-i440fx-resolute) >
pc-i440fx-resolute Ubuntu 26.04 PC (i440FX + PIIX, 1996) (default) >
ubuntu-q35 Ubuntu 26.04 PC (Q35 + ICH9, 2009) (alias of pc-q35-resolute) >
pc-q35-resolute Ubuntu 26.04 PC (Q35 + ICH9, 2009)
```

```
$MACHINE_TYPE-$DISTRO-RELEASE[-v*]
```

arch-capabilities bug¹⁰⁶⁵

```
pc-i440fx-questing-v2 Ubuntu 25.10 PC v2 (i440FX + PIIX, + 10.1 machine, 1996)
pc-i440fx-noble-v2     Ubuntu 24.04 PC v2 (i440FX + PIIX, arch-caps fix, 1996)
pc-q35-noble-v2       Ubuntu 24.04 PC v2 (Q35 + ICH9, arch-caps fix, 2009)
```

Machine type availability

-

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¹⁰⁶⁴ <https://bugs.launchpad.net/ubuntu/+source/qemu/+bug/1294823>

¹⁰⁶⁵ <https://bugs.launchpad.net/bugs/2131822>

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Upgrading the machine type of your VM

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How does this work?

Backing up guest definitions

Upgrade the machine type

Shut down the VM

```
$ virsh shutdown <your_machine>
```

```
$ virsh list --inactive
```

Edit the guest definition

```
$ virsh edit <your_machine>
```

```
...  
<type arch='x86_64' machine='pc-i440fx-resolute'>hvm</type>  
...
```

```
$ kvm -M ?  
  
pc-i440fx-resolute  Ubuntu 26.04 PC (i440FX + PIIX, 1996)  
...  
pc-i440fx-stonking  Ubuntu 26.10 PC (i440FX + PIIX, 1996) (default)  
...
```

Restart the guest

```
$ virsh start <your_machine> > # check from host, via dumping the active xml  
definition > virsh dumpxml <your_machine> | xmllint --xpath  
"string(//domain/os/type/@machine)" - > # or from the guest via dmidecode (if  
supported) > sudo dmidecode | grep Product -A 1 > Product Name: Standard PC
```

```
(i440FX + PIIX, 1996) > Version: pc-i440fx-bionic
```

Containers

- *Container tools overview*
- *About rock images*
- *Docker storage, networking, and logging*

Container tools in the Ubuntu space

LXC

Linux Containers, or LXC¹⁰⁶⁶

started¹⁰⁶⁸

LXC Introduction¹⁰⁶⁷

LXC Getting

LXD

Linux Containers Daemon, or LXD¹⁰⁶⁹

¹⁰⁶⁶ <https://linuxcontainers.org/>

¹⁰⁶⁷ <https://linuxcontainers.org/lxc/introduction/>

¹⁰⁶⁸ <https://linuxcontainers.org/lxc/getting-started/>

¹⁰⁶⁹ <https://canonical.com/lxd>

how to get started with LXD
recommend this tutorial¹⁰⁷⁰

chines¹⁰⁷¹

used to create virtual ma-

Docker

guide for sysadmins

Docker

runc

runc¹⁰⁷²

Open Container Initiative (OCI) runtime specification¹⁰⁷³

crun¹⁰⁷⁴

¹⁰⁷⁰ https://canonical.com/lxd/docs/latest/tutorial/first_steps/

¹⁰⁷¹ https://canonical.com/lxd/docs/latest/howto/instances_create/#create-a-virtual-machine

¹⁰⁷² <https://github.com/opencontainers/runc>

¹⁰⁷³ <https://github.com/opencontainers/runtime-spec>

¹⁰⁷⁴ <https://github.com/containers/crun>

containerd

containerd¹⁰⁷⁵

Docker storage, networking, and logging

Canonical's Kubernetes platform¹⁰⁷⁶

Skopeo

Skopeo¹⁰⁷⁷

Skopeo guide for sysadmins

About rock images

Ubuntu-based container image¹⁰⁷⁸

OCI-compliant¹⁰⁷⁹
Docker

created using Rockcraft¹⁰⁸⁰

uses Chisel¹⁰⁸¹

our how-to guide
official

Rockcraft documentation¹⁰⁸²

¹⁰⁷⁵ <https://containerd.io/>

¹⁰⁷⁶ <https://canonical.com/kubernetes>

¹⁰⁷⁷ <https://github.com/containerd/skopeo>

¹⁰⁷⁸ <https://documentation.ubuntu.com/rockcraft/latest/explanation/rocks/>

¹⁰⁷⁹ <https://opencontainers.org/>

¹⁰⁸⁰ <https://documentation.ubuntu.com/rockcraft/latest/explanation/rockcraft/>

¹⁰⁸¹ <https://documentation.ubuntu.com/rockcraft/latest/explanation/chisel/>

¹⁰⁸² <https://documentation.ubuntu.com/rockcraft/latest/>

Docker storage, networking, and logging

Storage

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Volumes

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Bind mounts

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Tmpfs

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- mentation¹⁰⁸³
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Docker docu-

The containerd image store

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-
- stargz¹⁰⁸⁴
- nydus¹⁰⁸⁵
- dragonfly¹⁰⁸⁶

¹⁰⁸³ <https://docs.docker.com/engine/storage/tmpfs/#options-for---tmpfs>

¹⁰⁸⁴ <https://github.com/containerd/stargz-snapshotter>

¹⁰⁸⁵ <https://github.com/containerd/nydus-snapshotter>

¹⁰⁸⁶ <https://github.com/dragonflyoss/dragonfly>

storage drivers¹⁰⁸⁷

```
{  
  "features": {  
    "containerd-snapshotter": true  
  }  
}
```

Choosing a snapshotter

```
$ ctr plugins ls | grep snapshot
```

Docker for System Administrators

```
{  
  "storage-driver": "your-chosen-snapshotter"  
}
```

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¹⁰⁸⁷ <https://docs.docker.com/engine/storage/drivers/>

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deduplica-

Managing disk space

```
version = 2  
root = "/mnt/my-extremely-large-drive"
```

Networking

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Logging

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- **journal**
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- Logentries¹⁰⁸⁸
- **gelf** *Graylog*
Graylog¹⁰⁸⁹ Logstash¹⁰⁹⁰ Fluentd¹⁰⁹¹
- Amazon CloudWatch Logs¹⁰⁹²
-
- **fluentd** *Fluentd*¹⁰⁹³
- **gcplogs** Google Cloud Logging¹⁰⁹⁴
- HTTP Event Collector¹⁰⁹⁵

Resources

- *Docker for system admins*

¹⁰⁸⁸ <https://www.rapid7.com/products/siem/>

¹⁰⁸⁹ <https://graylog.org/>

¹⁰⁹⁰ <https://www.elastic.co/logstash>

¹⁰⁹¹ <https://www.fluentd.org>

¹⁰⁹² <https://aws.amazon.com/cloudwatch/features/>

¹⁰⁹³ <https://www.fluentd.org>

¹⁰⁹⁴ <https://docs.cloud.google.com/logging/docs>

¹⁰⁹⁵ <https://dev.splunk.com/enterprise/docs/devtools/httpeventcollector/>

Other virtualisation tools

- *About OpenStack*

About OpenStack

QEMU/KVM

libvirt

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OpenStack components¹⁰⁹⁶

OpenStack installation

Canonical

OpenStack¹⁰⁹⁷

- OpenStack Charms¹⁰⁹⁸

¹⁰⁹⁶ <https://www.openstack.org/software/project-navigator/openstack-components#openstack-services>

¹⁰⁹⁷ <https://canonical-openstack.readthedocs-hosted.com/en/latest/tutorial/get-started-with-openstack/>

¹⁰⁹⁸ <https://docs.openstack.org/project-deploy-guide/charm-deployment-guide/latest/>

- [OpenStack Ansible](#)¹⁰⁹⁹
- [Manual Installation](#)¹¹⁰⁰
- [DevStack](#)¹¹⁰¹

See also

- [Virtualisation](#)
- [Containers](#)

clouds section

Clouds

Cloud-init

Introduction to cloud-init

Cloud-init¹¹⁰²

¹⁰⁹⁹ <https://docs.openstack.org/project-deploy-guide/openstack-ansible/latest/>

¹¹⁰⁰ <https://docs.openstack.org/install-guide/>

¹¹⁰¹ <https://docs.openstack.org/devstack/latest/>

¹¹⁰² <https://docs.cloud-init.io/en/latest/>

How does it work?

hostname

Related products

source¹¹⁰³

the data-

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-
-

check out their tutorials¹¹⁰⁴

Cloud images

- *Cloud images overview*

Public Cloud documentation¹¹⁰⁵

- Amazon EC2¹¹⁰⁶ Google Compute Engine (GCE)¹¹⁰⁷ Microsoft Azure¹¹⁰⁸

¹¹⁰³ <https://docs.cloud-init.io/en/latest/reference/datasources.html>

¹¹⁰⁴ <https://docs.cloud-init.io/en/latest/tutorial/index.html>

¹¹⁰⁵ <https://documentation.ubuntu.com/public-cloud/>

¹¹⁰⁶ <https://documentation.ubuntu.com/aws/aws-how-to/instances/find-ubuntu-images/>

¹¹⁰⁷ <https://documentation.ubuntu.com/gcp/google-how-to/gce/find-ubuntu-images/>

¹¹⁰⁸ <https://documentation.ubuntu.com/azure/azure-how-to/instances/find-ubuntu-images/>

Cloud images

<https://cloud-images.ubuntu.com>

public-cloud documentation.¹¹⁰⁹

Public clouds

Compute offerings

- Amazon Elastic Compute Cloud (EC2)¹¹¹⁰
- Google Compute Engine (GCE)¹¹¹¹
- IBM Cloud¹¹¹²
- Microsoft Azure¹¹¹³
- Oracle Cloud¹¹¹⁴

Container offerings

- Amazon Elastic Kubernetes Service (EKS)¹¹¹⁵
- [release of GKE](#)¹¹¹⁶

Private clouds

cloud-images.ubuntu.com¹¹¹⁷

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¹¹⁰⁹ <https://documentation.ubuntu.com/public-cloud/>

¹¹¹⁰ <https://documentation.ubuntu.com/aws/aws-how-to/instances/find-ubuntu-images/>

¹¹¹¹ <https://documentation.ubuntu.com/gcp/google-how-to/gce/find-ubuntu-images/>

¹¹¹² <https://canonical-ibm.readthedocs-hosted.com/ibm-how-to/find-ubuntu-images/>

¹¹¹³ <https://documentation.ubuntu.com/azure/azure-how-to/instances/find-ubuntu-images/>

¹¹¹⁴ <https://documentation.ubuntu.com/oracle/oracle-how-to/find-ubuntu-images/>

¹¹¹⁵ <https://cloud-images.ubuntu.com/docs/aws/eks/>

¹¹¹⁶ <https://docs.cloud.google.com/kubernetes-engine/docs/release-schedule#schedule-for-release-channels>

¹¹¹⁷ <https://cloud-images.ubuntu.com>

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Release support

lifecycle of an Ubuntu release¹¹¹⁸

High Availability

High Availability

Introduction

- *Introduction to High Availability*

Key concepts

- *Pacemaker resource agents*
- *Pacemaker fence agents*

Introduction to High Availability

from Wikipedia:¹¹¹⁹

¹¹¹⁸ <https://ubuntu.com/about/release-cycle>

¹¹¹⁹ https://en.wikipedia.org/wiki/High-availability_cluster

High Availability clusters

High-availability clusters (also known as **HA clusters** , **failover clusters** or **Metroclusters Active/Active**) are groups of **computers**¹¹²⁰ that support **server**¹¹²¹ **applications**¹¹²² that can be reliably utilised with a **minimum amount of down-time**¹¹²³. They operate by using **high availability software**¹¹²⁴ to harness **redundant**¹¹²⁵ computers in groups or **clusters**¹¹²⁶ that provide continued service when system components fail. Without clustering, if a server running a particular application crashes, the application will be unavailable until the crashed server is fixed. HA clustering remedies this situation by detecting hardware/software faults, and immediately restarting the application on another system without requiring administrative intervention, a process known as **failover**¹¹²⁷. As part of this process, clustering software may configure the node before starting the application on it. For example, appropriate file systems may need to be imported and mounted, network hardware may have to be configured, and some supporting applications may need to be running as well.

HA clusters are often used for critical **databases**¹¹²⁸, file sharing on a network, business applications, and customer services such as **electronic commerce**¹¹²⁹ **websites**¹¹³⁰.

¹¹²⁰ <https://en.wikipedia.org/wiki/Computer>

¹¹²¹ [https://en.wikipedia.org/wiki/Server_\(computing\)](https://en.wikipedia.org/wiki/Server_(computing))

¹¹²² https://en.wikipedia.org/wiki/Application_software

¹¹²³ https://en.wikipedia.org/wiki/High_availability

¹¹²⁴ https://en.wikipedia.org/wiki/High_availability_software

¹¹²⁵ [https://en.wikipedia.org/wiki/Redundancy_\(engineering\)](https://en.wikipedia.org/wiki/Redundancy_(engineering))

¹¹²⁶ https://en.wikipedia.org/wiki/Computer_cluster

¹¹²⁷ <https://en.wikipedia.org/wiki/Failover>

¹¹²⁸ https://en.wikipedia.org/wiki/Database_management_system

¹¹²⁹ https://en.wikipedia.org/wiki/Electronic_commerce

¹¹³⁰ <https://en.wikipedia.org/wiki/Websites>

High Availability cluster heartbeat

HA cluster implementations attempt to build redundancy into a cluster to eliminate single points of failure, including multiple network connections and data storage which is redundantly connected via **storage area networks**¹¹³¹.

HA clusters usually use a **heartbeat**¹¹³² private network connection which is used to monitor the health and status of each node in the cluster. One subtle but serious condition all clustering software must be able to handle is **split-brain**¹¹³³, which occurs when all of the private links go down simultaneously, but the cluster nodes are still running. If that happens, each node in the cluster may mistakenly decide that every other node has gone down and attempt to start services that other nodes are still running. Having duplicate instances of services may cause data corruption on the shared storage.

¹¹³¹ https://en.wikipedia.org/wiki/Storage_area_network

¹¹³² [https://en.wikipedia.org/wiki/Heartbeat_\(computing\)](https://en.wikipedia.org/wiki/Heartbeat_(computing))

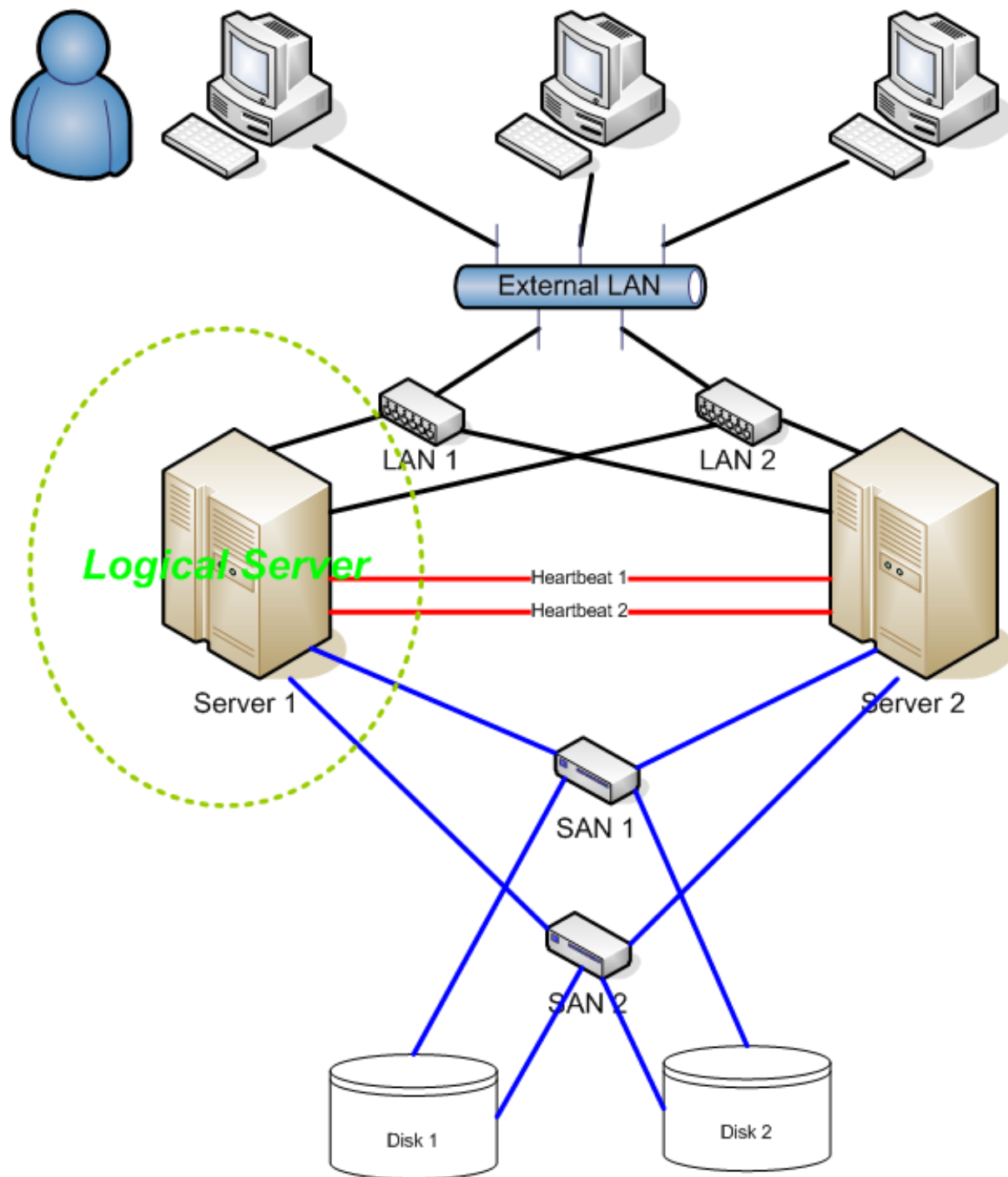
¹¹³³ [https://en.wikipedia.org/wiki/Split-brain_\(computing\)](https://en.wikipedia.org/wiki/Split-brain_(computing))

High Availability cluster quorum

HA clusters often also use **quorum**¹¹³⁴ witness storage (local or cloud) to avoid this scenario. A witness device cannot be shared between two halves of a split cluster, so in the event that all cluster members cannot communicate with each other (e.g., failed heartbeat), if a member cannot access the witness, it cannot become active.

¹¹³⁴ [https://en.wikipedia.org/wiki/Quorum_\(distributed_computing\)](https://en.wikipedia.org/wiki/Quorum_(distributed_computing))

Example of HA cluster quorum



Fencing

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- *GFS2*

Linux High Availability projects

Main Ubuntu HA packages

Package	URL
libqb	libqb¹¹³⁵ Upstream¹¹³⁶
kronosnet	kronosnet¹¹³⁷ Upstream¹¹³⁸
corosync	corosync¹¹³⁹ Upstream¹¹⁴⁰
pacemaker	pacemaker¹¹⁴¹ Upstream¹¹⁴²
resource-agents	resource-agents¹¹⁴³ Upstream¹¹⁴⁴
fence-agents	fence-agents¹¹⁴⁵ Upstream¹¹⁴⁶
crmsh	crmsh¹¹⁴⁷ Upstream¹¹⁴⁸
pcs*	pcs¹¹⁴⁹ Upstream¹¹⁵⁰
cluster-glue	cluster-glue¹¹⁵¹ Upstream¹¹⁵²
drbd-utils	drbd-utils¹¹⁵³ Upstream¹¹⁵⁴
dlm	dlm¹¹⁵⁵ Upstream¹¹⁵⁶
gfs2-utils	gfs2-utils¹¹⁵⁷ Upstream¹¹⁵⁸
keepalived	keepalived¹¹⁵⁹ Upstream¹¹⁶⁰

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•

¹¹³⁵ <https://launchpad.net/ubuntu/+source/libqb>

¹¹³⁶ <http://clusterlabs.github.io/libqb/>

¹¹³⁷ <https://launchpad.net/ubuntu/+source/kronosnet>

¹¹³⁸ <https://kronosnet.org/>

¹¹³⁹ <https://launchpad.net/ubuntu/+source/corosync>

¹¹⁴⁰ <http://corosync.github.io/corosync/>

¹¹⁴¹ <https://launchpad.net/ubuntu/+source/pacemaker>

¹¹⁴² <https://clusterlabs.org/projects/pacemaker/>

¹¹⁴³ <https://launchpad.net/ubuntu/+source/resource-agents>

¹¹⁴⁴ <https://github.com/ClusterLabs/resource-agents>

¹¹⁴⁵ <https://launchpad.net/ubuntu/+source/fence-agents>

¹¹⁴⁶ <https://github.com/ClusterLabs/fence-agents>

¹¹⁴⁷ <https://launchpad.net/ubuntu/+source/crmsh>

¹¹⁴⁸ <https://github.com/ClusterLabs/crmsh>

¹¹⁴⁹ <https://launchpad.net/ubuntu/+source/pcs>

¹¹⁵⁰ <https://github.com/ClusterLabs/pcs/>

¹¹⁵¹ <https://launchpad.net/ubuntu/+source/cluster-glue>

¹¹⁵² <https://github.com/ClusterLabs/cluster-glue>

¹¹⁵³ <https://launchpad.net/ubuntu/+source/drbd-utils>

¹¹⁵⁴ <https://linbit.com/drbd/>

¹¹⁵⁵ <https://launchpad.net/ubuntu/+source/dlm>

¹¹⁵⁶ <https://pagure.io/dlm>

¹¹⁵⁷ <https://launchpad.net/ubuntu/+source/gfs2-utils>

¹¹⁵⁸ <https://pagure.io/gfs2-utils>

¹¹⁵⁹ <https://launchpad.net/ubuntu/+source/keepalived>

¹¹⁶⁰ <https://www.keepalived.org/>

Totem Redundant Ring Protocol¹¹⁶¹

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- *GNU*

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- age system¹¹⁶²

distributed replicated stor-

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- *filesystem*

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Linux Virtual Server (IPVS)¹¹⁶³

VRRP¹¹⁶⁴

¹¹⁶¹ <https://discourse.ubuntu.com/t/corosync-and-redundant-rings/11627>

¹¹⁶² https://en.wikipedia.org/wiki/Distributed_Replicated_Block_Device

¹¹⁶³ <http://www.linux-vs.org/>

¹¹⁶⁴ <https://datatracker.ietf.org/wg/vrrp/documents/>

Ubuntu HA community packages

Package	URL
pcs*	Ubuntu ¹¹⁶⁵ Upstream ¹¹⁶⁶
csync2	Ubuntu ¹¹⁶⁷ Upstream ¹¹⁶⁸
corosync-qdevice	Ubuntu ¹¹⁶⁹ Upstream ¹¹⁷⁰
fence-virt	Ubuntu ¹¹⁷¹ Upstream ¹¹⁷²
sbd	Ubuntu ¹¹⁷³ Upstream ¹¹⁷⁴
booth	Ubuntu ¹¹⁷⁵ Upstream ¹¹⁷⁶

•

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Note:

pcs was added to the main repository in Ubuntu Lunar Lobster (23.04).

¹¹⁶⁵ <https://launchpad.net/ubuntu/+source/libqb>

¹¹⁶⁶ <https://github.com/ClusterLabs/pcs>

¹¹⁶⁷ <https://launchpad.net/ubuntu/+source/csync2>

¹¹⁶⁸ <https://github.com/LINBIT/csync2>

¹¹⁶⁹ <https://launchpad.net/ubuntu/+source/corosync-qdevice>

¹¹⁷⁰ <https://github.com/corosync/corosync-qdevice>

¹¹⁷¹ <https://launchpad.net/ubuntu/+source/fence-virt>

¹¹⁷² <https://github.com/ClusterLabs/fence-virt>

¹¹⁷³ <https://launchpad.net/ubuntu/+source/sbd>

¹¹⁷⁴ <https://github.com/ClusterLabs/sbd>

¹¹⁷⁵ <https://launchpad.net/ubuntu/+source/booth>

¹¹⁷⁶ <https://github.com/ClusterLabs/booth>

Ubuntu HA deprecated packages

Package	URL
ocfs2-tools	Ubuntu¹¹⁷⁷ Upstream¹¹⁷⁸

Ubuntu HA related packages

Package	URL
multipath-tools	Ubuntu¹¹⁷⁹ Upstream¹¹⁸⁰
open-iscsi	Ubuntu¹¹⁸¹ Upstream¹¹⁸²
sg3-utils	Ubuntu¹¹⁸³ Upstream¹¹⁸⁴
tgt OR targetcli-fb*	Ubuntu¹¹⁸⁵ Upstream¹¹⁸⁶
lvm2	Ubuntu¹¹⁸⁷ Upstream¹¹⁸⁸

•

LVM

lvmlockd(8)¹¹⁸⁹

¹¹⁷⁷ <https://launchpad.net/ubuntu/+source/ocfs2-tools>

¹¹⁷⁸ <https://github.com/markfasheh/ocfs2-tools>

¹¹⁷⁹ <https://launchpad.net/ubuntu/+source/multipath-tools>

¹¹⁸⁰ <https://github.com/opensvc/multipath-tools>

¹¹⁸¹ <https://launchpad.net/ubuntu/+source/open-iscsi>

¹¹⁸² <https://github.com/open-iscsi/open-iscsi>

¹¹⁸³ <https://launchpad.net/ubuntu/+source/sg3-utils>

¹¹⁸⁴ http://sg.danny.cz/sg/sg3_utils.html

¹¹⁸⁵ <https://launchpad.net/ubuntu/+source/tgt>

¹¹⁸⁶ <https://github.com/fujita/tgt>

¹¹⁸⁷ <https://launchpad.net/ubuntu/+source/lvm2>

¹¹⁸⁸ <https://sourceware.org/lvm2/>

¹¹⁸⁹ <https://manpages.ubuntu.com/manpages/resolute/man8/lvmlockd.8.html>

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Note:

`targetcli - fb` (Linux LIO) will likely replace `tgt` in future Ubuntu versions.

Upstream documentation

- - [Clusters From Scratch](#)¹¹⁹⁰
 - [Managing Pacemaker Clusters](#)¹¹⁹¹
 - [Pacemaker Configuration Explained](#)¹¹⁹²
- - [Ubuntu Bionic HA in Shared Disk Environments \(Azure\)](#)¹¹⁹³

A very special thanks and all the credit to [ClusterLabs Project](#)¹¹⁹⁴ for their detailed documentation.

¹¹⁹⁴ <https://clusterlabs.org/>

Pacemaker resource agents

Resource agents are the abstraction that allows Pacemaker to manage services it knows nothing about. They contain the logic for what to do when the cluster wishes to start, stop or check the health of a service. This particular set of agents conform to the Open Cluster Framework (OCF) specification.

¹¹⁹⁰ https://clusterlabs.org/projects/pacemaker/doc/3.0/Clusters_from_Scratch/html/

¹¹⁹¹ https://clusterlabs.org/projects/pacemaker/doc/3.0/Pacemaker_Administration/html/

¹¹⁹² https://clusterlabs.org/projects/pacemaker/doc/3.0/Pacemaker_Explained/html/

¹¹⁹³ <https://discourse.ubuntu.com/t/ubuntu-high-availability-corosync-pacemaker-shared-disk-environments/>

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Note:

There are two well known tools used to manage fence agents, they are `crmsh` and `pcs`. Here we will present examples with both, since `crmsh` is the recommended and supported tool until Ubuntu 22.10 Kinetic Kudu, and `pcs` is the recommended and supported tool from Ubuntu 23.04 Lunar Lobster onwards. For more information on how to migrate from `crmsh` to `pcs` [refer to this migration guide](#) (page 851).

IPaddr2

This Linux-specific resource manages IP alias IP addresses. It can add an IP alias, or remove one. In addition, it can implement Cluster Alias IP functionality if invoked as a clone resource.

```
$ crm configure primitive $RESOURCE_NAME ocf:heartbeat:IPaddr2 \ >  
ip=$IP_ADDRESS \ > cidr_netmask=$NET_MASK \ > op monitor interval=30s
```

```
$ pcs resource create $RESOURCE_NAME ocf:heartbeat:IPaddr2 \ > ip=$IP_ADDRESS  
\ > cidr_netmask=$NET_MASK \ > op monitor interval=30s
```

iscsi

Manages a local iSCSI initiator and its connections to iSCSI targets.

```
$ crm configure primitive $RESOURCE_NAME ocf:heartbeat:iscsi \ >  
target=$TARGET \ > portal=$PORTAL
```

```
$ pcs resource create $RESOURCE_NAME ocf:heartbeat:iscsi \ > target=$TARGET \  
> portal=$PORTAL
```

*to its manual page*¹¹⁹⁶

iSCSILogicalUnit

Manages iSCSI Logical Unit. An iSCSI Logical unit is a subdivision of an SCSI Target, exported via a daemon that speaks the iSCSI protocol.

```
$ crm configure primitive $RESOURCE_NAME ocf:heartbeat:iSCSILogicalUnit \ >  
implementation=lio-t \ > target_iqn=$IQN_TARGET \ > path=$DEVICE \ > lun=$LUN
```

```
$ pcs resource create $RESOURCE_NAME ocf:heartbeat:iSCSILogicalUnit \ >  
implementation=lio-t \ > target_iqn=$IQN_TARGET \ > path=$DEVICE \ > lun=$LUN
```

¹¹⁹⁵ https://manpages.ubuntu.com/manpages/resolute/man7/ocf_heartbeat_IPaddr2.7.html

¹¹⁹⁶ https://manpages.ubuntu.com/manpages/resolute/man7/ocf_heartbeat_iscsi.7.html

page¹¹⁹⁷

to its manual

iSCSITarget

Manages iSCSI targets. An iSCSI target is a collection of SCSI Logical Units (LUs) exported via a daemon that speaks the iSCSI protocol.

```
$ crm configure primitive $RESOURCE_NAME ocf:heartbeat:iSCSITarget \ >  
implementation=lio-t \ > iqn=$IQN_TARGET
```

```
$ pcs resource create $RESOURCE_NAME ocf:heartbeat:iSCSITarget \ >  
implementation=lio-t \ > iqn=$IQN_TARGET
```

to its manual page¹¹⁹⁸

LVM-activate

This agent manages LVM activation/deactivation work for a given volume group.

```
$ crm configure primitive $RESOURCE_NAME ocf:heartbeat:LVM-activate \ >  
vgname=$VOLUME_GROUP \ > vg_access_mode=system_id
```

¹¹⁹⁷ https://manpages.ubuntu.com/manpages/resolute/man7/ocf_heartbeat_iSCSILogicalUnit.7.html

¹¹⁹⁸ https://manpages.ubuntu.com/manpages/resolute/man7/ocf_heartbeat_iSCSITarget.7.html

```
$ pcs resource create $RESOURCE_NAME ocf:heartbeat:LVM-activate \ >  
vgname=$VOLUME_GROUP \ > vg_access_mode=system_id
```

*to its manual page*¹¹⁹⁹

Systemd

```
$ crm configure primitive $RESOURCE_NAME systemd:$SERVICE_NAME
```

```
$ pcs resource create $RESOURCE_NAME systemd:$SERVICE_NAME
```

Further reading

- [ClusterLabs website](#)¹²⁰⁰
- [The OCF resource-agent developer's guide](#)¹²⁰¹

Pacemaker fence agents

A **fence agent** (or **fencing agent**) is a **stonith**-class resource agent.

The fence agent standard provides commands (such as off and reboot) that the cluster can use to fence nodes. As with other resource agent classes, this allows a layer of abstraction so that Pacemaker doesn't need any knowledge about specific fencing technologies — that knowledge is isolated in the agent.

¹¹⁹⁹ https://manpages.ubuntu.com/manpages/resolute/man7/ocf_heartbeat_LVM-activate.7.html

¹²⁰⁰ <https://clusterlabs.org/>

¹²⁰¹ <https://github.com/ClusterLabs/resource-agents/blob/master/doc/dev-guides/ra-dev-guide.asc>

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Note:

There are two well known tools used to manage fence agents, they are `crmsh` and `pcs`. Here we present examples with both, since `crmsh` is the recommended and supported tool until Ubuntu 22.10 Kinetic Kudu, and `pcs` is the recommended and supported tool from Ubuntu 23.04 Lunar Lobster onwards. For more information on how to migrate from `crmsh` to `pcs` [refer to our migration guide](#) (page 851).

fence_ipmilan

`fence_ipmilan` is an I/O Fencing agent which can be used with machines controlled by IPMI. This agent calls support software `ipmitool`. WARNING! This fence agent might report success before the node is powered off. You should use `-m/method onoff` if your fence device works correctly with that option.

```
$ crm configure primitive $RESOURCE_NAME stonith:fence_ipmilan \ > ip=$IP \ >  
ipport=$PORT \ > username=$USER \ > password=$PASSWD \ > lanplus=1 \ >  
action=$ACTION
```

```
$ pcs stonith create $RESOURCE_NAME fence_ipmilan \ > ip=$IP \ > ipport=$PORT \ > username=$USER \ > password=$PASSWD \ > lanplus=1 \ > action=$ACTION
```

hostname

to its manual page¹²⁰²

fence_mpath

fence_mpath is an I/O fencing agent that uses SCSI-3 persistent reservations to control access multipath devices. Underlying devices must support SCSI-3 persistent reservations (SPC-3 or greater) as well as the “preempt-and-abort” subcommand. The fence_mpath agent works by having a unique key for each node that has to be set in /etc/multipath.conf. Once registered, a single node will become the reservation holder by creating a “write exclusive, registrants only” reservation on the device(s). The result is that only registered nodes may write to the device(s). When a node failure occurs, the fence_mpath agent will remove the key belonging to the failed node from the device(s). The failed node will no longer be able to write to the device(s). A manual reboot is required.

```
$ crm configure primitive $RESOURCE_NAME stonith:fence_mpath \ > pcmk_host_map="$NODE1:$NODE1_RES_KEY;$NODE2:$NODE2_RES_KEY;$NODE3:$NODE3_RES_KEY" \ > pcmk_host_argument=key \ > pcmk_monitor_action=metadata \ > pcmk_reboot_action=off \ > devices=$MPATH_DEVICE \ > meta provides=unfencing
```

```
$ pcs stonith create $RESOURCE_NAME fence_mpath \ > pcmk_host_map="$NODE1:$NODE1_RES_KEY;$NODE2:$NODE2_RES_KEY;$NODE3:$NODE3_RES_KEY" \ > pcmk_host_argument=key \ > pcmk_monitor_action=metadata \ > pcmk_reboot_action=off \ > devices=$MPATH_DEVICE \ > meta provides=unfencing
```

its manual page¹²⁰³

¹²⁰² https://manpages.ubuntu.com/manpages/resolute/man8/fence_ipmilan.8.html

¹²⁰³ https://manpages.ubuntu.com/manpages/resolute/man8/fence_mpath.8.html

fence_sbd

fence_sbd is I/O Fencing agent which can be used in environments where SBD can be used (shared storage).

```
$ crm configure primitive $RESOURCE_NAME stonith:fence_sbd devices=$DEVICE
```

```
$ pcs stonith create $RESOURCE_NAME fence_sbd devices=$DEVICE
```

to its manual page¹²⁰⁴

fence_scsi

fence_scsi is an I/O fencing agent that uses SCSI-3 persistent reservations to control access to shared storage devices. These devices must support SCSI-3 persistent reservations (SPC-3 or greater) as well as the “preempt-and-abort” subcommand. The fence_scsi agent works by having each node in the cluster register a unique key with the SCSI device(s). Reservation key is generated from “node id” (default) or from “node name hash” (RECOMMENDED) by adjusting “key_value” option. Using hash is recommended to prevent issues when removing nodes from cluster without full cluster restart. Once registered, a single node will become the reservation holder by creating a “write exclusive, registrants only” reservation on the device(s). The result is that only registered nodes may write to the device(s). When a node failure occurs, the fence_scsi agent will remove the key belonging to the failed node from the device(s). The failed node will no longer be able to write to the device(s). A manual reboot is required.

```
$ crm configure primitive $RESOURCE_NAME stonith:fence_scsi \ >  
pcmk_host_list="$NODE1 $NODE2 $NODE3" \ > devices=$SCSI_DEVICE \ > meta  
provides=unfencing
```

```
$ pcs stonith create $RESOURCE_NAME fence_scsi \ > pcmk_host_list="$NODE1  
$NODE2 $NODE3" \ > devices=$SCSI_DEVICE \ > meta provides=unfencing
```

¹²⁰⁴ https://manpages.ubuntu.com/manpages/resolute/man8/fence_sbd.8.html

to its manual page¹²⁰⁵

fence_virsh

fence_virsh is an I/O Fencing agent which can be used with the virtual machines managed by libvirt. It logs via SSH to a dom0 and there run virsh command, which does all work. By default, virsh needs root account to do properly work. So you must allow SSH login in your sshd_config.

```
$ crm configure primitive $RESOURCE_NAME stonith:fence_virsh \ >  
ip=$HOST_IP_ADDRESS \ > login=$HOST_USER \ > identity_file=$SSH_KEY \ >  
plug=$NODE \ > ssh=true \ > use_sudo=true
```

```
$ pcs stonith create $RESOURCE_NAME fence_virsh \ > ip=$HOST_IP_ADDRESS \ >  
login=$HOST_USER \ > identity_file=$SSH_KEY \ > plug=$NODE \ > ssh=true \ >  
use_sudo=true
```

to its manual page¹²⁰⁶

```
$ crm configure location fence-$NODE-location $RESOURCE_NAME -inf: $NODE
```

```
$ pcs constraint location $RESOURCE_NAME avoids $NODE
```

Further reading

- [ClusterLabs website¹²⁰⁷](#)
- [Fence agents API documentation¹²⁰⁸](#)

¹²⁰⁵ https://manpages.ubuntu.com/manpages/resolute/man8/fence_scsi.8.html

¹²⁰⁶ https://manpages.ubuntu.com/manpages/resolute/man8/fence_virsh.8.html

¹²⁰⁷ <https://clusterlabs.org/>

¹²⁰⁸ <https://github.com/ClusterLabs/fence-agents/blob/master/doc/FenceAgentAPI.md>

See also

- [Set up a Distributed Replicated Block Device \(DRBD\)](#)
- [High availability](#)

Our system tuning

Performance

Note:

Disclaimer - To tune you need to know your system and workload

Almost all tunable parameters can be good for one and bad for another type of workload or environment. If the system could do it for you, it probably would be the default setting already.

Not even the goal of tuning is the same for everyone; do you want to improve latency, throughput, thermal or work-unit-per-power? But those default settings generally have to aim for a good compromise on all of these aspects, no matter what you will do with your system.

Therefore you have to know your workload, your system and the important metrics of what you want to achieve. The more you know, the more you'll be able to improve your system it to suit your needs by:

- Identifying bottlenecks as unblocking them usually has the biggest impact for your workload
- Identifying where your needs are different from the generic assumptions to change related tunables
- Identifying architectural mismatches between the solution and your needs to allow adapting

- [PGO](#)
- [hwloc and lstopo](#)
- [CPU Power states](#)
- [cpupower](#)
- [with TuneD](#)
- [CPU affinity, NUMA placement, and isolation](#)

Profile-Guided Optimization

Profile-guided optimization¹²⁰⁹

Caveats

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perf and AutoFDO

GCC¹²¹⁰

autofdo¹²¹¹

¹²⁰⁹ https://en.wikipedia.org/wiki/Profile-guided_optimization

¹²¹⁰ <https://gcc.gnu.org>

¹²¹¹ <https://github.com/google/autofdo>

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AMD

for Ubuntu Noble¹²¹²

upstream repository¹²¹³

A simple PGO scenario: openssl speed

Running OpenSSL tests

¹²¹² <https://launchpad.net/~sergiodj/+archive/ubuntu/autofdo>

¹²¹³ <https://github.com/google/autofdo>

```
$ openssl speed -seconds 60 -evp md5 sha512 rsa2048 aes-256-cbc
```

```
The 'numbers' are in 1000s of bytes per second processed.
type          16 bytes    64 bytes    256 bytes    1024 bytes    8192 bytes    16384
bytes
sha512         91713.74k   366632.96k   684349.30k   1060512.22k   1260359.00k
1273277.92k
aes-256-cbc    1266018.36k  1437870.33k  1442743.14k  1449933.84k   1453336.84k
1453484.99k
md5            140978.58k   367562.53k   714094.14k   955267.21k   1060969.81k
1068296.87k
              sign      verify    sign/s      verify/s
rsa 2048 bits 0.000173s 0.000012s  5776.8    86791.9
```

Building OpenSSL for profiling

```
$ file /usr/bin/openssl
```

```
/usr/bin/openssl: ELF 64-bit LSB pie executable, x86-64, version 1 (SYSV),
dynamically linked, interpreter /lib64/ld-linux-x86-64.so.2,
BuildID[sha1]=42a5ac797b6981bd45a9ece3f67646edded56bdb, for GNU/Linux 3.2.0, with
debug_info, not stripped
```

```
$ eu-readelf -S /usr/bin/openssl
```

```
[...]
Section Headers:
[Nr] Name                               Type            Addr            Off             Size             ES Flags
Lk Inf Al
[...]
[31] .debug_abbrev                          PROGBITS        0000000000000000 001bcbf3 000110c4 0
0 0 1
[...]
```

Using perf to obtain profile data

```
$ echo -1 > /proc/sys/kernel/perf_event_paranoid
```

```
$ sudo perf record \> -e br_inst_retired.near_taken \> --branch-any \> --all-cpus \> --output openssl-nonpgo.perfdata \> -- openssl speed -mr -seconds 60 -evp md5 sha512 rsa2048 aes-256-cbc
```

```
$ sudo perf record \> -e 'cpu/event=0xc4,umask=0x0,name=ex_ret_brn_tkn/' \> --branch-any \> --all-cpus \> --output openssl-nonpgo.perfdata \> -- openssl speed -mr -seconds 60 -evp md5 sha512 rsa2048 aes-256-cbc
```

```
$ sudo chown ubuntu:ubuntu openssl-nonpgo.perfdata
```

Converting perfdata to gcov with autotfdo

```
$ create_gcov \> --binary /usr/bin/openssl \> --gcov openssl-nonpgo.gcov \> --profile openssl-nonpgo.perfdata \> --gcov_version 2 \> -use_lbr false
```

```
[WARNING:[...]/perf_reader.cc:1322] Skipping 200 bytes of metadata: HEADER_CPU_
TOPOLOGY
[WARNING:[...]/perf_reader.cc:1069] Skipping unsupported event PERF_RECORD_ID_INDEX
[WARNING:[...]/perf_reader.cc:1069] Skipping unsupported event PERF_RECORD_EVENT_
UPDATE
[WARNING:[...]/perf_reader.cc:1069] Skipping unsupported event PERF_RECORD_CPU_MAP
[WARNING:[...]/perf_reader.cc:1069] Skipping unsupported event UNKNOWN_EVENT_17
[WARNING:[...]/perf_reader.cc:1069] Skipping unsupported event UNKNOWN_EVENT_18
[...] many more lines [...]
```

```
$ dump_gcov openssl-nonpgo.gcov

cpu_get_tb_cpu_state total:320843603 head:8557544
2: 8442523
7: 8442518
7.1: 8442518
7.4: 9357851
8: 9357851
10: 9357854
19: 0
21: 0
[...] many more lines [...]
```

Rebuilding OpenSSL with PGO

```
-fauto-profile=/path/to/openssl-nonpgo.gcov
```

Our results

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- [blog post](#)¹²¹⁴

Alternative scenario: profiling the program using other approaches

[profiled QEMU internally](#)¹²¹⁵

¹²¹⁴ <https://ubuntu.com/blog/profile-guided-optimization-a-case-study>

¹²¹⁵ <https://ubuntu.com/blog/profile-guided-optimization-a-case-study>

```
$ sudo perf record \ > -e br_inst_retired.near_taken \ > --branch-any \ >  
--all-cpus \ > -p 1234 \ > --output myprogram.perfdata  
  
-- sleep 2m
```

Conclusion

hwloc

Note:

System tuning tools are either about better understanding the system's performance, or applying such knowledge to improve it. See our common [system tuning thoughts](#) (page 1078) for some generally applicable considerations about that.

GUI

hwloc-info - hierarchy and details

```
$ hwloc-info
```

```
depth 0:      1 Machine (type #0)
depth 1:      1 Package (type #1)
  depth 2:    1 L3Cache (type #6)
    depth 3:  4 L2Cache (type #5)
      depth 4: 4 L1dCache (type #4)
        depth 5: 4 L1iCache (type #9)
          depth 6: 4 Core (type #2)
            depth 7: 4 PU (type #3)
Special depth -3: 1 NUMANode (type #13)
Special depth -4: 3 Bridge (type #14)
Special depth -5: 4 PCIDev (type #15)
Special depth -6: 3 OSDev (type #16)
```

```
$ hwloc-info -p core:0
```

```
Core L#0
type = Core
full type = Core
logical index = 0
os index = 0
gp index = 2
depth = 6
sibling rank = 0
children = 1
memory children = 0
i/o children = 0
misc children = 0
cpuset = 0x00000001
complete cpuset = 0x00000001
allowed cpuset = 0x00000001
nodeset = 0x00000001
complete nodeset = 0x00000001
allowed nodeset = 0x00000001
symmetric subtree = 1
cpukind = 0
cpukind efficiency = 0
cpukind info FrequencyMaxMHz = 4000
cpukind info FrequencyBaseMHz = 1800
```

*hwloc-info(1)*¹²¹⁶

¹²¹⁶ <https://manpages.ubuntu.com/manpages/resolute/man1/hwloc-info.1.html>

hwloc-ls - system topology in CLI

```
$ hwloc-ls

Machine (31GB total)
  Package L#0
    NUMANode L#0 (P#0 31GB)
      L3 L#0 (8192KB)
        L2 L#0 (256KB) + L1d L#0 (32KB) + L1i L#0 (32KB) + Core L#0 + PU L#0 (P#0)
        L2 L#1 (256KB) + L1d L#1 (32KB) + L1i L#1 (32KB) + Core L#1 + PU L#1 (P#1)
        L2 L#2 (256KB) + L1d L#2 (32KB) + L1i L#2 (32KB) + Core L#2 + PU L#2 (P#2)
        L2 L#3 (256KB) + L1d L#3 (32KB) + L1i L#3 (32KB) + Core L#3 + PU L#3 (P#3)
      HostBridge
        PCI 00:02.0 (VGA)
        PCIBridge
          PCI 04:00.0 (Network)
            Net "wlp4s0"
        PCIBridge
          PCI 40:00.0 (NVMExp)
            Block(Disk) "nvme0n1"
        PCI 00:1f.6 (Ethernet)
          Net "enp0s31f6"
```



```
Package L#0
  NUMANode L#0 (P#0 378GB)
  L3 L#0 (32MB)
    L2 L#0 (1024KB) + L1d L#0 (48KB) + L1i L#0 (32KB) + Core L#0
    PU L#0 (P#0)
    PU L#1 (P#128)
  ...
    L2 L#63 (1024KB) + L1d L#63 (48KB) + L1i L#63 (32KB) + Core L#63
    PU L#126 (P#63)
    PU L#127 (P#191)
  HostBridge
    PCIBridge
      PCI 12:00.0 (NVMEExp)
      Block(Disk) "nvme1n1"
  HostBridge
    PCIBridge
      PCI 51:00.0 (NVMEExp)
      Block(Disk) "nvme0n1"
    PCIBridge
      PCI 52:00.0 (Ethernet)
      Net "enp82s0"
    PCIBridge
      PCI 54:00.0 (VGA)
Package L#1
  NUMANode L#1 (P#1 378GB)
  L3 L#8 (32MB)
    L2 L#64 (1024KB) + L1d L#64 (48KB) + L1i L#64 (32KB) + Core L#64
    PU L#128 (P#64)
    PU L#129 (P#192)
  ...
    L2 L#127 (1024KB) + L1d L#127 (48KB) + L1i L#127 (32KB) + Core L#127
    PU L#254 (P#127)
    PU L#255 (P#255)
  HostBridge
    PCIBridge
      2 x { PCI f2:00.0-1 (SATA) }
  Misc(MemoryModule)
```

hwloc-ls - system topology in GUI

```
$ hwloc-ls
```

Keyboard shortcuts:

Zooming, scrolling and closing:

```

Zoom-in or out ..... + -
Reset scale to default ..... 1
Try to fit scale to window ..... F
Resize window to the drawing ..... r
Toggle auto-resizing of the window .. R
Scroll vertically ..... Up Down PageUp PageDown
Scroll horizontally ..... Left Right Ctrl+PageUp/Down
Scroll to the top-left corner ..... Home
Scroll to the bottom-right corner ... End
Refresh the topology ..... F5
Show this help ..... h H ?
Exit ..... q Q Esc

```

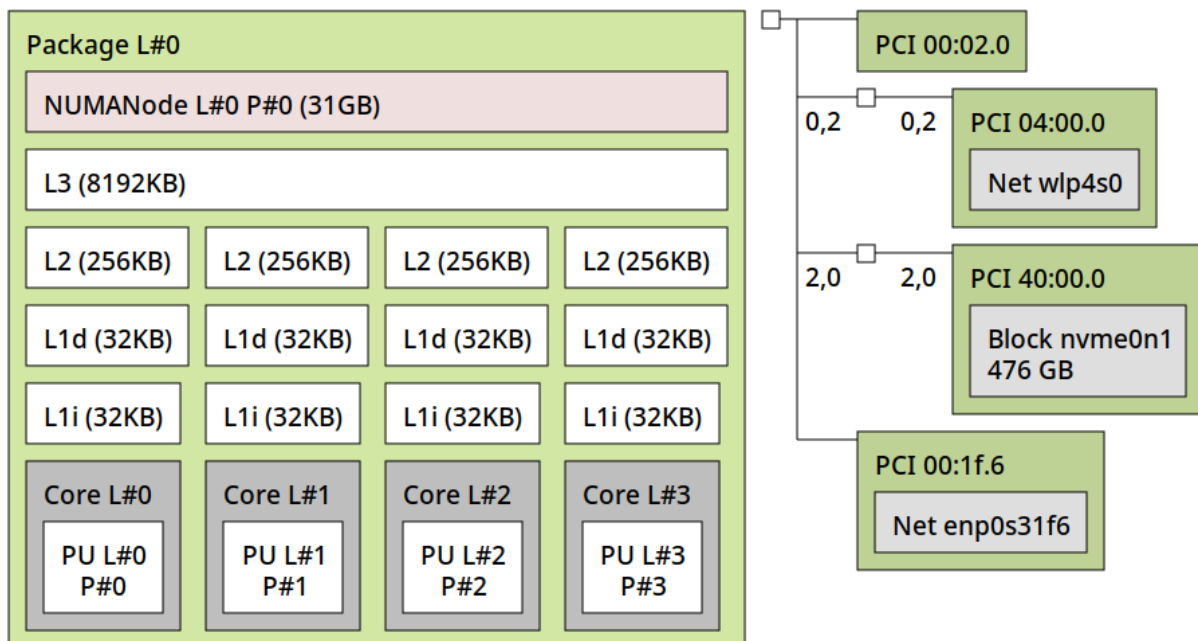
Configuration tweaks:

```

Toggle factorizing or collapsing .... f
Switch display mode for indexes ..... i
Toggle displaying of object text .... t
Toggle displaying of obj attributes . a
Toggle displaying of CPU kinds ..... k
Toggle color for disallowed objects . d
Toggle color for binding objects .... b
Toggle displaying of legend lines ... l
Export to file with current config .. E

```

Machine (31GB total)



Further reading

- `lstopo(1)`¹²¹⁷
- `hwloc(7)`¹²¹⁸
- `hwloc-diff(1)`¹²¹⁹

CPU Power states

Note:

System tuning tools are either about better understanding the system's performance, or applying such knowledge to improve it. See our common *system tuning thoughts* (page 1078) for some generally applicable considerations about that.

Power Management¹²²⁰

The

cpupower tool

¹²¹⁷ <https://manpages.ubuntu.com/manpages/resolute/man1/lstopo.1.html>

¹²¹⁸ <https://manpages.ubuntu.com/manpages/resolute/man7/hwloc.7.html>

¹²¹⁹ <https://manpages.ubuntu.com/manpages/resolute/man1/hwloc-diff.1.html>

¹²²⁰ <https://docs.kernel.org/admin-guide/pm/working-state.html>

CPU governors

Governor	Design philosophy
ondemand	This sets the CPU frequency depending on the current system load. This behavior is usually a good balance between the more extreme options.
conservative	Similar to ondemand, but adapting CPU speed more gracefully rather than jumping to max speed the moment there is any load on the CPU. This behavior is more suitable in a battery-powered environment.
performance	This sets the CPU statically to the highest frequency. This behavior is best to optimize for speed and latency, but might waste power if being under-used.
powersave	Sets the CPU statically to the lowest frequency, essentially locking it to P2. This behavior is suitable to save power without compromises.
userspace	Allows a user-space program to control the CPU frequency.

[Linux Governors Documentation](#)¹²²¹

Setting CPU Governors

```
$ cat /sys/devices/system/cpu/cpu0/cpufreq/scaling_governor
```

-
-

```
$ cat /sys/devices/system/cpu/cpu0/cpufreq/scaling_available_governors
```

¹²²¹ <https://www.kernel.org/doc/Documentation/cpu-freq/governors.txt>

```
$ echo performance | sudo tee  
/sys/devices/system/cpu/cpu0/cpufreq/scaling_governor
```

Energy Performance Preference

Caveats

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- *BIOS*

EPP values on Ubuntu

```
$ cat /sys/devices/system/cpu/cpu0/cpufreq/energy_performance_preference
```

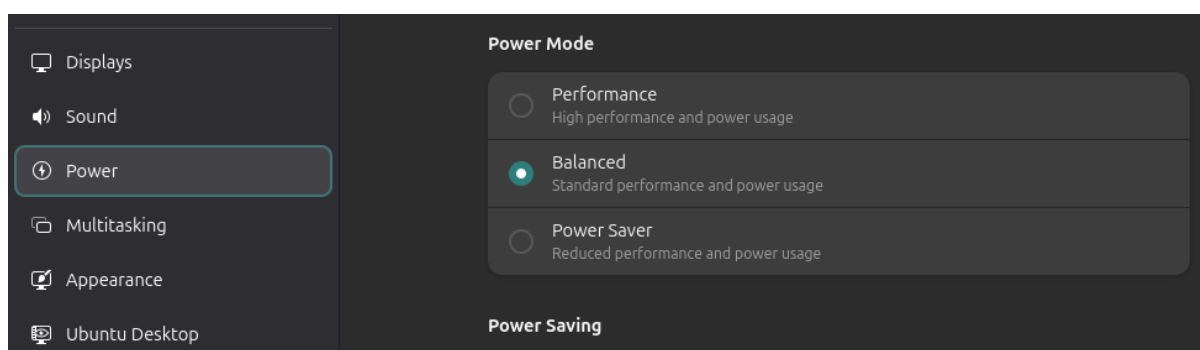
-
-
-
-
-

```
$ cat
/sys/devices/system/cpu/cpu0/cpufreq/energy_performance_available_preferences

default performance balance_performance balance_power power
```

Setting EPP values

```
$ echo performance | sudo tee
/sys/devices/system/cpu/cpu0/cpufreq/energy_performance_preference
```



Power Mode setting	EPP profile
Performance	performance
Balanced	balance_performance
Power Saver	power

The cpupower tool

Note:

System tuning tools are either about better understanding the system's performance, or applying such knowledge to improve it. See our common [system tuning thoughts](#) (page 1078) for some generally applicable considerations about that.

CPU Power states

Monitor CPU frequency

```
$ sudo cpupower monitor -l

Monitor "Nehalem" (4 states) - Might overflow after 922000000 s
C3      [C] -> Processor Core C3
C6      [C] -> Processor Core C6
PC3     [P] -> Processor Package C3
PC6     [P] -> Processor Package C6
Monitor "Mperf" (3 states) - Might overflow after 922000000 s
C0      [T] -> Processor Core not idle
Cx      [T] -> Processor Core in an idle state
Freq    [T] -> Average Frequency (including boost) in MHz
Monitor "RAPL" (4 states) - Might overflow after 8640000 s
pack    [M] ->
dram    [M] ->
core    [M] ->
unco    [M] ->
Monitor "Idle_Stats" (9 states) - Might overflow after 4294967295 s
POLL    [T] -> CPUIDLE CORE POLL IDLE
C1      [T] -> MWAIT 0x00
C1E     [T] -> MWAIT 0x01
C3      [T] -> MWAIT 0x10
C6      [T] -> MWAIT 0x20
C7s     [T] -> MWAIT 0x33
C8      [T] -> MWAIT 0x40
C9      [T] -> MWAIT 0x50
C10     [T] -> MWAIT 0x60
```

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```
$ sudo cpupower monitor -i 60 -m Mperf,RAPL
```

	Mperf			RAPL			
CPU	C0	Cx	Freq	pack	dram	core	unco
0	61,83	38,17	1850	616950936	145911797	375373063	71556823
1	62,03	37,97	1848	616950936	145911797	375373063	71556823
2	65,51	34,49	1852	616950936	145911797	375373063	71556823
3	62,04	37,96	1852	616950936	145911797	375373063	71556823

Get details about the boundaries for the CPU frequency

```
$ cpupower frequency-info
```



```
analyzing CPU 8:
  driver: amd-pstate-epp
  CPUs which run at the same hardware frequency: 8
  CPUs which need to have their frequency coordinated by software: 8
  energy performance preference: balance_performance
  hardware limits: 400 MHz - 5.14 GHz
  available cpufreq governors: performance powersave
  current policy: frequency should be within 1.10 GHz and 5.14 GHz.
                   The governor "powersave" may decide which speed to use
                   within this range.
  current CPU frequency: 2.63 GHz (asserted by call to kernel)
  boost state support:
    Supported: yes
    Active: yes
  amd-pstate limits:
    Highest Performance: 196. Maximum Frequency: 5.14 GHz.
    Nominal Performance: 145. Nominal Frequency: 3.80 GHz.
    Lowest Non-linear Performance: 42. Lowest Non-linear Frequency: 1.10 GHz.
    Lowest Performance: 16. Lowest Frequency: 400 MHz.
    Preferred Core Support: 1. Preferred Core Ranking: 220.
```

Get details about the idle states

Idle states¹²²²

```
$ cpupower idle-info
```

¹²²² <https://docs.kernel.org/admin-guide/pm/cpuidle.html>

CPUidle driver: intel_idle

CPUidle governor: menu

analyzing CPU 0:

Number of idle states: 9

Available idle states: POLL C1 C1E C3 C6 C7s C8 C9 C10

POLL:

Flags/Description: CPUIDLE CORE POLL IDLE

Latency: 0

Usage: 26053898

Duration: 695768311

C1:

Flags/Description: MWAIT 0x00

Latency: 2

Usage: 263751626

Duration: 21296361635

C1E:

Flags/Description: MWAIT 0x01

Latency: 10

Usage: 1071864698

Duration: 122465703132

C3:

Flags/Description: MWAIT 0x10

Latency: 70

Usage: 941753727

Duration: 117177626397

C6:

Flags/Description: MWAIT 0x20

Latency: 85

Usage: 2580936435

Duration: 1258804567087

C7s:

Flags/Description: MWAIT 0x33

Latency: 124

Usage: 2946723

Duration: 1783856599

C8:

Flags/Description: MWAIT 0x40

Latency: 200

Usage: 1580297534

Duration: 1234136981613

C9:

Flags/Description: MWAIT 0x50

Latency: 480

Usage: 2015405

Duration: 3198208930

C10:

Flags/Description: MWAIT 0x60

Latency: 890

Usage: 511786893

Duration: 1546264384800

```
$ sudo cpupower monitor
```

```

      | Nehalem          || Mperf          || RAPL
      || Idle_Stats
CPU| C3  | C6  | PC3  | PC6  || C0  | Cx   | Freq || pack | dram | core | unco
   || POLL | C1  | C1E  | C3   | C6   | C7s  | C8   | C9   | C10
   0|  2,99| 11,92|  0,00|  0,00|| 70,98| 29,02|
1991||13733058|2706597|7438396|3080986||  0,05|  1,84|  5,01|  3,87| 14,05|  0,06|
3,81|  0,00|  0,04
   1|  3,58| 14,84|  0,00|  0,00|| 67,65| 32,35|
1991||13733058|2706597|7438396|3080986||  0,07|  1,87|  5,42|  4,46| 17,21|  0,36|
2,73|  0,00|  0,00
   2|  3,99|  7,15|  0,00|  0,00|| 73,25| 26,75|
1990||13733058|2706597|7438396|3080986||  0,09|  1,95|  8,76|  5,20|  9,44|  0,01|
1,12|  0,04|  0,00
   3|  3,86| 13,68|  0,00|  0,00|| 68,40| 31,60|
1990||13733058|2706597|7438396|3080986||  0,03|  2,52|  6,35|  4,92| 15,97|  0,00|
1,52|  0,00|  0,00

```

What should I do with all of this?

-
-

Does it match what you expect?

-
-
-

Control the CPU governors and CPU frequency

```
$ cpupower frequency-set -g performance
```

```
$ cpupower -c 0-7 frequency-set -g conservative
```

Powertop

```
$ sudo powertop
```

-
-
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-
-
-

Further reading

- [Intel Frequency scaling](#)¹²²³
- [Idle states](#)¹²²⁴
- [C-states](#)¹²²⁵

¹²²³ https://docs.kernel.org/admin-guide/pm/intel_uncore_frequency_scaling.html

¹²²⁴ <https://docs.kernel.org/admin-guide/pm/cpuidle.html>

¹²²⁵ https://docs.kernel.org/admin-guide/pm/intel_idle.html

TuneD

Note:

System tuning tools are either about better understanding the system's performance, or applying such knowledge to improve it. See our common *system tuning thoughts* (page 1078) for some generally applicable considerations about that.

TuneD¹²²⁶

-
-
-

Static vs. dynamic tuning

-
-

¹²²⁶ <https://tuned-project.org/>

Profiles

Environment	Default profile
Compute nodes	throughput-performance
Virtual Machines	virtual-guest
Containers	default
Other cases	balanced

Anatomy of a profile

Note:

In TuneD version 2.23, available in Ubuntu 24.04 LTS and earlier releases, the location of the profiles was different. An upgrade would have migrated any custom profiles, however for reference the paths changes were:

- Predefined profiles:
/usr/lib/tuned/<profile-name> -> /usr/lib/tuned/profiles/<profile-name>
- User defined profiles:
/etc/tuned/<profile-name> -> /etc/tuned/profiles/<profile-name>

```
[main]
include=PROFILE # if inheriting from another profile
summary=A short summary
description=A short description
```

```
[plugin instance]
type=<TYPE like cpu, scheduler, ...>
replace=<REPLACE = true | false>
enabled=<ENABLED = true | false>
devices=<DEVICES like /dev/sda, /dev/sdb>
```

```
[another plugin instance]
type=<TYPE like cpu, scheduler, ...>
replace=<REPLACE = true | false>
enabled=<ENABLED = true | false>
```

(continues on next page)

(continued from previous page)

```
devices=<DEVICES like /dev/sda, /dev/sdb>  
... other plugin-specific options ...  
  
...
```

-
-
-
-
-

*tuned.conf(5)*¹²²⁷

```
[fast_storage]  
type=disk  
devices=nvme0n*  
... options for these devices  
  
[slow_storage]  
type=disk  
devices=sda, sdb  
... options for these devices
```

Available profiles and plugins

```
$ tuned-adm list profiles
```

¹²²⁷ <https://manpages.ubuntu.com/manpages/resolute/man5/tuned.conf.5.html>

Available profiles:

- accelerator-performance - Throughput performance based tuning with disabled higher latency STOP states
- atomic-guest - Optimize virtual guests based on the Atomic variant
- atomic-host - Optimize bare metal systems running the Atomic variant
- (...)

-
-
-

```
cpu
  load_threshold
  latency_low
  latency_high
  force_latency
  governor
  sampling_down_factor
  energy_perf_bias
  min_perf_pct
  max_perf_pct
  no_turbo
  pm_qos_resume_latency_us
  energy_performance_preference
  boost
```

Customizing a profile

-

-

-

```
[main]
include=postgresql

[cpu]
load_threshold=0.5
latency_low=10
latency_high=10000
```

Merging profiles

```
$ sudo tuned-adm profile cpu-partitioning intel-sst
```

```
$ sudo tuned-adm profile virtual-guest powersave
```

Warning:

Just because tuned-adm accepted to merge two profiles doesn't mean it makes sense. There is no checking done on the resulting merged parameters, and the second profile could completely revert what the first profile adjusted.

An example profile: hpc-compute

```
#
# tuned configuration
#

[main]
summary=Optimize for HPC compute workloads
description=Configures virtual memory, CPU governors, and network settings for HPC compute
workloads.
include=latency-performance

[vm]
# Most HPC application can take advantage of hugepages. Force them to on.
transparent_hugepages=always

[disk]
# Increase the readahead value to support large, contiguous, files.
readahead=>4096

[sysctl]
# Keep a reasonable amount of memory free to support large mem requests
vm.min_free_kbytes=135168

# Most HPC applications are NUMA aware. Enabling zone reclaim ensures
# memory is reclaimed and reallocated from local pages. Disabling
# automatic NUMA balancing prevents unwanted memory unmapping.
vm.zone_reclaim_mode=1
kernel.numa_balancing=0

# Busy polling helps reduce latency in the network receive path
# by allowing socket layer code to poll the receive queue of a
# network device, and disabling network interrupts.
# busy_read value greater than 0 enables busy polling. Recommended
# net.core.busy_read value is 50.
# busy_poll value greater than 0 enables polling globally.
# Recommended net.core.busy_poll value is 50
net.core.busy_read=50
net.core.busy_poll=50

# TCP fast open reduces network latency by enabling data exchange
# during the sender's initial TCP SYN. The value 3 enables fast open
# on client and server connections.
net.ipv4.tcp_fastopen=3
```

- *HPC*
-
-

*1: This is a universe package Ubuntu ships this software as part of its [universe repository](#)¹²²⁸, which is maintained by volunteers from the Ubuntu community. Canonical also offers [Ubuntu Pro](#)¹²²⁹ – a free-for-personal-use subscription that provides a 10 year [security maintenance commitment](#)¹²³⁰.

¹²²⁸ [https://documentation.ubuntu.com/pro-client/en/latest/explanations/about_esm/](https://documentation.ubuntu.com/pro-client/en/latest/explanations/about_esm/#what-are-main-and-universe)

#what-are-main-and-universe

¹²²⁹ <https://ubuntu.com/pro>

¹²³⁰ <https://ubuntu.com/security/esm>

Further reading

- [TuneD website](#)¹²³¹
- *tuned-adm(8)*¹²³²
- *TuneD profiles*¹²³³
- *TuneD daemon*¹²³⁴
- *TuneD configuration*¹²³⁵

CPU affinity, NUMA placement, and isolation

Note:

System tuning tools are either about better understanding the system's performance, or applying such knowledge to improve it. See our common [system tuning thoughts](#) (page 1078) for some generally applicable considerations about that.

CPU

NUMA

¹²³¹ <https://tuned-project.org/>

¹²³² <https://manpages.ubuntu.com/manpages/resolute/man8/tuned-adm.8.html>

¹²³³ <https://manpages.ubuntu.com/manpages/resolute/man7/tuned-profiles.7.html>

¹²³⁴ <https://manpages.ubuntu.com/manpages/resolute/man8/tuned.8.html>

¹²³⁵ <https://manpages.ubuntu.com/manpages/resolute/man5/tuned.conf.5.html>

CPU and memory topology

*lscpu(1)*¹²³⁶

*numactl(8)*¹²³⁷

*tat(8)*¹²³⁹

*taskset(1)*¹²³⁸

numas-

```
$ lscpu -e=CPU,NODE,SOCKET,CORE,ONLINE
```

CPU	NODE	SOCKET	CORE	ONLINE
0	0	0	0	yes
1	0	0	1	yes
2	1	0	2	yes
3	1	0	3	yes

```
$ numactl --hardware
```

```
available: 2 nodes (0-1)
node 0 cpus: 0 1
node 0 size: 1930 MB
node 0 free: 1614 MB
node 1 cpus: 2 3
node 1 size: 1965 MB
node 1 free: 1710 MB
node distances:
node      0      1
  0:    10    20
  1:    20    10
```

¹²³⁶ <https://manpages.ubuntu.com/manpages/resolute/man1/lscpu.1.html>

¹²³⁷ <https://manpages.ubuntu.com/manpages/resolute/man8/numactl.8.html>

¹²³⁸ <https://manpages.ubuntu.com/manpages/resolute/man1/taskset.1.html>

¹²³⁹ <https://manpages.ubuntu.com/manpages/resolute/man8/numastat.8.html>

hardware

topology with hwloc

CPU affinity

```
$ taskset -c 2-3 ./workload
```

```
$ taskset --all-tasks --pid --cpu-list 1864
```

```
pid 1864's current affinity list: 0,1  
pid 1866's current affinity list: 0,1  
pid 1867's current affinity list: 0,1  
pid 1868's current affinity list: 0,1
```

```
$ taskset --all-tasks --pid --cpu-list 2-3 1864
```

```
pid 1864's current affinity list: 0,1
pid 1864's new affinity list: 2,3
pid 1866's current affinity list: 0,1
pid 1866's new affinity list: 2,3
pid 1867's current affinity list: 0,1
pid 1867's new affinity list: 2,3
pid 1868's current affinity list: 0,1
pid 1868's new affinity list: 2,3
```

*nice(1)*¹²⁴⁰ *renice(1)*¹²⁴¹

NUMA memory policies

page fault

¹²⁴⁰ <https://manpages.ubuntu.com/manpages/resolute/man1/nice.1.html>

¹²⁴¹ <https://manpages.ubuntu.com/manpages/resolute/man1/renice.1.html>

Option	Allocation behavior	Trade-off
<code>--localalloc</code>	Prefer the node of the CPU making the allocation, with fallback to other nodes.	Locality follows the allocating thread, which may move between nodes.
<code>--preferred=0</code>	Prefer node 0, with fallback to other nodes.	Keeps a preferred location without making it a strict requirement.
<code>--mbind=0</code>	Allocate only from node 0.	Allocation can fail when that node lacks memory, even if other nodes have free memory.
<code>--interleave=all</code>	Distribute allocations round-robin across the allowed nodes, with fallback under memory pressure.	Can spread memory-bandwidth demand but increases remote accesses for threads running on one node.

```
$ numactl --cpunodebind=0 --preferred=0 -- ./workload
```

```
$ numactl --physcpubind=2-3 --mbind=1 -- ./workload
```

*numa(3)*¹²⁴²

```
$ numastat -p 1864
```

```
Per-node process memory usage (in MBs) for PID 1864 (workload)
      Node 0      Node 1      Total
-----
Huge          0.00          0.00          0.00
Heap          0.00          0.00          0.00
Stack         0.02          0.00          0.02
Private       578.89        190.78        769.67
-----
Total         578.91        190.78        769.69
```

¹²⁴² <https://manpages.ubuntu.com/manpages/resolute/man3/numa.3.html>

Placement for systemd services

*exec(5)*¹²⁴³

systemd.

```
[Service]
CPUAffinity=2-3
NUMAPolicy=preferred
NUMAMask=1
```

```
$ systemctl show -p CPUAffinity -p NUMAPolicy -p NUMAMask workload.service

CPUAffinity=2-3
NUMAPolicy=preferred
NUMAMask=1
```

*control(5)*¹²⁴⁴

systemd.resource-

CPU isolation

```
isolcpus=domain,2-3
```

¹²⁴³ <https://manpages.ubuntu.com/manpages/resolute/man5/systemd.exec.5.html>

¹²⁴⁴ <https://manpages.ubuntu.com/manpages/resolute/man5/systemd.resource-control.5.html>


```
$ cat /proc/cmdline
```

```
BOOT_IMAGE=/vmlinuz-7.0.0-30-generic root=UUID=90d99162-58b5-439f-b459-a3d205abce11  
ro console=tty1 console=ttyS0 isolcpus=domain,2-3
```

```
$ cat /sys/devices/system/cpu/isolated
```

```
2-3
```

*irqbalance(1)*¹²⁴⁵

IRQ

Warning:

Reserve enough housekeeping CPUs for operating-system tasks, device drivers, and SSH access. Remember SMT siblings: an unconstrained task running on a sibling thread will still contend for execution pipelines on the same core. Always verify boot isolation on a test host with remote console access before modifying production bootloaders.

Placement for virtual machines and packet processing

Libvirt

vCPU

¹²⁴⁵ <https://manpages.ubuntu.com/manpages/resolute/man1/irqbalance.1.html>

Domain setting	Purpose
cputune/vcpupin	Map each guest vCPU to an allowed set of host logical CPUs.
cputune/emulatorpin	Place emulator threads separately from vCPU threads.
cputune/iothreadpin	Place configured input/output (I/O) threads.
numatune/memory	Set the host NUMA memory policy and node set for the guest.
cpu/numa	Describe the NUMA topology visible inside the guest.

Libvirt

Checking the effect

*pidstat(1)*¹²⁴⁶

```
$ pidstat -t -u -w -p 1864 1 1
```

Linux 7.0.0-30-generic (numa2)				09/15/26		_x86_64_		(4 CPU)	
18:49:51	UID	TGID	TID	%usr	%system	%guest	%wait	%CPU	CPU
Command									
18:49:52	0	1864	-	200.00	0.00	0.00	0.00	200.00	1
workload									
18:49:52	0	-	1864	0.00	0.00	0.00	0.00	0.00	1
__workload									
18:49:52	0	-	1866	69.00	0.00	0.00	31.00	69.00	2
__workload									
18:49:52	0	-	1867	65.00	0.00	0.00	36.00	65.00	2
__workload									
18:49:52	0	-	1868	68.00	0.00	0.00	33.00	68.00	3
__workload									
18:49:51	UID	TGID	TID	cswch/s	nvcswh/s	Command			
18:49:52	0	1864	-	0.00	0.00	workload			
18:49:52	0	-	1864	0.00	0.00	__workload			
18:49:52	0	-	1866	0.00	115.00	__workload			
18:49:52	0	-	1867	0.00	129.00	__workload			
18:49:52	0	-	1868	0.00	115.00	__workload			

¹²⁴⁶ <https://manpages.ubuntu.com/manpages/resolute/man1/pidstat.1.html>

Further reading

- [`sched\_setaffinity\(2\)`](#)¹²⁴⁷
- [`numa\(7\)`](#)¹²⁴⁸ [`set\_mempolicy\(2\)`](#)¹²⁴⁹
- [Linux kernel CPU isolation documentation](#)¹²⁵⁰
- [Linux kernel parameters](#)¹²⁵¹
- [Control group v2 cpuset controller](#)¹²⁵²
- [Libvirt CPU tuning](#)¹²⁵³ [NUMA node tuning](#)¹²⁵⁴

Docs¹²⁵⁶

hosted in GitHub¹²⁵⁵

rendered on Read the

options

that don't require any coding

please let us know

Canonical Open Documentation Academy¹²⁵⁷

¹²⁴⁷ https://manpages.ubuntu.com/manpages/resolute/man2/sched_setaffinity.2.html

¹²⁴⁸ <https://manpages.ubuntu.com/manpages/resolute/man7/numa.7.html>

¹²⁴⁹ https://manpages.ubuntu.com/manpages/resolute/man2/set_mempolicy.2.html

¹²⁵⁰ <https://www.kernel.org/doc/html/latest/admin-guide/cpu-isolation.html>

¹²⁵¹ <https://www.kernel.org/doc/html/latest/admin-guide/kernel-parameters.html>

¹²⁵² <https://www.kernel.org/doc/html/latest/admin-guide/cgroup-v2.html#cpuset>

¹²⁵³ <https://libvirt.org/formatdomain.html#cpu-tuning>

¹²⁵⁴ <https://libvirt.org/formatdomain.html#numa-node-tuning>

¹²⁵⁵ <https://github.com/canonical/ubuntu-server-documentation>

¹²⁵⁶ <https://ubuntu.com/server/docs/>

¹²⁵⁷ <https://github.com/canonical/open-documentation-academy>

directly in the GitHub web interface

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Local development

The outline of proposed work has not already been agreed by a maintainer.

follow the outlined procedures

The pull request far exceeds the scope of what was agreed.

The contribution makes changes without value.

We deem the contribution to be AI generated.

The issue the pull request is addressing was assigned to someone else.

Acceptable use of AI

-
-
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1.

2.

3.

4.

AI tooling

our contributors

Types of contributions

issues list¹²⁵⁸

Technical

-
-

Find technical issues to work on¹²⁵⁹

Coding options

-
-
-
-

using Mermaid¹²⁶⁰

Find coding issues to work on¹²⁶¹

¹²⁵⁸ <https://github.com/canonical/ubuntu-server-documentation/issues>

¹²⁵⁹ <https://github.com/canonical/ubuntu-server-documentation/issues?q=is%3Aissue+is%3Aopen+label%3A%22code%3A+technical%22+no%3Aassignee>

¹²⁶⁰ <https://mermaid.live/>

¹²⁶¹ <https://github.com/canonical/ubuntu-server-documentation/issues?q=is%3Aissue+is%3Aopen+label%3A%22code%3A+coding%22+no%3Aassignee>

Low-code options

-
-
-
-
-

online generator¹²⁶²

Find low-code issues to work on¹²⁶³

No-code options

Creating issues

User-experience testing

¹²⁶² <https://www.mermaidchart.com/app/dashboard>

¹²⁶³ <https://github.com/canonical/ubuntu-server-documentation/issues?q=is%3Aissue+is%3Aopen+label%3A%22code%3A+low-code%22+no%3Aassignee>

Find no-code issues to work on¹²⁶⁴

Local development

Prerequisites

Code of Conduct

Code of Conduct¹²⁶⁵

GitHub account

GitHub account¹²⁶⁶

Workflow overview

- *Find an issue to work on*
- *Pick your workflow*
- *get help*
- *Test your contribution*
- *Submit it for review*

Find issues to work on

checking our issues list¹²⁶⁷

use this link¹²⁶⁸

¹²⁶⁴ <https://github.com/canonical/ubuntu-server-documentation/issues?q=is%3Aissue+is%3Aopen+label%3A%22code%3A+non-code%22+no%3Aassignee>

¹²⁶⁵ <https://ubuntu.com/community/docs/ethos/code-of-conduct>

¹²⁶⁶ <https://github.com/>

¹²⁶⁷ <https://github.com/canonical/ubuntu-server-documentation/issues>

¹²⁶⁸ <https://github.com/canonical/ubuntu-server-documentation/issues?q=is%3Aissue+is%3Aopen+no%3Aassignee>

Claim an issue

Create a new issue

-
-
-

Choose your workflow

-
-

quickstart guide

Local setup

Small changes directly in GitHub

GitHub's web interface¹²⁶⁹

1. in the documentation¹²⁷⁰
- 2.
- 3.
- 4.
- 5.
- 6.

¹²⁶⁹ <https://docs.github.com/en/repositories/working-with-files/managing-files/editing-files>

¹²⁷⁰ <https://ubuntu.com/server/docs>

Local setup

Create a fork

Ubuntu Server GitHub repository¹²⁷¹

with git¹²⁷²

working

Get the documentation

```
$ git clone git@github.com:canonical/ubuntu-server-documentation.git
```

```
$ cd ubuntu-server-documentation
```

Create a new branch

```
$ git pull
```

Note:

This won't have any effect if you have only just cloned the repository (since you're already up-to-date).

```
$ git checkout -b my-new-branch
```

¹²⁷¹ <https://github.com/canonical/ubuntu-server-documentation>

¹²⁷² https://documentation.academy/docs/howto/get-started/using_git/

Build and preview

```
$ make run
```

<http://127.0.0.1:8000/>

```
$ export SPHINX_HOST=0.0.0.0
```

Note:

If you have problems getting the documentation to run on your machine, reach out to the team or leave a comment on your issue to get additional support.

Other build commands

-
-
-

Writing guidance

guidance for writing

Test your changes

Automatic testing

```
$ make spelling > make linkcheck
```

Manual testing

Creating redirects

Internal redirects

```
redirect/path/from/ redirect/path/to/
```

Sphinx Rediraffe docs¹²⁷³

¹²⁷³ <https://sphinxext-rediraffe.readthedocs.io/en/latest/>

External redirects

```
redirects = {  
  "example/source": "https://exampletarget.org",  
  "how-to/containers/lxc-containers": "https://linuxcontainers.org/lxc/documentation/",  
}
```

<https://ubuntu.com/server/docs>

Submitting your pull request

Commit your changes

-
-
-

Note:

Try to group your changes logically. For example, if you have one set of changes that modifies spelling in 10 files, and another set of changes that modifies formatting in 10 different files, you can group them into two commits (one for spelling, and one for formatting). You don't need a separate commit for every file.

Push and create a pull request

```
$ git push <your username> <branch name>
```

[create a pull request](#)¹²⁷⁴

¹²⁷⁴ <https://docs.github.com/en/pull-requests/collaborating-with-pull-requests/proposing-changes-to-your-work-with-pull-requests/creating-a-pull-request>

Read the Docs preview

Reviews

Note:

The team has adopted the [Conventional Comments](https://conventionalcomments.org/)¹²⁷⁵ approach with the intention of making feedback easier to parse.

¹²⁷⁵ <https://conventionalcomments.org/>

Guidance for writing

Files and structure

support¹²⁷⁶

MyST

Diátaxis¹²⁷⁷

¹²⁷⁶ <https://myst-parser.readthedocs.io/en/latest/intro.html>

¹²⁷⁷ <https://diataxis.fr/>

Tip:

For example, if a page appears in the How-to section about Virtualisation, the landing page you need will be `how-to/virtualisation.md`.

Versioning policy

Versioning by admonition

```
 :::{note}
For Ubuntu 24.04 LTS (Noble) onwards, the recommended method for ...
 :::
```

Versioning by tabs

helpful demonstration¹²⁷⁸

the QEMU page

¹²⁷⁸ <https://mystmd.org/guide/dropdowns-cards-and-tabs#tabs>

Style guide

Canonical documentation style guide¹²⁷⁹

Language

-
-

Acronyms

Links

¹²⁷⁹ <https://docs.ubuntu.com/styleguide/en>

Writing style

Markdown elements

Sections and headings

```
# Heading level 1
## Heading level 2
### Heading level 3
#### Heading level 4
```

Semantic markup

MyST

Roles¹²⁸⁰

```
Keyboard keys: {kbd}`Ctrl`, {kbd}`C`
```

```
Manpages: {manpage}`package-name(section)` (e.g. {manpage}`dpkg(1)`)
```

*dpkg(1)*¹²⁸¹

```
Matrix: {matrix}`channel-name` (e.g. {matrix}`server`)
```

#server:ubuntu.com¹²⁸²

```
Launchpad source: {lpsrc}`package-name` (e.g. {lpsrc}`bash`)
```

bash¹²⁸³

¹²⁸⁰ <https://myst-parser.readthedocs.io/en/latest/syntax/roles-and-directives.html>

¹²⁸¹ <https://manpages.ubuntu.com/manpages/resolute/man1/dpkg.1.html>

¹²⁸² <https://matrix.to/#/#server:ubuntu.com>

¹²⁸³ <https://launchpad.net/ubuntu/+source/bash>

Launchpad bug: {lpbug}`bug-number` (e.g. {lpbug}`1852389`)

LP: #1852389¹²⁸⁴

Lists

```
1. This is the first item
1. This is the second
1. This is the third
```

The {terminal} directive

```
```{terminal}
:copy:
:user:
:host:
:dir:
command

output
...`
```

```
$ command
```

```
output
```

- 
- 
- 
- 

---

<sup>1284</sup> <https://bugs.launchpad.net/bugs/1852389>

```
'''{terminal}
:copy:
:user: username
:host: ubuntu
:dir: ~/src
command
```

```
output
'''
```

```
username@ubuntu:~/src$ command
```

```
output
```

```
'''{terminal}
:copy:
:user: username
:host: ubuntu
:dir: ~/src
command
'''
```

Some additional info for the reader.

```
'''{terminal}
:output-only:
The output
'''
```

```
username@ubuntu:~/src$ command
```

```
The output
```

## Code blocks

```
```yaml
Some code block here
```
```

## Admonition blocks

allowed by the MyST syntax<sup>1285</sup>

## AI tools in this project

*AI policy*

## Where the tooling lives

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- 
- 
- 

---

<sup>1285</sup> <https://myst-parser.readthedocs.io/en/stable/syntax/admonitions.html>

## Prompts you can invoke

`new-page`

**What it does**

`review-page`

**What it does**

`capture-learnings`

**What it does**

## Instructions that guide the AI

- 

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**Getting help**

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<sup>1286</sup> <https://discourse.ubuntu.com/c/project/server/17>

<sup>1287</sup> <https://discourse.ubuntu.com/c/community/open-documentation-academy/166>

## Chat to us

- To chat to the Server team<sup>1288</sup>
- To talk to technical authors about documentation<sup>1289</sup>

## Our contributors

## Content updates

| PR number               | Contributed by      | Contribution made                                                                               |
|-------------------------|---------------------|-------------------------------------------------------------------------------------------------|
| PR #202 <sup>1290</sup> | network-charles     | Updated instructions for building docs locally                                                  |
| PR #200 <sup>1291</sup> | Zetway-Technologies | Clarified instructions in <code>install-lxc-dhcp-server.md</code>                               |
| PR #192 <sup>1292</sup> | minaelee            | Updated CODA links in our contributing guide                                                    |
| PR #190 <sup>1293</sup> | kenyon              | Removed no-longer-needed reference to a recently closed bug                                     |
| PR #181 <sup>1294</sup> | davidekete          | Deprecated the <code>backup-with-shell-scripts.md</code> page                                   |
| PR #130 <sup>1295</sup> | Sophie-Pages        | Deprecated the <code>lxc-containers.md</code> page                                              |
| PR #119 <sup>1296</sup> | Sophie-Pages        | Added introduction to <code>install-openldap.md</code> page                                     |
| PR #115 <sup>1297</sup> | minaelee            | Updated and refreshed content on <code>lxd-containers.md</code> page                            |
| PR #113 <sup>1298</sup> | teward              | Updated <code>apparmor.md</code> to explain how to totally enable/disable AppArmor              |
| PR #54 <sup>1299</sup>  | khbecker            | Added details about <code>kdump</code> enablement in 24.10 to <code>kernel-crash-dump.md</code> |

<sup>1288</sup> <https://matrix.to/#/#server:ubuntu.com>

<sup>1289</sup> <https://matrix.to/#/#documentation:ubuntu.com>

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<sup>1290</sup> <https://github.com/canonical/ubuntu-server-documentation/pull/202>  
<sup>1291</sup> <https://github.com/canonical/ubuntu-server-documentation/pull/200>  
<sup>1292</sup> <https://github.com/canonical/ubuntu-server-documentation/pull/192>  
<sup>1293</sup> <https://github.com/canonical/ubuntu-server-documentation/pull/190>  
<sup>1294</sup> <https://github.com/canonical/ubuntu-server-documentation/pull/181>  
<sup>1295</sup> <https://github.com/canonical/ubuntu-server-documentation/pull/130>  
<sup>1296</sup> <https://github.com/canonical/ubuntu-server-documentation/pull/119>  
<sup>1297</sup> <https://github.com/canonical/ubuntu-server-documentation/pull/115>  
<sup>1298</sup> <https://github.com/canonical/ubuntu-server-documentation/pull/130>  
<sup>1299</sup> <https://github.com/canonical/ubuntu-server-documentation/pull/54>



## Copyediting and proofreading

| PR number              | Contributed by  | Contribution made                                                                                                     |
|------------------------|-----------------|-----------------------------------------------------------------------------------------------------------------------|
| PR #184 <sup>130</sup> | thorn-shadow99  | Improved the sssd.md page                                                                                             |
| PR #183 <sup>130</sup> | thorn-shadow99  | Improved the security.md page                                                                                         |
| PR #167 <sup>130</sup> | POS-ToakDEV     | Improved the libvirt.md page                                                                                          |
| PR #155 <sup>130</sup> | thorn-shadow99  | Improved the multipath.md page                                                                                        |
| PR #149 <sup>130</sup> | thorn-shadow99  | Improved the high-availability.md page                                                                                |
| PR #148 <sup>130</sup> | thorn-shadow99  | Improved the ebpf.md page                                                                                             |
| PR #147 <sup>130</sup> | bird-stare      | Improved the introduction-to-backups.md, introduction-to-mail-services.md and introduction-to-virtualisation.md pages |
| PR #146 <sup>130</sup> | thorn-shadow99  | Improved the crypto-libraries.md page                                                                                 |
| PR #141 <sup>130</sup> | gregbell        | Improved the qemu-microvm.md page                                                                                     |
| PR #132 <sup>130</sup> | bird-stare      | Substantial copyediting on the networking.md page                                                                     |
| PR #131 <sup>131</sup> | network-charles | Improved the ebpf.md page                                                                                             |
| PR #123 <sup>131</sup> | davideket       | Converted "note" blocks into MyST admonitions (explanation section)                                                   |
| PR #122 <sup>131</sup> | YanisaHS        | Improved the install-dns.md page                                                                                      |
| PR #120 <sup>131</sup> | davideket       | Converted "note" blocks into MyST admonitions (reference section)                                                     |
| PR #83 <sup>1314</sup> | daveedoo        | Improved the serve-ntp-with-chrony.md page                                                                            |
| PR #81 <sup>1315</sup> | peat-psuwit     | Improved the report-a-bug.md page                                                                                     |
| PR #67 <sup>1316</sup> | pnhuy           | Improved the configur-nginx.md page                                                                                   |

## Error fixes

| PR number                              | Contributed by | Contribution made                                                         |
|----------------------------------------|----------------|---------------------------------------------------------------------------|
| <a href="#">PR #97</a> <sup>1317</sup> | yukihane       | Corrected an error in <code>install-openvpn.md</code>                     |
| <a href="#">PR #66</a> <sup>1318</sup> | Pexers         | Removed duplicate text block in <code>peer-to-site-on-router.md</code>    |
| <a href="#">PR #52</a> <sup>1319</sup> | sbworth        | Corrected an error in <code>WireGuard VPN on-an-internal-system.md</code> |
| <a href="#">PR #48</a> <sup>1320</sup> | nikolas        | Corrected an error in <code>install-postfix.md</code>                     |

## Functionality and enhancements

| PR number                               | Contributed by  | Contribution made                                                 |
|-----------------------------------------|-----------------|-------------------------------------------------------------------|
| <a href="#">PR #193</a> <sup>1321</sup> | network-charles | Added hover-over referencing functionality                        |
| <a href="#">PR #116</a> <sup>1322</sup> | activus-d       | Created an issue template to help readers provide better feedback |
| <a href="#">PR #112</a> <sup>1323</sup> | goodylili       | Created a pull request template                                   |

<sup>1300</sup> <https://github.com/canonical/ubuntu-server-documentation/pull/184>

<sup>1301</sup> <https://github.com/canonical/ubuntu-server-documentation/pull/183>

<sup>1302</sup> <https://github.com/canonical/ubuntu-server-documentation/pull/167>

<sup>1303</sup> <https://github.com/canonical/ubuntu-server-documentation/pull/155>

<sup>1304</sup> <https://github.com/canonical/ubuntu-server-documentation/pull/149>

<sup>1305</sup> <https://github.com/canonical/ubuntu-server-documentation/pull/148>

<sup>1306</sup> <https://github.com/canonical/ubuntu-server-documentation/pull/147>

<sup>1307</sup> <https://github.com/canonical/ubuntu-server-documentation/pull/146>

<sup>1308</sup> <https://github.com/canonical/ubuntu-server-documentation/pull/141>

<sup>1309</sup> <https://github.com/canonical/ubuntu-server-documentation/pull/132>

<sup>1310</sup> <https://github.com/canonical/ubuntu-server-documentation/pull/131>

<sup>1311</sup> <https://github.com/canonical/ubuntu-server-documentation/pull/123>

<sup>1312</sup> <https://github.com/canonical/ubuntu-server-documentation/pull/122>

<sup>1313</sup> <https://github.com/canonical/ubuntu-server-documentation/pull/120>

<sup>1314</sup> <https://github.com/canonical/ubuntu-server-documentation/pull/83>

<sup>1315</sup> <https://github.com/canonical/ubuntu-server-documentation/pull/81>

<sup>1316</sup> <https://github.com/canonical/ubuntu-server-documentation/pull/67>

<sup>1317</sup> <https://github.com/canonical/ubuntu-server-documentation/pull/97>

<sup>1318</sup> <https://github.com/canonical/ubuntu-server-documentation/pull/66>

<sup>1319</sup> <https://github.com/canonical/ubuntu-server-documentation/pull/52>

<sup>1320</sup> <https://github.com/canonical/ubuntu-server-documentation/pull/48>

## Glossary

| PR number               | Contributed by  | Contribution made                         |
|-------------------------|-----------------|-------------------------------------------|
| PR #194 <sup>1324</sup> | network-charles | Contributed terms to the glossary         |
| PR #187 <sup>1325</sup> | network-charles | Contributed terms to the glossary         |
| PR #186 <sup>1326</sup> | network-charles | Contributed terms to the glossary         |
| PR #185 <sup>1327</sup> | network-charles | Contributed terms to the glossary         |
| PR #182 <sup>1328</sup> | network-charles | Contributed terms to the glossary         |
| PR #174 <sup>1329</sup> | network-charles | Contributed terms to the glossary         |
| PR #162 <sup>1330</sup> | network-charles | Contributed terms to the glossary         |
| PR #161 <sup>1331</sup> | kaushikjayant   | Contributed terms to the glossary         |
| PR #160 <sup>1332</sup> | network-charles | Contributed terms to the glossary         |
| PR #159 <sup>1333</sup> | network-charles | Contributed terms to the glossary         |
| PR #158 <sup>1334</sup> | network-charles | Contributed terms to the glossary         |
| PR #156 <sup>1335</sup> | network-charles | Contributed terms to the glossary         |
| PR #152 <sup>1336</sup> | Ibom99          | Contributed "J" terms to the glossary     |
| PR #134 <sup>1337</sup> | RomainDust      | Contributed terms to the glossary         |
| PR #124 <sup>1338</sup> | activus-d       | Added the glossary page and functionality |

<sup>1321</sup> <https://github.com/canonical/ubuntu-server-documentation/pull/193>

<sup>1322</sup> <https://github.com/canonical/ubuntu-server-documentation/pull/116>

<sup>1323</sup> <https://github.com/canonical/ubuntu-server-documentation/pull/112>

## PDF improvements

| PR number                     | Contributed by | Contribution made                          |
|-------------------------------|----------------|--------------------------------------------|
| <b>PR #10</b> <sup>1339</sup> | SecondSkoll    | Added the ability to build the PDF locally |
| <b>PR #9</b> <sup>1340</sup>  | SecondSkoll    | Fixed our PDF build issues                 |

<sup>1324</sup> <https://github.com/canonical/ubuntu-server-documentation/pull/194>  
<sup>1325</sup> <https://github.com/canonical/ubuntu-server-documentation/pull/187>  
<sup>1326</sup> <https://github.com/canonical/ubuntu-server-documentation/pull/186>  
<sup>1327</sup> <https://github.com/canonical/ubuntu-server-documentation/pull/185>  
<sup>1328</sup> <https://github.com/canonical/ubuntu-server-documentation/pull/182>  
<sup>1329</sup> <https://github.com/canonical/ubuntu-server-documentation/pull/174>  
<sup>1330</sup> <https://github.com/canonical/ubuntu-server-documentation/pull/162>  
<sup>1331</sup> <https://github.com/canonical/ubuntu-server-documentation/pull/161>  
<sup>1332</sup> <https://github.com/canonical/ubuntu-server-documentation/pull/160>  
<sup>1333</sup> <https://github.com/canonical/ubuntu-server-documentation/pull/159>  
<sup>1334</sup> <https://github.com/canonical/ubuntu-server-documentation/pull/158>  
<sup>1335</sup> <https://github.com/canonical/ubuntu-server-documentation/pull/156>  
<sup>1336</sup> <https://github.com/canonical/ubuntu-server-documentation/pull/152>  
<sup>1337</sup> <https://github.com/canonical/ubuntu-server-documentation/pull/134>  
<sup>1338</sup> <https://github.com/canonical/ubuntu-server-documentation/pull/124>  
<sup>1339</sup> <https://github.com/canonical/ubuntu-server-documentation/pull/10>  
<sup>1340</sup> <https://github.com/canonical/ubuntu-server-documentation/pull/9>